An Introduction to Object-Oriented Programming in C++
with Applications in Computer Graphics

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Graham M. Seed

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with Applications in Computer Graphics

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To the memory of
Tom Bannister (1909–1992)
what a guy!

Preface

Why Another Book on C++ and why Programming *and* Graphics?

Anyone who has browsed through the 'Computing' section of a bookshop (assuming it has one) will not need much convincing that there are a lot of C++ books out there. So why add yet another to the shelf?

This book attempts to introduce you to the C++ language via computer graphics because the object-oriented programming features of C++ naturally lend themselves to graphics. Thus, this book is based around a central theme: computer graphics and the development of 'real' object-oriented tools for graphical modelling. This approach is adopted (as opposed to learning by small, unrelated, often hypothetical, examples) because I didn't want to introduce C++ as a collection of language features. While introducing the syntax and features of C++, it is just as important to demonstrate simultaneously the reason for such features and when to apply them – in other words, language and design are given equal priority. Also, a key objective in writing this book is to present you with a comprehensive introductory text on programming in the C++ language.

The first part of the book introduces the fundamentals of C++ programming. After the foundations have been laid, the object-oriented programming features of C++ are introduced via the development of classes such as `Point` and `Line`. These *primitives* are then put to good use in developing powerful two- and three-dimensional graphics classes such as `Circle`, `Triangle`, `Quadrilateral`, `Polygon` and `Tetrahedra`. Along the way, *tool classes* are also developed, such as `Vector`, `Vector3D`, `Matrix`, `Complex`, `String`, `LinkedList`, `ReadAndWriteFile`, `World`, `DIBitmap` and many more.

What you Need to Know

No previous programming experience in C or C++ is required for this book. If you are considering learning C before C++, then my advice is: don't. When I first started to learn C++ I asked myself the exact same question and unfortunately made the wrong decision and developed a good understanding of C before attempting to learn C++. I wouldn't say that this proved to be a complete waste of time, but it is certainly not necessary to learn C in order to learn C++.

As a C++ programmer you will develop programs and think about program design in a completely different manner than the traditional procedural techniques in C. C++ offers a programmer object-oriented features such as encapsulation, abstraction, inheritance and multiple inheritance, and polymorphism through the use of classes and objects, member functions, operator overloading, virtual functions, templates, exception handling, run-time type information and so on. In addition, C++ offers a lot more. C++ provides many new features (for instance **`const`**, **`new`** and

delete), stronger type checking and minor, but nevertheless important, modifications to C.

If you do know another procedural language, such as Fortran, Basic, Pascal or C, then great. Some skills that you have learned with such languages will undoubtedly prove useful (such as the meaning of variables, conditions and loops), whereas other skills will appear to conflict with the C++ way of programming.

This book does assume that you have had some contact with computers before and are familiar with either Microsoft's Windows or GNU/Linux operating systems.

Hardware and Software Requirements

It is assumed that readers of this book have access to a computer and a compiler, such as Borland C++, Microsoft Visual C++, Symantec C++ or GNU/Linux C++ for a Windows environment. It is difficult to specify general minimum hardware and software requirements that cover all compilers, but as an example, Borland C++ version 5.5 for Windows requires Windows 95/98/NT, a hard disk with 175 Mbyte free disk space for a full installation, at least 16 Mbyte RAM, a 3.5"/1.44 Mbyte floppy drive or CD-ROM and a Windows-compatible mouse. An Intel Pentium or higher processor is recommended.

Obtaining the Program Files

All of the source code, including solutions to chapter exercises, presented throughout the book can be downloaded from the author's web page:

```
http://www.hw.ac.uk/mecWWW/research/staff/gms.htm
```

The program files have been compressed in to a single PKZIP file CPP_PROG.ZIP. To preserve the chapter directory structure of the files use the `-d` option when unzipping the files. For instance:

```
c:\cpp_prog>pkunzip cpp_prog.zip -d
```

All of the programs have been tested with the Borland C++ BCC32 (version 5.5) compiler running on Microsoft Windows NT (version 4) and the GNU C++ g++ (version 2.91) compiler running on Red Hat Linux (version 6.2). Two versions of program files are to found: .cpp and .cc file extensions corresponding to the Borland and GNU compilers respectively.

Preparation of the Book

The text of this book was written with the aid of Microsoft's Word for Windows 2000. The C++ program code was written with the Borland Integrated Development Environment (IDE) editor. The program code was transferred from IDE to Word using

Copy and *Paste*. Tables and figures were completed using Word and CorelDRAW version 7.0. The final typesetting of the book was performed using Corel Ventura version 8.0.

All program code developed for the book was developed exclusively on a desktop PC consisting of an Intel Pentium Pro 200 MHz CPU, 256 Mbyte RAM, 8 Mbyte video memory and 10 Gbyte hard disk space.

How this Book is Organised

It would be great to dive straight into the object-oriented programming (OOP) features of C++. Unfortunately, this is simply not possible for an introductory text. Therefore, the initial chapters lay the foundations of the C++ language by examining the 'traditional' procedural C features of C++.

If you already have experience of another programming language, particularly C, it may be tempting to skip the first few chapters and jump straight into the object-oriented features of C++. If you choose to do this then be careful. C++ has many differences from and additions to C, some of which you may miss if you jump for OOP. The object-oriented features *really* begin in Chapters 8 and 9, when structures and classes are introduced.

The book begins by presenting a general overview of the C++ programming language and object-oriented programming and what is meant by *an object*. We take a look at the history of C++, its relationship to C and what makes C++ an object-oriented programming language. Chapter 2 briefly discusses your programming environment and what support to expect from a C++ compiler.

Chapter 3 introduces several key elements of program construction in C++ via the simplest of programs. Each line of a simple C++ program is introduced in detail to help the reader who is learning C++ for the first time feel comfortable with the basics of program construction. The first elements of a C++ program discussed are user comments, preprocessor directives, the use of header files, a function definition, program statements, strings, the `cout` standard output stream object, the `endl` manipulator and the `<<` insertion operator. The importance of developing a consistent programming style is stressed and several different popular styles are presented.

The fundamental data types of the C++ language, and their corresponding data type modifiers, are introduced in Chapter 4. The notion of a constant or variable identifier is demonstrated with the aid of numerous examples for each fundamental data type, and the distinction between an identifier declaration and definition is noted. Once data types have been introduced, statements, operators, expressions, type conversions and casting are covered. Chapter 4 concludes by discussing storage class specifiers and declaring assembly language program code directly within a C++ program.

Chapter 5 examines the process of making decisions and performing repetition in C++. Decisions are implemented using either **`if-else`** or **`switch`** statements, whereas repetition is performed using a **`for`**, **`while`** or **`do-while`**-loop. In addition, C++ supports keywords such as **`break`** and **`default`** for **`if-else`** and **`switch`** statements and the keywords **`break`** and **`continue`** for loops which allow a programmer to interrupt program flow. Both decision-making and repetition require the use of the relational and logical operators, and Chapter 5 examines these operators and the bitwise and shift operators in detail.

The function is the basic tool of the majority of programming languages for modularising a large complex program. Chapter 6 discusses the C++ function: its construction, definition, scope and application. Chapter 6 also discusses the powerful technique of function overloading, which C++ supports, and **inline** functions. The chapter concludes with an application of functions to the popular geometric problem of the intersection of two straight lines. Arrays are covered in Chapter 7. The chapter discusses what an array is, how the elements of an array are indexed, the dimension of an array and typical applications of arrays.

Chapter 8 presents structures, unions and enumerations for the creation of user-defined data types and typedefs for user-defined synonyms of data types. The **struct** statement allows a programmer to create a new data type which is a collection of similar or different data types. A **union** data type is similar to **struct** except that a **union** can hold only one of its data members at any given time. The user-defined **enum** data type is a collection of named data members which have equivalent integer values. A **typedef** is a synonym for an existing data type.

The **class** is the key object-oriented programming feature of the C++ language, and Chapter 9 covers the **class** in detail via a **class** called `Point`. Similar to **struct**, the **class** allows a programmer to create new data types by encapsulating data members, but the **class** also allows a programmer to encapsulate member functions which operate on a **class**'s data members. Operators and overloading operators for user-defined data types are examined in Chapter 10. Overloading operators for a user-defined **class** can significantly enhance the level of abstraction of the **class**.

Friend functions and classes are introduced in Chapter 11. Basically, a **friend** function or **class** has access to the **private** data members of an object of a given **class**, although the **friend** is not a member of that **class**.

Chapter 12 comprehensively examines the tricky subject of pointers. The dynamic memory allocation and deallocation operators **new** and **delete** and the **this** pointer are examined. Useful `Vector` and `Matrix` classes are developed which make good use of pointer manipulation and the **new** and **delete** operators, and help apply many of the topics covered in previous chapters.

Generic functions and parametrised data types are introduced in Chapter 13 through the **template** language-based feature. The exact type of object(s) that a function or **class** operates on is left unspecified in the function definition or **class** declaration until a function is called or an object is defined. The C++ **try-catch-throw** termination mechanism for handling runtime errors is presented in Chapter 14 and demonstrated for various classes.

The topic of **class** inheritance is covered in Chapter 15, which allows a derived **class** to inherit the features of one or more other base classes. Several properties of inheritance are discussed with reference to a `Shape` **class** hierarchy, including polymorphic classes which contain one or more **virtual** functions. Chapter 16 examines the run-time type information mechanism, which allows a programmer to obtain the type of an object at run-time, and the related topic of casting.

Both C and C++ streams are discussed in Chapter 17. Standard input and output and file streams are covered and an overview is given of the C function-based and C++ **class**-based stream libraries. Chapter 18 discusses the preprocessor, although it is not strictly part of the C++ language.

Chapter 19 discusses *namespaces*, one of the newest major features to be added to the C++ language. Namespaces are a language-based solution to the problem of name clashes resulting from C++'s single name space for a translational unit. An overview of the Standard Template Library (STL) for C++ is presented in Chapter 20.

The STL provides a comprehensive collection of efficient and robust generic data structures and algorithms that have been approved by the ANSI committee for the standardisation of the C++ language. This means that the STL must be provided by all ANSI C++ compatible compilers in an analogous manner to the C-style global functions library.

Each chapter concludes with an *Exercises* section with seven exercises which are chosen to reflect the key topics covered in the associated chapter. The solutions to all chapter exercises can be found with the accompanying chapter program files in the subdirectory \XERCISES.

Graham M. Seed

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About the Author

The author completed a B.Eng. degree at Leeds University, followed by an M.Phil. and a Ph.D. at Sheffield University in Mechanical Engineering. He has been programming in C and C++ for over ten years and lectures in programming techniques and applications in engineering to undergraduate and postgraduate students. His primary research interests involve computational mechanics and geometry and object-oriented methods. He is presently a lecturer at the Department of Mechanical and Chemical Engineering at Heriot-Watt University, Edinburgh. Dr Seed is also the author of *Strength of Materials: An Undergraduate Text* published by Saxe-Coburg Publications.

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List of Programs

Programs are listed in the order in which they appear.

Chapter 3

MAIN.CPP	the simplest of C++ programs.
INFO.H	header file of information.
CAPITALM.CPP	illustrates that C++ is a case sensitive language.
MAIN_STD.CPP	illustrates the `std` namespace.
MAIN_STD1.CPP	illustrates placing `std` in to the global namespace.
MAIN_CON.CPP	pausing output for GUI programs.

Chapter 4

CHAR.CPP	illustrates the **`char`** data type.
INT.CPP	illustrates the **`int`** data type.
FLOAT.CPP	illustrates the **`float`** data type.
DOUBLE.CPP	illustrates the **`double`** data type.
BOOL.CPP	illustrates **`bool`**.
LIMITS.CPP	examines LIMITS.H.
FLOATS.CPP	examines FLOAT.H.
SIZES.CPP	examines type sizes.
MANIP.CPP	illustrates the stream manipulators.
OFILE.CPP	output of strings to a file.
OFILE1.CPP	output of integers to a file.
IFILE.CPP	get integers from a file.
AUTO.CPP	illustrates the **`auto`** storage class specifier.
FILE.CPP	some variable definitions.
EXTERN.CPP	illustrates the **`extern`** storage class specifier.
REGISTER.CPP	illustrates the **`register`** storage class specifier.
FILE1.CPP	some **`static`** variable definitions.
STATIC.CPP	illustrates the **`static`** storage class specifier.
ASM.CPP	illustrates the **`asm`** keyword.

Chapter 5

IF.CPP	illustrates the **`if`** statement.
IFIF.CPP	illustrates the **`if`** statement.
IF_ELSE.CPP	illustrates the **`if-else`** statement.
IF_COMP.CPP	illustrates the **`if-else`** compound statement.
IF_MANIP.CPP	illustrates the **`if-else`** statement and the `dec`, `hex` and `oct` stream manipulators.
IF_NEST.CPP	illustrates nested **`if`**s.
LOG_AND.CPP	illustrates the logical AND operator (`&&`).
LOG_OR.CPP	illustrates the logical OR operator (`||`).
LOG_NOT.CPP	illustrates the logical NOT operator (`!`).
SWITCH.CPP	illustrates the **`switch`** statement.
B_SWITCH.CPP	illustrates the **`switch`** statement and the keyword **`break`**.

DEFAULT.CPP	illustrates the **switch** statement, **break** and **default**.
COND.CPP	illustrates the conditional operator (? :).
FOR.CPP	illustrates the **for**-loop.
WHILE.CPP	illustrates the **while**-loop.
DO_WHILE.CPP	illustrates the **do-while** loop.
B_FOR.CPP	illustrates the **for**-loop and **break** statement.
CONTINUE.CPP	illustrates the **continue** statement.
BIT_AND.CPP	illustrates the bitwise AND operator (&).
EVEN_ODD.CPP	illustrates the bitwise AND operator (&) by determining whether a number is odd or even.
BIT_OR.CPP	illustrates the bitwise OR operator (|).
BIT_XOR.CPP	illustrates the bitwise XOR operator (^).
BIT_ONE.CPP	illustrates the bitwise one's complement operator (~) by finding the two's complement of a number.
SHIFT.CPP	illustrates the left and right shift operators << and >>.
BS_ASSGN.CPP	illustrates the bitwise and shift assignment operators.

Chapter 6

FUNC.CPP	illustrates a C++ function.
FUNC_DEC.CPP	illustrates a C++ function declaration.
L_SCOPE.CPP	illustrates local scope.
G_SCOPE.CPP	illustrates global scope.
RES_OP.CPP	illustrates the scope resolution operator : :.
P_VALUE.CPP	illustrates the passing of arguments by value.
P_REF.CPP	illustrates the passing of arguments by reference.
P_CONST.CPP	illustrates the use of **const** for passing arguments by reference.
RETURN2.CPP	illustrates returning more than 1 value.
R_REF.CPP	illustrates returning by reference.
MY_LIB.CPP	implementation file for my library of functions.
MY_LIB.H	header file for MY_LIB.CPP.
NEW_FUNC.CPP	demonstrates the use of implementation and header files.
OVER.CPP	illustrates function overloading.
GET_SET.CPP	illustrates function overloading.
DEF_ARG.CPP	illustrates default function arguments.
STATIC.CPP	illustrates local **static** variables.
INLINE.CPP	illustrates an **inline** function.
FILE.CPP	a function definition.
EXTERN.CPP	illustrates the **extern** storage class specifier.
EXTERN1.CPP	further illustrates the **extern** storage class specifier.
LI_LIB.CPP	implementation of clockwise/anticlockwise and line intersection functions.
LI_LIB.H	header file for LI_LIB.CPP.
LI.CPP	intersection of two two-dimensional line segments.

Chapter 7

ARRAY.CPP	illustrates a simple one-dimensional array.
STATIC.CPP	illustrates that a C++ array must be defined constant.
A_SIZE.CPP	illustrates the **sizeof** operator for determining the number of elements and size of an array.
INDEXING.CPP	illustrates indexing an array out of bounds.

INITIAL.CPP	illustrates initialising arrays.
ARRAY_2D.CPP	illustrates two-dimensional arrays.
ARRAY_3D.CPP	illustrates three-dimensional arrays.
V_STRING.CPP	illustrates variable strings.
CIN_GET.CPP	illustrates the `istream::get()` function.
C_STRING.CPP	illustrates constant strings.
INIT_STR.CPP	illustrates initialising a 2D array of strings.
FUNC_ARG.CPP	illustrates passing of array to functions.
ADDRESS.CPP	examines an array's address.
F_ARG_2D.CPP	illustrates passing two-dimensional arrays as function arguments.
STR_ARG.CPP	illustrates passing strings as function arguments.
ARG.CPP	illustrates that a function does not know the size of an array passed as an argument.

Chapter 8

STRUCT.CPP	illustrates structures.
SCOPE.CPP	illustrates a structure's scope.
SIZEOF.CPP	illustrates determining the size of a structure.
ASSIGN.CPP	illustrates assignment of structure variables.
ARRAY.CPP	illustrates an array of structures.
FUNC_ARG.CPP	illustrates a structure function argument.
NEST.CPP	illustrates nested structures.
POINT.CPP	illustrates a `Point` structure.
DAT_FUN.CPP	illustrates a structure declaration with data function members.
PLT.CPP	`Point`, `Line` and `Triangle` structures.
BIT_FLD.CPP	illustrates bit fields.
UNION.CPP	illustrates **unions**.
ENUM.CPP	illustrates **enum**.
TYPEDEF.CPP	illustrates the **typedef**.

Chapter 9

CLASS_1.CPP	introduces the **class**.
PRI_PUB.CPP	illustrates **private** and **public** data members.
PUBPOINT.CPP	a `Point` **class** with **public** data members.
MEMPOINT.CPP	illustrates **class** member functions.
MEM_OVER.CPP	illustrates overloaded **class** member functions.
OUTSIDE.CPP	a `Point` **class** with member functions defined outside the **class** declaration.
PT_LIB.H	header file for `Point` **class**.
PT_LIB.CPP	implementation file for `Point` **class**.
MY_PROG.CPP	application of `Point` **class**.
STATIC.CPP	illustrates **static** data and function members of a **class**.
AU_EX_RE.CPP	illustrates that the storage class specifiers **auto**, **extern** and **register** cannot be used for declaring **class** data members.
CON_MUT.CPP	illustrates **const** and **mutable**.
VOL.CPP	illustrates **volatile** data members.
BIT_FLD.PP	illustrates bit fields.
CONSTR.CPP	illustrates constructors.
OVER_CON.CPP	illustrates overloaded constructors.
DEF_CON.CPP	illustrates the use of default constructors.

CON_INIT.CPP	illustrates data member initialisation.
COPY_CON.CPP	illustrates the copy constructor.
OVER_CC.CPP	overloads `Point`'s copy constructor.
COPY_FUN.CPP	illustrates the copy constructor with function arguments and return values.
CON_CV.CPP	illustrates constructors, **`const`** and **`volatile`**.
DESTR.CPP	illustrates destructors.
DES_CV.CPP	illustrates destructors, **`const`** and **`volatile`**.
INLINE.CPP	illustrates **`inline`** member functions.
AUTO_IN.CPP	illustrates automatic **`inline`** member functions.
CON_MEM.CPP	illustrates **`const`** member functions.
CON_MEM1.CPP	further illustrates **`const`** member functions.
VOL_MEM.CPP	illustrates **`volatile`** member functions.
DEF_ARG.CPP	illustrates default member function arguments.
ARG_RET.CPP	illustrates passing object arguments to functions and returning an object from a function.
ARG_RET2.CPP	illustrates passing object arguments to functions and assigning values to an object's data members.
RET_REF.CPP	illustrates returning by reference for both constant and non-constant objects.
NO_OBJ.CPP	illustrates a constructor and member function call in a single statement.
ARRAY_OB.CPP	illustrates an array of objects.
LOCAL.CPP	illustrates local classes.
NESTED.CPP	illustrates nested classes.
NESTED1.CPP	further illustrates nested classes.
LI_CLASS.CPP	`Point` and `Line` classes with `Point` clockwise/anticlockwise and `Line` intersection routines.

Chapter 10

ADD.CPP	attempts to add two `Point` objects using the addition operator + before overloading.
OVER_AS.CPP	overloads the arithmetic (+) and (-) operators for **`class`** `Point`.
ARITH.CPP	overloads the arithmetic and arithmetic assignment operators.
INC_DEC.CPP	overloads the prefix and postfix increment and decrement operators.
COMPARE.CPP	overloads the relational operators.
ASSIGN.CPP	illustrates overloading the assignment operator.
NON_MEM.CPP	illustrates non-member overloaded operator functions.
SUBSCRPT.CPP	illustrates overloading the subscript operator.
COMPOSIT.CPP	illustrates overloading an array indexing operator.
CONVERT.CPP	illustrates conversions between different types.
EXPLICIT.CPP	illustrates the **`explicit`** specifier.
STRING.CPP	illustrates a `String` **`class`**.
OVER_IO.CPP	overloads C++'s insertion, <<, and extraction, >>, operators.
OVER_IO1.CPP	further illustrates overloading the insertion and extraction operators.
OVER_IO2.CPP	another approach to overloading the insertion and extraction operators.

Chapter 11

FRI_FUNC.CPP	illustrates **`friend`** functions.

ABC.CPP	illustrates a function which is a **friend** of three classes A, B and C.
FR_CLASS.CPP	illustrates **friend** classes.
FR_TRAN.CPP	illustrates that friendship is not transitive.
FR_OP.CPP	illustrates operator overloading using **friend** functions.
FR_CPLX0.CPP	illustrates overloaded operators using **friend** functions via a Complex **class**.
FR_CPLX1.CPP	illustrates overloaded operators using **friend** functions via a Complex **class** and a conversion function.
FR_IO.CPP	illustrates overloading C++'s insertion and extraction operators for **class** Complex.

Chapter 12

PTR_1ST.CPP	illustrates pointer variables.
AOO.CPP	illustrates the address-of operator &.
PTR.CPP	illustrates the pointer variables and the address-of operator &.
PTR_NEXT.CPP	another look at pointers.
NONVALID.CPP	illustrates on uninitialised pointer.
NULL.CPP	illustrates the 0 pointer.
MIX.CPP	illustrates the mixing of types with pointers.
PTR_VOID.CPP	illustrates pointers to **void**.
CONST_P.CPP	illustrates the use of **const** with pointers.
FUNC_ARG.CPP	illustrates passing function arguments by pointer.
P_ADD.CPP	illustrates pointer addition.
P_SUB.CPP	illustrates pointer subtraction.
P_INC.CPP	illustrates the increment operator and pointers.
PARRAY.CPP	illustrates pointers and arrays.
P_INC1.CPP	illustrates incrementing a pointer.
P_A_2D.CPP	illustrates pointers and two-dimensional arrays.
PAAFA.CPP	illustrates pointers, arrays and function arguments.
F_P_A_2D.CPP	illustrates passing a general two-dimensional array to a function.
PTR_PTR.CPP	illustrates a pointer to a pointer.
ARRAY_P.CPP	illustrates an array of pointers.
A_P_STR.CPP	illustrates arrays of pointers to strings.
P_ARRAY.CPP	illustrates pointers to arrays.
PTR_STR.CPP	illustrates pointers to string constants.
NR.H	header file for Newton–Raphson method.
NR.CPP	implementation of Newton–Raphson method.
P_FUNC.CPP	illustrates pointers to functions.
F_RET_P.CPP	illustrates functions that return a pointer.
CAST_P.CPP	illustrates casting pointers.
P_OBJ.CPP	illustrates pointers to objects.
PTR_MEM.CPP	illustrates the pointer-to-member operators .* and ->*.
ND_OBJ.CPP	illustrates the **new** and **delete** operators applied to objects.
DELETE.CPP	illustrates the **delete** operator.
ND_STR.CPP	illustrates the use of the **new** and **delete** operators with a String **class**.
A_P_STR1.CPP	illustrates arrays of pointers to String objects.
VECT.CPP	a Vector **class** that illustrates the use of the **new** and **delete** operators for dynamically allocating and deallocating memory for arrays.

VECT1.CPP	a `Vector` **`class`** that caters for an increasing/decreasing number of elements.
NEW_INIT.CPP	illustrates the combined use of the **`new`** operator with a **`class`**'s constructor data member initialisation list.
OVER_ND.CPP	illustrates overloading the **`new`** and **`delete`** operators.
WIN_MEM.CPP	illustrates a global memory **`class`** for programming in a Windows environment.
PLACEMNT.CPP	illustrates placement.
RET_PROB.CPP	illustrates a problem with returning an object from a function.
PASS_PROB.CPP	illustrates a problem with passing an object by value to a function.
THIS.CPP	illustrates the **`this`** pointer.
THIS_PT.CPP	illustrates the **`this`** pointer applied to **`class`** `Point`.
THIS_STR.CPP	illustrates the use of the **`this`** pointer with the **`operator`**`=()` function for **`class`** `String`.
THIS_EXP.CPP	experiments with the **`this`** pointer.
MTRX.CPP	a `Matrix` **`class`**.
VECTOR.H	header file for `Vector` **`class`**.
VECTOR.CPP	implementation file for `Vector` **`class`**.
MATRIX.H	header file for `Matrix` **`class`**.
MATRIX.CPP	implementation file for `Matrix` **`class`**.
MTRX_TST.CPP	tests the modified `Matrix` **`class`** which constructs an array of `Vector` objects and allows C++-style subscripting of `Matrix` elements.

Chapter 13

FUN_OVER.CPP	implements a *`Max()`* function with different argument types by function overloading.
TMP_FUN.CPP	implements a *`Max()`* function as a **`template`** function.
TMP_DEC.CPP	illustrates a **`template`** function declaration.
MULT_ARG.CPP	illustrates multiple type arguments for **`template`** functions.
DIF_TYPE.CPP	illustrates that **`template`** functions do not perform implicit casting.
TMP_OVER.CPP	illustrates that **`template`** functions can be overloaded.
TMP_O1.CPP	illustrates overloading a **`template`** function.
PT_TMP.CPP	illustrates a `Point` **`template class`**.
PT_TMP.H	**`template class`** `Point`.
TMP_TST.CPP	tests the **`template class`** `Point` which is declared and defined in the header file PT_TMP.H.
FRI_FUNC.CPP	illustrates a **`friend`** function of a **`template class`**.
MTCA.CPP	multiple **`template class`** arguments.
EXP_TMP.CPP	illustrates that a **`template class`** can have an explicit **`template class`** declaration.
TMP_C_F.CPP	illustrates the use of a **`template class`** as an argument to a **`template`** function.
NESTED.CPP	illustrates nested **`template`** classes.
TYPENAME.CPP	illustrates **`typename`**.
SMRT_PTR.CPP	illustrates smart pointers.
WIN_TMP.CPP	illustrates a global memory **`template class`** for programming in a Windows environment.

VEC_TMP.H	**template class** Vector.
MTX_TMP.H	**template class** Matrix.
MTMP_TST.CPP	tests the Point, Vector and Matrix **template** classes.
ASSOC_A.CPP	illustrates associative arrays.
V_IT_F.H	**template class** Vector with iterators.
IT_F_TST.CPP	tests the **template class** Vector with iterators.
V_IT_C.H	**template class** VectorIterator.
IT_C_TST.CPP	tests the **template class** VectorIterator.
LL.H	**template** classes LinkedList and LinkedListIterator.
LL_TST.CPP	tests the LinkedList and LinkedListIterator **template** classes.
STACK.H	**template class** Stack.
POLY.H	**class** Polygon.
POLY.CPP	Polygon implementation file.
PT_LINE.CPP	classes Point and Line.
PT_LINE.CPP	Point and Line implementation file.
POLY_TST.CPP	test **class** Polygon.

Chapter 14

TCT.CPP	introduces the **try**, **catch** and **throw** keywords.
TERM.CPP	illustrates abnormal program termination.
TYPES.CPP	illustrates different handler types.
MULT.CPP	illustrates multiple **catch** statements.
CATCH_AL.CPP	illustrates the **catch**-all statement.
INT_ALL.CPP	illustrates both **catch**-**int** and **catch**-all statements.
OUT_TRY.CPP	illustrates throwing an exception from outside a **try** block.
TCT_FUNC.CPP	illustrates the **try**-**catch**-**throw** setup from within a function.
EX_SPEC.CPP	illustrates exception specifications.
RETHROW.CPP	illustrates re-throwing an exception.
SET_TERM.CPP	illustrates the *set_terminate()* function.
SET_UNX.CPP	illustrates the *set_unexpected()* function.
STD_EXC.CPP	illustrates the C++ standard exceptions.
SET_HAND.CPP	illustrates the *set_new_handler()* function.
CON_DES.CPP	illustrates calls to constructors, copy constructors and destructors with exception handling.
VEC_X.H	**template class** Vector with exception handling.
VEC_TST.CPP	tests the exception handling features of the **template class** Vector.
VEC_X1.H	**template class** Vector with exception handling.
VEC_TST1.CPP	tests the exception handling features of the **template class** Vector having modified the Memory, Size and Range exception classes.
ASSERT.H	*Assert()* **template** function.
RANGE.H	**class** Range.
INDEX.H	**class** Index.
RNG_INDX.CPP	illustrates the use of Range and Index classes with a Vector **template class**.
X_NEW.CPP	illustrates exception handling and the **new** operator.
NOTHROW.CPP	illustrates the *nothrow* **new** operator.

Chapter 15

B_D.CPP	introduces base and derived classes.
ACC_SPEC.CPP	examines the **public** and **private** access specifiers for derived classes.
CHNG_ACC.CPP	illustrates changing an access specification associated with data members of a derived **class**.
PROPS.CPP	illustrates certain properties of inheritance.
DER_DER.CPP	illustrates deriving from a derived **class**.
PROT.CPP	illustrates the **protected** keyword.
PRO_ACCS.CPP	examines the **protected** access specifier for derived classes.
NO_D_CD.CPP	illustrates a derived **class** with no constructor or destructor.
D_CD.CPP	illustrates a derived **class** with constructor and destructor.
D_CD_D.CPP	illustrates a derived **class** with constructor, destructor and data members.
ODD_CD.CPP	further illustrates a derived **class** with constructor and destructor.
D_TO_B.CPP	illustrates passing constructor argument objects from a derived **class** constructor to a base **class** constructor.
D_TO_B1.CPP	further illustrates passing constructor argument objects from a derived **class** constructor to a base **class** constructor.
CC_INH.CPP	illustrates a base **class** copy constructor and inheritance.
CC_INH1.CPP	illustrates the use of a both base and derived **class** copy constructors and inheritance.
OP_INH.CPP	illustrates the use of a base **class** overloaded assignment operator and inheritance.
OP_INH1.CPP	illustrates the use of both base and derived **class** overloaded assignment operators and inheritance.
OP_INH2.CPP	illustrates base and derived **class** overloaded assignment operator functions and inheritance.
OP_INH3.CPP	illustrates the use of a copy constructor and overloaded assignment operator function for both base and derived classes.
ND_INH.CPP	illustrates overloading the **new** and **delete** operators.
OVERRIDE.CPP	illustrates derived **class** member functions which override base **class** member functions.
OO.CPP	illustrates overriding base **class** overloaded member functions.
SCPE_RES.CPP	illustrates the use of the scope resolution operator to access an overridden base **class** member function from a derived **class**.
FR_INH.CPP	illustrates friendship and inheritance.
FR_INH1.CPP	further illustrates friendship and inheritance.
POINT.H	**class** `Point`.
POINT.CPP	implementation file for **class** `Point`.
NUM_PT.CPP	a numbered `Point` **class**.
VEC.H	header file for `Vector` **class**.
VEC.CPP	implementation file for `Vector` **class**.
VEC3D.H	header file for **class** `Vector3D`.
VEC3D.CPP	implementation file for `Vector3D` **class**.
VEC_TST.CPP	tests base and derived classes `Vector` and `Vector3D`.
PT.H	**class** `Point`.
PT.CPP	implementation file for `Point` **class**.
LINE.H	header file for `Line` **class**.

LINE.CPP	implementation file for **class** Line.
TRI_CON.H	header file for Triangle **class** to illustrate containment.
TRI_CON.CPP	implementation file for Triangle **class**.
TET_CON.H	header file for Tetrahedra **class** to illustrate containment.
TET_CON.CPP	implementation file for Tetrahedra **class**.
TTT_CON.CPP	tests the Triangle and Tetrahedra classes.
MULT_INH.CPP	illustrates multiple inheritance.
M_NO_CD.CPP	illustrates a **class** with no constructor and destructor derived from two base classes.
M_CD.CPP	illustrates a **class** with a constructor and destructor derived from two base classes.
D_TO_BS.CPP	illustrates passing constructor argument objects from a derived **class** constructor to base **class** constructors.
M_PROB.CPP	illustrates ambiguous accessing of data members by a derived **class**.
M_PROB1.CPP	further illustrates ambiguous accessing of data members by a derived **class**.
V_B_CLSS.CPP	illustrates **virtual** base classes.
NON_VIRT.CPP	illustrates the use of pointers to non-**virtual** member functions.
VIRT.CPP	illustrates the use of **virtual** member functions.
V_HIER.CPP	illustrates that **virtual** functions are hierarchical.
V_DEST.CPP	illustrates the need for **virtual** destructors.
PURE_ABS.CPP	illustrates pure **virtual** functions and abstract classes.
SHAPE.H	header file for Shape **class**.
PLANE_V.H	header file for Plane **class**.
PLANE_V.CPP	implementation file for Plane **class**.
POLY_V.H	header file for Polygon **class**.
POLY_V.CPP	implementation file for Polygon **class**.
TRI_V.H	header file for Triangle **class**.
TRI_V.CPP	implementation file for Triangle **class**.
QUAD_V.H	header file for Quadrilateral **class**.
QUAD_V.CPP	implementation file for Quadrilateral **class**.
CIRC_V.H	header file for Circle **class**.
CIRC_V.CPP	implementation file for Circle **class**.
TET_V.H	header file for Tetrahedra **class**.
TET_V.CPP	implementation file for Tetrahedra **class**.
SHAPES.CPP	tests abstract base **class** Shape and derived classes.
TMP_INH.CPP	illustrates the use of **template** classes and inheritance.
RV_TMP.H	**template class** Vector.
RV_TST.CPP	tests the **template class** RVector, which is inherited from **template class** Vector.
CONVERT.CPP	illustrates inheritance, **template** classes and conversion.
VEC_PTR.H	Vector of pointers **template class**.
PTRV_TST.CPP	tests the **template class** PtrVector, which is inherited from **template class** Vector.
VEC_SORT.H	**template class** SortedVector.
SORT_TST.CPP	tests the **template class** SortedVector, which is inherited from **template class** Vector.
WIN_LGM.CPP	illustrates local and global memory **template** classes for programming in a Windows environment.
VEC_TMP.H	**template class** Vector.

MTX_TMP.H	**template class** Matrix and floating-point classes DoubleMatrix and FloatMatrix.
MTX_TST.CPP	tests the Point, Vector and Matrix **template** classes and the DoubleMatrix and FloatMatrix classes.
STACK.H	**template class** Stack.
QUEUE.H	**template class** Queue.
S_Q_TST.CPP	tests the LinkedList **template class**.
ITER.H	**template class** Iterator.
VEC_ITER.H	**template** classes Vector and VectorIterator.
LL_ITER.H	**template** classes LinkedList and LinkedListIterator.
LL.H	**template class** LinkedList.
ITER_TST.CPP	tests the iterator classes VectorIterator and LinkedListIterator.
EXP_INH.CPP	illustrates exception handling and inheritance.
BD_X_INH.CPP	illustrates base and derived classes, exception handling, and inheritance.
B_PRI_D.CPP	illustrates base and privately derived classes, exception handling, and inheritance.
BD_CATCH.CPP	illustrates catching a derived **class** exception.
X_SP_INH.CPP	illustrates function exception specifications and inheritance.

Chapter 16

WHO.CPP	illustrates detecting the type of an object through the use of an abstract base **class** code and type casting an abstract base **class** pointer.
DBL_DIS.CPP	illustrates double dispatch.
DYN_CSTP.CPP	illustrates the dynamic_cast<T*>() operator.
DYN_CSTR.CPP	illustrates the dynamic_cast<T&>() operator.
V_B_C.CPP	illustrates the use of the dynamic_cast<T*>() operator with **virtual** base classes.
TYPEID.CPP	illustrates the use of the typeid() operator.
TYPEINFO.CPP	illustrates the Type_info **class**.
STAT_CST.CPP	illustrates the static_cast<>() operator.
REIN_CST.CPP	illustrates the reinterpret_cast<>() operator.
CNST_CST.CPP	illustrates the const_cast<>() operator.

Chapter 17

COUT.CPP	illustrates the cout stream object and insertion operator <<.
CIN.CPP	illustrates the cin stream object and extraction operator >>.
USR_TYPE.CPP	illustrates overloading the insertion and extraction operators for a user-defined **class**.
IN_OUT.CPP	further illustrates overloading the insertion and extraction operators for **class** Point.
MANIP.CPP	illustrates the C++ stream manipulators.
MY_MANIP.CPP	illustrates user-defined non-parametrised stream manipulators.
P_MANIP.CPP	illustrates user-defined parametrised stream manipulators.
P_MANIP1.CPP	an alternative approach to user-defined parametrised stream manipulators.
IOS_F_MF.CPP	illustrates the format flags and the *setf()*, *unsetf()* and *flags()* member functions of **class** ios_base.
WIDTH.CPP	illustrates the ios::*width()* member function.

LOCALE.CPP	illustrates **class** `locale`.
IFSTR.CPP	illustrates the `ifstream` stream **class** for handling file input.
IFSTR1.CPP	illustrates the `ifstream` stream **class** for handling file input and the `istream::getline()` member functions.
NO_LINES.CPP	illustrates the use of the `istream::getline()` member function to determine the number of lines of a file.
OFSTR.CPP	illustrates the `ofstream` stream **class** for handling file output.
FSTR.CPP	illustrates the `fstream` stream **class** for handling file input and output.
IOS_STAT.CPP	illustrates the `ios` stream status flags and associated member functions.
EOF.CPP	illustrates the `ios::eof()` member function for detecting the end of a file.
OP_CL.CPP	illustrates the `open()` and `close()` stream functions.
OP_OR_CL.CPP	illustrates the `is_open()` function for determining whether a file is open or closed.
RW.CPP	illustrates the `read()`, `write()` and `gcount()` member functions.
CGP.CPP	illustrates the `seekg()` and `tellg()` member functions of **class** `istream` for extracting the current get position.
RW1.CPP	further illustrates the `read()`, `write()` and `gcount()` member functions.
OB_FILE.CPP	illustrates the use of user-defined objects of **class** `NumberedPoint` and an input file stream.
PT_TMP.H	**template class** `Point`.
NPT_TMP.H	a numbered `Point` **template class**.
OB_FILE1.CPP	further illustrates the use of user-defined objects of **class** `NumberedPoint` and an input file stream.
RW_FILE.H	`ReadFile`, `WriteFile` and `ReadAndWriteFile` **class** declarations.
RW_FILE.CPP	implementation file for classes `ReadFile`, `WriteFile` and `ReadAndWriteFile`.
RW_TST.CPP	tests the `ReadFile`, `WriteFile` and `ReadAndWriteFile` classes.
WORLD.H	a `World` **class**.
WORLD.CPP	implementation file for **class** `World`.
READ_OBJ.CPP	tests the `World` **class** by reading a set of objects from a disk file.
SHAPE_V.H	header file for `Shape` **class**.
POLY_V.H	header file for `Polygon` **class**.
ISTRSTR.CPP	illustrates the C++ input string stream **class** `istrstream`.
CONSTR.CPP	illustrates the C++ console stream **class** `constream`.
REDIRECT.CPP	illustrates redirection.
HELLO.CPP	illustrates command line arguments and the passing of strings.
SQ_ROOT.CPP	illustrates command line arguments and the passing of numerics.
CPP_CPY.CPP	illustrates command line arguments.
PRINTF0.C	illustrates the `printf()` function.
PRINTF1.C	further illustrates the `printf()` function.
PRINTF2.C	further illustrates the `printf()` function.
SPRINTF.C	illustrates the `sprintf()` function.

SCANF0.C	illustrates the *scanf()* function.
SCANF1.C	further illustrates the *scanf()* function.
SCANF2.C	further illustrates the *scanf()* function.
SCANF3.C	further illustrates the *scanf()* function.
SCANF4.C	further illustrates the *scanf()* function.
SCANF5.C	illustrates the scanset feature of the *scanf()* function.
SCANF6.C	further illustrates the scanset feature of the *scanf()* function.
SCANF7.C	further illustrates the scanset feature of the *scanf()* function.
SSCANF.C	illustrates the *sscanf()* function.
ORDC.C	illustrates opening, reading, displaying and closing a file using the C *fopen()*, *fgetc()* and *fclose()* file handling functions.
ORDC1.C	illustrates opening, reading, displaying and closing a file using the C *fopen()*, *fgets()* and *fclose()* file handling functions.
C_CPY.C	illustrates the *fputc()* function.
C_CPY1.C	illustrates the *fputs()* function.
FRFW.C	illustrates the *fread()* and *fwrite()* functions.
C_RND_AC.C	illustrates random file access in C using the *fseek()*, *ftell()*, *fgetpos()* and *fsetpos()* functions.
DIB.H	device independent bitmap **class** for Windows.
DIB.CPP	implementation file.
DIB_TST.CPP	a DIB displayer for Windows.
DIB_TST.RC	resource script file.
DIB_TST.RH	resource script header file.
DIB_TST.DEF	module definition file.

Chapter 18

DEFINE.CPP	illustrates the `#define` preprocessor directive.
CPP_VER.CPP	a C++ version of DEFINE.CPP.
ERROR.CPP	illustrates the `#error` preprocessor directive.
INC.CPP	illustrates the `#include` preprocessor directive.
LINE.CPP	illustrates the `#line` preprocessor directive.
UNDEF.CPP	illustrates the `#undef` preprocessor directive.
IF.CPP	illustrates the `#if` preprocessor directive.
IFNDEF.CPP	illustrates the `#ifndef` preprocessor directive.
#&##.CPP	illustrates the # and ## preprocessor operators.
PREDEF.CPP	illustrates the predefined macros.
ASSERT.CPP	illustrates the *assert()* macro.

Chapter 19

NCLASH.CPP	illustrates name clashes.
N_CLASH1.CPP	illustrates name clashes.
NSPACE.CPP	illustrates the **namespace** declaration.
NS_NEST.CPP	illustrates nested namespaces.
NO_CLASH.CPP	modified N_CLASH.CPP using namespaces.
POLYGON.H	**class** Polygon.
SPHERE.H	**class** Sphere.
USE_DEC.CPP	illustrates the **using** declaration.

USE_DEC1.CPP	**using** declaration applied to classes.
USE_DIR.CPP	illustrates the **using** directive.
GLOBAL.CPP	namespaces and global space.
OVERLOAD.CPP	overloading namespaces.
INHERIT.CPP	illustrates namespaces and inheritance.
UN_HIDE.CPP	a namespaces solution to the problem of a derived **class** member hiding a base **class** member.
MULTIPLE.CPP	namespaces and multiple inheritance.

Chapter 20

IT_PTR.CPP	introduces iterators.
IT_FORW.CPP	illustrates forward iterators.
IT_REVER.CPP	illustrates reverse iterators.
IT_RAND.CPP	illustrates random access iterators.
FUNC_OBJ.CPP	illustrates function objects.
ALGO_NM.CPP	illustrates the STL non-mutating functions.
ALGO_M.CPP	illustrates the STL mutating functions.
SORT.CPP	illustrates the STL sorting functions.
SET_ALGO.CPP	illustrates the STL set functions.
HEAP.CPP	illustrates the STL heap functions.
PERM.CPP	illustrates the STL permutation functions.
NUMERIC.CPP	illustrates the STL numeric functions.
VECTOR.CPP	illustrates the STL `vector` and `deque` classes.
LIST.CPP	illustrates the STL `list` **class**.
SET_STL.CPP	illustrates the STL `set` **class**.
MAP.CPP	illustrates the STL `map` **class**.
COMPLEX.CPP	illustrates the STL `complex` **class**.
VALARRAY.CPP	illustrates the STL `valarray` **class**.
IN_IT.CPP	illustrates the insert iterator adaptors.
RAW_IT.CPP	illustrates `raw_storage_iterator`.
STACK.CPP	illustrates the STL `stack` adaptor **class**.
QUEUE.CPP	illustrates the STL `queue` adaptor **class**.
P_QUEUE.CPP	illustrates the STL `priority_queue` adaptor **class**.
BS.CPP	illustrates the STL `basic_string` **class**.

Chapter 1

Overview

I'm an explorer, okay? I like to find out.
Richard Feynman (Sykes, 1994)

This chapter reviews C++: what it is, its history and its future. Alternative object-oriented programming languages are reviewed and we examine what is actually meant by object-oriented programming and what we should expect from an object-oriented programming language.

1.1 Why C++?

In the words of the original designer of C++, B. Stroustrup (1991, Preface to the first Edition): 'C++ is a general purpose programming language designed to make programming more enjoyable for the serious programmer. Except for minor details, C++ is a superset of the C programming language. In addition to the facilities provided by C, C++ provides flexible and efficient facilities for defining new types.... When used well, these techniques result in shorter, easier to understand, and easier to maintain programs.' Note the emphasis placed on the programmer and the program.

C++'s *major* features are:

- A superset of the C language.
- Stronger type checking than C.
- Support for data abstraction and object-oriented programming.
- Classes and abstract classes that encapsulate data and functions which operate on a **`class`**'s data, defining a given structure and behaviour.
- Inheritance and multiple inheritance, enabling the creation of hierarchies of classes.
- Support for run-time polymorphism via **`virtual`** functions, allowing a **`class`** in an arbitrary inheritance hierarchy to redefine its parent/s member functions.
- Overloading. Overloaded functions and member functions allow a function with the same name but with a different number or type of arguments, or both, to be overloaded. A **`virtual`** function of a base **`class`** can be overridden by a function in a derived **`class`** with the same name and the same number or type of arguments, or both, and return type. Overloaded operators allow you to give new functionality to existing operators when defining a **`class`**.

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
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- Support for the creation of parametrised types, generics or templates and generic functions.
- Exception handling or error handling techniques.
- Run-time type information (RTTI) to obtain run-time identification of types and expressions.
- Namespaces to solve the problem of a single global namespace or scope.
- The Standard Template Library (STL) which provides a collection of generic data structures and algorithms.

Why choose C++ rather than C? C++ is built on C and is thus a superset of C. The C++ language adds more than just object-oriented programming capabilities to C. For example, compare the C++ **new** and **delete** operators with 'equivalent' C user-defined macros *NEW()* and *DELETE()*. The C version is:

```
#include <stdio.h>      // printf()
#include <stdlib.h>     // malloc(), free(), exit()

#define NEW(p, ptype)\
  if ((p=ptype*) malloc (sizeof (ptype)) == NULL)\
    {\
    printf ("Out of memory\n") ;\
    exit (0) ;\
    }
#define DELETE(p)
  if (p)\
    {\
    free ((char*)p ) ;\
    p = NULL ;\
    }
//...
struct X
  { /* ... */ };
//...
void main ()
  {
  struct X* xptr ;
  //...
  NEW (xptr, X);
  //...
  DELETE (xptr) ;
  //...
  }
```

requiring the inclusion of two header files (STDIO.H and STDLIB.H) and four function calls *malloc()*, *printf()*, *exit()* and *free()* in the definition of *NEW()* and *DELETE()*. *NEW()* and *DELETE()* are defined as macros rather than functions because C will not permit the manipulation of a variable without strict regard to type. Thus, it would be necessary to define separate functions for each different data type used, e.g. *New_Int(p, **int**)*, *New_X(p X)*,.... Therefore, to avoid this, macros (as opposed to functions) are used, which are expanded inline prior to compilation, which can be prone to errors.

The equivalent C++ version is:

```
class X
   { /* ... */ };
   //...
void main ()
   {
   X* xptr = new X ;
   //...
   delete xptr ;
   //...
   }
```

No header files or function calls are required and **new** incorporates an implicit memory allocation check. The **new** and **delete** operators allow a programmer to perform memory allocation/deallocation in a structured and consistent language-based manner. In addition, the **new** operator supports a *placement syntax* to allow an object to be placed at a particular location in memory rather than relying on the operating system to determine where an object is to be stored. Thus, the **new** and **delete** operators should be viewed as part of a more general and comprehensive memory management system and not simply allocation/deallocation mechanisms. This example clearly illustrates the compactness and elegance of C++. Although certain features in C are still available in C++, they are seldom used.

The object-oriented programming features of C++ will be discussed shortly, but first let us take a look at the history of C++.

1.2 C++'s History

For a unique insight to the history of C++ refer to *The Design and Evolution of C++* by Bjarne Stroustrup (1994)[1]. In addition, this book is an invaluable text for the reader who wishes to develop a greater understanding of the C++ language, and I would certainly recommend reading it after you have developed a reasonable knowledge of C++.

The origin of C++ begins with the languages ALGOL 60, CPL and BCPL. In 1970, Ken Thompson of AT&T's Bell Laboratories began developing a language called B, which was based on an existing typeless language, BCPL, originally developed by Martin Richards in the mid-1960s (Richards and Whitney-Strevens, 1979). BCPL (Basic CPL) was a derivative of the CPL programming language developed at both Cambridge and London Universities. Shortly afterwards, it became apparent that B was not suitable for implementing the Unix operating system, also originally designed and implemented by Ken Thompson. At the same time, Dennis Ritchie[2], also of Bell Laboratories, was developing a successor to B that was to be a compact and robust language called C. The C language was so successful that approximately 90% of the Unix operating system was rewritten in C. Today, most of Unix is written in C.

The C language quickly became very popular, particularly in the university environment, due to the availability of inexpensive compilers and to the success of Unix. In 1978, Kernighan and Ritchie published the book *The C Programming Language* (frequently referred to as the *white book*) which contained a C reference manual as an appendix. Following the publication of this key book, many compilers were referred to as 'K&R compliant'. Incidentally, the Borland C++

1 See also the articles by Ritchie (1993) and Stroustrup (1993a) on the development of C and C++ in the ACM SIGPLAN Second History of Programming Languages Conference.

2 Dennis Ritchie's home page is at `http://www.cs.bell-labs.com/who/dmr/` and contains some fascinating articles about the development of the C language.

Table 1.1 Key events in C++'s history.

1979	C with Classes development commences
1982	First paper on C with Classes (Stroustrup, 1982)
1983	C++ implementation in use; C++ named
1984	First C++ manual
1985	First commercial release (Cfront Release 1.0)
1986	*The C++ Programming Language*, 1st edn (Stroustrup, 1986)
1987	First USENIX C++ conference
1988	Zortech C++ Release
1989	Cfront Release 2.0
1990	Borland C++ Release *The Annotated C++ Reference Manual* (Ellis and Stroustrup, 1990)
1991	Cfront Release 3.0 *The C++ Programming Language*, 2nd edn (Stroustrup, 1991)
1992	DEC, Microsoft and IBM C++ Releases
1994	Draft ANSI/ISO standard
1998	ANSI/ISO standard finalised *The C++ Programming Language*, 3rd edn (Stroustrup, 1997)

compiler (version 5.x) still supports K&R language compliance. By the mid-1980s, there were more than 20 C compilers for MS-DOS, and C was in danger of being fragmented into several dialects. As a result, in 1982 an American National Standards Institute (ANSI) standardisation committee was formed to produce a standard definition of the C language. Seven years later, the ANSI X3.159-1989 standard was produced. This standard is frequently referred to as the 'ANSI C' standard. The latest C standard is *Programming Languages-C*: ANSI/ISO/IEC 9899-1999.

As a consequence of languages such as Simula and Smalltalk[3], in the late 1970s object-oriented programming was becoming an increasingly popular style of programming. In 1979, Bjarne Stroustrup, of AT&T's Computing Science Research Centre of Bell Laboratories in Murray Hill, New Jersey, began developing a language called *C with Classes*, which essentially added Simula classes to C via a preprocessor called Cpre. Initially, C with Classes was used exclusively by employees of AT&T. By 1982, C with Classes was sufficiently successful for it to be redesigned in to a new language, which was to be called C++. Also, around this time C++ received a new compiler front-end implementation (Cfront). The first commercial release of C++ was in 1985 (Cfront Release 1.0). From 1989 onwards, the emphasis was placed by the ANSI/ISO standards committee, X3J16, on developing a C++ standard, and a draft standard was first issued in 1994. In 1998 the standard was finalised, details of which can be found in Stanrdard (1998). Refer to Table 1.1 for an overview of key events in C++'s history.

C++ got its name from Rick Mascitti in 1983. The C language provides the ++ unary operator for incrementing a variable or pointer. For example, a common operation is incrementing a variable, say i, by adding the constant 1; i.e. $i=i+1$, or, using the ++ operator, i++. Thus, C++ indicates the 'next' C.

1.3 C++'s Future

It has been stated above and throughout the literature that C++ is a superset of C. As C++ continues to develop this is gradually becoming less and less the case. C++ is becoming a separate

3 For a fascinating discussion of the early history of Smalltalk refer to Kay (1993).

language, gradually abandoning its complete compatibility with C. C++ is a continually evolving language, a *living language*. Fortunately, it is evolving to meet the needs of users.

1.4 There are Other OOP Languages

C++ is not the only programming language that supports object-oriented programming features. For instance, two languages that support OOP which have greatly influenced the development of C++ are Simula and Smalltalk. Three good works covering Simula and Smalltalk are Kirkerud (1989), Goldberg and Robson (1983) and Pinson and Weiner (1988).

Other object-oriented programming languages are Common Lisp Object System (CLOS) (Bobrow *et al.*, 1988), Eiffel (Meyer, 1991), Ada (Ada Reference Manual, 1983) and Oberon-2 (Reiser and Wirth, 1992; Mössenböck, 1993). Figure 1.1 illustrates the development of several of the more well-known object-oriented programming languages to date. For the interested reader, Saunders (1989) provides a survey of more than 80 object-oriented programming languages.

Although not the first object-oriented language, Smalltalk was instrumental in the development of object-oriented programming by demonstrating that object-oriented programming is a feasible solution to software development. Smalltalk's strengths are its excellent user interface and class library. Smalltalk is a dynamically typed language, and as a result suffers from static typing. For an introduction to Smalltalk, refer to, for example, Budd (1987).

Oberon-2 evolved from Oberon, which in turn evolved from Pascal and Modula-2. In Oberon-2, classes are represented as records which encapsulate both data and procedures, rather than by introducing an additional class construct. Oberon-2 integrates with the Oberon operating system, which provides run-time support for Oberon-2 programs. Refer to Mössenböck (1993) for details of how to obtain the Oberon-2 compiler and Oberon system.

Eiffel, although not as well-known as C++, is nevertheless a very powerful and industrially popular language and is more of a *pure* object-oriented language than C++, having not evolved primarily from a non-object-oriented language. Even though Eiffel is an object-oriented programming language, it does not forsake the ability to generate efficient program code such

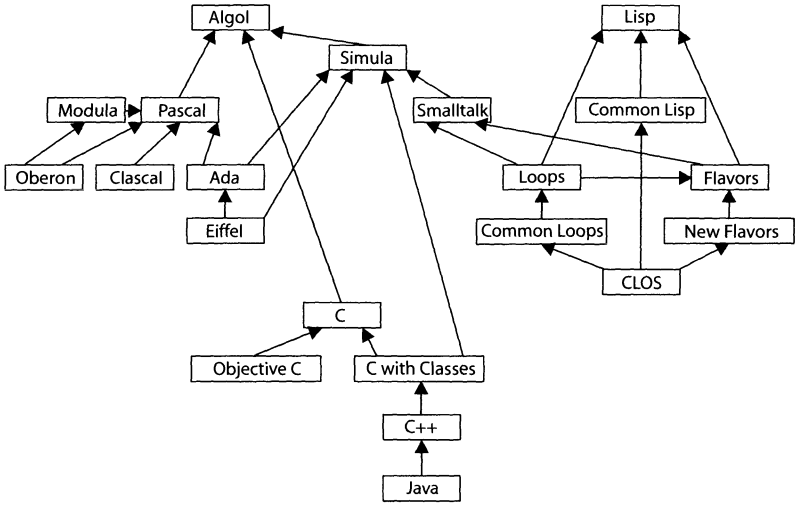

Fig. 1.1 Evolution of several object-oriented programming languages.

as that generated by C. Eiffel is a strong statically typed language and is particularly strong with regard to inheritance.

New object-oriented programming languages are continually emerging. Of the more popular ones to recently emerge are Java and Python. Java was developed by Sun Microsystems and derived from C++. Java is a simple, compact implementation of C++, but with several C++ features removed and with a few important additional features. Some of the major features removed are pointers, templates, multiple inheritance and operator overloading. Java also eliminates the preprocessor and header files. Multithreading is integral in the Java language, so that applications can be developed with multithreads; this is an important feature when developing applications for the Internet, where Java has been particularly successful. A Java language compiler generates architecture-neutral object files by generating bytecode instructions which are architecture-independent. This means that a single Java application can be developed which will execute on several operating systems. Java also supports automatic memory management. For further details of the Java programming language refer to the Java web site, Sun (2000), and Anuff (1996). Python (2000) is an interpreted, interactive, object-oriented programming language that runs on a variety of operating systems. Like Java, Python is copyrighted but freely available.

The language you choose (if you have this choice) to work with is ultimately up to you. However, I would like to add that C++ is a relatively new language and still very much growing. Personally, one of the most exciting things about the language is watching the language grow and mature. That's not to say that because the language is young it has the quirky characteristics of early versions of software. Remember that C++ has been built on C, with numerous object-oriented programming features *borrowed* from existing languages.

1.5 Programming Paradigms

There are a number of different programming styles that programmers generally adopt. These styles are mainly a result of the programming language used, just as different people trained in different disciplines, such as mathematics, physics, computer science or engineering, will tackle a given problem completely differently according to their own 'subject language', with which they are familiar or feel most comfortable. There are five main categories of programming style (Booch, 1994):

- procedure-oriented: algorithms
- object-oriented: classes and objects
- logic-oriented: goals
- rule-oriented: if–then rules
- constraint-oriented: invariant relationships

The following sections examine the current two most popular styles of programming adopted, namely procedural and data abstraction/object-oriented.

1.6 Procedural and Modular Programming

Examples of procedural (sometimes referred to as structured) programming languages are COBOL, Fortran 77/90, Pascal and C. An example of procedural programming in C is:

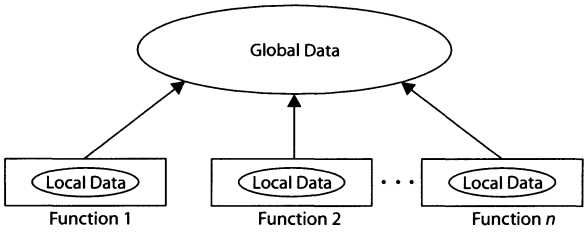

Fig. 1.2 Procedural approach.

```
#include <stdio.h>
/* get/print your name */

void main ()
  {
  char name[21] ;
  printf ("enter your name: ") ;
  gets (name) ;
  printf ("\nhello, %s", name) ;
  }
```

which simply gets a person's name and displays a message based on the name entered. A procedural program is a sequential list of statements or instructions, focusing on processing. Provided the program size remains fairly small, this approach is bearable. However, for large programs a list of statements becomes incomprehensible. Thus the use of functions came about. Each function performs a specified operation, is clearly defined and interfaces with other functions and the program; see Fig. 1.2. In the above example, *gets()* and *printf()* are functions, and each performs a different input/output operation. Consider the following hypothetical header and implementation files, which declare and define several related mathematical functions.

```
/* my_math.h, declarations of floating point math routines */
//...
double cos  (double x) ;
double sin  (double x) ;
double sqrt (double x) ;
//...

/* my_math.c, implementation of floating point math routines */
#include "my_math.h"
//...
double cos (double x)
  {
  /* code for calculating the cosine */
  }
//...
```

This grouping together of a number of related functions is often referred to as a *module*. It is worth mentioning that the concept of a module enables data, variables and functions to be hidden within the module.

A typical use of the module could be:

```
#include <stdio.h>
#include <stdlib.h>
#include <my_math.h>
/* print the square root of your age */

void main ()
  {
  char   buffer[9] ;
  double sqrt_age ;
  printf ("enter your age: ") ;
  sqrt_age = sqrt (atoi (gets (buffer)) ) ;
  printf ("\nsquare root of your age: %f", sqrt_age) ;
  }
```

In general, functions perform operations on supplied data. The emphasis is placed on doing things: do this, do that, did it work? Thus, the idea of functional abstraction enhances an algorithmic problem-solving design approach rather than a feature-based abstraction approach.

1.7 Data Abstraction

Data abstraction is the programming technique of being able to create new data types. The word *abstract* is used to reflect how a programmer abstracts features and concepts from a complex system into new data types. A programmer-defined data type is often referred to as an Abstract Data Type (ADT) to distinguish it from a built-in Fundamental Data Type (FDT), such as characters and integers. Apart from making a programmer think in terms of features, design and program organisation, data abstraction also makes programs more flexible, shorter and conceptually easier to understand. ADTs are implemented in C++ by declaring *classes*. Classes allow both data and functions to be associated with the **`class`** name. A **`class`** is similar to the C **`struct`** and the **`RECORD`** (combined with procedures) of Pascal and Oberon-2.

As an example of data abstraction, consider a complex system such as the Sierpinski gasket (also known as Sierpinski's sieve or carpet, after Vaclav Sierpinski) shown in Fig. 1.3. We could model the gasket solely in terms of integer and floating fundamental data types. Alternatively, we take the data abstraction route by observing that the gasket is composed purely of triangles placed in some kind of self-replicating fractal order. Similarly, each triangle can be decomposed into vertices or points, edges and a face. A face is composed of three edges, an edge consists of two end-points and a point is three floating-point coordinates.

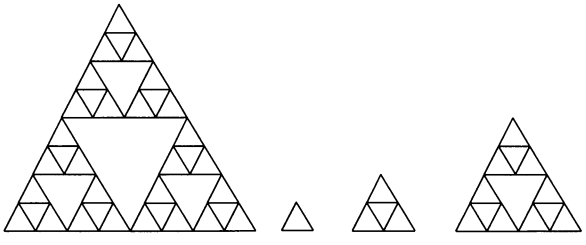

Fig. 1.3 Sierpinski's gasket and the first three stages of construction.

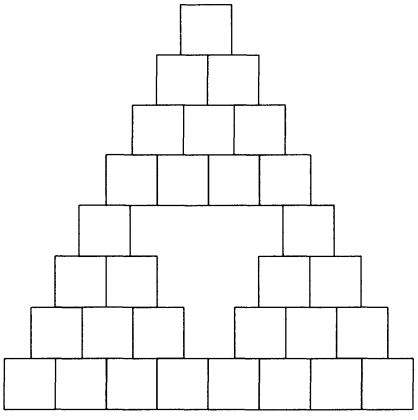

Fig. 1.4 Alternative Sierpinski gasket.

An advantage of the abstract data approach is that our point, edge and face data types can now be applied to alternative geometrical constructions, such as the similar construction shown in Fig. 1.4, which comprises squares rather than triangles. Thus, data abstraction has allowed us to generate generic types that possess common features that can be realised and utilised by a whole class of problems. It is worth pointing out that all abstract data types, however complex and abstract, can be degenerated into fundamental data types.

Data abstraction forms the foundation of object-oriented programming, and as a result a more detailed discussion of data abstraction is postponed till the next section.

1.8 Object-Oriented Programming

There are many different and diverse approaches to program development and object-oriented programming is just one of them. Object-oriented programming is not a new concept. The key ideas of object-oriented programming date back to as early as 1967, from the Simula programming language, and the early 1970s, from the Smalltalk language. Simula was developed by Dahl, Myrhaug and Nygaard (1970), whereas Smalltalk was developed at Xerox Palo Alto Research Center. Simula was based on the Algol language, with the addition of encapsulation and inheritance. C++ is an object-oriented programming language and has inherited many of the object-oriented features of its predecessors.

So what actually is object-oriented programming and how do we put this programming approach in to practice in C++? By way of a typical example we shall now examine object-oriented programming in C++. An object-oriented program consists of a number of different objects, as shown in Fig. 1.5. Objects are literally everywhere, including ourselves: see Fig. 1.6. Each object 'carries' its own set of characteristics or data and is continually sending messages to and receiving messages from other objects.

Initially, the best way to think of the object-oriented paradigm is via 'real' objects: everyday objects around us. Therefore, consider the development of a program to characterise a variety of different shapes. At this point you will probably be unfamiliar with the C++ implementation details, so try to concentrate on the object-oriented approach. At a later stage you may want to refer back to this chapter in order to fully understand the ideas presented.

If you are familiar with the procedural way of thinking about program construction it is tempting at this stage to be focusing your attention more on the details and algorithmics of

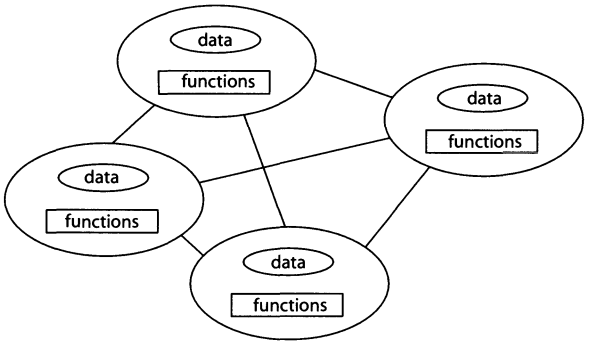

Fig. 1.5 Object approach. Objects talk to each other.

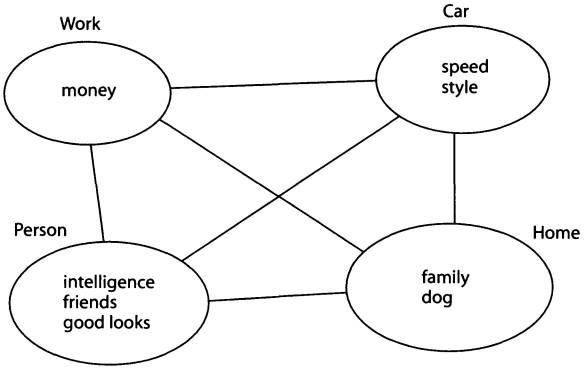

Fig. 1.6 World objects.

generating individual shapes, the division of such details into functions, and the flow of information through a shapes program into a series of steps. Let us take an alternative approach of decomposing the problem into key abstractions which embody their own unique structure and behaviour, each abstraction modelling an object in the real world.

What general properties do *all* objects around us have? They have colour, size, position relative to a given space, orientation and possibly taste, smell and so on. What kinds of object are there, or more importantly which of the infinite number of objects in the real world do we want to model? Let us limit our discussion to, say, a rectangle, a circle and a hexahedron (a three-dimensional object composed of quadrilateral faces, such as a cube). What specific properties do objects have? A circle has a centre, radius, line thickness and so on. These features are *our* abstractions, or the essential characteristics of objects as we perceive them to be. Of course[4], there is no right or wrong set of abstractions. Different sets of people choose different sets of abstractions. However, if there is a general rule – and there isn't – try to choose abstractions that are as general as possible so that they do not become too specialised and known only to a small group of people.

A **`class`**, `FuzzyShape`, that defines the general properties of all shapes is:

4 Bob Reuben

```
class FuzzyShape
  {
  public:
    Colour   colour ;
    Size     size ;
    Position position ;
    //...
    FuzzyShape () ;
    //...
  };
```

assuming that we already have classes which define position, colour and size:

```
class Position
  {
  public:
    double x, y, z ;        // (x, y, z) coordinates in space
    Position () ;
    //...
  };

class Colour
  {
  public:
    int red, green, blue ; // 3 primary components of colour
    //..
  };

class Size
  { /* ... */ };
```

The `FuzzyShape` **`class`** captures our abstraction of the characteristic features of all shapes. The **`class`** is called `FuzzyShape` rather than `Shape` to emphasise that the **`class`** does not give a clear or *crisp* definition of any particular shape.

To create an object or an instance of `FuzzyShape` so that we can do things to the object we typically write:

```
FuzzyShape fuzzy_object ;
//...
cout << "fuzzy object's position: " << fuzzy_object.position ;
cout << "fuzzy object's colour:   " << fuzzy_object.colour ;
//...
```

just as you might define a variable `var` of type **`int`**:

```
int var ;
//...
var = 7 ;
cout << "value of var: " << var ;
//...
```

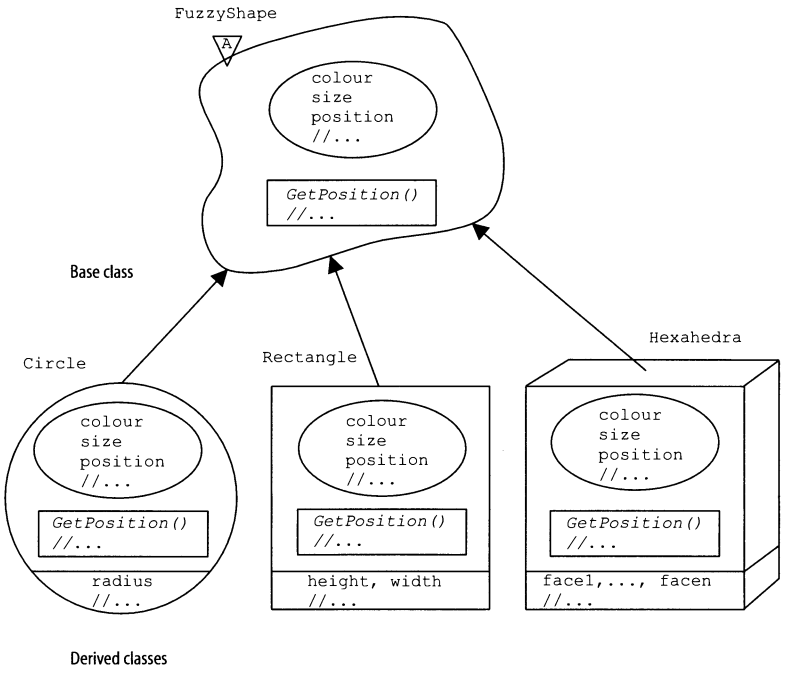

Fig. 1.7 Inheritance of shapes. The inverted triangle encompassing the uppercase letter A indicates an abstract base class in accordance with the Booch (1994) notation.

At present, our `FuzzyShape` **`class`** is nothing more than an elaborate list of properties which are difficult to operate on and are of limited use. Also, you may have noticed that the `colour`, `size` and `position` data members were declared with **`public`** access, which allows access to all clients. Generally, we are interested in separating an object's data from its client interface by *data-hiding*. Data hiding allows an object complete control in specifying what data a client needs to have access to and what data is, or should be, of no interest to a client. Hiding the internal details of an object can greatly reduce the possibility of accidental object-data corruption.

In the above example we were able to create an object of `FuzzyShape` **`class`** when such an object is meaningless in the real world, since `FuzzyShape` models general properties of shapes but is not an abstraction of any real shape. The inheritance mechanism of C++ enables us to define `FuzzyShape` as a base **`class`** and shape classes such as `Rectangle`, `Circle` and `Hexahedra` to be *derived* from **`class`** `FuzzyShape`; see Fig. 1.7[5]. In fact, `FuzzyShape` is made an *abstract* base **`class`**, since we don't want to define any objects of this **`class`**. In other words `FuzzyShape` acts only as a base **`class`** for other classes. Also, in the following modified example the data members of `FuzzyShape` are now **`protected`** to enforce data-hiding and restrict access to the **`class`** itself, derived classes and **`friend`**s:

```
class FuzzyShape
  {
  protected:
    Colour   colour ;
    Size     size ;
```

5 Where possible, the Booch (1994) notation has been adopted throughout this book for diagrammatically illustrating classes, objects and their interactions.

```
    Position position ;
    //...
  public:
    FuzzyShape () ;
    // member functions
    Position GetPosition () const ;
    void     SetPosition (const Position& p) ;
    //...
    // pure virtual member functions
    virtual void Draw () = 0 ;
    //...
  };

class Rectangle : public FuzzyShape
  {
  protected:
    double height, width ;
  public:
    Rectangle () ;
    void Draw () ;
    //...
  };

class Circle : public FuzzyShape
  {
  protected:
    double radius ;
  public:
    Circle () ;
    //...
  };

class Hexahedra : public FuzzyShape
  {
  public:
    Hexahedra () ;
  //...
  };
```

The implementation file might be:

```
FuzzyShape::FuzzyShape ()
  {
  position = 0.0 ;
  //..
  }

Position FuzzyShape::GetPosition ()
  { return position ; }

void FuzzyShape::SetPosition (const Position& p)
```

```
    { position = p ; }
  //...
```

A typical use of the above implementation could be:

```
  // error: can't create an instance of an abstract class
  FuzzyShape fuzzy_object ;

  Rectangle rectangle_object ;
  Circle    circle_object ;
  //...
  // error: not accessible
  cout << "circle position: " << circle_object.position ;

  // O.K., access thro' access member function
  cout << "circle position: " << circle_object.GetPosition () ;
  //...

  FuzzyShape* fs_array[N] ;           // array of pointers to
                                      // FuzzyShapes
  //...
  fs_array[0] = &circle_object ;      // place addresses in
                                      // pointer array
  fs_array[1] = &rectangle_object ;
  //...
  // draw all shapes using a single function call!
  for (int i=0; i<N; i++)
        fs_array[i]->Draw () ;
  //...
```

noting that `position` now has restricted access.

The code above illustrates that after an array of pointers to `FuzzyShape` is created and the object addresses are assigned to the pointer array we are then able to draw a variety of shapes using a single function call. This example of calling completely different functions with a single function call is an example of *polymorphism*, made possible with the aid of inheritance and **`virtual`** functions.

It should be clear from the discussion above that object-oriented programming takes a great deal more planning than traditional procedural or modular programming, but is a more natural way of solving problems.

To be considered an object-oriented programming language it is often stated that a language must support the following key elements: *abstraction*, *encapsulation*, *modularity*, *inheritance* and *polymorphism*; see, for example, Meyer's ten key OO concepts (Meyer, 1995). A brief description of each follows.

1.8.1 Abstraction

Abstraction is concerned with formulating the *essential* characteristics of an object. In the above example we characterised a circle purely in terms of a radius as well as the inherited characteristics of colour, size and position. Real objects have an infinite number of features, and it is simply not possible to model and comprehend all of them and the complex interactions between different objects. Alternatively, a complex object is broken down into a finite set

of simplified, yet essential, characteristics. Because of the infinite number of object characteristics it is important to focus on the essential characteristics when formulating an object abstraction. Formulating an abstraction is generally not as straightforward a process as it might at first appear. Different people inherently view the same object differently. What is required is a set of characteristics which are considered essential by the majority of users for a given problem domain.

1.8.2 Encapsulation

While abstraction addresses the characteristics of an object from an external viewer's perspective, encapsulation focuses on the implementation of an object. The implementation encapsulates the details about an abstraction into separate elements. The implementation should be considered a secret of the abstraction and hidden from most clients. For instance, a car is composed of literally thousands of different components, but can be viewed as an engine, chassis, body, controls etc., with the specific details concerning the implementation and operation of these different abstractions hidden from the driver. The driver is supplied with *just enough* information about the different abstractions to be able to drive the car.

Encapsulation is another word for the *black box* paradigm. Given an input, the black box generates a respective output – this is all the information we know about the box. The black box can perform any number of other operations, but we are given no clues as to the nature of the internal details of the black box.

Note that the abstraction of an object should precede its implementation. The abstraction of an object should not comprise any implementation details.

1.8.3 Modularity

Modularity is simply the decomposition of an abstraction into separate, discrete cells. It is an intuitive process to break a large complex system problem down into a manageable number of components. When programming in the large, organising a program into separate coherent modules is essential to the design of an application. Individual modules can be tested reasonably independently of the overall application. Modularity helps in designing an application to be both extendable and reusable.

1.8.4 Inheritance

Inheritance is concerned with placing separate but related abstractions into a structured hierarchy which will enable the passing of shared characteristics. Figure 1.8 illustrates a schematic **`class`** hierarchy in C++. A *derived* **`class`** inherits properties of a *base* **`class`**. Figure 1.8 also illustrates the concept of multiple inheritance, in which a **`class`** is derived from more than one base **`class`**. Multiple inheritance is plausible when you remember that we all have two parents.

The most important property of inheritance from a programming perspective is that it allows the reuse of program code, and consequently reduces the repetition of code.

1.8.5 Polymorphism

Polymorphism is the ability of a child object in an inheritance hierarchy to exhibit different behaviour based on the type of the object. This feature enables an object to respond to a common set of operations in different ways.

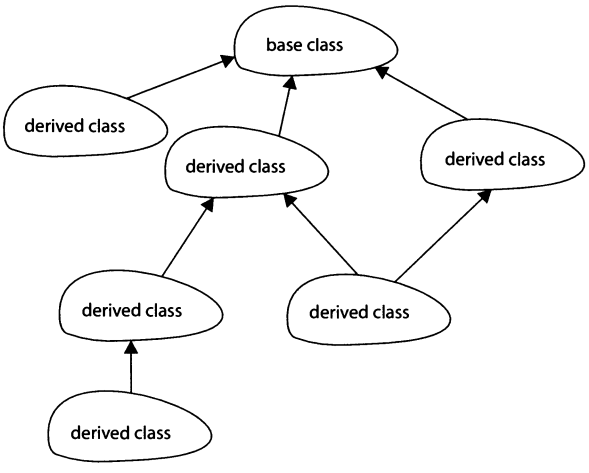

Fig. 1.8 Class hierarchy illustrating single and multiple inheritance.

In addition to the above five key elements, there are some minor elements of object-oriented programming which warrant mentioning: *typing*, *automatic memory management*, *concurrency*, *persistence* and *operator overloading*.

1.8.6 Typing

The distinction between type and **class** is very confusing, and we will assume that the two terms are equivalent. It suffices to say that *a class implements a type.*

Programming languages are referred to as untyped, weakly typed or strongly typed. A strongly typed language requires that a strict conformance of type is enforced at all times. Operations cannot be performed on an object unless the object's **class** possesses an exact signature of the operation or member function. For strongly typed languages such problems can be dealt with during compilation. If violations of type are detected at compilation then a language is referred to as *strongly statically typed.* An untyped language, such as Smalltalk, allows an object to send any message to the object's **class** even if the **class** does not know how to respond to the message. Such errors generally go unnoticed until program execution. If violations of type are allowed to pass compilation and type checking is postponed until run-time then a language is referred to as *dynamically typed.*

Regarding classes, C++ is strongly typed. Consider the shapes program (introduced above) once more:

```
class FuzzyShape
  { /* ... */ };

class Rectangle : public FuzzyShape
  {
  protected:
    double height, width ;
    //...
  public:
    //...
```

```
    double Area () ;          // compute rectangle's area
    //...
  };

class Hexahedra : public FuzzyShape
  {
  protected:
    Rectangle* face_array ;
    //...
  public:
    //...
    double SurfaceArea () ;  // compute surface area
    //...
  };
```

The following assignment is illegal in C++:

```
Hexahedra hex ;
double    area ;
//...
area = hex.Area () ;
```

because the member function *Area()* is not defined for the **class** or any base classes of object hex. However, the following assignments are legal:

```
Rectangle rect ;
Hexahedra hex ;
double    r_area, h_area ;
Position  h_pos ;
//...
r_area = rect.Area () ;
h_area = hex.SurfaceArea () ;
h_pos  = hex.GetPosition () ;
```

The more static typing performed and the more bugs found and fixed at the time of compilation the better. However, static typing alone is too constricting for object-oriented programming. The ideal solution to typing is a mixture of static typing, dynamic binding and polymorphism, which are so essential to object-oriented programming. C++ supports both strong static typing and dynamic binding, but in a controlled manner. By categorising similar abstractions into class hierarchies with a common base class and using polymorphism, a base class object is allowed to point to objects of classes within its own class hierarchy, but is excluded from pointing to objects outside of its respective hierarchy.

1.8.7 Automatic Memory Management

A large object-oriented application will create and destroy numerous objects. The manual task of allocating and freeing memory is a difficult task for a programmer involved in a large object-oriented application. Automatic memory management mechanisms track down memory used by objects that are no longer accessible to an application. Automatic memory management is an important support feature for object-oriented programming, which not only makes a pro-

grammer's life easier but can also reduce programmer errors associated with manual memory management.

C++ does not support automatic memory management, although optional memory management is being considered (Stroustrup, 1994). Several other object-oriented programming languages, such as Oberon-2 and Java (which is a derivative of C++), do support automatic memory management.

1.8.8 Concurrency

Concurrency allows objects to operate simultaneously. The Microsoft Windows NT non-preemptive multi-tasking operating system is a good example of different objects being *alive* at the same time and interacting with one another.

1.8.9 Persistence

Objects in a program require a certain amount of space in memory. Objects also exist for a certain length of time. For example, the lifetime of temporary objects in C++ can be very short, whereas certain objects can outlive the execution lifetime of a program.

An object is referred to as persistent if it outlives its creator or program execution and/or an object survives a move from its original memory address.

1.8.10 Operator Overloading

Although C++ is frequently referred to as an object-oriented programming language because of its support for classes, abstraction, encapsulation, inheritance and polymorphism, an equally important feature of C++ is operator overloading. Having created classes and abstract data types, C++ allows us to overload operators to manipulate objects just as if they were objects of the fundamental data types. Operator overloading is a particularly attractive feature when performing operations on numeric objects. For example, consider objects `m1`, `m2` and `m3` of a `Matrix` **`class`** which models mathematical matrices. If we are required to add the two `Matrix` objects `m1` and `m2` and assign the result to object `m3`, then using **`class`** member functions we could write:

```
//...
Matrix m1, m2, m3 ;
//...
m3 = Add (m1, m2) ;
```

However, operator overloading allows us to overload the addition operator (+) specifically for objects of the `Matrix` **`class`**. Then it is possible to write our `Matrix` object addition in a more natural form:

```
//...
Matrix m1, m2, m3 ;
//...
m3 = m1 + m2 ;
```

1.8.11 The Unified Modeling Language

The Unified Modeling Language (UML) is a standard object-oriented design language for specifying, visualising and documenting the components of software systems. Over the past

few years the UML has been standardised by the Object Management Group (OMG) and has now gained universal acceptance among software developers. The UML represents a collection of best engineering practices that have proven successful in the modelling of large and complex systems. The primary objectives of the UML are to (1) encourage and integrate best practices via a formal, expressive visual modelling language so they can develop and exchange meaningful models, (2) be independent of particular programming languages and development processes and (3) support higher-level development concepts such as frameworks and components. The UML is not part of the C++ language but it is worth being aware of the UML, particularly as you become a more experienced programmer. The best Web site to keep up to date on the UML is that of Rational Rose (`http://www.rational.com/uml/`) who were pioneers in the development of the UML. Three key texts on the UML are Booch *et al.* (1999), Jacobson *et al.* (1999) and Rumbaugh *et al.* (1999).

1.9 Objects

It can be seen from the discussions in the previous sections that an object is a single entity with a well-defined structure and behaviour defined in the object's **`class`**. An object does things and we do things to an object by *sending messages*, via member functions. Thus an object has a state and an identity which can change with the lifetime of the object. You should be aware that an object is also referred to as an *instance*, and member functions are called *methods* or *operations* in other object-oriented programming languages.

Thinking in terms of messages, consider again an object, `c`, of the geometric **`class`** `Circle`, which describes a circle. To determine the area of `c` we send an area message (in the form of a call to the `Circle::Area()` member function) to object `c`:

```
Circle c ;
//...
double area = c.Area () ;  // compute area of object c
```

The object itself processes the area message. If we send the same area message to other objects, such as `Rectangle` and `Triangle`, then the *message–object–process* is identical and independent of the type of object. Each object processes the area message appropriately. This is contrary to procedural programming, which would place emphasis on the area procedure rather than the objects.

1.10 Syntax, Semantics and Pragmatics

Frequently in programming we hear the words *syntax*, *semantics* and *pragmatics*. The following three sections describe what these buzzwords mean.

1.10.1 Syntax

The syntax of a programming language consists of the rules for the correct use of the language. These involve the correct grammatical construction and arrangement of the language, correct spelling, hyphenation, inflection and so on. Although the syntactical rules of a programming

language can be difficult at first, they are far simpler than the syntactical rules of a natural language. The syntax of a programming language has to be strictly adhered to.

1.10.2 Semantics

The semantics of a programming language deal with the meanings given to syntactically correct constructs of the language. Semantics also deal with the relationships between symbols and the ideas given to the symbols.

1.10.3 Pragmatics

The pragmatics of a programming language deal with the practical application of the language, not the theory. Computer programs are designed and written to solve problems, and their ultimate success depends on how they solve the problem to which they were designed. Large, complex programs are continually changing according to the changing demands placed upon them. Thus, pragmatics also deals with the evolution of a program.

1.11 Obtaining a C++ Compiler

Several commercial C++ compilers exist. To name a few of the more popular compilers there are Borland C++/C++ Builder (`http://www.borland.com/`), Microsoft Visual C++ (`http://www.microsoft.com/`), Symantec C++ (`http://www.symantec.com/`) and KAI C++ (`http://www.kai.com/`). At the time of going to press a free compiler (version 5.5) can be downloaded from the Borland Web site. Furthermore, the free GNU C/C++ compiler (gcc/g++) for the Linux operating system can be obtained from the Free Software Foundation (`http://www.gnu.org/` or specifically `http://gcc.gnu.org/`).

1.12 Applications of C++

Although my suggestions of typical applications of C++ are biased and some what limited, here are a few books which illustrate the diverse fields to which C++ is now applied.

Masters, T. (1993) *Practical Neural Network Recipes in C++*, Academic Press, New York.
Porter, A. (1993) *C++ Programming for Windows*, Osborne–McGraw-Hill, New York.
Rao, V.B. and Rao, H.V. (1993) *C++ Neural Networks and Fuzzy Logic*, MIS Press, New York.
Stevens, R.T. (1992) *Fractal Programming and Ray Tracing with C++*, Prentice Hall, Englewood Cliffs NJ.
Wilt, N. (1994) *Object-Oriented Ray Tracing in C++*, John Wiley & Sons, New York.

1.13 References for C++

If you are interested in referring to alternative or additional published work on C++, then a glance through the references at the end of the book should supply you with more than enough material to get going. Of the works referenced, my favourites are:

Lafore, R. (1991) *Object-Oriented Programming in Turbo C++*, Waite Group Press, Mill Valley CA.

You would be hard pushed to find a better introductory text to C++ than this, which is now in its second edition. Robert Lafore has an excellent style of writing and presentation, typical of authors for the Waite Group Press.

Adams, J., Leestma, S. and Nyhoff, N. (1995) *C++: An Introduction to Computing*, Prentice Hall, Englewood Cliffs NJ.

This book simultaneously introduces the reader to computing and C++. This book is an excellent text for student learning and well supported with numerous examples and programming tips.

Schildt, H. (1994) *C++ from the Ground Up* and Schildt, H. (1995) *C++: The Complete Reference*, both published by Osborne–McGraw-Hill, New York.

A list of recommended reading on C++ would not be complete without a couple of references to Schildt's many C and C++ programming books. These two, of his more recent books, are excellent first books to learn C++. When it comes to writing C++ books, Schildt is simply a magician!

Lippman, S.B. (1991) *C++ Primer*, Addison-Wesley, Reading MA.

For a more in depth study of C++, this book is a must.

Barton, J.J. and Nackman, L.R. (1994) *Scientific and Engineering C++: An Introduction with Advanced Techniques and Examples*, Addison-Wesley, Reading MA.

For those of you interested in the scientific and engineering applications of C++, this well-written book is essential reading. It helps scientific and engineering programmers in the transition from Fortran and C to C++, covering a wide range of applications of C++.

Booch, G. (1994) *Object-Oriented Analysis and Design with Applications*, Benjamin/Cummings, New York.

Although not specifically a text on C++ programming, this book provides an invaluable text on object-oriented programming, design and analysis. All example code in the book is implemented in C++.

Swan, T. (1999) *Tom Swan's GNU C++ for Linux*, MacMillan.

If your interests are specifically C++ for the GNU/Linux operating system then Swan's book is a must.

If you require a standard on the C++ language:

Ellis, M.A. and Stroustrup, B. (1990) *The Annotated C++ Reference Manual*, Addison-Wesley, Reading MA.

The first reference manual on C++ including expected future extensions of the language.

Stroustrup, B. (1997) *The C++ Programming Language*, 3rd edn, Addison-Wesley, Reading MA.

Although not primarily a reference manual, one is included.

Details of the latest working paper for the ANSI/ISO C++ standard can be found at Standard (1998).

If your interests are more up to date then some key journals that feature C++ are:

The C++ Report
The C++ Journal
The C/C++ Users' Journal
The Journal of Object-Oriented Programming (JOOP)
Object-Oriented Systems
Computer Language

USENIX (Unix Users' association) conference proceedings: the first C++ Workshop conference in November 1987 marked the start of a series of conferences held by USENIX in October 1988, April 1990, April 1991, August 1992 and July 1993. Refer to USENIX in the references section for further details. Also refer to the proceedings of the OOPSLA, ECOOP and C++ At Work conferences.

C++ Internet newsgroups: `comp.lang.c++`, `comp.std.c++` and `comp.object` for discussions on the C++ language, the Standard Template Library (STL) and object-oriented programming respectively. For example, the C++ newsgroup can be accessed by entering `news:comp.lang.c++` in the Location field of an Internet browser.

1.14 References for Graphics

Mortenson, M.E. (1989) *Computer Graphics: An Introduction to the Mathematics and Geometry*, Industrial Press.

This is a good first text on computer graphics. The book covers points, lines, planes, curves, surfaces, projections, coordinate systems and transformations.

I also recommend the works of Glaeser (1994), O'Rourke (1994), Preparata and Shamos (1985), Stevens (1994), Watt and Watt (1992) and Wilt (1994). More recently, Laszlo (1996) has addressed the topics of computational geometry and computer graphics in C++.

1.15 Notation

The notational convention used throughout this book is as follows:

`Courier`	User-defined identifiers, objects, classes, etc.
`Courier Italic`	User comments; C/C++ and user-defined macro and function names

`Courier Bold`	C++ keywords
CAPITALS	Disk directories and filenames; user-defined constants
File | Save	Save command from the File menu
.CPP, .H	Filename extensions for C++ implementation and header files respectively
`//...`	More program code considered not essential to the present discussion.

1.16 Summary

C++ is a superset of the C language. C++ not only adds object-oriented programming capabilities to C, but also adds numerous other features which make programming safer, easier and more enjoyable. Object-oriented programming helps focus a programmer's attention more on the design and organisation of a program than on the details. The fundamental concept of object-oriented programming is the object. Objects possess their own properties and operations – *objects do things.*

Exercises

1.1 What do you understand by object-oriented programming? Highlight the key features which define a programming language to be object-oriented.

1.2 What is meant by an *object*?

1.3 If you do not already possess or have access to the ANSI/ISO C++ standard, then try getting hold of a copy – it's an impressive document! For further details of how to obtain the C++ standard refer to Standard (1998).

1.4 What are the main differences between abstraction, encapsulation and inheritance?

1.5 Obtain a copy of and read the following paper: Parnas, D.L. (1972) On the criteria to be used in decomposing systems into modules, *Communications of the ACM*, **15**(12), 1059–62. This paper was instrumental in introducing the concept of separating a program's implementation form its interface which is at the heart of data-hiding and encapsulation.

1.6 What are the essential abstractions of a triangle and a polygon? Identify those abstractions that are common to both objects.

1.7 Try to find out more about other object-oriented programming languages. Is C++ the best object-oriented programming language? What are the advantages and disadvantages of C++ compared with other object and non-object-oriented programming languages?

Chapter 2

The Development Environment

Your working environment is very important. If you are to do a lot of programming, optimise your program development and enjoy your programming as much as possible, then you naturally need the best environment. This chapter attempts to highlight a few key requirements and tools that you should look for in your environment by providing an overview of the Borland C++ for Windows compiler and development suite.

2.1 Hardware, Software and Setup Used

All of the program code presented throughout this book was written and prepared with the help of the following hardware and software.

2.1.1 Hardware

An IBM-compatible desktop PC with an Intel Pentium Pro 200 MHz processor, 256 Mbyte RAM, 8 Mbyte video RAM, 10 Gbyte hard disk and an SVGA monitor was used exclusively for all development. This specification was found to be an excellent platform for all of the programs presented in this book, with quick compile and link times.

2.1.2 Software

The compiler used was Borland's Borland C++ for Windows, version 5.x (2000). Why Borland? Basically, because I have used the Borland C++ compiler since 1990, when it was (and still is, in my opinion), the leader in commercial C++ compilers for the PC. Incidentally, Borland first released their C++ compiler in May 1990. Although it is not for me to state which is the best C++ compiler on the market for your particular applications, you should, if you can, simply choose the best compiler and environment to suit your needs and style.

No keywords, libraries, container classes, ObjectWindows Library (OWL) classes or features specific to the Borland compiler have been used in this book and all programs provided on the

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supplied disk should (in theory) be portable to any preferred draft ANSI/ISO C++-compatible compiler.

2.2 The Borland Integrated Development Environment

Commercial compilers such as Borland C++, Microsoft Visual C++ and Symantec C++ for Windows all support excellent development environments. This section presents a brief overview of the Borland C++ compiler and development tools.

The Borland Integrated Development Suite is an excellent environment to work in (Fig. 2.1). Apart from the Compiler, Linker and Librarian, it includes such Windows tools as a customisable Editor, SpeedBar, Browser, Debugger, WinSight and WinSpector, AppExpert, ClassExpert, Project Manager, and Resource Workshop. Command-line tools such as MAKE, TLINK and TLIB are also available. Borland C++ also fully implements the ANSI C standard with numerous additions. The list of tools and features is endless. Let us take a brief look at a few of the key features offered by the Borland C++ Integrated Development Environment (IDE).

2.2.1 Editor

Figure 2.1 illustrates a typical editing session. Since the client area of the IDE is a Multiple Document Interface (MDI), we can have multiple files open simultaneously. The editor is completely customisable.

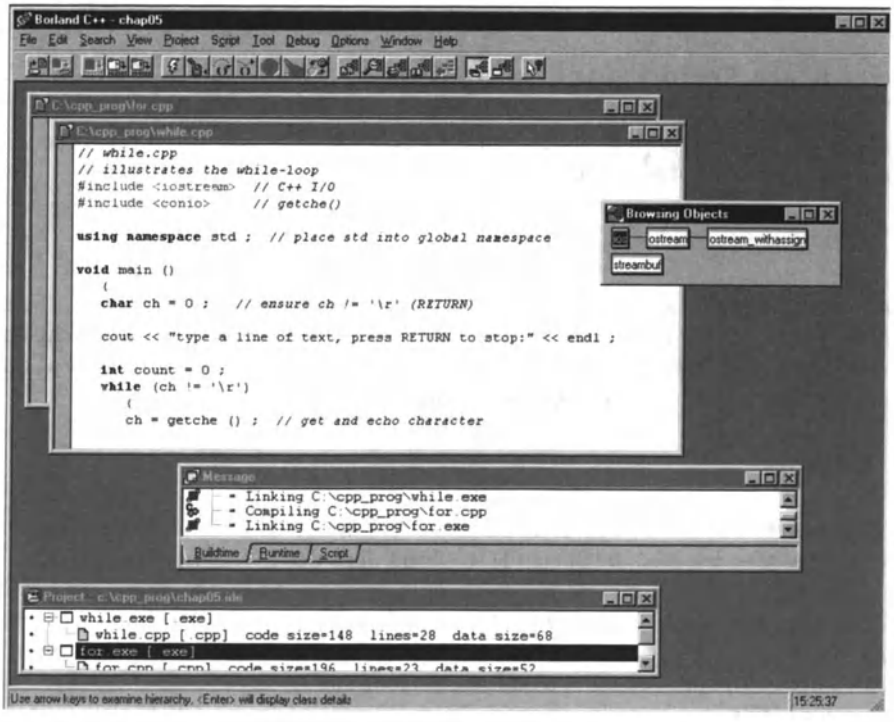

Fig. 2.1 The Borland IDE.

2.2.2 Text Highlighting

Syntax highlighting is a very useful facility. It allows the different elements of your code to have associated colour and font attributes. Keywords are, by default, shown in bold type, identifiers in plain type, comments in purple, strings in blue and so on. For me, syntax highlighting is not simply a nifty gimmick but greatly enhances my understanding of the code and helps to eliminate silly syntactical errors.

Syntax highlighting works by default on files with the extensions .CPP, .C, .H and .HPP. Additional file extensions can be added to the list.

2.2.3 Project Manager

The Project Manager manages applications that are built from many and varied components each requiring a different compilation. Figure 2.1 illustrates an example of the Project Manager window. The project tree shows the dependencies of each *node*. A project node represents the entire project, with all files required to build the project appearing under it. A target node (such as .EXE or .DLL) represents a file that is created when its dependants are built. A node (e.g. .CPP) represents a file used to build a target. A run-time node (e.g. .OBJ and .LIB) is a file used at run-time. The Project Manager relieves the programmer of having to generate a make file (.MAK). Project files (.IDE) can be converted to make files by the IDE with the 'Project | Generate makefile' option. Project Manager has an *autodependency* check which automatically translates all out-of-date nodes (.OBJ, .LIB) in the project. If a node is up-to-date then it is not re-translated.

The Borland IDE can be configured by the *ObjectScripting* feature, which uses the cScript language to make the IDE fully programmable.

2.2.4 Messages

Error and warning messages are displayed in a Message window when a program is compiled and linked. When a message is selected from the Message window the Editor automatically transfers to the Edit window and places the cursor on the line and column within your program code where the warning or error is most likely to have occurred. This is a useful feature that greatly speeds up the debugging phase of program development; see Fig. 2.1.

2.2.5 Browser

The Browser allows you to view graphically the **`class`** hierarchy, **`virtual`** and member functions, variables, constants, types, classes and objects of a program. Figure 2.1 illustrates a browsing window. The *Browsing Objects* window shows the **`class`** hierarchy of the `ios` C++ standard library **`class`**.

2.2.6 AppExpert

AppExpert lets you generate a program shell for Windows. The program is automatically generated and includes such features as MDI windows, a menu, a status line, a button bar, drag and drop support, on-line help and printing support. AppExpert offers a visual approach to Windows program development and interfaces with the Resource Workshop, ObjectWindows **`class`** library and the Project Manager. AppExpert also caters for Java-specific applications.

2.2.7 ClassExpert

The Borland ClassExpert assists the development of AppExpert programs by providing three window panes: *Edit*, *Events* and *Classes*. The Edit pane lets you edit your program code. The Classes pane lists the classes that ClassExpert manages. The Events pane lists Command-Notifications (e.g. CM_EDITCUT), Windows-Messages (e.g. WM_PAINT) and **`virtual`** functions of the selected **`class`**'s base classes. ClassExpert interfaces with the Resource Workshop to enable you to associate resources to classes.

2.2.8 Debugger

No matter how good a programmer you are the programs that you develop will always have their fair share of bugs in the early stages of development. The Borland Integrated Debugger assists you in locating bugs in your program which prevent it from executing correctly. The Borland CodeGuard 32/16 assists programmers in detecting program bugs at run-time for both 16- and 32-bit Windows applications. CodeGuard 32/16 validates both pointers and pointer arithmetic at the point of execution and is a useful feature for detecting memory and resource leaks.

2.2.9 Resource Workshop

Resources are the visual parts of a Windows program, such as dialog boxes, menus, bitmaps, icons, cursors, fonts and user-defined resources. Generally, resources are kept separate from a Windows program (in a resource compiler script file .RC, for example) so that program code is not altered when alterations are made to resources and also to enable resources to be used by other programs. The Borland Resource Workshop has both a text and a graphical resource editor.

2.2.10 Help

The Borland C++ IDE has an on-line help system that gives immediate access to detailed information about several key topics, such as C++ keywords and standard library class and function declarations. The Borland help system is subdivided into key topics such as Essentials, Tools, Tasks, Language Reference and Master Index. A search can be performed on any C++ language feature, keywords, macros, functions, classes, Windows API, Container and ObjectWindows **`class`** libraries and so on. The Master Index is a complete reference to language features in the Borland C++ supplied documentation. Also, examples are given for several key language features.

Figure 2.2 is an extract from Borland C++ help on the keyword **`delete`**. An underline indicates a jump to an additional, related topic.

2.2.11 Java Development Tools

The Borland C++ (version 5.x) compiler includes development tools for the Java programming language, including Sun's Java Development Kit, which is fully integrated with Borland's IDE. The Java compiler is supported with a GUI debugger.

delete

See Also Example Operators

Syntax

```
<::> delete <cast-expression>
<::> delete [ ] <cast-expression>
delete <array-name> [ ];
```

Description

The delete operator offers dynamic storage deallocation, deallocating a memory block allocated by a previous call to new. It is similar but superior to the standard library function free. You can also use the delete operator to remove arrays that you no longer need.

Fig. 2.2 An extract from Borland C++ help on the keyword **delete**.

2.3 Container Class and ObjectWindows Libraries

Borland C++ contains an extensive library of container classes (also known as the Borland International Data Structures (BIDS)) and ObjectWindows classes. Using such classes saves you the massive task of developing such classes yourself and thus prevents you from reinventing the wheel. In addition, these classes have been developed over a number of years by professional programmers and are, as a result, efficient and robust.

Containers are objects that implement common data structures. Borland containers are implemented using *templates*, allowing you to pass into the template any C++ or user-defined type or **class**, while ensuring correct data-hiding and providing the appropriate access to the container **class**'s data members via member functions. The Borland container **class** library is composed of two categories: Fundamental Data Structures (FDS) and Abstract Data Structures (ADS):

Fundamental Data Structures	Abstract Data Structures
Binary tree	Array
Hash table	Association
Linked list	Dequeue
Double-linked list	Dictionary
Vector	Queue
	Set
	Stack

The ObjectWindows library classes (version 5.x) are classes specifically developed to assist a programmer in developing applications for Windows 95, 98 and NT more easily and quickly, and in an object-oriented programming fashion. The library is extensive and I cannot do it justice here, but a few ObjectWindows classes are listed below:

`TApplication`	Application **`class`** supplying the basic behaviour of a Windows program.
`TDC`	Windows GDI device context **`class`** performing graphical operations.
`TDialog`	Modal and modeless dialog box **`class`**.
`TRect`	Encapsulates a two-dimensional rectangle.
`TPoint`	Encapsulates a two-dimensional point (x, y).
`TWindow`	Provides a generic window that can be manipulated as required. `TWindow` encapsulates many of the Windows API functions.

2.4 Summary

Your programming environment is very important. Spend some time getting to know it. Understanding your environment will make you a better C++ programmer and a happier one!

An environment such as the Borland C++ Integrated Development Environment for Windows is a world of features, tools and libraries that can reduce your development time. Becoming too dependent, however, on a particular programming environment can make the transition from one environment to another very difficult – don't get hooked.

Exercises

2.1 Get to know your compiler and your compiler's development environment.

2.2 Examine your compiler's documentation and on-line information regarding ANSI/ ISO C++ compatibility and portability with respect to function and class libraries.

2.3 Investigate which libraries your compiler is shipped with. Look for both function and **`class`** (and container classes) libraries for developing command-line and GUI programs.

2.4 See whether your compiler has several tutorial programs to help you become familiar with the processes of compiling and linking a typical program.

2.5 What program debugging tools does your compiler support?

2.6 If your compiler supports an on-line help facility, try to obtain some information about the *`main()`* function.

2.7 In the next chapter we shall examine a program file called MAIN.CPP. Create either a project or make file for this single program file.

Chapter 3

Getting Started

This chapter introduces us to programming in C++. As with any language (not just computer languages) there are several fundamentals that you have to acquire before you can do even the simplest of things. With the aid of the simplest of C++ programs, this chapter illustrates several of the key elements of program construction, such as language keywords, the main() function, program statements and directing program output to the display screen. The program also introduces the processes of compiling and linking a program, user comments, use of the preprocessor and header files and stresses the need for developing a consistent programming style. A good programming style will make your programs more readable and less prone to syntactical errors, and will help you to communicate better with fellow C++ programmers.

3.1 The Simplest of Programs

The simplest of C++ programs, which produces an output of *hello, object*, is:

```
// main.cpp
#include <iostream>

void main ()
  {
  cout << "hello, object" << endl ;
  }
```

This program follows the now traditional manner of beginning introductory books on C and C++ programming, after the book *The C Programming Language* by Kernighan and Ritchie (1978). See also the second edition (Kernighan and Ritchie, 1988).

Note that an even simpler (actually the minimal C/C++ program) is:

```
main ()
  {}
```


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So, what does the above program mean and what does it actually do?

You can either type the program into your computer yourself or use the supplied file MAIN.CPP (C:\CPP_PROG\CHAP03\MAIN.CPP, assuming that the file resides on the C drive). At this stage it is recommended that you type the program yourself to gain confidence with your editing environment and compiler, but ensure that you pay particular attention to the case of the characters in the code. The filename extension .CPP (C Plus Plus) is used by Borland C++ compilers. The extensions .CC, .CP and .C++ are frequently used by other C++ compilers.

Let us now examine each line of the program separately in slightly more detail. If certain material in what follows appears confusing, 'don't panic' – it will become clearer (believe me) as we gradually work through the chapters in order. Also, when you have successfully typed in, compiled and linked the program, don't be afraid to experiment with the program, particularly with the comment and the output message.

3.2 The Comment Line

The first line of the program is:

```
// main.cpp
```

The double forward slash comment characters (//) inform the compiler that what is to the right-hand side of // is a *comment* inserted within the program body and is to be ignored during compilation. The comment extends to the end of the line (which will depend on the editor that you are using) and no closing comment characters are required.

The original C style of explicitly terminated comments involves the use of the characters (/*) at the beginning of the comment followed by the characters (*/) at the end of the comment:

```
/* main.cpp */
```

No spaces are allowed between the characters of //, /* and */; i.e. / / or / * are illegal. For single-line comments, as is the case here, the (//) style is generally preferred, whereas the (/*...*/) style is preferred for multi-line comments:

```
/*
main.cpp
The simplest of programs

John Smith, 1st. September '00
(c) Smith & Jones Ltd.
All rights reserved
*/
```

In addition to the name of the program (MAIN.CPP), a title, name of programmer, date and copyright have been added as comments.

Comments are not just reserved for an entire line or a single block as above. The beauty of commenting is that it can be integrated neatly into the program code:

```
// main.cpp
#include <iostream> // header file for C++ I/O
void main ()
```

```
{
cout << "hello, object" << endl ;
} // end of main()
```

As you become more experienced as a programmer and the problems that you tackle naturally become more complex then the need for good, concise commenting becomes more important. As with programming style, to be discussed shortly, commenting a program as you develop the program is a simple skill well worth mastering for two main reasons. Firstly, the process of commenting helps us to concentrate and ask ourselves 'What am I actually trying to do here?', just as writing down ideas on paper by hand assists our concentration. Secondly, life is an awful lot easier for yourself, and particularly others, when reading your code on returning to a piece of code some time after it was originally written. Generally, if you have a good understanding of the problem at hand then commenting is straightforward.

A few problems can arise with commenting that you should be aware of. The (//) style of comment finishes at the end of a line, with the /*, */ and // characters being treated just like any other character within a // comment:

```
// main.cpp /* not main.c */ // or main.c++
```

Since the // style comment extends to the end of the line, be careful of comments of the form:

```
void function ()
  { // a function without a body }
```

in which case the closing brace, }, is just another comment character and not the end of the function body.

Nested comments are, strictly speaking, not allowed in C and C++. An example of nested comments is:

```
/*
main.cpp
/*
The simplest of programs
*/
*/
```

Pre-ANSI C compilers allowed nested comments, and as a result several commercial compilers possess an option of allowing programs to contain nested comments (e.g. Borland's Borland C++ for Windows (version 5.x): Project Options | Compiler | Source *Nested comments* checkbox). However, try not to use nested comments, because they are generally too confusing.

It is worth noting that comments anywhere within a program have no effect on the executable code size or on the speed of execution of a program. Comments are removed by the compiler before any program code is generated by the compiler. This can be tested by adding the comment line // end of program to the end of the program and comparing the execution size (.EXE) of this program with that of the original program. When I performed this comparison the execution size of both programs was 79,240 bytes. Your execution size may vary from this value depending upon differences in code generation for different compilers or different versions of the same compiler. So, if comments are free then make good use of them.

3.3 Preprocessor Directives

The second line of the program:

```
#include <iostream>
```

contains a *preprocessor directive* that informs the compiler to read a file named IOSTREAM.H into the program. A preprocessor directive is a command to the compiler's preprocessor that deals with instructions before compilation begins. A preprocessor directive begins with the number (or hash) symbol (#).

There are several preprocessor directives, such as `#define`, `#if` and `#endif`. The `#include` directive instructs the compiler to insert the *header* or *include* file IOSTREAM.H into the program at the point where the directive occurs. For example, suppose you have a file called INFO.H which contains relevant information:

```
/*
The simplest of programs

John Smith, 1st. September '00
(c) Smith & Jones Ltd.
All rights reserved
*/
```

If we rewrite the MAIN.CPP program as:

```
#include "info.h"
#include <iostream>

void main ()
  {
  cout << "hello, object" << endl ;
  }
```

then after the preprocessor pass the program, from the compiler's perspective, will be:

```
/*
The simplest of programs

John Smith, 1st. September '00
(c) Smith & Jones Ltd.
All rights reserved
*/
#include <iostream>

void main ()
  {
  cout << "hello, object" << endl ;
  }
```

The `#include` preprocessor directive has inserted the header file INFO.H into the program.

3.4 What is IOSTREAM.H?

IOSTREAM.H contains **class** declarations that are essential for programs that want to perform input and output; namely `basic_istream`, `basic_ostream` and `basic_iostream`. The file derives its name from input/output (io) stream. A *stream* is in fact an abstraction of data flow. In order to use the `cout` identifier and << operator (which we will examine shortly) the compiler needs to have their declarations by `include`ing the appropriate header file.

Header files should, in general, not contain any executable program code, and a glance through IOSTREAM.H confirms that the file consists entirely of **class** declarations for input and output operations. The implementation of these **class** member functions is restricted to an implementation file (.CPP). If you do intend to look through IOSTREAM.H then make sure that you *do not* alter the file in any way. Such standard C++ header files are essential to the correct operation of your programs.

3.5 " " or <>

You may have noticed above that the file INFO.H is enclosed between double quotes (" "), whereas IOSTREAM.H is enclosed between angle brackets (<>). The `#include` preprocessor directive may take one of two forms for header files:

```
#include <header_filename>
#include "header_filename"
```

`header_filename` must be a valid filename with an extension traditionally of the form .H for header. The pathname and delimiters are optional but if specified then the compiler searches only the specified location. For instance we could use the exact path for INFO.H:

```
#include "c:\cpp_prog\chap03\info.h"
```

Note that a single backslash character (\) is sufficient to indicate subdirectories for a preprocessor directive.

Ensure that there are no spaces between the filename and <> or " ". For example, the following include preprocessor directive will not be able to locate IOSTREAM.H:

```
#include < iostream >  // error: unable to locate file
```

(<>) indicates a *standard* C++ include file and the preprocessor looks for the file in the Include directories in the order that they are defined. For the Borland C++ compiler (see Options | Project | Directories) the Include directories resident on the C drive are:

```
C:\BC5\INCLUDE;C:\BC5\INCLUDE\OWL;C:\BC5\INCLUDE\CLASSLIB
```

(" ") indicates a user-defined include file, and the preprocessor first looks in the same directory as the file containing the `#include` directive. If the include file is not found then the search proceeds to the current directory which is usually the directory where the source file being compiled resides. If the file `included` between double quotes is not found, then the include directories are searched, as in the (<>) case.

Having these two different `#includes` for header files allows the standard C++ header files provided with your compiler to be kept separate from user-defined header files. Also, making the correct choice between (" ") and (<>) will speed up compilation.

C++ inherited the C-style of including header files by using the .H suffix; for example:

```
#include <iostream.h>
```

However, C++ now supports the new style of naming header files by name only:

```
#include <iostream>
```

This style does not mean that there are two versions of the input/output stream files (i.e. IOSTREAM.H and IOSTREAM), but simply that the .H suffix is redundant, since the angle brackets <> denote a standard library file. In fact, the majority of standard header files are stored using the .H suffix. The new style came about following C++'s recent standardisation because new header files were required for redefined versions of standard library routines. Simply using the old .H names would lead to a backward compatibility problem, so the decision was made to drop the .H suffix in standard header files.

As a final point, it is worth noting that some compilers interpret the inclusion of the .H suffix as a request for backward compatibility.

3.5.1 Other Header Files

If you browse through the subdirectories of your compiler you will probably find a directory called INCLUDE or HEADERS. For the Borland C++ (version 5.x) compiler the directory name is C:\BC5\INCLUDE. This directory contains several header files, such as _DEFS.H, IOSTREAM.H and WSNWLINK.H. We shall discuss header files in more detail at a later stage when we discuss libraries, but for now it is enough to know that each header file is an interface to a particular library.

3.6 Keywords

A keyword, or reserved word, such as **`void`**, is a feature of the C++ language which has associated with it some special meaning. Other keywords include **`char`**, **`int`** and **`class`**.

A keyword cannot be used as an identifier, object or function name:

```
//...
void void ()    // error
   {
   //...
   int char ;    // error
   //...
   }
```

Appendix A lists all the keywords of the C++ programming language.

3.7 The *main()* Function

Every C and C++ console-based program must call the *main()* function at program-startup:

```
void main ()
  { /* ... */ }
```

It is possible not to use *main()* and define an alternative function name:

```
void my_main ()
//...
```

and still have successful program compilation. However, if there is no *main()* function in your program the linker will produce an error depending on the memory model used.

Inter-word spaces are not allowed for keywords, function names, **class** names etc. Thus *vo id* and *my main()* are illegal and hence the reason for using the underscore: *my_ main()*.

The function name main is followed by a pair of parentheses (opening and closing). Since main is not a *keyword* in C++, the parentheses after the function name indicate to the compiler that *main()* is a function.

Note that *main()* is not a user-defined function. The convention used throughout this book is that user-defined function names begin with an uppercase letter. Thus, the equivalent user-defined *main()* function would be *Main()*. These two functions, *main()* and *Main()*, are two different functions because C++ is a case-sensitive language. Let us just emphasise this important point:

C++ is a case-sensitive language

The following program illustrates this:

```
// capitalm.cpp
// illustrates that C++ is a case-sensitive language
#include <iostream>

void main ()
  {
  cout << "hello, object" << endl ;
  }

void Main ()  // O.K.- Main() is different from main()
  {
  cout << "Hello, Object" << endl ;
  }
```

If the uppercase letter M in *Main()* is replaced with a lowercase m you will get a compilation error of the form 'function main() already defined'. Therefore *main()*, *Main()* and *MAIN()* all have different meanings in C++. Make sure that the *main()* you use is always in lowercase.

Functions, C++ and user-defined, will be discussed in more detail at a later stage, but for now the essential components of a function are:

```
return_type FunctionName (parameter_list)
  {  // opening brace
  /*

  function body

  */
  }  // closing brace
```

`return_type:`	The type of the object/value, or **`void`**, returned by the function.
`FunctionName:`	Name of the function.
`parameter_list:`	List of *parameters*, or **`void`**, passed to the function. Also known a *function arguments* or formal arguments.
`function body:`	The opening brace, `{`, indicates the start of the function, while the closing brace, `}`, indicates the end of the function body. Any code that appears between the pair of braces `{` and `}` is the function body.

Notice that *`main()`* has no parameter list. If a function does not take any function parameters you can use the **`void`** keyword to indicate this explicitly:

```
void main (void)
  { /* ... */ }
```

Generally, *`main()`* and *`main(void)`* are equivalent, but there are some C++ implementations where this is not the case. Therefore, if you use **`void`** for the parameter list be sure that the function has no parameters.

`void` as a function return type indicates that the function does not return an object or value. Although **`int`** is the default return type of *`main()`*, some compilers insist that *`main()`* returns an **`int`**. The **`int`** keyword (or *type specifier* to be more specific) will be discussed later, but if you are getting any warning messages concerning the return type of *`main()`* then use the following format:

```
int main ()
  {
  //...
  return 0 ;  // main() returns 0 to the operating system
  }
```

placing the **`return`** `0` just before the end brace of *`main()`*. The return value of the *`main()`* function indicates to the operating system the program's exit status. Generally, a return value of 0 indicates that the program terminated normally, whereas a return value of -1 indicates abnormal termination.

3.8 Program Statements

Program statements are the fundamental units of C++ programming. The program statement:

```
cout << "hello, object" << endl ;
```

instructs the computer to display the string constant *hello, object* followed by a new line. Try recompiling the program leaving out the semicolon (;) – this is an error. The semicolon indicates the end of the statement. The compiler ensures that a statement is completed before the next statement is executed. The semicolon is easily forgotten, so get into the habit early of ending with a semicolon. Fortunately, compilers are very good at detecting missing semicolons, so this is not a big deal.

The semicolon can work to your advantage sometimes by offering a more succinct form of assignment:

```
int x, y, z ;
//...
x = 1 ; y = 2 ; z = 3 ;
```

rather than:

```
//...
x = 1 ;
y = 2 ;
z = 3 ;
```

The minimal statement possible in C++ is the *null statement*, which simply consists of a semicolon:

```
//...
;
```

3.8.1 The `cout` Identifier/Object

`cout` (common `output`) is a standard output stream identifier or object. Output, by default, is directed to the screen, but can be redirected to alternative devices.

3.8.2 The << Insertion Operator

<< is the *insertion* or 'put-to' operator. It simply directs *what's on the right* of << to *what's on the left* of <<. In the present example << directs the string *hello, object* to `cout` for displaying.

The << operator can be *cascaded* or *chained*. Cascading is the multiple use of << within a single statement. For instance:

```
cout << "hello, object" << endl ;
```

first of all sends the string *hello, object* to `cout` and then sends `endl` to `cout`.

If you are familiar with C, the equivalent would be:

```
printf ("hello, object\n") ;  // need to: #include <stdio.h>
```

which is function-based and messy.

3.8.3 String Constants

The text *hello, object* between the double quotes is a string constant. Being a constant, the string does not alter throughout the entire lifetime of the program.

3.8.4 The `endl` Manipulator

The `endl` manipulator inserts a line feed into the stream. This ensures that any following output is on a new line and not a continuation of the present line. You are completely free to choose where you place `endl` and how many new lines you require; for example:

```
cout << endl << "hello, object" << endl << endl ;
```

which inserts a new line before the output and two new lines after the string. Note that an insertion operator is required for each `endl`. The following is illegal:

```
cout << "hello, object" << endl endl ;  // error
```

`endl` is equivalent to the C '`\n`' *escape sequence* – just more elegant.

3.9 The Standard Library Namespace

A detailed discussion of *namespaces* will be postponed until Chapter 19. However, it is worth briefly mentioning the C++ standard library namespace at this stage, since you may be experiencing compilation errors on building the MAIN.CPP program. Every function, class etc. of the standard library is defined in a **`namespace`** called `std`. This ensures that name clashes between similarly defined functions and classes in other libraries are prevented. As a result, for the MAIN.CPP program to adhere strictly to the `std` **`namespace`** we should have written:

```
// main_std.cpp
#include <iostream>

void main ()
  {
  std::cout << "hello, object" << std::endl ;
  }
```

The `std::` prefix to `cout` and `endl` informs the compiler that both the `cout` and `endl` objects are members of the `std` **`namespace`**. All C++ standard library functions and classes must be similarly prefixed. Aternatively, the `std` **`namespace`** can be placed in to the global namespace by way of the *using-directive*:

```
// main_std1.cpp
#include <iostream>

using namespace std ; // place std into global namespace

void main ()
  {
  cout << "hello, object" << endl ;
  }
```

with `cout` and `endl` now called as before. In general it is bad practice to place an entire namespace in to the global namespace, but for the sake of brevity the `std::` prefix will be used

sparingly throughout the book although present in the source code files. It was noted in Section 3.5 that some compilers interpret the inclusion of the .H suffix as a request for backward compatibility and hence a request to place `std` in to the global namespace.

3.10 Compiling and Linking

The program source code in MAIN.CPP is a text file. To generate an executable program there are two steps that need to be performed: *compiling* and *linking*.

3.10.1 Compiling

First, your source code is compiled into an object file (.OBJ), which is not to be confused with an object in object-oriented programming. An object file contains machine-language instructions which can be executed by a computer. Standard C++ and user-defined header files are `#included` in your source code by the preprocessor. The header files will contain information necessary for compilation, such as IOSTREAM.H for the `cout` identifier/object and << operator in the example above. Depending on the memory model used, the input/output functions, classes and objects are in a library routine (such as CWL.LIB for a large memory model).

The Borland C++ IDE lets you select either the menu option 'Project | Compile' or a button on the SpeedBar to compile a program. The resulting dialog box provides statistics on the number of warnings and errors and the number of lines generated.

3.10.2 Linking

Invariably, a program consists of a number of different object and library files. Linking simply combines the separate object and library files into a single executable program. Provided that the necessary object and library files exist and are correctly accessed, the linker will extract the appropriate information from the files and generate the executable program.

As with compiling, the Borland C++ IDE allows you to link a program either by selecting 'Project | Make all' or by the compile and link button on the SpeedBar. Similarly, linking statistics are generated.

Figure 3.1 schematically illustrates the general process of compile and link.

3.10.3 Note for GUI Compilers

If you are using an IDE compiler for Windows such as Borland C++ or C++Builder then you may experiencing difficulties in displaying the console application program MAIN.CPP. When the program executes, the DOS box may just be displayed for a split second, making it impossible to read the program output. This is because at the end of *`main()`* the program terminates and the DOS console window immediately closes. This is easily remedied by adding a call to the C standard library function *`getch()`* which will cause the program execution to pause until any key from the keyboard has been entered. In order to use *`getch()`* the CONIO.H header file must be included:

```
// main_con.cpp
//...
#include <conio>  // getch()
```

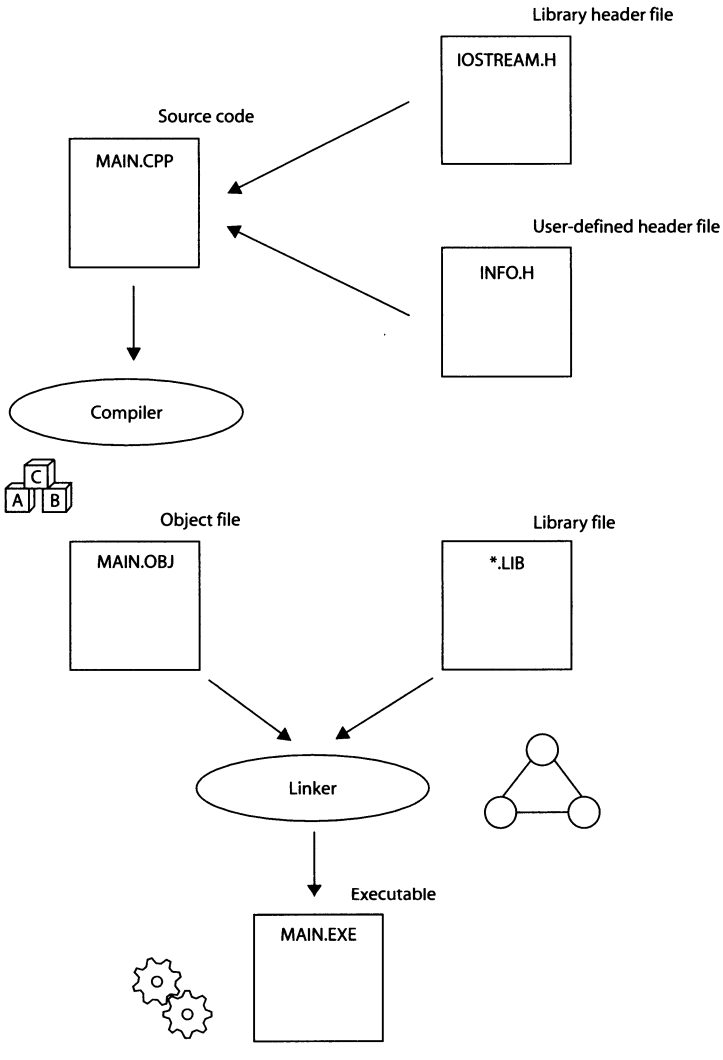

Fig. 3.1 Compilation and linking.

```
//...
void main ()
  {
  //...
  cout << "Press any key to continue..." ;
  getch () ;
  }
```

The output of the program will now read:

```
hello, object
Press any key to continue...
```

Alternatively, if you prefer a C++ object approach then the member function `get()` of the input stream object `cin` can be used instead of `getch()`:

```
//...
void main ()
  {
  //...
  cout << "Press any key to continue..." ;
  cin.get () ;
  }
```

This approach is also illustrated in MAIN_CON.CPP.

3.11 Make-ing Executables

Throughout this chapter we have examined the MAIN.CPP program. To build this C++ file in to an executable program we require the use of a compiler and linker. For the purposes of illustration we will consider the free Borland C++ (version 5.5) command-line 32-bit compiler BCC32[1]. The syntax of BCC32 is:

```
BCC32 [option [option...]] filename [filename...]
```

where items enclosed in brackets are optional. The `option` variable refers to the command-line options whereas `filename` refers to the source-code file/s that are to be compiled. For a summary of the compiler options then type BCC32 without any options or file names. Typical options are `-Ixxx` for specifying the include files directory, `-Lxxx` for specifying the library files directory and `-wxxx` for controlling warning messages; where `xxx` is an identifier used by the option. All options and filenames are preceded by either a hyphen (-) or a forward slash (/) and override settings in configuration files.

To compile and link MAIN.CPP at command-line we would enter:

```
bcc32 main.cpp
```

after which both .OBJ and .EXE files would be generated. To generate a .OBJ file only, with no link then the `-c` option would be used:

```
bcc32 -c main.cpp
```

To specify explicitly where the include and library files are located then we could write:

```
bcc32 -Ic:\Borland\bcc55\Include -Lc:\Borland\bcc55\Lib
  main.cpp
```

for include files located at c:\Borland\bcc55\Include and library files located at c:\Borland\bcc55\Lib.

1 For details of how to obtain the Borland C++ (version 5.5) compiler, Incremental Linker (version 5.0) and Debugger, refer to Borland's Web site at `http://www.borland.com/`.

Let's now consider building a program which consists of the two source files MAIN.CPP and FUNCTION.CPP and its corresponding header file FUNCTION.H. The main program file is now given by:

```
// main.cpp
#include "function.h"

void main ()
        {
        PrintString ("hello, object") ;
        }
```

and displays the string *hello, object* by passing the string to the function `PrintString()` which is defined in FUNCTION.CPP:

```
// function.cpp
#include "function.h"  // include file

void PrintString (string s)
        {
        cout << s ;
        }
```

with corresponding header file:

```
// function.h
#include <iostream>  // C++ I/O
#include <cstring>   // C++ string class

void PrintString (string s) ;
```

To build the program now requires both MAIN.CPP and FUNCTION.CPP to be passed to the compiler:

```
bcc32 main.cpp function.cpp
```

Clearly, if the application consisted of more files then this approach could be continued. However, as the number of files increase this approach becomes increasingly cumbersome. A simple solution to this problem is to use a batch file, but a more elegant solution is to use a *make file*. The majority of commercial compilers provide a command-line utility (Borland's MAKE or Microsoft's NMAKE) which assists in the management of compiling and linking an application. For the Borland compiler the general syntax for MAKE is:

```
MAKE [options...] [target[target]]
```

where `options` control the operation of MAKE and `target` is the name of the target to be built. Numerous options are supported and a summary is provided by the `-h` or `-?` option. MAKE expects an ASCII file called MAKEFILE.MAK located in the current directory, although a different file can be specified by using the `-f` option. In addition, the Borland MAKE utility refers to the make file BUILTINS.MAK for default make settings, which can be ignored using the `-r` option.

The make file consists of three key components: (1) explicit and implicit rules, (2) macros and (3) directives. Explicit rules specify instructions that MAKE must adhere to when building specific targets and are of the form:

```
target [target...]:[:][{path}] [dependent[s]...]
    [commands]
```

where `target` specifies the file name to be built, `path` lists the directories of the dependent files, `dependent` is the file/s whose date and time MAKE tests to see if it is newer than `target`, and `commands` are operating system commands and must be indented so that they are distinguished from a target. For example, the make file for MAIN.CPP could be of the form:

```
# explicit rules
main.obj: main.cpp
  bcc32 -c main.cpp
main.exe: main.obj
  bcc32 main.obj
```

with # indicating user comments. Thus, the following command-line instruction would generate the MAIN.OBJ file:

```
make main.obj
```

whereas the target MAIN.EXE would generate the executable MAIN.EXE:

```
make main.exe
```

Implicit rules specify a general rule of how to build files ending with a specific extension and have the following general syntax:

```
[{source_dir}].source_ext[{target_dir}].target_ext:
  [commands]
```

where `source_dir` specifies the directory/s of the dependent files, `source_ext` specifies the dependent file extension, `target_dir` indicates the directory where MAKE places the target files, `target_ext` specifies the target file name extension and `commands` are operating system commands and must be indented so as to distinguish them from a target. Using implicit rules a make file suitable for MAIN.CPP is:

```
# implicit rules
.cpp.obj:
  bcc32 -c $<
.obj.exe:
  bcc32 $<
```

where `$<` is a default MAKE macro which expands the path, file name and extension of a file.

The general syntax for defining a macro is as follows:

```
MacroName = expansion_text
```

where `MacroName` is the name of the macro and `expansion_text` may consist of alphanumeric, punctuation and space characters. A macro is called by placing `MacroName` within parentheses, preceeded by a `$` sign: `$(MacroName)`. For example, the following defines and calls the macro `MAIN_CPP`:

```
MAIN_CPP = main.cpp

MAIN.OBJ: $(MAIN_CPP)
  bcc32 -c $(MAIN_CPP)
//...
```

The final components of make files are directives, which are analogous to preprocessor directives. MAKE directives begin with either an exclamation mark or period and include such directives as `!if` for conditional statements, `!include`, which acts like the C++ preprocessor directive `#include`, and `.autodepend` which turns on autodependency checking. For example, to include the file MAKEINC.INC:

```
!include makeinc.inc
//...
```

You've probably gathered from the above discussion that command-line MAKE utility programs are comprehensive since they have to cater for every scenario. Thus, if you intend to use a command-line compiler extensively it is worth while getting to know both the compiler and make utility options in some detail. A good starting point is to examine the make files found in each of the chapter and exercises source code directories.

3.12 A Note on Programming Style and Notation

C++ is a *free-format* language. Provided the code is syntactically correct, the compiler almost completely ignores *white space*. White space is defined as spaces, tabs, new lines, line feeds, form feeds and carriage returns. Thus, there are no restrictions on the style to adopt in formatting your program code. There are several styles that can be adopted when programming. Three frequently used alternative formats for the *`main()`* function body are:

```
void main ()
{
cout << "hello, object" << endl ;
}

void main () {
...
}

void main () { cout << "hello, object" << endl ; }
```

Clearly, the possibilities are endless. It is not my intention to enforce any particular style on the reader, but it is necessary to be aware of the style adopted in this book and to bear in mind the variety of styles around.

A couple of exceptions that prevent C++ from being a completely free-format language are lines that begin with the preprocessor directive # and continuation lines. For example, the following is illegal:

```
#include
<iostream>
```

A preprocessing directive and any other line may be extended on to the next line by the insertion of a backslash character, \, at the end of the line to be continued:

```
#include \
<iostream>
```

You may be wondering why you would want to do this, but there are occasions when an expression or a string of text is too long to fit on one line:

```
cout << "this is a very long string of text \
     that extends beyond the width of my screen" << endl ;
```

You may have noticed that I place spaces between function names and parentheses and between the end of statement semicolon and the last character on the line. This is a style that I picked up from Charles Petzold's classic book *Programming Windows 3.1* (Petzold, 1992). The following lines are equivalent:

```
//...
cout<<"Petzold's book:"<<book.Petzold()<<endl;

cout << "Petzold's book: " << book.Petzold () << endl ;
```

In my opinion the latter adds clarity to the program code.

Also, I adopt a style of aligning types, function declarations and assignments whenever possible (don't worry about the specifics of the code for the moment):

```
class Person
  {
  protected:
    char*  name ;
    int    age_in_years ;
    double height_in_metres ;
    //...
  public:
    Person () ;
    void          OutputStatistics () ;
    virtual void Occupation () ;
    //...
  };
```

rather than:

```
class Person
  {
```

```
protected:
  char* name ;
  int age_in_years ;
  double height_in_metres ;
  //...
public:
  Person () ;
  void OutputStatistics () ;
  virtual void Occupation () ;
  //...
};
```

The above example also illustrates the convention adopted for naming classes and variables/objects. A **class** name always begins with an uppercase letter, such as `Person`. This is to distinguish **class** names from variables and objects, which are in lowercase, e.g. `name`. In the case of multi-word **class** and function names, uppercase letters are used (`EmployedPerson` and *`OutputStatistics()`*), while an underscore is used for variables/objects (`age_in_years`). These conventions are clearly more readable than, say, `employedperson` or `ageinyears`. More importantly, such a convention helps to distinguish between classes, functions, identifiers, objects etc. in a complex program listing. Borland C++ adopts the convention of preceding a **class** name by the letter T (T for type), e.g. `TPerson` or `TPoint`.

Constants – things that don't change throughout the entire program once defined and initialised – are defined entirely in uppercase:

```
const int    I_CONST    = 0 ;
const double FEIGENBAUM = 4.6692016 ;
//...
```

There is no limit to the number of characters used to define an identifier/object. However, the previous restriction was 32 characters and some compilers may still only recognise the first 32 characters.

Some of the above styles may appear to be fairly trivial and subtle mannerisms, but they are nevertheless worth thinking about. Choose a formatting style that best suits you and enhances your programming style. However, think of other programmers who will have to read your program code, particularly if you work in a programming group.

3.12.1 Be Consistent!

The style that you adopt will continue to change as you gradually become a more experienced programmer, but try to consciously develop your style and be consistent. A consistent style of writing code will help your own readability of a program.

Consider the following cases:

```
// main.cpp
#include <iostream>
void main () { cout << "hello, object" << endl ; }
```

or:

```
//...
void main () {
```

```
cout <<
"hello, object" << endl ;
}
```

which are (to me) a bit confusing.

The style that I use will become clearer as we proceed through the book.

3.13 Summary

This chapter has introduced us to a few basics of a C++ program. We have seen two types of user *comment*, namely `//` and `/*...*/`. The *preprocessor directive* `#include` informs the compiler to insert a *header* file at the point specified. Every console-based program must have a *`main()` function* which marks the start and end points of a C++ program. Every function is delimited by a pair of braces or curly brackets: `{` and `}`. We noted that C++ is a case-sensitive language so that the functions *`main()`* and *`Main()`* have two different meanings. Output in C++ is commonly performed using the `cout` object/identifier and the << *insertion* operator. The `endl` manipulator inserts a line feed into the stream and is equivalent to the C `'\n'` *escape sequence*. A string of text is placed within double quotes (`"..."`). Spend some time experimenting with different programming styles, particularly if you intend to do a lot of programming or work in a programming group.

Exercises

3.1 To help you get started with the C++ programming language and your compiler, write a program that outputs your name, address and personal details (whatever they might be!).

3.2 Place two `endl` manipulators before the *hello, object* string of MAIN.CPP.

3.3 Debug the following program:

```
// 03_03.cpp
a C++ program with a few errors

#include <iostream  // C++ I/O

void Main ()
  {
  cout << "The C++ Programming Language < endl ;
  // } main()
```

3.4 Locate and open the IOSTREAM.H header file and find the declarations for the `cout` stream object and the `endl` stream manipulator.

3.5 Include a header file in MAIN.CPP which includes details about yourself, program information, date of editing and a program copyright.

3.6 Try changing the angle brackets with double quotes for `include`ing IOSTREAM.H in MAIN.CPP. Include the exact path of IOSTREAM.H.

3.7 Modify the *`main()`* function in MAIN.CPP so that it returns type **`int`**.

Chapter 4

Fundamental Data Types, Declarations, Definitions and Expressions

live with the language
g. w.

This chapter begins by describing the different kinds of Fundamental Data Type (FDT) in C++. The chapter then introduces identifiers (constant and variable) and the declaration and definition of identifiers. This is followed by discussing assignment; initialisation of identifiers, expressions and statements; operators; and type conversions. Basic input from the keyboard and output to the screen is covered, along with the use of manipulators for formatting input and output. The topic of casting (the process of converting an object of one data type to an object of another data type) is discussed. The chapter concludes by examining the auto, extern, register and static storage class specifiers for specifying exactly how a variable is to be stored, and the asm declaration for directly integrating assembly language code into C++ code.

4.1 Fundamental Data Types

There are four *fundamental*, *basic* or *primitive* data types in C++:

char	Character: letter, symbol, punctuation and digit
int	Integer: whole number
float	Floating: point number
double	Floating: point number

char, **int**, **float** and **double** are keywords or, more specifically, *type specifiers*. Each data type has its own *attribute list* which defines the characteristics of the type and may vary from one machine system to another. The types **char**, **int** and **double** each have variations, or *data type modifiers*, such as **short**, **long**, **signed** and **unsigned**, to enable more efficient use of the data types.

An additional character type is **wchar_t**:

wchar_t	Wide character constant

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
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There are also two additional types in C++ that I feel warrant mention at this point, namely **enum** and **bool**:

enum	Enumeration constant
bool	False–True enumeration

Compared with the fundamental types, **enum** and **bool** may appear to be oddballs, but just bear them in mind at this stage until we discuss them in more detail at a later date.

The characteristics of each data type will now be discussed separately.

4.1.1 The char Data Type

The **char** data type is just large enough to store one character of your machine's character set. Variables of type **char** can be used to store numbers (–128 to 127 for a signed **char** and 0 to 255 for an unsigned **char**), but are generally used to hold the ASCII characters. Most C++ compilers use the ASCII character set, which has defined characters from 0 to 126. However, most compilers extend this range from 0 to 255 so as to accommodate the extended IBM character set. Appendix B contains a complete listing of the ASCII character set. For example, the decimal integers 0 to 9 are stored as decimal constants 48 to 57 and the lowercase letters a, b, ... z are stored as decimal constants 97 to 122. Symbols such as +, - and # all have an equivalent ASCII character constant. At first it can be a little confusing to find out that characters are stored as integers which in turn are stored as binaries. Appendix B also lists the binary equivalents of the character set. Note that characters in the ASCII character set from 0 to 127, inclusive, could be represented as seven data bits instead of eight bits. However, since for most computers the byte (8 bits) is the primary unit, a variable of type **char** occupies one byte of memory.

For MS-DOS, characters 0–31 are reserved for control characters, 32–126 represent keys on the keyboard and 127–255 are the IBM extended characters. It is worth noting that the IBM extended character set is not part of the ASCII character set and may not be fully portable. MS-DOS also includes character sets or code pages. For example, the English and multilingual character sets are designated by the numbers 437 and 850 respectively. Note also that the Windows character sets differ from the MS-DOS character sets.

Character Constants

Character constants are written in C++ as a single character enclosed in apostrophes (or single quotes): `'C'`, `'p'`, `'+'`. Character constants are different from string constants enclosed within double quotes:

```
//...
char char_const = 'C' ;                    // character constant
//...
cout << "this is a string constant" ;   // string constant
//...
```

When a character constant is encountered, the compiler converts it into its respective numeric code. The following example program illustrates character constants and variables:

```
// char.cpp
// illustrates the char data type
#include <iostream>  // C++ I/O
```

```
void main ()
  {
  cout << "'A': " << 'A' << endl << endl ; // character
                                           // constant

  char c_var1 ;        // definition of character variable
  cout << "c_var1 before assignment: " << c_var1 << endl ;
  c_var1 = 'C' ;       // assignment
  cout << "c_var1 after assignment: " << c_var1 << endl ;

  //char c_var1 ;      // error: multiple definition for
                       // 'c_var1'

  char c_var2 = 'p' ; // define & initialise thro'
                       // assignment
  cout << "c_var2: " << c_var2 << endl ;
  c_var2 = 43 ;        // re-assignment
  cout << "c_var2: " << c_var2 << endl ;

  cout << << c_var1 << c_var2 << c_var2 << endl ;

  const char C_CONST = 'C' ;
  //const char C_CONST ; // error: must initialise a
                         // constant
  //C_CONST = 'p' ;     // error: can't modify a constant
  }
```

which generates the following output:

```
'A': A

c_var1 before assignment:
c_var1 after assignment: C
c_var2: p
c_var2: +
C++
```

Note that the more explicit definition of **signed char** could have been used in place of **char** in the above example program, since the two are equivalent.

The program first of all outputs the character constant 'A'. Next a variable, `c_var1`, of type **char** is defined but not initialised and then displayed. The output illustrates that uninitialised variables of type **char** are set to the null ASCII character or the decimal value of 0. This is directly followed by assigning the character 'C' to the variable `c_var1`, which is then displayed. Then a single line has been commented out which would cause a compilation error. The line defines `c_var1` a second time, and multiple definitions of a variable are illegal. In C++ all variables must be defined before they are used. The next line illustrates that C++ allows you to define a variable or object wherever it is required. The variable `c_var2` is defined and initialised (by assignment) in a single statement. You may be thinking 'big deal', but the traditional C style is to define all variables and objects at the top of a block before the first executable statement:

```
// C style:
char c_var1 ;
char c_var2 ;
//...
c_var2 = 112 ; // assignment
cout << "c_var2: " << c_var2 << endl ;
//...
```

In a small example like this, the C++ *define-anywhere* style may not appear any better than the C *define-at-top* style, but when program code becomes lengthy and complex you'll be grateful for the C++ style. The C++ style is also less error-prone.

After the value of `c_var2` is displayed, the next line assigns a new character to `c_var2` by assigning the decimal equivalent, 43, of the ASCII character '+'. Although it is possible to assign a character to a **`char`** variable or constant by assigning the character's decimal equivalent, it is good programming practice to assign the actual character constant, if possible. There are occasions, however, such as when displaying non-keyboard characters (e.g. the → character, with decimal equivalent 26), when the decimal equivalent of a ASCII character has to be used.

In the above program `c_var2` was declared as a variable, and variables can have variable values. You may be thinking, 'How do I, or the compiler, know that `c_var2` is a variable when I didn't explicitly define it as a variable?'. Basically, if an identifier is not defined as a constant it is assumed to be a variable. The program example above defines and initialises a **`char`** constant, `C_CONST`, to the character 'C'. Two commented lines are included within the program to indicate two key rules for constants: (1) a constant object must be initialised when defined and (2) a constant cannot be modified after initialisation. These rules apply to all **`const`** objects of an integral or user-defined data type and are not just restricted to objects of type **`char`**. The modifier **`const`** is discussed later. Furthermore, some languages (such as Pascal) require that objects are explicitly defined as either constants or variables:

```
// Pascal:
const
  PI    = 3.14159 ;
  BLACK = 0 ; WHITE = 1 ;
var
  x, y, z: real ;
  quit:    boolean ;
//...
```

but C++ adopts a more minimal style than this.

If we required several variables we could define each one on a separate line:

```
char c_var1 ;
char c_var2 ;
//...
char c_varn ;
```

An alternative approach is to use a comma (,) as a separator:

```
char c_var1, c_var2, c_var3 ;
```

and if we require the variables to be initialised then we can take either of the following two approaches:

```
char c_v1 = 'a' ; char c_v2 = 'b' ; char c_v3 = 'c' ;
char c_v1 = 'a', c_v2 = 'b', c_v3 = 'c' ;
```

The *comma* operator allows us to separate expressions but not separate statements. Use the comma operator when you have expressions with the same attributes; e.g. `c_v1`, `c_v2` and `c_v3` are all of type **`char`**.

Character String Constants

We saw above that character constants are different from string constants. Basically, a string constant consists of a series or *string* of characters. For example:

```
"C++ is a programming language"
```

Escape Sequences

In C++ there are certain character constants that have a special meaning. For instance, you may have seen the new line escape sequence '`\n`' used:

```
cout << "M. C. Escher\n" ;
```

instead of:

```
cout << "M. C. Escher" << endl ;
```

The backslash character (`\`) indicates a special sequence to the compiler. In the previous example '`\n`' is interpreted as a new line and not as the constant character 'n'. Table 4.1 lists the escape sequences available in C++. Note that the escape sequences are viewed by the compiler as a single character and thus occupy one byte and not two, as may have been expected.

Any character in the ASCII set can be alternatively described in terms of escape sequences by using the character's octal or hexadecimal code. For example `'\60'` and `'x30'` and `'48'` represent the digit 0 in octal, hexadecimal and decimal ASCII character constants respectively.

Table 4.1 Escape sequences.

Escape sequence	*Character*
`\a`	Alert. Audible (bell) or visible alert.
`\b`	Backspace. Move to previous position.
`\f`	Form feed. Advance to next page.
`\n`	New line (line feed). Advance to next line.
`\r`	Carriage return. Current position becomes 1st. position on next line.
`\t`	Horizontal tab. Advance to next horizontal tab position.
`\v`	Vertical tab. Advance to next vertical tab position.
`\\`	Backslash. Display backslash.
`\'`	Single quote (apostrophe). Display single quote.
`\"`	Double quote. Display double quote.
`\?`	Question mark. Display question mark.
`\0`	Null termination character.
`\o`	Octal code. `o` = string of 3 digits maximum.
`\xh`	Hexadecimal code. `h` = string of hexadecimal digits.

Table 4.2 Trigraphs.

#	[	]	{	}	\	^	|	~
??=	??(	??)	??<	??>	??/	??'	??!	??-

Escape sequences can be useful for string constants containing double quotes. For example, imagine you wanted to display the string *the man said "thank you."*:

```
cout << "the man said "thankyou"." << endl ;    // error
cout << "the man said \"thankyou\"." << endl ; // O.K. with
                                                // '\"' e.s.
```

The first statement is an error because the compiler interprets the character string as `"the man said "`.

Trigraphs and Digraphs

For several European computers the ASCII special characters `[`, `]`, `{`, `}`, `|` and `\` occupy alphabetical positions designated by the ISO standard. Also, certain European keyboards are without the number sign (`#`). The C *trigraphs* offer a solution to this problem; see Table 4.2. Thus, the following two program examples are equivalent:

```
//...
void main ()
  {
  //...
  }

//...
void main ()
  ??<
  //...
  ??>
```

Trigraphs are not very readable. An alternative to trigraphs is *digraphs*[1]; see Table 4.3. Note that there is not a complete overlap between trigraphs and digraphs. In terms of digraphs the above example is now:

```
//...
void main ()
  <%
  //...
  %>
```

Note also the keywords for the bitwise operators (`&`, `|`, `^`, `~` and `^=`), relational operators (`!` and `!=`) and logical operators (`&&`, `||`, `&=` and `|=`). Such operators will be discussed later.

1 A digraph is a token consisting of two characters.

Table 4.3 Digraphs and keywords for C++ special characters.

Digraphs		*Keywords*	
[	<:	&	**bitand**
]	:>	|	**bitor**
{	<%		**xor**
}	%>	~	**compl**
#	%:	^=	**xor_eq**
##	%:%:	!	**not**
		!=	**not_eq**
		&&	**and**
		||	**or**
		&=	**and_eq**
		|=	**or_eq**

4.1.2 The `int` Data Type

The **int** data type is used to store integer or whole numbers such as 0 or 21, but not numbers with a fractional part. The size of an **int** depends on the largest data register (16 or 32 bits) of the computer. For a 16-bit machine the high bit is used for the sign of the integer, so the range of values is from -32 768 (-2^{15}) to 32 767 ($2^{15}-1$). For a 32-bit machine an **int** has the range -2 147 483 648 (-2^{31}) to 2 147 483 647 ($2^{31}-1$), with the high bit similarly reserved for the sign of the number. To combat the portability problem of transferring a program from one machine to another, C++ allows the **int** data types to be modified through the use of the **short** and **long** data modifiers, to be discussed later.

The following example illustrates a few experiments with the **int** type.

```
// int.cpp
// illustrates the int data type
#include <iostream>  // C++ I/O

void main ()
  {
  int i ;              // definition of i

  i = 4 ;              // assignment
  cout << "i: " << i << endl ;

  const int CI = 90 ;
  //CI = 11 ;         // error: can't modify a constant
  cout << "CI: " << CI << endl ;

  i = i + CI ;         // addition and assignment
  cout << "i + CI: " << i << endl ;

  int j = 1900 ;       // definition and assignment of j

  cout << "i + j: " << (i + j) << endl ;
```

```
    int const CJ = 91 ; // 'int const' instead of 'const int'
    }
```

with output:

```
i before assignment: 0
i after assignment: 4
CI: 90
i + CI: 94
i + j: 1994
```

An uninitialised **int** variable is assigned the value 0. Most of the statements are self-explanatory, but it is worth pointing out that, after the statement `i=i+CI;`, `i` is assigned a new value of 94. Note, as with the character example program, a **const** object cannot be altered after it has been initialised. Experiment with the program until you feel more comfortable with the **int** data type.

The library header file LIMITS.H defines the implementation-specific limits for both the **char** and **int** data types.

4.1.3 The `float` Data Type

The **float** type is used to represent floating-point numbers. A floating-point number has an integer part, a decimal point and a fraction, e.g. 28.75. Type **float** stores a number within the approximate range 1.2×10^{-38} to 3.4×10^{38}. Most compilers conform to the IEEE standard, which enables you to represent a floating-point number in terms of an exponent (e or E), e.g. 1.25e10 ($=1.25\times10^{10}$).

If you require the exact specification of a floating-point number that your compiler uses then refer to the FLOAT.H header file. The minimum and maximum floating-point numbers associated with the **float** type are FLT_MIN and FLT_MAX, respectively. The number of digits of precision is set by the constant FLT_DIG, and for a **float** it is generally set to 6. Most compilers allocate four bytes of memory to a **float**.

The following example program (FLOAT.CPP) experiments with the **float** data type:

```
// float.cpp
// illustrates the float data type
#include <iostream>    // C++ I/O

void main ()
   {
   float f ;    // definition

   f = 5.6 ;    // assignment
   cout << "f: " << f << endl ;

   float       fi = 0.0000001 ;
   const float CF = 1.25e10 ;

   cout << "fi: " << fi << "; CF: " << CF << endl ;
   }
```

with corresponding output:

```
f before assignemt: 1.4013e-45
f after assignment: 5.6
fi: 1e-07; CF: 1.25e+10
```

and illustrates that unitialised floating-point variables are not assigned the value 0; see Section 4.2.7 for a discussion of the numeric limits of the fundamental data types. Notice that the output is automatically converted into IEEE format with explicit sign indication. Note also the chained output using the `cout` object and insertion operator <<.

4.1.4 The `double` Data Type

The **`double`** data type is also used to store floating-point numbers. Most compilers allocate eight bytes of memory to store a **`double`**. Type **`double`** stores a number within the approximate range 2.2×10^{-308} to 1.8×10^{308}. As with type **`float`**, if you require the exact specification of a **`double`** floating-point number that your compiler uses then refer to the FLOAT.H header file. The minimum and maximum floating-point numbers associated with the **`double`** type are `DBL_MIN` and `DBL_MAX`, respectively. The number of digits of precision is set by the constant `DBL_DIG` and for a **`double`** it is generally set to 15. In addition to **`float`** and **`double`**, C++ offers us an additional floating-point type, **`long double`**, which has a range of 3.4×10^{-4932} to 1.2×10^{4932} and generally occupies 10 bytes (but refer to your C++ compiler's documentation for the exact size).

The following program uses **`double`** exclusively to prompt for the user's height in metres, convert the height to feet and output the person's height in both metres and feet:

```
// double.cpp
// illustrates the double type
#include <iostream>  // C++ I/O

void main ()
  {
  const double m_to_ft  = 3.281 ;    // conversion factor
  double       height_m = 0.0 ;      // initialise to zero

  cout << "enter your height in metres: " ;
  cin  >> height_m ;

  double height_ft = m_to_ft * height_m ;  // perform
                                           // conversion

  cout << "your height: " << height_m << "m; "
                          << height_ft << "ft" << endl ;
  }
```

A sample run is:

```
enter your height in metres: 1.8
your height: 1.8m; 5.9058ft
```

The above program introduces the `cin` *stream object*, which lets you input a value into the variable `height_m`. The user-value entered is placed in the `cin` object, and then the *extraction*

operator (>>) transfers the value from the left-hand side of the operator to the right-hand side. The object `cin` is defined in the header file IOSTREAM.H.

Having now introduced both integer and floating-point numbers, consider the following statement:

```
//...
double x = 5 ;
```

Is x a **double** or an **integer**? Some old compilers would have created the constant 5 as an **int** and then cast this into x, making x an **int**. Modern compilers do not require a decimal point after a numeric constant and hence x is stored, as defined, as a **double**. However, I think it is clearer to define a floating-point number explicitly as a floating-point number and not as an integer:

```
//...
double x = 5.0 ;
```

Epsilon Error

An epsilon error is a small floating-point number such that if it were added to the number 1.0 then the resulting number (number+epsilon error) would be detected by the compiler as *not* being equal to 1.0. Basically this is equivalent to the number of digits of precision offered by the compiler for each floating-point type. The exact values of the epsilon error for **float**, **double** and **long double** can be found in the FLOAT.H header file, defined as `FLT_EPSILON`, `DBL_EPSILON` and `LDBL_EPSILON` respectively. For the Borland C++ (version 5.x) compiler these are:

```
#define FLT_EPSILON      1.19209290E-07F
#define DBL_EPSILON      2.2204460492503131E-16
#define LDBL_EPSILON     1.084202172485504434E-19L
```

4.1.5 The `wchar_t` Type

wchar_t is a fundamental data type that can represent a character that will not fit into the storage space of one byte allocated by type **char** and is typically used to store characters of a 16-bit character set such as Unicode. Strictly speaking **wchar_t** is defined as a **typedef** in the header files STDLIB.H, STDDEF.H and WCHAR.H as:

```
typedef unsigned short wchar_t ;
```

and thus is stored in two bytes.

A character constant that is preceded by an L is a wide character constant of type **wchar_t**:

```
wchar_t wcc = L'12' ;
```

which is a two-character constant.

A string constant that is similarly preceded by an L is a wide character string of type **wchar_t**:

```
wchar_t wcs[] = L"abcd" ;
```

which is a multi-byte string constant with a memory allocation of two bytes per character. In the case of a wide character string constant the string is required to be defined as an array, `wcs[]`. Note that an identifier or string of type **`wchar_t`** is a constant.

It is worth mentioning that the C++ standard defines **`wchar_t`** as a keyword and that a `wstring` **`class`** has been added to the standard C++ library for generating strings of wide characters:

```
typedef basic_string<wchar_t> wstring ;   // STRING.H
```

Arrays, typdefs and classes will be discussed in more detail in Chapters 7, 8 and 9.

4.1.6 The enum Type

At first, **enum**s can appear a little confusing. Basically **enum** is short for enumeration. No clearer! To enumerate means to count one by one. Thus an **enum** is a user-defined type with named constants of type integer (including negative values) and can be used in any expression where integer constants are valid. Contrary to the previous four *integral* types, **enum** is a user-defined type.

Suppose we wanted to define three constants RED, GREEN and BLUE and assign the values 0, 1 and 2 to them respectively. One way would be:

```
const int RED   = 0 ;
const int GREEN = 1 ;
const int BLUE  = 2 ;
```

An equivalent, but more elegant, way would be to define the three constants as enumerators:

```
enum
  {
  RED, GREEN, BLUE
  };
```

Note the use of the parentheses and a semicolon. By default the assigned enumerator values incrementally increase from 0 (i.e. 0, 1, 2, 3,...), and hence RED=0, GREEN=1 and BLUE=3.

Why use the enumerators RED, GREEN and BLUE? Well, most people find it a lot easier to use constants as mnemonics rather than using numbers like 0, 1 and 2. It is simply easier to work with a descriptive identifier than a number.

An enumeration can be named:

```
enum Colour
  {
  RED, GREEN, BLUE
  };
```

The **enum** declaration can also incorporate a variable definition, `colour`:

```
enum Colour
  {
  RED, GREEN, BLUE
  } colour ;
```

The enumerators can also be assigned explicit values during declaration:

```
enum Time
  {
  SEC_PER_MIN     = 60,
  MIN_PER_HOUR    = 60,
  HOUR_PER_DAY    = 24
  };
//...
cout << "minutes per hour: " << MIN_PER_HOUR << endl ;
```

which is a little strange, because assigning explicit, arbitrary (non-incrementing) values appears to contradict the definition of an enumeration. One might have expected the **enum** syntax to allow us to define an initial enumerator value explicitly, after which successive enumerators incrementally increase or decrease, but not to allow enumerators to be arbitrarily initialised. However, explicitly assigning values to enumerators *is* legal.

Note that the value of an enumerator is of type **int**, and it is possible to assign an **int** to an enumerator:

```
//...
Colour colour ;
//...
colour = GREEN ;   // ok
colour = 1 ;       // ok but warning issued!
```

This is a good example of C++ being weakly typed, because a user-defined type, such as `Colour`, is indistinguishable from the fundamental type **int**. Although not strictly a compilation error, it does introduce a contradiction between **enum** and the **struct** or **class**. When **enum**, **struct** and **class** are described in more detail later, it will become evident that **enum** is similar to a **class**. With regard to classes, C++ is a strongly typed language, whereas with regard to enumerations C++ is weakly typed. Another example of weak typing in C++ is the use of **typedef**s. A **typedef** is not a new type but a synonym for an alternative type.

```
typedef int Int ;
//...
int i = 1 ;
Int j = 2 ;
//...
j = i ;  // mixing of types int and Int
```

Although **enum**, **typedef**, **struct** and **class** will be discussed in Chapters 8 and 9, they have been mentioned here to highlight possible contradictions in compiler warnings that you may be experiencing. The Borland C++ (version 5.x) compiler issues a warning of the form 'Initializing Colour with int' for the above **enum** example, but no warning for the **typedef** example – so be careful!

4.1.7 The `bool` Type

bool is an integral type in C++ with literals **true** (1) and **false** (0). Frequently in mathematical expressions we require a *boolean* data type which consists entirely of the two values

logical-true (1) and *logical-false* (0). In C++ the value 1 is used to represent logical-true and 0 to represent logical-false. In fact, any non-zero value can be used to represent logical-true, but the constant 1 is generally used.

Before the **bool** type was introduced to the C++ language, we frequently saw a declaration of the following form in C++ programs:

```
enum Boolean { FALSE, TRUE };
```

which makes `Boolean` a distinct user-defined type with literals (constants values) `TRUE` and `FALSE`. However, C++ now has an integral type called **bool** with literals **true** and **false**. Thus it is now possible to define an appropriately descriptive variable, `boolean`, of type **bool**, which is useful for logical tests:

```
bool boolean ;
//...
if (boolean)  // condition is true
  {
  // do something
  }
else          // condition is false
  {
  // do something else
  }
//...
```

or to use **bool** as the return type of a function:

```
//...
bool Function ()
  { /* ... */ }
//...
if (Function ())  // condition is true
  {
  // do something
  }
```

The following program examines type **bool**:

```
// bool.cpp
// illustrates bool
#include <iostream>  // C++ I/O

void main ()
  {
  bool b1 ;          // uninitialised
  bool b2 = false ;
  bool b3 (true) ;
  int i1 = true ;    // true to int
  int i2 = false ;   // false to int
  int i3 = b3;       // bool to int
  int i4 = 10 ;
```

```
bool b4 = i4 ;     // int to bool
bool b5 = -i4 ;    // int to bool

// error: can't assign to true-false literals
// true  = 2 ;
// false = -1 ;

cout << "b1: " << b1 << endl ;
//...
}
```

with output:

```
b1: 0
b2: 0
b3: 1
i1: 1
i2: 0
i3: 1
b4: 1
b5: 1
```

The output illustrates that an uninitialised **bool** variable (e.g. b1) is assumed to be logical-false. The program illustrates that it is possible to convert between types **int** and **bool**. A zero **int** value sets **false** to zero, whereas any non-zero **int** value sets **true** to one. Because the **true** and **false** literals are rvalues[2] they cannot have their values altered by assignment.

Don't worry if the above examples appear confusing at this point – all will become clear later. For the moment, just note that the use of type **bool** is more descriptive than a similar use of type **int** for an identifier or function that we want to hold or return (respectively) one of two logical values.

4.2 The short, long, unsigned, signed, const and volatile Data Type Modifiers

C++ would be somewhat limited if we only had at our disposal the four fundamental data types. Therefore, C++ allows us to effectively 'create' new data types by the use of *modifiers* to alter the attributes of a given data type. The six modifiers are **short**, **long**, **unsigned**, **signed**, **const** and **volatile**. It is worth noting that the order of modifier and type is irrelevant. Thus **signed int** or **int signed** are equivalent, although it is conventional to precede the data type by the modifier(s), e.g. **signed int**, **signed short int** or **const int**. As mentioned above, the **short** and **long** data modifiers also tackle the problem of portability between 16-bit and 32-bit computers.

2 The rvalue (right value) of a variable is the actual value of the variable which is stored at the variable's lvalue (left value) memory address.

4.2.1 The `short` Modifier

The **`short`** modifier modifies the **`int`** data type only. A 16-bit **`short int`** variable is guaranteed to have the range -32 768 (-2^{15}) to 32 767 ($2^{15}-1$), independent of the machine.

4.2.2 The `long` Modifier

The **`long`** modifier modifies the **`int`** and **`double`** types only. A long variable is guaranteed to have the range -2 147 483 648 (-2^{31}) to 2 147 483 647 ($2^{31}-1$). The suffix L or l attached to any constant forces that constant to be represented as a **`long`**, e.g. 101L.

4.2.3 The `unsigned` Modifier

Some numbers don't need a negative value: for example, speed or a millionaire's bank balance. Thus, to avoid wasting the signed bit of such numbers C++ allows a number to have no sign or be unsigned. Thus, a 16-bit **`unsigned int`** or **`unsigned short int`** ranges from 0 to 65 535 ($2^{16}-1$), whereas 32-bit **`unsigned int`** and **`unsigned long int`** assume a maximum value of 4 294 967 295 ($2^{32}-1$). The **`unsigned`** keyword cannot be used with the **`float`** and **`double`** types. The suffix U or u attached to any constant forces that constant to be represented as an **`unsigned`**, e.g. 101U.

4.2.4 The `signed` Modifier

The **`signed`** modifier allows a programmer to indicate explicitly that a data type uses a sign bit. A **`signed int`** variable has the identical negative to positive range of values as the **`int`** type.

4.2.5 The `const` Modifier

There is another modifier that warrants mention here: **`const`**. This modifier (frequently called an *access* modifier) prevents any assignments to an object after the object has been initialised. Also, a **`const`** object cannot be incremented (++) or decremented (--). The following are example uses of **`const`**:

```
const char   C_CONST                = 'C' ;
const        C                      = 0 ;  // default int
const int    I_CONST                = 9 ;
const float  TWENTY_TWO_OVER_SEVEN  = 3.1428571 ;
const double PI                     = 3.1415927 ;
```

After these definitions the following are illegal:

```
//...
C_CONST = 'A' ;
C = C + 1 ;
C++ ;
//...
```

Since the notation adopted is that all constant identifiers are in uppercase, we know that (`C=C+1`) is illegal before compilation.

4.2.6 The `volatile` Modifier

The **`volatile`** modifier, also known as an *access* modifier, is a peculiar thing and rarely used. **`volatile`** is necessary because the majority of commercial compilers will generally try to optimise an expression which contains a variable whose value appears to remain unaltered from one statement to the next. The compiler optimisation may take the form of using a register copy of a variable rather than reloading the variable from memory. Basically, **`volatile`** informs the compiler that a variable's value may be altered by mechanisms unknown to the compiler and therefore the compiler must not perform any optimisations on a **`volatile`** variable.

As an example of **`volatile`**, you may define a variable, `interrupt`, which receives data from an external device such as a serial port:

```
volatile int interrupt ;
```

The keyword **`volatile`** in the above statement informs the compiler that expressions involving the variable `interrupt` should not be optimised. Note that a variable can be both **`const`** and **`volatile`**.

4.2.7 Choosing a Data Type

Table 4.4 lists the valid combinations of types for both 16-bit and 32-bit systems. Combinations such as **`short char`**, **`long char`**, **`signed double`**, **`unsigned double`** and **`signed float`** are illegal. If you are using 32-bit architecture then check your compiler's documentation. For example, **`long double`** is implemented as a 80-bit value (10 bytes) for the Borland C++ compiler (version 5.x) but can be a 96-bit or 128-bit value on some systems. The C++ implementation of each of the data types can be found in the header files FLOAT.H and LIMITS.H. The header file FLOAT.H deals with floating-point types. The following is an excerpt from FLOAT.H for the Borland C++ compiler:

```
//...
#define DBL_DIG  15
#define FLT_DIG  6
#define LDBL_DIG 18
//...
```

which `defines` the number of digits of precision for **`double`**, **`float`** and **`long double`** respectively. The header file LIMITS.H contains implementation details for the limits of data type values, such as

```
//...
#define LONG_MAX  2147483647L
#define ULONG_MAX 4294967295UL
//...
```

for the maximum **`signed`** and **`unsigned long`** data types.

The above `#defines` are macros that C++ inherited from C. In addition, C++ has its own style of defining limits on fundamental data types and their modifiers. The C++ approach is based on the `numeric_limits` template class and its explicit class declarations of LIMITS.H:

Table 4.4 Range and size of data types.

Type specifier	*Range (16/32 bits)*	*Size 16-bit (bytes)*	*Size 32-bit (bytes)*
char	–128:127	1	1
int	–32 768:32 767/-2 147 483 648:2 147 483 647	2	4
float	±(3.4e–38:3.4e+38)	4	4
double	±(1.7e–308:1.7e+308)	8	8
wchar_t	same as **unsigned int**	1	2
bool	`true` (1) or `false` (0)	1	1
short	same as **int**	2	2
short int	same as **int**	2	2
long	–2 147 483 648:2 147 483 647	4	4
long int	same as **long**	4	4
long double	±(3.4e–4932:1.2e+4932)	10	10
unsigned	same as **unsigned int**	2	4
unsigned char	0:255	1	4
unsigned int	0:65 535/0:4 294 967 295	2	4
unsigned short	same as **unsigned int**	2	4
unsigned short int	same as **unsigned int**	2	4
unsigned long	0:4 294 967 295	4	4
unsigned long int	same as **unsigned long**	4	4
signed	same as **int**	2	4
signed char	same as **char**	1	1
signed int	same as **int**	2	4
signed short	same as **int**	2	2
signed short int	same as **int**	2	2
signed long	same as **long**	4	4
signed long int	same as **long**	4	4

```
template <class T>
class numeric_limits { /* ... */ };
//...
class numeric_limits<char> // explicit class declaration
  {
  public:
    //...
    static inline char min () { return CHAR_MIN ; }
    //...
  };
```

Explicit class declarations are provided for all of the fundamental types **char**, **int**, **float**, **double**, **bool** and **wchar_t** and the modified types **signed char**, **unsigned char**, **short**, **unsigned short**, **unsigned int**, **long**, **unsigned long** and **long double**.

For example, to determine the epsilon error for type **double** then we would call the *epsilon()* member function of class numeric_limits<**double**>:

```
double ep_error = numeric_limits<double>::epsilon () ;
```

Template classes will be discussed in Chapter 13 and the numeric_limits classes will be discussed further in Chapter 20 in connection with the C++ Standard Template Library.

Try running the provided program files LIMITS.CPP, FLOATS.CPP to test the implementation-specific details of the character, integer and floating-point data types for your compiler. These programs use both approaches of the C macros and C++ numeric_limits classes. An additional program, SIZES.CPP, outputs the size of all the supported built-in types and uses the **sizeof** *()* operator.

The data type or modified data type that you choose will undoubtedly depend on the application and the precision required. Although C++ gives a programmer a large degree of flexibility in choosing a date type, always try to select the smallest data type for the job at hand. It is seen from Table 4.4 that the respective sizes in bytes of **float**, **double** and **long double** are 4, 8 and 10 respectively. As a consequence, **double** and **long double** not only consume more memory than **float**, but are also slower in arithmetic operations. However, there are occasions when the exact range of values that an identifier or object will hold are not known by the programmer during development.

Another consideration, particularly when deciding between **float** and **double**, is the order of precision. The types **float**, **double** and **long double** have 7-, 15- and 19-digit precision, respectively. If you are performing accurate numerical computational work then **double** is the best compromise between size and precision.

The following sections continue our present discussion of the data types and help recap on the material covered above.

4.3 Constants

C++ provides four different types of constant (also referred to as *literals*): integer, floating point, character and string, and enumeration.

4.3.1 Integer Constants

Integer constants can be decimal (base 10), octal (base 8) or hexadecimal (base 16) in C++. The suffixes (L, l) or (U, u) can be assigned to a constant to force it to be represented as a **long** or an **unsigned**, respectively; e.g. 1U or 2l. Combinations of L, l, U and u can be used on a given constant in any one of the eight permutations: ul, uL, Ul, UL, lu, lU, Lu or LU; e.g. 3Ul, 4UL or 5lu.

Decimal Constants

A digit sequence of base 10 is a decimal. Decimal constants can vary from 0 to 4 294 967 295 ($2^{31}-1$) for an **unsigned long int** data type. A decimal constant cannot begin with a zero in C++ or it will be assumed to be an octal constant. An example of a decimal constant is $17_{10}=1\times10^1+7\times10^0$.

Table 4.5 Decimal, octal and hexadecimal constants.

Decimal	*Octal*	*Hexadecimal*
0	0	0x0
1	01	0x1
2	02	0x2
3	03	0x3
4	04	0x4
5	05	0x5
6	06	0x6
7	07	0x7
8	010	0x8
9	011	0x9

Octal Constants

A digit sequence of base 8 is an octal. Octal constants in C++ begin with a zero. Note that an octal constant cannot contain the digits 8 or 9. An example of an octal constant is $17_8=1\times8^1+7\times8^0=15_{10}$.

Hexadecimal Constants

A digit sequence of base 16 is a hexadecimal. Hexadecimal constants in C++ begin with 0x or 0X. An example of a hexadecimal constant is $17_{16}=1\times16^1+7\times16^0=23_{10}$.

Table 4.5 illustrates decimal, octal and hexadecimal constants for the decimal digits 0 to 9.

4.3.2 Floating-Point Constants

We generally think of a floating-point number as, say, 343.6 – that is, a number with a decimal integer, a decimal point and a decimal fraction. However, in C++ we can also represent floating-point numbers in terms of a signed integer exponent or a type suffix or both; e.g. 2.998e08 which has the value 2.998×10^8. Thus a floating-point constant can consist of:

- decimal integer
- decimal point
- decimal fraction
- signed integer exponent (e or E)
- suffix (F, f, L or l)

The suffix F or f forces a constant to be of type **`float`**, whereas the suffix L or l forces a constant to be of type **`long double`**, e.g. 2.51f and 2.52L. Note that combinations of both suffixes is not possible.

4.3.3 Character and String Constants

A character constant can be either one or two characters depending on whether the constant is defined as type **`char`** or **`wchar_t`**. A character string constant is simply a string of characters. There are 'special' character constants, *escape sequences*, that are preceded by a backslash (\) such as '\n' for inserting a new line.

4.3.4 Enumeration Constants

Enumeration constants are identifiers defined in an **enum** type declaration. For example:

```
enum TelephoneDigits
  {
  ZERO, ONE, TWO, THREE, FOUR, FIVE, SIX, SEVEN, EIGHT, NINE
  };
```

which defines the identifiers ZERO, ONE, TWO, ..., NINE with the integer constants 0, 1, 2, ..., 9 respectively. There are several alternative formats of the **enum** declaration which will not be discussed here but left for further discussion until Chapter 8.

4.4 const or #define?

We saw above that the keyword **const** ensures that an identifier cannot be inadvertently altered throughout a program:

```
const int I_CONST = 5 ;
//...
I_CONST = 6 ;  // error: attempt to modify a const
```

Constants can also be specified in C++ with the #define preprocessor directive:

```
#define I_DEFINE 1
//...
```

Although it is feasible to use the #define directive, I don't recommend it. The use of #define stemmed from C before **const** was introduced into the C++ language. There are several problems with #define, but in the definition of symbolic constants the main problem is that the type of the constant is not specified when defined. Also, with no type defined no type checking can be performed by the compiler.

4.5 The sizeof Operator

In Table 4.4 we saw the size, in bytes, of all of the available integral data types in C++. If you would like to determine the size, in bytes, of a type or an identifier, use the **sizeof** unary operator. Applied to an identifier the general form is:

```
sizeof operand
```

where the operand can be either a constant or a variable. Alternatively, if you would like to apply the **sizeof** operator to a type (integral or user-defined) then parentheses must be used:

```
sizeof (type_specifier)
```

For example, **sizeof** (**int**) would return the value 4 for a 32-bit system. Note that the type specifier cannot be **enum**, **struct** or **class**.

4.6 Keywords

Keywords are words that form the language or grammar of C++ and have a special meaning to the compiler, such as **char**, **int** and **enum**. As a consequence, a keyword cannot be used as the name of an identifier. Some compilers, such as Borland C++, also have additional non-ANSI keywords, which are generally underscored, e.g. **_asm** and **__asm**. However, if you require ANSI-compatibility and portability then always use ANSI-specified keywords.

It is fascinating to compare the C++ language, which comprises approximately 50 keywords, with, say, the English language, which is composed of literally millions of words. Computer languages have a long way to go before they reach the sophisticated level of abstraction of traditional spoken-written languages[3].

All of the keywords in C++ are listed in Appendix A.

4.7 Declaration or Definition?

The difference between *declaration* and *definition* is a subtle one. A declaration introduces a variable's or object's name and associates a type with the variable/object. A definition is a declaration that simultaneously allocates memory for the variable/object.

The following statement is in fact a definition, because it not only declares the variable's name, x, and type, **double**, but also allocates 8 bytes of memory for x:

```
double x ;
```

To help emphasise the distinction between declaration and definition, consider the **enum** Colour once more. The following is an **enum** declaration:

```
enum Colour
  {
  RED, GREEN, BLUE
  };
```

whereas the statement:

```
Colour colour ;
```

defines a variable, colour, of type Colour.

If, throughout the text, I forget the difference between declaration and definition, then please forgive me!

3 The traditional written and spoken languages are often referred to as *natural* languages so as to distinguish them from *artificial* programming languages.

In the above example programs a variety of constant and variable names have been chosen. So what constitutes a valid and sensible name? Basically, C++ allows you complete control to dream up whatever names you like, except for three main exceptions:

- keywords: You cannot use a C++ keyword as an identifier name.
- digits: You cannot use a digit as the first character of an identifier name, e.g. `2i`.
- operators: You cannot use an operator as part of an identifier name, e.g. `c+p`.

C++ allows an unlimited number of characters to define an identifier, although some compilers may still adhere to the pre-standardisation limit of 32 characters. Don't get carried away dreaming up too-descriptive identifiers, which can be tedious to work with. In the following, an example of 'Hungarian notation' (`hPrevInstance`) is used to shorten an identifier and make the identifier more informative (Petzold, 1992). For example, all of the following are legal:

```
int       i2 ;
const int I_CONST = 1 ;
int       handle_to_previous_instance ; // OK but lengthy
                                        // name!
int       hPrevInstance ;               // manageable
                                        // identifier
```

whereas the following are illegal:

```
int    2i ;
float  f+ ;
double at@ ;
```

For a more detailed discussion of the style that I adopt throughout this book, refer to Section 3.12.

4.8 Assignment

Statements such as:

```
int i ;       // definition
//...
i = 99 ;      // assignment
//...
int j = 0 ;   // definition and initialisation thro'
              // assignment
```

assign values to variables. Assignment is performed with the aid of the assignment operator =.

4.9 Initialisation

We have seen above that variables *can be* initialised when they are defined, whereas constants *must be* initialised when they are defined:

```
char       c_var ;
const char C_CONST = 'c' ;
//...
```

Previous sections have illustrated that the fundamental data types are initialised by default to zero (0), although some older compilers may still adopt the pre-standardisation style of leaving uninitialised variables undefined. Zero (0) is an **int** and can be used as a constant for any fundamental data type. Strictly speaking, if no initialiser is specified then a global, namespace or local static object is initialised to 0.

The following defines a variable `i` of type **int** but does explicitly initialise `i`:

```
int i ;  // defined but not explicitly initialised!
//...
```

Be careful of uninitialised variables – don't assume that they are equal to zero. For instance, local objects (objects created on the free store) are not initialised by default:

```
void Function ()
  {
  int i ;  // initialisation of i is not well defined
  }
```

If you want a variable to be initialised to zero then it is good practice to do so explicitly:

```
int i = 0 ;  // defined and initialised to zero
//...
```

A neater and more formally correct form of initialisation, which will become clearer when we discuss classes in Chapter 9, is to use the variable's type/**class** constructor:

```
int i (0) ;  // also defined and initialised to zero
//...
```

4.10 Expressions

An *expression* is a collection of operators and identifiers (or operands) used in a specified order to evaluate a computation. An example expression is:

```
z = x + y ;
```

Note the distinction between an expression and a statement. The above is in fact a statement which assigns the addition of `x` and `y` to `z`. The semicolon informs the compiler that the expression is complete and that the code to evaluate the expression can be generated. Without the semicolon the compiler would assume that the expression was incomplete.

Note also that the order of evaluation of sub-expressions within an expression is undefined. For example, in the following statement it is undefined whether function *F()* or *G()* is called first:

```
int value = F(i) + G(j) ;
```

Since such behaviour is undefined, do not enforce restrictions of evaluation order.

4.11 Operators

C++ contains a variety of operators; see Appendix C. Operators are generally categorised in terms of *unary*, *binary* and *ternary* (the *arity* of the operator). A unary operator can be applied to a single operand; e.g. –45. A binary operator has two operands (e.g. 40 + 45), and a ternary operator has three operands. 'A ternary operator?', I hear you ask. C++ supports only one ternary operator (`? :`, the conditional operator), which is used in **`if-else`** statements and is examined in the next chapter.

An *operand* is an identifier or object which an operator acts on. In the expression:

```
a = b ;
```

the operator is the assignment sign and the operands are `a` and `b`. There are two operands because = is a binary operator.

4.11.1 Arithmetic Operators

The addition, subtraction, multiplication and division arithmetic binary operators (`+`, `-`, `*` and `/`) form the basis of mathematical computations. There is another arithmetic binary operator, `%` (the *modulus* operator), which is a bit different from the others. For example, `4%3=1`, `4%4=0` and `4%5=4`. The modulus operator gives the remainder of `operand1` divided by `operand2`:

```
operand1 % operand2
```

Naturally, the modulus operator is restricted to integer operands.

4.11.2 Chaining Operators

Operators can be *chained* or *cascaded*. Chaining is simply using an operator repeatedly in a single statement. The following example code illustrates chaining with the insertion, addition and assignment operators:

```
cout << "x = " << x << "; y= " << y << "; z = " <<  z ;
i = 2 + 3 + 4 ;
x0 = y0 = z0 = 0.0 ;
//...
```

The last statement is equivalent to the following three statements:

```
x0 = 0.0 ;
y0 = 0.0 ;
z0 = 0.0 ;
```

Chaining can also be performed on the extraction operator (>>):

```
//...
```

```
cout << "enter two positive or negative numbers: " ;
cin  >> number1 >> number2 ;
```

The extraction operator is chained to allow the user to enter two numbers separated by a space.

4.11.3 Operator Shortcuts

In computing, the following statement is frequently seen:

```
sum = sum + value ;
```

which at first appears slightly confusing and contradictory because `sum` appears on both sides of the = operator. Since the precedence of + is greater than =, the expression `sum+value` is first evaluated and then the result assigned to `sum`, thus overwriting `sum`'s previous value. Because such statements are so popular in C++, an alternative notation is offered: the *arithmetic assignment operators* (+=, -=, *=, /= and %=). Thus, the above can be written more concisely as:

```
sum += value ;
```

Appendix C also shows other bitwise and bit-shift assignment operators: &=, |=, ^=, <<= and >>=, which will be covered in Chapter 5.

4.11.4 The Increment (++) and Decrement (--) Operators

In computing, we frequently need to increment a variable by the constant 1:

```
int count = 0 ;
//...
count = count + 1 ;
```

With our 'new' arithmetic assignment operator (+=) we could rewrite this as:

```
//...
count += 1 ;
```

But there exists a more concise representation in C++ with the aid of the *increment* operator (++):

```
count++ ;
```

Actually, there are two versions of the increment operator: *prefix* and *postfix*. The following is prefix:

```
++count ;
```

while

```
count++ ;
```

is postfix.

To help explain the difference between prefix and postfix let us consider the two separately, prefix first:

```
int c = 1 ;
//...
value = ++c ;  // increment before assignment
//...
// value = 2, c = 2
```

where c is *incremented before assignment.* In the case of the postfix operator:

```
int c = 1 ;
//...
value = c++ ;  // increment after assignment
//...
// value = 1, c = 2
```

where c is *incremented after assignment.*

You may be thinking that this is a subtle difference, and you'd be right. However, there are occasions where the distinction between prefix and postfix can be useful.

Similarly, we have the *decrement* operator (--). Therefore, for the prefix decrement operator:

```
int c = 1 ;
//...
value = --c ;  // decrement before assignment
//...
// value = 0, c = 0
```

while the postfix decrement operator case gives:

```
int c = 1 ;
//...
value = c-- ; // decrement after assignment
//...
// value = 1, c = 0
```

4.11.5 Operator Precedence, Associativity and Arity

Consider the following expression:

```
10 - 4 * 3 ;  // = -2
```

Is this expression to be interpreted as `(10-4)*3` or `10-(4*3)`? Who knows? However, the computer has to apply some kind of logic to a situation like this, and the system that it adopts is to associate a precedence level with each of C++'s operators. Precedence, in the present context, means that an order of processing is associated for each operator. *Sub-expressions* are evaluated according to the following rule: *highest precedence wins.* Therefore, in the expression above the sub-expression `(4*3)` is evaluated first, followed by the sub-expression `(10-12)`. This is be-

cause the precedence level of multiplication (*) is 13 and that of subtraction (-) is 12, so multiplication wins. Hence, the above expression is evaluated to -2.

If you think that the above expression should equal 18, then the way around this problem is to use parentheses, (and), which have a higher precedence level (16) than the multiplication sign:

```
(10 - 4) * 3 ;   // = 18
```

What if an expression contains operators with the same precedence? For instance, the multiplicative (*, / and %) operators all have the same precedence, so how is the following expression evaluated?

```
12 / 2 * 3 ;   // = 18
```

The answer lies in the *associativity* of the operators. An expression involving operators of the same precedence is evaluated according to the associativity of the operators. All of the multiplicative operators are left-associative, so that the above expression is evaluated from left to right: `(12 / 2) * 3`. If you are ever in doubt about the exact order of evaluation of an expression, use parentheses. Parentheses help to clarify the intended order of evaluation in an expression.

Associativity is also helpful in clarifying the order of assignment. Consider the following expression, which was introduced above:

```
a = b ;
```

Is `b` assigned to `a` or `a` assigned to `b`? To us the answer is obvious, but the compiler has to apply the associativity rule to the assignment operator. From Appendix C we see that the assignment operator is right-associative, and therefore the right-hand operand, `b`, is assigned to the left-hand operand, `a`.

Consider the following use of the arithmetic (+) operator with the insertion (<<) operator:

```
cout << 1 + 2 ;
```

The addition operator is evaluated first because it has a higher precedence than the insertion operator. Thus the output is 3. However, certain compilers may issue a warning in such a situation, indicating that the operators need parentheses.

The arity of an operator is simply the operator's number of operands. For example, the insertion and addition operators are both binary because they operate on two operands. C++ supports the unary, binary and ternary arities.

We shall return to precedence, associativity and arity in Chapter 10, when we discuss operator overloading, but for now Appendix C lists all of the operators in the C++ language with their associated level of precedence, associativity and arity.

4.12 Manipulators

Manipulators are function-like operators that enable us to change the format of a stream. Table 4.6 lists the available stream manipulators. To be able to use the `setw()` and `setprecision()` manipulators you will have to include the IOMANIP.H header file.

Table 4.6 C++ stream manipulators.

Manipulator	*Description*
`dec`	Set decimal conversion base format flag
`endl`	Insert a new line and flush stream
`ends`	Insert a null termination in a string
`flush`	Flush an ostream
`hex`	Set hexadecimal conversion format flag
`oct`	Set octal conversion base format flag
`resetiosflags (long f)`	Clear the format bits specified by `f`
`setfill (int c)`	Set the fill character to `c`
`setiosflags (long f)`	Set the format bits specified by `f`
`setprecision (int n)`	Set the floating-point precision to `n`
`setw (int n)`	Set the field width to `n`
`ws`	Extract the white space characters

The use of manipulators and other functions or objects for formatting both program input and output is essential to good programming. Good formatting of program output can make a program more readable. To illustrate this, consider the formatted stream output (with the aid of the *`setw()`* and *`setprecision()`* manipulators) for the **`float`**, **`double`** and **`long double`** floating-point data types:

```
// manip.cpp
// illustrates the stream manipulators
#include <iostream> // C++ I/O
#include <iomanip>  // C++ manipulators

void main ()
  {
  // exponential
  const float       F_E  = 2.7182818 ;              //  7 d.p.
  const double      D_E  = 2.718281828459045 ;      // 15 d.p.
  const long double LD_E = 2.7182818284590452354 ; // 19 d.p.

  // 'straight' output
  cout << "float e: "       << F_E  << endl ;
  cout << "double e: "      << D_E  << endl ;
  cout << "long double e: " << LD_E << endl ;
  cout << endl ;

  // setw() 'manipulated' output
  cout << setw (15) << "float e: "
       << setw (7) << F_E  << endl
       << setw (15) << "double e: "
       << setw (7) << D_E  << endl
       <<              "long double e: "
       << setw (7) << LD_E << endl ;
  cout << endl ;

  // setw(), setprecision() 'manipulated' output
```

```
    cout << setw (15) << "float e: "
         << setw (21) << setprecision (8)  << F_E  << endl
         << setw (15) << "double e: "
         << setw (21) << setprecision (16) << D_E  << endl
         <<              "long double e: "
         << setw (21) << setprecision (20) << LD_E << endl ;
    }
```

which produces the following output:

```
float e: 2.71828
double e: 2.71828
long double e: 2.71828

      float e: 2.71828
     double e: 2.71828
long double e: 2.71828

      float e:            2.7182817
     double e:    2.718281828459045
long double e:  2.71828182845904509
```

The output is divided into three parts to illustrate the *setw()* and *setprecision()* manipulators. The first part is simply the default output provided by the output stream object `cout`. Observe that the output is unformatted and the number of decimal places is five. The second part uses *setw()* to give the text strings and the three constant numbers a field width. Finally, we use the *setprecision()* manipulator, passing the number of decimal places, to output the three constant numbers to the required levels of precision. Note that the **`float`** and **`long double`** constants are not displayed as accurately as expected!

The following sections describe several manipulators in more detail.

4.12.1 The `endl` Manipulator

The program above and many of the previous programs have used the `endl` manipulator with the `cout` stream object. The `endl` manipulator inserts a new line into the stream output and then *flushes* the stream.

4.12.2 The `flush` Manipulator

If you need to flush an output stream then use the `flush` manipulator, e.g.

```
cout << flush ;
```

4.12.3 The *setw()* Manipulator

Each output value has its own field width. The default field is just wide enough to hold the value or string being displayed. For instance, the string `"float e: "` has a field width of 8. The *setw()* manipulator allows you to specify your own field width. Clearly, *setw()* has to occur before the string.

4.12.4 The *setprecision()* Manipulator

setprecision() allows you to specify the number of decimal places of a displayed value.

4.12.5 The *setiosflags()* Manipulator

The C++ ios_base **class** (equivalent **typedef** ios for type **char**) provides operations that are common to both input and output. The ios_base **class** is a base **class** for the stream classes basic_istream, basic_ostream and basic_iostream. For the moment you don't need to know the details of the ios_base **class**, only that it contains format flags such as dec, oct, hex, showpoint and fixed. For example, with the aid of the *scope resolution* operator (: :) we are able to set the showpoint and fixed flags so that the output will always contain a decimal point and use a fixed (rather than exponential) decimal point for floating-point numbers:

```
cout << setiosflags (ios::showpoint)
     << setiosflags (ios::fixed)
     << setprecision (2)
     << 456.00f ;
```

Refer to your compiler's documentation for a complete list of the format flags (**enum** fmt_flags) available with the ios_base **class**, or alternatively refer to Section 17.6.2.

4.12.6 The *resetiosflags()* Manipulator

If any format flags have been set using the *setiosflags()* manipulator they can be unset using the *resetiosflags()* manipulator.

4.12.7 The dec, hex and oct Manipulators

To display a number as a decimal, hexadecimal or octal number use the dec, hex and oct manipulators respectively. For example:

```
int i = 9 ;
//...
cout << hex << i ;   // O/P integer variable i as a
                     // hexadecimal number
```

Note that the decimal base is the default.

The dec, hex and oct manipulators can also be used to manipulate input:

```
int number = 0 ;
//...
cout << "enter a hexadecimal integer: " ;
cin  >> hex >> number ;
```

The hex manipulator is used to configure the input to accept a hexadecimal integer.

4.13 Basic File Input and Output

In the previous section we examined output and the use of manipulators to the standard output device (the display screen) using the `cout` object and the insertion operator, `<<`. Although we will examine file input and output in detail in Chapter 17, let's take a brief look at basic file streams. Consider the following program, which writes some strings of text to a file called STRINGS.TXT[4]:

```
// ofile.cpp
// output of strings to a file
#include <fstream>  // C++ file I/O

void main ()
  {
  // file object and open file
  ofstream file_out ("STRINGS.TXT") ;

  file_out << "Jack's mother said, 'We're stony broke!"  << endl ;
  file_out << "'Go out and find some wealthy bloke"      << endl ;
  file_out << "Who'll buy our cow. Just say she's sound" << endl ;
  file_out << "'And worth at least a hundred pound. "    << endl ;
  file_out << "But don't you dare to let him know"       << endl ;
  file_out << "'That she's as old as billy-o'"           << endl ;

  file_out.close () ;  // close file
  }
```

Rather than outputting the strings of text to the display screen using `cout`, an output file stream object, `file_out`, is defined of the C++ library **`class`** `ofstream` (*o*utput *f*ile *stream*) and initialised to the text file STRINGS.TXT. Upon defining `file_out`, the file STRINGS.TXT is created if it does not already exist; it will be overwritten if it does already exist. Apart from operating on `file_out` rather than `cout` there is syntactically no difference in writing to a file rather than to the display screen. When we are finished writing to the output file, the file is closed by a call to the *`close()`* function. The peculiar looking `file_out` definition and initialisation and *`close()`* function call are in fact calls to **`class`** `ofstream`'s *constructor* and *member function* *`close()`* respectively. In order to define an object of **`class`** `ofstream`, the header FSTREAM.H must be `#included`.

Consider now writing integer numbers to a file:

```
// ofile1.cpp
// output of integers to a file

#include <fstream>  // C++ file I/O

void main ()
  {
  // file object and open file
```

4 Jack and the Beanstalk, *Roald Dahl's Revolting Rhymes*, Picture Puffins, London.

```
    ofstream file_out ("INTS.DAT") ;

    file_out << 0 << endl ;
    file_out << 1 << endl ;
    file_out << 2 << endl ;

    file_out.close () ;  // close file
    }
```

After executing OFILE1.EXE the contents of the file INTS.DAT are:

```
0
1
2
```

The similarity of programs OFILE.CPP and OFILE1.CPP indicates that the process of writing to a file using an `ofstream` object and the insertion operator is independent of the type of data written to the file.

So how about reading data from a file? Let's take a look at the following program, which reads the integers in file INTS.DAT generated by OFILE1.CPP and then proceeds to display the numbers to the display screen:

```
// ifile.cpp
// get integers from a file
#include <fstream>  // C++ file I/O

void main ()
  {
  // in-file object and open file
  ifstream file_in ("INTS.DAT") ;

  int i, j, k ;

  // get from file
  file_in >> i ;
  file_in >> j ;
  file_in >> k ;

  // O/P to screen
  cout << i << endl
       << j << endl
       << k << endl ;

  file_in.close () ;  // close file
  }
```

with output to the display screen:

```
0
1
2
```

To read a file we define a file object, `file_in`, of the C++ library **`class`** `ifstream` and initialise `file_in` to the file INTS.DAT. Then, instead of the standard input stream object `cin` and the extraction operator >>, the input file stream object `file_in` and >> are used to extract the three integers from file INTS.DAT. After displaying the integers on the screen (using `cout` and <<), the file INTS.DAT is closed.

Strictly speaking classes `ofstream` and `ifstream` are typedefs of classes `basic_ofstream` and `basic_ifstream` for type **`char`**:

```
typedef basic_ofstream<char,char_traits<char> > ofstream ;
typedef basic_ifstream<char,char_traits<char> > ifstream ;
```

Typedefs `ofstream` and `ifstream` are used for convenience. If the above discussion appears slightly confusing, all will become clear in Chapter 17 when we discuss file streams in more detail.

4.14 Type Conversion and Casting

Before discussing the storage class specifiers in the next section, let us consider automatic and user-specified type conversions. Consider what happens in the program:

```
int   working_day = 8 ;
float hourly_rate = 7.70 ;

double days_pay = working_day * hourly_rate ;
cout << "days pay: £" << days_pay ;
```

An **`int`** is multiplied by a **`float`** and assigned to a **`double`**! This is legal, but bad practice, and a compiler will perform the type conversion automatically and not issue a warning message.

Occasionally you need to be able to perform your own type conversions. C++ offers the programmer two alternative cast styles:

```
int i = 1 ;
//...
long j = (long) i ;   // C style
long k = long (i) ;   // C++ style
```

which both cast, or coerce, integer `i` to type **`long`**. The first of the two forms, **`(long)`**, is the C syntax, whereas **`long`** *`()`* is the generally adopted syntax in C++. If you are familiar with functions, then the syntax **`long`** *`()`* may at first appear to indicate that **`long`** is a function. However, as you become more familiar with programming in C++ the reasons for adopting the *functional notation* **`long`** *`()`* will make more sense.

Anyone who has programmed for Windows will be used to type casting to avoid error messages during compilation. For example, the Windows API (Application Program Interface) function *`SendMessageW()`* has the following signature `WINUSER.H`:

```
LRESULT SendMessageW (HWND hWnd, UINT uMsg,
                      WPARAM wParam, LPARAM  lParam) ;
```

The *SendMessageW()* function sends a specified message (uMsg) to the given window (hwnd) procedure function and does not return until that window procedure has processed the message. The relevant defined types in WINDOWS.H/WINDEF.H are:

```
typedef unsigned int WORD ;
typedef unsigned int UINT ;
typedef signed long  LONG ;
typedef UNIT         WPARAM ;
typedef LONG         LPARAM ;
typedef LONG         LRESULT ;
//...
```

A typical use of *SendMessageW()* could be:

```
//...
char szBuffer[MAX] ;
WORD n ;
//...
n = WORD (SendMessageW (hwndList, LB_GETTEXT, n,
                        LONG (LPSTR(szBuffer)))) ;
//...
```

which casts the string szBuffer first to type LPSTR (long pointer to string) and then to a LONG and finally casts the *SendMessage()* function return type to WORD – nasty!

4.15 The Storage Class Specifiers auto, extern, register and static

C++ supports four[5] storage class specifiers, called **auto**, **extern**, **register** and **static** (each a keyword), which inform the compiler of exactly how a variable is to be stored. The following subsections discuss each specifier separately.

4.15.1 The auto Specifier

The **auto** storage class specifier explicitly declares a variable as local or *automatic*. Since local variables are declared **auto** by default, **auto** is essentially redundant and hardly ever used:

```
// auto.cpp
// illustrates the auto storage class specifier
#include <iostream>  // C++ I/O

void main ()
  {
  int       i = 0 ;
  auto int ai = 0 ;
```

5 Discounting **mutable**, which will be discussed in Chapter 9.

```
cout << i  << endl ;
cout << ai << endl ;
}
```

with output:

```
0
0
```

4.15.2 The `extern` Specifier

Computer programs generally consist of several separate different source files or modules. Consider that you are working with a given file, FILE.CPP, in which you declare a number of variables that you would like to access from another file. Declaring the variables with the **`extern`** specifier gives the variables *external linkage* and can be accessed from another file. Here is FILE.CPP:

```
// file.cpp
// some variable definitions

char ch = 'a' ;
int  i  = 1 ;
```

If the file FILE.CPP is linked with another file, say EXTERN.CPP, then EXTERN.CPP has access to the variables `ch` and `i` if the **`extern`** specifier is used for each variable:

```
// extern.cpp
// illustrates the extern storage class specifier
#include <iostream>  // C++ I/O

extern char ch ;
extern int  i ;

void main ()
  {
  cout << "ch: " << ch << "; i: " << i << endl ;
  }
```

with output:

```
ch: a; i: 1
```

Although the variables `ch` and `i` were defined and initialised in FILE.CPP they are accessible in EXTERN.CPP with the help of the **`extern`** *linkage specification* declaration. It is worth noting that **`extern`** is a declaration and not a definition, and therefore does not allocate memory; see the previous discussion on declarations and definitions. Try removing the **`extern`** keyword from EXTERN.CPP and relink:

```
// extern.cpp
//...
char ch ;
int  i ;
```

```
void main ()
  {
  cout << "ch: " << ch << "; i: " << i << endl ;
  }
```

A linker error of the form 'variables _ch and _i defined in module FILE.CPP are duplicated in module EXTERN.CPP' will be produced. This error occurs because in a C++ program a variable can be defined only once but may be declared any number of times:

```
int i = 1 ;
//...
int i ;  // error: multiple definition for i
```

Alternatively, if variables `ch` or `i` had been declared **`extern`** but simultaneously initialised, then the **`extern`** specifier would be ignored because an initialised declaration is a definition:

```
// extern.cpp
//...
extern char ch = 'b' ;
extern int  i  = 2 ;

void main ()
  {
  cout << "ch: " << ch << "; i: " << i << endl ;
  }
```

This will again result in a linker error because of the multiple definitions of `ch` and `i`.

The **`extern`** specifier is also applicable to functions, which are discussed in Chapter 6.

4.15.3 The `register` Specifier

The **`register`** specifier is a request to the compiler that a variable be allocated a *register* of the processor or cache memory. Thus, **`register`** variables are accessed more quickly than non-register variables. The keyword **`register`** is only a request, and the compiler has the last say on whether or not a variable is allocated a register. If a register is not available, a local **`auto`** variable is allocated. The **`register`** specifier can be applied to any variable type.

The following program illustrates **`register`**:

```
// register.cpp
// illustrates the register storage class specifier
#include <iostream>  // C++ I/O

void main ()
  {
  register char   ch  = 'a' ;
  register int    i   = 0 ;
  register double d   = 0.0 ;

  cout << "ch: "   << ch
       << "; i: "  << i
```

```
        << "; d: "  << d << endl ;
}
```

with output:

```
ch: a; i: 0; d: 0
```

The **register** specifier is best reserved for variables that are used frequently. A good case for using a **register** variable is as the control variable in a loop:

```
for (register int i=0; i<MAX; i++)
  { /* ... */ }
```

The **for**-loop is discussed in the next chapter.

4.15.4 The `static` Specifier

The **extern** specifier, discussed above, illustrated external linkage. The **static** specifier declares a variable with *internal linkage*; that is, local to a source file. To illustrate **static**, let us modify the file FILE.CPP, presented above, to include **static** definitions of variables `ch` and `i`:

```
// file1.cpp
// some static variable definitions

static char ch = 'a' ;
static int  i  = 1 ;
```

noting that **static** variables can be simultaneously defined and initialised, just as non-**static** variables can. A second file, STATIC.CPP, also contains **static** definitions of **char** and **int** variables with the same names `ch` and `i`:

```
// static.cpp
// illustrates the static storage class specifier
#include <iostream>  // C++ I/O

static char ch = 'b' ;
static int  i  = 2 ;

void main ()
  {
  cout << "ch: " << ch << "; i: " << i << endl ;
  }
```

On linking and executing the two files we have the following output:

```
ch: b; i: 2
```

Although both files define `ch` and `i` the program will link correctly because the **static** specifier ensures that each file has its own `ch` and `i`, therefore eliminating name clashes and multiple definitions.

The **static** specifier in connection with functions will be discussed in Chapter 6.

4.16 The asm Declaration

The C++ language allows assembly language instructions to be placed in a C++ program rather than being assembled in separate assembly modules. The keyword **asm** is used to inform the compiler of assembly instructions. Although **asm** is implementation-dependent, compilers generally support the following **asm** syntax:

```
asm instruction instruction_operands ;

asm instruction instruction_operands newline

asm
   {
   // multiple instructions
   }
```

The first **asm** statement contains an assembly language instruction followed by possible operand(s) for the instruction and terminated with a semicolon. The second form is identical to the first except that the end of an **asm** statement is terminated by a new line rather than a semicolon. The final **asm** declaration includes multiple **asm** instructions between the opening, {, and closing, }, braces.

A discussion of assembly language programming is beyond the scope of this book, but to illustrate the three types of **asm** declaration discussed above consider the following program:

```
// asm.cpp
// illustrates the asm keyword
#include <iostream>  // C++ I/O

void main ()
   {
   int i = 0 ;

   // call to interrupt procedure
   asm INT 0x05 ;

   // move registers
   asm MOV i, 1

   // push and pop
   asm
      {
      PUSH AX ;
      PUSH CX ;
      //...
      POP AX ;
      POP CX ;
      }
   }
```

4.17 Summary

It is important to have a very good understanding of the fundamental data types, simply because they are so fundamental to C++ programming. This chapter has introduced you to the four primitive types: **char**, **int**, **float** and **double**. Variations of these exist through the use of the modifiers **short**, **long**, **unsigned**, **signed**, **const** and **volatile**. Also, C++ possesses three other data types that have been introduced: **wchar_t**, **enum** and **bool**.

Standard input and output has been introduced through the use of the stream objects `cin` and `cout` and the operators >> and <<, and input received from the keyboard and output sent to the display screen. The manipulation of files has been briefly introduced by defining file stream objects of the C++ library classes `basic_ofstream/ofstream` and `basic_ifstream/ifstream`. Input and output can be formatted by the use of *manipulators*, which operate on the stream objects.

The C and C++ syntax styles of *casting* have been discussed for converting an object of one data type to an object of another data type. Casting will be discussed in much more detail in Chapter 16 when the C++ *run-time type information* language feature is covered.

The four storage class specifiers **auto**, **extern**, **register** and **static** were introduced. The **auto** specifier explicitly declares a variable as local, but is rarely used because variables are local by default. The **extern** specifier attaches external linkage to a variable, whereas **static** attaches internal linkage to a variable. A register can be allocated to a variable using the **register** specifier so that a frequently used variable can be accessed more quickly.

The **asm** keyword allows assembly language code to be integrated directly into C++ code.

We have also been introduced to the basics of programming in C++ and program construction. Different types of operators and expressions have been examined for the four primitive types. If there are any points that you don't fully understand in this chapter, give them a second visit.

Exercises

4.1 Run the LIMITS.CPP, FLOATS.CPP and SIZES.CPP program files to test your compiler's implementation of the integral type ranges.

4.2 Write a program that gets four characters from a user via the keyboard using the `cin` input stream object and the extraction operator, >>.

4.3 Write a program that reads in values entered by a user from the keyboard for the coefficients a, b and c of the quadratic equation $ax^2 + bx + c = 0$ and then proceeds to find the two roots of this general equation, which are given by:

$$x = \frac{-b \pm (b^2 - 4ac)^{1/2}}{2a}$$

Test your program with several values for a, b and c for which the discriminant ($b^2 - 4ac$) remains positive.

Hint: Include the header file MATH.H in your program to enable you to use the square root function:

```
double sqrt (double) ; // #include <cmath>
```

4.4 Display the following numbers to the display screen ensuring that the formatting and precision are preserved:

```
3.14159
2.71828
1.41421
1.293
9.998e+02
11370.00
```

Use the *setw()*, *setprecision()* and *setiosflags()* manipulators.

4.5 What are the results of the following expressions?

```
12 * 4 + 3 * 2
6  * 4 / 2 % 2
```

4.6 A file called EXP.DAT contains three integer numbers:

```
142
857
285
```

Write a program which reads in the above three numbers and displays the average to the display screen.

4.7 Cast a floating-point number to an integer number.

Chapter 5

Making Decisions and Repetition

Second Thoughts. Don't have them. Switch over.
Gerard Gilbert, TV Guide 23/9/94, *The Independent*

Programming is mainly concerned with getting a computer program to demonstrate a little intelligence or to make decisions. The basis of a decision in computing lies ultimately in testing whether an expression is true or false. This chapter introduces the C++ tools available for making decisions: the if, if-else and switch statements and the conditional operator, ?:, which is a specialised form of the if-else statement.

The great power of computers is their ability to perform a task over and over again (namely repetition) with phenomenal speed, accuracy and reliability. Repetition is something that we humans find very boring and difficult to do. In this chapter we introduce the three facilities offered by C++ for performing repetition: the for loop, the while loop and the do-while loop.

The chapter concludes by examining the bitwise and shift operators.

5.1 Decisions

A program frequently requires the capability to make a decision based on some *known*[1] information. For example, if a car's speed is below the speed limit then accelerate; otherwise, if the car's speed is above the speed limit then decelerate. Given the speed of the car and the speed limit we can make this kind of accelerate or decelerate decision. Decisions occur in a variety of disguises: *on–off*, *yes–no*, *true–false* etc.

C++ offers us two key ways of making decisions: the **`if-else`** and **`switch`** statements. We shall begin by examining the **`if`** statement alone, followed by the **`if-else`** decision statement, after which we shall discuss the **`switch`** statement, finishing by examining the *conditional* operator (`? :`), which is a specialised form of the **`if-else`** statement.

1 Note the emphasis placed on known information. We can always make a purely random or a *blind* decision, but we shall not be concerned with such decisions.

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

5.1.1 The `if` Statement

Consider the program:

```
// if.cpp
// illustrates the if statement
#include <iostream> // C++ I/O

void main ()
  {
  float number ;

  // get user-entered number
  cout << "enter a positive or negative number: " ;
  cin  >> number ;

  // compare number to zero
  if (number > 0)
    cout << number << " is greater than zero" << endl ;
  }
```

The following shows a typical program–user interaction:

```
enter a positive or negative number: 10.12
10.12 is greater than zero
```

Upon execution the program extracts a `number` from the user and then proceeds to test whether the `number` is greater than zero. The keyword **`if`** directly followed by the required expression in parentheses (`number>0`) performs the comparison. The general syntax of the **`if`** statement is:

```
if (test expression)
  statement ;  // single statement if-body
```

Figure 5.1 illustrates the **`if`** statement schematically. Think of the **`if`** statement as a box in which we have only an input and an output. If the input 'passes the test' then the box *fires* and the box produces an action. Alternatively, if the box 'fails the test' then the box produces no output.

Try running the above program but enter a negative number. What happens? Nothing. The program is so simple that it can only test whether the number entered is greater than zero.

OK: let's make the above program a bit more intelligent:

```
// ifif.cpp
// illustrates the if statement
#include <iostream>  // C++ I/O

void main ()
  {
  float number ;
```

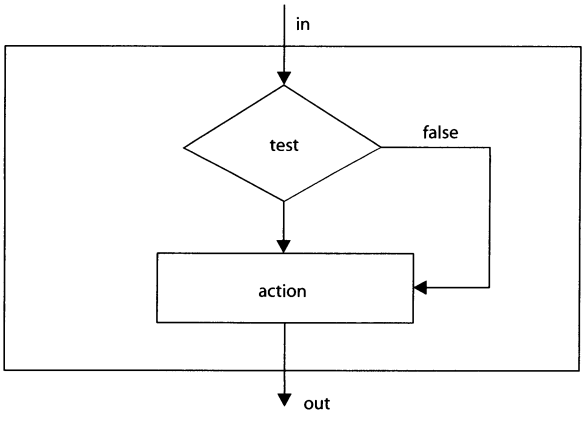

Fig. 5.1 `if` statement.

```
// get user-entered number
cout << "enter a positive or negative number: " ;
cin  >> number ;

// compare number to zero
if (number > 0)
  cout << number << " is greater than zero" << endl ;

if (number < 0)
  cout << number << " is less than zero" << endl ;

if (number == 0)
  cout << number << " is equal to zero" << endl ;
}
```

This program simply adds another **`if`** statement which tests whether the number entered is less than zero. Actually, a third **`if`** statement is also added, which tests whether the number is in fact equal to zero[2]. This has been performed to give the program *closure*. A program without closure (in the present case, the possibility of a user entering zero) is prone to bugs. Thinking of special closure cases such as these to make a program more watertight is the bane of programming.

Using multiple **`if`**s works fine, but there is a more elegant way: the **`if-else`** statement.

5.1.2 The `if-else` Statement

If we examine each of the above three test expressions in the **`if`** statements, we observe that in each case the test expression has to be true for the single statement in the **`if`** body to be executed. However, in making decisions we are frequently interested in performing a true–false comparison rather than a series of true conditions. The **`if-else`** statement is designed for just such a comparison. The following program illustrates the **`if-else`** version:

2 Is zero positive or negative or both?

```
// if_else.cpp
// illustrates the if-else statement
#include <iostream>  // C++ I/O

void main ()
  {
  float number ;

  // get user-entered number
  cout << "enter a positive or negative number: " ;
  cin  >> number ;

  // compare number to zero
  if (number > 0)
    cout << number << " is greater than zero" << endl ;

  else if (number < 0)
    cout << number << " is less than zero" << endl ;

  else
    cout << number << " is equal to zero" << endl ;
  }
```

Note that if the program did not have to test whether the entered number is equal to zero we could simply use an **if-else** statement rather than an **if-else if-else**:

```
//...
if (number > 0)  // test if true
  cout << number << " is greater than zero" << endl ;
else             // else false
  cout << number << " is less than zero" << endl ;
}
```

The general syntax of the **if-else** selection statement is:

```
if (test expression)
  statement ;
else if (test expression)
  statement ;
//...
else              // none of the above
  statement ;
```

and can have any number of **else if** statements. Figure 5.2 illustrates the operation of the **if-else** statement. If the test is either true or false then a respective action is generated. The flow through a multiple **if-else** statement is shown in Figure 5.3.

To date, we have only used **if** and **if-else** statements with a single statement. If we require *multiple* or *compound* statements within an **if** or **if-else** block then we have to enclose (or delimit) the block of statements by a pair of braces, { and }:

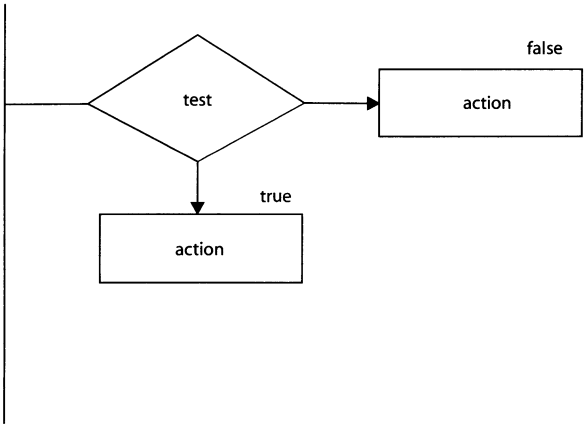

Fig. 5.2 `if-else` statement.

```
// if_comp.cpp
// illustrates the if-else compound statements
#include <iostream>  // C++ I/O

void main ()
  {
  float number ;

  // get user-entered number
  cout << "enter a positive or negative number: " ;
  cin  >> number ;

  // compare number to zero
  if (number > 0)
    {
    cout << number << " is greater than zero" << endl ;
    cout << "try again entering a negative number" << endl ;
    }
  else if (number < 0)
    {
    cout << number << " is less than zero" << endl ;
    cout << "try again entering a positive number" << endl ;
    }
  else
    {
    cout << number << " is equal to zero" << endl ;
    cout << "why don't you enter a nonzero number?" << endl ;
    }
  }
```

The general syntax of the **`if-else`** statement with compound body statements is:

```
if (test expression)
  {
```

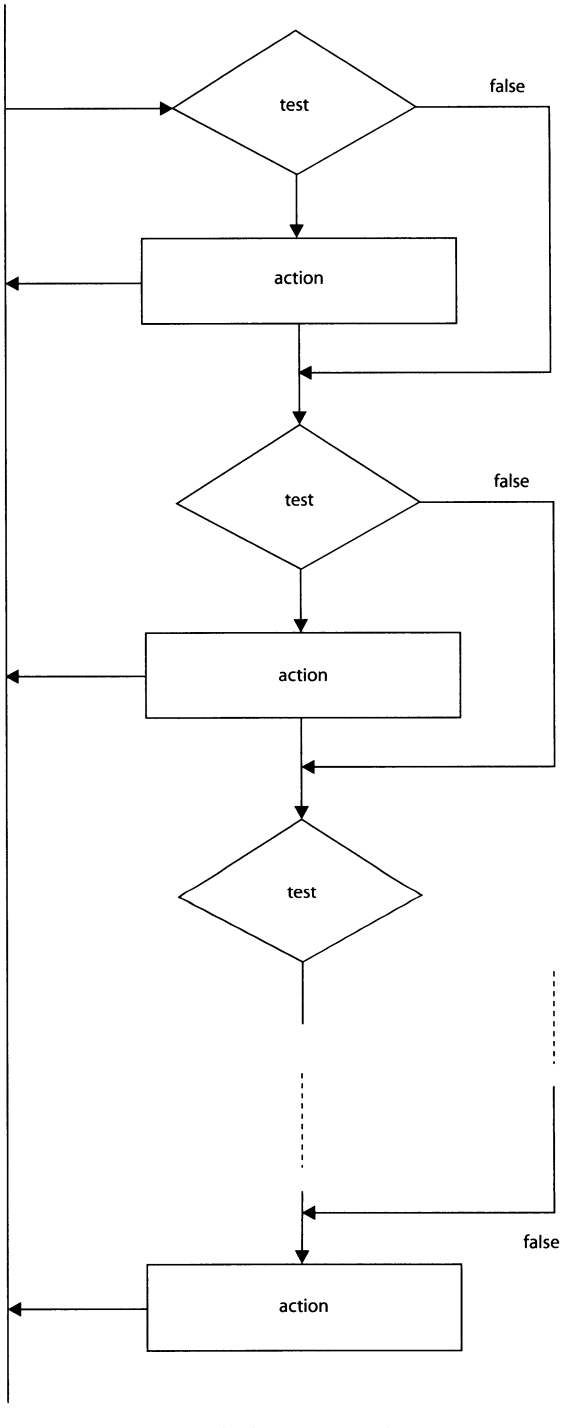

Fig. 5.3 Multiple `if-else` selection.

```
    statement1 ;  // compound statements if-body
    statement2 ;
    //...
    statementn ;
    }
  else if (test expression)
    {
    statement1 ;
    statement2 ;
    //...
    }
  else
    {
    statement1 ;
    statement2 ;
    //...
    }
```

Note the style adopted of indenting the compound statement delimiters by one tab stop. This style makes it easier to read the compound statements. Several programmers adopt the following style:

```
if (test expression) {
  statements ;
  //...
}
```

which, although it occupies one line fewer than the previous style, in my view does not help to clarify the **if** statement body.

A more useful illustration of the compound **if-else** statement is the following program:

```
// if_manip.cpp
// illustrates the if-else statement and the
// dec, hex and oct stream manipulators
#include <iostream>   // C++ I/O
#include <iomanip>    // dec, hex, oct

void main ()
  {
  char ch     = 'x' ;
  int  number = 0 ;

  cout << "enter a format; 'd' decimal,
            'h' hexadecimal, 'o' octal: " ;
  cin  >> ch ;

  if (ch == 'd')
    {
    cout << "enter a decimal integer: " ;
    cin  >> dec >> number ;
    cout << "hexadecimal: " << hex << number << endl
```

```
        << "octal      : " << oct << number << endl ;
    }
  else if (ch == 'h')
    {
    cout << "enter a hexadecimal integer: " ;
    cin  >> hex >> number ;
    cout << "decimal: " << dec << number << endl
         << "octal  : " << oct << number << endl ;
    }
  else
    {
    cout << "enter an octal integer: " ;
    cin  >> oct >> number ;
    cout << "decimal    : " << dec << number << endl
         << "hexadecimal: " << hex << number << endl ;
    }
}
```

The above program prompts the user to enter one of three characters: `d`, `h` or `o`, which respectively inform the program whether the user is entering an integer in decimal, hexadecimal or octal base. The **`if-else`** statement then directs program flow to prompt the user to enter an integer in the requested base, and then displays the integer entered in two alternative bases. Note that both the input and output are manipulated using the `dec`, `hex` and `oct` stream manipulators; refer to Chapter 4. An example of some user interaction is:

```
enter a format; 'd' decimal, 'h' hexadecimal, 'o' octal: h
enter a hexadecimal integer: ffa
decimal: 4090
octal  : 7772
```

5.1.3 Nested `if`s

Nested **`if`**s are **`if`** statements within **`if`** statements within.... The following program illustrates nested **`if`**s:

```
// if_nest.cpp
// illustrates nested ifs
#include <iostream>  // C++ I/O

void main ()
  {
  float number ;

  // get user-entered number
  cout << "enter a positive or negative number: " ;
  cin  >> number ;

  // compare number to zero
  if (number > 0)
    {
    cout << number << " is greater than zero" << endl ;
```

```
    if (number < 100)   // nested if
      {
      cout << "0 < " << number << " < 100" ;
      }
    }
  else
    {
    cout << number << " is less than zero" << endl ;
    if (number > -100)   // nested if
      {
      cout << "-100 < " << number << " < 0" << endl ;
      }
    }
}
```

You must be getting familiar with this `number` program by now! This version uses nested **`if`**s to test whether the number entered is within the range (0 < `number` < 100) if the number is positive or in the range (-100 < `number` < 0) if the entered number is negative. Note that it wasn't actually necessary to use the delimiters { and } for the nested **`if`**s, since both **`if`**-bodies are single statements, but the delimiters increase the clarity of the program.

Similarly, nested **`if-else`** statements are allowed:

```
if  (number > 0)
  {
  //...
  }
else
  {
  if (/*...*/)
    {
    //...
    }
  else
    {
    if (/*...*/)
      {
      //...
      }
    //...
    }
  //...
  }
```

Precedence of Scope

Before we examine relational operators, let us take a quick look at the *scope* or *visibility* of variables defined within different levels of **`if`** statements:

```
void main ()
  {
  int i = 10 ;
```

```
//...
if (/*...*/)  // outer if
  {
  int i = 20 ;
  //...
  if (/*...*/) // nested if
    {
    int i = 30 ;
    //...
    }
  //...
  }
//...
}
```

The scope of a variable within an **if** statement is local to the statement block in which it is defined. In other words, the variable dies or is not visible from outside the statement block. The above program illustrates three variables defined with the same name, i. In each **if** statement block the variable with the most current scope takes precedence.

Chapter 6 examines *local* and *global* scope more closely for function definitions.

5.1.4 Relational Operators

Relational operators are used to test the *relationship* between two expressions. In the programs above we have seen the three relational operators less than (<), greater than (>) and equal to (==). Table 5.1 lists the relational operators in C++.

Comparing these operators with the operators in Appendix C we observe that the relational operators are all binary operators and left-associative. Strictly speaking, the equality and inequality operators are referred to as *equality* operators, but for most purposes it suffices to consider them as relational operators. In addition, we see from Appendix C that the precedence of the == and != operators is one less than the other four. A general expression that uses the relational operators is of the form:

```
expression1 relational_operator expression2
```

The result of the comparison will be either logical-true or logical-false. In C++ logical-false is indicated by 0, whereas logical-true can be any non-zero value (e.g. 1, 21 or -11). The following **if** statement is therefore always true:

Table 5.1 Relational operators.

Operator	*Description*
<	less than
<=	less than or equal to
>	greater than
>=	greater than or equal to
==	equal to
!=	not equal to

```
if (-11)
  cout << "this if-body statement will always be executed!" ;
//...
```

Although any non-zero value indicates a logical-true, it is general practice to use the constant 1.

In the example program IF.CPP above we saw the following use of the greater than (>) relational operator:

```
if (number > 0)
//...
```

Also note that the equal to (==) operator uses two = signs. Don't make the mistake of using a single = sign (assignment operator) in a relational operation:

```
if (number = 0)   // don't!
//...
```

This expression means that `number` is assigned the value zero and therefore the conditional test will always equate to logical false. This is easy to do and is not a compilation error, although the Borland C++ (version 5.x) compiler issues a 'possibly incorrect assignment' warning message.

There is a strange way around this problem in C++, although I don't recommend using it. Simply flip round the expression:

```
if (0 == number)   // O.K. but weird!
```

This works – try it. If you now omit one of the = signs then a compilation error is generated:

```
if (0 = number)    // error: lvalue required
```

The reason why a compilation error is issued is because a constant does not have an *lvalue* or a memory address, so the compiler has nowhere it can store the variable `number`.

5.1.5 Logical Operators

Logical operators allow us to combine boolean (true–false) values logically. There are three logical operators provided by C++, which are listed in Table 5.2.

A general expression that uses the binary logical operators `&&` and `||` is of the form:

```
expression1 logical-operator expression2
```

The logical operators `&&` and `||` are both binary operators and left-associative. The logical operator `!` is a unary operator and right-associative and has the general form:

Table 5.2 Logical operators.

Operator	*Description*
`&&`	logical AND
`||`	logical OR
`!`	logical NOT

```
! expression
```

Let's now examine each of the three logical operators in turn.

Logical AND Operator (&&)

The logical AND operator (&&) is used to test whether two expressions are logically true or false. Consider the following program that prompts a user for two numbers (yes two!):

```
// log_and.cpp
// illustrates the logical AND operator (&&)
#include <iostream>  // C++ I/O

void main ()
  {
  float number1, number2 ;

  // get user-entered numbers
  cout << "enter two positive or negative numbers: " ;
  cin  >> number1 >> number2 ;

  // compare both numbers to zero using logical AND (&&)
  if (number1 > 0 && number2 > 0)
    {
    cout << number1 << " and " << number2
         <<  " are greater than zero" << endl ;
    }
  else
    {
    cout << number1 << " and " << number2
         <<  " are not greater than zero" << endl ;
    }
  }
```

The **if** statement is now of the form:

```
if (number1 > 0 && number2 > 0)
```

which will be logically true only if *both* `number1` and `number2` are greater than zero. Only if both expressions for the logical AND operator are true will the **if**-body statement be displayed. Since we are now testing two expressions, each of which can be true or false, we have four possible permutations. These four permutations are usually presented in the form of a *truth table*; see Table 5.3.

Table 5.3 illustrates that only when both expressions are logical-true is the logical result true. The other three permutations each produce a logical-false result.

You may be wondering why the **if** statement wasn't written as:

```
if (number1 && number2 > 0)
```

Try it and see what happens. If you enter the two numbers 10 and 11 then the string '`10 and 11 are greater than zero`' is correctly displayed. However, if you enter the two num-

Table 5.3 Logical AND truth table.

expression1	*expression2*	*Logical result*
false	false	false
false	true	false
true	false	false
true	*true*	*true*

bers -12 and 13 then the string '`-12 and 13 are greater than zero`' is incorrectly displayed. This form of condition test is not testing `number1` but only testing `number2`. In addition, the **`if`**-body statement is executed because in C++ any non-zero (positive or negative) number is logical-true. So, be careful of this common error!

Also, why don't we need parentheses around the two separate expressions to distinguish between the logical (`&&`) and relational (`>`) operators? For example:

```
if ((number1 > 0) && (number2 > 0))
```

You've probably guessed that the answer lies with the precedence of the operators. A quick check with Appendix C shows that the precedence of the relational operator (`>`) is greater than the precedence of the logical operator (`&&`). Thus the relational comparisons are performed before the logical comparison. Although in the present example the use of the parentheses is redundant, do use parentheses if you feel that they add to the clarity of a program.

Logical OR Operator (| |)

The logical OR operator (`| |`) is used to test whether *either* of two expressions is logically true; if so, the result is logical-true. Logical OR is a binary operator and is left-associative. The truth table for the logical OR operator is shown in Table 5.4.

Note that there are three permutations of logical-true for the logical OR operator. If either or both of the two expressions are true then the result is true.

Consider the `numbers` program again:

```
// log_or.cpp
// illustrates the logical OR operator (||)
#include <iostream>  // C++ I/O

void main ()
  {
  float number1, number2 ;

  // get user-entered numbers
  cout << "enter two positive or negative numbers: " ;
```

Table 5.4 Logical OR truth table.

expression1	*expression2*	*Logical result*
false	false	false
false	*true*	*true*
true	*false*	*true*
true	*true*	*true*

```
cin  >> number1 >> number2 ;

// compare both numbers to zero using logical OR (||)
if (number1 > 0 || number2 > 0)
  {
  cout << number1 << " or " << number2
       <<  " are greater than zero" << endl ;
  }
else  // note: else-executed only if both numbers are
      // negative!
  {
  cout << number1 << " and " << number2
       <<  " are not greater than zero" << endl ;
  }
}
```

Since the **if**-body statement will be executed if either or both of the two numbers entered is positive, we know that if the **else**-body statement is executed then both numbers are negative – that's logic for you.

Logical NOT Operator (!)

The logical NOT operator (!) reverses the logical value of its operand. Thus a true expression is reversed to false and a false expression is reversed to true. The logical NOT operator is a unary operator and is right-associative. Consider the program:

```
// log_not.cpp
// illustrates the logical NOT operator (!)
#include <iostream>  // C++ I/O

void main ()
  {
  float number ;

  // get user-entered number
  cout << "enter zero or a positive/negative number: " ;
  cin  >> number ;

  if (!number)  // if number=0 (false) then !number !=0 (true)
    {
    cout << number << " is zero" << endl ;
    }
  else
    {
    cout << number << " is nonzero" << endl ;
    }
  }
```

Admittedly, we could simply use **if** (number) rather than **if** (!number):

```
//...
if (number)
  {
  cout << number << " is nonzero" << endl ;
  //...
  }
```

5.1.6 The switch Statement

The **switch** statement is a convenient way of dealing with a large, complex series of similar decisions in a program.

The general syntax of the **switch** statement is:

```
switch (test expression)
  {
  case constant_expression1:
    // statements ;

  //...

  case constant_expressionn:
    // statements ;
  }
```

The expression is typically of type **int**, **char**, **long** or **unsigned**, and cannot be of a floating-point type such as **float** or **double**. The body of the **switch** statement is delimited by a pair of braces. Within the **switch** statement flow is controlled by the keyword **case**. The keyword **case** is directly followed by a constant expression, which is in turn followed by a colon (:). Let us consider an example which prompts a user to enter an integer number between 0 and 2, inclusive, and displays a message according to the number entered:

```
// switch.cpp
// illustrates the switch statement
#include <iostream>  // C++ I/O

void main ()
  {
  int number ;

  cout << "enter an integer number in the range (0:2): " ;
  cin  >> number ;

  switch (number)
    {
    case 0:
      cout << "number entered: zero"  << endl ;
    case 1:
      cout << "number entered: one"   << endl ;
    case 2:
      cout << "number entered: two"   << endl ;
```

```
    }
}
```

Some typical user interaction, entering the number 2, is:

```
enter an integer number in the range (0:2): 2
number entered: two
```

That works fine. Now try entering the number 0:

```
enter an integer number in the range (0:2): 0
number entered: zero
number entered: one
number entered: two
```

When the number 0 is entered we see that the program *falls through* the remaining **case** statements. Thus, we need a way out of the **switch** statement when we have finished executing the statement(s) associated with a particular **case**. Fortunately, C++ provides the keyword **break** to solve this problem. The **break** keyword terminates a **case** statement(s) block after the last statement. Upon exiting the **switch** statement control is directed to the first statement after the closing brace of the **switch** statement. Consider the same program as above, but now modified by the keyword **break**:

```
// b_switch.cpp
// illustrates the switch statement and the keyword break
#include <iostream>  // C++ I/O

void main ()
  {
  int number ;

  cout << "enter an integer number in the range (0:2): " ;
  cin  >> number ;

  switch (number)
    {
    case 0:
      cout << "number entered: zero"  << endl ;
      break ;
    case 1:
      cout << "number entered: one"   << endl ;
      break ;
    case 2:
      cout << "number entered: two"   << endl ;
      // note: no break required here!
    }
}
```

Note that no **break** for **case** 2 is necessary since **case** 2 is the last **case** statement of the **switch** statement.

There are occasions when you may require your program to fall through the remaining **case** statements, but generally programs **break** out after the **case** statement.

What happens in the above program if a user enters a number other than 0, 1 or 2? Nothing. The **switch** statement at present does not cater for entered numbers that are outside the range [0:2]. C++ provides another keyword, **default**, that will catch all of the unexpected user input. The following program uses **default** to catch exceptions:

```
// default.cpp
// illustrates the switch statement, break and default
#include <iostream>  // C++ I/O

void main ()
  {
  int number ;

  cout << "enter an integer number in the range (0:2): " ;
  cin  >> number ;

  switch (number)
    {
    case 0:
      cout << "number entered: zero"  << endl ;
      break ;
    case 1:
      cout << "number entered: one"   << endl ;
      break ;
    case 2:
      cout << "number entered: two"   << endl ;
      break ;
    default:
      cout << "enter either 0, 1 or 2: " << endl ;
    }
  }
```

There are a few points to note about **default**. Notice that no expression or value is attributed to **default**. The **default** keyword can appear anywhere within the **switch** statement but can appear only once. Naturally, **default** is usually placed at the bottom of the **switch** statement. A good reason for placing **default** at the bottom of the **switch** statement is that no **break** statement is required. If you place **default** elsewhere, then a **break** will be required to prevent fall-through.

If it is required that a variable is defined local to a **case** statement block then { and } braces are required:

```
switch (number)
  {
  //...
  case 2:
    {
    int i = 0 ;  // variable local to case
    //...
    }
```

```
//...
}
```

Finally, if you have programmed for Windows in C, using a non-object-oriented approach, then you will probably have noticed that the majority of Windows programs use a **switch-case** statement construction to process the window procedure messages:

```
long FAR PASCAL _export WndProc (HWND hwnd, UINT message,
                                 /*...*/)
  {
  //...
  switch (message)
    {
    case WM_CREATE:
      // process WM_CREATE message
      return 0 ;
    case WM_PAINT:
      // process WM_PAINT message
      return 0 ;
    case WM_KEYDOWN:
      switch (wParam)
        {
        case VK_HOME:
          // process VK_HOME key
          break ;
        //...
        }
    //...
    case WM_DESTROY:
      // process WM_DESTROY message
      return 0 ;
    }
  return DefWindowProc (hwnd, message, //...
  }
```

Notice the **switch** within a **switch** or *nested* **switch** statements. Windows provides over 200 messages, and the **switch** statement for a large Windows application can be unmanageable.

5.1.7 The Conditional Operator (? :)

The following typical **if-else** statement:

```
//...
if (a > b)
  // statements ;
else
  // other statements ;
```

is so popular in C++ that a *conditional* operator is available which is more concise. The conditional operator is a ternary operator and is left-associative. The conditional operator is of the general syntax:

```
expression1 ? expression2 : expression3
```

The second expression (`expression2`) is evaluated if `expression1` is logical-true. Otherwise, if `expression1` is logical-false then the third expression (`expression3`) is evaluated.

The following program demonstrates the conditional operator:

```
// cond.cpp
// illustrates the conditional operator (?:)
#include <iostream>  // C++ I/O

void main ()
  {
  float n1, n2 ;

  cout << "enter two positive or negative numbers: " ;
  cin  >> n1 >> n2 ;

  // if-else
  cout << endl << "if-else: " ;
  if (n1 > n2)
    cout << n1 << ">" << n2 ;
  else
    cout << n1 << "<" << n2 ;

  // conditional operator
  cout << endl << "conditional: " ;
  n1 > n2 ? cout << n1 << ">" << n2
          : cout << n1 << "<" << n2 ;
  }
```

5.2 Repetition

The general technique for performing repetition in C++ is through the use of a loop. C++ offers us three alternatives for constructing a loop: **`for`**, **`while`** and **`do-while`**. We shall consider each of these separately in the following subsections.

5.2.1 The `for`-Loop

Of the three alternative forms of loop, the **`for`**-loop is by far the most frequently used. The general syntax of the **`for`**-loop is:

```
for (expression1; expression2; expression3)
  {
  // statement body
  }
```

There are three parts to a **`for`**-loop:

- initialisation — initialises a variable that controls the **for**-loop
- test — determines whether another iteration should be performed
- control — controls the **for**-loop variable

Each of the three separate parts is contained within the pair of parentheses (and) and separated by a semicolon. The **for**-loop body begins with the opening brace, {, and ends with the closing brace, }. Let's consider an example:

```
// for.cpp
// illustrates the for-loop
#include <iostream>  // C++ I/O

void main ()
  {
  double a, x_old, x_new ;

  cout << "enter const. (0<a<=4) and initial (0<x<1) values: " ;
  cin  >> a >> x_old ;

  cout << endl << x_old << endl ;

  for (int i=0; i<10; i++)
    {
    x_new = a * x_old * (1.0 - x_old) ;
    x_old = x_new ;
    cout << x_new << endl ;
    }
  }
```

This program prompts a user for a constant a and an initial value x_0 of a variable x. The initial x value is first displayed and then the **for**-loop of the program computes the following iteration 10 times:

$$x_{n+1} = ax_n(1 - x_n)$$

Each new iteration (x_new) uses the previous x value (x_old) determined by the formula. The following shows the output for $a = 2.65$ and $x_0 = 0.8$:

```
enter const. (0<a<=4) and initial (0<x<1) values: 2.65 0.8

0.8
0.424
0.647194
0.605085
0.633236
0.615457
0.627174
0.619641
0.624568
0.621379
0.623458
```

and if enough iterations were performed then x would converge to 0.622 642[3].

The above program illustrates that the **for**-loop performs a fixed number of iterations determined by the test expression:

```
for (int i=0; i<10; i++)
  { /* ... */ }
```

Note that I don't adopt the end of statement style of placing a space between the first two expressions and the semicolons and a space between the operators and operands:

```
for (int i = 0 ; i < 10 ; i++)
  { /* ... */ }
```

The former style is adopted purely for reasons of compactness. Choose a style that suits you.

Let us examine each of the three parts of the **for**-loop separately

for Loop Initialisation

The first expression statement in the **for**-loop initialises the **for**-loop variable `i`:

```
int i=0;
```

The variable is initialised to zero when the **for**-loop is entered. Note that in C++ we are able to define a variable when we require it, so we don't have to use the following inelegant C-style **for**-loop:

```
int i ;
for (i=0; i<10; i++)
  { /* ... */ }
```

The variable `i` can be initialised to any value that you require and is initialised only once, when the **for**-loop begins.

The scope of the loop variable is restricted to the scope of the **for**-loop statement block such that the following two **for**-loops each define their own local variable `i`:

```
for (int i=0; i<10; i++)
  { /* ... */ }
//...
for (int i=0; i<20; i++)
  { /* ... */ }
```

Generally, the **for**-loop has only one variable, but it is possible to have multiple initialisations:

```
for (int i=0, int j=9; i<10; i++)
  { /* ... */ }
//...
```

---

3 $x_{n+1} = ax_n(1 - x_n)$ is often referred to as the *quadratic* or *logistic* law and models a simple feedback system that can have very complicated behaviour. Try entering the values $a = 4.0$ and $x_0 = 0.8$. The behaviour of x becomes *chaotic* and convergence will not occur. For an excellent discussion of this logistic law and chaos in general refer to Peitgen *et al.* (1992).

Each variable (`i` and `j`) is separated by the comma operator (`,`).

`for`-Loop Test

The second expression statement in the **`for`**-loop is the test expression:

```
i<10;
```

This expression determines the number of times the **`for`**-loop statement body will be executed. From the previous section, `i` is initialised to 0, and as long as `i` remains less than 10 the test expression is true and the **`for`**-loop statement body will continue to be processed. In the present case the test expression involves the less than relational operator (<). Relational operators are generally used for the test expression. When `i` equals 10 the **`for`**-loop terminates because the test expression is now false. Since the first and last values of `i` are 0 and 9 respectively, the **`for`**-loop body is executed 10 times.

Always ensure that the test expression can be logical-false; otherwise the **`for`**-loop will iterate endlessly. For example, a common error is to use the assignment operator:

```
for (int i=0; i=10; i++)
  { /* ... */ }
```

When the value of `i` is incremented to 10, the test expression remains logical true and the **`for`**-loop will iterate forever. We shall take a look at *forever-loops* later.

`for` Loop Control

The third and last expression in the **`for`**-loop is:

```
i++
```

This expression is used to increment the variable `i`. Remember that the C++ increment operator (++) offers a more concise syntax of incrementing a variable `i` by the constant 1 than `i=i+1` or `i+=1`. It was noted above that multiple variable initialisations are allowed in a **`for`**-loop. Similarly, multiple control expressions are allowed:

```
for (int i=0, int j=0; i<10; i++, j++)
  { /* ... */ }
```

Again, each control expression is separated by the comma operator. While multiple expressions are syntactically correct and increase conciseness, they simultaneously decrease readability.

The test expression is not limited to increments of 1. C++ allows us to use any incrementing or decrementing step size. The following illustrates two other possible **`for`**-loops:

```
for (int i=0; i<10; i=i+2)   // incrementing step size of 2
  { /* ... */ }

for (int j=10; j>0; j--)     // decrementing step size of 1
  { /* ... */ }
```

Also, it is worth noting that function return values can be integrated with **for**-loop initialisation and test expressions. If two functions exist, called *StartValue()* and *NumberOfIterations()*, which return the initialisation value and the maximum number of iterations, respectively, then the following **for**-loop construction is possible:

```
int StartValue ()
  {
  //...
  return start_value ;
  }

int NumberOfIterations ()
  {
  //...
  return number_of_iterations ;
  }
//...
for (int i=StartValue (); i<NumberOfIterations (); i++)
  { /* ... */ }
```

5.2.2 The while-Loop

The general syntax of the **while**-loop is:

```
while (test expression)
  {
  //  statement body
  }
```

Provided the test expression is logical-true then the loop will continue to iterate. The difference between the **while**-loop and the **for**-loop is that the **while**-loop can be used when you do not know in advance how many iterations of the loop are required. The following program illustrates this point by prompting a user to type in some text:

```
// while.cpp
// illustrates the while-loop
#include <iostream>  // C++ I/O
#include <conio>     // getche()

void main ()
  {
  char ch = 0 ;    // ensure ch != '\r' (RETURN)

  cout << "type a line of text, press RETURN to
           stop:" << endl ;

  int count = 0 ;
  while (ch != '\r')
    {
    ch = getche () ;  // get and echo character
```

```
    if (ch == 'a' || ch == 'e' || ch == 'i'
                  || ch == 'o' || ch == 'u')
      {
      count++ ;
      }
    }
  cout << endl << endl << "no. of vowels entered: " << count ;
  }
```

At the time of writing the program we do not know how much or how little text a user will type in. The **while**-loop is ideal for this situation if we terminate entering the text by a carriage return. Each character that is typed in is tested against the carriage return escape sequence character '\r'. If the user has pressed carriage return then the program terminates; otherwise, the program will continue indefinitely. Note that although there is no initialisation within the loop statement, as with the **for**-loop, initialisation still has to be performed, and before the loop begins. Thus, the **while**-loop is not as concise as the **for**-loop, and this is its main disadvantage.

The **char** variable ch is initialised to the zero character, 0, to ensure that ch is not equal to carriage return at the start of program execution. Within the **while**-loop a character is obtained from the user and echoed to the screen by the function *getche()*. Each character is tested against the five vowels, and if a vowel is entered by the user a vowel count is incremented. Finally, the total number of vowels entered is displayed. For example:

```
type in a line of text, press RETURN to stop:
triangles and tetrahedra are tricky to teach

no. of vowels entered: 14
```

The *getche()* function *gets* a *ch*aracter read from the keyboard and echoes it to the text screen using the BIOS or direct video. The header file CONIO.H must be included to use *getche()*. The signature of *getche()* is:

```
int getche () ;
```

5.2.3 The do-while Loop

The general syntax of the **do-while** loop is:

```
do
  {
  // statement body
  } while (test expression) ;
```

Note the semicolon following the **do-while** loop test expression – this is easily forgotten. Essentially, the difference between the **while** and **do-while** loops is that the **do-while** loop executes the loop body statements before the test expression is evaluated. If the test expression is logical-true, then the statement body is executed again, and so on until a logical-false value is evaluated by the test expression. This is contrary to the **while**-loop, which evaluates the test expression before the statement body. The **do-while** loop is particularly useful when you need to execute a set of statements at least once.

The following program modifies the FOR.CPP program presented above. The **do-while** loop enables the **for**-loop iteration to be performed at least once even if a user enters values for *a* and *x* that are outside the specified range:

```
// do_while.cpp
// illustrates the do-while loop
#include <iostream>  // C++ I/O

void main ()
  {
  double a, x_old, x_new, x0 ;

  do
    {
    cout << "enter const. (0<a<=4) and initial
              (0<x<1) values " << endl ;
    cout << "program terminates if (a) or (x)
              outside range: " ;
    cin >> a >> x0 ;
    x_old = x0 ;

    cout << endl << x_old << endl ;
    // iterate for 10 iterations
    for (int i=0; i<10; i++)
      {
      x_new = a * x_old * (1.0 - x_old) ;
      x_old = x_new ;
      cout << x_new << endl ;
      }
    } while (a>0 && a<=4 && x0>0 && x0<1) ;
  }
```

This program is far from ideal, since it continues to perform the **for**-loop iteration even if a user enters values for *a* and *x* that are outside the specified range. To modify the above program to prevent this would require the use of an **if** statement and the *exit()* or *abort()* C++ library functions, which are discussed later.

5.3 Forever Loops

An example was presented above where a **for**-loop could iterate indefinitely. This was accidentally achieved by incorrectly using the assignment operator in the test expression instead of a relational operator. A better and more intentional method of achieving a *forever loop* is to omit the three expressions altogether:

```
for (; ;)    // forever for-loop
  {
  //...  // for-loop body statements executed indefinitely
  }
```

The null program statements are legal in C++.

To achieve similar infinite **while** and **do-while** loops we simply use a non-zero test condition:

```
//...
while (1)  // infinite while-loop
   {
   //...
   }
//...
do         // infinite do-while-loop
   {
   //...
   } while (1)
```

5.4 Nested Loops

A nested loop is a loop within a loop. The following is an example of a **for**-loop within a **for**-loop:

```
for (int i=0; i<MAX_I; i++)
   {
   //...
   for (int j=0; j<MAX_J; j++)  // nested for-loop
      {
      //...
      }
   }
```

Variable `i` controls the outer **for**-loop, while variable `j` controls the inner **for**-loop. Nested **for**-loops are mostly used with arrays, to be discussed in Chapter 7.

5.5 The `break` Statement

When we examined the **switch** statement we also saw the **break** statement, which allowed us to exit from a **case** statement (preventing fall-through) and ultimately to exit from the **switch** statement. The **break** statement can also be used to exit from a loop to give premature termination; that is before the loop test condition is logical-false. Consider the program:

```
// b_for.cpp
// illustrates the for-loop and break statement
#include <iostream>  // C++ I/O
#include <cstdlib>   // rand(), srand(), exit()
#include <ctime>     // time(), time_t
```

```
void main ()
  {
  int n_iterations = 0 ;
  cout << "enter the number of iterations (1:100): " ;
  cin  >> n_iterations ;

  // test number of iterations entered
  if (n_iterations <= 0 || n_iterations > 100)
    {
    cout << "a non-valid number of iterations!" ;
    exit (EXIT_SUCCESS) ;  // quit execution
    }

  // initialise random number generator
  time_t t ;
  srand (unsigned (time(&t))) ;

  // for-loop to display random numbers
  for (int i=0; i<n_iterations; i++)
    {
    int random_number = rand () ;
    cout << random_number << endl ;

    if (random_number > 25000)
      {
      cout << "stop: last random number > 25,000" ;
      break ;
      }
    }
  }
```

Some user interaction is:

```
enter the number of iterations (1:100): 25
291
5446
22902
7252
19967
7115
31344
stop: last random number > 25,000
```

This program generates random numbers and displays them on the screen. If a random number is greater than 25 000 the program terminates as a result of the **break** statement. The **break** statement directs program control to the first statement immediately *outside the loop* that contains the **break** statement. Note that control is not directly sent outside the **if** statement, as you may have expected, but outside the loop. This is an important distinction and is in contradiction to using **break** and the **switch** statement that we saw previously. If you recall, a **break** within a **switch** statement sends control to the first statement directly after the **switch** statement's closing brace.

Thus, in the **for**-loop above the **break** statement sends control to the closing brace of the **for**-loop, upon which we have reached the end of the program.

The random numbers are generated using two functions:

```
int  rand () ;
void srand (unsigned seed) ;
```

which require the inclusion of the STDLIB.H header file.

rand() generates a pseudo-random number in the range 0 to RAND_MAX (defined in STDLIB.H) and returns the number as type **int**. The function *srand()* initialises the random number generator. The random number generator is reinitialised by passing as an argument to *srand()* the function *time()*:

```
time_t time (time_t* timer) ;
```

time() returns the time of day in seconds elapsed since 00:00:00 Greenwich Mean Time (GMT) on 1 January 1970. The TIME.H header file has to be #included in order to use *time()*.

If we don't reinitialise the random number generator each time the program is executed, the same series of random numbers will be generated.

Note also the test performed on n_iterations in the program to ensure that the number of iterations entered by the user is in the specified range 1 to 100. This is necessary to prevent a user from entering a negative number, zero or a large number that will take all day to iterate through:

```
//...
// test number of iterations entered
if (n_iterations <= 0 || n_iterations > 100)
  {
  cout << "a non-valid number of iterations!" ;
  exit (EXIT_SUCCESS) ;  // quit execution
  }
//...
```

Note the use of the relational operators (<= and >) and the logical OR operator. The function *exit()* terminates program execution immediately. For console-based programs *exit()* is the recommended way of initiating program shutdown. The header file STDLIB.H must be #included for *exit()*, whose signature is:

```
void exit (int status) ;
```

Before termination all open files are closed and any buffered output is written. The program above passes the EXIT_SUCCESS status identifier, which indicates normal program termination, to the operating system. Abnormal program termination is signalled by the status identifier EXIT_FAILURE. Refer to your compiler's documentation for a more detailed description of the *exit()* and *atexit()* functions.

There is another termination function, called *abort()*, which is similar to *exit()*:

```
void abort () ;
```

The signature indicates that *abort()* is not passed any arguments and does not return a status value to the operating system. The function *abort()* does not close open files; it simply terminates program execution. *abort()* requires the inclusion of the STDLIB.H header file.

The **break** statement can be used with **for**, **while** or **do-while** loops. For example:

```
//...
count = 0 ;
while (count < MAX_COUNT)
   {
   //...
   if (number > MAX_NUMBER)
      {
      //...
      break ;  // break while loop
      }
   //...
   count++ ;
   }
```

5.6 The continue Statement

We noted above that the **break** statement, when used in conjunction with a loop, directs control to the first statement directly after the closing brace of the loop. Sometimes this is not what we want. C++ gives us the keyword **continue**, which allows you to continue with the next iteration in the loop but to jump over any remaining code in the loop. Control is transferred to the control expression in a **for**-loop and to the test condition in **while** and **do-while** loops. An example program that illustrates the **continue** statement in a **for**-loop is:

```
// continue.cpp
// illustrates the continue statement
#include <iostream>   // C++ I/O
#include <cmath>      // sqrt()

void main ()
   {
   double number = 0.0 ;
   for (int i=0; i<3; i++)
      {
      cout << "enter a number: " ;
      cin  >> number ;

      if (number < 0)
         {
         cout << "number < 0" << endl << endl ;
         continue ;
         }
      cout << "sqrt(" << number << "): "
           << sqrt (number) << endl << endl ;
      }
   }
```

A typical output is:

```
enter a number: 2
sqrt(2): 1.41421

enter a number: -1
number < 0

enter a number: 10
sqrt(10): 3.16228
```

This program prompts the user to enter a number three times. If the user enters zero or a positive number the program simply displays the square root of the number entered. If the number entered is negative the **if** condition is true and a message 'number < 0' is displayed and the **continue** statement is executed. The **continue** statement directs control to the third expression (i++) in the **for**-loop rather than displaying the **if**-body statement and then continuing to evaluate the square root. Thus, the **continue** statement offers a way of jumping over the following loop body statements but continuing with the next loop iteration. Try commenting out the **continue** statement and entering a negative number. A run-time error of the form 'sqrt: domain error' or 'floating-point error' will be issued.

5.7 The goto Statement

There is another method of creating a loop that I didn't bother to mention above: the **goto** statement. This statement is so confusing (because of the potential it generates for writing code that jumps from one place to another indiscriminately) that if you can forget I ever mentioned it then please do. If not, then read on. The general syntax of the **goto** statement is:

```
goto label ;
//...
label:
```

The **goto** statement consists of the keyword **goto** followed by a label. The label can occur before or after the **goto** statement. The label name is followed by a colon. For example:

```
//...
start:
cout << "enter a positive number: " ;

float n ;
cin >> n ;
if (n < 0)
  {
  goto start ;
  }
```

This example is analogous to a **while**-loop, in that if a user persists in entering a negative number the **if**-condition will be logical-true and the **goto** statement will transfer control back to the **goto** label (start) until the **if**-condition is logical-false.

5.8 Loop Body Visibility

Before we bring this chapter to a close by discussing the bitwise and shift operators, let us discuss the visibility of objects within a loop body. Constants, variables or objects that are defined within the loop body (between the opening, {, and closing, }, braces) have *local scope*, which limits their availability to the loop body in which they are defined. In other words, such objects are not *visible* from outside the loop body. For example:

```
for (int i=0; i<10; i++)
   {
   double mean ;
   //...
   }
//...
cout << "mean of numbers: " << mean ;   // error: 'mean'
                                        // undefined
```

The notion of local scope is intuitive – a program would be chaos without it!

Scope and visibility will be discussed further when we introduce functions, structures and classes.

5.9 The Bitwise and Shift Operators

In addition to the operators discussed above, C++ also supports a set of operators called the *bitwise* and *shift* operators, which allow a programmer to manipulate the bits of a variable; see Table 5.5. These operators were not discussed above because the bitwise AND and OR operators are similar to the logical AND and OR operators, and if you are new to C++'s operators then the logical and bitwise operators are best discussed separately.

Note that the bitwise and shift operators can only be used on C++ integral data types.

Before we discuss the bitwise operators, let us first have a brief reminder of binary numbers. Decimal numbers are in base 10, meaning that the digits 0 to 9 are used to represent a decimal number. Binary numbers are in base 2, which means that only the digits 0 and 1 are used to represent a binary number.

Figure 5.4 illustrates the eight-bit binary representation of the decimal number 19. Shown also in Fig. 5.4 are the bit positions ranging from 0 to 7, with their corresponding powers of two and decimal values. The high bit is position 7 and the low bit is position 0. A value of 1 turns on a bit, whereas a value of 0 turns off a bit. The decimal equivalent of a bit string is the sum of the decimal values of all turned-on bits. As shown in Fig. 5.4, the decimal equivalent of 00010011 is $2^4+2^1+2^0$, or 19.

5.9.1 Bitwise AND Operator (&)

The bitwise AND operator (&) is a binary operator and has the following general syntax:

```
operand1 & operand2
```

If the same bit position for *both* operands is on, the resultant bit is on, else if either or both of the same bit positions for both operands is off the resultant bit is off. Therefore, the truth table for

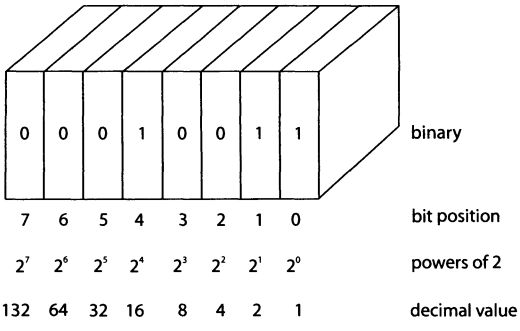

Fig. 5.4 Eight-bit representation of a decimal number. The decimal equivalent of 0010011 = $2^4 + 2^1 + 2^0 = 16 + 2 + 1 = 19$.

bitwise AND is identical to that of logical AND (Table 5.3), except that the bitwise AND operator operates on bits rather than expressions. The bitwise AND truth table is shown in Table 5.6.

As an example, consider the bitwise AND operator operating on the decimal numbers 19 and 37:

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
0	0	0	1	0	0	1	1	19
0	0	1	0	0	1	0	1	37
0	0	0	0	0	0	0	1	19&37

Only bit position 0 is turned on as a result of the bitwise AND operator, since that is the only bit position in which both operand bits are turned on. The result of 19&37 is therefore equal to the decimal value of 1. This example is illustrated in the following program:

```
// bit_and.cpp
// illustrates the bitwise AND operator (&)
#include <iostream>  // C++ I/O

void main ()
  {
  int i (19), j (37) ;

  cout << "i&j: " << (i & j) << endl ;
  }
```

with output:

```
i&j: 1
```

Table 5.6 Bitwise AND truth table.

operand1	*operand2*	*Bitwise result*
0	0	0
0	1	0
1	0	0
1	*1*	*1*

A popular illustration of the bitwise AND operator is determining whether an integer number is odd or even. If a number is odd the zero position bit is turned on, else if a number is even the zero position bit is turned off. By combining this property of odd and even numbers with a number which is known to be odd, say 1, we have a method of determining whether a number is odd or even:

```
// even_odd.cpp
// illustrates the bitwise AND operator (&)
// by determining whether a number is odd or even
#include <iostream>  // C++ I/O

void main ()
  {
  int number (0) ;

  cout << "enter an integer number: " ;
  cin  >> number ;
  cout << number << " is " ;

  number & 1 ? cout << "odd" : cout << "even" ;

  cout << endl ;
  }
```

With some user interaction:

```
enter an integer number: 2
2 is even
```

Because the logical AND (&&) and the bitwise AND (&) differ only by one &, always ensure that you are using the correct operator.

5.9.2 Bitwise OR Operator (|)

The bitwise OR operator (|) is a binary operator and has the following general syntax:

```
operand1 | operand2
```

If the same bit position for *either* operands is on the resultant bit is on, else if both of the same bit positions for both operands are off the resultant bit is off. Therefore, the truth table for bitwise OR is identical to that of logical OR (Table 5.4), except that the bitwise OR operator operates on bits rather than expressions. The bitwise OR truth table is shown in Table 5.7.

Table 5.7 Bitwise OR truth table.

operand1	*operand2*	*Bitwise result*
0	0	0
0	1	1
1	0	1
1	1	1

As before, consider the bitwise OR operator operating on the decimal numbers 19 and 37:

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
0	0	0	1	0	0	1	1	19
0	0	1	0	0	1	0	1	37
0	0	1	1	0	1	1	1	19|37

The result of 19|37 is therefore equal to the decimal value of 55. This example is illustrated in the following program:

```
// bit_or.cpp
// illustrates the bitwise OR operator (|)
#include <iostream>  // C++ I/O

void main ()
   {
   int i (19), j (37) ;

   cout << "i|j: " << (i | j) << endl ;
   }
```

with output:

```
i|j: 55
```

5.9.3 Bitwise Exclusive OR Operator (^)

The bitwise exclusive OR (XOR) operator (^) is a binary operator and has the following general syntax:

```
operand1 ^ operand2
```

The operation of XOR is similar to that of OR, in that if the same bit position for either operands is on the resultant bit is on, except that if both bit positions are turned on the resultant bit is off. The bitwise XOR truth table is shown in Table 5.8.

Sticking with the numbers 19 and 37, we have for XOR:

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
0	0	0	1	0	0	1	1	19
0	0	1	0	0	1	0	1	37
0	0	1	1	0	1	1	0	19^37

The result of 19^37 is therefore equal to the decimal value of 54. This example is illustrated in the following program:

```
// bit_xor.cpp
// illustrates the bitwise XOR operator (^)
#include <iostream>  // C++ I/O
```

Table 5.8 Bitwise XOR truth table.

operand1	*operand2*	*Bitwise result*
0	0	0
0	*1*	*1*
1	*0*	*1*
1	1	0

```
void main ()
  {
  int i (19), j (37) ;

  cout << "i^j: " << (i ^ j) << endl ;
  }
```

with output:

```
i^j: 54
```

5.9.4 Bitwise One's Complement Operator (~)

The bitwise one's complement operator (~) is a unary operator and has the following general syntax:

```
~ operand
```

The one's complement operator reverses the bits of its operand. Thus, turned-on bits are turned off and vice versa. Consider the one's complement operator operating on the 8-bit number 19:

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
0	0	0	1	0	0	1	1	19
1	1	1	0	1	1	0	0	~19

The result of ~19 is therefore equal to the decimal value of 236.

To illustrate the one's complement operator let's take a look at the two's complement of a number, which simply adds 1 to the one's complement of a number. The two's complement is frequently used to obtain the negative of a number:

```
// bit_one.cpp
// illustrates the bitwise one's complement operator (~)
// by finding the two's complement of a number
#include <iostream>  // C++ I/O

void main ()
  {
  int number (0) ;
```

```
    cout << "enter an integer number: " ;
    cin  >> number ;

    cout << "two's complement of number: "
             << (~number + 1) << endl ;
    }
```

With some user interaction:

```
enter an integer number: 19
two's complement of number: -19
```

5.9.5 The Shift Operators

The left (<<) and right (>>) shift operators are binary operators and have the following general syntax:

```
operand << number_of_bits_to_shift
operand >> number_of_bits_to_shift
```

`operand` is the variable to be shifted `number_of_bits_to_shift` bit positions to the left or right, respectively.

To illustrate the left and right shift operators, let them operate on the number 19 with one bit shift position:

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
0	0	0	1	0	0	1	1	19
0	0	1	0	0	1	1	0	19<<1
0	0	0	0	1	0	0	1	19>>1

As a bit is shifted off either the high or low bit positions, a 0 is automatically placed on the opposite end of the bit string. This example is illustrated in the following program:

```
// shift.cpp
// illustrates the left and right shift operators << and >>
#include <iostream>  // C++ I/O

void main ()
  {
  int i (19) ;

  cout << "i<<1: " << (i << 1) << endl
       << "i>>1: " << (i >> 1) << endl ;
  }
```

with output:

```
i<<1: 38
i>>1: 9
```

Try not to confuse the left and right shift operators with the insertion and extraction stream operators.

5.9.6 The Bitwise and Shift Assignment Operators

Chapter 4 illustrated the arithmetic assignment operators for writing concise expressions. Similarly, both the bitwise and shift operators have assignment versions, i.e. &=, |=, ^=, <<= and >>=. These are demonstrated in the following program:

```
// bs_assgn.cpp
// illustrates the bitwise and shift assignment operators
#include <iostream>  // C++ I/O

void main ()
  {
  int i (19), j (37), k(73), l (145), m (289) ;

  // bitwise assignment
  k &= j ;
  cout << "k&=j: " << k << endl ;
  l |= j ;
  cout << "l|=j: " << l << endl ;
  m ^= j ;
  cout << "m^=j: " << m << endl ;

  // shift assignment
  i <<= 1 ;
  cout << "i<<=1: " << i << endl ;
  i >>= 1 ;
  cout << "i>>=1: " << i << endl ;
  }
```

with output:

```
 k&=j: 1
 l|=j: 181
 m^=j: 260
i<<=1: 38
i>>=1: 19
```

5.10 Summary

The relational operators (<, >, <=, >=, == and !=) allow us to compare two expressions to see if they are equal, not equal, one is greater than the other and so on. The logical result of the relational comparison is either true or false. The logical operators AND (&&) and OR (||) operate on two expressions to test whether they are logically true or false. The logical NOT operator (!) reverses an expression's logical value from true to false or from false to true.

The **if** and **if-else** statements form the basis of decision-making in C++. The relational and logical operators give the **if-else** statement great flexibility. The **switch** statement provides a simple tree structure for dealing with a large number of decisions. With the help of the additional keywords **break** and **default** the **switch** statement is very powerful. The keyword **break** prevents fall-through and sends control to the statement directly after the **switch** statement's closing parenthesis, while **default** enables a **switch** statement to deal with unexpected exceptions. The conditional operator (? :) provides us with a concise form of the **if-else** statement.

This chapter has also introduced three loops for performing repetition: **for**-loop, **while**-loop and **do-while**-loop. The **for**-loop is undoubtedly the most frequently used of the three loops. The **for**-loop's popularity lies in the fact that the initialisation, test and control expressions are all in one place. The **while**-loop must initialise its control variable before the loop commences and the **do-while**-loop similarly initialises its control variable before the loop starts but evaluates the test expression at the end of the loop. Thus, the **while** and **do-while**-loops are used when the test condition arises within the loop. The **for**-loop executes a predetermined number of iterations, whereas the **while** and **do-while**-loops are generally used to execute an unknown number of iterations, with the **while**-loop not necessarily executing at all and the **do-while** loop always executing at least once. The use of the **break** statement within a loop directs control to the closing brace of the innermost loop, whereas the **continue** statement sends control to the start of the loop for the next iteration. The **goto** statement redirects program control to a specified label.

The chapter concluded by discussing the bitwise and shift operators. Although not as frequently used as the relational and logical operators, the bitwise and shift operators are an essential part of the C++ language because they allow a programmer to manipulate the individual bits of a variable of an integral type. The bitwise operators are AND (&), OR (|), XOR (^) and one's complement (~). The bitwise AND (&) and OR (|) are not to be confused with the logical AND (&&) and OR (| |). The left and right shift operators, << and >>, shift bits by a specified number of bit positions left and right respectively.

Exercises

5.1 Debug the following program:

```
#include <iostream>  // C++ I/O

void main ()
  {
  for (int i=0; i<10; i++)
    {
    if (i = 3)
      cout << "i=3" << endl ;
    }
  }
```

5.2 Write a program that prompts the user to enter a number which can be positive or negative and displays on the screen the absolute value of the number entered.

5.3 Using a loop of your choice, write a program that displays the first 10 terms of the Fibonacci series:

$$f(x_n, x_{n+1}) = x_n + x_{n+1}, \qquad n = 1, 2, 3$$

5.4 Write a program that writes the ASCII character set to a file with the equivalent decimal, hexadecimal and octal values for each character. Refer to Chapter 4 for basic file input and output.

5.5 Using a loop of your choice, write a program that calculates π using Gregory's series:

$$\tan^{-1} x = \sum_{m=0}^{\infty} \frac{(+1)^m x^{2m+1}}{2m+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots$$

If we set x=1 in the series then:

$$\tan^{-1}(1) = \frac{\pi}{4}$$

Note the slow convergence of this series.

Hint: Use the *pow()* function, which calculates x to the power of y:

```
double pow (double x, double y) ;  // #include <cmath>
```

5.6 Write a program (using a **switch** statement) that converts three numbers specifying an HSV (Hue–Saturation–Value) colour to three numbers specifying an RGB (Red–Green–Blue) colour.

Hint: The logic of the program is shown in the following pseudocode:

```
double h, s, v, r, g, b ;
int    i ;
double f, p, q, t ;

if s=0
  r = v ; g = v ; b = v ;
else
  if h=1
    h = 0 ;

  h = h / 6 ;
  i = h ;
  f = h - i ;
  p = v(1-s) ;
  q = v(1-sf) ;
  t = v(1-s(1-f))) ;

  case i
    case 0:
      r = v ; g = t ; b = p ;
    case 1:
      r = q ; g = v ; b = p ;
    case 2:
      r = p ; g = v ; b = t ;
    case 3:
      r = p ; g = q ; b = v ;
    case 4:
      r = t ; g = p ; b = v ;
```

```
        case 5:
          r = v ; g = p ; b = q ;
```

which assumes that both HSV and RGB are in the range [0,1].

5.7 Write a program that prompts the user for an integer number and then proceeds to divide and multiply the number by 2 using the right and left shift operators.

Chapter 6

Functions

If you think writing software is difficult, try rewriting software.
B. Meyer (1995)

A function is a collection of program statements. Functions are an essential tool for creating modularity within a program by discretising a large complex program into a series of separate units. The function helps a programmer to organise a program and at the same time to reduce program size and develop reusable code. A programmer's life without the function would be unbearable.

We begin by taking a look at the characteristics of all C++ functions, such as the function name, return type, arguments and function body. Functions are called by name and can return a value or object following the execution of the function's program statements. Information is passed to a function via the function arguments and can be passed either by value or by reference (memory location). Functions define their own scope. In other words, an object defined within a function is local to that function.

The chapter spends some time considering why functions are so important to programming and how similar functions can be arranged into declaration and implementation files so that function libraries can be built and easily reused. Programming without the extensive ANSI C and C++ library functions would be very difficult indeed. An overview of the functions supplied with all ANSI-compatible compilers is presented.

C++ allows functions to be overloaded so that the same name can be given to functions which differ according to the number of objects and/or type of arguments. Function overloading eliminates the need for multiple different function names when one will suffice. Functions can be assigned default arguments so that a function can be called without explicitly passing all of the function arguments, and this is a useful feature when it is desired that a function has a default behaviour. For small functions that are frequently called, the inline feature allows a function to be expanded inline and eliminates the need to define a macro. C++ supports external linkage with non-C++ functions so that C++ and non-C++ functions can communicate.

The chapter concludes by examining the popular geometric problems of the orientation of three points and the intersection of two lines.

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
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6.1 A Look at Functions

Let's begin our discussion of functions by examining the following program:

```
// func.cpp
// illustrates a C++ function
#include <iostream>  // C++ I/O

// returns the maximum of two numbers
double MyMax (double a, double b)
  {
  return a > b ? a : b ;
  }

void main ()
  {
  double number1, number2 ;

  cout << "enter two positive or negative numbers: " ;
  cin  >> number1 >> number2 ;

  double largest = MyMax (number1, number2) ;

  cout << "largest of the two numbers: " << largest ;
  }
```

This program contains two functions: *MyMax()* and *main()*. You should be familiar with the form of the *main()* function by now; we'll examine *main()* in more detail later. The other function, *MyMax()*, returns the maximum value of two numbers that are passed to the function. Note the combined use of the relational (>) and conditional (? :) operators to test for the largest of two numbers.

Some user interaction with the above program is:

```
enter two positive or negative numbers: 1 25
largest of the two numbers: 25
```

6.2 The Body of a Function

A function in C++ has the following general syntax:

```
return_type FunctionName (function_argument_list)
  {
  // function body
  //...
  return return_type ;
  }
```

We shall examine each part of a C++ function in more detail in the following sub-sections.

6.2.1 Function Name

Any name can be given to a function. The name can contain the letters a to z and A to Z, an underscore '_' and the digits 0 to 9. However, there is the restriction that the first character must be either a letter or an underscore. The number of characters used to define a function name is unlimited although some older compilers may only recognise the first 32 characters. These rules are the same as for naming identifiers, constants, variables, objects, classes etc.

The style that I adopt for function naming was highlighted in Chapter 3 and combines lower and upper case letters rather than being entirely in lowercase and using the underscore to separate words (which is the reserved style for identifiers):

```
NumberOfWindows () ;     // yes
number_of_windows () ;   // no
NumWin () ;              // ok
NoW () ;                 // too cryptic!
```

Try to choose function names that are as descriptive as possible rather than as cryptic as possible and known only to you.

Also, try to avoid choosing function names that are standard C++ library functions. The following example defines a function called *sqrt()* which has an identical signature to the C++ library function declared in the header file MATH.H:

```
#include <cmath>  // C++ sqrt()
//...
// user-defined square root function
double sqrt (double a)
   {
   double value ;
   //...
   return value ;
   }

void main ()
   {
   double sq_root = sqrt (2) ;  // calls user-defined sqrt()
   //..
   }
```

The user-defined *sqrt()* function takes precedence over the C++ library *sqrt()* function.

In fact, if you adopt the style of naming user-defined functions using upper- and lower-case letters (e.g. *Sqrt()*) you will never accidentally override a C++ library function, because they are all named in lowercase.

6.2.2 Function Body

The function body is the program code that actually performs a function's task. In the above program, the function body of *MyMax()* is a single statement:

```
return a > b ? a : b ;
```

The function body can be as small or as large as is necessary. Within the body of a function it is perfectly legal to define new constants, variables, objects etc., and to call other C++ or user-defined functions:

```
double AnyOldFunction ()
  {
  //...
  const double SIZE  = 100 ;
  int          count = 0 ;
  //...
  double any_old_value = AnyOtherOldFunction () ;
  //...
  return any_old_value ;
  }
```

The start of the function body begins with the opening brace: `{`. The end of the function definition is marked by the closing brace: `}`. There is no difference between the body of a user-defined function and the *`main()`* function.

The Fortran programming language uses a `FUNCTION` header statement and an `END` statement to indicate the start and finish, respectively, of a function body. Similarly, in Basic, Pascal and a Windows resource script file the keywords `BEGIN` and `END` are used to delimit a function body or a block of statements. C++ adopts a more minimal approach to function definition by simply requiring the delimiting braces `{` and `}`.

6.2.3 Function Return Type

The last part of a function definition is generally the function return type. The *`MyMax()`* function above returns type **`double`**. A function can return any C++ type or user-defined type, but ensure that the function returns the correct type. For example, change the return type of *`MyMax()`* from **`double`** to **`int`**:

```
int MyMax (double a, double b)
  { /*...*/ }
//...
```

A user interaction with the program will now produce the output:

```
enter two positive or negative numbers: 23.45 67.89
largest of two numbers: 67
```

`MyMax()` has automatically cast the largest of two **`double`** numbers in to an **`int`**.

6.2.3.1 `void` Return Type

You can define functions that do not return a value. An example is a function that outputs certain information:

```
void PrintCircleInfo ()
  {
  cout << "circle radius: " << radius << endl ;
```

```
    cout << "circle centre: " << centre << endl ;
    }
```

The *PrintCircleInfo()* function takes no arguments and returns no type. The keyword **void** explicitly specifies to the compiler that the function does not return a value. This function would be called purely by name and empty parentheses:

```
PrintCircleInfo () ;
//...
```

Since the return value of *PrintCircleInfo()* is not used, why be so specific and use the keyword **void**? Using the keyword **void** prevents any incorrect use of the function, and incorrect usage will be detected by static type checking:

```
int i = PrintCircleInfo (); // error:incorrect assignment of
                            // type void
```

6.2.3.2 Default Return Type

What if you forget or want to use the default return type of the function definition? The default function return type is **int**. The following function definition will return type **int**:

```
MyMax (int a, int b)
   {
   return a > b ? a : b ;
   }
```

It works, but it's not very informative to a user. If a function returns type **int**, then explicitly indicate so.

6.3 The `return` Statement

We saw above that a function that does not return a value has return type **void**. If a function returns a value of a given type, the function has to instruct the compiler the type of the value that it wants to return. This is performed via the **return** statement, which returns a *single* value whose type must be compatible with the specified return type in the function declarator. For instance:

```
// mixes type of return value (int)
// with return type specifier (double)
double Number ()
   {
   int i = 10 ;
   return i ;
   }
```

Most commercial compilers will not issue a warning or error compilation message for the above function definition, so be careful of making such errors.

After a **return** statement has been executed, control returns to the function caller. In the case of the function *MyMax()* in the program FUNC.CPP, the return value is copied to the variable `largest`, defined in *main()*. The variable `largest` is then used to output the largest of two numbers through the use of the `cout` stream object.

The **return** statement in the *MyMax()* function combines the relational and conditional operators into a single statement:

```
double MyMax (double a, double b)
   {
   return a > b ? a : b ;
   }
```

Alternatively, for increased clarity, we could have written the function as:

```
double MyMax (double a, double b)
   {
   double largest = a > b ? a : b ;
   return largest ;
   }
```

The above two functions are equivalent. In the first version of *MyMax()*, a temporary variable is created by the compiler to which the value of either `a` or `b` is assigned before returning the value. In the second version of *MyMax()* we have explicitly indicated this procedure by defining the additional variable `largest`. If you find the intention of the second version of *MyMax()* clearer then use this approach. More experienced programmers tend to try to write a program as concisely as possible without a significant loss of clarity, and as a result the technique of eliminating unnecessary variables is a common practice.

It is also worth mentioning that when an identifier or object is returned by a function the compiler generates a temporary object to hold a copy of the object returned by the function. The temporary object simply goes out of scope when the object has been returned to the calling function. This creation of a temporary object when a function returns will be discussed in more detail when we discuss *copy constructors* and *pointers* in Chapters 9 and 12.

6.4 Function Calling

The function *MyMax()* is called in the *main()* function by the statement:

```
double largest = MyMax (number1, number2) ;
```

In the present case the function call involves assignment of the value returned by *MyMax()* to the **double** variable `largest`. The function call passes the two user-entered numbers `number1` and `number2` as arguments to the function and ends with a semicolon to indicate the end of statement.

Since *MyMax()* was designed and defined to return a value of type **double**, its return value is assigned directly to a variable. Alternatively, *MyMax()* can simply be called with no assignment:

```
MyMax (number1, number2) ;
```

but such a call is meaningless since we are still left wondering which of the two numbers is the largest! The compiler will not issue a warning or error message for such a call. If such a function call is what you require, then a redesign of *MyMax()* would have to be made:

```
void MyMaxPrint (double number1, double number2)
  {
  double largest = number1 > number2 ? number1 : number2 ;
  cout << endl << "largest: " << largest ;
  }
```

Such a function definition greatly restricts the possible applications of *MyMaxPrint()*.

Notice that the function call allows a program to do things without worrying about the details of the function body. All the function caller needs to know is the *signature* of the function. The function signature provides the necessary information to call the function correctly. In the program above the function signature of *MyMax()* is:

```
double MyMax (double a, double b) ;
```

which informs a programmer of the function name, number and type of arguments, and return type. Function libraries should be accompanied by sufficiently informative documentation to enable a user to call and apply the function in the manner that the function designer intended.

6.5 Function Definition

The function definition of *MyMax()* is:

```
double MyMax (double a, double b)
  {
  return a > b ? a : b ;
  }
```

The first line is referred to as the *declarator*; we shall see why in the next section. After the declarator we have the function body, which comprises the function program statements. That's it – no big deal.

One point to note is that function definitions cannot be nested:

```
void Function ()
  {
  //...
  void Function ()
    {
    //...
    }
  }
```

6.6 Function Declaration

In the above program, FUNC.CPP, the definition of *MyMax()* preceded the *main()* function. Try flipping these two function definitions round:

```
// func_dec.cpp
// illustrates a C++ function declaration
#include <iostream>  // C++ I/O

// prototype (function declaration)
double MyMax (double a, double b) ;

void main ()
  {
  double number1, number2 ;

  cout << "enter two positive or negative numbers: " ;
  cin  >> number1 >> number2 ;

  double largest = MyMax (number1, number2) ;

  cout << "largest of the two numbers: " << largest ;
  }

// returns the maximum of two numbers
double MyMax (double a, double b)
  {
  return a > b ? a : b ;
  }
```

Note the inclusion of the *function declaration* or *prototype*:

```
double MyMax (double a, double b) ;
```

Without the function declaration the compiler will issue an error message of the form 'call to an undefined function'. This error message sums it up. In the program FUNC_DEC.CPP a call is made to *MyMax()*, but the function has not been defined at this point. The function declaration indicates to the compiler that the function has not been defined yet, but is defined else where – so go away and find it, Mr Compiler. Note that the function declaration ends with a semicolon since it is a statement.

The function declaration can be of the form above, which includes in the declaration the names of the function arguments (a and b), or alternatively, the function declaration can simply include the argument types:

```
double MyMax (double, double) ;
```

Choose the type of function declaration that suits you, but remember that it is good programming practice to include the argument identifiers in the function declaration, simply because the declaration is more informative with the argument identifiers. Also, it is a simple matter to

generate the function declaration by a *copy and paste* of the function declarator to the top of the program, provided you remember to add on a semicolon.

Whichever style you adopt, the most important thing is to ensure that the function declaration agrees with the declarator. Clearly, the return type, number of arguments, and type and argument order must be identical for definition and declarator.

6.7 Function Scope

We noticed above that it is possible to define identifiers within a function body. Are these identifiers accessible from outside the function body? The answer is no – they are local to the function in which they are defined. Accessibility is generally discussed in terms of *scope*. Scope simply refers to the visibility, availability and lifetime of an identifier within a program.

6.7.1 Local Scope

An identifier (constant, variable or object) has *local scope* if it is defined within a statement block or function body. For example, consider the program:

```
// l_scope.cpp
// illustrates local scope
#include <iostream>  // C++ I/O

void Function ()
  {
  int i = 20 ;
  cout << "Function (): i= " << i << endl ;
  }

void main ()
  {
  int i = 10 ;
  cout << "main (): i= " << i << endl ;

  Function () ;
  }
```

with output:

```
main (): i= 10
Function (): i= 20 ;
```

The output illustrates that the defined `i` variables in `main()` and `Function()` are not the same. The variable `i` defined in `main()` has local scope that is confined to the `main()` function body and is not visible outside `main()`. The variable `i` in `Function()` is local to `Function()`, is not visible from outside `Function()` and its lifetime exists between definition and the closing brace of `Function()`'s function body. Figure 6.1 shows schematically the scope of the variables `i`.

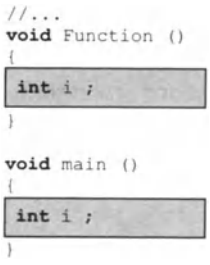


Fig. 6.1 Local scope of variables `i` defined in the functions *`main()`* and *`Function()`*.

6.7.2 Global Scope

Let's now take a look at global scope:

```
// g_scope.cpp
// illustrates global scope
#include <iostream>  // C++ I/O

int i = 10 ;

void Function ()
  {
  cout << "Function (): i= " << i << endl ;
  }

void main ()
  {
  cout << "main (): i= " << i << endl ;

  Function () ;
  }
```

In the above program, variables `i` have been removed from the function body of *`main()`* and *`Function()`* and placed outside of any statement block. Variable `i` now has global scope. Figure 6.2 shows the global scope of `i`, which encompasses both *`main()`* and *`Function()`*. Thus both *`main()`* and *`Function()`* now have access to `i`.

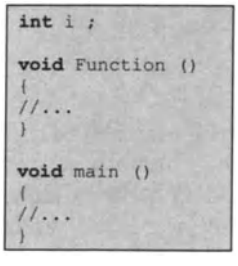


Fig. 6.2 Global scope of variable `i` defined outside the functions *`main()`* and *`Function()`*.

6.7.3 Precedence of Scope

Consider the program below, which defines a variable `i` within the function body of *`Function()`*. Although `i` is defined globally, the definition within *`Function()`* does not redefine `i`. This is because when two variables are defined with the same name within the same scope, the variable with the most current scope takes the highest precedence:

```
//...
int i = 10 ;

void Function ()
   {
   int i = 20 ;
   cout << "Function (): i= " << i << endl ;
   }
//...
```

and thus *`Function()`* outputs the value 20 for `i`. Precedence of levels of scope was also noted in Chapter 5 for nested **`if`** statements. The rules for precedence of scope are the same for **`if`** and **`if-else`** statements, loops, and functions.

6.7.4 Pros and Cons of Local and Global Scope

On comparing the above two programs (L_SCOPE.CPP and G_SCOPE.CPP), we observe that defining the variable `i` as global eliminates the necessity of passing `i` as an argument to *`Function()`*. However, the main disadvantage (and a serious one) of global identifiers is that they are not private. Any function has access to a global identifier. Consider the program:

```
//...
int i = 10 ;

void Function ()
   {
   i = 20 ;
   cout << "Function (): i= " << i << endl ;
   }

void main ()
   {
   cout << "main (): i= " << i << endl ;

   Function () ;
   cout << "main (): i= " << i << endl ;
   }
```

The output of this program is:

```
main (): i= 10 ;
Function (): i= 20 ;
main (): i= 20 ;
```

which illustrates that the global variable `i` has been altered by *`Function()`*. This is the danger of global identifiers. Any function can change a global identifier. It is good programming practice to make as few global identifiers as possible because of the accessibility problem. When we introduce classes in Chapter 9, the subjects of scope and data access will be re-examined, and it will be shown that classes give us greater data protection and access than do functions.

6.7.5 Scope Resolution Operator

What if we require access to a global identifier instead of a local identifier from within a function body? For such cases C++ provides the scope resolution operator (: :). As the name suggests, this operator resolves the problem of an identifier's scope. The scope resolution operator is illustrated in the following program:

```
// res_op.cpp
// illustrates the scope resolution operator ::
#include <iostream>  // C++ I/O

int i = 10 ;  // global i

void main ()
  {
  int i = 20 ;  // local i

  cout << "main () i: " << i   << endl ;
  cout << "global  i: " << ::i << endl ;
  }
```

Within the *`main()`* function the statement:

```
::i ;  // global variable i
```

accesses the global variable `i` rather than the local variable.

6.8 Function Arguments

Generally, arguments are passed to a function so that the function can operate. The function argument list follows directly after the function name and is enclosed within the parentheses (and) . Each argument in the list is a type specifier (e.g. **`int`** or **`double`**) followed by an identifier name. If the function argument list contains more than one argument, the comma operator is used to separate the arguments. For example:

```
double MyMax (double a, double b)
  { /* ... */ }
```

Function arguments are generally referred to as the *formal arguments.*

Similar to the evaluation order of sub-expressions within an expression, the order of evaluation of argument expressions is undefined:

```
Function (array[i], i++) ;
```

Since the evaluation order of function arguments is undefined then it is bad practice to impose an order.

Occasionally you may come across a style of function definition in which the argument type is placed outside the parentheses and between the function declarator and the opening brace, {, of the function body:

```
double MyMax (a, b)
  double a ; double b ;
  { /* ... */ }
```

This syntax was defined by Kernighan and Ritchie (1978) and is not the ANSI C/C++ standard for function definitions. The K&R standard of function definitions is now considered obsolete by many compilers, which will issue a compilation warning of the form 'function definition syntax is now obsolete'.

6.8.1 No Argument List

If the function has no argument list (i.e. the function does not require arguments to operate correctly) then there are two ways to indicate a zero arguments list. The first is the use of empty parentheses, as we have seen with *main()*, and the second is to use the keyword **void**. Thus we can use either of the following:

```
return_type Function ()
```

or

```
return_type Function (void)
```

6.8.2 By Value or by Reference?

We have left the discussion of function arguments until now because the passing of arguments can be broken down into two separate categories: *passing by value* and *passing by reference.*

Passing by Value

Let's look at a program:

```
// p_value.cpp
// illustrates the passing of arguments by value
#include <iostream>  // C++ I/O

void Function (int i)
  {
  cout << endl << "Function(): rvalue of i: " << i
               << ", lvalue of i: " << &i ;
  }
```

```
void main ()
  {
  int i = 10 ;
  cout << endl << "main()     : rvalue of i: " << i
               << ", lvalue of i: " << &i ;

  Function (i) ;
  }
```

Note the use of the *address of* operator (&). The output of the program is:

```
main()     : rvalue of i: 10, lvalue of i: 0x470721e8
Function(): rvalue of i: 10, lvalue of i: 0x470721dc
```

When you run the program, the lvalues that you will see displayed will probably be different, depending on where the variables are stored in memory. The *main()* program calls the function *Function()*, passing the **int** constant 10 as an argument. The output illustrates that although the i variables in *main()* and *Function()* have the same value (rvalue) the variable i defined in *main()* is stored at a different memory address (lvalue) than the i defined in *Function()*.

When *main()* calls *Function()* the compiler places the rvalue of the i defined in *main()* on the stack. Since the rvalue of i is passed to *Function()*, a copy of i is passed as the argument to *Function()*, and not the variable itself. Thus, the compiler has generated a new variable (which is an exact copy of the passed value) to hold the passed value to *Function()*. Passing arguments in this manner is referred to as *passing by value*, since a copy of the argument is made. An important consequence of passing by value is that there is no way that the function can alter the original variable, because a copy of the variable was sent and the function does not have access to where the variable is stored in memory. This returns us to the previous discussion of local scope. A variable i defined in *main()* is a different variable from a variable i defined in *Function()*.

Passing by value ensures that a function cannot alter the function-caller argument in any manner. For this security we have to pay the price of allocating additional memory, since a function argument is copied. We shall shortly see that **const** reference arguments give us the best of both worlds: the ability to define an identifier as constant, thus ensuring that the function-caller argument is not altered, and the use of reference arguments, which eliminate any copying.

Passing by Reference

The above section illustrated how function arguments are passed by value, which requires a copy of the argument to be made. Alternatively, *passing by reference* passes an argument to a function by reference or memory address, and does not make a copy of the function argument. Thus, passing by reference accesses the function-caller argument.

The following program illustrates passing function arguments by reference:

```
// p_ref.cpp
// illustrates the passing of arguments by reference
#include <iostream>  // C++ I/O

void Function (int& ir)
```

```
    {
    cout << endl << "Function(): rvalue of ir: " << ir
         << ", lvalue of ir: " << &ir ;
    // change alias ir
    ir = 20 ;
    }

  void main ()
    {
    int i = 10 ;

    cout << endl << "main()     : rvalue of i : " << i
         << ", lvalue of i : " << &i ;

    Function (i) ;

    cout << endl << "main()     : rvalue of i : " << i
         << ", lvalue of i : " << &i ;
    }
```

The program output is:

```
main()     : rvalue of i : 10, lvalue of i : 0012FF88
Function(): rvalue of ir: 10, lvalue of ir: 0012FF88
main()     : rvalue of i : 20, lvalue of i : 0012FF88
```

This time, the lvalues of the two variables `i` and `ir` are the same. However, when the alias `ir` is changed within *`Function()`*, variable `i` in *`main()`* is correspondingly modified.

Reference arguments are indicated by an ampersand (&) after the data type (e.g. **`int`**`& ir`). The ampersand indicates that `ir` is an alias for the variable passed by the function caller. Let us highlight this important point:

A reference is an alias for an identifier

Note that there is no difference between the function call of *`Function()`* in the two programs P_VALUE.CPP and P_REF.CPP:

```
Function (i) ;
```

Thus, from a user's point of view, there is no difference in the function call.

Before we discuss the use of the **`const`** modifier with reference arguments, let us just mention that the ampersand can be attached to either the type specifier or the identifier:

```
Function (int& ir)  // general style
```

or

```
Function (int &ir)
```

Which style you choose is completely up to you, and makes no difference to compilation, but the general convention is the former style of attaching the ampersand to the type specifier.

6.8.3 `const` Reference Arguments

Passing by reference eliminates the copying of argument parameters, through the use of an alias, but introduces the danger of accidentally altering the value of a function-caller variable. To prevent this from occurring we define the function reference arguments as type **`const`** (e.g. **`const`** `T&`). The program below illustrates this:

```
// p_const.cpp
// illustrates the use of const for passing arguments by
// reference
#include <iostream>  // C++ I/O

void Function (const int& i)
  {
  // error: can't alter i since defined constant
  i = i + 10 ;
  }

void main ()
  {
  int i = 10 ;

  Function (i) ;
  }
```

This program will not compile because *`Function()`* attempts to modify the **`const`** variable `i` within its function body.

As a further example, the following illustrates the overloaded `basic_ostream` function *`write()`*, declared in the header file OSTREAM.H:

```
ostream_type& write (const char_type* s, streamsize n) ;
```

Note the use of **`const`**, although with a pointer (`*`) instead of a reference, to prevent alteration of the caller's data. Pointers will be discussed in Chapter 12.

6.8.4 `const` Return Value

The **`const`** keyword can be placed after a **`class`** member function argument list:

```
class Circle
  {
  private:
    double centre_x, centre_y, radius ;
  public:
    Circle () ;
    //...
    double Radius () const ;
    double CentreX () const ;
    //...
  };
```

```
//...
double CentreX () const
  { return centre_x ; }
```

The **const** keyword is restricted to **class** member functions and will therefore be discussed in more detail in Chapter 9, when we introduce classes. The **const** function declarator indicates to the compiler that the **class** member access function *CentreX()* is allowed access to the data member centre_x but should not modify it.

6.9 Returning More than One Value

You may be wondering how we return more than one value from a function in C++. Programming languages such as Fortran, Pascal and QBasic provide the *subroutine, procedure* and *subprogram*, respectively, to enable multi-return values from a function. C++ adopts a more minimal approach, in that it refrains from having two function types where one will do.

The **return** statement only allows us to return a single value. However, passing by reference allows a function to return more than one value to its caller. The following program illustrates multiple return values for a *Circle()* function:

```
// return2.cpp
// illustrates returning more than 1 value
#include <iostream>  // C++ I/O
#include <cmath>     // atan()

// computes surface area and circumference of a circle
// returns logical-true if ok, logical-false if not ok
bool Circle (const double& radius,
             double& area, double& circ)
  {
  bool ok (true) ;
  if (radius <= 0)
    {
    return !ok ;  // return logical-false: error
    }
  else
    {
    const double PI = 4.0 * atan (1.0) ;

    area  = PI * radius * radius ;
    circ  = 2.0 * PI * radius ;
    return ok ; // return logical-true: ok
    }
  }

void main ()
  {
  double circle_rad, circle_area, circle_circ = 0.0 ;
```

```
cout << "enter the radius of a circle: " ;
cin  >> circle_rad ;

bool test = Circle (circle_rad, circle_area, circle_circ) ;

if (test)
  {
  cout << "circle area: " << circle_area << endl ;
  cout << "circle circ: " << circle_circ << endl ;
  }
else
  {
  cout << "illegal radius entered!" << endl ;
  }
}
```

Some typical user interaction is:

```
enter the radius of a circle: 5
circle area: 78.5398
circle circ: 31.4159
```

Through the use of reference arguments we are able to return two values from *Circle()*, namely the area and circumference of a circle. This procedure is more convenient than returning single values and using two separate functions:

```
double CircleArea (const double& radius) ;
double CircleCirc (const double& radius) ;
//...
```

Similarities will be observed between the present discussion of returning more than one value from a single function call and the previous discussions of local and global scope and returning by value or reference.

6.10 Return by Reference

In the sections above we have seen passing by value and reference. It is also possible to return by reference. Returning by reference really comes into its own when we examine overloading operators. For example, the following overloads the insertion (<<) operator for use with a user-defined **class**/type `Point`:

```
friend ostream& operator << (ostream& s, const Point& p) ;
```

For now, it suffices to say that returning by reference allows a function to be placed on the left-hand side of an assignment operator:

```
// r_ref.cpp
// illustrates returning by reference
```

```
#include <iostream>  // C++ I/O

int area = 0 ;  // define global variable

int& Area ()
  {
  return area ;
  }

void main ()
  {
  Area () = 10 ;
  cout << Area() ;
  }
```

The return type specifier of *Area()* is **int**&. The function *Area()* simply returns the value of the global variable area. Within the function body of *main()* the global variable area is assigned a value of 10 through use of the *Area()* function on the left-hand side of the assignment expression. After assignment, *Area()* is used to output the value of area. Note that area has to be made a global variable since you cannot return a reference to a local variable. For instance, the following is illegal:

```
int& Area ()
  {
  int area = 0 ;
  return area ;
  }
```

because the lifetime of the local variable area expires at the closing brace of *Area()* and it is therefore not possible for *Area()* to return a memory address

6.11 Why Use Functions?

In the program FUNC.CPP above the single program statement within the *MyMax()* function is simple enough not to have been placed in a function body. We could have simply incorporated the statement that tests two numbers into the *main()* function:

```
//...
void main ()
  {
  double number1, number2 ;

  cout << "enter two positive or negative numbers: " ;
  cin  >> number1 >> number2 ;

  double largest = number1 > number2 ? number1 : number2 ;

  cout << "largest of the two numbers: " << largest ;
  }
```

This is quite legitimate, since the *MyMax()* function accomplishes a small task. However, as function size increases, this procedure of placing program statements within the *main()* function clearly becomes impractical. Functions also hide away the gory details of a function body, which are of no importance and a distraction to the function-caller. For instance, when a user calls the C++ standard library *rand()* function to return a pseudo-random number, it is not necessary for the user to know every program detail of the *rand()* function. The *rand()* user is simply interested in generating pseudo-random numbers for a particular application in mind. Documentation can enhance a user's understanding of the *rand()* function, but a more detailed knowledge of random number generation will require the user to examine in some detail random numbers and possibly to implement his or her own routine(s).

Functions are reusable, and not only by the programmer(s) who developed the functions. A spin-off from reuseability is that functions dramatically reduce program size. Consider developing a library of functions which determine the maximum and minimum of two numbers, the absolute value of a positive or negative number and the square root of a positive number. We shall call these functions appropriately *MyMax()*, *MyMin()*, *MyAbs()* and *MySqrt()*. We have already developed one of these functions above, *MyMax()*, and we now need to place this function and the three additional functions in a form that makes them suitably accessible to other programs – hopefully any C++ program. How do we go about doing this? Well, one way is simply to add our new functions to the above program, FUNC.CPP:

```
// func.cpp
//...
double MyMax (double a, double b)
  { /*...*/ }
double MyMin (double a, double b)
  { /*...*/ }
double MyAbs (double n)
  { /*...*/ }
double MySqrt (double n)
  { /*...*/ }
//...
void main ()
  { /* ... */ }
```

This is OK if the *My* functions are only to be called by *main()* in FUNC.CPP.

What we really need to do is to separate the *My* functions from the *main()* function. We shall place the *My* function *definitions* into a separate *implementation* file called MY_LIB.CPP, and at the same time place the *My* function *declarations* in a *header* file called MY_LIB.H.

The form of the implementation file is:

```
// my_lib.cpp
//...
double MyMax (double a, double b)
  {
  return a > b ? a : b ;
  }

double MyMin (double a, double b)
  {
  return a < b ? a : b ;
```

```
    }
//...
```

while the declaration header file is:

```
// my_lib.h
//...
double MyMax  (double a, double b) ;
double MyMin  (double a, double b) ;
double MyAbs  (double n) ;
double MySqrt (double n) ;
```

`include`-ing the header file MY_LIB.H in the main program enables us to use our library of functions:

```
// new_func.cpp
//...
#include "my_lib.h"

void main ()
  { /* ... */ }
```

With the library and main program structure out of the way, let's now examine the implementation details of the *My* functions in MY_LIB.CPP.

The *MyMin()* function is a simple modification of the *MyMax()* function; just flip round the relational operator. The function *MyAbs()* returns the absolute value of a number. If a number is negative then the negative sign is removed by *MyAbs()*; otherwise if the number is positive the number remains the same. The trickiest function of the four is *MySqrt()*. The square root is calculated using the ancient[1] square root formula:

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right), \quad n = 0, 1, 2, \ldots$$

This formula calculates the square root of number a (> 0) by an iterative process: $x_0, x_1, x_2, \ldots, xn$. As n increases the above formula can be shown to tend to the square root of the number a. To start, the iterative process makes an arbitrary guess, x_0, of the square root (assume for computational purposes that $x_0 = a$). The initial value of x_0 is placed in the above formula and a value of x_1 is determined. Then we pass this new value for x, x_1, back into the formula and so on. As an example, consider determining the square root of the constant 2:

$$x_1 = \frac{1}{2}\left(x_0 + \frac{2}{x_0}\right) = \frac{1}{2}\left(2 + \frac{2}{2}\right) = 1.5$$

$$x_2 = \frac{1}{2}\left(x_1 + \frac{2}{x_1}\right) = \frac{1}{2}\left(1.5 + \frac{2}{1.5}\right) = 1.4166\ldots$$

$$x_3 = \ldots$$

1 When I use the word *ancient*, I mean older than I am – 4000 years old in this case.

Convergence is very fast. The number of correct digits approximately doubles with each iteration. If convergence has occurred, to the required accuracy, before the maximum number of iterations has been reached, the function **break**s out of a **for**-loop. The function *MySqrt()* requires only six iterations to obtain an accuracy of 15 decimal places. *MySqrt()* calculates the square root of a positive number, and as a result the function has to perform an internal check to see whether the user of the function has passed a negative number. If a negative number is passed the function returns -1 to indicate an error. Here is the complete MY_LIB.CPP file:

```
// my_lib.cpp
// implementation file for my library of functions

#include "my_lib.h"  // my_lib header file

// returns the maximum of two numbers
double MyMax (double a, double b)
  {
  return a > b ? a : b ;
  }

// returns the minimum of two numbers
double MyMin (double a, double b)
  {
  return a < b ? a : b ;
  }

// returns the absolute value of a number
double MyAbs (double n)
  {
  return n < 0 ? -n : n ;
  }

// returns the square root of a positive number
// return -1 if negative number is passed
double MySqrt (double n)
  {
  const int MAX_NO_ITERATIONS (50) ;
  double x_old (n), x_new (0.0) ;
  double sqrt (0.0) ;

  if (n < 0)  // if negative number
    {
    sqrt = -1.0 ;
    }
  else  // ok
    {
    // loop thro'
    for (int i=0; i<MAX_NO_ITERATIONS; i++)
      {
      x_new = (x_old + n / x_old) / 2.0 ;
```

```
        // test current accuracy
        if (fabs (x_new - x_old) < TOLERANCE)
          {
          sqrt = x_new ;
          break ;
          }
        else
          x_old = x_new ;
        }
      }
    return sqrt ;
    }
```

Note the use of the C++ library function *fabs()*[2], which is declared in MATH.H and returns the absolute value of a floating-point number. The use of *fabs()* is required in the above **if**-statement because (x_new-x_old) could be less than zero but not necessarily less than TOLERANCE. The accompanying header file is:

```
// my_lib.h
// header file for my_lib.cpp

#ifndef _MY_LIB_H  // prevent multiple includes
#define _MY_LIB_H

#include <cmath>  // fabs()

const double TOLERANCE = 1e-06 ;

double MyMax  (double a, double b) ;
double MyMin  (double a, double b) ;
double MyAbs  (double n) ;
double MySqrt (double n) ;

#endif // _MY_LIB_H
```

The header file is fairly straightforward. It contains a declaration for each of the four functions and a defined constant, TOLERANCE, which specifies the required level of accuracy of the square root value returned by *MySqrt()*.

Note the conditional compilation statement in the header file:

```
#ifndef _MY_LIB_H  // prevent multiple includes
#define _MY_LIB_H

// include any header files that header file needs
#include <cmath>
//...
// function declarations
```

2 Equally, the *MyAbs()* function could have been used instead of the C++ library function *fabs()*. The *fabs()* function is used to illustrate include-ing header files within a user-defined header file.

```
double max (double a, double b) ;
//...
#endif  // _MY_LIB_H
```

This statement uses the preprocessor directives `#ifndef` (`if` not `defined`), `#define` and `#endif` to prevent including a header file several times. As function libraries get more complex it is easy to include the same header file several times in an implementation file, because each header file generally includes other header files. Including a header file more than once is in effect equivalent to redefining a function and will generate a compilation error. One way around this problem is simply to ensure that each header file is included just the once, but this can be difficult and tedious to do. The simplest way to avoid this problem is the above use of the preprocessor directives. If the identifier `_MY_LIB_H` is already defined then the compiler will skip over the subsequent source code up to the `#endif` statement. Alternatively, if `_MY_LIB_H` is not defined then the source code is compiled. Note that the identifier `_MY_LIB_H` directly after `#endif` is commented out. Some commercial compilers accept:

```
#endif _MY_LIB_H
```

but this may not be portable. Therefore, adopt the style of commenting out the identifier.

As a further example consider the declaration of two classes `Line` and `Face`, which include a `Point` **class** declared in the header file POINT.H:

```
// point.h
// header file for Point class
#ifndef _POINT_H  // prevent multiple includes
#define _POINT_H
//...
class Point
  {
  private:
    double x, y, z ;
  public:
    Point ()
      { x = y = z = 0.0 ; }
    //...
  };
#endif  // _POINT_H
```

The `Point` **class** will be discussed in detail in Chapter 9. The `Line` and `Face` **class** declarations, in LINE.H and FACE.H respectively, could be of the form:

```
// line.h
// header file for Line class
//...
#include "point.h"  // include Point class header file

class Line
  {
  private:
    Point p1, p2 ;
  public:
```

```
    Line ()
      { }
    //...
  };

// face.h
// header file for Face class
//...
#include "point.h"  // include Point class header file
#include "line.h"   // include Line class header file

class TriangularFace
  {
  private:
    Point p1, p2, p3 ;
    Line  l1, l2, l3 ;
  public:
    TriangularFace ()
      { }
    //...
  };
```

The header file POINT.H is included in LINE.H and FACE.H, and if the conditional preprocessor directive was not used then **class** `Point` would be declared twice.

Ensure that the identifiers used for the `#ifndef` and `#define` preprocessor directives are identical and that all source code and `#include` statements are within the conditional compilation statement. I always adopt the notation of `_FILENAME_H` for naming conditional compilation identifiers, where `FILENAME` is the name of the header file. Preceding `FILENAME` with an underscore ensures that there is no confusion with uppercase `#defined` or **const** constants.

The example program below illustrates typical applications of the *My* functions:

```
// new_func.cpp
// demonstrates the use of implementation and header files
// my_lib.cpp and my_lib.h
#include <iostream>  // C++ I/O

#include "my_lib.h"  // my_lib header file

void main ()
  {
  double number1, number2 ;
  cout << "enter a positive and a negative number: " ;
  cin  >> number1 >> number2 ;

  cout << "MyMax (" << number1 << ", " << number2 << "): "
       << MyMax (number1, number2) << endl ;
  cout << "MyMin (" << number1 << ", " << number2 << "): "
       << MyMin (number1, number2) << endl ;
  cout << "MyAbs (" << number2 << "): "
       << MyAbs (number2) << endl ;
```

```
    cout << "MySqrt (" << number1 << "): "
         << MySqrt (number1) << endl ;
    }
```

Note that the header file MY_LIB.H is `included` within double quotes (" ") rather than angle brackets (<>) because the header file resides in the same directory as the main program and is a user-defined file. The distinction between " " and <> was discussed in Chapter 3. The following shows some user interaction:

```
enter a positive and a negative number: 30 -10.4
MyMax (30, -10.4): 30
MyMin (30, -10.4): -10.4
MyAbs (-10.4): 10.4
MySqrt (30): 5.47723
```

6.12 Overloaded Functions

Overloading allows a function to perform different actions depending on the data types that it is sent[3]. A function is said to be overloaded when different function declarations exist for the same function name in the same scope. Consider a program that requires three different functions to output a message that is dependent on the type of argument passed to the function. One way is simply to name each function differently:

```
//...
void FunctionChar (char ch)
  { /*...*/ }

void FunctionInt (int i)
  { /*...*/ }

void FunctionDouble (double d)
  { /*...*/ }
//...
```

This works fine, but requires a programmer to know that three differently named functions exist for types **char**, **int** and **double**. C++ allows a function name to be overloaded:

```
// over.cpp
// illustrates overloaded functions
#include <iostream>  // C++ I/O

void Function (char ch)
  {
  cout << "Function (char): " << ch << endl ;
  }
```

3 Overloaded functions are an example of compile-time polymorphism in which a single function can have multiple different operations.

```
void Function (int i)
  {
  cout << "Function (int): " << i << endl ;
  }

void Function (double d)
  {
  cout << "Function (double): " << d << endl ;
  }

void main ()
  {
  Function ('c') ;  // char
  Function (1) ;    // int
  Function (1.0) ;  // double
  }
```

with output:

```
Function (char): c
Function (int): 1
Function (double) 1
```

The above example illustrates overloading the same function name with the same number of arguments (in this case one argument). Multiple arguments are also valid:

```
//...
void Function (int a)
  { /* ... */ }

void Function (int a, int b)
  { /* ... */ }

void Function (int a, double b, char c)
  { /* ... */ }
//...
```

How does the compiler know which function to choose? The correct function is selected by comparing the number and types of the actual function arguments with the formal arguments. However, an overloaded function cannot only differ in its return type. For instance, consider the following program code:

```
void   Function (int i) ;
int    Function (int i) ;
double Function (int i) ;
//...
void main ()
  {
  //...
  Function (0) ;  // call which Function()?
  //...
  }
```

For the above call to *Function()* within *main()* the compiler would not know which version of *Function()* to choose and would correspondingly issue a compilation error.

Frequently in programming, particularly with classes that we shall observe in Chapter 9, we are required to *get* and *set* a data value (say `position`). One way to do this is simply to define two functions:

```
//...
double GetPosition () ;              // gets position
void   SetPosition (double pos) ;    // sets position
```

A more concise way is to overload a more general *Position()* function that can both get and set the value of `position`:

```
// get_set.cpp
// illustrates function overloading
#include <iostream>  // C++ I/O

double position (10.0) ;  // some global variable: position

// gets position
double Position ()
  { return position ; }

// sets position
void Position (double pos)
  { position = pos ; }

void main ()
  {
  Position (20) ;
  cout << "get position: " << Position () << endl ;
  }
```

Overloading *Position()* allows us to get and set a data value with a single function name.

Overloaded functions are clearly a very powerful feature of C++, but there are a few exceptions that you should be aware of. The first case illustrates arguments of type `T` and `T&`. Since types `T` and `T&` accept the same set of initialisers, the following two functions are indistinguishable from one another:

```
//...
void Func (int i)
  {
  cout << endl << "Func (int)" ;
  }

// error: Func(int&) can't be distinguished from Func(int)
void Func (int& i)
  {
  cout << endl << "Func (int&)" ;
  }
```

This example code is extracted from OVERLOAD.CPP. Another case, also extracted from OVERLOAD.CPP, which requires some careful consideration is the following:

```
//...
void Func (int i)
  {
  cout << endl << "Func (int)" ;
  }

void Func (float f)
  {
  cout << endl << "Func (float)" ;
  }

void main ()
  {
  // error: Func(float) can't be distinguished from Func(int)
  Func (1) ;
  Func (1.0) ;
  }
```

The compiler is unable to distinguish between *Func (**int**)* and *Func (**float**)* in the function calls. The solution is to use the suffix `f` or `F` on the constant of type **`float`** that is passed as an argument:

```
//...
void main ()
  {
  Func (1) ;
  Func (1.0f) ;  // suffix f removes ambiguity
  }
```

The following is also legal:

```
//...
void Func (int i)
  {
  cout << endl << "Func (int)" ;
  }

void Func (double d)
  {
  cout << endl << "Func (double)" ;
  }

void main ()
  {
  Func (1) ;          // int
  Func (1.0) ;        // double
  }
```

The following program code eliminates any ambiguity, but the function call *Func(1.0)* will call the **double** version:

```
//...
void Func (int i)
  {
  cout << endl << "Func (int)" ;
  }

void Func (float f)
  {
  cout << endl << "Func (float)" ;
  }

void Func (double d)
  {
  cout << endl << "Func (double)" ;
  }

void main ()
  {
  Func (1) ;    // int
  Func (1.0) ; // double
  }
```

Thus, to eliminate any confusion or ambiguity in function overloading when the argument(s) are both floating-point numbers of type **float** and **double**, use the suffix `f` or `F` for arguments of type **float**.

We shall examine typedefs and enumerations in Chapter 8, but it is worth highlighting that a **typedef** is a synonym for an existing type and not a new type, and therefore functions with the same name and number of arguments cannot be overloaded using **typedef**s:

```
//...
void Func (int i) ;
//...
typedef int Integer ;

// error: re-defines Func (int)
void Func (Integer ti) ;
```

The above is an error because *Func(Integer)* redefines *Func(**int**)*. Enumerations, however, are distinct types, and functions involving enumerations can be overloaded:

```
//...
void Func (int i)

enum Enum
  {
  a
  };

void Func (Enum e) ;
```

There are several other subtle distinctions to be found in function overloading, so feel free to experiment with the OVERLOAD.CPP program.

6.13 Default Arguments

C++ allows a function to be called without specifying all of its arguments. How is this possible? If a *function declaration* supplies default values for certain or all arguments then a function can be called without values for these arguments:

```
// def_arg.cpp
// illustrates default function arguments
#include <iostream>  // C++ I/O

// prototype with default argument values
double RectangleArea (const double& width=12.0,
                      const double& height=5.0) ;

// returns area of a rectangle
double RectangleArea (const double& width,
                      const double& height)
  { return width * height ; }

void main ()
  {
  cout << "area (12, 5): " << RectangleArea ()      << endl ;
  cout << "area (8,  5): " << RectangleArea (8)     << endl ;
  cout << "area (2, 10): " << RectangleArea (2, 10) << endl ;
  }
```

which produces the following output:

```
area (12, 5): 60
area (8,  5): 40
area (2, 10): 20
```

The first statement in `main()` uses both default argument values of `RectangleArea()`. The second statement just uses one default argument (`height=5.0`), and the last statement uses no default arguments. In the case of the second statement, how does the compiler know which of the two default arguments to choose? The system adopted by the compiler is that only trailing arguments in an argument list can be assigned default values. For instance:

```
Triangle (const Point& p1, const Point& p2, const Point& p3,
          const Surface& surface=TWOD, const Colour&
          colour=BLACK) ;
```

The first three function arguments of `Triangle()` define the vertices of a triangle and the last two arguments set the default surface and colour attributes of a triangle to two-dimensional and black, respectively.

It is important to observe that default arguments are restricted to the declaration of a function. Therefore, in the above example of *RectangleArea()* the following is an illegal use of default arguments:

```
double RectangleArea (const double& width=12.0,
                      const double& height=5.0)
  { /* ... */ }
```

Can a default argument expression use a local function variable? In the above program, DEF_ARG.CPP, we may alternatively require that *RectangleArea()* defaults to the area of a square if the height of a rectangle is unspecified:

```
double RectangleArea (const double& width=12.0,
                      const double& height=width) ;  // error
```

Regrettably, the use of local variables in default argument expressions is illegal in C++.

6.14 Local `static` Variables

Chapter 4 examined the use of the **static** storage class specifier for global variables. Let us now examine local **static** variables. The following program compares two functions containing non-**static** and **static** local function variables:

```
// static.cpp
// illustrates local static variables
#include <iostream>  // C++ I/O

// non-static local variable
int Function0 ()
  {
  int ns_count = 0 ;
  return ++ns_count ;
  }

// static local variable
int Function1 ()
  {
  static int s_count = 0 ;
  return ++s_count ;
  }

void main ()
  {
  cout << "non-static local variable Function0(): " ;
  for (int i=0; i<5; i++)
    cout << Function0 () << " " ;
  cout << endl ;

  cout << "    static local variable Function1(): " ;
```

```
    for (int j=0; j<5; j++)
      cout << Function1 () << " " ;
    cout << endl ;

    //ns_count = 0 ;  // error: local to Function0()
    //s_count  = 0 ;  // error: local to Function1()
    }
```

with output:

```
non-static local variable Function0(): 1 1 1 1 1
    static local variable Function1(): 1 2 3 4 5
```

Function0() defines and initialises to zero an **int** non-**static** local variable called ns_count and then returns ns_count's prefix increment value. As expected, each time *Function0()* is called from the **for**-loop in *main()* the value of 1 is returned by *Function0()* because upon returning from *Function0()* the value of ns_count is lost. However, the program output illustrates that this is not the case for **static** local variables. Each time *Function1()* is called s_count maintains its value from the last call to *Function1()*.

To understand why s_count maintains its value between function calls, it helps to note that s_count is defined and initialised to zero at the start of *Function1()* – so why isn't s_count zeroed upon each call to *Function1()*? The answer lies in the fact that local **static** variables are identical to global variables in that they are defined and initialised only once and that their lifetime extends to the end of the program and not when they go out of scope when a function returns. Where local **static** variables differ from global variables is that a local **static** variable's scope is only from its point of definition to the end of the function block in which it resides. The scope of the local non-**static** and static variables ns_count and s_count is illustrated at the end of the *main()* function above.

6.15 `inline` Functions

Functions save memory space because all function calls use the same executable code. The disadvantage of function calls is that they take time. Consider the *MyMax()* function call:

```
//...
double MyMax (double a, double b)
  {
  return a > b ? a : b ;
  }

void main ()
  {
  //...
  double largest = MyMax (number1, number2) ;
  //...
  }
```

When *MyMax()* is called in *main()* then control goes to *MyMax()* and when *MyMax()* returns control goes back to *main()*. This transfer of control can involve large memory jumps. If we simply expand the *MyMax()* function code inline in *main()*, no transfer of control is required:

```
void main ()
  {
  //...
  double largest = a > b ? a : b ;
  //...
  }
```

Clearly, if the function is large this is not feasible, and at the same time we have lost a neat function *MyMax()*.

C programmers would implement the *MyMax()* function **inline** by defining a macro:

```
#define MYMAX(a, b) ( ((a) > (b)) ? (a) : (b) )
```

Macros don't generate any function overhead code. All of a macro's code is generated inline, and thus executes faster than its function counterpart because memory access is sequential. However, macros generate inline code, which means that each time you call the macro the compiler generates the macro code, whereas the compiler generates function code just the once. Also, a major disadvantage of macros is that they receive no static type checking, and the use of parentheses can be confusing and the source of difficult-to-find errors.

C++ offers us the **inline** keyword to make a function inline. The INLINE.CPP program illustrates the use of **inline** with the *MyMax()* function:

```
// inline.cpp
// illustrates an inline function
#include <iostream>  // C++ I/O

// returns the maximum of two numbers
inline double MyMax (double a, double b)
  { return a > b ? a : b ; }

void main ()
  {
  double number1, number2 ;
  //...
  double largest = MyMax (number1, number2) ;
  //...
  }
```

The keyword **inline** inserts the function code directly **inline** in the code of the function caller. Ensure that the function definition appears before an **inline** function call. This is simply because it is not enough for the compiler to see just the function declaration before a function call: the compiler has to insert the inline function code in the function-caller program.

So which functions do you make **inline**? Generally, reserve **inline** functions for small functions that are frequently called. Inline functions are usually defined with their declaration in their corresponding header file in order to prevent the problem mentioned above of a function call occurring before the function definition. Actually, the compiler ultimately decides

what action to take with **inline** functions. The compiler can ignore the **inline** request if it considers that the function is unsuitable to be made **inline**.

6.16 The *main()* Function

In Chapter 3 the *main()* function was introduced, and we have seen every program so far use the *main()* function. So, before we discuss external linkage and C++ library functions let's take a closer look at the *main()* function.

Every C++ console-based program must call the *main()* startup function. The *main()* function indicates the start and end of program execution via the opening, {, and closing, }, braces of the *main()* function body. If *main()* is not defined within your program you will not get correct linkage. Where you place *main()* within your program is completely up to you. Programmers generally place *main()* either at the start or the end of a program. There appears to be no general consensus about the placement of *main()*, but I always place *main()* at the end of the program, so I know where to find it.

All of the programs that we have seen so far have defined *main()* in the general form:

```
//...
void main ()
   { /* ... */ }
```

which indicates that the return type of *main()* is **void** and that it does not return a value. It was mentioned above that we could explicitly indicate that *main()* has no argument list as:

```
void main (void)
   {
   // ...
   }
```

The value returned by *main()* is the status code of the program, and is in fact an integer. Thus, it is legal to define *main()*'s return type specifier as **int**:

```
int main ()
   {
   //...
   return 0 ;   // normal program termination
   }
```

Type **int** is the default return type of the *main()* function for the majority of C++ compilers and hence the reason why the return type specifier of *main()* can be **void**. The value returned by *main()* is used to indicate to the operating system the program's exit status. Most operating systems indicate normal exit by a zero and abnormal exit by a non-zero value. For instance, the *exit()* function, which terminates program execution, uses two error status signals (defined in STDLIB.H): EXIT_SUCCESS (0) and EXIT_FAILURE (1), which indicate normal and abnormal program termination respectively.

6.16.1 Command Line Arguments

The *main()* function can be defined with *command line arguments*. A typical use of command line arguments is in executing Microsoft's DOS screen editor. The editor can simply be

executed by typing the editor's executable program name (EDIT) followed by a carriage return and then proceeding to open a text file by way of File | Open. Alternatively, the text file to edit can be passed as a command line argument:

```
EDIT \CPP_PROG\CHAP06\CHAP06.TXT
```

which incorporates the path and filenames of the text file as a command line argument. We shall continue our discussion of command line arguments further when we discuss input and output, files, and streams in Chapter 17, but for now just bear in mind that they are possible.

6.16.2 *WinMain()*

Windows programs call the *WinMain()* function at startup instead of *main()*:

```
int PASCAL WinMain (HANDLE hInstance, HANDLE hPrevInstance,
                    LPSTR lpszCmdLine, int nCmdShow)
  {
  //...
  if (!hPrevInstance) // if first instance
    {
    //...
    }
  hwnd = CreateWindow (/*...*/) // create window procedure
  //...
  ShowWindow (hwnd, nCmdShow) ; // show & update
  UpdateWindow (hwnd) ;
  //...
  while (GetMessage (&msg, NULL, 0, 0) // message loop
    {
    //...
    }
  return msg.wParam ; // return messages to operating system
  }
```

If you have programmed for Windows using the Borland ObjectWindows **class** library then you may have seen the *OwlMain()* function.

6.17 External Linkage

Chapter 4 examined the use of the **extern** storage class specifier for global variables. We are now in a position to discuss **extern** as applied to functions. Let us first define a function, *Function()*, in FILE.CPP:

```
// file.cpp
// a function definition
#include <iostream>  // C++ I/O

void Function ()
  {
```

```
    cout << "Function() called" << endl ;
    }
```

If we would like to use *Function()* in a separate file, say EXTERN.CPP, then we can use the **extern** specifier in a *linkage specification* to declare *Function()* and inform the compiler that external linkage is required to obtain the definition of *Function()*:

```
// extern.cpp
// illustrates the extern storage class specifier

extern Function () ;

void main ()
   {
   Function () ;
   }
```

with output:

```
Function() called
```

It is perfectly legal to link the above two files FILE.CPP and EXTERN.CPP, even though they both contain a reference to *Function()*, because EXTERN.CPP contains an **extern** declaration of *Function()* and not a definition. As highlighted in Chapter 4, a program can contain multiple variable or function declarations but only one definition of the same named variable or function.

The **extern** specifier can also be used to inform the compiler of linkage between C++ and non-C++ variables and functions. C++/non-C++ linkage is achieved by placing a string literal in the linkage specification which can have one of the two following general forms:

```
extern string_literal declaration;

extern string_literal
   {
   // multiple declarations
   }
```

where *declaration* is any legal variable or function declaration. The second form of the **extern** declaration includes multiple **extern** declarations between the open, {, and close, }, braces. The `string_literal` indicates the kind of linkage, such as C++, C or Fortran. C and C++ linkage are guaranteed by all ANSI/ISO compatible compilers, with C++ linkage being the default, whereas with other languages linkage is implementation-dependent. To illustrate the above two **extern** C and C++ linkage specifications, consider the program:

```
// extern1.cpp
// further illustrates the extern storage class specifier
#include <iostream>  // C++ I/O

extern double fabs (double x) ;  // default C++ linkage

extern "C"   double sqrt (double x) ;
```

```
extern "C++" double pow  (int x, int y) ;

extern "C"
  {
  double sin (double x) ;
  double cos (double x) ;
  double tan (double x) ;
  }

void main ()
  {
  cout << "tan(1): " << tan (1.0) << endl ;
  }
```

with output:

```
tan(1): 1.55741
```

Note that the C++ standard library header file MATH.H is not included in the above program.

6.18 Library Functions

Aside from learning the syntax, semantics and pragmatics of a programming language, it is equally important to learn what tools are shipped with the language. These tools are generally in the form of run-time function and **class** libraries. This section presents an overview of functions supplied with C++, whereas C++'s standard **class** library will be covered from Chapter 9 onwards, when classes are discussed in more detail.

Attempting to program in C++ without getting to know the function and **class** libraries would be nearly impossible and clearly impractical. The libraries save you from reinventing the wheel every time you want to do the simplest of things. Also, the shipped libraries are generally robust and efficient and have been developed over a number of years by professionals.

At a first glance, the C++ libraries can appear overwhelming and incomprehensible. Don't attempt to learn every function and every detail of each function. This is impossible and not necessary. Most experienced programmers develop a general knowledge of the C++ libraries and become familiar with most of the functions in the libraries. Try to tailor your understanding of the C++ libraries according to your needs. If you are heavily involved in numerical and computational work, the mathematical functions in the MATH.H header will be extremely useful.

For example, say that you are required to determine the square root of a number. Do you spend a week developing your own square root function? Maybe, but a quicker way is to check your compiler's documentation. A function called *sqrt()* is part of the ANSI library functions and is declared in the header file MATH.H:

```
double sqrt (double x) ;
```

A check with my C++ compiler's documentation confirms that this function calculates the square root of a positive number of type **double**, which is passed as an argument. If the argu-

ment is negative, the global variable `errno` is set to the identifier `EDOM`, which indicates a domain error.

Another related non-ANSI function is:

```
long double sqrtl (long double x) ;
```

which takes a **`long double`** argument and returns a **`long double`** square root result.

Further, we note that the *`sqrt()`* function can be used with `bcd` (binary-coded decimal) and `complex` types. Class `bcd` is declared in the BCD.H header file and the *`sqrt()`* function is declared as:

```
friend inline bcd sqrt (const bcd& x) ;
```

while the `complex` square root function is declared in the header file COMPLEX.H as:

```
template <class T>
inline complex<T> sqrt (const complex<T>& x) ;
```

This example has illustrated that determining the square root of a number in C++ is well covered. The compiler documentation has provided us with the necessary implementation details and indicated that *`sqrt()`* can operate also on `bcd` and `complex` types.

6.18.1 The ANSI C Standard Library

When ANSI formed a standard for the C language it also defined a standard library of functions. Any compiler that is an ANSI C-compliant compiler has to supply the ANSI C standard library functions, and each of these functions is completely independent of the compiler and hardware used. Table 6.1 lists all macros and functions that are ANSI C-compatible. The table is subdivided into the following routines: classification, conversion, diagnostic, input/output, international locale API, manipulation, mathematical, memory, process control, time and date, and variable arguments.

6.19 Header Files

Header files (*.H) provide function and **`class`** declarations. This section divides the ANSI C and C++ header files into two subsections: ANSI C and C++. Each subsection gives a complete list of the header filenames and a brief description of each header file.

6.19.1 ANSI C Header Files

ASSERT.H	*`assert()`* macro
CTYPE.H	character classification and conversion
ERRNO.H	constant mnemonics for error codes
FLOAT.H	implementation-specific floating-point number routines
LIMITS.H	implementation-specific limits on type values
LOCALE.H	country and language specific routines
MATH.H	mathematical functions and error handlers

Table 6.1 ANSI C macros and functions.

Function	*Description*	*Header file*
Classification		
`isalnum`	tests for an alphanumeric character	CTYPE.H
`isalpha`	tests for letters	CTYPE.H
`iscntrl`	tests for control characters	CTYPE.H
`isdigit`	tests for digits	CTYPE.H
`isgraph`	tests for printing characters	CTYPE.H
`islower`	tests for lowercase letters	CTYPE.H
`isprint`	tests for printable characters	CTYPE.H
`ispunct`	tests for punctuation characters	CTYPE.H
`isspace`	tests for white space	CTYPE.H
`isupper`	tests for uppercase letters	CTYPE.H
`isxdigit`	tests for hexadecimal digits	CTYPE.H
Conversion		
`atof`	converts a string to a floating-point number	STDIO.H
`atoi`	converts a string to an integer	STDIO.H
`atol`	converts a string to a long	STDIO.H
`strtod`	converts a string to a double	STDIO.H
`strtol`	converts a string to a long	STDIO.H
`strtoul`	converts a string to an unsigned long	STDIO.H
`tolower`	converts characters to lowercase	STDIO.H
`toupper`	converts characters to uppercase	STDIO.H
Diagnostic		
`assert`	tests a condition and possibly aborts	ASSERT.H
`perror`	prints a system error message	ERRNO.H
Input/output		
`clearerr`	resets error indication	STDIO.H
`fclose`	closes a stream	STDIO.H
`feof`	detects an end-of-file on a stream	STDIO.H
`ferror`	detects errors on a stream	STDIO.H
`fflush`	flushes a stream	STDIO.H
`fgetc`	gets a character from a stream	STDIO.H
`fgetpos`	gets the current file pointer	STDIO.H
`fgets`	gets a string from a stream	STDIO.H
`fopen`	opens a stream	STDIO.H
`fprintf`	writes formatted output to a stream	STDIO.H
`fputc`	puts a character on a stream	STDIO.H
`fputs`	outputs a string on a stream	STDIO.H
`fread`	reads data from a stream	STDIO.H
`freopen`	associates a new file with an open stream	STDIO.H
`fscanf`	scans and formats input from a stream	STDIO.H
`fseek`	repositions a file pointer on a stream	STDIO.H
`fsetpos`	positions the file pointer of a stream	STDIO.H
`fstat`	gets open file information	SYS\STAT.H

Table 6.1 *(contd.)*

Function	*Description*	*Header file*
`ftell`	returns the current file pointer	STDIO.H
`fwrite`	writes to a stream	STDIO.H
`getc`	gets a character from a stream	STDIO.H
`getchar`	gets a character from stdin	STDIO.H
`getenv`	gets a string from the environment	STDIO.H
`gets`	gets a string from stdin	STDIO.H
`longjmp`	performs a non-local goto	SETJMP.H
`printf`	writes formatted output to stdout	STDIO.H
`putc`	outputs a character to a stream	STDIO.H
`putchar`	outputs a character to stdout	STDIO.H
`puts`	outputs a string to stdout	STDIO.H
`remove`	removes a file	STDIO.H
`rename`	renames a file	STDIO.H
`rewind`	repositions the file pointer to the start of a stream	STDIO.H
`scanf`	scans and formats input from stdin	STDIO.H
`setbuf`	assigns buffering to a stream	STDIO.H
`setjmp`	sets up for non-local goto	SETJMP.H
`setvbuf`	assigns buffering to a stream	STDIO.H
`sprintf`	writes formatted output to a stream	STDIO.H
`sscanf`	scans and formats input from a string	STDIO.H
`stat`	gets information about a file	SYS\STAT.H
`tmpfile`	opens a temporary file for binary mode	STDIO.H
`tmpnam`	creates a unique filename	STDIO.H
`ungetc`	pushes a character back into an input stream	STDIO.H
`vfprintf`	write formatted output to a stream	STDIO.H
`vprintf`	write formatted output to stdout	STDIO.H
`vsprintf`	write formatted output to a string	STDIO.H
International locale API		
`localeconv`	gets the current locale structure	LOCALE.H
`setlocale`	selects or queries a locale	LOCALE.H
Manipulation		
`bsearch`	binary search of an array	STDLIB.H
`memchr`	searches for a character	MEM.H, STRING.H
`memcmp`	compares two blocks	MEM.H, STRING.H
`memcpy`	copies a block	MEM.H, STRING.H
`memmove`	copies a block	MEM.H, STRING.H
`memset`	sets bytes of a block to a specified value	MEM.H, STRING.H
`qsort`	sorts according to the quicksort algorithm	STDLIB.H
`strcat`	appends one string to another	STRING.H
`strchr`	scans a string for a character	STRING.H
`strcmp`	compares two strings	STRING.H
`strcoll`	compares two strings	STRING.H
`strcpy`	copies one string into another	STRING.H
`strcspn`	scans a string for a subset of characters	STRING.H

Table 6.1 *(contd.)*

Function	*Description*	*Header file*
`strerror`	gets an error message string	STRING.H
`strlen`	gets the length of a string	STRING.H
`strncat`	appends a portion of one string to another	STRING.H
`strncmp`	compares a portion of one string with another	STRING.H
`strncpy`	copies bytes from one string to another	STRING.H
`strnbrk`	scans a string for a character	STRING.H
`strrchr`	scans a string for a character	STRING.H
`strspn`	scans a string for a subset of characters	STRING.H
`strstr`	scans a string for a substring	STRING.H
`strtok`	searches a string for tokens	STRING.H
`strxfrm`	transforms a portion of a string	STRING.H
Mathematical		
`abs`	absolute value of an integer	COMPLEX.H, STDLIB.H
`acos`	arc cosine	COMPLEX.H, MATH.H
`asin`	arc sine	COMPLEX.H, MATH.H
`atan`	arc tangent	COMPLEX.H, MATH.H
`atan2`	arc tangent of y/x	COMPLEX.H, MATH.H
`ceil`	rounds up	MATH.H
`cos`	cosine	COMPLEX.H, MATH.H
`cosh`	hyperbolic cosine	COMPLEX.H, MATH.H
`div`	divides two integers	MATH.H
`exp`	exponential	COMPLEX.H, MATH.H
`fabs`	absolute value of a floating-point number	MATH.H
`floor`	rounds down	MATH.H
`fmod`	x modulo y	MATH.H
`frexp`	divides a number into a mantissa and exponent	MATH.H
`log`	natural logarithm	COMPLEX.H, MATH.H
`log10`	logarithm to base	COMPLEX.H, MATH.H
`pow`	x to the power y	COMPLEX.H, MATH.H
`rand`	random number generator	STDLIB.H
`sin`	sine	COMPLEX.H, MATH.H
`sinh`	hyperbolic sine	COMPLEX.H, MATH.H
`sqrt`	square root	COMPLEX.H, MATH.H
`srand`	random number initialiser	STDLIB.H
`tan`	tangent	COMPLEX.H, MATH.H
`tanh`	hyperbolic tangent	COMPLEX.H, MATH.H
Memory		
`calloc`	allocates main memory	ALLOC.H, STDLIB.H
`free`	frees an allocated block of memory	ALLOC.H, STDLIB.H
`malloc`	allocates main memory	ALLOC.H, STDLIB.H
`realloc`	reallocates main memory	ALLOC.H, STDLIB.H
Process control		
`abort`	abnormally terminates a program	PROCESS.H

Table 6.1 *(contd.)*

Function	*Description*	*Header file*
`atexit`	registers termination function	STDLIB.H
`exit`	terminate program	PROCESS.H
`raise`	send software signal to program	SIGNAL.H
`signal`	specifies signal handling actions	SIGNAL.H
Time and date		
`asctime`	converts date and time to ASCII	TIME.H
`clock`	processor time	TIME.H
`ctime`	converts date and time to a string	TIME.H
`difftime`	difference between two times	TIME.H
`gmtime`	converts date and time to Greenwich Mean Time	TIME.H
`localtime`	converts date and time to a structure	TIME.H
`mktime`	converts time to calendar format	TIME.H
`strftime`	formats time for output	TIME.H
`time`	gets the time of day	TIME.H
Variable arguments		
`va_arg`	access to variable argument list	STDARG.H
`va_end`	access to variable argument list	STDARG.H
`va_start`	access to variable argument list	STDARG.H

SETJMP.H	defines a **`typedef`** `jmp_buf` and declares the functions `setjmp()` and `longjmp()`
SIGNAL.H	signal and raise functions
STDARG.H	definitions for accessing parameters in functions that accept a variable number of arguments, such as `vprintf()`
STDDEF.H	defines several common data types and macros
STDIO.H	standard input and output
STDLIB.H	definitions for common types, variables and functions
STRING.H	string and memory manipulation
TIME.H	time and time conversion

6.19.2 C++ Header Files

ALGORITHM.H	general algorithms
BCD.H	**`class`** `bcd`
BITSET.H	vector of Booleans
CASSERT.H	C-style `assert()` macro support
CCTYPE.H	C-style character classification and conversion
CERRNO.H	C-style error handling
CFLOAT.H	C-style floating-point implementation-specific limits
CHECKS.H	diagnostics
CLIMITS.H	C-style implementation-specific limits
CLOCALE.H	C-style representations of cultural differences
CMATH.H	C-style mathematical routines
COMPLEX.H	`complex` math **`class`**

CONSTREA.H	console output support
CSETJMP.H	C-style stack unwinding
CSIGNAL.H	C-style signal handling
CSTDARG.H	C-style support for variable length function arguments
CSTDDEF.H	C-style library language support
CSTDIO.H	C-style I/O
CSTDLIB.H	C-style definitions for common types, variables and functions
CSTRING.H	C-style string support
CTIME.H	C-style data and time support
CWCHAR.H	C-style wide character support
CWTYPE.H	wide character classification
EXCEPT.H	exceptions
FSTREAM.H	file input and output stream classes
FUNCTION.H	function objects
IOMANIP.H	input and output stream manipulators
IOS.H	standard library basic stream I/O
IOSFWD.H	forward declarations for the I/O streams
IOSTREAM.H	standard input and output stream classes
ISTREAM.H	input streams
ITERATOR.H	iterators
LIST.H	doubly-linked `list` container class of type T
LOCALE.H	representing cultural differences
MAP.H	associative container classes `map` and `multimap`
MEMORY.H	allocators for classes
NEW.H	`new_handler` and `set_new_handler`
NUMERIC.H	generalised numerical operations
OSTREAM.H	output streams
REF.H	reference counting support for **`class`** `string`
REGEXP.H	regular expression searching
QUEUE.H	`queue` and `priority_queue` adaptor classes of type T
SET.H	associative container classes `set` and `multiset`
SSTREAM.H	I/O string streams
STACK.H	`stack` adaptor class of type T
STDEXCEPT.H	standard Library exception classes
STREAMBUF.H	stream buffers
STRING.H	`basic_string` and `string` classes
STRSTREA.H	stream classes for use with byte arrays in memory
TYPEINFO.H	run-time type identification
UTILITY.H	operators and data structure `pair`
VALARRAY.H	numerical vector class `valarray`
VECTOR.H	one-dimensional `vector` and `deque` container classes of type T

6.19.3 Additional Header Files

This section lists header files that are not specifically defined by ANSI C or C++.

ALLOC.H	memory management functions
BIOS.H	IBM PC ROM BIOS routines
CONIO.H	operating system console input and output routines
DIR.H	directory and path names
DIRECT.H	directory and path names

DIRENT.H	POSIX directory operations
DOS.H	MS-DOS and 80x86 specific calls
EXCPT.H	C exceptions support
FCNTL.H	specific constants for the `open()` function
GENERIC.H	macros for generic class declarations.
IO.H	low-level input and output routines.
SYS\LOCKING.H	definitions for `mode` parameter of `locking()` function
MALLOC.H	memory management
MEM.H	memory manipulation
PROCESS.H	process control
SEARCH.H	searching and sorting
SYS\STAT.H	opening and creating files
SYS\TIMEB.H	structure and function declarations for `ftime()`
SYS\TYPES.H	declares **`typedef`**s used with time functions
UTIME.H	declares `utimbuf` structure and `utime()` function
VALUES.H	constants for Unix compatibility.
VARARGS.H	variable number of arguments for Unix compatibility.

Note that for every C-style <X.H> header file there is an equivalent <cX.H> C++-style header file. The C routines in <X.H> are placed in to global scope whereas the <cX.H> header file places the equivalent routines in to the standard library namespace `std`. Refer to Chapter 19 for a discussion on namespaces.

6.20 Not in the Library

If you require a function that is not provided by your compiler, there are the following main options to consider:

1. Do the compiler's libraries provide a similar function?
2. Can a new function be developed which comprises several existing functions?
3. Is a function available in another language which can be called from C++ using external linkage?
4. Is there a suitable library function floating around on the Internet that I can appropriate?
5. Is there a vendor software library available that does the job, and is it one that I can afford?

or finally

6. Oh, no: I'm actually going to have write my own function!

6.21 Intersection of Two Two-Dimensional Line Segments

Before we bring this chapter to a close, let's examine the intersection of two straight line segments[4] in a two-dimensional space (Fig. 6.3). This problem is particularly interesting in geom-

4 Note the distinction between a line and a line segment. A line can, in theory, be of infinite length, whereas a line segment has finite length and has two distinct end-points. Therefore, any straight line segment can be extended indefinitely in a straight line; Euclid, 300 BC.

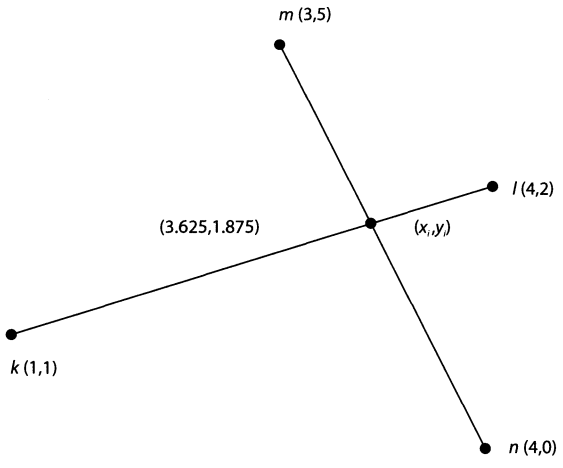


Fig. 6.3 Two intersecting straight line segments $l_1(k, l)$ and $l_2(m, n)$. The point of intersection is denoted by (x_i, y_i)

etry, sufficiently well covered in the literature, and a good realistic problem to illustrate more complex applications of functions.

At first glance, the intersection of two straight line segments appears to be a straightforward problem, but the more you examine the problem the more complicated it becomes. There are a number of alternative methods that can be used for determining whether two lines intersect or not, the first of which is the most straightforward and direct.

6.21.1 Parametric Lines

The procedure outlined in this section follows the method of Bowyer and Woodwark (1983, pp. 48–53).

It is possible to determine whether two line segments intersect or not from the parametric form of the two lines. Figure 6.4 shows two lines l_1 (k, l) and l_2 (m, n), whose end-points are (k, l) and (m, n) respectively. The parametric form of line l_1 is:

$$x = x_k + (x_l - x_k)s$$
$$y = y_k + (y_l - y_k)s$$

so that $s = 0$ at the start point, k, and $s = 1$ at the terminal point, l. Similarly, if we represent line l_2 in parametric form, using the parametric variable t, we have:

$$x = x_m + (x_n - x_m)t$$
$$y = y_m + (y_n - y_m)t$$

Eliminating x and y from the above parametric equations:

$$x_m + (x_n - x_m)t = x_k + (x_l - x_k)s$$
$$y_m + (y_n - y_m)t = y_k + (y_l - y_k)s$$

Hence:

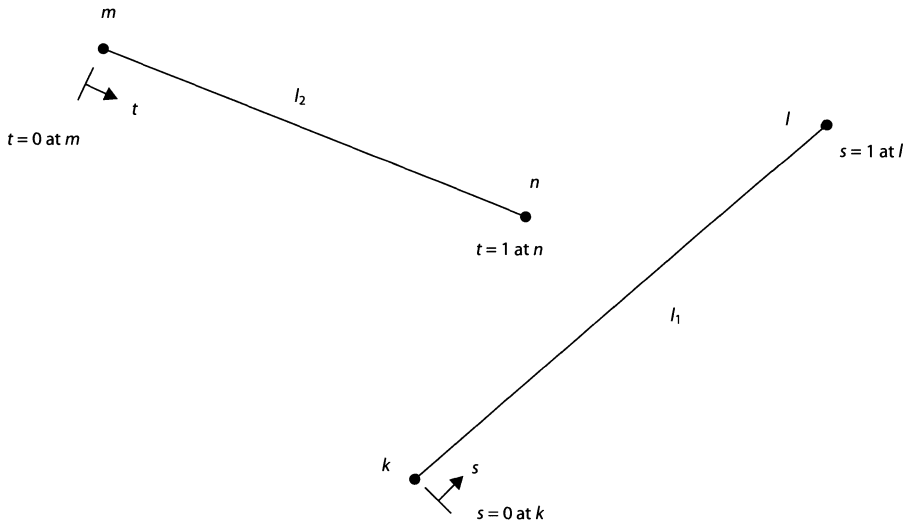


Fig. 6.4 Parametric representation of two lines.

$$s = \frac{(x_m - x_k) + (x_n - x_m)t}{(x_l - x_k)}, \quad t = \frac{(y_k - y_m) + (y_l - y_k)s}{(y_n - y_m)}$$

Solving these equations for s and t we find:

$$s = \frac{(x_n - x_m)(y_m - y_k) - (x_m - x_k)(y_n - y_m)}{(x_n - x_m)(y_l - y_k) - (x_l - x_k)(y_n - y_m)}$$

$$t = \frac{(x_l - x_k)(y_m - y_k) - (x_m - x_k)(y_l - y_k)}{(x_n - x_m)(y_l - y_k) - (x_l - x_k)(y_n - y_m)}$$

If both s and t are within the range $0 \leq s, t \leq 1$ then the lines intersect; else the lines do not intersect.

If the line segments do intersect then the (x_i, y_i) point of intersection can be found from the parametric equations of either of the lines. Using line l_1 (k, l):

$$x_i = x_k + (x_l - x_k)s$$
$$y_i = y_k + (y_l - y_k)s$$

An implementation of the above procedure is presented in program LI_LIB.CPP by means of a function called `ParametricIntersection()`:

```
// li_lib.cpp
// implementation of clock/anticlockwise
// and line-intersection functions
//...
bool ParametricIntersection (const double& xk,
                             const double& yk,
                             const double& xl,
                             const double& yl,
```

```
                                const double& xm,
                                const double& ym,
                                const double& xn,
                                const double& yn,
                                double& xi, double& yi)
{
// reduce terms
double xlk = xl - xk ;
double xnm = xn - xm ;
double xmk = xm - xk ;
double ylk = yl - yk ;
double ynm = yn - ym ;
double ymk = ym - yk ;

double denom = xnm*ylk - xlk*ynm ;

if (fabs (denom) < TOLERANCE)  // parallel lines (denom=0)
  return false ;
else
  {
  double s = (xnm*ymk - xmk*ynm) / denom ;
  double t = (xlk*ymk - ylk*xmk) / denom ;

  if (s<0 || t<0 || s>1 || t>1)  // no intersection
    return false ;
  else  // intersection
    {
    xi = xk + xlk*s ;  // use either s or t
    yi = yk + ylk*s ;  // pt. of intersection (use s)
    return true ;
    }
  }
}
```

The header file LI_LIB.H contains the function declaration of *ParametricIntersection()*:

```
// li_lib.h
// header file for li_lib.cpp
//...
bool ParametricIntersection (const double& xk, /*...*/) ;
```

and the file LI.CPP tests the implementation:

```
// li.cpp
// intersection of two two-dimensional line segments

#include <iostream>  // C++ I/O
#include "li_lib.h"  // line intersection

void main ()
```

```
{
// intersection of 2 lines:
double l1x1, l1y1, l1x2, l1y2, l2x1, l2y1, l2x2, l2y2 = 0.0 ;

// intersecting lines:
l1x1 = 1.0 ; l1y1 = 1.0 ; l1x2 = 4.0 ; l1y2 = 2.0 ;
l2x1 = 3.0 ; l2y1 = 5.0 ; l2x2 = 4.0 ; l2y2 = 0.0 ;
//...
double xi, yi = 0.0 ;

bool intersect = ParametricIntersection
                    (l1x1, l1y1, l1x2, l1y2,
                     l2x1, l2y1, l2x2, l2y2, xi, yi) ;
if (intersect)
  cout << "lines intersect at point: ("
       << xi << ", " << yi << ")" << endl ;
else
  cout << "lines do not intersect" << endl ;
//...
}
```

with output:

```
lines intersect at point: (3.625, 1.875)
```

The implementation of function `ParametricIntersection()` is fairly straightforward except for the test of parallel lines. The test case given in program LI.CPP is for the two lines shown in Fig. 6.3.

6.21.2 Bounding Boxes and Parametric Intersection

If the two line segments do not intersect then the function `ParametricIntersection()` is inefficient. The procedure of determining whether or not two lines intersect can be improved by introducing a *rejection* stage (Cormen *et al.*, 1990, pp. 886–90). The rejection stage involves testing whether the bounding boxes of the two lines intersect or not. The bounding box of a line segment is the smallest rectangle, with sides parallel to the x- and y-axes, that encompasses the line. Figure 6.5 illustrates the bounding boxes of lines l_1 and l_2. Rectangle $(\hat{k},\hat{l})$ is defined with lower left point $\hat{k}(\hat{x}_k, \hat{y}_k)$ and upper right point $\hat{l}(\hat{x}_l, \hat{y}_l)$, where $\hat{x}_k = \min(x_k, x_l)$, $\hat{y}_k = \min(y_k, y_l)$, $\hat{x}_l = \max(x_k, x_l)$ and $\hat{y}_l = \max(y_k, y_l)$. By experimenting with the two bounding boxes $(\hat{k},\hat{l})$ and $(\hat{m},\hat{n})$ it is found that intersection occurs only if the following test is logical-true:

$$(\hat{x}_l \geq \hat{x}_m)\&\&(\hat{x}_n \geq \hat{x}_k)\&\&(\hat{y}_l \geq \hat{y}_m)\&\&(\hat{y}_n \geq \hat{y}_k)$$

The implementation of the line bounding box function is:

```
bool LineBBoxIntersection (const double& xk,
                           const double& yk,
```

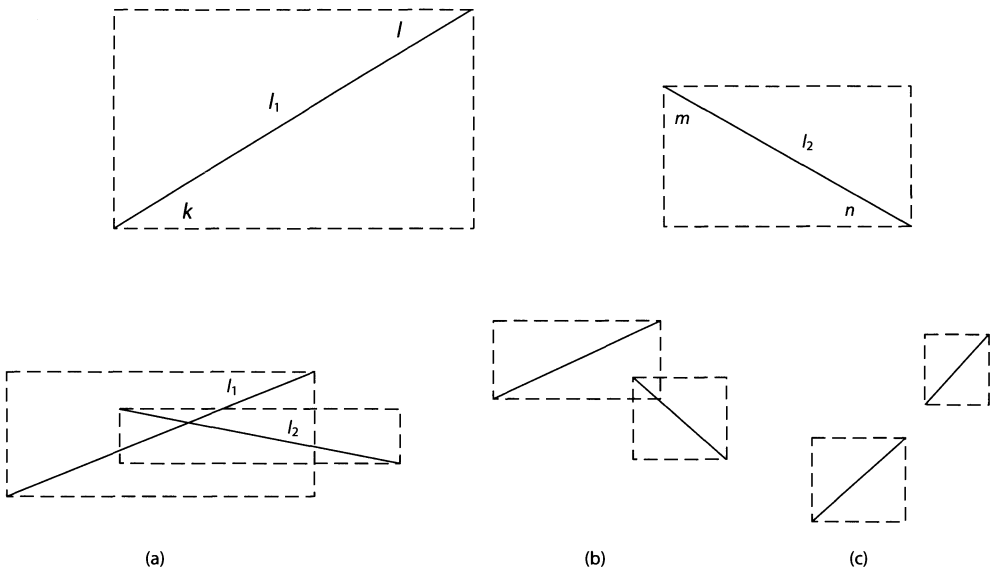


Fig. 6.5 Bounding boxes $(\hat{k}, \hat{l})$ and $(\hat{m}, \hat{n})$ of lines $l_1(k, l)$ and $l_2(m, n)$ respectively. (a) Intersecting lines, intersecting bounding boxes. (b) Non-intersecting lines, intersecting bounding boxes. (c) Non-intersecting collinear lines, non-intersecting bounding boxes.

```
                              const double& xl,
                              const double& yl,
                              const double& xm,
                              const double& ym,
                              const double& xn,
                              const double& yn)
{
// line l1(k,l)
double xk_hat = Min (xk, xl) ; double yk_hat = Min (yk, yl) ;
double xl_hat = Max (xk, xl) ; double yl_hat = Max (yk, yl) ;
// line l2(m,n)
double xm_hat = Min (xm, xn) ; double ym_hat = Min (ym, yn) ;
double xn_hat = Max (xm, xn) ; double yn_hat = Max (ym, yn) ;

if (xl_hat >= xm_hat && xn_hat >= xk_hat &&
    yl_hat >= ym_hat && yn_hat >= yk_hat)
  return true ;
return false ;
}
```

`LineBBoxIntersection()` allows us now to perform a quick rejection test when testing whether two lines intersect:

```
//...
bool ParBBoxIntersection (const double& xk, /*...*/)
  {
```

```
// rejection test (line bounding boxes do not intersect)
if (!LineBBoxIntersection (xk, yk, xl, yl, xm, ym, xn, yn))
  return false ;
//...
}
```

6.21.3 Clockwise or Anticlockwise?

Alternatively, it is possible to determine whether two lines intersect by considering the direction of rotation of three points (Cormen *et al.*, 1990, pp. 888–9; Sedgewick, 1988, pp. 349–51). Figure 6.6 illustrates two sets of points in a two-dimensional plane. On travelling from points p_0 to p_1 to p_2 in Fig. 6.6(a) we turn anticlockwise, whereas the set of points (p_0, p_1, p_2) in Fig. 6.6(b) make a clockwise rotation.

There are several different techniques that can be applied to the problem of whether three points make a left or right turn. The first method considers the magnitude of the angles between a fixed global axis (x-axis in the present case) and the line segments (p_0, p_1) and (p_0, p_2) (Fig. 6.7). The following function implements this method:

```
int CACPolar (double x0, double y0, double x1, double y1,
              double x2, double y2)
  {
  int cac = 0 ;

  // use point 0 as 'local' origin, compute polar angles
  double angle_01 = atan ((y1-y0)/(x1-x0)) ;
  double angle_02 = atan ((y2-y0)/(x2-x0)) ;

  if (angle_01 < angle_02)        // anti-clockwise
    cac = 1 ;
  else if (angle_01 > angle_02)   // clockwise
    cac = -1 ;
  else                            // collinear
    cac = 0 ;
  return cac ;
  }
```

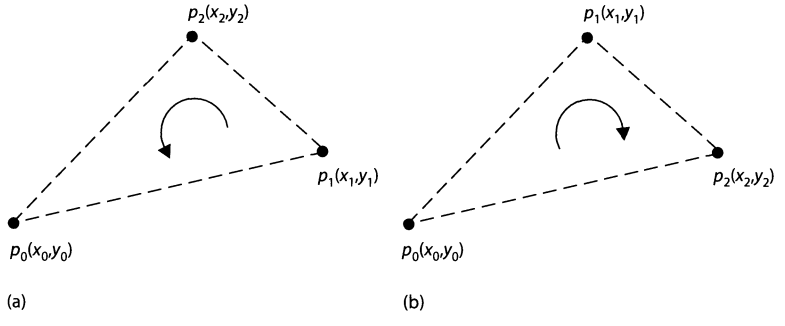


Fig. 6.6 Direction of rotation of three points p_0, p_1 and p_2. (a) Points (p_0, p_1, p_2) make an anticlockwise rotation. (b) Points (p_0, p_1, p_2) make a clockwise rotation.

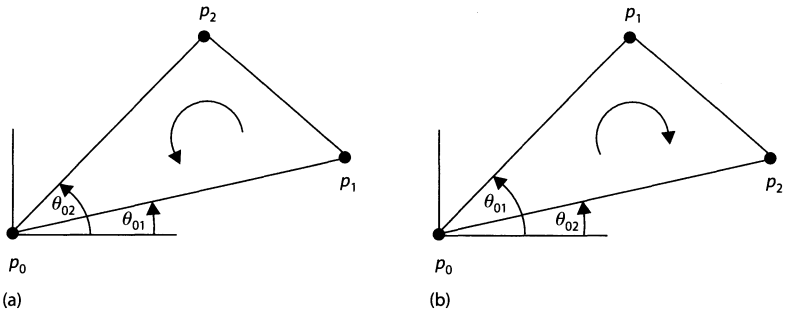


Fig. 6.7 Direction of rotation of three points p_0, p_1 and p_2 in terms of the magnitude of the angles between the line segments (p_0, p_1) and (p_0, p_2) and the x-axis. (a) Anticlockwise rotation. (b) Clockwise rotation.

Note that this function is far from perfect, since it does not account for vertical points $((x_1-x_0)=0$ or $(x_2-x_0)=0)$ or angles which are negative or greater than 90°. Also, the `atan()` function is a computationally expensive function to use for this simple function.

A more general method is to consider the determinant of the cross product of the two vectors[5] $\mathbf{v}_{01}(p_1-p_0)$ and $\mathbf{v}_{02}(p_2-p_0)$:

$$|\mathbf{v}_{01} \times \mathbf{v}_{02}| = |(p_1 - p_0) \times (p_2 - p_0)| = (x_1 - x_0)(y_2 - y_0) - (x_2 - x_0)(y_1 - y_0)$$

If the determinant of the cross product of $\mathbf{v}_{01}$ and $\mathbf{v}_{02}$ is positive then points (p_0, p_1, p_2) make an anticlockwise rotation; if the cross product is negative then (p_0, p_1, p_2) make a clockwise rotation, and if the cross product is zero then the points (p_0, p_1, p_2) are collinear:

```
int CACCrossProduct (const double& x0, const double& y0,
                     const double& x1, const double& y1,
                     const double& x2, const double& y2)
  {
  int cac = 0 ;

  double cp = (x1-x0)*(y2-y0) - (x2-x0)*(y1-y0) ;

  if (cp > 0)          // anticlockwise
    cac = 1 ;
  else if (cp < 0)     // clockwise
    cac = -1 ;
  else                 // collinear
    cac = 0 ;
  return cac ;
  }
```

The *norm*[6] of the cross product $\mathbf{v}_{01} \times \mathbf{v}_{02}$ is equal to the area of the parallelogram, $A(p_0, p_1, p_2, p_1+p_2)$, determined by $\mathbf{v}_{01}$ and $\mathbf{v}_{02}$ (Fig. 6.8). The area A is given by:

5 An excellent introduction to the properties of vectors and linear algebra can be found in Anton (1991).

6 The norm of a vector $\mathbf{v}(v_1, v_2, v_3)$ is the length or magnitude of $\mathbf{v}$ and is denoted by $||\mathbf{v}||$. It follows by Pythagoras' theorem that $||\mathbf{v}|| = (v_1^2, v_2^2, v_3^2)^{1/2}$.

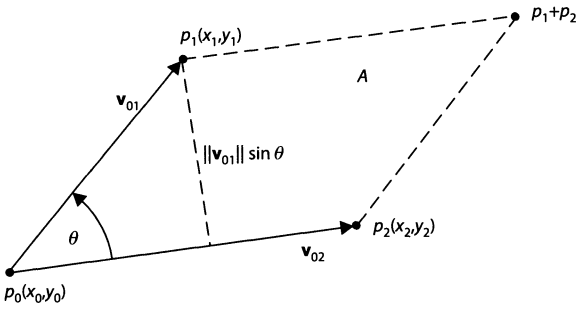


Fig. 6.8 Area of a parallelogram ($p_0, p_1, p_2, p_1 + p_2$).

$$A(p_0, p_1, p_2, p_1 + p_2) = \|\mathbf{v}_{01}\| \|\mathbf{v}_{02}\| \sin\theta = \|\mathbf{v}_{01} \times \mathbf{v}_{02}\|$$

The signed (+, –) area of the parallelogram indicates the direction of rotation of points (p_0, p_1, p_2).

Alternatively, we can examine the signed area of a triangle formed by the three vertices (p_0, p_1, p_2) (Fig. 6.9). The area of the triangle (i, j, k) is determined from the area of the three trapeziums (i, l, m, k), (k, m, n, j) and (i, l, n, j):

$$A(i, j, k) = \text{Area}(ilmk + kmnj - ilnj)$$

where

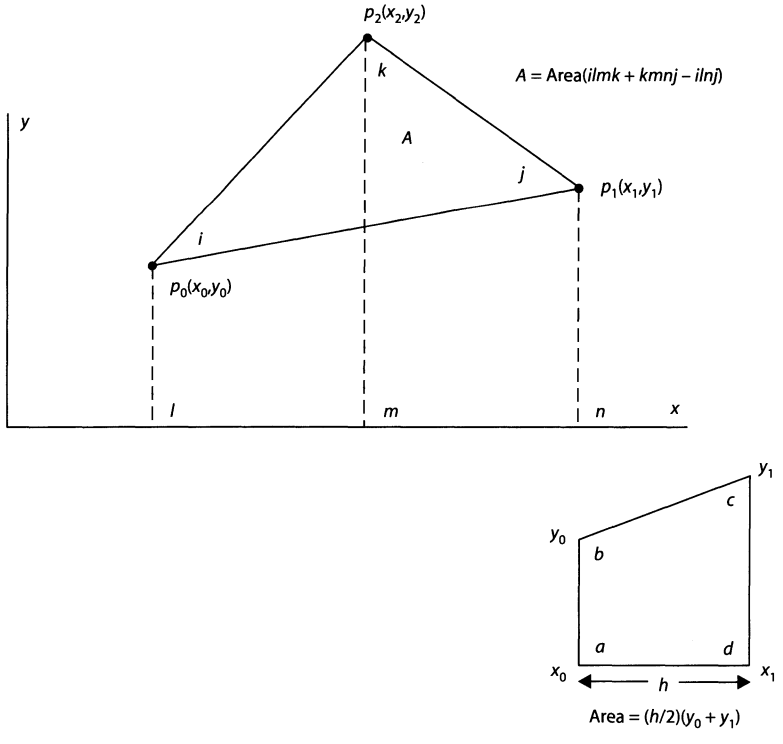


Fig. 6.9 Area of a triangle (p_0, p_1, p_2) and a trapezium (a, b, c, d).

$$A(i,l,m,k) = \frac{(x_2 - x_0)}{2}(y_0 + y_2); \quad A(k,m,n,j) = \frac{(x_1 - x_2)}{2}(y_1 + y_2);$$

$$A(i,l,n,j) = \frac{(x_1 - x_0)}{2}(y_0 + y_1)$$

Hence:

$$A(i,j,k) = \tfrac{1}{2}(x_0 y_1 + x_1 y_2 + x_2 y_0 - x_0 y_2 - x_1 y_0 - x_2 y_1)$$

The area, A, of triangle (p_0, p_1, p_2) is therefore the following determinant:

$$A(p_0, p_1, p_2) = \frac{1}{2}\begin{vmatrix}(x_j - x_i) & (x_k - x_i)\\ (y_j - y_i) & (y_k - x_i)\end{vmatrix} = \begin{vmatrix}1 & x_i & y_i\\ 1 & x_j & y_j\\ 1 & x_k & y_k\end{vmatrix}$$

If the determinant is positive then (p_0, p_1, p_2) make an anticlockwise rotation. If the determinant is negative then (p_0, p_1, p_2) make a clockwise rotation. If the determinant is zero then (p_0, p_1, p_2) are collinear. The function `CACSignedAreaOfTriangle()` given in LI_LIB.CPP implements the method outlined.

The final algorithm that we will consider is taken from Sedgewick (1988, pp. 349–51); refer also to Sedgewick (1992). This algorithm is particularly elegant and allows a simple implementation of a line intersection function. Figure 6.10 illustrates a set of three points (p_0, p_1, p_2) with p_0 representing the *local* origin. The gradients of lines $l_1(p_0, p_1)$ and $l_2(p_0, p_2)$ are denoted by $m_{01}(=dy_1/dx_1)$ and $m_{02}(dy_2/dx_2)$ respectively. If $m_{02} > m_{01}$ then the direction of rotation is anticlockwise; if $m_{01} < m_{02}$ the direction of rotation is clockwise; and if $m_{01} = m_{02}$ the three points are collinear. To prevent division by zero, if either dx_1 or dx_2 is zero the inequality $m_{02} > m_{01}$ is multiplied by $dx_1 dx_2$:

$$m_{02} > m_{01} \Rightarrow \frac{dy_2}{dx_2} > \frac{dy_1}{dx_1} \Rightarrow dx_1 dy_2 > dx_2 dy_1$$

The function implementation is given below:

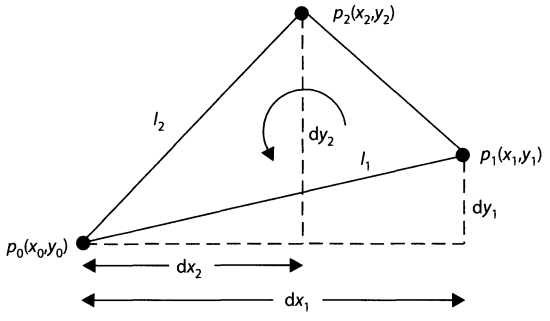


Fig. 6.10 Three points (p_0, p_1, p_2). The gradients of lines l_1 and l_2 are m_{01} $(= dy_1/dx_1)$ and m_{02} $(= dy_2/dx_2)$ respectively. If $m_{02} > m_{01}$ then (p_0, p_1, p_2) make an anticlockwise rotation. If $m_{01} > m_{02}$ then (p_0, p_1, p_2) make a clockwise rotation. If $m_{01} = m_{02}$ then (p_0, p_1, p_2) are collinear.

```
int CAC (const double& x0, const double& y0,
         const double& x1, const double& y1,
         const double& x2, const double& y2)
  {
  int cac = 0 ;  // return var. indicating direction

  // use point 0 as 'local' origin
  double dx1 = x1 - x0 ; double dy1 = y1 - y0 ;
  double dx2 = x2 - x0 ; double dy2 = y2 - y0 ;
  // eliminate infinite gradients by multiplying gradients
  // between points 0-1 and 0-2 by (dx1*dx2)
  if (dx1*dy2 > dx2*dy1)  // anti-clockwise (m02>m01)
    cac = 1 ;

  if (dx1*dy2 < dx2*dy1)  // clockwise (m02<m01)
    cac = -1 ;

  if ((dx1*dy2 - dx2*dy1)< TOLERANCE)  // same gradients
    {
    if ((dx1*dx2 < 0) || (dy1*dy2 < 0))
      cac = -1 ;
    else if ((dx1*dx1 + dy1*dy1) >= (dx2*dx2 + dy2*dy2))
      cac = 0 ;
    else
      cac = 1 ;
    }
  return cac ;
  }
```

Rather than designing the *CAC()* function to return either logical-true or false to indicate the direction of rotation, the function returns one of three values: –1, 1 or 0. *CAC()* returns 1 or –1 if the three points make an anticlockwise or clockwise order, respectively. If the points are collinear then *CAC()* returns –1 if p_0 is between p_1 and p_2, 1 if p_1 is between p_0 and p_2 and 0 if p_2 is between p_0 and p_1. Adopting this convention allows a simple implementation of the line intersection function.

So how do we apply the notion of rotating either clockwise or anticlockwise from one point to a second to a third to the intersection of two lines? Figure 6.11 illustrates two sets of lines. Figure 6.11(a) shows two lines, l_1 and l_2, intersecting, whereas l_1 and l_2 in Fig. 6.11(b) do not intersect. If l_1 and l_2 intersect then the direction of rotation reverses when we consider the three points $(l_1.p_1, l_1.p_2, l_2.p_1)$ and $(l_1.p_1, l_1.p_2, l_2.p_2)$ separately, using l_1 as a reference line. The exception to the rule is shown in Fig. 6.12, with reference line l_1. In this case the directions of rotation are reversed, but the two lines do not intersect. However, if we also use l_2 as a reference line then the points $(l_2.p_1, l_2.p_2, l_1.p_1)$ and $(l_2.p_1, l_2.p_2, l_1.p_2)$ are in the same order. Therefore, the intersection algorithm should compare the directions of rotation with lines l_1 and l_2 as reference lines. The implementation of the intersection function is:

```
bool LineIntersection (const double& l1x1,
                       const double& l1y1,
    const double& l1x2, const double& l1y2,
    const double& l2x1, const double& l2y1,
    const double& l2x2, const double& l2y2)
```

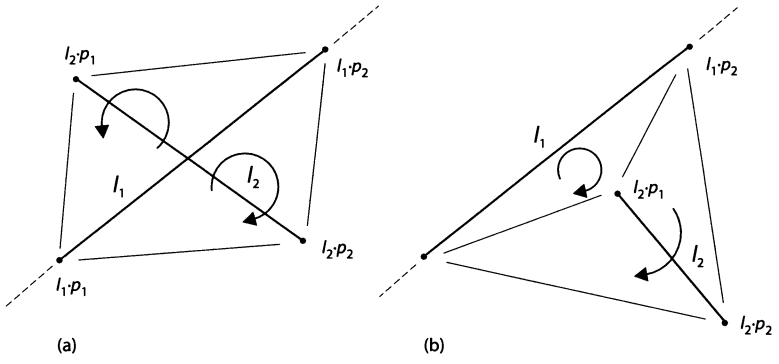


Fig. 6.11 (a) Intersecting lines. (b) Non-intersecting lines.

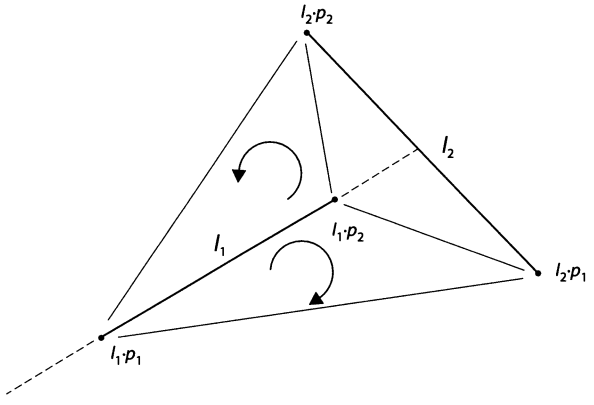


Fig. 6.12 Non-intersecting lines.

```
{
// intersection
if ( ((CAC (l1x1, l1y1, l1x2, l1y2, l2x1, l2y1) *
   CAC (l1x1, l1y1, l1x2, l1y2, l2x2, l2y2)) <= 0) &&
     ((CAC (l2x1, l2y1, l2x2, l2y2, l1x1, l1y1) *
   CAC (l2x1, l2y1, l2x2, l2y2, l1x2, l1y2)) <= 0) )
  return true ;
// no intersection
else
  return false ;
}
```

Although the functions `CAC()` and `LineIntersection()` are relatively simple, they are nevertheless very powerful and able to deal with a variety of line intersections and non-intersections; see Fig. 6.13.

The declarations and definitions of all the above line-intersection functions are to be found in LI_LIB.H and LI_LIB.CPP respectively. LI_LIB.H can now be conveniently included in additional geometric libraries or application programs.

We shall return to the problem of line segment intersection in Chapter 9 when we introduce the C++ **`class`** and member functions.

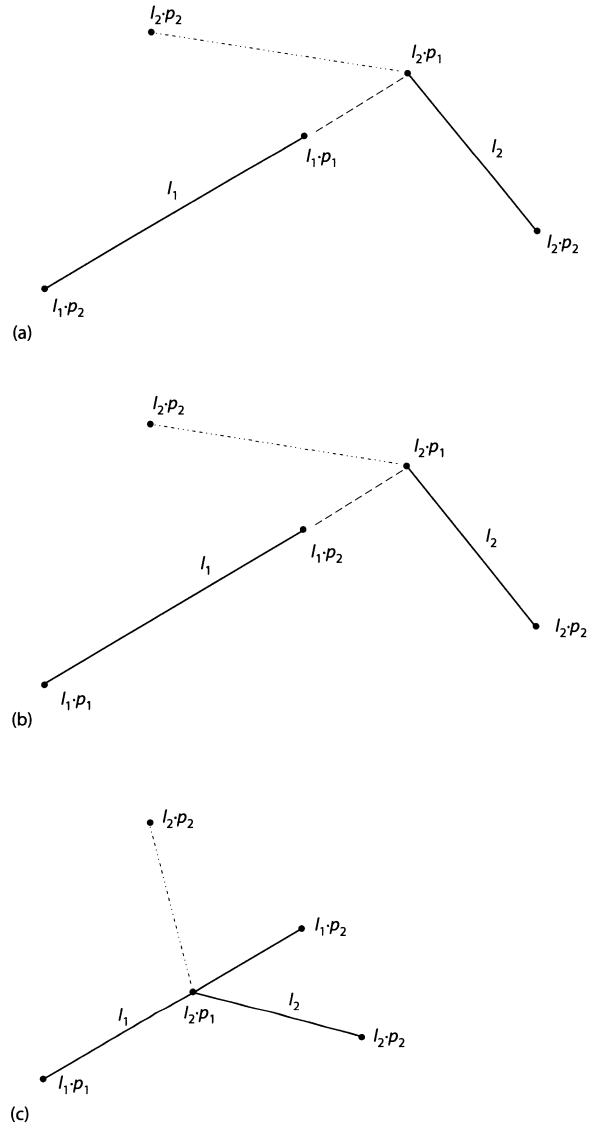


Fig. 6.13 Line intersections and non-intersections.

6.22 Summary

We have dealt with the key elements of a function in this chapter. Functions are a cornerstone of programming in that they help a programmer to conceptualise, break down and organise a large program. Functions are reusable, have their own local scope and hide their implementation details from a function-caller.

The function definition is the body of a function, or a function's implementation. The function declaration or prototype is the signature of the function and indicates the function's name, return type specifier and number and type of arguments.

Functions can be overloaded. Function overloading is an example of compile-time polymorphism in which a single function name can have multiple different operations. We shall continue the discussion of function overloading and, in particular, run-time polymorphism, when we discuss inheritance and **virtual** functions in Chapter 15.

Functions can have default arguments, so that functions can be called without passing all of the arguments. Default arguments are particularly useful for functions with large argument lists. We have also seen that functions can be **inline**. Inline functions execute faster and are generally restricted to small functions that are frequently called.

We have seen that arguments can be passed to a function either by value or by reference. Passing by value makes a copy of the argument, whereas passing by reference does not make a copy but accesses the caller-argument. Reference arguments can be passed as **const** to ensure that the function-caller argument is not altered during the function call.

C++ compilers are shipped with extensive function and **class** libraries, some of which are ANSI standard-compatible and some of which are not. For instance, the Borland C++ compiler supplies it own Container and ObjectWindows classes, whereas the Microsoft Visual C++ compiler supports the Microsoft Foundation Classes (MFC) library. Spend some time getting to know the libraries – you could save yourself a great deal of blood, sweat and tears by not developing a routine that already exists.

Exercises

6.1 Within the *main()* function add together the global and *main()* g variables for the following program:

```
int g = 1 ;

void main ()
  {
  int g = 2 ;
  //...
  }
```

6.2 Develop two functions that are passed a character and return an integer value indicating whether the character is lowercase or uppercase. The signature of the two functions should be of the following form:

```
int IsLowerCase (char c) ;
int IsUpperCase (char c) ;
```

Also, develop two functions which are passed a character and which will convert the character to lowercase and uppercase and have the following signatures:

```
char ToLowerCase (char c) ;
char ToUpperCase (char c) ;
```

6.3 Write two functions which convert from degrees to radians and vice versa. Place the function declarations in a header file MY_MATH.H and the function implementations in MY_MATH.CPP. Test your functions by include-ing MY_MATH.H in a test program.

6.4 Develop three **inline** overloaded `Abs()` functions for types **int**, **float** and **double** which return the absolute value of a number.

6.5 Implement a function called `FractionalPart()` which returns the fractional part of a floating-point number. The return value should always be positive. In your implementation of `FractionalPart()` make use of the C++ library function `fmod()`:

```
double fmod (double x, double y) ;  // MATH.H
```

which returns x modulo y and is the same sign as x.

6.6 Develop a function that models the following damped, rectified sine curve:

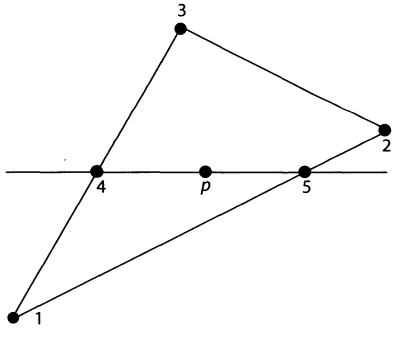
$$y(x) = A|\sin(\omega x + \theta_0)|e^{-kx}$$

where A is the initial amplitude, ω is the angular frequency, θ_0 is the phase angle and k is the damping constant.

6.7 Develop a function which performs interpolation using the Gouraud shading algorithm for a two-dimensional (x,y; z=0) straight-sided triangle. The signature of the function should be of the form:

```
int GouraudInterpolation (double i1, double i2, double i3,
                          double x1, double y1,
                          double x2, double y2,
                          double x3, double y3,
                          double px, double py,
                          double& ip) ;
```

where i1, i2 and i3 are the intensities at the triangle vertices 1, 2 and 3 whose (x,y) coordinates are denoted by (x1,y1), (x2,y2) and (x3,y3) respectively, and (px,py) denotes a point, p, which is interior to the triangle with interpolated intensity ip:



The intensity at point 4 is linearly interpolated from the intensities at vertices 1 and 3 and the intensity at point 5 is linearly interpolated from the intensities at vertices 1 and 2. The intensity at point p is linearly interpolated from the intensities at points 4 and 5. The Gouraud shading algorithm can be found in several computer graphics textbooks, and a typical reference is Hearn and Baker (1994, pp. 523–5).

Chapter 7

Arrays

Arrays are aggregates of a given fundamental or user-defined data type. An array element is accessed by an index, rather than by name, within a consecutive list of elements. Arrays are the basic tool for handling large amounts of similar data.

The elements of an array can be of any valid type, such as int, double or a user-defined type. We shall see that an array of characters defines a string. Arrays can be of any dimension, although one- and two-dimensional arrays are the most frequently used when programming.

C++ arrays are frequently referred to as static because their size is set at the time of compilation and remains fixed throughout the lifetime of a program. Irrespective of the dimension of an array, the elements of an array are stored contiguously in memory, and this means that accessing the elements of an array is random; i.e. it takes the same time to access the last element as to access the first element.

Arrays of variable size are possible and frequently used when programming in C++, but they require the use of pointers, which will be discussed in Chapter 12.

7.1 An Array

Let's begin our introduction to arrays with a simple program:

```
// array.cpp
// illustrates a simple one-dimensional array
#include <iostream>  // C++ I/O

void main ()
  {
  // define a 1D array of 5 elements of type int
  int array[5] ;

  // get numbers
  for (int i=0; i<5; i++)
```


G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

```
    {
    cout << "enter an integer number: " ;
    cin  >> array[i] ;
    }
  cout << endl ;
  // output numbers
  for (int j=0; j<5; j++)
    {
    cout << "number " << (j+1) << ": " << array[j] << endl ;
    }
}
```

Here's some user interaction:

```
enter an integer number: 1
enter an integer number: 12
enter an integer number: 23
enter an integer number: 34
enter an integer number: 45

number 1: 1
number 2: 12
number 3: 23
number 4: 34
number 5: 45
```

So, what is an array? An array is a group of values or objects of *identical* fundamental or user-defined data types in which each *element* of the array can be referenced as a single unit. Note that all the elements of an array are of the same type. In the program above, an array, `array`, is defined of type **`int`** which has five elements:

```
int array[5] ;
```

A **`for`**-loop is used to prompt the user to enter five integer numbers. Each number is assigned to be a different, unique element of the array:

```
//...
cin  >> array[i] ;
```

When all of the numbers (five in total) have been entered the program outputs the entered numbers to the screen:

```
//...
cout << "number " << (j+1) << ": " << array[j] << endl ;
```

The general syntax of an array is:

```
type_specifier array_name[number_of_elements] ;
```

`type_specifier` is the type of each of the elements of the array (e.g. a fundamental data type, such as **`int`** or **`double`**, or a user-defined type, such as `Complex` or `Triangle`).

The name of the array is denoted by `array_name`, and the usual syntax of defining identifiers also applies to the naming of arrays. The number of elements in an array can be as small or as large (within limits) as the programmer specifies and is denoted by `number_of_elements` within the square brackets `[` and `]`. The array size must be supplied since there is no default array size in C++. Defining an uninitialised array without a size is illegal in C++.

7.2 The Size of an Array

The size of the array defined above is fixed at compile time to be equal to five elements of type **int** and is thus referred to as a *fixed-size* or *static* array. A common confusion when defining an array is to use a variable array size. The following program illustrates an array defined as above, but now the array size is determined at run-time by the user:

```
// static.cpp
// illustrates that a C++ array must be defined constant
#include <iostream>  // C++ I/O

void main ()
  {
  int array_size ;

  cout << "enter an array size: " ;
  cin  >> array_size ;

  // define a 1D array of a size determined by a user
  int array[array_size] ; // error: const. expression
                          // required
  //...
  }
```

This program will not compile because the array size must be a constant and defined at compilation, since the compiler has to determine how much memory to allocate for the array. In Chapter 12, we shall examine *dynamic* arrays, whose size can vary at run-time, but for now we shall restrict ourselves to static arrays.

Rather than defining the array as:

```
int array[5] ;
```

we could define the array size as a constant identifier:

```
int const ARRAY_SIZE = 5 ;  // array size
//...
int array[ARRAY_SIZE] ;
```

which increases the generality of a program.

In conclusion, once the size of an array has been set and the program compiled, the array size is fixed for the entire lifetime of the program.

7.3 The `sizeof` Operator

Remember the **sizeof** operator introduced in Chapter 4? We noted that the **sizeof** operator is a unary operator and thus has one operand. For an identifier the syntax is:

```
sizeof operand
```

and for a type specifier:

```
sizeof (type_specifier)
```

A type specifier must be enclosed in parentheses, but the parentheses are optional for an identifier. The **sizeof** operator can also be used to determine the size, in bytes, of an array. The following program, A_SIZE.CPP, illustrates determining the size of several arrays:

```
// a_size.cpp
// illustrates the sizeof operator for determining
// the number of elements and size of an array
#include <iostream>  // C++ I/O

void main ()
  {
  char   c_array[10] ;
  int    i_array[10] ;
  float  f_array[10] ;
  double d_array[10] ;

  cout << "sizeof c_array[]: " << sizeof c_array << endl ;
  cout << "sizeof i_array[]: " << sizeof i_array << endl ;
  cout << "sizeof f_array[]: " << sizeof f_array << endl ;
  cout << "sizeof d_array[]: " << sizeof d_array << endl ;

  double big_array[50][50] ;

  cout << endl << "sizeof big_array[][]: "
       << sizeof big_array << endl ;

  // no. of elem.= (total array size) / (size of single
  //                element)
  cout << "number of elements in big_array[][]: "
       << sizeof big_array / sizeof big_array[0][0] ;
  }
```

and generates the output:

```
sizeof c_array[]: 10
sizeof i_array[]: 40
sizeof f_array[]: 40
sizeof d_array[]: 80
sizeof big_array[][]: 20000
```

```
number of elements in big_array[][]: 2500
```

which outputs sizes of 10, 40, 40 and 80 bytes for four arrays with 10 elements of type **char**, **int**, **float** and **double** respectively. Also, the output displays the size of a large two-dimensional array (50 × 50), `big_array`, which eats up 20 kbyte of memory. The program outputs the total number of elements in the array by evaluating the expression:

```
sizeof big_array / sizeof big_array[0][0] ;
```

which divides the total storage required for the array by the storage requirements of a single element. In the present case the single element chosen was the first element, although any element could be chosen since all the elements of an array are of the same type. This technique of determining the number of elements in an array is frequently employed, since it is more general than:

```
cout << "number of elements in big_array[][]: 2500 " ;
```

For instance, if we now changed the definition of `big_array` and recompiled the program the correct number of elements would still be displayed if we use the **sizeof** operator.

7.4 Array Indexing

7.4.1 First Element

You may have noticed from the above program (ARRAY.CPP) that both **for**-loops begin indexing the array of numbers entered from zero and increase incrementally from `i=0` to `i=4`, i.e. `array[0]`, `array[1]`, ..., `array[4]`. C++ arrays start indexing from element 0. The majority of other programming languages start array indexing from element 1. Also, for the majority of people, it is intuitive to begin counting with 1, so array indexing from 0 is a common source of confusion with arrays in C++. So, let's have a note:

C++ arrays start indexing from element 0

Figure 7.1 schematically (as a block of ice-cream) illustrates the 'numbers' array of the ARRAY.CPP program stored consecutively somewhere in memory. Element 0 of the array is the value 1, element 2 is value 12 and so on. The array elements are stored in memory adjacent to one another. Since each element of the array is of type **int** they occupy four bytes of memory for a 32-bit system. There are five elements in the array, so the total memory allocated to the array is twenty bytes.

7.4.2 Setting and Getting an Element

In ARRAY.CPP the numbers entered by a user are inserted into `array` by the statement:

```
cin >> array[i] ;
```

which involves the `cin` input stream object and the extraction operator (>>). The elements of `array` are extracted by the statement:

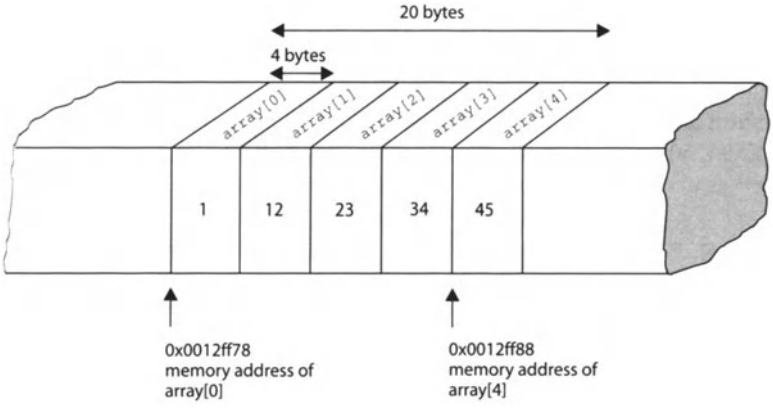


Fig. 7.1 An array somewhere in memory.

```
cout << "number " << (j+1) << ": " << array[j] << endl ;
```

Thus an element of an array is indexed by the array name and the element index number.

7.4.3 Last Element

C++ does not have array bound checking. Let's say that again:

There is no bound checking in C++

When accessing the elements of an array, C++ assumes that you know what you are doing. Let's look at a program which indexes elements outside of the memory allocated for an array:

```
// indexing.cpp
// illustrates indexing an array out of bounds
#include <iostream>  // C++ I/O

void main ()
  {
  // define a 1D array of 5 elements of type int
  int array[5] ;

  // get numbers
  for (int i=0; i<5; i++)
    {
    cout << "enter an integer number: " ;
    cin  >> array[i] ;
    }
  cout << endl ;
  // out of bounds
  for (int j=0; j<10; j++)
    {
    cout << "number " << (j+1) << ": " << array[j] << endl ;
    }
  }
```

and some user interaction:

```
enter an integer number: 1
enter an integer number: 2
enter an integer number: 3
enter an integer number: 4
enter an integer number: 5

number 1: 1
number 2: 2
number 3: 3
number 4: 4
number 5: 5
number 6: 1245112
number 7: 4302062
number 8: 1
number 9: 8200656
number 10: 8201524
```

The output illustrates that although we only defined an array of five elements, we are able to index elements `array[5]` to `array[9]`, which are filled with garbage. The compiler does not guarantee that out-of-bounds array elements are initialised to zero or any other value. They are undefined except for the index beyond the high end of the array.

Stepping out of bounds isn't as common as you might at first expect, but bound checking becomes more important when dealing with dynamic arrays, which we will see later.

7.5 Initialisation of One-Dimensional Arrays

Just as we can initialise a variable when it is defined, we can similarly initialise arrays when they are defined:

```
// initial.cpp
// illustrates initialising arrays
#include <iostream>   // C++ I/O
#include <iomanip>    // setprecision(), ...

void main ()
  {
  const int NUM_BOOKS = 10 ;
  float     computer_book_prices[NUM_BOOKS] =
    { 18.99, 27.45, 15.95, 26.95, 21.95,
      24.95, 29.95, 30.50, 16.95, 26.95 } ;

  float total_price (0.0) ;
  for (int i=0; i<NUM_BOOKS; i++)
    {
    total_price += computer_book_prices[i] ;
    }
```

```
    cout << setiosflags (ios::fixed)
         << setprecision (2)
         << "total price: " << total_price << endl
         << "average price of my 10 favourite computer books: "
         << total_price / NUM_BOOKS ;
}
```

which displays to the screen the total and average prices of ten books.

The above program defines an array, `computer_book_prices`, whose elements are the retail prices (£) of my 10 favourite computer books. The individual elements of the array are separated by the comma operator. This list of initial values for the array elements is referred to as the *initialiser list*. The initialiser list is enclosed between the opening, `{`, and closing, `}`, braces, not forgetting the semicolon indicating the end of the statement. The general syntax for initialising an array is:

```
type_specifier array_name[num_elements] { elem0, elem1, ...
                                          elemn } ;
```

If too many initialisers are supplied then a compilation error will be issued, whereas if too few initialisers are supplied then 0 is assigned to the remaining elements:

```
char ca[3] = { 'a', 'b', 'c', 'd' } ; // error: too many
                                      // initialisers
int ia[4] = { 1, 2 } ; // OK: ia={1, 2, 0, 0}
```

It is worth noting that in the initialisation of the one-dimensional array in the program INITIAL.CPP it is not necessary to define the array size explicitly as 10. The following is equally valid and legal:

```
float computer_book_prices[] =
  { 18.99, 27.45, 15.95, 26.95, 21.95,
    24.95, 29.95, 30.50, 16.95, 26.95 } ;
```

The compiler is able to work out for itself how many elements are in the array simply by counting the number of initialisers in the initialiser list.

7.6 Two-Dimensional Arrays

Let's take a look at a two-dimensional array of integers:

```
// array_2d.cpp
// illustrates two-dimensional arrays
#include <iostream>  // C++ I/O

void main ()
  {
  const int ROWS = 2 ;  // no. of rows and columns
  const int COLS = 5 ;
```

```
// define & initialise a 2D array
int array_2d[ROWS][COLS] = { {0, 1, 2, 3, 4},
                             {5, 6, 7, 8, 9} };
// O/P nested for loop
for (int i=0; i<ROWS; i++)
  {
  for (int j=0; j<COLS; j++)
    {
    cout << array_2d[i][j] << " " ;
    }
  cout << endl ;  // newline after each row
  }
}
```

which generates the output:

```
0 1 2 3 4
5 6 7 8 9
```

A two-dimensional array is defined by the following general syntax:

```
type_specifier array_name[rows][columns] ;
```

and allocates sufficient memory for (rows by columns) values of type `type_specifier`. Note that indexing the elements of an array is (row by column) rather than (column by row).

Figure 7.2 illustrates the array `array_2d` in the above program. To index the integer 7 we access row 1 and column 2; namely `array_2d[1][2]`.

Pay particular attention to the use of the nested **`for`**-loop in order to access the elements of a two-dimensional array correctly. Note also the placement of the output stream object `cout` and the new line manipulator `endl` to output the array as a (2×5) array rather than displaying all of the array elements on a single line.

Visualising two-dimensional arrays can be difficult at first, but provided you remember the (row by column) rule then, given time, the penny should drop!

Although it is convenient for us to visualise two-dimensional arrays as a (row by column) table of values, the values are in fact stored linearly in memory, as shown in Fig. 7.3. The compiler generates an equivalent pointer referencing system to access the array elements. The

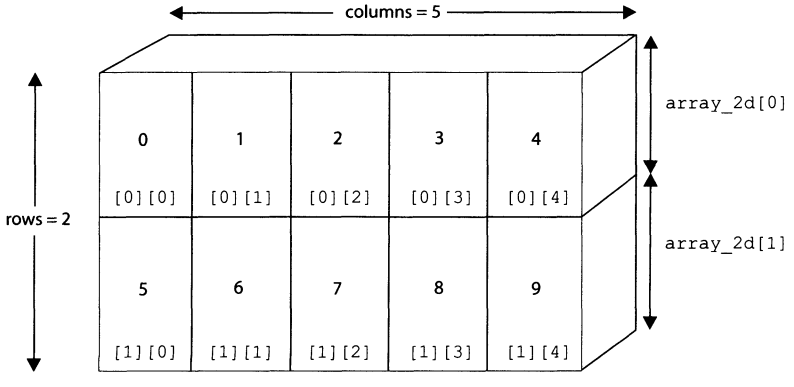


Fig. 7.2 A two-dimensional array of integers.

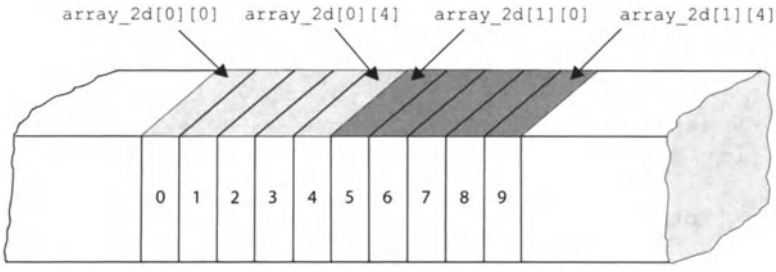


Fig. 7.3 Memory representation of a two-dimensional array.

array subscript operator [] is used for user clarity. When pointers are discussed later, the equivalence between array indexing by means of the subscript operator, [], and by a pointer, *, will become clearer.

7.7 Initialisation of Two-Dimensional Arrays

The program above, ARRAY2D.CPP, also demonstrates initialisation of two-dimensional arrays:

```
int array_2d[ROWS][COLS] = { {0, 1, 2, 3, 4},
                             {5, 6, 7, 8, 9} } ;
```

Each element of each row of the array is separated by the comma operator, just as in a one-dimensional array. The difference between initialisation of one- and two-dimensional arrays is the use of the inner braces to indicate explicitly each row of the array. Imagine if the inner braces were not present:

```
int array_2d[ROWS][COLS] = { 0, 1, 2, 3, 4,
                             5, 6, 7, 8, 9 } ;
```

Clearly, the compiler would view the right-hand side of the expression as the initialisation of a one-dimensional array. However, if you have a clever compiler, such as the Borland C++ (version 5.x) compiler, such an array definition and initialisation will compile without an error message, but will generate a warning message of the form 'array initialisation is partially bracketed'.

As with one-dimensional arrays it is possible to neglect the row dimension of an array – the compiler is clever enough to figure out what you mean. The contrary, neglecting the column but including the row dimension, is not true:

```
int array_2d[][COLS] = { {0, 1, 2, 3, 4},
                         {5, 6, 7, 8, 9} } ;   // ok

int array_2d[ROWS][] = { {0, 1, 2, 3, 4},
                         {5, 6, 7, 8, 9} } ;   // NOT ok

int array_2d[][]     = { {0, 1, 2, 3, 4},
                         {5, 6, 7, 8, 9} } ;   // definitely
                                               // NOT ok
```

However, always try to indicate your intentions as explicitly as possible (for programming languages anyway) and define both the row and column dimensions.

7.8 How are Two-Dimensional Arrays Used?

Two-dimensional arrays have several applications. One of the more well-known applications is the solution of systems of simultaneous linear equations. A system of m simultaneous linear equations in n unknowns $x_1, x_2, ..., x_n$ is of the form:

$$\begin{aligned} a_{11} + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21} + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1} + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{aligned}$$

where the coefficients a_{ij} and b_i are known values. This system of linear equations can be written as a single vector equation:

$$\mathbf{Ax} = \mathbf{b}$$

where the coefficient matrix $=[a_{ij}]$ is an $m \times n$ matrix. The column vector **b** represents the known right-hand side values and **x** is the solution column vector:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2}n \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix}, \quad \mathbf{b} = \begin{Bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{Bmatrix}$$

For example, the system of equations

$$\begin{aligned} 2x + y + z &= 1 \\ x + 4y + 6z &= 2 \\ 3x + 8y + z &= 1 \end{aligned}$$

can be written in matrix form as:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 4 & 6 \\ 3 & 8 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{Bmatrix} x \\ y \\ z \end{Bmatrix}, \quad \mathbf{b} = \begin{Bmatrix} 1 \\ 2 \\ 1 \end{Bmatrix}$$

The method of solution of such a system of equations can be either by Cramer's rule for small systems, direct methods, such as Gaussian elimination, or iterative methods, such as Gauss–Siedel iteration. Chapter 12 presents an implementation of the Gaussian elimination method.

7.9 Three- and Higher-Dimensional Arrays

Arrays of dimensions higher than two are possible in C++ and are of the general form:

```
type_specifier array_name[dimension1]...[dimensionn] ;
```

A three-dimensional array is defined in the program below:

```
// array_3d.cpp
// illustrates three-dimensional arrays
#include <iostream>  // C++ I/O

void main ()
  {
  const int X_SIZE = 2 ;  // no. of rows, columns, depth
  const int Y_SIZE = 2 ;
  const int Z_SIZE = 2 ;

  // define & initialise a 3D array
  int array_3d[X_SIZE][Y_SIZE][Z_SIZE] =
          { {{10, 11}, {12, 13}},
            {{20, 21}, {22, 23}} } ;

  // O/P nested loop
  for (int i=0; i<X_SIZE; i++)
    {
    for (int j=0; j<Y_SIZE; j++)
      {
      for (int k=0; k<Z_SIZE; k++)
        {
        cout << array_3d[i][j][k] << " " ;
        }
      cout << endl ;
      }
    cout << endl ;
    }
  }
```

which defines and initialises a three-dimensional array of 8 elements of type **int**. The output of the program is:

```
10 11
12 13

20 21
22 23
```

which displays the 'front' layer of elements first and then the 'back' layer. Clearly, since the display screen is two-dimensional, there is a problem displaying three-dimensional arrays! To assist our visualisation of the above array, `array_3d`, Fig. 7.4 illustrates an exploded view of the

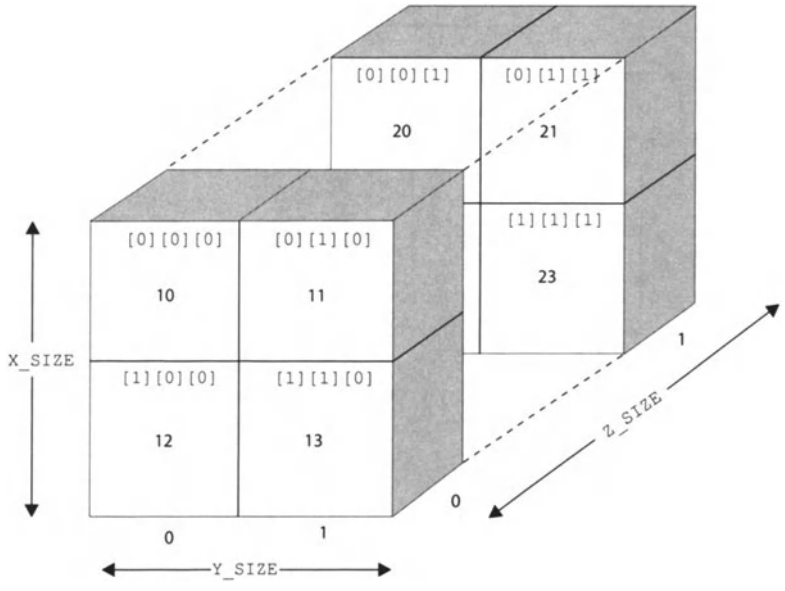


Fig. 7.4 An exploded three-dimensional array.

three-dimensional array. Note the addition of the third **for**-loop to access the third-dimensional elements of the array and the addition of braces in the array initialiser list to indicate dimensions 1, 2 and 3.

Three-dimensional and higher order arrays are more difficult to comprehend than are two-dimensional arrays, and are consequently rarely used in practice.

7.10 Strings

Strings are one-dimensional arrays of type **char**. Strings, like other types, can be either constant or variable. Let's start by examining variable strings.

7.10.1 Variable Strings

Let's maintain our usual format of introducing a new topic with the aid of a program:

```
// v_string.cpp
// illustrates variable strings
#include <iostream>  // C++ I/O

void main ()
  {
  // string variable: array of char
  char name[10] ;

  cout << "enter your forename: " ;
  cin  >> name ;
```

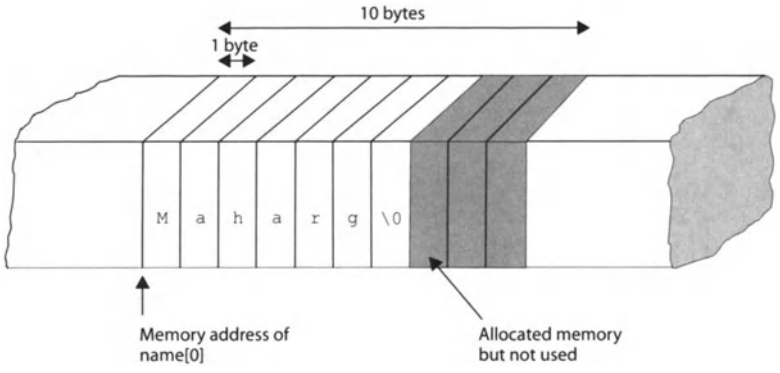


Fig. 7.5 A string in memory is composed of an array of characters.

```
cout << "hello " << name ;
}
```

Some output:

```
enter your forename: Maharg
hello Maharg
```

The definition of the variable string, `name`, is:

```
char name[10] ;
```

which defines `name` to be an array of 10 characters, each occupying one byte of memory. Things look the same as with the program ARRAY.CPP for integer arrays, except for one important difference: the *null termination character.*

A C++ string must terminate with the null termination character '`\0`' or the value 0. Figure 7.5 illustrates the above-entered name in memory. Note the null termination character at the end of the variable string `name`. Why is the null terminator placed at the end of a string? Simply because the null terminator indicates the end of a string. Try running the above program V_STRING.CPP with the following user input:

```
enter your forename: abcdefghij
```

This *strange* name will cause a system run-time error because the user entered a variable string of exactly ten characters in length, and one more character for the null terminator adds up to eleven. Remember that C++ does not perform bound checking on arrays.

7.10.2 Variable Strings with Embedded Spaces

The insertion operator (<<) displays a string simply by displaying each element of the string until it encounters the null terminator. This is the reason for the null terminator: it indicates the end of a string. Without the null terminator a stream object would not know when the end of string had been encountered. You may be thinking that a space should indicate the end of a string; if so, then try running V_STRING.CPP and enter two names that are separated by a space:

```
enter your forename: John Dan
hello John
```

The problem with the extraction operator (>>) is that it also considers white space to be a null terminator. The reason for this is that, as you may remember, the extraction operator can be chained:

```
//...
cout << "enter two numbers: " ;
cin  >> number1 >> number2 ;
```

With typical user input:

```
enter two numbers: 12 24
```

which illustrates that the space between 12 and 24 is treated as a null terminator and is necessary for the correct extraction of the two numbers entered.

A way around this problem is the *get()* member function of **class** basic_istream (of which cin is an object) which extracts a single character:

```
// cin_get.cpp
// illustrates the istream::get() function
// for extracting single characters from a string
#include <iostream>  // C++ I/O

void main ()
  {
  const int STRING_LEN = 80 ;
  // string variable: array of char
  char string[STRING_LEN] ;

  cout << "enter a string: " ;
  cin.get (string, STRING_LEN) ;

  cout << string ;
  }
```

which could produce the output:

```
enter a string: always question the assumptions
always question the assumptions
```

This demonstrates that the function basic_istream::*get()* extracts the entire string entered by a user until a new line is entered by pressing carriage return. Note that the function *get()* is called using the direct member access operator (.), since it is a member function of the basic_istream **class**. This will become clearer when we examine classes.

As the last chapter discussed function overloading and default function arguments, it is informative to examine the signatures of the basic_istream::*get()* function more closely:

```
istream_type& get (char_type* buffer, streamsize n,
                   char_type delim) ;
```

```
istream_type& get (char_type* buffer, streamsize n) ;
```

where `char_type` is a **typedef** which represents types such as **char**, **signed char** and **unsigned char**. Member function *get()* extracts single characters from the input stream and places them in the buffer, `buffer`, until the delimiter, `delim`, is encountered, the end of file is reached or until (`length` - 1)[1] bytes have been read in. The second overloaded version of *get()* uses the new line character '\n' as the delimiter. Refer to your compiler's documentation or the ISTREAM.H header file for further information.

7.10.3 Constant Strings

In Chapter 4 we had a brief introduction to constant strings. As the name suggests, a constant string is an array of type **char** that is initialised to a string of characters. The following program illustrates constant strings:

```
// c_string.cpp
// illustrates constant strings
#include <iostream>  // C++ I/O

void main ()
  {
  char C_STRING1[30] = "C++ is a programming language" ;
  char C_STRING2[]   = "C++ is a programming language" ;

  cout << C_STRING1 << endl ;
  cout << C_STRING2 ;
  }
```

Note that the length of the first string is defined as 30, rather than 29, to account for a null terminator at the end of the string, while the second string is initialised with empty braces. As we saw with arrays of type **int** previously, the compiler can determine for itself the number of elements in a one-dimensional array.

Similarly, a two-dimensional array of constant strings can be initialised:

```
// init_str.cpp
// illustrates initialising a 2D array of strings
#include <iostream>  // C++ I/O

void main ()
  {
  const int MAX_STR_LEN = 9 ;
  const int NUM = 5 ;

  char names[NUM][MAX_STR_LEN] =
    { "Graham", "Ian", "Jim", "Jonathan", "Tony" } ;

  for (int i=0; i<NUM; i++)
```

1 (`length-1`) to leave room for the null terminator.

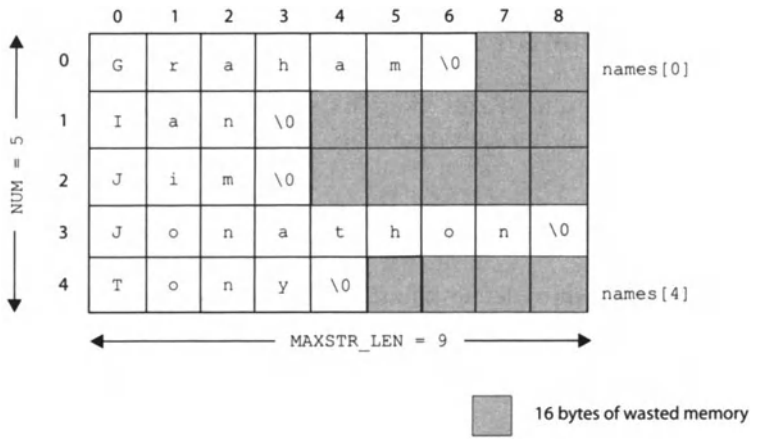


Fig. 7.6 A two-dimensional array of strings.

```
    {
    cout << names[i] << endl ;
    }
}
```

An array of strings is a two-dimensional array because a single string is a one-dimensional array of characters. Although `names` is a two-dimensional array, each string is accessed using only one subscript operator:

```
names[i]
```

which accesses the start of each string. If you need to access an individual element of the array then indexing is as usual. For example, to access the letter 'y' in 'Tony' we would index column 3 in row 4:

```
names[4][3]
```

Figure 7.6 illustrates the five names in the array. Note the 16 bytes of unused memory.

7.11 Wasted Space

Returning to Fig. 7.6, we note that 16 bytes of the allocated 45 bytes of memory is not used because four of the five names are shorter in length than the maximum string length of nine. In order to remove this inefficiency in allocated memory we will need pointers. Although static arrays can be inefficient with respect to allocated memory, this inefficiency can be compensated for by the property that access time is the same for accessing any element of an array. Because the elements of an array are stored contiguously in memory, accessing the last element takes no longer than accessing the first element.

7.12 Arrays as Function Arguments

In the previous chapter we examined functions in C++ and the passing of arguments to functions. Arrays can be passed as function arguments.

7.12.1 One-Dimensional Arrays

Let us consider a program which defines a function that is passed a one-dimensional array of numbers of type **float** and returns the average of the numbers. This program is similar to the INITIAL.CPP program presented above:

```
// func_arg.cpp
// illustrates passing of arrays to functions
#include <iostream>   // C++ I/O
#include <iomanip>    // setprecision(), ...

const int NUM_BOOKS = 10 ;

// returns the average of a 1D array of numbers
// passed as an argument
float Average (float array[NUM_BOOKS])
  {
  float total_price = 0.0 ;

  for (int i=0; i<NUM_BOOKS; i++)
    {
    total_price += array[i] ;
    }
  return total_price / NUM_BOOKS ;
  }

void main ()
  {
  float computer_book_prices[NUM_BOOKS] =
    { 18.99, 27.45, 15.95, 26.95, 21.95,
      24.95, 29.95, 30.50, 16.95, 26.95 } ;

  // call function passing array of numbers
  float av_price = Average (computer_book_prices) ;

  cout << setiosflags (ios::fixed)
       << setprecision (2)
       << "average price of my 10 favourite computer books: "
       << av_price ;
  }
```

The function declarator is given by:

```
float Average (float array[NUM_BOOKS])
```

indicating that a one-dimensional array of elements of type **float** is passed to the function *Average()* as an argument. We are able to pass the size of the array, since it is known at the time of compilation. However, it is possible to pass the array without including the array size:

```
float Average (float array[])
```

This is worth examining more closely, because it is a clue to what is actually passed to the function. When functions were discussed in Chapter 6 we noted that function arguments can be passed by *value*, *reference* or *pointer*; i.e. by *rvalue* or *lvalue*. In fact, an array is passed by lvalue, or by memory address. It is advantageous to pass arrays by address rather than by value because passing by value would involve copying each element of the passed array. Clearly, if the array was very large this would be highly inefficient from both memory and time considerations.

The following program examines the passing of an array from *main()* to a function *Func()*:

```
// address.cpp
// examines an array's address
#include <iostream>  // C++ I/O

const int MAX = 5 ;

void Func (int a_func[MAX])
  {
  cout << "rvalue a_func[0]: "   << a_func[0]
       << ", lvalue a_func[0]: " << &a_func[0]
       << endl ;
  }

void main ()
  {
  int a_main[MAX] = { 0, 1, 2, 3, 4 } ;

  cout << "rvalue a_main[0]: "   << a_main[0]
       << ", lvalue a_main[0]: " << &a_main[0]
       << endl ;

  // call function, passing array
  Func (a_main) ;

  cout << endl ;
  cout << "a_main    : "  << a_main    << endl ;
  cout << "&a_main[0]: "  << &a_main[0] << endl ;
  cout << "a_main[4] : "  << a_main[4]  << endl ;
  cout << "&a_main[4]: "  << &a_main[4] << endl ;
  }
```

which generates the following output:

```
rvalue a_main[0]: 0, lvalue a_main[0]: 0012ff78
rvalue a_func[0]: 0, lvalue a_func[0]: 0012ff78
```

```
a_main     : 0012ff78
&a_main[0]: 0012ff78
a_main[4]  : 4
&a_main[4]: 0012ff88
```

The memory addresses will probably be different when you execute the above program. The first two lines of the output indicate that when an array is passed as an argument to a function the array elements are *not* copied, and in fact the address of the array is passed (since the lvalues of `a_main[0]` and `a_func[0]` are identical). Although, the above program only illustrates that arrays are passed by address by considering element 0, you may want to confirm for yourself that this is the general case by experimenting with the other array elements.

The last four lines of output illustrate some important properties of arrays. The first of the four lines indicates that when the array name, `a_main`, is used alone it in fact points to the address at which the array is stored. In other words:

The name of an array is a synonym for the memory location of the array

The memory location of an array is at its zeroth element. Alternatively, and equivalently, the memory location at which an array resides can be accessed with the aid of the address-of operator (`&`). In terms of the example above this location is `&a_main[0]`, whereas the value of an array element at a particular index, `i`, from the zeroth element is given by `a_main[i]`. In the example above, we accessed the value of the fifth and last element of `a_main`, namely `a_main[4]`, and the memory address of this element: `&a_main[4]`.

7.12.2 Two-Dimensional Arrays

The following program illustrates passing two-dimensional arrays as function arguments:

```
// f_arg_2d.cpp
// illustrates passing two-dimensional
// arrays as function arguments

#include <iostream>  // C++ I/O

const int SIZE = 2 ;  // size of square arrays

// add two 2D arrays
void Add (int array1[][SIZE], int array2[][SIZE],
          int add[][SIZE])
  {
  for (int i=0; i<SIZE; i++)
    {
    for (int j=0; j<SIZE; j++)
      {
      add[i][j] = array1[i][j] + array2[i][j] ;
      }
    }
  }

// displays a 2D array
void Display (int array[][SIZE], char string[])
```

```
    {
    // display string
    cout << string << endl ;

    for (int i=0; i<SIZE; i++)
      {
      for (int j=0; j<SIZE; j++)
        {
        cout << array[i][j] << " " ;
        }
      cout << endl ;
      }
    }

  void main ()
    {
    // define and initialise 2 arrays
    int a1[SIZE][SIZE] = { {1, 2}, {3, 4} } ;
    int a2[SIZE][SIZE] = { {5, 6}, {7, 8} } ;

    // define placement array
    int a1pa2[SIZE][SIZE] ;

    // add
    Add (a1, a2, a1pa2) ;

    // display arrays
    Display (a1, "array a1:") ;
    Display (a2, "array a2:") ;
    Display (a1pa2, "array a1pa2:") ;
    }
```

with output:

```
array a1:
1 2
3 4
array a2:
5 6
7 8
array a1pa2:
6 8
10 12
```

The declarator of function *Add()* is:

```
void Add (int array1[][SIZE], int array2[][SIZE],
          int add[][SIZE])
```

which illustrates that three arrays are passed as arguments. The first two function arguments are the arrays to be added and the last argument is the resultant array. *Add()* is called from within *main()* with the call:

```
Add (a1, a2, a1pa2) ;
```

When the *Add()* function returns, the resultant matrix (a1+a2) is placed in matrix a1pa2.

A frequent mistake when returning an array from a function is to return (using a **return** statement) an array which is defined local to the function:

```
int Add (int array1[][SIZE], int array2[][SIZE])
  {
  int add[SIZE][SIZE] ;  // local to Add() function
  //...
  return add ;  // returning address of array
  }
```

Clearly, the return type specifier is in disagreement with the **return** statement. More importantly, the scope of array add is limited to the function definition, and when the function returns the memory allocated for array add is deleted, making it impossible for the function caller to access the resultant matrix.

Note that the *Display()* function is passed a string as a function argument so that the name of the array can be output by the function:

```
void Display (int array[][SIZE], char string[]) ;
```

7.13 Calling Functions with Arrays as Arguments

In the program above, FUNC_ARG.CPP, the function *Func()* was called passing an array, a_main, as an argument:

```
Func (a_main) ;
```

illustrating that when a function is called with an array as an argument, only the name of the array is passed. From the discussion in the previous section, the address of the array is passed, since the array name is in fact a synonym for the array's memory location.

7.14 Strings as Function Arguments

From above we saw that a string is an array of characters. Let's now examine the passing of strings to a function. The following program, STR_ARG.CPP, illustrates passing strings to functions:

```
// str_arg.cpp
// illustrates passing strings as function arguments
#include <iostream>  // C++ I/O

// compares two strings
// returns 1 if str1==str2, 0 if str1!=str2
int StrCompare (char str1[], char str2[])
```

```
    {
    int i = 0 ;
    while (str1[i] == str2[i])
      {
      if (str1[i++] == '\0')
        return 1 ;
      }
    return 0 ;
    }

  // copies str2 into str1
  void StrCopy (char str1[], char str2[])
    {
    int i = 0 ;
    while ( (str1[i] = str2[i]) != 0)
      i++ ;
    }

  void main ()
    {
    char string1[] = "this is a string" ;
    char string2[] = "this is a string" ;
    char string3[] = "another string" ;

    // compare string1 and string2
    int comp12 = StrCompare (string1, string2) ;

    if (comp12 == 1)
      cout << "string1 == string2" << endl ;
    else
      cout << "string1 != string2" << endl ;

    // compare string2 and string3
    int comp23 = StrCompare (string2, string3) ;

    if (comp23 == 1)
      cout << "string2 == string3" << endl ;
    else
      cout << "string2 != string3" << endl ;

    // copy string3 to string1
    StrCopy (string1, string3) ;

    cout << endl
         << "string1: " << string1 << endl ;
    }
```

StrCompare() performs the comparison of the two strings passed as arguments by simply comparing each character of the two strings that have the same array index and returns a value of 1 if the strings are identical or 0 if the strings are different. *StrCopy()* copies one string, character by character, into another string.

The functions `StrCompare()` and `StrCopy()` are similar to the library functions `strcmp()` and `strcpy()`, respectively, declared in the header file STRING.H. `strcmp()` compares two strings and returns zero if the two strings are identical, or a negative or positive integer, depending on whether or not the first string is lexicographically less than or greater than the second string. `strcpy()` performs a similar operation to `StrCopy()`.

7.15 Passing the Size of an Array as an Argument

Consider the program:

```
// arg.cpp
// illustrates that a function does not know the size
// of an array passed as an argument
#include <iostream>   // C++ I/O
#include <string>     // strlen()

// uses strlen() to find string length
void StrLen (char str[])
  {
  const int LENGTH = strlen (str) ;
  cout << "StrLen::LENGTH: " << LENGTH << endl ;
  }

// uses sizeof to find string length
void StrSizeOf (char str[])
  {
  const int LENGTH = sizeof str / sizeof str[0] ;
  cout << "StrSizeOf::LENGTH: " << LENGTH << endl ;
  }

// returns 1 if str1==str2, 0 if str1!=str2
int StrCompare1 (char str1[], char str2[], int length)
  {
  for (int i=0; i<length+1; i++)
    {
    if (str1[i] != str2[i])
      return 0 ;
    }
  return 1 ;
  }

// passed a (2x3) array
void Display1 (int array[2][3])
  {
  cout << "Display1: " << endl ;
  for (int i=0; i<2; i++)
    {
    for (int j=0; j<3; j++)
      {
```

```
      cout << array[i][j] << " " ;
      }
    cout << endl ;
    }
  }

// passed a (2x3) array and no. of rows
void Display2 (int array[][3], int rows)
  {
  cout << "Display2: " << endl ;
  for (int i=0; i<rows; i++)
    {
    for (int j=0; j<3; j++)
      {
      cout << array[i][j] << " " ;
      }
    cout << endl ;
    }
  }

// passed a (2x3) array and no. of rows & cols
void Display3 (int array[][], int rows, int cols)  // error!
  {
  cout << "Display2: " << endl ;
  for (int i=0; i<rows; i++)
    {
    for (int j=0; j<cols; j++)
      {
      cout << array[i][j] << " " ;
      }
    cout << endl ;
    }
  }

void main ()
  {
  // define & initialise 2 strings
  char string1[] = "string" ;
  char string2[] = "string" ;

  // length of strings
  StrLen    (string1) ;
  StrSizeOf (string1) ;

  // compare string1 and string2
  int comp = StrCompare1 (string1, string2, 6) ;

  if (comp == 1)
    cout << "string1 == string2" << endl ;
  else
    cout << "string1 != string2" << endl ;
```

```
cout << endl ;

// define & initialise a (2x3) array of int's
int A[2][3] = { {0, 1, 2}, {3, 4, 5} } ;

Display1 (A) ;          // pass A
Display2 (A, 2) ;       // pass A & rows
Display3 (A, 2, 3) ;    // pass A, rows, cols
}
```

When an array is passed as an argument to a function the size of the array is not known to the function. To illustrate this point, the above program, ARG.CPP, defines two functions, *StrLen()* and *StrSizeOf()*, which determine the length of a string argument by means of the *strlen()* library function and the **sizeof** operator, respectively. *StrLen()* outputs the correct string length (not including the null terminator). *StrSizeOf()* outputs the incorrect string length by using the **sizeof** operator because the size of the string array `string`, defined in *main()*, is not available to *StrSizeOf()*. If we don't want to call the *strlen()* library function to determine the length of a string argument, one way around this problem is to pass the string length as an argument. This is the technique exercised in the function *StrCompare1()*, which is a modification of *StrCompare()* in the program STRCMP.CPP.

However, since strings are null-terminated, knowing the exact length of a string argument is not a problem. For instance, *strlen()* determines the correct length of a string argument by means of the null terminator, just as the *StrCompare()* function in STRCMP.CPP compared two string arguments without knowing the length of either string. For the interested reader, the general form of *strlen()* is (Kernighan and Ritchie, 1978):

```
int strlen (char str[])
  {
  char* p = str ;

  while (*p != '\0')
    p++ ;
  return p - str ;
  }
```

which involves the use of a pointer to **char**, `p`.

However, two-dimensional arrays are slightly more difficult. The function *Display1()* in the ARG.CPP program listed above displays the integer elements of a (2×3) array passed as an argument, with the array sizes included in the function declarator. Since the first array size is not required by the compiler to determine the memory location of the first element of the array, *Display2()* passes the row size as an integer argument.

What we would really like to do is to define a function that is passed a two-dimensional array that is as general as possible, such as *Display3()*:

```
void Display3 (int array[][], int rows, int cols)   // error!
```

Unfortunately, this is illegal because the second size (number of columns) must be known by the compiler in order to locate the first element of the array. A solution to this problem requires the use of pointers, which will be discussed in Chapter 12.

Table 7.1 Memory models.

Memory model	*Data segments*	*Code segments*
Small	1	1
Medium	1	Multiple
Compact	Multiple	1
Large	Multiple	Multiple
Huge	Multiple	Multiple

7.16 Small, Medium, Compact, Large and Huge Memory Models

In the A_SIZE.CPP program a two-dimensional square array, `big_array`, was defined as 2500 (50 × 50) elements of type **`double`** (8 bytes) or 20 kbyte. How big can an array be? Build the program for a 16-bit system and increase the array size to (90 × 90), or 64 800 bytes in total. This array size should be OK. Now try increasing the array size to (91 × 91), or 66 248 bytes. The program will no longer compile, because the array size is too large. To see why this array size is too large we first need a brief introduction to memory management.

IBM-compatible PCs are based on the Intel 8088 and 80x86 family of microprocessors. The internal registers of the microprocessors are 16 bits in length. The Intel 8088 microprocessor has four internal registers, referred to as *segment registers*, which are characterised as *code*, *data*, *stack* and *extra* segment registers. In order to construct a complete memory address, the 8088 microprocessor shifts the contents of a segment register to create a 20-bit address, which can access up to 1 Mbyte of memory. This 1 Mbyte of memory is divided into the following categories: The highest 64 kbyte is reserved for ROM BIOS; the next 192 kbyte is reserved for ROM expansion; and the next 128 kbyte is reserved for the video display, giving a total of 384 kbyte. Subtracting this memory from the total memory allocation of 1024 kbyte (=1 Mbyte), we are left with 640 kbyte of read–write memory. Of these 640 kbyte, the lowest area is reserved for read–write data required by ROM BIOS, and above this area MS-DOS loads, followed by device drivers. The remaining memory is used by programs executing under MS-DOS.

The Microsoft and Borland C++ compilers support five different memory models: small, medium, compact, large and huge. Different memory models refer to different data and code segments; see Table 7.1. If a program requires more than 64 kbyte of code or data, it must contain two or more code or data segments respectively.

In Chapter 20 we will examine the C++ Standard Template Library (STL) solution to dealing with different memory models via the `allocator` class and functions such as *`allocate()`*.

7.16.1 Small and Medium Memory Models

The small memory model is the default memory model. The majority of small programs for Windows are compiled with the small memory model, with one code segment and one data segment. The compiler usually assumes that the stack segment (SS) is equal to the data segment (DS) for both small and medium memory models, although this can be overridden.

Functions for medium memory model programs generate far calls. Using the medium memory model for small programs is not necessary. Use the medium model when your program involves more than one source code module. The medium model allows each source code to be a different code segment, which can be both moveable and discardable. The space required to fit the program code into memory is the size of the largest code segment.

7.16.2 Compact, Large and Huge Memory Models

Compact and large memory models are available for Windows applications, but are not recommended. The additional data segments provided by these memory models must be fixed in memory and cannot be moveable. Also, a program that uses compact and large memory models is restricted to a single instance of a program.

Functions for compact and large memory model programs generate far calls and assume that they are passed long pointers to data. Limited support is offered for the huge memory model.

7.16.3 Libraries

Each of the five memory models has C and C++ libraries that are specific to a memory model (e.g. the CWS.LIB and CWM.LIB small and medium library files in the Borland C++ (version 5.x) library directory: \BC5\LIB). This is because a given function is called differently depending on the memory model. Medium and large memory model functions assume that they have been called from another segment, whereas the large and compact models assume that they have been passed long pointers to data. The huge model is similar to the large model, except that individual data items may be greater than 64 kbyte.

7.16.4 Windows Programming

Windows programs extensively use far calls to Windows functions, window procedures and call-back functions. The majority of data pointers exchanged between a Windows program and Windows are far pointers.

7.17 Summary

This chapter has introduced us to the array. An array is an aggregate of elements of a given fundamental or user-defined data type, and can have one, two, three or higher dimensions. The elements of an array are stored contiguously in memory, and an element of an array is accessed via the array name and an index, which is an integer. Indexing an array in C++ starts with element 0, not element 1. C++ performs no default bounds checking on an array. Arrays, like variables, can be initialised when they are defined. Arrays can be either static or dynamic. The size of a static array is constant and is defined at compilation. An array can be passed, by address, as an argument to a function. When a function is called that passes an array as an argument, only the array name is passed to the function, because the name of an array is a synonym for the array's memory location.

A string is a one-dimensional array of characters. Strings can be either constant or variable.

The elements of an array are stored consecutively in memory. This has the important consequence that when an array is defined at the time of compilation the size of the array remains constant throughout the lifetime of a program. Static arrays can be inefficient if all the allocated memory is not used, but this inefficiency can be compensated for by the fact that array indexing is random. To remove the inefficiency of wasted memory allocation we use an array which allocates memory *dynamically* as and when it requires it. A solution to the problem of dynamically expanding or reducing the size of an array in response to the insertion or deletion of elements is to use data structures such as linked lists, queues, stacks and trees. Linked lists provide a neat way of creating a dynamic list of elements or nodes which respond well to the

insertion and deletion of elements, but suffer from the point of view of searching. Dynamic arrays and data structures will be discussed in later chapters.

An additional consequence of arrays storing their element data consecutively is that they do not lend themselves well to searching. Searching an array generally involves an exhaustive search of the entire array for a given element.

There is a direct correlation between arrays and pointers, which have yet to be discussed. Array indexing by means of the subscript operator `[ ]` can equally be performed with pointers, although with reduced clarity. A pointer *points* to the memory address of an identifier, just as a reference to an array element contains the memory address of the element.

Exercises

7.1 Write a program which gets a set of floating-point numbers from a user and places the numbers in an array.

7.2 Extend the program in Exercise 7.1 so that the array of floating-point numbers can be passed to a function that returns the average of the array of numbers.

7.3 Develop a function which transposes the elements of a two-dimensional array. The transpose of an $(m \times n)$ array $\mathbf{A}=[a_{ij}]$ is the $(n \times m)$ array $\mathbf{A}^{\mathrm{T}}=[a_{ji}]$.

7.4 Develop a function called `StrLen()` (without using `strlen()`) which has the same functionality as the C++ library function `strlen()` in that it is passed a string and returns the length of the string, not counting the null-terminator character.

7.5 Frequently, when working with data acquisition systems, data arrives from a one-dimensional array buffer. However, when manipulating the data it is often more convenient to manipulate a two-dimensional array. For example, the rows of a two-dimensional array may correspond to different channel numbers. Develop two functions which convert a (12×1) one-dimensional array to a (3×4) two-dimensional array, and vice versa. The signatures of the two functions should be of the form:

```
void OneDToTwoD (int array1D[3*4],  int array2D[3][4]) ;
void TwoDToOneD (int array2D[3][4], int array1D[3*4]) ;
```

7.6 Develop three functions which add, subtract and multiply two two-dimensional arrays of floating-point numbers. The functions must cater for matrices of (3×3) and (3×1) sizes and conform with the rules of matrix operations. Also, develop functions which display the elements of (3×3) and (3×1) two-dimensional arrays. Test your program on a variety of arrays that are initialised within your program. Place the function signatures and definitions in files called ARRAY.H and ARRAY.CPP, respectively, so that the next exercise can `include` ARRAY.H.

7.7 This exercise is concerned with the rotation of a point, $p(x,y,z)$, in three-dimensional space. For three-dimensional space it is convenient to use matrix algebra in which a rotation is defined by the three angular components θ,β and γ about the x-, y- and z-axes with respect to the x'-, y'- and z'-axes, respectively. A point $p(x,y,z)$ is transformed to a point $p'(x', y', z')$ by an application of a rotation matrix, **R**:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \mathbf{R} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

If we rotate about the z-axis ($\theta \neq 0$), the rotation matrix is given by:

$$\mathbf{R}_\theta = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

If we rotate about the y-axis ($\beta \neq 0$), the rotation matrix is:

$$\mathbf{R}_\beta = \begin{bmatrix} \cos\beta & 0 & -\sin\beta \\ 0 & 1 & 0 \\ \sin\beta & 0 & \cos\beta \end{bmatrix}$$

Finally, if we rotate about the x-axis ($\gamma \neq 0$), the rotation matrix is:

$$\mathbf{R}_\gamma = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\gamma & \sin\gamma \\ 0 & -\sin\gamma & \cos\gamma \end{bmatrix}$$

The total rotation matrix for an arbitrary rotation is $\mathbf{R}=\mathbf{R}_\theta\mathbf{R}_\beta\mathbf{R}_\gamma$ and can be shown to be given by:

$$\mathbf{R} = \begin{bmatrix} \cos\theta\cos\beta & \sin\theta\cos\gamma + \cos\theta\sin\beta\sin\gamma & \sin\theta\sin\gamma - \cos\theta\sin\beta\cos\gamma \\ -\sin\theta\cos\beta & \cos\theta\cos\gamma - \sin\theta\sin\beta\sin\gamma & \cos\theta\sin\gamma + \sin\theta\sin\beta\cos\gamma \\ \sin\beta & -\cos\beta\sin\gamma & \cos\beta\cos\gamma \end{bmatrix}$$

Develop four functions for $\mathbf{R}_\theta$, $\mathbf{R}_\beta$, $\mathbf{R}_\gamma$ and $\mathbf{R}$ of the form:

```
void RTheta (const double& theta, double array[3][3]) ;
```

Test your rotation matrices on various rotated points using your matrix manipulation functions developed in Exercise 7.6.

Chapter 8

Structures, Unions, Enumerations and Typedefs

This chapter examines structures, unions, enumerations and typedefs which allow a programmer to create new data types. The ability to create new types is an important and very powerful feature of C++ and releases a programmer from being restricted to the integral types offered by the language. Structures enable a programmer to form a collection of similar or different data types into a single user-defined data type, unlike arrays, which are collections of a single data type. The data members of a structure are, by default, publicly accessible. User-defined structures integrate exactly into the language. In other words, there is a single syntax for the definition and manipulation of objects of both integral C++ and user-defined types.

A union is similar to a structure except that a union can hold only one of its data members at any given time. Union data members share the same location in memory. An enumerated data type is a collection of named members which have equivalent integer values. A typedef is in fact not a new data type but merely a synonym for an existing type.

8.1 Structures

A structure is a user-defined data type that is a collection of similar or dissimilar data items or members.

The previous chapter illustrated the array, which enables us to form an aggregate of *identical* data items. If we are required to form a collection of different data types, we cannot use an array. The C++ keyword **`struct`** enables a programmer to create *new* structure data types which encompass a collection of similar or different data types. The ability to develop new user-defined data types means that a programmer is not restricted to the integral data types offered by the C++ language. The following program illustrates a `Person` structure:

```
// struct.cpp
// illustrates structures
```


G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

```
#include <iostream>  // C++ I/O

// declare a structure 'Person'
struct Person
  {
  char  name[20] ;  // 4name
  int   age ;       // yr
  float height ;    // m
  float mass ;      // kg
  };

void main ()
  {
  // define a structure variable 'person'
  Person person ;

  // prompt user for data and assign
  // values to struct Person members
  cout << "enter your forename: " ; cin >> person.name ;
  cout << "enter your age: " ;      cin >> person.age ;
  cout << "enter your height: " ;   cin >> person.height ;
  cout << "enter your mass: " ;     cin >> person.mass ;

  // O/P
  cout << endl << "hello "
       << person.name   << ", you are "
       << person.age    << " years old, "
       << person.height << "m tall and "
       << person.mass   << "kg in mass" ;

  // define and initialise a Person variable 'harry'
  Person harry = { "Harry", 25, 1.85, 80 } ;

  // O/P Harry:
  cout << endl << "hello "
       << harry.name   << ", you are "
       << harry.age    << " years old, "
       << harry.height << "m tall and "
       << harry.mass   << "kg in mass" ;
  }
```

Some user interaction with the above program is:

```
enter your forename: Tom
enter your age: 45
enter your height: 1.8
enter your mass: 74

hello Tom, you are: 45 years old, 1.8m tall and 74kg in mass
hello Harry, you are 25 years old, 1.85m tall and 80kg in mass
```

8.1.1 Declaring a Structure

The above program *declares* a structure called `Person`, shown again below:

```
struct Person
  {
  char  name[20] ;
  int   age ;
  float height ;
  float mass ;
  };
```

Note that this is a declaration rather than a definition. It was noted in Chapter 4 that a declaration does not allocate storage, whereas a definition is a declaration that simultaneously allocates memory for a data item. The structure declaration is a *mould* or a template that outlines the contents of a structure.

The general syntax of a structure declaration is of the form:

```
struct Name
  {
  type_specifier member_name ;    // structure members
  //...
  type_specifier member_name ;
  };
```

The structure declaration begins with the keyword **`struct`**. This is followed by the structure tag or name. Note that the style I adopt is to use an uppercase letter for the first letter of single or multiple-connected structure names (e.g. `Person`, `Triangle`, `TwoDTriangle`, `ThreeDTriangle`). Between the opening, `{`, and closing, `}`, braces of the structure declaration are the members of the structure. Finally (and it's easily forgotten!), there comes the semicolon, following directly after the closing brace. The semicolon after the closing brace helps to distinguish a structure declaration from a function definition.

The name of a **`struct`** is available for use immediately following the name and not after the closing brace. For instance, a popular application of this feature is in the design of a `Node` structure for a linked-list data structure:

```
struct Node
  {
  Node* next ; // ok: pointer to Node
  //...
  };
```

However, it is not possible to declare an object of a **`struct`** until the declaration is complete:

```
struct Name
  {
  Name object ; // error: object of Name
  //...
  };
```

We will examine in detail a linked-list class, `LinkedList`, in Chapter 13.

8.1.2 Scope of a Structure

In the above program, STRUCT.CPP, the structure `Person` was declared outside the *`main()`* function and thus has global scope. Thus, any function (including *`main()`*) can define a variable of type `Person`. However, if a structure is declared within the body of a function, then the structure declaration is local to that function. For example:

```
// scope.cpp
// illustrates a structure's scope
#include <iostream>  // C++ I/O

// prototype
void Function () ;

void main ()
  {
  // declare a structure 'MainPerson' within main()
  struct MainPerson
    {
    char  name[20] ;  // 4name
    int   age ;       // yr
    float height ;    // m
    float mass ;      // kg
    };

  MainPerson m_person ;
  }

void Function ()
  {
  MainPerson f_person ;  // error: undefined type MainPerson!
  }
```

This program declares a structure `MainPerson` within the function body of *`main()`* and is therefore not accessible outside *`main()`*.

8.1.3 Structure Tag or Name

The structure tag or name is not a variable or an object but rather a name to a collection of data types.

8.1.4 Defining a Structure Variable or Object

The program STRUCT.CPP defines a variable or an object of type `Person`:

```
//...
Person person ;
```

This statement allocates enough memory to create a single structure variable or object of type `Person`. This definition is similar to defining a variable of a fundamental data type:

```
//...
int    var ;
double x ;
```

Just as for an array, a structure variable can be initialised when it is defined. STRUCT.CPP illustrates that a variable `harry` is initialised by an *initialiser list*, whose elements correspond exactly to the members of the structure. Each element of the initialiser list is separated by the comma operator, and all elements are placed between the braces { and }:

```
Person harry = { "Harry", 25, 1.85, 80 } ;
```

8.1.5 Accessing a Structure's Data Members

Program STRUCT.CPP outputs to the screen the data members of a user-entered data structure by accessing each of the structure's data members independently through the use of the dot or period operator (.):

```
Person person ;
//...
// O/P
cout << endl << "hello "
     << person.name   << ", you are: "
     << person.age    << " years old, "
     << person.height << "m tall and "
     << person.mass   << "kg in mass" ;
```

The four members of structure `Person` (`name`, `age`, `height` and `mass`) are accessed using the dot operator, which is placed between the structure variable name and the structure member. The general syntax for accessing a structure's members is:

```
structure_name.structure_member
```

Such access is often referred to as *direct member access*, as opposed to *indirect member* access via the use of the -> operator for use with pointers (Chapter 12).

8.1.6 The Size of a Structure

The following program illustrates the use of the **`sizeof`** operator to determine the size of a structure:

```
// sizeof.cpp
// illustrates determining the size of a structure
#include <iostream>  // C++ I/O

// declare a structure 'Person'
struct Person
  {
  char  name[20] ;  // 4name
  int   age ;       // yr
  float height ;    // m
```

Table 8.1 Size of type `Person`.

***Person* member**	*Member data type*	*Size (bytes), 32-bit*
`name[20]`	**`char`** (1)	20
`age`	**`int`** (4)	4
`height`	**`float`** (4)	4
`mass`	**`float`** (4)	4
Total		32

```
  float mass ;        // kg
  };

void main ()
  {
  Person person ;

  cout << "sizeof (Person): " << sizeof (Person) ;
  }
```

which produces the output:

```
sizeof (Person): 30
```

Note the use of the parentheses (and) when the **`sizeof`** operator is applied to `Person`. This is because `Person` is a type.

To see how the size of type `Person` is determined, refer to Table 8.1. The size of `Person` is the sum of its individual members.

8.1.7 Declaring and Defining a Structure in a Single Statement

It is possible to declare and define a structure simultaneously in a single statement. The following program code illustrates this:

```
// declare and define a structure
struct Person
  {
  //...
  } person ;
```

which defines a structure variable name, `person`, between the closing brace, }, of the `Person` structure declaration and the semicolon. In addition, we can define several variables at the same time:

```
struct Person
  {
  //...
  } tom, dick, harry ;
```

The main problem with declaring and defining a structure in a single statement is that the structure variable(s) have the same scope as the structure declaration. It was noted above that

generally structure declarations are given global scope to enable all functions to access to the structure. Thus declaring and defining a structure simultaneously creates global variables, the number of which should be minimised if possible.

8.1.8 Defining a Structure Variable or Object Without a Structure Name

Structures can also be defined without a structure tag or name:

```
struct  // nameless structure!
  {
  //...
  } person ;
```

Such an *anonymous* structure definition is of limited use because we have no structure name to enable us to create a variable other than `person`.

8.1.9 Output of a Structure's Data Members

You may have noticed from the above program that in order to output the data members of a given structure variable we must explicitly output (by means of the `cout` stream object) each of the structure's data members:

```
//...
// O/P
cout << endl << "hello "
     << person.name   << ", you are: "
     << person.age    << " years old, "
     << person.height << "m tall and "
     << person.mass   << "kg in mass" ;
```

This can be a bit laborious and messy. Instead, why can't we simply write:

```
//...
cout << person ;  // error: illegal structure operation
```

This output statement would be nice, but unfortunately the `cout` stream object was not designed for our `Person` structure and hence doesn't know how to output it. Try modifying STRUCT.CPP to include the above single output statement. The compiler will issue an error message of the form 'illegal use of structure variable'.

We shall see later when we discuss classes and operator overloading in Chapters 9 and 10 that simple statements such as:

```
cout << person ;
```

are possible.

8.1.10 Assignment

One structure variable can be assigned to another. The following program illustrates this for two `Person` variables `susan` and `sarah`:

```
// assign.cpp
// illustrates assignment of structure variables
#include <iostream>  // C++ I/O

// declare a structure 'Person'
struct Person
  {
  char  name[20] ;  // 4name
  int   age ;       // yr
  float height ;    // m
  float mass ;      // kg
  };

void main ()
  {
  // define two structure variables
  Person susan = { "Susan", 31, 1.6, 58 } ;
  Person sarah = { "Sarah", 41, 1.7, 60 } ;

  // O/P:
  cout << "before assignment:" << endl
       << susan.name   << " is " << //...

  cout << sarah.name   << " is " << //...

  // assign 'sarah' to 'susan'
  susan = sarah ;

  cout << "after assignment:" << endl
       << susan.name   << " is " << //...
  }
```

The assignment statement is given by:

```
susan = sarah ;
```

and copies *all* of `sarah`'s data members to `susan`'s data members. This is the default operation of the assignment operator: it will copy all data members from an object on the right-hand side of the assignment operator, member-by-member, to an object on the left-hand side of the assignment operator.

Try adding the following statement to the above program, ASSIGN.CPP:

```
susan + sarah ;  // error: illegal use of structure
```

This will result in a compilation error of the form 'illegal operation on a structure object'. While the assignment operator has a default behaviour, other operators, such as the arithmetic operators, do not have default operations. To write such statements and expressions in which variables and objects of user-defined structures and classes are operands will require the use of overloaded operators. Overloading operators specifically for a user-defined type is covered in Chapter 10.

8.1.11 Arrays of Structures

The following program defines and initialises an array of five elements of type `Person`:

```
// array.cpp
// illustrates an array of structures
#include <iostream>  // C++ I/O

const int NUM_OF_PEOPLE = 5 ;

// declare a structure 'Person'
struct Person
  { /* ... */ };

void main ()
  {
  // define and initialise an array of Person's
  Person persons[NUM_OF_PEOPLE] =
        { { "Tom",   23, 1.71, 65 },
          { "Dick",  24, 1.74, 75 },
          { "Harry", 25, 1.85, 80 },
          { "Susan", 31, 1.6,  58 },
          { "Sarah", 41, 1.7,  60 } } ;

  // O/P
  for (int i=0; i<NUM_OF_PEOPLE; i++)
    {
    cout << persons[i].name << " is "
         << persons[i].age << " years old, "
         << persons[i].height << "m tall and "
         << persons[i].mass << "kg in mass" << endl ;
    }
  }
```

The array of type `Person` is defined as:

```
Person persons[NUM_OF_PEOPLE] ;
```

which has exactly the same syntax as an array of elements of a fundamental data type. Accessing the data members of the `Person` structure now uses the dot member access and array subscript operators:

```
persons[i].name ;
```

which accesses the `name` data member of `Person` for each array element.

8.1.12 Structures as Function Arguments

Provided a suitable structure declaration exists, function arguments of a user-defined structure type are perfectly feasible:

```
// func_arg.cpp
// illustrates a structure function argument
#include <iostream>  // C++ I/O

const int NUM_OF_PEOPLE = 5 ;

// declare a structure 'Person'
struct Person
  { /* ... */ };

// displays a Person
void Display (Person p)
  {
  cout << p.name   << " is "
       << p.age    << " years old, "
       << p.height << "m tall and "
       << p.mass   << "kg in mass" << endl ;
  }

void main ()
  {
  // define and initialise an array of Person's
  Person persons[NUM_OF_PEOPLE] =
    { { "Tom", 23, 1.71, 65 },
      { "Dick", 24, 1.74, 75 },
      { "Harry", 25, 1.85, 80 },
      { "Susan", 31, 1.6, 58 },
      { "Sarah", 41, 1.7, 60 } } ;

  // O/P
  for (int i=0; i<NUM_OF_PEOPLE; i++)
    {
    Display (persons[i]) ;
    }
  }
```

The above program modifies ARRAY.CPP by simply placing the `cout` stream output into a function called *`Display()`*. The signature of *`Display()`* is:

```
void Display (Person p) ;
```

which passes, by value, a variable of type `Person` as a function argument. We noted in Chapter 6 that function arguments can also be passed by reference. Passing by reference a `Person` function argument to function *`Display()`* could be written as either of the following:

```
void Display (Person& p) ;
```

or

```
void Display (const Person& p) ;
```

The latter ensures that the `Display()` function does not alter the function argument.

All of the properties of functions examined in Chapter 6 apply equally to user-defined structure types.

8.1.13 Nested Structures

Once a structure has been declared, it is perfectly feasible to use this structure just like a fundamental data type, and hence nest a structure within another structure. For instance, if we are required to extend our `Person` structure to include a date of birth data member, we could declare another structure called `Date`:

```
struct Date
  {
  unsigned int day ;
  unsigned int month ;
  unsigned int year ;
  };
```

Thus our original `Person` structure could be modified in order to include a `Person`'s date of birth:

```
struct Person
  {
  char          name[20] ;  // 4name
  unsigned int  age ;       // yr
  float         height ;    // m
  float         mass ;      // kg
  Date          dob ;       // date of birth
  };
```

In addition, an `EmployedPerson` structure could be declared which contains the necessary data members for the categorisation of an employed person:

```
struct EmployedPerson
  {
  Person        person ;           // Person details
  unsigned int  salary ;           // wages
  unsigned int  hours_per_week ;   // working week
  };
```

Note the use of **`unsigned int`** for data members which are known to be positive integers (assuming an employee does not pay an employer for working!). The following program demonstrates the use of nested structures using the `Date`, `Person` and `EmployedPerson` structures declared above.

```
// nest.cpp
// illustrates nested structures
#include <iostream>  // C++ I/O

struct Date
  {
```

```
  unsigned int day ;
  unsigned int month ;
  unsigned int year ;
  };

struct Person
  {
  char          name[20] ; // 4name
  unsigned int  age ;      // yr
  float         height ;   // m
  float         mass ;     // kg
  Date          dob ;      // date of birth
  };

struct EmployedPerson
  {
  Person        person ;           // Person details
  unsigned int  salary ;           // wages
  unsigned int  hours_per_week ;   // working week
  };

void main ()
  {
  // define an EmployedPerson variable
  EmployedPerson ep ;

  // set data members
  cout << "enter your name: " ;    cin >> ep.person.name ;
  cout << "enter your age: " ;     cin >> ep.person.age ;
  cout << "enter your height: " ; cin >> ep.person.height ;
  cout << "enter your mass: " ;    cin >> ep.person.mass ;
  cout << "enter your date of birth: " ;
  cin  >> ep.person.dob.day
       >> ep.person.dob.month
       >> ep.person.dob.year ;
  cout << "enter your salary: " ;
  cin  >> ep.salary ;
  cout << "enter no. of hr's/wk. at work: " ;
  cin  >> ep.hours_per_week ;

  // O/P
  cout << endl
       << "name: "    << ep.person.name    << endl << //...
  }
```

To access the data members of nested structures requires the use of multiple dot operators. For instance, to access the day of the month of an employed person's date of birth, we use the dot operator three times as follows:

```
ep.person.dob.day
```

The `EmployedPerson` structure contains a `Person` data member which contains a `Date` data member which contains a `day` data member. Structures can be nested to any degree.

Finally, it is possible to initialise nested structures upon definition. The following example initialises a variable `john` of type `Person`:

```
//...
Person john = { "John", 30, 1.675, 81, {12, 1, 65} } ;
```

Note the parentheses within the initialiser list, encompassing the fifth data member, `dob`, of type structure `Date`.

8.1.14 A `Point` Structure

In geometric modelling we frequently require the coordinates of a point relative to a given reference point or origin. Figure 8.1(a) illustrates an arbitrary point P in a two-dimensional domain, such as a display screen, while Fig. 8.1(b) illustrates a point Q in a general three-dimensional space. The position of P is categorised by the Cartesian coordinate system (x, y) with origin at the point O (0,0). If the x- and y-coordinates of P are P_x and P_y, respectively, then point P is given by P (P_x, P_y). Similarly, the point Q is given by the triple points (Q_x, Q_y, Q_z); where z is the out-of-plane dimension.

A general three-dimensional point could simply be modelled as three separate variables:

```
//...
// define & initialise 3 coordinate positions of a point p
double p_x, p_y, p_z = 0.0 ;
//...
// move point, p, to (1,2,0)
p_x = 1.0 ; p_y = 2.0 ; p_z = 0.0 ;
//...
```

Alternatively, point `p` could be represented as an array, since each of its three coordinates are of the same type, **`double`**:

```
//...
const int 3D = 3 ;
//...
// define & initialise 2 point arrays
double p[3D] = { 0.0, 0.0, 0.0 } ;
double q[3D] = { 0.0, 0.0, 0.0 } ;
//...
// move point, p, to (1, 2, 0)
p[0] = 1.0 ; p[1] = 2.0 ; p[2] = 0.0 ;
//...
```

Both of the above two representations of a given point work, but are very clumsy and do nothing to increase the generality of a program and hide the details of a point from a user. A point in a given space is composed of three individual coordinates x, y and z, but it is instructive to view a point as a single entity. Thus a point naturally lends itself to a structure with the three coordinates being the structure's members:

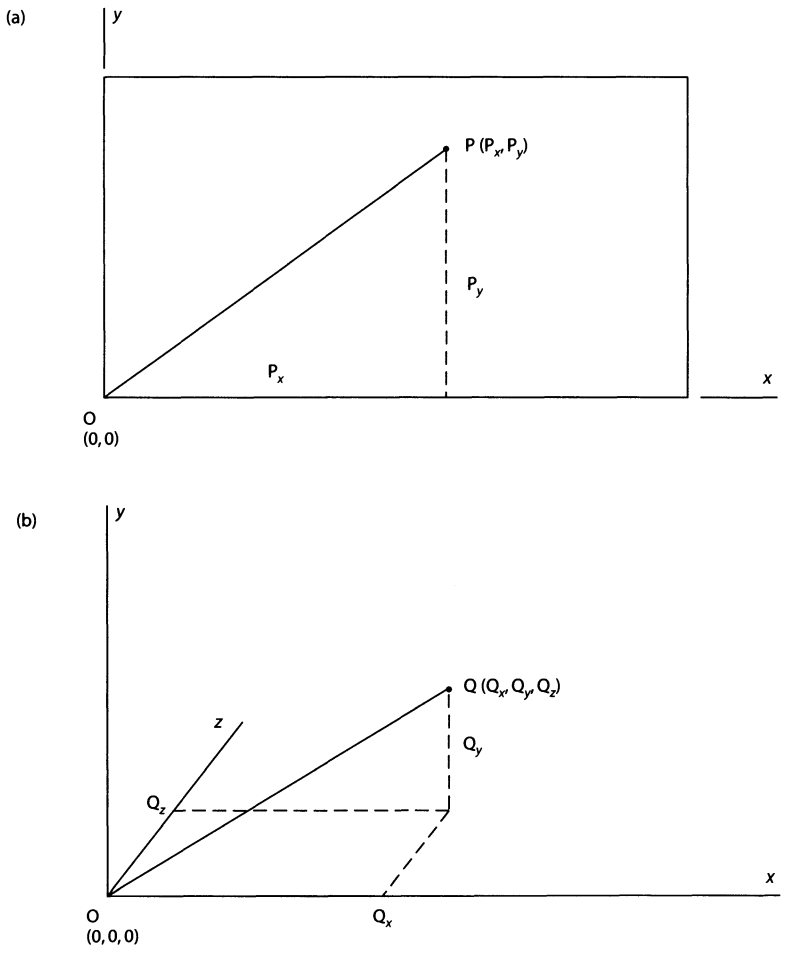


Fig. 8.1 (a) Two-dimensional and (b) three-dimensional points P and Q.

```
struct Point
  {
  double x, y, z ;
  };
```

This structure is used in the following program, which defines a function `Distance()` that returns the distance between two `Point`s:

```
// point.cpp
// illustrates a Point structure
#include <iostream>   // C++ I/O
#include <cmath>      // sqrt()

// a Point structure for representing a 3D point
struct Point
  {
```

```
  double x, y, z ;
  };

// returns the distance between two Points
double Distance (const Point& p1, const Point& p2)
  {
  double x = p2.x - p1.x ;
  double y = p2.y - p1.y ;
  double z = p2.z - p1.z ;

  double r_sq = x * x + y * y + z * z ;

  return sqrt (r_sq) ;
  }

void main ()
  {
  // define & initialise two Points in the plane z=0
  Point p = {1.0, 1.0, 0.0 } ;
  Point q = {3.0, 2.0, 0.0 } ;

  cout << "distance between p and q: " << Distance (p, q) ;
  }
```

The distance, r, between two points p (p_x, p_y, p_z) and q (q_x, q_y, q_z) is given by:

$$r = [(q_x - p_x)^2 + (q_y - p_y)^2 + (q_z - p_z)^2]^{1/2}$$

At present our `Point` structure is beautifully simple, but also fairly limited in its applications. However, the `Point` type is something that we shall return to again and again throughout subsequent chapters (when we visit classes) because it is so fundamental to geometry and computer graphics.

8.1.15 Data and Functions

In C++, structures are typically used to hold different data types, while classes are used to hold both data and functions. In fact, C++ also allows structures to have data and function members. For instance, we could extend the `Point` structure to provide two member functions *`Add()`* and *`Subtract()`*, which add and subtract, respectively, two objects of type `Point`:

```
// dat_fun.cpp
// illustrates a structure declaration
// with data and function members
#include <iostream>  // C++ I/O

struct Point
  {
  // data members:
```

```
    double x, y, z ;

    // function members:
    void Add (const Point& p1, const Point& p2)
      {
      x = p1.x + p2.x ;
      y = p1.y + p2.y ;
      z = p1.z + p2.z ;
      }

    void Subtract (const Point& p1, const Point& p2)
      {
      x = p1.x - p2.x ;
      y = p1.y - p2.y ;
      z = p1.z - p2.z ;
      }
  };

void main ()
  {
  Point p1, p2, p3 ;

  p1.x = 1.0 ; p1.y = 2.0 ; p1.z = 3.0 ;
  p2.x = 4.0 ; p2.y = 5.0 ; p2.z = 6.0 ;
  p3.x = 0.0 ; p3.y = 0.0 ; p3.z = 0.0 ;

  // O/P before Add():
  cout << "before Add(): " << endl ;
  cout << "p3.x: " << p3.x << endl ;
  cout << "p3.y: " << p3.y << endl ;
  cout << "p3.z: " << p3.z << endl ;

  p3.Add (p1, p2) ;

  // O/P after Add():
  cout << "after Add(): " << endl ;
  cout << "p3.x: " << p3.x << endl ;
  //...
  }
```

The declaration of type `Point` now includes two member functions *`Add()`* and *`Subtract()`*. These member functions are passed two arguments of type `Point` and manipulate these two `Point`s by adding or subtracting their respective (`x`, `y` and `z`) data members. The result of the addition or subtraction is assigned to the data members of the object that calls the member functions. If this seems unclear at the moment then don't worry – the next chapter will discuss in detail **`class`** member functions.

In C++, structures are generally reserved as a data encapsulation mechanism that holds only data members, whereas the **`class`** is used to define types that hold both data and functions. Thus a more detailed discussion of member functions is left to the next chapter.

8.1.16 Private or Public?

We saw above that, if a structure is declared with global scope, its data members are accessible from any function. In other words, its members are *public*. This may be what we require, but it can be the source of problems. The members of a structure are by default public, and are therefore accessible to an object.

In the previous section two member functions were added to the structure `Point` for adding and subtracting its data members. However, the structure declaration did not correspondingly restrict the access of the structure's data members to the member functions only. Thus, with both data and function members declared public by default, we are still able to manipulate `Point`'s data members from outside the structure declaration:

```
//...
// define and initialise two Points:
p1.x = 1.0 ; p1.y = 2.0 ; p1.z = 3.0 ;
p2.x = 4.0 ; p2.y = 5.0 ; p2.z = 6.0 ;
p3.x = 0.0 ; p3.y = 0.0 ; p3.z = 0.0 ;
//...
Point p3 ;
// p3 = p1 + p2:
p3.x = p1.x + p2.x;
p3.y = p1.y + p2.y ;
p3.z = p1.z + p2.z ;
// or:
p3.Add (p1, p2) ;
//...
```

A further discussion of private and public accessibility will be delayed until the next chapter, when the C++ **`class`** is introduced.

8.1.17 `Point`, `Line` and `Triangle` Structures

Let us make use of the `Point` structure by modelling a line segment and a triangle in a three-dimensional space. A straight line segment, `Line`, can be represented as start and terminal `Points`. A triangle, `Triangle`, can be composed of three vertices and three edges. The following program, PLT.CPP, declares `Point`, `Line` and `Triangle` structures and defines three functions *`Distance()`*, *`TrianglePerimeter()`* and *`TriangleCentroid()`* which calculate the minimum distance between two `Points`, the perimeter of a `Triangle` and the centroid of a `Triangle`, respectively. The perimeter, p, of a triangle is simply the sum of the triangle's edge lengths. The centroid, c, of a triangle (i, j, k) is:

$$c(c_x, c_y, c_z) = \left(\frac{x_i + x_j + x_k}{3}, \frac{y_i + y_j + y_k}{3}, \frac{z_i + z_j + z_k}{3} \right)$$

Figure 8.2 illustrates a triangle in the plane z=0 for which the perimeter and centroid are:

$$p = \frac{2\sqrt{5} + \sqrt{17} + \sqrt{45}}{2}, \quad c = \left(\frac{13}{6}, \frac{7}{3}, 0 \right)$$

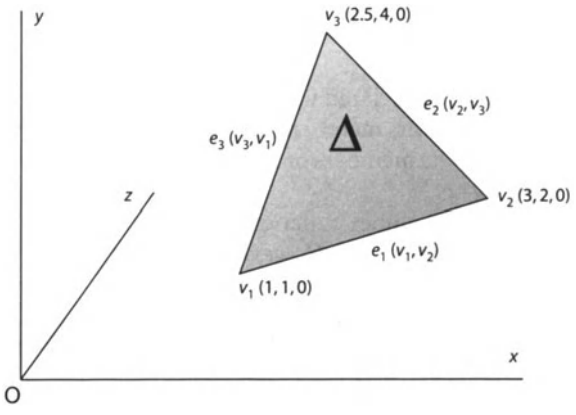


Fig. 8.2 A triangle in the plane $z = 0$.

Program PLT.CPP tests the `Point`, `Line` and `Triangle` structures for the triangle shown in Fig. 8.2:

```
// plt.cpp
// Point, Line and Triangle structures

#include <iostream>  // C++ I/O
#include <cmath>     // sqrt()

struct Point
  {
  double x, y, z ;     // triple coordinates of Point
  };

struct Line
  {
  Point p1, p2 ;       // two end Points of line
  };

struct Triangle
  {
  Point v1, v2, v3 ;  // three vertices of triangle
  Line  e1, e2, e3 ;  // three edges of triangle
  };

// returns the distance between two Points
double Distance (const Point& p1, const Point& p2)
  {
  double x = p2.x - p1.x ;
  double y = p2.y - p1.y ;
  double z = p2.z - p1.z ;

  double r_sq = x * x + y * y + z * z ;
```

```
  return sqrt (r_sq) ;
  }

// returns the perimeter of a triangle
double TrianglePerimeter (const Triangle& t)
  {
  double e1_length = Distance (t.e1.p1, t.e1.p2) ;
  double e2_length = Distance (t.e2.p1, t.e2.p2) ;
  double e3_length = Distance (t.e3.p1, t.e3.p2) ;

  return e1_length + e2_length + e3_length ;
  }

// returns the centroid of a triangle
Point TriangleCentroid (const Triangle& t)
  {
  Point cent ;

  cent.x = (t.v1.x + t.v2.x + t.v3.x) / 3.0 ;
  cent.y = (t.v1.y + t.v2.y + t.v3.y) / 3.0 ;
  cent.z = (t.v1.z + t.v2.z + t.v3.z) / 3.0 ;

  return cent ;
  }

void main ()
  {
  // define a Triangle object tri
  Triangle tri ;

  // set tri's data members:
  // vertices
  tri.v1.x = 1.0 ; tri.v1.y = 1.0 ; tri.v1.z = 0.0 ;
  //...
  // edges
  tri.e1.p1.x = 1.0 ; tri.e1.p1.y = 1.0 ; tri.e1.p1.z = 0.0 ;
  tri.e1.p2.x = 3.0 ; tri.e1.p2.y = 2.0 ; tri.e1.p2.z = 0.0 ;
  //...

  double perimeter = TrianglePerimeter (tri) ;
  Point  cent      = TriangleCentroid  (tri) ;

  // O/P
  cout << "perimeter of triangle: "  << perimeter << endl ;
  cout << "centroid  of triangle: (" << cent.x << ", "
                                     << cent.y << ", "
                                     << cent.z << ")"
                                                  << endl ;
  }
```

with output:

```
perimeter of triangle: 7.65172
centroid of  triangle: (2.16667, 2.33333, 0)
```

The declarations of the `Line` and `Triangle` structures are:

```
struct Line
  {
  Point p1, p2 ;       // two end Points of line
  };

struct Triangle
  {
  Point v1, v2, v3 ;   // three vertices of triangle
  Line  e1, e2, e3 ;   // three edges of triangle
  };
```

Since a `Line` incorporates two `Point`s the `Triangle` structure could have been declared with only three `Line` data members rather than three `Point` and `Line` data members. However, the encapsulation of three `Point` and `Line` data members greatly assists the use of the `Triangle` structure.

Since all of the data members of structures `Point`, `Line` and `Triangle` are **`public`**, by default, the `Point` and `Line` data members of a `Triangle` object, `tri`, are typically accessed by:

```
void main ()
  {
  //...
  Triangle tri ;       // Triangle object tri

  tri.v1.x = 1.0 ;     // x coor. of Point vertex v1 of tri
  //...
  tri.e1.p1.x = 1.0 ; // x coor. of Line e1, Point p1 of tri
  //...
  }
```

8.1.18 Bit Fields

Before discussing the **`union`**, let's take a brief look at bit fields, which are similar in syntax to structures except that they allow a programmer to specify the number of bits for each data member. Consider the following program:

```
// bit_fld.cpp
// illustrates bit fields
#include <iostream>  // C++ I/O

struct X
  {
  unsigned int mem0 : 2 ;
  unsigned int mem1 : 6 ;
```

```
    int mem2 : 2 ;
    };

  void main ()
    {
    X x ;

    cout << "sizeof x: " << sizeof x << endl ;

    // assign values to bit fields of x and display
    x.mem0 = 2 ; x.mem1 = 21 ; x.mem2 = -1 ;
    cout << "x.mem0: "   << x.mem0
         << "; x.mem1: " << x.mem1
         << "; x.mem2: " << x.mem2 << endl ;

    // assign values to bit fields of x and display
    x.mem0 = 4 ; x.mem1 = 64 ; x.mem2 = 4 ;
    cout << "x.mem0: "   << x.mem0
         << "; x.mem1: " << x.mem1
         << "; x.mem2: " << x.mem2 << endl ;
    }
```

with output:

```
sizeof x: 2
x.mem0: 2; x.mem1: 21; x.mem2: -1
x.mem0: 0; x.mem1: 0; x.mem2: 0
```

The syntax of a bit field is similar to that of a structure, except that a colon and bit length are placed between the member name and the semicolon:

```
struct X
  {
  unsigned int mem0 : 2 ;
  unsigned int mem1 : 6 ;

  int mem2 : 2 ;
  };
```

which defines three bit fields to structure X. Bit field mem0 is allocated two bits, mem1 six bits and mem2 two bits. The number of bits allocated to each bit field is completely arbitrary and dependent on the particular application. The total number of bits in the three bit fields of **struct** X is ten (i.e. 1 byte and 2 bits), but note that the **sizeof** operator rounds up the size of object x.

A bit field must be of an integral type. Bit fields mem0 and mem1 are both of type **unsigned int**, whereas bit field mem2 is of type **int**. Because mem0 is allocated two bits and is **unsigned** it can hold values from 0 to 3 (2^4–1), while mem1 can hold values from 0 to 63 (2^6–1), since it is allocated six bits and **unsigned**. Although bit field mem2 is allocated two bits, it is of type **int** and can hold values from –1 to +1 (not 0 to 3), because a bit is reserved for the ± sign. If a value is assigned to a bit field which is outside the range allocated to the field, the result is implementation-dependent. The above program illustrates assigning values to the

three bit fields of x that are outside the allocated ranges. The decision of the compiler used for the above program was to assign a value of 0 to the **unsigned int** and **int** bit fields.

The ability to specify exactly the number of bits for each member of a structure is very important when memory allocation is to be minimised.

8.2 Unions

A **union** is similar to a **struct** in that it is a collection of similar or dissimilar data types, but it holds only one of its members at any one time. The size of a **union** is the size of its largest data member. The general syntax of a **union** declaration is:

```
union Name
  {
  type_specifier member_name ;  // union members
  //...
  type_specifier member_name ;
  };
```

A program example will help explain the **union**:

```
// union.cpp
// illustrates unions
#include <iostream>  // C++ I/O

// declare a union 'UnionTypes'
union UnionTypes
  {
  char   c ;
  int    i ;
  float  f ;
  double d ;
  };

void main ()
  {
  // define a union variable 'type'
  UnionTypes type ;

  // set type's data members
  type.c = 'C' ; type.i = 2 ; type.f = 4.0 ; type.d = 8.0 ;

  // O/P union type as initialised
  cout << "type.c= "   << type.c << ", type.i= " << type.i
       << ", type.f= " << type.f << ", type.d= " << type.d
       << endl ;

  cout << endl ;

  type.c = 'A' ;
```

```
    cout << "type.c= "   << type.c << ", type.i= " << type.i
         << ", type.f= " << type.f << ", type.d= " << type.d
         << endl ;

    type.i = 12 ;
    cout << "type.c= "   << type.c << ", type.i= " << type.i
         << ", type.f= " << type.f << ", type.d= " << type.d
         << endl ;

    type.f = 14.0 ;
    cout << "type.c= "   << type.c << ", type.i= " << type.i
         << ", type.f= " << type.f << ", type.d= " << type.d
         << endl ;

    type.d = 18.0 ;
    cout << "type.c= "   << type.c << ", type.i= " << type.i
         << ", type.f= " << type.f << ", type.d= " << type.d
         << endl ;
    }
```

which generates the output:

```
type.c=  , type.i =0, type.f= 0, type.d= 8

type.c= A, type.i= 65, type.f= 9.10844e-44, type.d= 8
type.c=  , type.i= 12, type.f= 1.68156e-44, type.d= 8
type.c=  , type.i= 1096810496, type.f= 14, type.d= 8
type.c=  , type.i= 0, type.f= 0, type.d= 18
```

The output illustrates that the **union** variable `type` can be used to hold either a **char**, **int**, **float** or **double**, but only one of these four types at any given time. Each of the last four displayed lines has been output directly after the data members `c`, `i`, `f` and `d` of the **union** `UnionTypes` have been assigned the values 'A', 12, 14.0 and 18.0 respectively. The data members that have not been assigned a value hold garbage.

Note that a **union** can only be initialised through the **union**'s first data member:

```
UnionTypes type = { 'Z' } ;
```

8.2.1 sizeof a union

The following program code fragment outputs the size of a **struct** and a **union** with identical data members:

```
//...
// declare a union 'UnionTypes'
union UnionTypes
  {
  char   c ;
  int    i ;
```

```
  float  f ;
  double d ;
  };

// declare a structure 'StructTypes'
struct StructTypes
  {
  char   c ;
  int    i ;
  float  f ;
  double d ;
  };
//...
cout << "sizeof (StructTpyes) : "
     << sizeof (StructTypes) << endl ;
cout << "sizeof (UnionTpyes)  : "
     << sizeof (UnionTypes) << endl ;
```

which would output the following:

```
sizeof (StructTypes): 17
sizeof (UnionTypes) : 8
```

The size of a **struct** is the sum of all of its data members (1+4+4+8=17 for a 32-bit system), whereas the size of a **union** is the size of its largest data member (8). When the UnionTypes variable type holds a **double** no bytes are unused. When type holds a **float** 4 bytes are unused, 6 bytes are unused when type holds an **int** and 7 bytes are unused when type holds a **char**.

Similar to a **struct**, a **union** can be anonymous and can contain bit fields, but only one at any given time. A **union** cannot have **static** data members, although a **union** can have member functions (including constructors and destructors). Unions cannot be used in a **class** hierarchy in that they cannot act as a base **class** or be derived from. The **class**, constructors, destructors and inheritance will be covered in later chapters.

8.3 Enumerations

An **enum** is a user-defined type with named constants of type integer. For example:

```
enum
  {
  RED, GREEN, BLUE
  };
```

defines three constants RED, GREEN and BLUE equal to 0, 1 and 2 respectively. The data members of an **enum** are called enumerators and the default integer constant of the first enumerator in the list of data members is equal to 0. Note that, unlike **struct** and **union**, the members of an **enum** are separated by the comma operator. The above enumeration is equivalent to defining the three constants, RED, GREEN and BLUE, as:

```
const int RED   = 0 ;
const int GREEN = 1 ;
const int BLUE  = 2 ;
```

An enumeration can be named:

```
enum WeekDays
  {
  SUNDAY, MONDAY, TUESDAY, WEDNESDAY,
  THURSDAY, FRIDAY, SATURDAY
  };
```

Alternatively, the enumerators can be assigned constant values or constant expressions during the declaration:

```
enum Hexahedron
  {
  VERTICES = 8,    // no. of vertices
  EDGES    = 12,   // no. of edges
  FACES    = 6,    // no. of faces
  };
```

or, using a constant expression[1]:

```
enum Hexahedron
  {
  VERTICES = 8,                        // no. of vertices
  EDGES    = 12,                       // no. of edges
  FACES    = 2 + EDGES - VERTICES,     // no. of faces
  };
```

The general syntax of the **enum** declaration is:

```
enum Name
  {
  enumerator1, enumerator2, ..., enumeratorn
  };
```

or

```
enum Name
  {
  enumerator1 = constant_expression1,
  enumerator2 = constant_expression2,
  ...,
  enumeratorn = constant_expressionn
  };
```

1 The `Hexahedron` enumeration makes use of Euler's formula $V - E + F = 2$ for a *simple* polyhedron, where V is the number of vertices, E the number of edges and F the number of faces.

Undoubtedly, the two most common uses of **enum** are the implementation of `Switch` and `Boolean` types:

```
enum Switch
  {
  OFF, ON  ;
  };

enum Boolean
  {
  FALSE, TRUE ;
  };
```

`Switch` and `Boolean` lend themselves well to the **enum**, since they are essentially binary operations: on (1)–off (0) or true (1)–false (0). The following program illustrates the implementation and use of the `Boolean` enumeration:

```
// enum.cpp
// illustrates enum
#include <iostream>  // C++ I/O
#include <conio>     // getche()

// Boolean enumeration
enum Boolean
  {
  FALSE, TRUE
  };

// returns TRUE if a vowel passed to function, else FALSE
Boolean Vowel (char c)
  {
  switch (c)
    {
    case 'a':
    case 'e':
    case 'i':
    case 'o':
    case 'u':
      return TRUE ;
    default:
      return FALSE ;
    }
  }

void main ()
  {
  char ch = 'x' ;  // initialise ch

  do
    {
    cout << endl << "enter a letter (non-vowel to stop): " ;
```

```
    ch = getche () ;
    } while (Vowel (ch)) ;
}
```

and user interaction:

```
enter a letter (non-vowel to stop): a
enter a letter (non-vowel to stop): e
enter a letter (non-vowel to stop): z
```

Note the use of *fall-through* in the **switch** statement. The function *Vowel()* returns either TRUE (1) or FALSE (0), depending on whether a user enters a vowel or a consonant. The use of Boolean is more informative than simply using an integer:

```
Boolean Vowel (char c) ;
```

This function signature clearly indicates to a user that the function returns one of two possible values (0 or 1).

Finally, note that since **enum** is a 'type of' **struct** I adopt the same notation of using upper-case for the first letter in the enumeration name, e.g. Colour.

8.4 typedefs

A **typedef** allows a programmer to create a synonym for an existing integral or user-defined data type. **typedef**s are *not* new types.

The following program code illustrates a use of **typedef** to declare a new name, Length, for the integral data type **double**:

```
//...
typedef double Length ;
//...
Length Distance (const Point& p1, const Point& p2)
  {
  //...
  Length length = sqrt (r_sq) ;
  return length ;
  }
```

The **typedef** Length allows a programmer to give a more descriptive name to the return type of a *Distance()* function, which returns the straight distance (a **double**) between the two Points p1 and p2.

typedefs can offer a programmer a convenient shorthand notation for lengthy types or types that are frequently used:

```
typedef char*       String ;   // pointer to char
typedef const char* CString ;  // pointer to const char
```

The disadvantage of **typedef**s is that they introduce additional type names and can make existing types more abstract.

The following program illustrates a use of **typedef** with the Point structure:

```
// typedef.cpp
// illustrates the typedef
#include <iostream>  // C++ I/O

const int NUM_VERTICES = 3 ;

// a Point structure for representing a 3D point
struct Point
  {
  double x, y, z ;
  };

void main ()
  {
  // declare a typedef Polygon array
  // with 3 elements of type Point
  typedef Point Polygon[NUM_VERTICES] ;

  // define a Polygon variable
  Polygon poly ;

  // assign coordinate value to vertices
  poly[0].x = 1.0 ; poly[0].y = 1.0 ; poly[0].z = 0.0 ;
  poly[1].x = 3.0 ; poly[1].y = 2.0 ; poly[1].z = 0.0 ;
  poly[2].x = 3.0 ; poly[2].y = 4.0 ; poly[2].z = 0.0 ;

  // O/P poly vertices
  for (int i=0; i<NUM_VERTICES; i++)
    {
    cout << "poly[" << i << "]: ("
         << poly[i].x << ", "
         << poly[i].y << ", "
         << poly[i].z << ")"
         << endl ;
    }
  }
```

The following statements declare Polygon a synonym for an array of Points and defines a variable poly of type Polygon:

```
typedef Point Polygon[NUM_VERTICES] ;
//...
Polygon poly ;
```

8.4.1 typedefs for Windows

If you have experience with programming for Windows you may have noticed that virtually every data type is a **typedef**. The Windows **typedef**s are defined in the header file WINDOWS.H. A few are listed below:

```
// windows.h/windef.h
//...
typedef unsigned char  BYTE ;
typedef unsigned short WORD ;
typedef unsigned long  DWORD ;
typedef unsigned int   UINT ;
//...
```

These alternative data types were introduced to ease the porting of Windows applications to different architectures. As mentioned in Chapter 4, certain C++ data types are machine-specific and the alternative data types were developed to keep programs as independent as possible of machine architecture. This point is particularly important when developing both 16-bit and 32-bit Windows applications.

WINDOWS.H also contains several applications of the combined use of **typedef** and **struct**. The following program code illustrates two **typedef**s, RECT and POINT, which are structures that encapsulate the integer dimensions of a two-dimensional planar rectangle and the position of a two-dimensional point:

```
// windows.h/windef.h
//...
typedef struct tagRECT
  {
  int left ;
  int top ;
  int right ;
  int bottom ;
  } RECT ;

typedef struct tagPOINT
  {
  int x ;
  int y ;
  } POINT ;
```

The above also serve to illustrate the **typedef struct** construct:

```
typedef struct tagName
  {
  //...
  } Name ;
```

8.4.2 typedefs Offer Little Type Safety

Since a **typedef** is not a new type but a synonym for an alternative type, **typedef**s are weakly typed in C++ and offer little type safety. An example illustrates this:

```
typedef int Int ;
//...
int i = 1 ;
Int j = 2 ;
//...
```

```
j = i ;  // mixing of types int and Int
```

This example illustrates that it is perfectly legal to mix an integral type with a **typedef**. This mixing of data types can be confusing and is prone to errors that are difficult to detect. Other languages, such as Ada and Eiffel, are more strict regarding the typing of fundamental data types than C++.

8.4.3 typedef Style

Several programmers adopt the style of using entirely uppercase letters for **typedef**s. The majority of programmers reserve uppercase identifiers exclusively for constants, and thus it is easy to confuse a **typedef** with a constant. The style that I adopt for naming a **typedef** is the same as for structures, unions, enumerations and classes, and reserves uppercase identifiers for constants.

8.5 Summary

A structure allows data members of the same or different types to be encapsulated in a single implementation, unlike arrays, which are aggregates of a single data type. The data members of a structure are, by default, publicly accessible, although a structure can have different member access rights other than **public**. In addition, a structure can in fact encapsulate functions which operate on the structure's data, but such capabilities are generally reserved for the **class**.

An important feature of user-defined structure types is their equivalence with C++'s integral data types. In the C programming language the name of a structure must be preceded by the **struct** keyword:

```
struct Person p ;  // C style
double d ;
```

C++ neglects the **struct** keyword:

```
Person p ;  // C++ style
double d;
```

which illustrates that there is no syntactical difference made between integral and user-defined data types. Similarly, user-defined structures have the same capabilities as integral types with regard to definition, initialisation, arrays, function arguments and return type etc. In other words, and as will become apparent in the following chapters, a user-defined type can be designed to be as powerful and flexible as a C++ integral type.

The **union** data type is similar to **struct** in that it is a collection of data members of similar or different types, but unlike a **struct** definition, which allocates sufficient memory to hold all data members, a **union** can hold only one data member at any given time, and the size of a **union** is therefore the size of its largest member.

An enumerated list of integers can be generated in C++ by the enumeration keyword **enum**. Enumerated types are useful in assisting a programmer to attribute identifiers to constant values when the members of a data type map exactly to a set of integers, such as the `Boolean` **enum** with literals `FALSE` (0) and `TRUE` (1). Associating meaningful identifier names with

integer values can increase the clarity of programs, as illustrated by the combined use of **enum**s and functions which return a `Boolean` logical value rather than an integer.

A **typedef** is not a new data type but a synonym for an existing type. The use of **typedef**s can introduce convenient, more descriptive, names to types.

Exercises

8.1 Declare `Address` and `Telephone` structures to model a person's address and telephone number. Modify the `Person` structure presented in this chapter to include both `Address` and `Telephone` data members.

8.2 Declare a `City` structure which encapsulates the name, county, country and population of a city.

8.3 Debug the following program:

```
#include <iostream>  // C++ I/O

struct Point
  {
  double x, y, z ;
  };

void main ()
  {
  Point p ;

  p.x = 1.1 ; p.y = 2.3 ; p.z = 0.0 ;

  cout << "Point p: " << p << endl ;
  }
```

8.4 Declare a `Circle` structure to represent a circle. Use structure `Point` in your declaration of `Circle`.

8.5 Develop a `Line` structure which uses a `Point` structure to model a straight line segment. Also, develop a `Triangle` structure which uses both `Point` and `Line` structures to model a triangle. Develop three functions which calculate the perimeter, centroid and area of a general three-dimensional triangle which is described in terms of three edges (`Lines`) and three vertices (`Points`). Test your program on a triangle whose three vertices are (1, 1, 0), (3, 2, 0) and (2.5, 4, 0).

The centroid, $c\,(c_x, c_y, c_z)$, of a triangle defined by the three points $i\,(x_i, y_i, z_i)$, $j\,(x_j, y_j, z_j)$ and $k\,(x_k, y_k, z_k)$ is given by:

$$c = \left(\frac{x_i + x_j + x_k}{3}, \frac{y_i + y_j + y_k}{3}, \frac{z_i + z_j + z_k}{3} \right)$$

Hint: Use the *`Distance()`* function in program POINT.CPP to enable you to calculate the perimeter length of the triangle:

```
double Distance (const Point& p1, const Point& p2) ;
```

8.6 Declare an **enum** called `Colours` which encapsulates various colours. Compare your `Colours` with the `COLORS` **enum** in the C++ library header file CONIO.H.

8.7 Declare a **typedef** called `Percentage` for representing percentages.

Chapter 9

The C++ Class

```
class Foo { /* ... */ };
```

The class is the key element for object-oriented programming in C++. Classes enable a programmer to create new types that have all the features of integral types. In fact, by applying the full features of classes, user-defined types can be designed to be as powerful, well-behaved and 'concrete' as integral types such as int.

The class is analogous to the structure, in that a class is a data encapsulation mechanism, but with the additional support of functions specific to the class which operate on the class's data. Both class data and member functions can have associated access rights which allow the designer of a class to either allow or restrict access to specific data members to a non-class member or an object of the class. Thus, a class can be designed with an interface to non-class members which encapsulates the functionality of the class and at the same time hides the inner details of the class.

A class defines a mould from which objects or instances can be created, just as Jean is a particular instance of the class Person. An object is a single, unique entity that comprises both data and member functions which operate on their data.

Throughout this chapter, emphasis is placed on the development of a single class called Point which characterises a point in three-dimensions. As the chapter progresses we shall steadily see the functionality of class Point grow, so that when the chapter concludes Point will be a simple yet powerful class which reflects the majority of features that are characteristic of all C++ classes.

It is important to develop a good understanding of the class because it is the cornerstone of object-oriented programming in C++ and will play an increasing role in subsequent chapters.

9.1 A `Point` `class`

This chapter introduces the C++ **`class`**, which is used in the creation of new user-defined types. Classes are introduced with the help of a `Point` **`class`** which is an excellent **`class`** to illustrate the key features of classes and is essential for graphics programming in C++.

`Point`-style classes are widely used in computer graphics. For example, in programming for Windows use is frequently made of the `POINT` structure defined in the WINDOWS.H header file:

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

```
// windows.h/windef.h
//...
typedef struct tagPOINT
  {
  int x ;
  int y ;
  } POINT ;
```

The **typedef** POINT is extensively used to model a point on a two-dimensional display screen. Note that tagPOINT's data members are of type integer.

The Borland C++ (version 5.x) compiler supports an ObjectWindows library **class** TPoint. The **class** TPoint inherits the tagPOINT x and y data points and provides a sufficient number of overloaded operators to enable a natural use of TPoint objects:

```
// ...\WINSYS\GEOMETRY.H
//...
class  TPoint : public tagPOINT
  {
  public:
    // constructors
    TPoint () {}
    //...
  };
```

Both tagPOINT and TPoint x and y data members are **public**. The distinction between **public** and **private** will be discussed later.

In this chapter we will be interested in a more accurate representation of a Point for general three-dimensional geometric modelling. Therefore, there are three data members for our Point **class**, each of type **double**. Type **double** can at first appear rather memory-expensive when compared with type **float**, but type **double** is necessary for the required level of accuracy. When we examine the C++ **template** we will observe that a generic Point **class** can be declared with data members whose types are determined when an object of the Point **class** is defined.

Figure 9.1 illustrates an arbitrary point, p (p_x,p_y,p_z), in a three-dimensional space. The position of point p is given relative to a reference point or origin O. The origin O can have any assigned position, but it is generally assumed that O has coordinates (0, 0, 0). Point O is the origin of a three-dimensional coordinate system that consists of three mutually perpendicular straight lines or axes. Such a rectangular coordinate system is referred to as a *Cartesian* coordinate system.

Let's begin our examination of the Point **class** by copying the Point structure presented in the program POINT.CPP in Chapter 8 and simply replacing the keyword **struct** in the Point structure declaration by the keyword **class**:

```
// class_1.cpp
// introduces the class
#include <iostream>  // C++ I/O

// Point class declaration
class Point
  {
  double x, y, z ;
```

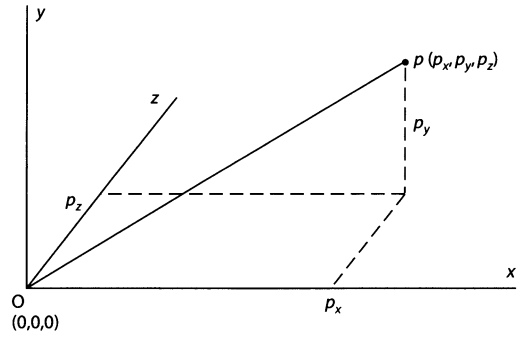


Fig. 9.1 A point *p* in a three-dimensional Cartesian coordinate system (*x*, *y*, *z*).

```
};

void main ()
  {
  // object p of class Point
  Point p ;

  p.x = 1.0 ;  // error: Point's data members not accessible
  p.y = 2.0 ;
  p.z = 3.0 ;
  }
```

Unfortunately, the above program will not compile. So why include a program that doesn't even compile? This program does not compile because the data members of **class** `Point` are not accessible to an object, `p`, of **class** `Point`. To see why let's first look at the **class** declaration of `Point`:

```
class Point
  {
  double x, y, z ;
  };
```

It was noted in Chapter 8 that the data members of a **struct** are by default **public**. Thus an object of **struct** `Point` has access to `Point`'s three data members. The data members of a **class** are by default **private**, which means that a **class**'s data members can only be accessed from within the **class**. Although **private** is the default, it is good programming practice to indicate explicitly which members are **private**:

```
class Point
  {
  private:
    double x, y, z ;
  };
```

Note the colon (:) directly after the keyword **private**.

private and **public** are keywords that signal which data members are only accessible from within a **class** or have general outside access, respectively. For example, the following

class declaration illustrates that the data member `pri_data` is not accessible to `object` but `pub_data` is:

```
// pri_pub.cpp
// illustrates private and public data members
#include <iostream>  // C++ I/O

// Name class declaration
class Name
  {
  private:              // private data member
    int pri_data ;
  public:               // public data member
    int pub_data ;
  };

void main ()
  {
  // define object of class Name
  Name object ;

  object.pub_data = 1 ; // ok:    public data member
                        //        is accessible

  object.pri_data = 1 ; // error: private data member
                        //        is not accessible
  }
```

Within the delimiting braces { and } of the **class** declaration, all data members following the keyword **private** are **private** unless the keyword **public** is encountered, and vice versa. In addition, several **private** or **public** keywords can be placed within a **class** declaration:

```
class Name
  {
  private:
    int pri_data1 ;
    int pri_data2 ;
  public:
    int pub_data1 ;
  //...
  private:
    int pri_data3 ;
    //...
  //...
  public:
    int pub_data2 ;
    //...
  //...
  };
```

This can be a useful feature when dealing with a large **class** declaration, and is in agreement with C++'s *define-anywhere* style of defining an identifier wherever it is required, rather than the C *define-at-top* style.

So why the emphasis on **private** and **public**? The keyword **private** enables a programmer to hide certain data or algorithms specific to a **class**'s implementation from external variables, objects, classes, functions etc. Data hiding is one of the key features of the C++ **class** and of object-oriented programming and restricts the access of data that is prescribed as **private**.

9.2 `class` Declaration Syntax

As a recap let's look at the general syntax of a **class** declaration. Later, this syntax will be extended to incorporate member functions, constructors and destructors, but for now here it is:

```
class ClassName
  {
  type_specifier pri_member_name ;    // class private members
  //...
  private:
    type_specifier pri_member_name ; // class private members
    //...
  public:
    type_specifier pub_member_name ; // class public members
    //...
  } object_list ;
```

The **class** declaration begins with the keyword **class**. This is directly followed by the user-defined name of the **class**. Data members are by default **private**, but can be explicitly indicated to be **private** by the keyword **private**. Public data members are similarly preceded by the **public** keyword. Both of the **private** and **public** keywords have an associated colon (:). **private** or **public** data members can be of either an integral or a valid user-defined type. The syntactical rules of naming **private** or **public** data members are identical to those of naming *normal* variables and objects. The body of a **class** is delimited by the braces { and } with the closing brace, }, directly followed by a semicolon (;). It was noted in Chapter 8, when we discussed declaring and defining a structure in a single statement, that we could define an object between the closing brace of a structure declaration and the semicolon. Similarly, it is possible to declare and define a **class** in a single statement:

```
class Point
  {
  //...
  } p, q, r ;
```

which defines the objects p, q and r of **class** Point.

The style adopted for **class** declarations is similar to that of **struct** declarations. The **class** body braces are indented a single horizontal tab position from the **class** keyword. Similarly, the **private** and **public** members of a **class** are indented a single tab position from the **private** and **public** keywords. If both of the keywords **private** and **public**

appear only once within a **class** declaration, I always place the **class**'s **private** data members at the top of the declaration. Different people adopt different styles, and one of the more popular alternatives is:

```
class Name {
  int pri_data ;
  //...
public:
  int pub_data ;
  //...
};
```

As always, choose what's best for you.

9.3 Objects and Instances

Above we saw the following definition:

```
Point p ;
```

This defines an *object* or *instance* p of **class** Point. Object p is a particular instance of **class** Point. The class Point is a type of *mould* that encapsulates the essential features of a three-dimensional point in space, whereas we act on an object p that has Point's characteristics but holds its own unique set of data.

9.4 A public Point class

Having discussed the keywords **private** and **public**, let's now return to the Point **class** presented in program CLASS_1.CPP, but now make **class** Point's data members **public** so that they are accessible to objects of Point:

```
// pubpoint.cpp
// a Point class with public data members
#include <iostream>  // C++ I/O

class Point
  {
  public:              // public data members
    double x, y, z ;
  };

void main ()
  {
  // define objects p and q
  Point p, q ;
```

```
    // assign values to objects p and q data members
    p.x = 1.0 ; p.y = 2.0 ; p.z = 3.0 ;
    q.x = 4.0 ; q.y = 5.0 ; q.z = 6.0 ;

    // O/P objects
    cout << "object p ("
         << p.x << ", " << p.y << ", " << p.z << ")" << endl
         << "object q ("
         << q.x << ", " << q.y << ", " << q.z << ")" << endl ;
    }
```

with output:

```
object p (1, 2, 3)
object q (4, 5, 6)
```

The output of program PUBPOINT.CPP illustrates that the **public** data members x, y and z of **class** Point are now accessible to the objects p and q, which are both of type Point. Thus, the **class** Point now acts as a **struct** since data members of a **struct** are **public** by default.

Objects p and q are defined in the statement:

```
Point p, q ;
```

and then values are assigned to each object's data members:

```
p.x = 1.0 ; p.y = 2.0 ; p.z = 3.0 ;
q.x = 4.0 ; q.y = 5.0 ; q.z = 6.0 ;
```

Figure 9.2 schematically illustrates the two objects p and q. Each object has its own x, y and z data members. A common source of confusion to programmers who are new to object-oriented programming is thinking that several different objects of a given **class** all access the same data, as shown in Fig. 9.3. This incorrect conceptualisation is analogous to the idea of different functions accessing the same global data; this is also illustrated in Fig. 9.3.

It must be reiterated that:

Each object carries its own unique set of data.

An object is a single entity with its own characteristics and not simply a pointer to some global data bank. To illustrate this important point further, try adding the following two lines to the above program PUBPOINT.CPP:

```
cout << "sizeof p: " << sizeof p << endl
     << "sizeof q: " << sizeof q << endl ;
```

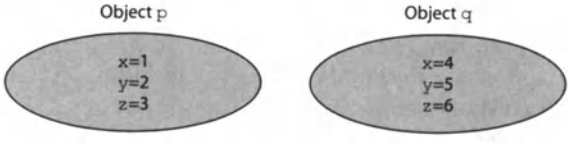


Fig. 9.2 Two objects p and q of **class** Point.

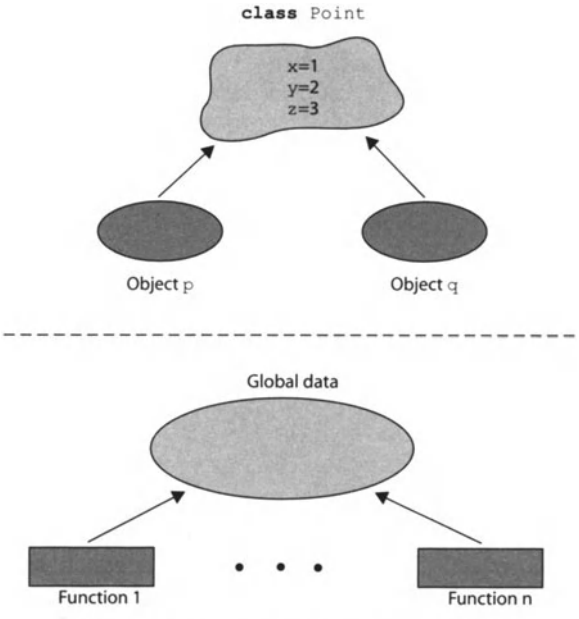


Fig. 9.3 Incorrect conceptualisation of objects.

which will generate the output:

```
sizeof p: 24
sizeof q: 24
```

This output illustrates that the size of each object is 24 (3 × 8) bytes, since the size of type **double** is 8 bytes and **class** `Point` has three **double** data members. Thus each object is allocated sufficient storage to hold its own data.

Also, when objects `p` and `q` are defined to be of type `Point`, the **class** `Point` declaration does not alter. The **class** `Point` declaration is a mould or a specification for `Point` objects.

9.5 Member Functions

In the previous section, the data members of **class** `Point` were made **public** so that an object has access to the data members. This enabled us to assign values and to access (via the `cout` output stream object) each object's data members. Thus, our present design of the `Point` **class** appears to be working well, but there is a problem! Assigning **public** access to `Point`'s data members unfortunately makes them easily accessible to objects of **class** `Point` from outside the **class** declaration. This leaves `Point`'s data members vulnerable to accidental alteration.

When learning a programming language, via small programming exercises, for the first time it is difficult to appreciate the importance of data hiding. As programming projects become larger, with a corresponding increase in the number of object interactions, it becomes increasingly important to reduce, if not eliminate entirely, the risk of an object being accidentally altered or altered in ways not intended by the designer of the object's **class**. By making

certain **class** data members **private** we can prescribe the manner in which an object's data is to be manipulated by specifying a **public** interface for the object.

So how do we solve the dilemma of assigning **private** access to a **class**'s data members while at the same time making them accessible to an object? The answer lies in **class** *member functions*[1]. Assigning **private** access to a **class**'s data members and simultaneously assigning **public** access to member functions allows an object *indirect* access to its data members while ensuring data-member safety. Another reason for the hiding of data members is to ensure that the details of a **class**'s implementation are not disclosed. To illustrate this let's extend the `Point` **class** by making the `x`, `y` and `z` data members **private** and introducing *set* and *get* member functions of **class** `Point`:

```
// mempoint.cpp
// illustrates class member functions
#include <iostream>  // C++ I/O

class Point
  {
  private:              // private data members
    double x, y, z ;
  public:               // public member functions
    void SetX (double x_arg)
      { x = x_arg ; }
    void SetY (double y_arg)
      { y = y_arg ; }
    void SetZ (double z_arg)
      { z = z_arg ; }
    double GetX ()
      { return x ; }
    double GetY ()
      { return y ; }
    double GetZ ()
      { return z ; }
  };

void main ()
  {
  // define objects p and q
  Point p, q ;

  // assign values to objects p and q data members
  // by aid of Point's member functions
  p.SetX (1.0) ; p.SetY (2.0) ; p.SetZ (3.0) ;
  q.SetX (4.0) ; q.SetY (5.0) ; q.SetZ (6.0) ;

  // O/P objects
  // access objects data members
  // by aid of Point's member functions
  cout << "object p ("
```

1 Member functions are frequently referred to as *methods*, *operations* or *messages* in the literature.

```
        << p.GetX () << ", " << p.GetY () << ", "
        << p.GetZ () << ")" << endl
        << "object q ("
        << q.GetX () << ", " << q.GetY () << ", "
        << q.GetZ () << ")" << endl ;
}
```

There are two types of member function in **class** Point, *Set?()* and *Get?()*, where ? is equal to X, Y or Z. The following illustrates these for setting and getting the x data member:

```
void SetX (double x_arg)
  { x = x_arg ; }
//...
double GetX ()
  { return x ; }
```

Note that all member functions are, at present, defined within the **class** declaration, i.e. within the delimiting braces of the **class** declaration. The member functions are function definitions, and defined within the **class** declaration are *inline* functions; refer to Chapter 6. For small member functions, such as the *set* and *get* functions above, it is convenient to define them within a **class** declaration, but for functions with larger function bodies a member function definition is generally placed outside the **class** declaration. Out-of-line member function definitions are described later.

The *SetX()* member function sets the value of the x data member of **class** Point to the function argument, x_arg, that is passed by value. The *GetX()* member function simply returns an object's x data member.

If you recall from Chapter 6, which dealt with *normal* functions, C++ allows us to overload functions. Thus, we are able to assign different operations to the same function name. This is illustrated in the following program:

```
// mem_over.cpp
// illustrates overloaded class member functions
#include <iostream>  // C++ I/O

class Point
  {
  private:              // private data members
    double x, y, z ;
  public:               // public member functions
    void X (double x_arg)
      { x = x_arg ; }
    void Y (double y_arg)
      { y = y_arg ; }
    void Z (double z_arg)
      { z = z_arg ; }
    double X ()
      { return x ; }
    double Y ()
      { return y ; }
    double Z ()
      { return z ; }
```

```
};

void main ()
  {
  // define objects p and q
  Point p, q ;

  // assign values to objects p and q data members
  // by aid of Point's member functions
  p.X (1.0) ; p.Y (2.0) ; p.Z (3.0) ;
  q.X (4.0) ; q.Y (5.0) ; q.Z (6.0) ;

  // O/P objects
  // access objects data members
  // by aid of Point's member functions
  cout << "object p (" << p.X () << //...
  }
```

Overloading the member functions `X()`, `Y()` and `Z()` has now eliminated the need for the *set* and *get* member function names.

You may be thinking that using member functions is a long-winded way of accessing a **`class`**'s data members. However, for the increased safety that member functions offer a programmer they are well worth it. In fact, let's compare the two opposing views of making `Point`'s data members **`private`** and **`public`**:

```
// public class data members:
Point p ;
//...
p.x = 1.0 ;
//...
cout << "object p (" << p.x //...
```

or

```
// private class data members
Point p ;
//...
p.X (1.0) ;
//...
cout << "object p (" << p.X() //...
```

Put this way, we see that there is little extra effort required from a user's perspective by making `Point`'s data members **`private`**, although greater implementation is required. Further, when we examine constructors shortly we shall see that objects can be defined and initialised more cleanly:

```
// private class data members
Point p (1.0, 2.0, 3.0) ;
//...
cout << "object p (" << p.X() //...
```

9.6 Naming of Member Functions

The convention adopted for the naming of **class** member functions is identical to the naming of *normal* functions, in that the first letter(s) of single or multi-word names are in uppercase. For example:

```
p.GetX () ;
```

9.7 Calling Member Functions

In the program MEM_OVER.CPP an object p of **class** Point called the member function *X()* with the following statement:

```
Point p ;
//...
p.X () ;
```

Notice that object p is connected to **class** Point's member function *X()* by the *dot operator.* It makes no sense to write the following statement:

```
//...
X () ;
```

because the function *X()* is a member of **class** Point.

The **public** member function *X()* is accessible only to objects of **class** Point, and thus the function call is associated with an object p of **class** Point by the dot operator or the *member access operator* (.). In general a **class**'s member function, *MemberFunction()*, is accessed by an object of the **class** by:

```
object.MemberFunction (parameter_list) ;
```

The function call:

```
p.X (1.0) ;
```

causes the x data member of object p to be assigned the value of 1.

9.8 Defining Member Functions

The program MEM_OVER.CPP defined Point's member functions *X()*, *Y()* and *Z()* within the **class** declaration:

```
// mem_over.cpp
//...
class Point
   {
```

```
  private:              // private data members
    double x, y, z ;
  public:               // public member functions
    void X (double x_arg)
      { x = x_arg ; }
    //...
  };
```

This is OK, but as the number of member functions increases and the size of certain member functions increases due to increased complexity it is desirable to define member functions outside a **class** declaration. The following program illustrates this for the `Point` **class** presented in the file MEM_OVER.CPP:

```
// outside.cpp
// a Point class with member functions
// defined outside of the class declaration
#include <iostream>  // C++ I/O

class Point
  {
  private:              // private data members
    double x, y, z ;
  public:               // public member functions
    void X (double x_arg) ;
    //...
    double X () ;
  };

// class Point member function definitions:

void Point::X (double x_arg)
  { x = x_arg ; }
//...
double Point::X ()
  { return x ; }
//...

void main ()
  {
  // define objects p and q
  Point p, q ;

  // assign values to objects p and q data members
  // by aid of Point's member functions
  p.X (1.0) ; p.Y (2.0) ; p.Z (3.0) ;
  q.X (4.0) ; q.Y (5.0) ; q.Z (6.0) ;

  // O/P objects
  // access objects data members
  // by aid of Point's member functions
```

```
cout << "object p (" << p.X () << //...
}
```

Observe how elegant our `Point` **`class`** declaration is in the above program now that the member functions are defined outside the **`class`** declaration. The member function *`X()`* is now declared within the `Point` **`class`** declaration:

```
class Point
  {
  //...
  public:              // public member functions
    void X (double x_arg) ;
    //...
  };
```

Such a member function declaration indicates to the compiler that the member function *`X()`* is defined outside the **`class`** declaration. In the program OUTSIDE.CPP the member function definition of *`X()`* is placed after the **`class`** declaration:

```
//...
void Point::X (double x_arg)  // X() definition
  { x = x_arg ; }
```

The syntax of a member function declarator defined outside a **`class`** declaration is:

```
return_type ClassName::MemberFunction (parameter_list)
```

The first part of the declarator is the return type of the member function and is then followed by the **`class`** name to which the member function belongs. Two colons placed together (`::`) are called the *scope resolution operator.* If `ClassName::` is removed from the member function declarator we return to a normal function declarator.

In addition, a **`class`** declaration is conventionally placed in a header file, .H, and the member function definitions are placed in an implementation file, .CPP. Consider the `Point` **`class`** declaration placed in the header file PT_LIB.H:

```
// pt_lib.h
// header file for Point class

#ifndef _PT_LIB_H  // prevent multiple includes
#define _PT_LIB_H

class Point
  {
  private:             // private data members
    double x, y, z ;
  public:              // public member functions
    void X (double x_arg) ;
    //...
  };
```

```
#endif // _PT_LIB_H
```

and the **class** member functions defined in the file PT_LIB.CPP:

```
// pt_lib.cpp
// implementation file for Point class
#include "pt_lib.h"  // Point class

// class Point member function definitions:

void Point::X (double x_arg)
  { x = x_arg ; }
//...
```

We can now simply include the header file PT_LIB.H in a program MY_PROG.CPP which defines objects p and q of **class** Point and assigns and accesses their respective x, y and z data members:

```
// my_prog.cpp
// application of Point class
#include <iostream>  // C++ I/O
#include "pt_lib.h"  // Point class header file

void main ()
  {
  // define objects p and q
  Point p, q ;

  // assign values to objects p and q data members
  // by aid of Point's member functions
  p.X (1.0) ; p.Y (2.0) ; p.Z (3.0) ;
  q.X (4.0) ; q.Y (5.0) ; q.Z (6.0) ;

  // O/P objects
  // access objects data members
  // by Point's member functions
  cout << "object p (" << p.X () //...
  }
```

The separation of **class** declarations and implementations into separate files is important for large programming projects which contain numerous separate program components developed independently. Although different classes and program components will interact in a program, they are generally developed independently, either one at a time by a single programmer or simultaneously by a group of programmers. The development of classes from problem domain abstractions can be similarly reflected in the organising of program files arranged in **class** hierarchies. Placing **class** declarations in header files not only assists in the reuse of classes, but eliminates entire program recompilations (which can be a time-consuming process for large programming projects) resulting from a change in a single **class**. Separate program components can be compiled and tested largely independently and brought together into a single executable program with Link and Make utilities.

9.9 `static` Data Members and Member Functions

A data member or member function can be declared as **`static`** in a **`class`**'s declaration. To illustrate the implications of **`static`** data and function members consider the following program, which is a modification of program OUTSIDE.CPP:

```
// static.cpp
// illustrates static data and function members of a class
//...
class Point
  {
  private:              // private static data members
    static double x, y, z ;
  public:               // public member functions
    //...
    // static member function
    static void Member () ;
  };

// class Point member function definitions:
//...
void Point::Member ()
  {
  cout << "(" << x << ", " << y << ", " << z << ")" << endl ;
  }

// global definitions of Point's data members
double Point::x ;
double Point::y ;
double Point::z ;

void main ()
  {
  // define objects p and q
  Point p, q ;

  // O/P objects
  cout << "object p (" << p.X () //...

  // assign values to object p
  p.X (1.0) ; p.Y (2.0) ; p.Z (3.0) ;

  // O/P objects
  cout << "object p (" << p.X () //...

  p.Member () ;
  }
```

with output:

```
object p (0, 0, 0)
object q (0, 0, 0)
object p (1, 2, 3)
object q (1, 2, 3)
(1, 2, 3)
```

The x, y and z data members of **class** Point are declared as **static**:

```
class Point
  {
  private:              // private static data members
    static double x, y, z ;
  //...
  };
```

The implication of declaring a **class**'s data members **static** is that **static** data members of a **class** are shared by *all* objects of the **class**. The **static** data members of an uninitialised object are initialised to zero. Thus, the above program output illustrates that following the uninitialised definitions of objects p and q in *main()*, the x, y and z data members of objects p and q are zero. Next, floating-point values are assigned to the data members of object p in *main()*:

```
//...
p.X (1.0) ; p.Y (2.0) ; p.Z (3.0) ;
```

but the output illustrates that the x, y and z data members of both objects p and q have been set to 1, 2 and 3, respectively. This is because, as mentioned above, a **class**'s **static** data members are shared by all objects of a **class**. Hence, if we alter the data members of object p we also alter the data members of object q.

It is important to note that when a **class**'s data members are declared as **static** they are not defined. Global definitions must be provided for each **static** data member of a **class**. This is performed in program STATIC.CPP by the following global definitions:

```
// global definitions of Point's data members
double Point::x ;
double Point::y ;
double Point::z ;
```

The scope resolution operator (::) associates each **static** data member with its respective **class**.

Program STATIC.CPP also declares a **static** member function of **class** Point:

```
static void Member () ;
```

This function simply displays an object's data members. Static member functions only have access to **static** data members, cannot access non-**static** data members and cannot call non-**static** member functions.

Static data and function members are rarely used, but there are occasions when they prove to be very useful. For instance, **static** data members can eliminate the use of global variables.

If you have wondered whether you can use the **auto**, **extern** and **register** storage class specifiers as **class** data members, the answer is simply no:

```
// au_ex_re.cpp
// illustrates that the storage class specifiers
// auto, extern and register cannot be used for
// declaring class data members

class X
  {
  private:
    auto     int data0 ;  // error
    extern   int data1 ;  // error
    register int data2 ;  // error
  };
```

9.10 const and mutable Data Members

The data members of a **class** can be declared as **const**:

```
class X
  {
  private:
    const int c_data ;
  public:
    X ()
      : c_data (0) {}
    //...
  };
```

Note, however, that a data member declared as **const** must be initialised in the constructor declarator and cannot be initialised via an assignment statement in the body of the constructor.

The keyword **mutable** is used to allow a member of an object to cast away **const**:

```
// con_mut.cpp
// illustrates const and mutable data members

class X
  {
   public:
      int          data ;
      const int    c_data ;  // const data member
      mutable int m_data ;  // mutable data member

      mutable const  int mc_data ;  // error: mutable const
      mutable static int ms_data ;  // error: mutable static
      X ()
        : data (0), c_data (0), m_data (0) {}
```

```
};

void main ()
  {
  X       x ;   // non-const object
  const X cx ;  // const object

  x.data    = 1 ;  // ok: non-const object, non-const member
  x.c_data  = 2 ;  // error: non-const object, const member
  cx.data   = 3 ;  // error: const object, non-const member
  cx.c_data = 4 ;  // error: const object, const member

  // ok: mutable member of const object can be modified
  cx.m_data = 5 ;
  }
```

The above program illustrates that a **const class** data member or **const** object cannot be altered. However, if a data member is declared **mutable** then the data member of a **const** object is not **const** and can therefore be modified. The **mutable** keyword is available because when a **const** object is defined this should not necessarily imply that the object's data members are **const**. The above program also illustrates that the **mutable** specifier cannot be applied to **const** or **static class** data members.

Note in the above **class** declaration that since `X` declares a **const** data member (`c_data`) and **const** objects must be initialised when they are defined, a constructor is required to initialise the `X::c_data` data member explicitly. **const** member functions are discussed separately in a later section.

9.11 `volatile` Data Members

The data members of a **class** can be declared **volatile**:

```
// vol.cpp
// illustrates volatile data members
#include <iostream>  // C++ I/O

class X
  {
  public:
    int                data ;
    volatile int       v_data ;
    const volatile int cv_data ;
    X ()
      : data (0), v_data (0), cv_data (0) {}
  };

void main ()
  {
  X x ;
  }
```

The above program also illustrates that a data member can be both **const** and **volatile**. **volatile** member functions are discussed in a later section.

9.12 Bit Fields

Before discussing constructors in the next section let us just note that a **class** declaration, similarly to a **struct** declaration, can declare bit fields. The following program is the **class** version of BIT_FLD.CPP presented in Chapter 8:

```
// bit_fld.cpp
// illustrates bit fields

class X
  {
  private:
    unsigned int mem0 : 2 ;
    unsigned int mem1 : 6 ;
  public:
    X ()
      : mem0 (0), mem1 (0) {}
  };

void main ()
  {
  X x ;
  }
```

9.13 Constructors

To date, we have defined objects of **class** `Point` and then assigned values to each object's data members by using member functions. For instance, the following is a typical extract from OUTSIDE.CPP:

```
//...
// define objects p and q
Point p, q ;

p.X (1.0) ; p.Y (2.0) ; p.Z (3.0) ;
q.X (4.0) ; q.Y (5.0) ; q.Z (6.0) ;
```

C++ provides *special* member functions called constructors[2] which allow us to initialise an object's data members when an object is defined. Let us illustrate the use of a constructor on our `Point` **class**:

2 Member functions are frequently referred to as *instance methods*, since they operate on objects of a **class**, whereas constructors are *class methods*.

```
// constr.cpp
// illustrates constructors
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructor
    Point (double x_arg, double y_arg, double z_arg) ;
    // public member functions
    double X () ;
    //...
  };

// constructor
Point::Point (double x_arg, double y_arg, double z_arg)
  {
  x = x_arg ;
  y = y_arg ;
  z = z_arg ;
  }

// class Point member function definitions:
double Point::X ()
  { return x ; }
//...

void main ()
  {
  // define objects p and q
  Point p (1.0, 2.0, 3.0) ;
  Point q (4.0, 5.0, 6.0) ;

  // O/P objects
  cout << "object p (" << p.X () //...
  }
```

The `Point` constructor declaration is:

```
Point (double x_arg, double y_arg, double z_arg) ;
```

which has three arguments that initialise the three data members of **`class`** `Point`. The constructor name is the same as the **`class`** name – this is how the compiler knows which member functions are constructors. Therefore, a member function cannot have the same name as the **`class`** name unless it is a constructor. In general, a **`class`** and constructor declaration are of the form:

```
class ClassName
  {
```

```
  private:
    //...
  public:
    // constructor
    ClassName (parameter_list) ;
    //...
  };
```

The `Point` **class** declaration above indicates that constructors do not return a value, since there is no return type. An attempt to declare a constructor with the keyword **void** will result in a compilation error of the form 'constructor cannot have a return type':

```
void Point () ;  // error: constructor cannot have a return
                 // type
```

Similarly, it makes no sense to call a constructor as if it is an *ordinary* member function:

```
cout << p.Point() ;  // error: Point() member function does
                     // not exist
```

Such a call will generate a compilation error of the form 'member function does not exist'.

Since the three-argument constructor of **class** `Point` defines an object, the set *X()*, *Y()* and *Z()* member functions are not required. The get *X()*, *Y()* and *Z()* access member functions are still required to access the `x`, `y` and `z` data members of a `Point` object.

Constructors must generally be declared as **public**. To illustrate why, consider the following:

```
class Point
  {
  private:
    double x, y, z ;
    // private constructor
    Point (double x_arg, double y_arg, double z_arg) ;
    //...
  public:
    //...
  };
//...
Point q (4.0, 5.0, 6.0) ;  // error: constructor not
                           // accessible
```

The definition of object `q` will produce a compilation error of the form '`Point::Point()` constructor is not accessible'. Since object `q` is defined outside the `Point` **class** declaration, the three-argument constructor has to be **public**ly accessible.

Note that constructors can be defined outside the **class** declaration. Constructors, like member functions, can be defined inside or outside a **class**'s declaration.

9.13.1 Constructors are Called Automatically

Provided **class** `Point` implements the appropriate constructor, a definition of an object `p` of **class** `Point`, given by:

```
Point p (1.0, 2.0, 3.0) ;
//...
```

will instruct the compiler to call the respective constructor when an object is created. The general syntax for defining an object is:

```
ClassName object (parameter_list) ;
```

Thus, constructors are called automatically and there is no need to define additional member functions explicitly for object initialisation.

Alternatively, object p could be defined via assignment:

```
Point p = Point (1.0, 2.0, 3.0) ;
```

However, such object definitions are rarely used, since the previous form offers a more concise definition.

It is worth noting that the above object definition–assignment statement:

```
Point p = Point (1.0, 2.0, 3.0) ;
```

is actually converted to:

```
Point p (1.0, 2.0, 3.0) ;
```

We shall continue the discussion of implicit conversions when we examine conversion functions and converting constructors in Chapter 10.

9.13.2 Default Initialisation

The above use of a three-argument constructor for the `Point` **`class`** illustrated how an object can be initialised at the same time as it is defined. It would be nice for a user of our `Point` **`class`** to be able simply to define an object of **`class`** `Point` and be assured of the values of its data members. This is possible if we define a no-argument constructor (using empty parentheses or parentheses with the keyword **`void`**) and assign values to the **`class`** data members:

```
//...
Point::Point ()
  {
  x = y = z = 0.0 ;
  }
```

This program code initialises the `x`, `y` and `z` data members of `Point` to zero. Note that initialisation can assign any value(s) to a **`class`**'s data member(s), and not necessarily zero. It is tempting to try to initialise a **`class`**'s data members in the **`class`** declaration rather than in a **`class`** constructor:

```
class Point
  {
  private:
    // error: can't initialise data members in declaration
```

```
    double x, y, z = 0.0 ;
    //...
  };
```

This is illegal in C++ and initialisation of data members is restricted to constructors or member functions.

Once a default constructor of **class** Point has been defined, an object of **class** Point can be defined as follows:

```
void main ()
  {
  //...
  Point p ;  // no parentheses or arguments
  //...
  }
```

Object p is defined without parentheses or arguments, and not as follows:

```
//...
Point p () ;  // function declaration
```

which declares a function called p that has no arguments and returns a Point object.

9.13.3 Overloaded Constructors

We have already seen that member functions can be overloaded, and since constructors are merely *special* member functions it is natural to overload constructors. Overloading constructors enables an object of a given **class** to be initialised differently when defined. Combining the above observations of default initialisation and the Point **class** in program CONSTR.CPP we have the program:

```
// over_con.cpp
// illustrates overloaded constructors
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point () ;
    Point (double x_arg, double y_arg, double z_arg) ;
    //...
  };

// constructors:

// no arg. constructor (default)
Point::Point ()
  {
```

```
  x = y = z = 0.0 ;
  }

// 3 arg. constructor
Point::Point (double x_arg, double y_arg, double z_arg)
  {
  x = x_arg ;
  y = y_arg ;
  z = z_arg ;
  }
//...
void main ()
  {
  // define objects p and q
  Point p (1.0, 2.0, 3.0) ;
  Point q ;

  // O/P objects
  cout << "object p (" << p.X () //...
  }
```

In this program an object q is defined to be of **class** Point, but it is not explicitly initialised:

```
Point q ;
```

The default no-argument constructor is called for such an object definition of **class** Point. The output of the program is:

```
object p (1, 2, 3)
object q (0, 0, 0)
```

which illustrates that each of q's three data members is assigned the value 0.

9.13.4 Default Constructors

If constructors are not defined for a given **class** then objects of the **class** can be defined, but the values of an object's data members are not guaranteed. The following program is similar to OVER_CON.CPP except that no constructors for **class** Point are defined:

```
// def_con.cpp
// illustrates the use of default constructors
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // public member functions
    //...
  };
```

```
//...
void main ()
  {
  // define objects p and q
  Point p, q ;

  // O/P objects
  cout << "object p (" << p.X () //...
  }
```

which generated the following output when I ran this program:

```
object p (1.79937e-307, 2.64187e-308, 2.50321e-308)
object q (1.85866e-307, 2.64205e-308, 1.84104e-307)
```

Try adding the three-argument `Point` constructor declaration to the **`class`** `Point` declaration and the outside constructor definition listed in OVER_CON.CPP to the above program DEF_CON.CPP:

```
class Point
  {
  //...
  Point (double x_arg, double y_arg, double z_arg) ;
  //...
  };
//...
Point::Point (double x_arg, double y_arg, double z_arg)
  {
  //...
  }
//...
void main ()
  {
  Point p, q ;  // error: no match found for Point::Point()
  //...
  }
```

Such a program will generate a compilation error of the form 'No match found for the constructor `Point::Point()`'. This program helps illustrate that either no constructors can be defined or constructors must be explicitly defined if a constructor is defined with one or more arguments. Generally, it is good programming practice always to define at least a no-argument constructor for each **`class`** that you develop.

9.13.5 Data Member Initialisation

Program OVER_CON.CPP illustrated that `Point`'s three data members can be assigned values when an object of **`class`** `Point` is defined by means of a three-argument constructor:

```
//...
Point::Point (double x_arg, double y_arg, double z_arg)
  {
```

```
    x = x_arg ;
    y = y_arg ;
    z = z_arg ;
    }
```

There is a neater way to initialise `Point`'s data members:

```
// con_init.cpp
// illustrates data member initialisation
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point () ;
    Point (double x_arg, double y_arg, double z_arg) ;
    //...
  };

// constructors:

// no arg. constructor (default)
Point::Point ()
  : x(0.0), y (0.0), z (0.0)
  {}

// 3 arg. constructor
Point::Point (double x_arg, double y_arg, double z_arg)
  : x (x_arg), y (y_arg), z(z_arg)
  {}
//...
void main ()
  {
  // define objects p and q
  Point p (1.0, 2.0, 3.0) ;
  Point q ;
  //...
  }
```

`Point`'s three-argument constructor initialises the `x`, `y` and `z` data members as follows:

```
Point::Point (double x_arg, double y_arg, double z_arg)
  : x (x_arg), y (y_arg), z(z_arg)
  {}
```

Rather than initialising the data members by assignment within the constructor body, initialisation takes place in the constructor declarator. Note the use of the colon (:) after the constructor argument list and the use of the comma operator (,) between data members and the empty

constructor body, since initialisation is now performed outside the constructor body. This data member initialisation is very neat, in that it uses the one-argument constructor of type **double** for each data member! For instance, in earlier chapters we have seen **int** variables defined and initialised as follows:

```
int i = 1 ;
```

when in fact we could initialise variable `i` using **int**'s one argument constructor:

```
int i (1) ;
```

The use of data member constructors with **class** constructors helps to emphasise the similarity between C++ integral and user-defined types and classes.

The general syntax is:

```
ClassName::ClassName (type_specifier arg1, ...,
                      type_specifier argn)
  : data_member1 (arg1), ..., data_membern (argn)
  { /* ... */ }
```

9.13.6 Copy Constructor

The previous section illustrated a three-argument constructor for initialising the three data members of a `Point` object:

```
Point::Point (double x_arg, double y_arg, double z_arg)
  : x (x_arg), y (y_arg), z(z_arg)
  {}
```

In fact, for each of the three data members `x`, `y` and `z` a *copy constructor* has been called to perform data member initialisation. To help illustrate what a copy constructor actually is, consider the following program:

```
// copy_con.cpp
// illustrates the copy constructor
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point () ;
    Point (double x_arg, double y_arg, double z_arg) ;
    //...
  };

// constructors:

// no arg. constructor (default)
```

```
Point::Point ()
  : x(0.0), y (0.0), z (0.0)
  {}

// 3 arg. constructor
Point::Point (double x_arg, double y_arg, double z_arg)
  : x (x_arg), y (y_arg), z(z_arg)
  {}
//...
void main ()
  {
  // define objects p, q and r
  Point p (1.0, 2.0, 3.0) ;  // define, 3 arg. con.
  Point q (p) ;              // copy constructor
  Point r = p ;              // copy initialised

  // O/P objects
  cout << "object p (" << p.X () //...
  }
```

with output:

```
object p (1, 2, 3)
object q (1, 2, 3)
object r (1, 2, 3)
```

The following two statements, from program COPY_CON.CPP, illustrate defining two objects `q` and `r` which are both initialised to object `p`:

```
Point q (p) ;              // copy constructor
Point r = p ;              // copy initialised
```

The first statement calls the C++ default copy constructor, which performs an exact copy of `p`'s data members to `q`'s respective data members, while the second statement performs a similar initialisation. Since copying and assigning one object to another are frequent requests, the compiler provides default routines to perform these tasks. The above two statements are slightly different from assignment:

```
Point s, t ;
//...
s = t ;
```

since *pure* assignment assumes that the objects `s` and `t` have previously been defined, whereas the copy constructor defines a new object as well as assigning one object's data members to another object.

Let's overload the default copy constructor for **`class`** `Point`:

```
// over_cc.cpp
// overloads Point's copy constructor
#include <iostream>  // C++ I/O
```

```
class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    Point (const Point& p) ;
    //...
  };
//...
// copy constructor
Point::Point (const Point& p)
  {
  x = p.x ; y = p.y ; z = p.z ;
  cout << "copy constructor called" << endl ;
  }
//...
void main ()
  {
  Point p (1.0, 2.0, 3.0) ;  // define, 3 arg. con.
  cout << "object p (" << p.X () //...

  Point q ;                  // define, no arg. con.
  cout << "object q (" << q.X () //...

  Point r (p) ;              // copy constructor
  cout << "object r (" << r.X () //...

  Point s = p ;              // copy initialised
  cout << "object s (" << s.X () //...
  }
```

which generates the output:

```
object p (1, 2, 3)
object q (0, 0, 0)
copy constructor called
object r (1, 2, 3)
copy constructor called
object s (1, 2, 3)
```

The output illustrates that the copy constructor is called when objects `r` and `s` are defined. The three-argument constructor is still called for object `p`, and the no-argument constructor for object `q`.

The copy constructor declarator is:

```
Point (const Point& p)
```

The object `p` is passed to the copy constructor by reference and **`const`** modified to ensure that `p` is not accidentally altered. Why wasn't the copy constructor declarator of the form:

```
Point (Point p)  // error:  X (X) not allowed
```

which passes object p by value? Such a copy constructor will produce a compilation error of the form '*Point (Point)* is not a valid copy constructor'. A copy constructor which passes an object by value is not allowed, because when the object is passed a copy of the object is made, which calls the copy constructor, which copies the object – and so on, until there is no memory left. Thus, pass by reference to a copy constructor!

The general syntax of the copy constructor is therefore:

```
ClassName (const ClassName& object)
  { /* ... */ }
```

Copy constructors are also invoked when an object is passed by value to a function and when a temporary object is created for the return object of a function. To illustrate this, consider the program:

```
// copy_fun.cpp
// illustrates the copy constructor with
// function arguments and return values
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point () ;
    Point (double x_arg, double y_arg, double z_arg) ;
    Point (const Point& p) ;
    // public member functions
    //...
    void  Function1 (Point p) ;
    Point Function2 () ;
  };
//...
// copy constructor
Point::Point (const Point& p)
  {
  x = p.x ; y = p.y ; z = p.z ;
  cout << "copy constructor called" << endl ;
  }
//...
void  Function1 (Point p)
  {
  cout << "Function1 () called" << endl ;
  }

Point Function2 ()
  {
  cout << "Function2 () called" << endl ;
```

```
    Point temp ;
    return temp ;
    }

  void main ()
    {
    Point p (1.0, 2.0, 3.0) ;  // define, 3 arg. con.

    Function1 (p) ;
    Function2 () ;

    Point q ;  // no arg. con.

    q = Function2 () ;
    }
```

with output:

```
copy constructor called
Function1 () called
Function2 () called
copy constructor called
Function2 () called
copy constructor called
```

When *Function1()* is called a `Point` object is passed as an argument:

```
void  Function1 (Point p)
  {
  cout << "Function1 () called" << endl ;
  }
```

Since the object is passed by value to *Function1()*, a copy is made. Copying the object invokes the copy constructor. If a `Point` object is passed by reference to *Function1()*:

```
void Function1 (Point& p)
  { /* ... */ }
```

the copy constructor is not called since the object is passed by reference.

Calling *Function2()* calls the copy constructor when the function returns a `Point` object:

```
Point Function2 ()
  {
  cout << "Function2 () called" << endl ;
  Point  temp ;
  return temp ;
  }
```

When an object is returned by a function, a temporary object is created by the compiler to hold a copy of the object. This copying of an object to a temporary object invokes the copy constructor. The statement:

```
q = Function2 () ;
```

in the above program similarly invokes the copy constructor when a temporary object is returned by *Function2()*. However, assigning this object to object q will invoke the assignment operator and not the copy constructor.

The copy constructor is very useful for ensuring that an object allocates its own memory when it is created. This is necessary in certain circumstances, because when an object is returned from a function the object goes out of scope, thus causing the object's destructor to be called. If the copy constructor is not adequately defined, the object's destructor can delete memory allocated to an object in the calling function. We shall cover this topic when we discuss pointers.

9.13.7 Constructors, const and volatile

A **class** constructor can be called by a **const** or **volatile** object, but cannot be declared **const** or **volatile**:

```
// con_cv.cpp
// illustrates constructors, const and volatile

class X
  {
  public:
    X () {}
  };

class Y
  {
  public:
    Y () const {}          // error: const constructor
    Y (int) volatile {}    // error: volatile constructor
  };

void main ()
  {
  const    X cx ;  // ok: const object
  volatile X vx ;  // ok: volatile object
  }
```

9.14 Destructors

If a **class** constructor creates an object, then what destroys an object? In C++ the complement of the constructor is the *destructor*. The program DESTR.CPP illustrates a destructor for the Point **class**:

```
// destr.cpp
// illustrates destructors
#include <iostream>  // C++ I/O
```

```
class Point
  {
  private:
    double x, y, z ;
  public:
    // constructor
    Point () ;
    // destructor
    ~Point () ;
    // public member functions
    //...
  };

// constructor
Point::Point ()
  : x (0.0), y (0.0), z (0.0)
  {
  cout << "constructor called" << endl ;
  }

// destructor
Point::~Point ()
  {
  cout << "destructor called" << endl ;
  }
//...
void main ()
  {
  // define object p
  Point p ;

  // O/P object
  cout << "object p (" << p.X () //...
  }
```

which produces the output:

```
constructor called
object p (0, 0, 0)
destructor called
```

The `Point` **`class`** destructor is declared within the **`class`** declaration as:

```
class Point
  {
  //...
  public:
    //...
    ~Point () ;
    //...
  };
```

In general, a **class** and destructor declaration are of the following syntax:

```
class ClassName
  {
  private:
    //...
  public:
    // destructor
    ~ClassName () ;
    //...
  };
```

The **class** destructor has the same name as the **class** name, and hence the constructor name, but is preceded by a tilde (~). A destructor does not return a type and has no function arguments. Destructors are not passed arguments, since it makes no sense to pass arguments to an object that is just about to be destroyed.

It was noted above that constructors generally have to be **public** so that they are accessible to objects of a **class**. At first you may think that it doesn't matter whether a destructor is **private** or **public** – well, it does. Just as a constructor does, a **class**'s destructor also needs to be **public**ly accessible to an object. The following program code has declared `Point`'s destructor as **private**, and a compilation error will be generated of the form 'class `Point`'s destructor is not accessible to object `p`':

```
class Point
  {
  private:
    //...
    // destructor
    ~Point () ;
    //...
  public:
    //...
  };
//...
void main ()
  {
  //...
  Point p ;  // error: destructor class Point not accessible
  //...
  }
```

Destructors are frequently used to deallocate (using the **delete** operator) memory which was dynamically allocated using the **new** operator. Thus, a more detailed discussion of destructors will be left until we discuss pointers and the **new** and **delete** operators in Chapter 12. For now, just bear in mind that a destructor destroys an object that was created by a constructor.

9.14.1 Destructors, `const` and `volatile`

As with constructors, a destructor can be invoked by a **const** or **volatile** object, but cannot be declared **const** or **volatile**:

```
// des_cv.cpp
// illustrates destructors, const and volatile

class X
  {
  public:
    ~X () {}
  };

class Y
  {
  public:
    ~Y () const {}       // error: const destructor
    ~Y () volatile {}    // error: volatile destructor
  };

void main ()
  {
  const    X cx ;  // ok: const object
  volatile X vx ;  // ok: volatile object
  }
```

9.15 `inline` Member Functions

Chapter 6 discussed **`inline`** functions with reference to *normal* functions. This section now examines **`inline class`** member functions. The following program illustrates **`inline class`** `Point` *`X()`*, *`Y()`* and *`Z()`* access member functions:

```
// inline.cpp
// illustrates inline member functions
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point () ;
    Point (double x_arg, double y_arg, double z_arg) ;
    // public member functions
    double X () ;
    double Y () ;
    double Z () ;
  };
//...
// class Point member function definitions:

inline double Point::X ()
```

```
  { return x ; }
//...
void main ()
  { /* ... */ }
```

An **inline** member function is simply a function that is expanded inline at the point at which the member function is called. The general syntax of defining a function to be **inline** is:

```
inline return_type MemberFunction (parameter_list)
  { /* ... */ }
```

which uses the **inline** keyword modifier before the member function return type. It was mentioned in Chapter 6 that inline functions are generally reserved for small functions that are frequently called. Thus, the access member functions *X()*, *Y()* and *Z()* are good candidates for making **inline**, since they simply return data members.

9.15.1 Automatic `inline` Member Functions

If we move the constructor and member function definitions from outside the **class** declaration in the above program, INLINE.CPP, and place them within **class** `Point`'s declaration, these functions are automatically made **inline**:

```
// auto_in.cpp
// illustrates automatic inline member functions
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point ()
      : x (0.0), y (0.0), z (0.0) {}
    Point (double x_arg, double y_arg, double z_arg)
      : x (x_arg), y (y_arg), z (z_arg) {}
    //...
  };

void main ()
  { /* ... */ }
```

It was discussed in Chapter 6 that the use of the keyword **inline** is in fact a request, and the compiler ultimately decides whether the member function will be made **inline** or not. There are several circumstances in which the compiler may issue warning messages regarding **inline** member functions. For example, the following `Point::`*`Divide()`* member function tests whether an integer number, n, is not equal to zero and if so proceeds to return a `Point` object that is `Point` object `p1` divided by n:

```
inline Point Divide (const Point& p1, const int& n)
  {
```

```
  if (n != 0)
    return Point (p1.x/n, p1.y/n, p1.z/n) ;
  }
```

The Borland C++ (version 5.x) compiler generates a compilation warning message of the form 'A function containing a missing return statement is not expanded inline'. In addition, member functions containing **for**, **while** and **do-while**-loops, **switch** and **goto** statements, and **static** variables are not expanded **inline**.

9.15.2 `inline` Constructors

Constructors can also be defined as **inline** functions:

```
class Point
  {
  //...
  public:
    // constructors
    Point () ;
    Point (double x_arg, double y_arg, double z_arg) ;
    //...
  };

// constructors:

// no arg. constructor (default)
inline Point::Point ()
  : x (0.0), y (0.0), z (0.0)
  {}

// 3 arg. constructor
inline Point::Point (double x_arg, double y_arg,
                     double z_arg)
  : x (x_arg), y (y_arg), z (z_arg)
  {}
```

Since the two constructors for **class** Point simply initialise the three data members of **class** Point, it is feasible to make them **inline**.

9.16 `const` Member Functions

It was mentioned in Chapter 6 that a **class** member function declarator can include the keyword **const** modifier. This allows a member function access to a **class**'s data members, but prevents the member function from altering the value of a data member. The following program illustrates the use of **const** for **class** Point's member functions:

```
// con_mem.cpp
// illustrates const member functions
```

```
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    double X () const ;
    //...
  };
//...
// class Point member function definitions:

inline double Point::X () const
  { return x ; }
//...
void main ()
  { /* ... */ }
```

The above program illustrates the use of the **const** keyword modifier in the member functions *X()*, *Y()* and *Z()* declarators:

```
inline double Point::X () const
  { return x ; }
//...
```

Note that the keyword **const** must appear in both the declaration and definition of a **const** member function. The use of **const** in a member function declarator indicates to the compiler that the function is to have access to the **class**' data members only, and should not alter the values of the data members. Thus, if the member function `Point::X()` attempted to alter the data member `Point::x` before returning, a compilation error would result:

```
inline double Point::X () const
  {
  x = x + 1 ;  // error: cannot modify a constant object
  return x ;
  }
```

You must be careful when operating on **const** member functions and **const** objects. A **const** member function can be called for both non-**const** and **const** objects, because the **const** member function declaration guarantees that the data members of the object that the member function operates on cannot be altered. However, a non-**const** member function cannot (or should not) be called for a **const** object because the non-**const** member function could alter the data members of a **const** object. For example:

```
// con_mem1.cpp
// further illustrates constant member functions
#include <iostream>  // C++ I/O

class X
```

```
    {
    private:
      int data ;
    public:
      X ()
        : data (0) {}
      int NonConstFunction ()
        { return data++ ; }
      int ConstFunction () const
        { return data ; }
    };

  void main ()
    {
    X     non_const_obj ;
    const X const_obj ;

    // non-const object
    non_const_obj.NonConstFunction () ;  // ok
    non_const_obj.ConstFunction () ;     // ok

    // const object
    const_obj.NonConstFunction () ;  // warning/error
    const_obj.ConstFunction () ;     // ok
    }
```

The use of **const** member functions makes data member access watertight.

9.17 volatile Member Functions

Member functions can be declared **volatile** in a similar manner to **const** member functions:

```
// vol_mem.cpp
// illustrates volatile member functions
#include <iostream>  // C++ I/O

class X
  {
  private:
    int data ;
  public:
    X ()
      : data (0) {}
    void DataV () volatile
      { cout << data << endl ; }
    void DataCV () const volatile
      { cout << data << endl ; }
  };
```

```
void main ()
  {
  X          x ;
  volatile X vx ;  // volatile object

  x.DataV ()  ; x.DataCV () ;  // warning/error
  vx.DataV () ; vx.DataCV () ;
  }
```

volatile member functions can only be called for **volatile** objects. The above program also illustrates that **const volatile** member functions are also legal.

9.18 Default Member Function Arguments

Chapter 6 illustrated that function arguments can be assigned default values to enable a function call in which not all of the function arguments are passed to the function. Function arguments are assigned default values within the function declaration. To illustrate this for our `Point` **class**, let us assign a default value of zero to the `z` data member for the three-argument constructor declaration. This is particularly useful when applying the `Point` **class** to two-dimensional points (x and y) that lie in a plane $z = 0$, such as a display screen. Program DEF_ARG.CPP illustrates default member function arguments:

```
// def_arg.cpp
// illustrates default member function arguments
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point () ;                          // default arg: z=0
    Point (double x_arg, double y_arg, double z_arg=0.0) ;
    // public member functions
    //...
  };

// constructors:

// no arg. constructor (default)
inline Point::Point ()
  : x (0.0), y (0.0), z (0.0)
  {}

// 3 arg. constructor
inline Point::Point (double x_arg, double y_arg,
                     double z_arg)
  : x (x_arg), y (y_arg), z (z_arg)
```

```
  {}
//...
void main ()
  {
  // define objects p and q, neglecting z data member
  Point p (1.0, 2.0) ;
  Point q (3.0, 4.0) ;
  //...
  }
```

Objects p and q are defined as:

```
Point p (1.0, 2.0) ;
Point q (3.0, 4.0) ;
```

The use of the default argument for the z data member in the declaration of the three-argument constructor of **class** Point relieves the user from having to pass a value of zero to the third argument of the constructor each time a two-dimensional Point object is created:

```
Point p (1.0, 2.0, 0.0) ;
Point q (3.0, 4.0, 0.0) ;
//...
```

It is worth noting that a default argument only appears in a constructor or member function declaration, and not in the definition.

What about default arguments for user-defined classes? Consider a Line **class** which encapsulates two Point objects:

```
class Line
  {
  private:
    Point p1, p2 ;
  public:
    // constructors
    Line ()
      : p1 (), p2 () {}
    Line (Point p, Point q)
      : p1 (p), p2 (q) {}
    //...
  };
```

If it is required that the default behaviour of the two argument constructor is to define a Line object which has one of its two Point objects at the origin of coordinates (0,0,0), then simply use Point's constructor in the default argument expression:

```
class Line
  {
  //...
  public:
    //...
    Line (Point p, Point q=Point())
```

```
      : p1 (p), p2 (q) {}
    //...
};
```

The `Line` **`class`** is discussed in more detail later in Section 9.26.

9.19 Functions with Object Arguments and Functions that Return Objects

When a user-defined **`class`** has been declared, there is no difference between passing objects of the user-defined **`class`** and passing identifiers or objects of a C++ integral type or **`class`** as arguments to a function. Similarly, returning objects of a user-defined **`class`** from a function follows the same rules as encountered in Chapter 6.

To illustrate passing object arguments to a function and returning an object from a function, let us extend our `Point` **`class`** by adding two member functions *`Add()`* and *`Subtract()`*, which perform the addition and subtraction of two objects of **`class`** `Point`:

```
// arg_ret.cpp
// illustrates passing object arguments to functions
// and returning an object from a function
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    Point Add      (const Point& p) ;
    Point Subtract (const Point& p) ;
  };
//...
// p1 + p2
inline Point Point::Add (const Point& p)
  {
  return Point (x+p.x, y+p.y, z+p.z) ;
  }

// p1 - p2
inline Point Point::Subtract (const Point& p)
  {
  return Point (x-p.x, y-p.y, z-p.z) ;
  }

void main ()
  {
  // define objects p1 and p2
  Point p1 (3.0, 4.0, 5.0) ;
```

```
    Point p2 (1.0, 2.0, 3.0) ;
    //...

    Point p3, p4 ;

    // add & subtract p1 & p2
    p3 = p1.Add (p2) ;
    p4 = p1.Subtract (p2) ;
    //...
    }
```

This program defines the *Add()* member function as:

```
inline Point Point::Add (const Point& p)
  {
  return Point (x+p.x, y+p.y, z+p.z) ;
  }
```

which enables two `Point` objects, `p1` and `p2`, to be added, and the result assigned to another object, `p3`, by a statement of the form:

```
//...
p3 = p1.Add (p2) ;
```

The *Add()* member function declarator indicates that the member function return type is **class** `Point`. Note that *Add()*'s function argument, `p`, is passed by reference, and **const** modified to ensure that `p`'s data members are not accidentally altered in the member function call. The statement:

```
return Point (x+p.x, y+p.y, z+p.z) ;
```

adds the respective `x`, `y` and `z` data members of objects `p1` and `p2` and returns a *new* object. This returned object is then assigned to the object `p3` in *main()*. Note that the object returned from *Add()* is created using the three-argument constructor. Alternatively, we could have defined *Add()* as:

```
Point Point::Add (const Point& p)
  {
  Point p_temp ;
  p_temp.x = x + p.x ;
  p_temp.y = y + p.y ;
  p_temp.z = z + p.z ;

  return p_temp ;
  }
```

The above two versions of *Add()* are equivalent, but the former is more concise.

Adding two `Point` objects `p1` and `p2` and assigning the result to object `p3` using the *Add()* function above:

```
//...
```

```
p3 = p1.Add (p2) ;
```

is rather messy. At present, the *Add()* member function is acting on the data members of object p1 and returns the result. A recommended alternative design for *Add()* is to remove the **return** statement and let the member function act on the data members of the object calling *Add()*:

```
void Point::Add (const Point& p1, const Point& p2)
  {
  x = p1.x+p2.x ; y = p1.y+p2.y ; z = p1.z+p2.z ;
  }
```

Note that a **return** statement is not required and two arguments are now passed to *Add()*. Thus, to add two Points p1 and p2 we now write:

```
//...
p3.Add (p1, p2) ;
```

which no longer requires the use of the assignment operator to assign p1+p2 to p3.

This approach is illustrated in the program ARG_RET2.CPP:

```
// arg_ret2.cpp
// illustrates passing object arguments to functions
// and assigning values to an object's data members
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    void Add      (const Point& p1, const Point& p2) ;
    void Subtract (const Point& p1, const Point& p2) ;
  };
//...
// p1 + p2
inline void Point::Add (const Point& p1, const Point& p2)
  {
  x = p1.x+p2.x ; y = p1.y+p2.y ; z = p1.z+p2.z ;
  }
//...

void main ()
  {
  // define objects p1 and p2
  Point p1 (3.0, 4.0, 5.0) ;
  Point p2 (1.0, 2.0, 3.0) ;
  //...

  Point p3, p4 ;
```

```
    // add & subtract p1 & p2
    p3.Add (p1, p2) ;
    p4.Subtract (p1, p2) ;
    //...
    }
```

Note that the *Add()* and *Subtract()* member functions are now **inline** functions, since they are small functions.

9.19.1 Returning by Reference

The member functions *X()*, *Y()* and *Z()* at present only allow us to access the **private** data members of **class** Point. Once an object of **class** Point has been defined, it is not possible to alter the x, y and z data members of the object using the *X()*, *Y()* and *Z()* member functions. However, if we make the return type of each member function a reference to a **double** (rather than simply a **double**), then the return value of each function will refer to its respective data member. This is demonstrated in the following program:

```
// ret_ref.cpp
// illustrates returning by reference for both
// constant and non-constant objects
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    // public member functions
    double& X () ;
    //...
    const double& X () const ;
    //...
  };
//...
// class Point member function definitions:

// non-const objects
inline double& Point::X ()
  { return x ; }
//...
// const objects
inline const double& Point::X () const
  { return x ; }
//...
void main ()
  {
  // define objects p1 and p2
```

```
Point p (1.0, 2.0) ;
//...
p.X () = 3.0 ;
p.Y () = 4.0 ;
//...
}
```

Two sets of the `X()`, `Y()` and `Z()` member functions are given, one for non-constant objects and one for constant objects.

9.20 Combining Constructor and Member Function Calls

The previous section demonstrated that an object can be returned from a function without explicitly defining the object:

```
Point Point::Add (const Point& p)
  {
  return Point (x+p.x, y+p.y, z+p.z) ;
  }
```

The compiler generates a temporary object which is returned from the function. Similarly, C++ allows a member function of a given **class** to be called without explicitly defining an object. In general:

```
ClassConstructor ().MemberFunction () ;
```

The following program illustrates such a member function call for **class** `Point`:

```
// no_obj.cpp
// illustrates a constructor and member function
// call in a single statement
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point () ;                          // default arg: z=0
    Point (double x_arg, double y_arg, double z_arg=0.0) ;
    // public member functions
    double& X () ;
    //...
  };
//...
void main ()
  {
```

```
double d = Point (1.0, 2.0, 3.0).X () ;

cout << "double d: " << d << endl ;
}
```

The statement:

```
double d = Point (1.0, 2.0, 3.0).X () ;
```

illustrates the combined use of `Point`'s constructor and the access member function `Point::X()`.

This concise notation may appear a little strange at first, but it can be useful in certain circumstances. A popular application of the above syntax is in executing dialog boxes in a Windows environment:

```
// application class
class MyApplication: public TApplication
  {
  private:
    TWindow* Client ;
    //...
  protected:
    void CmHelpAbout () ;
  public:
    MyApplication () ;
    virtual ~MyApplication () ;
    //...
    virtual void InitMainWindow () ;
    //...
  };
//...
// About dialog class
class AboutDialog: public TDialog
  { /* ... */ };
//...
// initialise main window of application
void MyApplication::InitMainWindow ()
  {
  Client = new TWindow (/*...*/) ;
  TFrameWindow* frame = new TFrameWindow (/*...*/) ;
  //...
  MainWindow = frame ;
  //...
  }
//...
// executes Help|About dialog box
void MyApplication::CmHelpAbout ()
  {
  AboutDialog (MainWindow).Execute () ;
  }
//...
```

The above program code incorporates the two **class** declarations `MyApplication` and `AboutDialog`. The **class** `MyApplication` is derived from the Borland *ObjectWindows* `TApplication` **class**. A `TApplication`-derived object encapsulates a Windows application. The **class** `AboutDialog` is derived from the Borland ObjectWindows `TDialog` **class**. An `AboutDialog`-derived object and an about-dialog box resource enable a dialog box to pop up which displays program information. The function *`CmHelpAbout()`* associates the pull-down menu option Help | About with the about-dialog box. The *`CmHelpAbout()`* function demonstrates the combined use of calling the one-argument constructor of **class** `AboutDialog` to create a temporary object, and in the same statement calling the *`Execute()`* function to display the about-dialog box.

9.21 Arrays of Objects

The following program defines an array of three objects of **class** `Point`:

```
// array_ob.cpp
// illustrates an array of objects
#include <iostream>  // C++ I/O

class Point
  {
  //...
  };
//...
void main ()
  {
  // define an array of 3 Point objects
  Point tri_vertices[3] ;

  // initialise array objects
  tri_vertices[0] = Point (1.0, 1.0) ;
  tri_vertices[1] = Point (4.0, 2.0) ;
  tri_vertices[2] = Point (3.0, 5.0) ;

  // O/P objects
  for (int i=0; i<3; i++)
    {
    cout << "tri_vertices[" << i << "]: "
         << "("  << tri_vertices[i].X ()
         << ", " << tri_vertices[i].Y ()
         << ", " << tri_vertices[i].Z ()
         << ")"  << endl ;
    }
  }
```

In the *`main()`* function an array `tri_vertices` of three elements of **class** `Point` is defined:

```
Point tri_vertices[3] ;
```

This definition is identical in format to an array of elements of a C++ integral type:

```
int i_array[3] ;
```

The operations on an array whose elements are of a user-defined **class** are identical to those of a fundamental or integral C++ type.

Similarly, arrays can be used as **class** data members. `Point`'s `x`, `y` and `z` data members could have been represented as an array with three elements:

```
class Point
  {
  private:
    double pt[3] ;
  public:
    Point () { pt[0]=0.0 ; pt[1]=0.0 ; pt[2]=0.0 ; }
    //...
  };
```

The `x`, `y` and `z` data members of `Point` are now represented as `pt[0]`, `pt[1]` and `pt[2]`, respectively.

9.22 Local Classes

The C++ programming language allows a **class** to be declared local to a function:

```
// local.cpp
// illustrates local classes
#include <iostream>  // C++ I/O

void Function (int i)
  {
  class X
    {
    private:
      int data ;
    public:
      //X (int i) ;
      X (int i)
        : data (i) {}
      void Set (int i) { data = i ; }
      int  Get () { return data ; }
    };

  //X::X (int i) // error: non-inline function of local class
  //   : data (i) {}

  X x (i) ;     // function argument i
  cout << x.Get () << endl ;
```

```
    int j (11) ; // local function variable j
    x.Set (j) ;
    cout << x.Get () << endl ;
    }

  void main ()
    {
    Function (10) ;
    }
```

with output:

```
10
11
```

The above program illustrates a **class** X declared within the scope of a function. LOCAL.CPP illustrates that objects of a local **class** can access function arguments and local function variables, but non-inline functions of a local **class** are illegal. Local classes are rarely used in practice, but nevertheless exist if the need arises!

9.23 Nested Classes

Before we summarise the syntax of a **class** declaration in the next section, let us briefly discuss nested classes. Consider the following, somewhat impractical, **class** declaration of Point:

```
// nested.cpp
// illustrates nested classes
#include <iostream>  // C++ I/O

class Point
  {
  private:
    // nested class
    class DataMembers
      {
      public:
        double x, y, z ;
      }dm ;
  public:
    // constructors
    Point () ;
    //...
    // public member functions
    double& X () ;
    //...
  };
```

```
// constructors:

// no arg. constructor (default)
inline Point::Point ()
  {
  dm.x = dm.y = dm.z = 0.0 ;
  }
//...
// class Point member function definitions:

inline double& Point::X ()
  { return dm.x ; }
//...
void main ()
  {
  Point p (1.0, 2.0) ;
  //...
  }
```

This program illustrates a **class** called DataMembers declared within the declaration of **class** Point. DataMembers is declared a **private** nested **class** of **class** Point and is therefore inaccessible to objects of **class** Point. The x, y and z data members of **class** DataMembers are declared **public** and are therefore accessible to **class** Point. An object, dm, of **class** DataMembers is declared as a **private** data member of **class** Point, and the program illustrates that access to the x, y and z data members is now via the dm object.

As a further illustration of nested classes, consider the following program:

```
// nested1.cpp
// further illustrates nested classes
#include <iostream>  // C++ I/O

class A  // class A
  {
  public:
    int adata ;
    A () ;
    class B  // nested class B
      {
      public:
        int bdata ;
        B () ;
        class C  // nested-nested class C
          {
          public:
            int cdata ;
            C () ;
          }; // C
      }; // B
  }; // A
```

```
// class A constructor
A::A ()
  { adata = 0 ; }

// nested class B constructor
A::B::B ()
  { bdata = 1 ; }

// nested-nested class C constructor
A::B::C::C ()
  { cdata = 2 ; }

void main ()
  {
  A a ;
  A::B b ;
  A::B::C c ;

  cout << "A::adata: " << a.adata << endl ;
  cout << "B::bdata: " << b.bdata << endl ;
  cout << "C::cdata: " << c.cdata << endl ;

  // error: explicit qualification required
  B b_noqual ;
  C c_noqual ;
  }
```

I am not suggesting that you adopt this programming style of nesting classes to the *n*th degree, but it does, more importantly, illustrate explicit qualification via the scope resolution operator for nested classes. The *main()* function above further illustrates that when a nested **class** is declared and there is no other **class** of the same name declared in the scope of the program, explicit qualification to the nested **class** is not essential, although most compilers will issue a compilation warning asking you to use explicit qualification.

9.24 Another `class` Declaration

A more comprehensive **class** declaration syntax which includes **class** constructors, destructor and member functions is of the form:

```
class ClassName
  {
  // private data members
  type_specifier pri_member_name ;
  //...
  type_specifier PrivateMemberFunction (parameter_list) ;
  //...
  private:
    // private data members
    type_specifier pri_member_name ;
```

```
    //...
    // private member functions
    type_specifier PrivateMemberFunction (parameter_list) ;
    //...
  public:
    // public data members
    type_specifier pub_member_name ;
    //...
    // constructors
    ClassName () ;
    ClassName (parameter_list) ;
    //...
    // destructor
    ~ClassName () ;
    // public member functions
    type_specifier PublicMemberFunction () ;
    //...
  //...
  } object_list ;
```

9.25 `struct` and `class`

We have now seen **`struct`** and **`class`** and it is instructive to remind ourselves of the similarities and differences between them. The data members of a **`struct`** are **`public`** by default, whereas **`class`** data members are **`private`** to the **`class`** by default. Access can be changed, for both **`struct`** and **`class`**, by the access specifiers **`private`** and **`public`**. Like the **`class`**, C++ **`struct`**s may have member functions, but in C++ programming the **`struct`** is generally reserved for data encapsulation, whereas the **`class`** is used for user-defined data types which encapsulate both data and member functions. In fact, the **`struct`** has the same scope and inheritance (Chapter 15) rules as the **`class`**, and differs only in the default access rights. There are instances in which the use of a **`struct`** is slightly more convenient than a **`class`** because of its default **`public`** access rights and in emphasising that a structure is a simple aggregation of data and not a concrete data type. For a discussion of the importance of maintaining a similarity between **`struct`** and **`class`** in the evolution of C++, refer to Stroustrup (1994, Section 3.5.1).

9.26 Intersection of Two Two-Dimensional Line Segments

When functions were covered in Chapter 6, we concluded the chapter with a discussion of the intersection of two two-dimensional line segments, and promised to return to the topic of line segment intersection in Chapter 9 – well, here we are. The following program implements the clockwise/anticlockwise and intersection functions, *`CAC()`* and *`LineIntersection()`*, of Chapter 6, but now from a **`class`** perspective:

```
// li_class.cpp
// Point and Line classes with Point clockwise/anticlockwise
```

```
// and Line intersection routines

#include <iostream>  // C++ I/O

const double TOLERANCE = 1e-06 ;

class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    void   Display (char msg[]) ;
    int    CAC (const Point& p1, const Point& p2) const ;
  };

// class Point member functions:

// passed 2 Points in 2D plane
// returns -1, 1 or 0 depending whether points are clockwise,
// anti-clockwise or collinear, ...
int Point::CAC (const Point& p1, const Point& p2) const
  {
  int cac = 0 ;  // return var. indicating direction

  // use point 0 as 'local' origin
  double dx1 = p1.X() - x ; double dy1 = p1.Y() - y ;
  double dx2 = p2.X() - x ; double dy2 = p2.Y() - y ;

  // eliminate infinite gradients by multiplying gradients
  // between points 0-1 and 0-2 by (dx1*dx2)

  if (dx1*dy2 > dx2*dy1)  // anti-clockwise (m02>m01)
    cac = 1 ;

  if (dx1*dy2 < dx2*dy1)  // clockwise (m02<m01)
    cac = -1 ;

  if ((dx1*dy2 - dx2*dy1)< TOLERANCE)  // same gradients
    {
    if ((dx1*dx2 < 0) || (dy1*dy2 < 0))
      cac = -1 ;
    else if ((dx1*dx1 + dy1*dy1) >= (dx2*dx2 + dy2*dy2))
      cac = 0 ;
    else
      cac = 1 ;
    }
  return cac ;
  }

void Point::Display (char msg[])
```

```
  {
  cout << msg << ": ("
       << x << ", " << y << ", " << z << ")" << endl ;
  }

class Line
  {
  private:
    Point p1, p2 ;
  public:
    // constructors
    Line ()
      : p1 (), p2 () {}
    Line (Point p, Point q)
      : p1 (p), p2 (q) {}
    // copy constructor
    Line (const Line& l)
      : p1 (l.p1), p2 (l.p2) {}
    // member functions
    Point& P1 () { return p1 ; }
    Point& P2 () { return p2 ; }
    const Point& P1 () const { return p1 ; }
    const Point& P2 () const { return p2 ; }
    void  Display (char msg[]) ;
    bool  Intersection (const Line& l) ;
  };

// class Line member functions:

// intersection of 2 Lines
// returns false if no intersection, true if intersection
bool Line::Intersection (const Line& l)
  {
  // intersection
  if ( ((p1.CAC (p2, l.p1) *
         p1.CAC (p2, l.p2)) <= 0) &&
       ((l.p1.CAC (l.p2, p1) *
         l.p1.CAC (l.p2, p2)) <= 0) )
    return true ;
  // no intersection
  else
    return false ;
  }

void Line::Display (char msg[])
  {
  cout << msg << ": "
       << "p1: (" << p1.X() << ", " << p1.Y()
           << ", " << p1.Z() << "), "
       << "p2: (" << p2.X() << ", " << p2.Y()
           << ", " << p2.Z() << ")" << endl ;
```

```
}

void main ()
  {
  // define Points & Lines
  Point i (1, 1), j (4, 2), k (3, 5), l (3, 0) ;
  Line l1 (i, j), l2 (k, l) ;
  // O/P Point's
  i.Display ("i") ; //...
  // O/P Line's
  l1.Display ("l1") ; //...

  if (l1.Intersection (l2))
    cout << "lines intersect" << endl ;
  else
    cout << "lines do not intersect" << endl ;
  } // main()
```

The function *CAC()* is a member function of `Point` and *Intersection()* is a member of `Line`. Apart from the member functions *CAC()* and *Intersection()* implicitly acting on the `Point` and `Line` objects that call the member functions, the functions are identical to those outlined in Chapter 6. The declaration of `Line` encapsulates two `Point` data members, which represent the end-points of a line segment. The zero- and two-argument constructors initialise the `Point` data members of `Line`, while the one-argument copy constructor performs an exact copy of a `Line` object. The member functions *P1()* and *P2()* simply return a reference to the `Point` data members for both non-constant and constant `Point` objects, *Display()* displays a `Line` object and *Intersection()* tests whether two `Line` line segment objects intersect. The two-argument `Line` constructor allows an object of **class** `Line` to be defined by passing two `Point` objects:

```
Point i (1, 1), j (4, 2) ;
//...
Line l1 (i, j) ;
```

Two `Line` objects can now easily be tested to see whether they intersect or not:

```
bool intersection = l1.Intersection (l2) ;
```

A comparison of the above program, LI_CLASS.CPP, with the program LI.CPP of Chapter 6 clearly illustrates the benefits of adopting an object-oriented programming approach, particularly for a user of a line segment intersection routine. The `Point` and `Line` classes offer the user a more natural representation of geometric points and lines and clearly associate the clockwise/anticlockwise function *CAC()* with points and the line segment intersection function *Intersection()* with lines.

9.27 Summary

The **class** is the central feature of object-oriented programming in C++. A **class** encapsulates both data and functions that operate on the data. The C++ **class** enables the program-

mer to develop new data types and model more closely objects in the real world than a procedural style of programming.

This chapter has focused on a **class** called Point which models a point in three-dimensional space. A quick glance through this chapter will illustrate how the Point **class** has grown and developed into a powerful **class**. The final declaration of **class** Point developed in this chapter is:

```
class Point
  {
  private:
    // private data members
    double x, y, z ;
  public:
    // constructors
    Point () ;
    Point (double x_arg, double y_arg, double z_arg=0.0) ;
    // public member functions
    double& X () { return x ; }
    double& Y () { return y ; }
    double& Z () { return z ; }
    const double& X () const { return x ; }
    const double& Y () const { return y ; }
    const double& Z () const { return z ; }
    void Add      (const Point& p1, const Point& p2) ;
    void Subtract (const Point& p1, const Point& p2) ;
  };
```

The Point **class** declaration above is just 16 lines long (discounting comment lines) and yet possesses the essential characteristics of the majority of classes that programmers develop. A **class** declaration begins with the keyword **class**, followed by the name of the **class**. The body of the **class** is within opening and closing braces and the declaration is completed with a semicolon. Class data members can be either **private** or **public**. By default, data members of a **class** are **private**, and if **public** data members are required they must be explicitly indicated as such. The keywords **private** and **public** give the developer of a **class** complete control over the access of a **class**'s data members.

Classes have *special* member functions called *constructors* and a single *destructor*. Both constructors and the destructor have the same name as the **class** name, but the destructor is preceded by a tilde. Constructors initialise an object's data members when the object is created and are called automatically. Default, no-argument, constructors can be defined which enable a user to define an object of a given **class** without passing any arguments to the constructor and still be guaranteed of the values of the object's data members. The destructor is automatically called when an object is destroyed.

A **class** can have member functions which operate specifically on the **class**'s data members. Member functions are not *normal* or *global* functions, but are specific to a given **class**. A member function is frequently referred to as a *message*, and it can be helpful to think of member functions as sending messages to an object. Member functions are frequently used as *access* functions to access an object's data members and to ensure that an object's data members are not accidentally altered. A **class**'s member functions can also be made **inline** to eliminate calling the function for small functions. The constructors and member functions of **class** Point are:

```
// constructors:
//...
inline Point::Point (double x_arg, double y_arg, double z_arg)
  : x (x_arg), y (y_arg), z (z_arg)
  {}

// class Point member function definitions:

inline double& Point::X ()
  { return x ; }
//...
inline void Point::Add (const Point& p1, const Point& p2)
  {
  x = p1.x+p2.x ; y = p1.y+p2.y ; z = p1.z+p2.z ;
  }
//...
```

The above illustrates that a **class**'s constructors and member functions can be defined outside a **class** declaration. Constructors and member functions can be passed an object of a user-defined **class** just as if it were an object of a C++ integral type or **class**. In addition, member functions can return an object of a user-defined **class**, illustrating that correctly implemented user-defined classes are equivalent to C++ integral types and classes.

Later chapters will extend the `Point` **class** even further by making the application and manipulation of `Point` objects more natural, particularly in the next chapter, which introduces the topic of operator overloading which allows a programmer to define a set of operators that act on objects of a user-defined type.

Exercises

9.1 Develop a `Window` **class** which models a two-dimensional rectangular window that is characterised by left, right, top and bottom integer values. Provide member functions for `Window` which return the width, height and origin of a `Window` object. It is required that the default constructor of **class** `Window` places a `Window` object at the origin and has width and height of 120 and 80 respectively.

9.2 Develop an `Int` **class** which mimics the C++ **int** data type. A default constructor of `Int` should initialise an object of **class** `Int` to zero. Provide display, increment, decrement and access member functions for `Int`.

9.3 Make the following `Node` **class** a **private** nested **class** of `LinkedList`:

```
class Node
  {
  public:
    int   value ;
    Node* next ;
    Node (int v, Node* next_ptr)
      : value (v), next (next_ptr) {}
  };
```

```
class LinkedList
  {
  private:
    Node* first ;
  public:
    //...
  };
```

9.4 Modify the following **class** so as to make use of the **inline** and **const** features of **class** member functions:

```
class Apple
  {
  private:
    Position       pos ;
    Size           siz ;
    SurfaceTexture tex ;
  public:
    // constructors
    Apple () ;
    Apple (const Position& p, const Size& s,
           const SurfaceTexture& st) ;
    // member functions
    Position       Location   () ;
    Size           Dimensions () ;
    SurfaceTexture Texture    () ;
  };

Apple::Apple ()
  : pos (), siz (), tex ()
  {}

Apple::Apple (const Position& p, const Size& s,
              const SurfaceTexture& st)
  : pos (p), siz (s), tex (st)
  {}

Position Apple::Location ()
  { return pos ; }

Size Apple::Dimensions ()
  { return siz ; }

SurfaceTexture Apple::Texture ()
  { return tex ; }
```

9.5 Rewrite the `Point`, `Line` and `Triangle` structures developed for Exercise 8.5 so that they are now classes. The perimeter, centroid and area functions should now be member functions of **class** `Triangle`. Test your program on a triangle whose three vertices are (1,1,0), (3,2,0) and (2.5,4,0). Compare the *main()* functions of Exercises 8.5 and 9.5.

9.6 Consider the development of a `Triangle` **class** which encapsulates three `Point` objects. Since several `Triangle` objects can share the same `Point` it is advantageous for

`Triangle` to contain references to its `Point` objects rather than the objects themselves.

9.7 Develop a **`class`**, `CartCoorSys`, to represent a three-dimensional Cartesian coordinate system which is defined in terms of an origin and three coordinate axes. The origin and axes are to be modelled as three-dimensional vectors. The default constructor of **`class`** `CartCoorSys` is to describe a coordinate system positioned at the origin, (0,0,0), with the unit vector axes (1,0,0), (0,1,0) and (0,0,1). Use your `CartCoorSys` **`class`** to declare a `CartPoint` **`class`** which now encapsulates a `CartCoorSys` data member as well as the **`double`** `x`, `y` and `z` data members. Consider the design of classes for cylindrical, polar and spherical coordinate systems.

Chapter 10

Operators and Overloading

01:10
01/11

C++ allows operators to be overloaded specifically for a user-defined class. Such a feature offers a programmer a more natural and concise syntax for performing operations on objects of a given class. For instance, consider a Matrix class which encapsulates an array of values or objects in to a single entity. If we are required to perform an addition of two Matrix objects, m1 and m2, then one solution would be for class Matrix to support an Add() member function which adds the respective data members of Matrix objects m1 and m2:

```
Matrix m3 ;
//...
m3.Add (m1, m2) ;
```

Such an approach works fine, but operator overloading offers a programmer a more elegant and powerful mathematical shorthand for performing operations on Matrix objects:

```
Matrix m3 = m1 + m2 ;
```

Overloading the addition operator, +, specifically for class Matrix allows a user of the Matrix class to write more abstract and at the same time more natural statements.

10.1 Overloading the Arithmetic Binary Operators

We saw in the last chapter that the `Point` **`class`** contained two member functions, *`Add()`* and *`Subtract()`*, for performing the addition and subtraction of two `Point` objects:

```
//...
// p1 + p2
inline void Point::Add (const Point& p1, const Point& p2)
  {
```


G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

```
    x = p1.x+p2.x ; y = p1.y+p2.y ; z = p1.z+p2.z ;
    }

  // p1 - p2
  inline void Point::Subtract (const Point& p1, const Point& p2)
    {
    x = p1.x-p2.x ; y = p1.y-p2.y ; z = p1.z-p2.z ;
    }

  void main ()
    {
    //...
    p3.Add (p1, p2) ;
    p4.Subtract (p1, p2) ;
    //...
    }
```

The `Point::Add()` member function is passed two objects, adds together their respective `x`, `y` and `z` data members, and assigns the results to the calling object's data members.

If we were performing the addition of two numbers of the integral type **`int`**, we could simply write:

```
int i1, i2, i3 = 0;
//...
i3 = i1 + i2 ;
```

which is clearly more intuitive than our present addition of two objects of **`class`** `Point`.

In C++, it is possible to write expressions of the form:

```
Point p1, p2, p3 ;
//...
p3 = p1 + p2 ;
```

To perform such natural mathematical expressions with objects of **`class`** `Point` requires the use of a C++ feature called *operator overloading*. Before we examine operator overloading in detail, let's take a look at a program which attempts to add two `Point` objects using the addition, +, operator. The following program is basically identical to the ARG_RET2.CPP program of Chapter 9:

```
// add.cpp
// attempts to add two Point objects using
// the addition operator + before overloading
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point ()
```

```
      : x (0.0), y (0.0), z (0.0) {}
    Point (double x_arg, double y_arg, double z_arg=0.0)
      : x (x_arg), y (y_arg), z (z_arg) {}
    // public member functions
    double& X () { return x ; }
    //...
    void Add      (const Point& p1, const Point& p2) ;
    void Subtract (const Point& p1, const Point& p2) ;
  };

// p1 + p2
inline void Point::Add (const Point& p1, const Point& p2)
  {
  x = p1.x+p2.x ; y = p1.y+p2.y ; z = p1.z+p2.z ;
  }

// p1 - p2
inline void Point::Subtract (const Point& p1,
                             const Point& p2)
  {
  x = p1.x-p2.x ; y = p1.y-p2.y ; z = p1.z-p2.z ;
  }

void main ()
  {
  // define objects
  Point p1 (3.0, 4.0, 5.0), p2 (1.0, 2.0, 3.0), p3, p4 ;

  // add p1 & p2
  p3 = p1 + p2 ;  //  error: illegal operation
  }
```

The statement:

```
p3 = p1 + p2 ;
```

will generate a compilation error of the form 'illegal operation on objects' because we have at present not informed the compiler of how to add objects of **class** Point. Thus, we need to instruct the compiler how to add two objects of **class** Point. To do this, we require the **operator** keyword.

10.1.1 The operator Keyword

C++ provides a keyword called **operator** which overloads a given operator specifically for a user-defined **class**. Let's overload the addition and subtraction arithmetic operators for **class** Point:

```
// over_as.cpp
// overloads the arithmetic (+) and (-) operators for Point
#include <iostream>  // C++ I/O
```

```
class Point
  {
  public:
    //...
    // overloaded operators
    Point operator + (const Point& p) ;
    Point operator - (const Point& p) ;
  };
//...
// overloaded operators:

// p1 + p2
inline Point Point::operator + (const Point& p)
  {
  return Point (x+p.x, y+p.y, z+p.z) ;
  }

// p1 - p2
inline Point Point::operator - (const Point& p)
  {
  return Point (x-p.x, y-p.y, z-p.z) ;
  }

void main ()
  {
  // define objects
  Point p1 (3.0, 4.0, 5.0), p2 (1.0, 2.0, 3.0), p3, p4, p5 ;

  // add p1 & p2
  p3 = p1 + p2 ;  // addition
  p4 = p1 - p2 ;  // subtraction
  p5 = p1 - p2 + p3 - p4 ;  // chained addition & subtraction

  // O/P
  cout << "p3 (" << p3.X () << ", " << p3.Y ()
       << ", "   << p3.Z () << ")"  << endl ;
  //...
  }
```

with output:

```
p3 (4, 6, 8)
p4 (2, 2, 2)
p5 (4, 6, 8)
```

The definition of the overloaded addition operator for **class** Point is:

```
inline Point Point::operator + (const Point& p)
  {
  return Point (x+p.x, y+p.y, z+p.z) ;
  }
```

The declarator is similar to member function declarators except for the keyword **operator**, which precedes the operator (in this case +). The keyword **operator** instructs the compiler that an operator is being overloaded. Note the use of **const** and **inline**.

The general syntax of an out-of-line overloaded operator member function is:

```
type_specifier ClassName::operator op (parameter_list)
  { /* ... */ }
```

where op is the overloaded operator, e.g. <, +, = or [ ]. At first, overloaded operator member functions can appear a little confusing, mainly because only a single member function argument is used for binary operators which operate on two operands:

```
p3 = p1 + p2 ;
```

The object p1, on the left-hand side of the + operator is an object of **class** Point to which the overloaded operator is a member function and is thus implicitly passed. The object p2, on the right-hand side of the + operator is also an object of **class** Point, but is passed as a member function argument and thus explicitly passed. This is reflected in the return statement of the overloaded + operator:

```
return Point (x+p.x, y+p.y, z+p.z) ;
```

which illustrates that the left-hand object's data members are directly accessed whereas the right-hand object's data members are accessed via the object function argument.

Returning a Point object in the overloaded + and - operator member functions allows a user to *chain* together operators to form complicated expressions. In the program above, OVER_AS.CPP, an example of chaining the + and - operators is given:

```
p5 = p1 - p2 + p3 - p4 ;
```

If the overloaded operator member functions did not return a Point object such expressions would not be possible.

Note that the previous functions *Add()* and *Subtract()* have been removed from the **class** Point declaration since they are now redundant.

Overloading the multiplication (*), division (/) and arithmetic assignment (+=, -=, *= and /=) binary operators follows a similar procedure to the + and - operators. The following program code is extracted from ARITH.CPP:

```
// arith.cpp
// overloads the arithmetic and
// arithmetic assignment operators
//...
class Point
  {
  //...
  public:
    //...
    Point operator *  (const Point& p) ;
    Point operator /  (const Point& p) ;
    Point operator += (const Point& p) ;
    Point operator -= (const Point& p) ;
```

```
    Point operator *= (const Point& p) ;
    Point operator /= (const Point& p) ;
  };
//...
// p1 * p2
inline Point Point::operator * (const Point& p)
  {
  return Point (x*p.x, y*p.y, z*p.z) ;
  }

// p1 / p2
inline Point Point::operator / (const Point& p)
  {
  return Point (x/p.x, y/p.y, z/p.z) ;
  }

// p1 += p2
inline Point Point::operator += (const Point& p)
  {
  x += p.x ; y += p.y ; z += p.z ;
  return Point (x, y, z) ;
  }

// p1 -= p2
inline Point Point::operator -= (const Point& p)
  {
  x -= p.x ; y -= p.y ; z -= p.z ;
  return Point (x, y, z) ;
  }

// p1 *= p2
inline Point Point::operator *= (const Point& p)
  {
  x *= p.x ; y *= p.y ; z *= p.z ;
  return Point (x, y, z) ;
  }

// p1 /= p2
inline Point Point::operator /= (const Point& p)
  {
  x /= p.x ; y /= p.y ; z /= p.z ;
  return Point (x, y, z) ;
  }
//...
```

It is now possible to write expressions of the form:

```
Point p, q, r ;
//...
r = p * q ;   // multiplication
//...
```

```
r = p += q ;  // chained assignment and
              // arithmetic assignment
//...
```

The last expression is legal because the arithmetic assignment operators return a `Point` object and thus can be chained to form complex expressions.

10.2 The Unary Increment (++) and Decrement (--) Operators

When a unary operator is overloaded it is not necessary to explicitly pass an object to an overloaded operator member function because the operation is implicitly performed on the object that calls the member function. To illustrate this, let us overload the prefix and postfix increment and decrement operators for **class** `Point`:

```
// inc_dec.cpp
// overloads the prefix and postfix
// increment and decrement operators
#include <iostream>  // C++ I/O

class Point
  {
  public:
    //...
    Point operator ++ () ;      // prefix
    Point operator ++ (int) ;   // postfix
    Point operator -- () ;
    Point operator -- (int) ;
  };
//...

// prefix increment operator (++p)
inline Point Point::operator ++ ()
  {
  x += 1.0 ; y += 1.0 ; z += 1.0 ;
  return Point (x, y, z) ;
  }

// postfix increment operator (p++)
inline Point Point::operator ++ (int)
  {
  x += 1.0 ; y += 1.0 ; z += 1.0 ;
  return Point (x-1.0, y-1.0, z-1.0) ;
  }

// prefix decrement operator (--p)
inline Point Point::operator -- ()
  {
  x -= 1.0 ; y -= 1.0 ; z -= 1.0 ;
  return Point (x, y, z) ;
```

```
    }

  // postfix decrement operator (p--)
  inline Point Point::operator -- (int)
    {
    x -= 1.0 ; y -= 1.0 ; z -= 1.0 ;
    return Point (x+1.0, y+1.0, z+1.0) ;
    }

  void main ()
    {
    // define objects p and q
    Point p (1.0, 2.0, 3.0), q (1.0, 2.0, 3.0), r ;

    // p
    cout << "p    (" << p.X () << ", " << p.Y ()
         << ", "    << p.Z () << ")"  << endl ;

    r = ++p ;  // prefix
    cout << "++p (" << r.X () << ", " << r.Y ()
         << ", "    << r.Z () << ")"  << endl ;

    // q
    cout << "q    (" << q.X () << ", " << q.Y ()
         << ", "    << q.Z () << ")"  << endl ;

    r = q++ ;  // postfix
    cout << "q++ (" << r.X () << ", " << r.Y ()
         << ", "    << r.Z () << ")"  << endl ;
    }
```

with output:

```
p    (1, 2, 3)
++p (2, 3, 4)
q    (1, 2, 3)
q++ (1, 2, 3)
```

The prefix overloaded increment operator is defined as:

```
inline Point Point::operator ++ ()  // prefix
  {
  x += 1.0 ; y += 1.0 ; z += 1.0 ;
  return Point (x, y, z) ;
  }
```

The overloaded prefix operator increments each of the three data members by the value of 1 and then returns the incremented object. This is in agreement with the output shown above for object p (++p).

We noted in Chapter 4 that for both the increment and decrement operators there exists both a prefix and postfix version. To overload the postfix increment operator C++ adopts a

similar style to the prefix operator, but uses a dummy member function argument of type **int**:

```
inline Point Point::operator ++ (int)  // postfix
  {
  x += 1.0 ; y += 1.0 ; z += 1.0 ;
  return Point (x-1.0, y-1.0, z-1.0) ;
  }
```

Note the return values of the object's data members. Each data member is reduced by the constant 1 before it is returned, thus returning the original object before incrementing occurred. This is in agreement with the C++ definition of a postfix increment operator for integral data types.

A similar procedure is exercised for the prefix and postfix decrement operators, but note that for the postfix decrement operator the constant 1 is added to each of the object's data members before the object is returned.

10.3 Physical Meaning of Overloaded `Point` Operators

Before we go crazy and overload every conceivable operator for our `Point` **class**, let's just stop and examine the physical meaning behind the operators that we have overloaded so far. The `Point` **class** defines the x-, y- and z-coordinate values of a point in a three-dimensional space. A vector is defined by a magnitude and a direction. Vectors require a start or initial point and a terminal point, which together define a vector's magnitude and direction. If we acknowledge the distinction between a point and a vector then some interesting contradictions occur when we perform otherwise simple operations on points and vectors. Consider the following cases of operating on a point, p, and a vector, $\mathbf{v}$ (Bowyer and Woodwark, 1993):

$$p - p = \mathbf{v}$$
$$\mathbf{v} + \mathbf{v} = \mathbf{v}$$
$$\mathbf{v} - \mathbf{v} = \mathbf{v}$$
$$p + \mathbf{v} = p$$
$$p - \mathbf{v} = p$$
$$p + p = ?$$

The above illustrates that a point minus a point is a vector, the addition or subtraction of two vectors is another vector, and the addition or subtraction of a vector and a point is a point, but the addition of two points is undefined! At present we have overloaded the subtraction operator for **class** `Point` as:

```
inline Point Point::operator - (const Point& p)
  {
  return Point (x-p.x, y-p.y, z-p.z) ;
  }
```

which returns a `Point`. From the above discussion, the subtraction of two points should return a vector, and later versions of **`class`** `Point` will overload the subtraction operator to return a `Vector3D`[1]:

```
inline Vector3D Point::operator - (const Point& p)
  {
  return Vector3D (x-p.x, y-p.y, z-p.z) ;
  }
//...
Point p, q ;
//...
Vector3D v = p1 - p2 ;
```

Similarly, we quickly overloaded the prefix and postfix increment and decrement operators for the `Point` **`class`**, without actually asking ourselves what does `p++` actually mean? Does such an operation have any physical interpretation?

These examples of overloading operators for the `Point` **`class`** should illustrate the danger of overloading operators without giving much thought to their actual application.

10.4 The Relational and Logical Operators

Let's now examine overloading the relational and logical operators (<, <=, >, >=, ==, != and !) for **`class`** `Point`. To begin with, consider the equality operators == and !=, which have clear physical interpretations for the `Point` **`class`**. Since a test expression involving a relational operator will be either logical-true or logical-false, it is advantageous to return a **`bool`** type rather than an **`int`**:

```
bool Point::operator == (const Point& p)
  {
  return (x==p.x && y==p.y && z==p.z) ? true : false ;
  }

bool Point::operator != (const Point& p)
  {
  return (x!=p.x && y!=p.y && z!=p.z) ? true : false ;
  }
```

The return type of **`bool`** for the above two overloaded operator member functions is clearly more informative to a user of the `Point` **`class`** than, say:

```
int Point::operator == (const Point& p)
```

which indicates that any valid integer value can be returned.

A more convenient implementation of the != operator is to use logical negation:

1 Class name `Vector3D` is used rather than `Vector` to help differentiate between a three-dimensional geometric vector and a general vector class.

```
bool Point::operator != (const Point& p)
  {
  if (*this == p)
    return false ;
  return true ;
  }
```

where ***this** is the dereferenced this pointer which refers to the implicit operated-on object; refer to Section 12.29 for a discussion of **this**.

Refer to program COMPARE.CPP for further details and the implementation of overloading the relational operators <, <=, > and >= for **class** Point.

10.5 The Assignment Operator

10.5.1 Default Assignment

By default, C++ will overload the assignment operator, =, for objects of user-defined classes. For instance, the following assignment:

```
p3 = p1 ;
```

will copy *exactly* (member by member) all of the data members from p1 to p3, without having to explicitly overload the assignment operator for the Point **class**. This will generally be what you require of the default assignment operator (thus its inclusion), but if you require an alternative action for the assignment operator then simply overload it.

When one object is assigned to another, a member by member (memberwise) copy is made as opposed to a bitwise copy. Beware of the default memberwise copy mechanism when pointer data members are declared.

If you do not desire a class to support the default assignment operator then overload the operator and make it **private**:

```
class X
  {
  private:
    void operator = (const X& x) ;
  //...
  };

void Function ()
  {
  X x1, x2 ;
  //...
  x1 = x2 ;  // error: operator = is private
  }
```

In addition, the address-of (&) and sequencing (,) operators are overloaded by default for user-defined types. If this is not desired then they can be similarly privately overloaded.

10.5.2 Overloaded Assignment Operator

The following program overloads the assignment operator for the `Point` **class** and simply displays a message when called:

```
// assign.cpp
// illustrates overloading the assignment operator
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point ()
      : x (0.0), y (0.0), z (0.0) {}
    Point (double x_arg, double y_arg, double z_arg=0.0)
      : x (x_arg), y (y_arg), z (z_arg) {}
    //...
    // overloaded operator
    Point operator = (const Point& p) ;
  };

// overloaded assignment operator
Point Point::operator = (const Point& p)
  {
  x = p.x ; y = p.y ; z = p.z ;
  cout << "overloaded = operator called" << endl ;
  return Point (x, y, z) ;
  }

void main ()
  {
  // define objects
  Point p1 (3.0, 4.0, 5.0), p2 ;

  cout << "p1(" << p1.X () << //...
  cout << "p2(" << p2.X () << //...

  p2 = p1 ;  // assignment

  cout << "p2(" << p2.X () << //...
  }
```

which generates the output:

```
p1(3, 4, 5)
p2(0, 0, 0)
overloaded = operator called
p2(3, 4, 5)
```

The overloaded assignment operator member function definition indicates that the three data members of a `Point` object on the right-hand side of the = operator are directly assigned to the data members of the object on the left-hand side. The return type of the member function is `Point`, thus enabling chaining of the assignment operator:

```
//...
p1 = p2 = p3  = p4 ;  // chained assignment
```

The overloaded assignment operator member function is passed a **`const`** object by reference rather than by value, and a `Point` object is returned. When we examine pointers, a better way of overloading the assignment operator will be presented which makes use of the **`this`** pointer. The **`this`** pointer enables a function to return the object which invokes the overloaded operator member function.

10.6 Non-Member Overloaded Operator Functions

Overloaded operator functions can be non-member functions:

```
// non_mem.cpp
// illustrates non-member overloaded operator functions
#include <iostream>  // C++ I/O

class Point
  { /* ... */ };

// non-member overloaded operators:

// p1 + p2
inline Point operator + (const Point& p1, const Point& p2)
  {
  return Point (p1.X()+p2.X(), p1.Y()+p2.Y(), p1.Z()+p2.Z()) ;
  }

// p1 - p2
inline Point operator - (const Point& p1, const Point& p2)
  {
  return Point (p1.X()-p2.X(), p1.Y()-p2.Y(), p1.Z()-p2.Z()) ;
  }

void main ()
  {
  // define objects
  Point p1 (3.0, 4.0, 5.0) ;
  Point p2 (1.0, 2.0, 3.0) ;
  Point p3, p4, p5 ;

  p3 = p1 + p2 ;  // addition
  p4 = p1 - p2 ;  // subtraction
```

```
// O/P
cout << "p3 (" << p3.X () << //...
cout << "p4 (" << p4.X () << //...
}
```

with output:

```
p3 (4, 6, 8)
p4 (2, 2, 2)
```

Since the overloaded binary ::**operator**+() and ::**operator**-() functions are non-member functions, they require two arguments with object data member access via member functions. Although non-member overloaded operators are legal in C++ they are rarely used.

10.7 The Array Subscript Operator

In C++ it is possible to overload the array subscript operator `[]`. To illustrate overloading the `[]` operator, let us represent **class** `Point`'s three data members as an array of type **double**:

```
class Point
  {
  double pt[3] ;
  //...
  };
```

where `pt[0]`, `pt[1]` and `pt[2]` represent the x-, y- and z-coordinates of a point. Such a representation will assist in accessing an object's data members via the overloaded subscript operator. The following program illustrates the overloaded subscript operator:

```
// subscrpt.cpp
// illustrates overloading the subscript operator
#include <iostream>   // C++ I/O
#include <cstdlib>    // exit()

class Point
  {
  private:
    double pt[3] ;
  public:
    // constructors
    Point ()
      { pt[0] = pt[1] = pt[2] = 0.0 ; }
    Point (double x_arg, double y_arg, double z_arg=0.0)
      { pt[0] = x_arg ; pt[1] = y_arg ; pt[2] = z_arg ; }
    // overloaded operator
    double& operator [] (int i) ;
  };
```

```
// overloaded subscript operator
double& Point::operator [] (int i)
  {
  if (i<0 || i>=3)
    {
    cout << "array index " << i
         << ": out of bounds" << endl ;
    exit (EXIT_SUCCESS) ;
    }
  return pt[i] ;
  }

void main ()
  {
  Point p (1.0, 2.0, 3.0) ;

  // access
  cout << "p (" << p[0] << ", " << p[1]
       << ", "  << p[2] << ")"  << endl ;

  // assign
  p[2] = 0.0 ;

      cout << "p (" << p[0] << ", " << p[1]
           << ", "  << p[2] << ")"  << endl ;

  // out of bounds index
  p[3] = 1.0 ;
  }
```

with output:

```
p (1, 2, 3)
p (1, 2, 0)
array index 3: out of bounds
```

The return type of the overloaded subscript operator member function is a reference to **double**. This allows the function to return a reference to an element of the `pt` array so that the subscript operator can be used on the left-hand side of an assignment statement:

```
p[2] = 0.0 ;
```

We noted in Chapter 7 when discussing arrays that C++ does not automatically perform bound checking of an indexed array. The beauty of overloading the subscript operator is that it allows us to perform our own bound checking. The above member function definition tests whether the index value, `i`, is not equal to 0, 1 or 2. If the index value is outside the acceptable range, a run-time error message is displayed and the program is successfully terminated.

We shall revisit overloading the `[]` operator when we develop `Vector` and `Matrix` classes in subsequent chapters.

10.8 Composite Operators

Unfortunately, at present C++ does not support overloading of composite operators, although the usefulness of such a feature is acknowledged (Stroustrup, 1994, pp. 251–2). This does have some important consequences for indexing the elements of a user-defined two-dimensional array **class**. For instance, we observed in Chapter 8 that to index an element of a two-dimensional array in C++ we use the general syntax:

```
array_name[row_index][col_index]
```

Thus, to overload the composite operators `[][]` for integer elements of a two-dimensional array of a **class** `Array2D` we would expect to use a member function declarator of the form:

```
int& Array2D::operator [][] (int i, int j)  // error
//...
Array2D array ;
//...
array[i][j] = x ;  // not possible!
```

This is illegal in C++ because composite operators cannot be overloaded.

In addition, it is not possible to use a single subscript operator in either of the two forms:

```
int& Array2D::operator [] (int i, int j)  // error
//...
Array2D array ;
//...
array[i][j] = x ;  // not possible!

array[i, j] = x ;  // not possible!
```

The first application of **operator**`[]` is illegal because its left-hand side operand (`array[i]`) is not an `Array2D` object. The second application of **operator**`[]`, which is similar to array indexing in other languages, such as Fortran, is illegal because this operation involves the use of three operands (`array`, `i` and `j`). We observe from Appendix C that `[]` is a binary operator, and attempting to use `[]` with three operands is an attempt to alter the arity[2] of the operator, which C++ does not allow.

Alternatively, the following approach must be adopted:

```
int& Array2D::operator () (int i, int j)
//...
Array2D array ;
//...
array(i, j) = x ;
```

2 For instance, the insertion operator << is a binary operator, and it is illegal to alter the arity of the operator from binary to unary. For a further discussion of the precedence, associativity and arity of the C++ operators, refer to Chapter 4.

This approach works, but it conflicts with C++'s philosophy of indexing two-dimensional arrays for integral types. In fact, indexing the elements of an object of **class** `Array2D` via the overloaded **operator** `()` member function resembles a function call more than an array index – an awful notation. The following program demonstrates an implementation of a two-dimensional **class** `Array2D` which overloads the `()` operator as a two-dimensional array subscript operator:

```
// composit.cpp
// illustrates overloading an array indexing operator
#include <iostream>   // C++ I/O
#include <cstdlib>    // exit()

class Array2D
  {
  private:
    int a[2][2] ;
  public:
    // constructor
    Array2D ()
      { a[0][0]=0 ; a[0][1]=1 ; a[1][0]=2 ; a[1][1]=3 ; }
    // overloaded operator
    int& operator () (int i, int j) ;
  };

// overloaded subscript operator
int& Array2D::operator () (int i, int j)
  {
  if (i<0 || j>=2 || i>=2 || j>=2)
    {
    cout << "array index " << i
         << ": out of bounds" << endl ;
    exit (EXIT_SUCCESS) ;
    }
  return a[i][j] ;
  }

void main ()
  {
  Array2D array ;

  for (int i=0; i<2; i++)
    {
    for (int j=0; j<2; j++)
      cout << array(i,j) << " " ;
    cout << endl ;
    }
  }
```

with output:

```
0 1
2 3
```

When pointers are discussed in Chapter 12 a `Matrix` **`class`** will be developed which allows element indexing in the C++ style:

```
Matrix m (5, 5) ;  // object m of class Matrix
//...
m[0][3] = 2.12 ;
```

The problem with indexing the elements of an object of the two-dimensional array **`class`**, `Array2D`, does not lie with the overloaded operator function **`operator`**`[] ()` but with the implementation of **`class`** `Array2D`.

Another frequently requested use of composite operators occurs when raising a number x to a power y, x^y. Such an operation in Fortran could make use of the exponentiation operator, `**`:

```
v = x ** y
```

whereas Basic uses the ^ operator:

```
v = x ^ y
```

Alternatively, C++ possesses a library function called *`pow()`*, which conveniently performs this operation:

```
v = pow (x, y) ;
```

The number of times that x^y is required when programming does not warrant the addition of a composite exponentiation operator.

10.9 Conversions

Chapter 4 illustrated data type conversions of the form:

```
int i ;
//...
double d = i ;
```

which implicitly casts an integer into a floating-point **`double`**. C++ allows a programmer to perform casting from one type to another explicitly using either of the following styles:

```
int i ;
//...
double d = (int) i ;  // C style
double d = int (i) ;  // C++ style
```

The first approach is the C syntax, while the second is the generally adopted style in C++, for reasons that will shortly become evident.

To illustrate conversions, let us add a constructor and a *conversion member function* to the `Point` **`class`**:

```
// convert.cpp
// illustrates conversions between different types
#include <iostream>   // C++ I/O
#include <cmath>      // sqrt()

class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    Point (double d) ;
    //...
    // conversion member function
    operator double () ;
  };

// constructor
// converts double to Point
Point::Point (double d)
  {
  x = y = z = d ;
  }

// conversion member function
// converts Point to double
Point::operator double ()
  {
  double r_sq = x*x + y*y + z*z ;
  return sqrt (r_sq) ;
  }

void main ()
  {
  Point p1 (5.0) ;                    // double to Point

  cout << "p1 (" << p1.X () << //...

  double distance1 = double (p1) ;  // Point to double

  cout << "distance1: " << distance1 << endl ;

  Point p2 = 10.0 ;                   // double to Point

  cout << "p2 (" << p2.X () << //...

  double distance2 = p2 ;             // Point to double

  cout << "distance2: " << distance2 << endl ;
  }
```

with output:

```
p1 (5, 5, 5)
distance1: 8.66025
p2 (10, 10, 10)
distance2: 17.3205
```

To convert from a **double** to a Point, the following one-argument *converting constructor* is defined:

```
Point::Point (double d)
  {
  x = y = z = d ;
  }
```

This constructor simply assigns the constructor argument to each data member of an object of **class** Point. Having defined such a constructor, the following object definitions are feasible:

```
Point p1 (5.0) ;
Point p2 = 10.0 ;
```

The second statement calls the one-argument constructor because a floating-point number is being assigned to a Point object, which requires converting. Since we have not overloaded the assignment operator in the present program, the compiler will use the one-argument constructor to convert from a **double** to a Point. If the assignment operator had also been overloaded, then the overloaded assignment operator member function would be called instead of the constructor.

To perform the contrary case of converting from a Point to a **double**, the following conversion member function is defined:

```
Point::operator double ()
  {
  double r_sq = x*x + y*y + z*z ;
  return sqrt (r_sq) ;
  }
```

This function can be used in statements of the form:

```
double distance1 = double (p1) ;
double distance2 = p2 ;
```

The conversion member function calculates and returns the minimum distance from a Point object to the origin. The conversion member function could have taken any number of forms. I simply chose the most intuitive way of determining a Point's distance from the origin.

Alternatively, you could use the C style of casting if preferred:

```
double distance = (double) q ;  // Point to double
```

Consider again the Point **class** in CONVERT.CPP, but with the **explicit** keyword specifier used in the declaration of the one-argument constructor:

```
// explicit.cpp
// illustrates the explicit specifier

class Point
  {
  //...
  public:
    Point ()
      : x (0.0), y (0.0), z (0.0) {}
    explicit Point (double d)  // explicit constructor
      : x (d), y (d), z (d) {}
    //...
  };

void main ()
  {
  Point p = 1.0 ;  // error: can't convert double to Point

  Point q (2.0) ;  // ok: explicit constructor call
  }
```

A non-converting constructor declared with the **explicit** specifier prevents the constructor from being used as a converting constructor unless the conversion is explicitly indicated by a constructor call. The **explicit** specifier can only be used in constructor declarations within a **class** declaration.

10.10 A String class

It was shown in Chapter 7 that C++ represents a string of characters as an array of **char**. The C++ standard library provides a class called string in the header file STRING.H for the manipulation of strings. In fact string is a **tyedef** of class basic_string:

```
template <class CharT, class Traits=char_traits<CharT>,
          class A=allocator<CharT> >
class basic_string
  { /* ... */ };
//...
typedef basic_string<char> string ;
```

Rather than using string, this section examines a String **class**, which is an excellent **class** to illustrate the usefulness of operator overloading. It was shown in Chapter 7 that C++ represents a string of characters as an array. The String **class** implemented below encapsulates a string data member with a maximum length of 80 characters:

```
// string.cpp
// illustrates a String class
#include <iostream>  // C++ I/O
#include <string>    // strcpy()
#include <cstdlib>   // exit()
```

```
const int STR_LEN = 80 ;  // maximum length of String

class String
  {
  private:
    char string[STR_LEN] ;
  public:
    // constructor
    String (char s[]="\0")
      { strcpy (string, s) ; }
    // member function
    void Show ()
      { cout << string << endl ; }
    // overloaded operators
    String operator = (const String& s) ;
    String operator = (char s[]) ;
    String operator + (const String& s) ;
    String operator + (char s[]) ;
  };

// overloaded = operator:

// String to String
String String::operator = (const String& s)
  {
  strcpy (string, s.string) ;
  return String (string) ;
  }

// C++ string to String
String String::operator = (char s[])
  {
  strcpy (string, s) ;
  return String (string) ;
  }

// overloaded + operator:

// String + String
String String::operator + (const String& s)
  {
  // don't overrun
  if ((strlen (string) + strlen (s.string))> STR_LEN)
    {
    cout << "concatenated strings too long" << endl ;
    exit (EXIT_SUCCESS) ;
    }
  String t ;

  strcpy (t.string, string) ;    // string to t
  strcat (t.string, s.string) ;  // add two strings
```

```
  return t ;
  }

// C++ string + String
String String::operator + (char s[])
  {
  // don't overrun
  if ((strlen (string) + strlen (s))> STR_LEN)
    {
    cout << "concatenated strings too long" << endl ;
    exit (EXIT_SUCCESS) ;
    }
  String t ;

  strcpy (t.string, string) ;     // string to t
  strcat (t.string, s) ;          // add two strings
  return t ;
  }

void main ()
  {
  String s1 ;                    // null String
  s1.Show () ;

  String  s2 ("operator ") ;     // initialised String
  s2.Show () ;

  String s3 = "overloading " ; // C++ string to String
  s3.Show () ;

  String s4 = s2 + s3 + "is cool "  ; // concatenate
  s4.Show () ;
  }
```

with output:

```
operator
overloading
operator overloading is cool
```

The one-argument constructor converts a C++ string to a `String` and assigns a null string as default:

```
String (char s[]="\0")
  { strcpy (string, s) ; }
```

The `String` **`class`** overloads the assignment operator so that one `String` object can be assigned to another `String` object or a C++ string can be assigned to a `String`. The `String::`**`operator`**`=()` declarations are:

```
String operator = (const String& s) ;
```

```
String operator = (char s[]) ;
```

which are overloaded, overloaded = operator member functions (i.e. doubly overloaded). Thus, C++ allows overloaded operator member functions to be overloaded just as if they were *normal* member functions.

With the above overloaded assignment operator member functions, statements of the form below are perfectly valid:

```
String s3 = "overloading " ; // C++ string to String
```

This statement converts a C++ string to a `String` object via the overloaded assignment operator rather than the one-argument constructor. Similarly, the addition operator is overloaded to enable the concatenation either of two objects of **`class`** `String` or of a C++ string and a `String` object:

```
String s4 = s2 + s3 + "is cool "  ; // concatenate
```

Since the overloaded + operator returns a `String`, it is possible to use chained + operators.

10.11 Overloading the C++ Input Extraction (>>) and Output Insertion (<<) Operators

All of the classes discussed so far have displayed the data members of a **`class`** either by using access member functions or a member function specifically designed for outputting an object's data members. For instance, the data members of an object `p` of **`class`** `Point` have typically been displayed using the following:

```
Point p ;
//...
cout << "Point p (" << p.X () << ", " << p.Y ()
     << ", "         << p.Z () << ")" << endl ;
```

Alternatively, a member function could be defined which performs a similar task:

```
class Point
  {
  public:
    void Display () ;
    //...
  };

void Point::Display ()
  {
  cout << "("  << x << ", " << y << ", " << z << ")"  << endl ;
  }

void main ()
```

```
{
Point p ;
//...
p.Display () ;
//...
}
```

Such approaches work fine, but we have seen throughout the text that C++ offers a more attractive form of output for integral types by using the insertion operator (<<):

```
char   ch ;
int    i ;
double d ;
//...
cout << "character ch: " << ch ;
cout << "integer i   : " << i ;
cout << "double d    : " << d ;
```

cout is an object of **class** basic_ostream, which provides standard output. The insertion operator is sufficiently overloaded to output all of C++'s integral types so that a user of the basic_ostream **class** does not have to specify the type of identifier or object when performing output. Typically, the form of the OSTREAM.H header file for the overloaded << operator is:

```
template <class CharT, class Traits=char_traits<CharT> >
class basic_ostream : virtual public basic_ios<CharT,Traits>
  {
  public:
    //...
    typedef basic_ostream<CharT,Traits> ostream_type ;
    //...
    ostream_type& operator << (short n) ;
    ostream_type& operator << (unsigned short n) ;
    ostream_type& operator << (int n) ;
    //...
  };
//...
typedef basic_ostream<char> ostream ;
```

which illustrates that the << operator is overloaded for all of the integral data types. If the << operator is not overloaded for a given **class**, T, it is the responsibility of the designer or user of **class** T to overload the << operator. For example, let's examine overloading the insertion and extraction operators for **class** Point:

```
// over_io.cpp
// overloads C++'s insertion, <<, and
// extraction, >>, operators
#include <iostream>  // C++ I/O

class Point
  {
```

```
  public:
    double x, y, z ;
    // constructors
    Point ()
      : x (0.0), y (0.0), z (0.0) {}
    Point (double x_arg, double y_arg, double z_arg=0.0)
      : x (x_arg), y (y_arg), z (z_arg) {}
  };

// overloaded operators:

inline ostream& operator << (ostream& s, const Point& p)
  {
  s << "("  << p.x << ", " << p.y << ", " << p.z
    << ")"  << endl ;
  return s ;
  }

inline istream& operator >> (istream& s, Point& p)
  {
  cout << "enter x, y and z coordinates of a Point: " ;
  s >> p.x >> p.y >> p.z ;
  return s ;
  }

void main ()
  {
  Point p (1.0, 2.0, 3.0), q ;

  // O/P p
  cout << p ;

  // I/P q
  cin >> q ;

  // O/P q
  cout << q ;
  }
```

Some user interaction is:

```
(1, 2, 3)
enter x, y and z coordinates of a Point: 4 5 6
(4, 5, 6)
```

It is worth noting that the overloaded << and >> operator functions are not member functions of **class** `Point` and that the `x`, `y` and `z` data members are declared **public**. `Point`'s data members are made **public** because the overloaded insertion and extraction operator functions are member functions of classes `ostream` and `istream`, respectively. As a result, the object on the left-hand side of the << or >> operators is implicitly assumed to be of **class**

ostream or istream, respectively, so overloaded insertion and extraction functions must be non-member functions of a user-defined **class**.

Let's examine the overloaded insertion operator:

```
inline ostream& operator << (ostream& s, const Point& p)
  {
  s << "("  << p.x << //...
  return s ;
  }
```

Two objects are passed as arguments to the function. The first argument is a reference to the output stream which occurs on the left-hand side of the << operator. The second argument is a **const** reference (since the function does not alter the data members of the object) to the object on the right-hand side of the << operator. The function returns a reference to an object of **class** ostream, which allows the overloaded operator to be chained.

Within the body of the above overloaded operator function the three data members of an object of **class** Point are output using the argument object s. The insertion operator, <<, can be applied to any specified stream, and thus using the argument stream object, s, instead of the cout object makes the overloaded insertion operator function more general. When we discuss file streams we shall see that the argument stream object can be used to direct output to the screen or a file without modification of the overloaded **operator**<< *()* member function.

A similar approach is taken when overloading the extraction operator, >>:

```
inline istream& operator >> (istream& s, Point& p)
  {
  cout << "enter x, y and z coordinates of a Point: " ;
  s >> p.x >> p.y >> p.z ;
  return s ;
  }
```

A user is first prompted to enter a Point's x, y and z data members, after which **double** values are assigned to the object's appropriate data members. The second argument, p, is not a **const** reference because p's data members are assigned values, and hence changed by the function. Unfortunately, in defining the overloaded **operator**>> *()* function we have inserted the standard output cout stream object within the function definition. This is a common mistake to make and greatly reduces the generality of the overloaded **operator**>> *()* function by restricting its use to standard input and output. A better definition is the simpler form:

```
inline istream& operator >> (istream& s, Point& p)
  {
  return s >> p.x >> p.y >> p.z ;
  }
```

Rather than declaring the data members of **class** Point **public** in OVER_IO.CPP the *X()*, *Y()* and *Z()* member functions can be used, provided they each return a reference to a **double**:

```
// over_io1.cpp
// further illustrates overloading
```

```
// the insertion and extraction operators
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    // member functions
    double& X () { return x ; }
    //...
  };

// overloaded operators:

inline ostream& operator << (ostream& s, const Point& p)
  {
  return s << "("  << p.X() << ", " << p.Y() << ", "
                                    << p.Z() << ")" << endl ;
  }

inline istream& operator >> (istream& s, Point& p)
  {
  return s >> p.X() >> p.Y() >> p.Z() ;
  }

void main ()
  {
  Point p (1.0, 2.0, 3.0), q ;

  // O/P p
  cout << p ;

  // I/P q
  cout << "enter x, y and z coordinates of a Point: " ;
  cin  >> q ;

  // O/P q
  cout << q ;
  }
```

If you prefer not to return a reference to a **double** for member functions *X()*, *Y()* and *Z()* while maintaining the **private** declaration of x, y and z, the Point three-argument constructor can be used in the definition of **operator**>> *()*:

```
// over_io2.cpp
// another approach to overloading the insertion
// and extraction operators
#include <iostream>  // C++ I/O
```

```
class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    Point (double x_arg, double y_arg, double z_arg=0.0)
      : x (x_arg), y (y_arg), z (z_arg) {}
    //...
  };

//...
inline istream& operator >> (istream& s, Point& p)
  {
  double x_temp, y_temp, z_temp ;

  s >> x_temp >> y_temp >> z_temp ;

  // use Point 3-arg. constructor
  p = Point (x_temp, y_temp, z_temp) ;

  return s ;
  }

void main ()
  {
  //... as before
  }
```

The overloaded extraction operator first places the x, y and z data members of a `Point` object into temporary **`double`** variables and then uses the three-argument constructor of `Point` to assign a `Point` object to argument p.

When we discuss **`friend`**s in the next chapter we shall revisit the overloaded insertion and extraction operators, so that the above techniques of declaring a `Point`'s data members **`public`**, using member functions and a constructor, are removed.

10.12 Operators Which Cannot be Overloaded

C++ is very generous as to which of the available operators can be overloaded. However, there are a few exceptions. Operators that cannot be overloaded are:

`::`	scope resolution
`.`	direct member access
`sizeof`	size, in bytes, of an object
`.*`	direct member pointer access
`?:`	conditional

An attempt to overload either of the above operators will result in a compilation error of the form 'Operator cannot be overloaded'.

10.13 Restrictions Attached to Operator Overloading

When a given operator is overloaded, the precedence, associativity and arity of the operator cannot be altered. Therefore, the usual rules attached to the use of operators in expressions are maintained:

```
//...
10 - 4 * 3 ;     // -2
//...
p1 - p2 * p3 ;   // objects of class Point
```

10.14 Choose Carefully Which Operators to Overload

When a programmer writes the following expression, which adds two variables of the integral type **int** and then proceeds to assign the result to another **int**:

```
int a, b, c ;
//..
c = a + b ;
```

the meaning attached to the addition and assignment operators is clear. However, if you overload the addition operator for the `Point` **class**, a user of the **class** will probably expect a similar operation of two `Point` objects as an addition of two integral types:

```
Point p, q, r ;
//...
r = p + q ;
```

Operator overloading, if used wisely, can give a **class** a more natural application. However, attaching an abstract meaning to an operator can be very confusing. For instance, if the `%` operator is overloaded for the `Point` **class** to return the distance between two `Point` objects, we could write:

```
double operator % (const Point& p)
  {
  return sqrt ((x-p.x)*(x-p.x) +
               (y-p.y)*(y-p.y) +
               (z-p.z)*(z-p.z)) ;
  }
//...
Point p1, p2 ;
//...
double distance = p1 % p2 ;
```

The usual meaning attached to the `%` operator is the arithmetic modulus. A user of the `Point` **class** would probably find the overloaded `%` operator confusing because it conflicts with the original meaning attached to the modulus operator. Therefore, when overloading operators try

to retain an operator's original meaning. For this reason, it would be better to define a member function instead of attaching an abstract meaning to an operator:

```
//...
double distance = p1.DistanceBetweenPoints (p2) ;
```

The ultimate test of whether overloading a given operator is worthwhile is how much the overloaded operator is actually used. If the operator is rarely used or frequently used incorrectly then I would seriously re-evaluate the decisions made for overloading an operator.

10.15 Summary

Operator overloading is not an essential feature in the definition of an object-oriented programming language and is not supported by object-oriented languages such as Eiffel, Oberon-2 and Java. Nevertheless, operator overloading is a powerful feature of C++ programming, particularly when manipulating objects of user-defined numeric classes, such as `Point`, `Vector` and `Matrix`.

Member functions provide a means of accessing and sending messages to a specific object's data members. Overloaded operators similarly allow a programmer to operate on an object's data members, but in a more natural manner. An overloaded operator member function declaration of a given **`class`** is similar to the **`class`**'s member functions except for the keyword **`operator`**, which indicates to the compiler that a given operator will have a specific, user-defined meaning when applied to objects of the **`class`**.

Both unary and binary operators can be overloaded, and all of the available operators can be overloaded except the following five: direct member access (`.`), scope resolution (`::`), direct member pointer access (`.*`), conditional operator (`?:`) and the **`sizeof`** operator. The assignment operator is overloaded by default for objects of a user-defined **`class`**. The default assignment operator will copy all of the data members from one object exactly into another object. At present, composite operators cannot be overloaded in C++.

One-argument constructors and conversion member functions can be designed to enable both implicit and explicit conversions from one type or **`class`** to another.

The C++ insertion and extraction operators can be overloaded for performing output and input, respectively, on objects of user-defined classes.

When we examine pointers in Chapter 12 it will be shown that in certain circumstances the **`this`** pointer offers us a favourable alternative for returning an object from an overloaded operator member function. The **`this`** pointer is a pointer to the object that invoked the overloaded operator member function and is passed implicitly when the function is called.

Exercises

10.1 Overload the `+=` operator for the `String` **`class`** presented in STRING.CPP.

10.2 Further develop the `Int` **`class`** of Exercise 9.2 so that `Int` supports an overloaded modulus operator `%`. In your implementation of **`operator`**`%()` use the C++ library function *`div()`*:

```
div_t div (int numer, int denom) ;  // STDLIB.H
```

```
struct div_t
  {
  int quot ;   // quotient
  int rem ;    // remainder
  };
```

div() returns numer divided by denom, placing the quotient and remainder in the structure div_t.

10.3 Design a **class** called Tensor which encapsulates the properties of a mathematical tensor of rank two, a_{ij} ($i,j = x,y,z$). Define for Tensor constructors and overloaded +, -, << and >> operators. Also design SymmetricTensor and AntiSymmetricTensor tensor classes which encapsulate a symmetric tensor, $a_{ij} = a_{ji}$, and antisymmetric tensor, $a_{ij} = -a_{ji}$, respectively. Provide adequate conversion functions for each of the three classes Tensor, SymmetricTensor and AntiSymmetricTensor, so that objects are implicitly converted from one **class** to another.

10.4 Provide overloaded equality (==) and inequality (!=) operators for the following **class**:

```
class Vector3D
  {
  private:
    double a, b, c ;
  public:
    Vector3D (double _a, double _b, double _c)
      : a (_a), b (_b), c (_c) {}
  };
```

10.5 Provide a type **double** one-argument non-converting constructor for the **class** Vector3D of Exercise 10.4 using the **explicit** keyword.

10.6 Develop a Time **class** which encapsulates hour, minute and second data members. Provide overloaded +, -, ++ and -- overloaded operators for Time.

10.7 For the Vector3D **class** of Exercise 10.4 overload the << and >> operators while maintaining that a, b and c are **private** members of Vector3D.

Chapter 11

Friends

e2:e4

Friends are so important in C++ that they deserve a chapter all to themselves. Both classes and functions can be made friends of a given class. A friend has access to a class's private data members, although it is not a member of the class.

This chapter discusses friend functions, classes and overloaded operators. Friend functions are particularly useful for overloading binary operators where the left-hand operand object is of a class other than the class in which the overloaded operator function is a member. An entire class can be made a friend of another class and consequently has access to the friend's private data members. The C++ insertion, <<, and extraction, >>, overloaded operator functions are frequently made friends to allow the C++ stream objects access to the data members of an object of a user-defined class.

11.1 `friend` Functions

To date, a **`class`**'s data members have been designated as either **`private`** or **`public`**. A **`private`** data member can only be accessed directly within the **`class`** or indirectly via a member function, whereas a **`public`** data member can be accessed directly by an object of the data member's respective **`class`**. However, a function which is made a **`friend`** of a **`class`** is allowed direct access to a **`class`**'s **`private`** data members. Let us consider the following program, which declares a **`class`** `Circle` and defines a function *`Area()`* that is a **`friend`** of **`class`** `Circle`:

```
// fri_func.cpp
// illustrates friend functions
#include <iostream>  // C++ I/O

class Circle
  {
  private:
    double radius ;
```


G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

```
  public:
    // constructors
    Circle ()
      : radius (0.0) {}
    Circle (double r)
      : radius (r) {}
    // friend
    friend double Area (const Circle& c) ;
  };

// friend function Area() of class Circle
double Area (const Circle& c)
  {
  // area=pi*r^2
  return 3.141593 * c.radius * c.radius ;
  }

void main ()
  {
  Circle circle (5.0) ;

  double area = Area (circle) ;

  cout << "area of circle: " << area << endl ;
  }
```

with output:

```
area of circle: 78.5398
```

The function *Area()* is not a member function of **class** `Circle` and yet has access to the `radius` data member of an object of **class** `Circle` that is passed to the function as an argument. *Area()* is declared within the body of **class** `Circle`'s declaration as:

```
friend double Area (const Circle& c) ;
```

The keyword **friend** precedes the return type of the function. In fact, it is irrelevant whether the keyword precedes the return type or vice versa, but **friend** is conventionally placed before the return type. In addition, it is irrelevant whether the **friend** function declaration is placed in the **private** or **public** sections of a **class** declaration, although **friend**s are generally placed in the **public** section. I always place **friend** declarations at the end of the **public** section (if one and only one exists!) of a **class** declaration. The general syntax of a **friend** function, *FunctionName()*, declaration is:

```
class ClassName
  {
  //...
  friend return_type FunctionName (parameter_list) ;
  //...
  };
```

The keyword **friend** is only used in a member function declaration and not in the respective declarator of the member function definition. A **friend** function has access to the **private** data members of the **class** in which the friendship is made. In the above program, *Area()* has access to the radius data member of an object of **class** Circle:

```
double Area (const Circle& c)
  {
  // area=pi*r^2
  return 3.141593 * c.radius * c.radius ;
  }
```

The radius data member of an object c is accessed via the member access operator (.) in the usual manner.

Area() is called with the statement:

```
double area = Area (circle) ;
```

illustrating that the member access operator (.) is not required for function calls since *Area()* is a *normal* global function. The **friend** function call is more in line with the calling of other global functions rather than the alternative **class** member function call:

```
double area = circle.Area () ;
```

Although the **friend** function *Area()* is in agreement with procedural function calls, it conflicts with an object-oriented approach, which clearly associates an area operation with **class** Circle.

11.1.1 So Why Use friend Functions?

In the preceding example, **class** Circle could clearly have been designed to include an *Area()* member function instead of a **friend** function. There are, however, circumstances in which **friend** functions can be very useful. We shall see later that **friend** functions can be useful for overloading operators and for giving a more general application to overloaded operators. In particular, **friend**s are important for overloading C++'s input and output stream insertion (<<) and extraction (>>) operators for use with user-defined **class** objects.

As an illustration of the usefulness of **friend**s, consider the following program. The program consists of three **class** declarations A, B and C. It is required that the data members of each **class** are simultaneously displayed. We could simply incorporate a member function in each **class** which appropriately displayed an object's data member:

```
class A
  {
  private:
    int a ;
  public:
    void Display ()
      { cout << "a: " << a << endl ; }
    //...
  };

class B
```

```
  {
  private:
    int b ;
  public:
    void Display ()
      { cout << "b: " << b << endl ; }
    //...
  };
//...
```

However, such an approach requires three different and separate member functions. If this overhead is unacceptable and the access speed of an object's data member is important, a better approach is to have a single *Display()* function which is a **friend** to each of the three classes A, B and C:

```
// abc.cpp
// illustrates a function which is a friend
// of three classes A, B and C
#include <iostream>  // C++ I/O

// class declarations
class B ;
class C ;

class A
  {
  private:
    int a ;
  public:
    A (int a_arg) { a = a_arg ; }
    // friend
    friend void Display (A a_disp, B b_disp, C c_disp) ;
  };

class B
  {
  private:
    int b ;
  public:
    B (int b_arg) { b = b_arg ; }
    // friend
    friend void Display (A a_disp, B b_disp, C c_disp) ;
  };

class C
  {
  private:
    int c ;
  public:
    C (int c_arg) { c = c_arg ; }
    // friend
```

```
    friend void Display (A a_disp, B b_disp, C c_disp) ;
  };

// friend function to classes A, B and C
void Display (A a_disp, B b_disp, C c_disp)
  {
  cout << "A::a = "   << a_disp.a << "; B::b = " << b_disp.b
       << "; C::c = " << c_disp.c << endl ;
  }

void main ()
  {
  A a_main (1) ;
  B b_main (2) ;
  C c_main (3) ;

  Display (a_main, b_main, c_main) ;
  }
```

with output:

```
A::a = 1; B::b = 2; C::c = 3
```

Thus, a single function is called in *main()* which has access to the data members of classes A, B and C.

Note the forward declarations of classes B and C before the **class** declaration of A:

```
class B ;
class C ;

class A
  { /* ... */ };
```

We have seen that C++ is strict concerning the definition of a variable or object before it is used and calling functions before they are defined. Similarly, in C++ a **class** must be declared before it is referenced. Within the declaration of **class** A the **friend** function *Display()* references classes B and C before they are declared:

```
class A
  {
  //...
  friend void Display (A a_disp, B b_disp, C c_disp) ;
  };

class B
  { /* ... */ };
//...
```

Therefore, it is necessary to indicate to the compiler that the **class** declarations B and C exist within the program.

11.1.2 `friend` Functions: Good, Bad or Ugly?

At first, **`friend`** functions seem to contradict completely C++'s philosophy of **`private`** data member access by allowing access to any old function. However, a closer look at **`friend`** functions illustrates that this is not the case. In the above program, FRI_FUNC.CPP, a **`class`** `Circle` was declared as follows:

```
class Circle
  {
  private:
    double radius ;
  public:
    //...
    // friend
    friend double Area (const Circle& c) ;
  };
```

The declaration indicates that the function *`Area()`* is a **`friend`** of **`class`** `Circle`. Therefore, the designer of **`class`** `Circle` specifically made *`Area()`* a **`friend`** and thus allowed *`Area()`* access to the `radius` data member of an object of **`class`** `Circle`. A user of **`class`** `Circle` who does not have access to `Circle`'s declaration cannot define arbitrary functions which have access to `Circle`'s **`private`** data members.

However, **`friend`** functions should be used with caution and reserved for special cases in which there is no other alternative or when the temptation of using a **`friend`** cannot be resisted. As we shall shortly see, **`friend`**s can give increased functionality to overloaded operators in certain cases.

11.2 `friend` Classes

It is also possible to make an entire **`class`** a **`friend`** of another **`class`**. A **`friend class`** has full access to the **`private`** data members of a **`class`** without being a member of that **`class`**. The following program helps illustrate **`friend`** classes:

```
// fr_class.cpp
// illustrates friend classes
#include <iostream>  // C++ I/O

// forward declare Vector class
// class Vector ;

class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point ()
      : x (0.0), y (0.0), z (0.0) {}
```

```
    Point (double x_arg, double y_arg, double z_arg=0.0)
      : x (x_arg), y (y_arg), z (z_arg) {}
    // public member functions
    double& X () { return x ; }
    //...
    // overloaded operator
    // Point operator + (const Vector& v) ;
    // friend class
    friend class Vector ;
  };

class Vector
  {
  private:
    double a, b, c ;
  public:
    // constructors
    Vector ()
      : a (0.0), b (0.0), c (0.0) {}
    Vector (double a_arg, double b_arg, double c_arg=0.0)
      : a (a_arg), b (b_arg), c (c_arg) {}
    // public member functions
    double& A () { return a ; }
    //...
    // overloaded operator
    Point operator + (const Point& p) ;
    //friend class Point ;
  };

// v + p
Point Vector::operator + (const Point& p)
  {
  return Point (a+p.x, b+p.y, c+p.z) ;
  }

/*
// p + v
Point Point::operator + (const Vector& v)
  {
  return Point (x+v.a, y+v.b, z+v.c) ;
  }
*/

void main ()
  {
  Point  p (1.0, 1.0) ;
  Vector v (2.0, 2.0) ;

  Point q = v + p ;

  cout << "Point q (" << q.X () << ", " << q.Y ()
```

```
        << ", "          << q.Z () << ")" << endl ;
}
```

This program declares two classes, `Point` and `Vector`. We shall briefly describe the fundamentals of a vector shortly, but for the moment let us concentrate on **`friend`** classes. The declaration of **`class`** `Point` indicates that a `Vector` **`class`** is made a **`friend`** of `Point`:

```
class Point
  {
  //...
  public:
    //...
    friend class Vector ;
  };
```

The declaration:

```
friend class Vector ;
```

can be placed anywhere within **`class`** `Point`'s declaration, in either the **`private`** or **`public`** sections. Such a declaration allows `Vector` full access to `Point`'s data members and member functions. Thus, we are able to define the overloaded `Vector::`**`operator`**`+()` member function as:

```
Point Vector::operator + (const Point& p)
  {
  return Point (a+p.x, b+p.y, c+p.z) ;
  }
```

which accesses object `p`'s **`private`** data members directly.

`friend` classes are strange things. We went to all the trouble of defining `Point`'s data members as **`private`** and providing adequate access member functions *`X()`*, *`Y()`* and *`Z()`*, and now a **`friend class`** has direct access to data members! If a **`class`**, such as `Point`, has implemented adequate member functions to access `Point`'s data members, there is simply no need for a **`friend class`**. The above overloaded + **`operator`** member function could easily have been written as:

```
Point Vector::operator + (const Point& p)
  {
  return Point (a+p.X (), b+p.Y (), c+p.Z ()) ;
  }
```

eliminating the need to make **`class`** `Vector` a **`friend`** of **`class`** `Point`.

The above program, FR_CLASS.CPP, also includes (commented out) a **`friend class`** declaration within `Vector`'s declaration, which makes `Point` a **`friend`** of `Vector`:

```
class Vector
  {
  //...
  public:
    //...
```

```
    friend class Point ;
  };
```

Similarly, an overloaded `Point::operator+()` member function definition is commented out which has direct access to `Vector`'s **`private`** data members. Note that if a **`class`** is forward declared it is possible to declare a **`friend class`** without the keyword **`class`**:

```
// forward declare
class Vector ;

class Point
  {
  //...
  public:
    //...
    friend Vector ;  // no class keyword required
  };
```

It is worth mentioning that **`class`** friendship is not transmitted through different classes:

```
// fr_tran.cpp
// illustrates that friendship is not transitive
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    // friend class
    friend class Vector ;
  };

class Vector
  {
  private:
    double a, b, c ;
  public:
    void PointFunction (Point& p)
      { p.x = p.y = p.z = 1.0 ; }  // ok: accessible
    // friend class
    friend class Matrix ;
  };

class Matrix
  {
  private:
    Vector array[3] ; // array of Vector's
  public:
    void VectorFunction (Vector& v)
```

```
    { v.a = v.b = v.c = 1.0 ; }   // ok: accessible
  void PointFunction (Point& p)
    { p.x = p.y = p.z = 1.0 ; }   // error: not accessible
};

void main ()
  {}
```

The above program illustrates three classes: `Point`, `Vector` and `Matrix`. `Vector` is a **`friend`** of `Point` and `Matrix` is a **`friend`** of `Vector`, and consequently **`class`** `Vector` has access to `Point`'s **`private`** `x`, `y` and `z` data members and **`class`** `Matrix` has access to `Vector`'s **`private`** `a`, `b` and `c` data members. However, just because `Vector` is a **`friend`** of `Point` and `Matrix` is a **`friend`** of `Vector` does not mean that `Matrix` has access to `Point`'s **`private`** data members.

In addition note that friendship is not inherited, as we shall see in Chapter 15.

11.2.1 Vectors

Let us now return to vectors. Figure 11.1 illustrates two points p and q in a three-dimensional Cartesian coordinate system with origin O. Consider a vector **v** with initial point p and terminal point q. Let (p_x, p_y, p_z) and (q_x, q_y, q_z) be the coordinates of points p and q respectively. The scalar *components* of vector **v**(a, b, c) are then defined as:

$$a = q_x - p_x, \quad b = q_y - p_y, \quad c = q_z - p_z$$

Vectors are frequently written in terms of the triple orthogonal unit vectors **i**, **j** and **k**; refer to Fig. 11.1. The vectors **i**, **j** and **k** each have positive directions in line with the positive directions of the coordinate axes. By making use of the unit vectors, vector **v** can be written:

$$\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$$

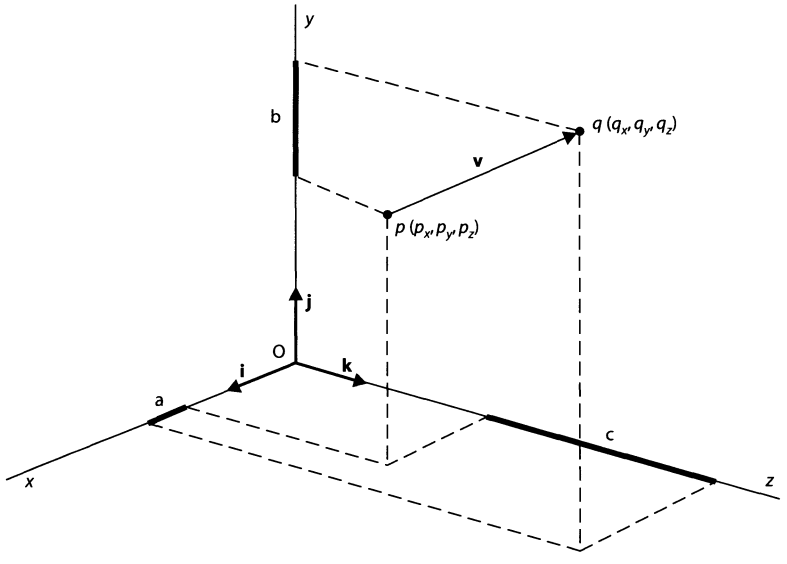


Fig. 11.1 A vector **v** in a three-dimensional Cartesian coordinate system with an initial point p and terminal point q. Also shown are the orthogonal unit vectors **i**, **j** and **k**.

The declaration of **class** Vector has three data members, a, b and c, which represent the three components of a vector:

```
class Vector
  {
  private:
    double a, b, c ;
  //...
  };
```

The form of Vector is similar to that of Point, except that the a, b and c data members represent the components of a vector, whereas the x, y and z data members of **class** Point represent a point.

11.3 Friends and Overloaded Operators

friends can be used in conjunction with overloading an operator. Let's overload the addition operator + for **class** Point using a **friend** function:

```
// fr_op.cpp
// illustrates operator overloading
// using friend functions
#include <iostream>  // C++ I/O

class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    // friend
    friend Point operator + (const Point& p1,
                             const Point& p2) ;
  };

// Point friend function (p + p)
inline Point operator + (const Point& p1, const Point& p2)
  {
  return Point (p1.x+p2.x, p1.y+p2.y, p1.z+p2.z) ;
  }

void main ()
  {
  Point  p (1.0, 1.0), q (-1.0, -1.0) ;

  Point r = p + q ;
  //...
  }
```

Note that **friend** functions can also be defined as **inline** and that no scope resolution operator (: :) is required for **friend** function declarators. Since a **friend** function does not operate implicitly on an object, both `Point` operands of a typical expression involving the binary operator + must be explicitly passed to the **friend** function. The overloaded + operator is declared a **friend** of **class** `Point` by the following declaration within the `Point` **class** declaration:

```
friend Point operator + (const Point& p1, const Point& p2) ;
```

Thus little seems to have been gained by using a **friend** overloaded operator. However, there is an important special case in operator overloading which makes **friend**s very useful. To illustrate this special case, let us consider a complex number **class**, `Complex`. Firstly, let us have a brief reminder of what a complex number is.

11.3.1 Complex Numbers

There are equations which cannot be solved purely in terms of real numbers:

$$x^2 + 2 = 0$$

A complex number, z, is a pair of real numbers (x, y) in which x is the *real part of* z and y is the *imaginary part of* z: Re $z = x$ and Im $z = y$. Complex numbers are generally represented in terms of the *imaginary unit*, i (0, 1):

$$z = x + iy$$

An important characteristic of i is that $i^2 = -1$: $((0,1)(0,1) = (-1,0) = -1)$.

Complex numbers are frequently represented as points in the *complex plane* or *Argand*[1] *diagram*. Figure 11.2 illustrates the complex plane, in which the x-axis is called the real axis and the y-axis is the imaginary axis.

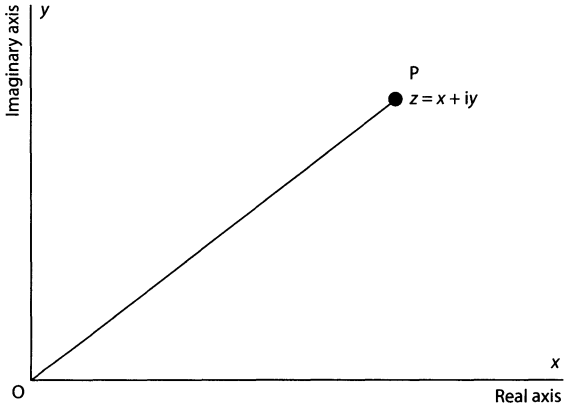


Fig. 11.2 The complex plane.

1 Jean Robert Argand (1768–1822)

The four basic arithmetical operations on two complex numbers, z_1 (x_1+iy_1) and z_2 (x_2+iy_2), are fairly straightforward and are given below.

Addition:

$$z_1 + z_2 = (x_1 + x_2) + \mathrm{i}(y_1 + y_2)$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + \mathrm{i}(y_1 - y_2)$$

Multiplication:

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + \mathrm{i}(x_1 y_2 + x_2 y_1)$$

Division:

$$\frac{z_1}{z_2} = \frac{(x_1 + \mathrm{i}y_1)(x_2 - \mathrm{i}y_2)}{(x_2 + \mathrm{i}y_2)(x_2 - \mathrm{i}y_2)} = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + \mathrm{i}\frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2}$$

The *conjugate* of a complex number z (= x + iy) is denoted by $\bar{z}$ (= x - iy).

11.3.2 Complex class

From the above discussion, a complex number **class** `Complex` should encapsulate both real and imaginary numbers. Therefore, the implementation of **class** `Complex` below contains two data members, `re` and `im`, which represent the real and imaginary parts of a complex number:

```
// fr_cplx0.cpp
// illustrates overloaded operators using
// friend functions via a Complex class
#include <iostream>  // C++ I/O

class Complex
  {
  private:
    double re, im ;
  public:
    // constructors
    Complex ()
      : re (0.0), im (0.0) {}
    Complex (double r, double i)
      : re (r), im (i) {}
    // access member functions
    double& Real () ;
    double& Imag () ;
    const double& Real () const ;
    const double& Imag () const ;
```

```
    // overloaded operators
    Complex operator + (const Complex& c) ;
    Complex operator + (double d) ;
    // overloaded operator/friend
    friend Complex operator + (double d, const Complex& c) ;
  };

// access member functions:

// non-const objects
inline double& Complex::Real ()
  { return re ; }

inline double& Complex::Imag ()
  { return im ; }
//...
// overloaded operators:

// (a+bi)+(c+di)=(a+c)+(b+d)i
inline Complex Complex::operator + (const Complex& c)
  {
  return Complex (re + c.re, im + c.im) ;
  }

// (a+bi)+d=(a+d)+(b)i
inline Complex Complex::operator + (double d)
  {
  return Complex (re + d, im) ;
  }

// overloaded operators/friends:

// d+(a+bi)=(a+d)+(b)i
inline Complex operator + (double d, const Complex& c)
  {
  return Complex (d + c.re, c.im) ;
  }

void main ()
  {
  Complex c1 (1.0, 2.0), c2 (3.0, 4.0) ;

  Complex c3 = c1   + c2 ;
  Complex c4 = c1   + 10.0 ;
  Complex c5 = 10.0 + c1 ;

  cout << "c3 (" << c3.Real () << ", "
               << c3.Imag () << ")" << endl ;
  //...
  }
```

The default and one-argument constructors both initialise the `re` and `im` data members of `Complex`. Two access member functions, *`Real()`* and *`Imag()`*, are defined which return, by reference, the `re` and `im` data members for non-constant and constant objects, respectively. The last three functions in the declaration of **`class`** `Complex` are functions that overload the + operator. The first of the three overloaded + operator functions is:

```
Complex operator + (const Complex& c) ;
```

which allows a user of **`class`** `Complex` to add together two `Complex` objects:

```
Complex c3 = c1 + c2 ;
```

When an overloaded operator is used in an expression, the object on the left-hand side of the operator must be a member of the **`class`** that overloads the operator. Therefore the above expression is valid, since object `c1` is an object of **`class`** `Complex` and object `c2` is passed as an argument to `Complex::`**`operator`**+*`()`*. The next overloaded operator function is:

```
Complex operator + (double d) ;
```

which allows a user of `Complex` to add a `Complex` object and a **`double`** floating-point number:

```
Complex c4 = c1 + 10.0 ;
```

Again, the operand on the left-hand side of the operator is an object of **`class`** `Complex`. However, what happens if a user attempts the following expression?

```
Complex c5 = 10.0 + c1 ;
```

The object on the left-hand side of the + operator is now of type **`double`**, not `Complex`. Type **`double`** was not defined with an operation involving the addition of a **`double`** and a `Complex`, and thus a compilation error will result. If a **`friend`** function is used in conjunction with the overloaded + operator and passed two arguments (instead of just the one, as above), a solution is found to the problem of adding a **`double`** and a `Complex`:

```
friend Complex operator + (double d, const Complex& c) ;
```

In this case, both operands are passed as arguments to the **`friend`** overloaded + operator function.

11.3.3 The Use of Conversion Functions

In the previous section, we had to explicitly define an overloaded operator function for the three cases `Complex+Complex`; `Complex+`**`double`** and **`double`**`+Complex`. Thus, for the four arithmetic operators (+, -, * and /) we would need 12 overloaded operator functions to perform basic arithmetical operations on `Complex` and **`double`** objects. It was noted in Chapter 10 that in C++ it is possible to define conversion member functions which enable conversion from one specified type or **`class`** to another. A conversion function which converts from **`double`** to `Complex` could be implemented as a one-argument constructor:

```
Complex::Complex (double d)
  { re = d ; im = 0.0 ; }
```

This conversion function simply assigns the **double** argument, d, to the re data member of **class** Complex and sets the im part to zero.

The conversion function now allows us to define **class** Complex with a single **friend** overloaded + operator function:

```
class Complex
  {
  //...
  public:
    // constructors
    Complex () ;
    Complex (double r, double i) ;
    // conversion func. (1 arg. constructor)
    Complex (double d) { re=d; im=0.0 ;}
    //...
    // overloaded operator/friend
    friend Complex operator + (const Complex& c1,
                               const Complex& c2) ;
  };
```

However, we can use a little trick in the case of the Complex **class**. At present, the conversion function is merely a one-argument constructor which assigns the constructor argument to the re data member and sets the im data member to zero. Thus, by assigning a default value to im in the two-argument constructor we are able to eliminate the one-argument constructor. This is illustrated in the program below, which also defines the overloaded operators -, * and /:

```
// fr_cplx1.cpp
// illustrates overloaded operators using friend functions
// via a Complex class and a conversion function
#include <iostream>   // C++ I/O
#include <cstdlib>    // exit()

const double TOLERANCE = 1e-06 ;

class Complex
  {
  private:
    double re, im ;
  public:
    // constructors
    Complex () ;          // default argument
    Complex (double r, double i=0.0) ;
    //...
    // overloaded operators/friends
    friend Complex operator + (const Complex& c1,
                               const Complex& c2) ;
    friend Complex operator - (const Complex& c1,
                               const Complex& c2) ;
```

```
    friend Complex operator * (const Complex& c1,
                               const Complex& c2) ;
    friend Complex operator / (const Complex& c1,
                               const Complex& c2) ;
  };

// constructors:

// no. arg. constructor:
// initialise Re & Im components of Complex no. to 0
inline Complex::Complex ()
  : re (0.0), im (0.0)
  {}

// 1 arg. constructor: set Re & Im components
inline Complex::Complex (double r, double i)
  : re (r), im (i)
  {}

// overloaded operators/friends:

// (a+bi)+(c+di)=(a+c)+(b+d)i
inline Complex operator + (const Complex& c1, const Complex& c2)
  {
  return Complex (c1.re + c2.re, c1.im + c2.im) ;
  }

// (a+bi)-(c+di)=(a-c)+(b-d)i
inline Complex operator - (const Complex& c1, const Complex& c2)
  {
  return Complex (c1.re - c2.re, c1.im - c2.im) ;
  }

// (a+bi)*(c+di)=(ac-bd)+(ad+bc)i
inline Complex operator * (const Complex& c1, const Complex& c2)
  {
  return Complex (c1.re*c2.re - c1.im*c2.im,
                  c1.re*c2.im + c1.im*c2.re) ;
  }

// (a+bi)/(c+di)=((ac+bd)/(c^2+d^2))+((bc-ad)/(c^2+d^2))i
inline Complex operator / (const Complex& c1, const Complex& c2)
  {
  double denom = c2.re*c2.re + c2.im*c2.im ;
  if (denom < TOLERANCE)
    {
    cout << "Complex:operator / :zero divide" << endl ;
    exit (EXIT_SUCCESS) ;
    }
  return Complex ((c1.re*c2.re + c1.im*c2.im)/denom,
                  (c1.im*c2.re - c1.re*c2.im)/denom) ;
```

```
    }

  void main ()
    {
    Complex c1 (1.0, 2.0), c2 (3.0, 4.0) ;

    Complex c3 = c1 + c2 ;
    Complex c4 = c1 - c2 ;
    Complex c5 = c1 * c2 ;
    Complex c6 = c1 / c2 ;

    cout << "c3 (" << c3.Real () << //...
    }
```

with output:

```
c3 (4, 6)
c4 (-2, -2)
c5 (-5, 10)
c6 (0.44, 0.08)
```

For further details of the declaration of the C++ standard library **class** complex the reader should refer to the header file COMPLEX.H. Class complex is a template class with specialisations complex<**float**>, complex<**double**> and complex<**long double**> so as to provide restricted conversions and allow for optimised implementations:

```
template <class T>
class complex
  {
  private:
    T re ,im;
    //...
  };

class complex<float>
  { /* ... */ };
//...
```

Template classes are examined in Chapter 13.

11.4 Overloading the C++ Input Extraction (>>) and Output Insertion (<<) Operators

The previous chapter illustrated that it is possible to overload the insertion, <<, and extraction, >>, operators so that we can output the data members of an object of a user-defined **class** in a manner similar to that of C++'s integral types:

```
Complex c ;
```

```
//...
cout << c ;
```

However, it was noted in Chapter 10 that the insertion and extraction operator functions are non-member functions of a user-defined **class**, and it is therefore necessary to make a **class**'s data members **public** so that the insertion and extraction operator functions have access to an object's data members. This goes deeply against the grain of programming in C++ in an object-oriented fashion.

friend functions can be used to allow the insertion and extraction operator functions to have access to an object's data members while specifying that an object's data is kept **private**. The following program illustrates the use of **friend** overloaded insertion and extraction functions for **class** Complex:

```
// fr_io.cpp
// illustrates overloading C++'s insertion
// and extraction operators for class Complex
//...
class Complex
  {
  private:
    double re, im ;
  public:
    //...
    friend ostream& operator << (ostream& s,
                                 const Complex& c) ;
    friend istream& operator >> (istream& s,
                                 Complex& c) ;
  };
//...
// overloaded operators/friend:
//...
// insertion operator
inline ostream& operator << (ostream& s, const Complex& c)
  {
  return s << "(" << c.re << ", " << c.im << ")" << endl ;
  }

// extraction operator
inline istream& operator >> (istream& s, Complex& c)
  {
  return s >> c.re >> c.im ;
  }

void main ()
  {
  Complex c1 (1.0, 2.0), c2 (3.0, 4.0) ;

  Complex c3 = c1 + c2 ;
  Complex c4 = c1 - c2 ;
  Complex c5 = c1 * c2 ;
  Complex c6 = c1 / c2 ;
```

```
    cout << "c3 " << c3 ;
    //...
    }
```

with output:

```
c3 (4, 6)
c4 (-2, -2)
c5 (-5, 10)
c6 (0.44, 0.08)
```

11.5 Summary

A **friend** function or **class** has access to the **private** data members of an object of a given **class**, although it is not a member of that **class**. Functions or classes are made **friend**s of a given **class** by use of the **friend** keyword in a **class**'s declaration. **friend**s can be very useful in certain cases, particularly when overloading binary operators in which the left-hand operand is not related to the **class** which overloaded the operator and when overloading the C++ insertion and extraction operators.

Exercises

11.1 For the following complex number **class**, `Complex`, define a function *`Sqrt()`* which returns the square root of a `Complex` object and is a **friend** of `Complex`:

```
class Complex
  {
  private:
    double re, im ;
  public:
    Complex (double real, double imag=0.0)
      : re (real), im (imag) {}
  };
```

11.2 Provide a single overloaded + operator, using **friend**, for the `String` **class** of Chapter 10, STRING.CPP.

11.3 Extend the `Tensor`, `SymmetricTensor` and `AntiSymmetricTensor` classes developed in Exercise 10.3 so that the overloaded +, -, << and >> operators are overloaded as **friend** functions.

11.4 For the `Int` **class** of Exercise 9.2 provide an overloaded + operator. Overload + as member, **friend**, and non-member and non-**friend** functions of **class** `Int`.

11.5 The following code compiles without errors, but results in a run-time error. Find the error. Why does it occur?

```
class Int
  {
  private:
    int data ;
  public:
    Int (int value)
    : data (value) {}
    //...
    friend Int operator + (const Int& i1, const Int& i2) ;
  };

Int operator + (const Int& i1, const Int& i2)
  {
  return Int (i1+i2) ;
  }

void main ()
  {
  Int i1 (1), i2 (2) ;

  Int i3 = i1 + i2 ;
  }
```

11.6 Overload the insertion and extraction operators for **class** `Point` without the use of **friend** functions.

11.7 The following program code lists two classes called `LinkedList` and `LinkedListIterator`:

```
class LinkedList
  {
  private:
    class Node
      {
      //...
      } * first ;
  public:
    LinkedList ()
      : first (0) {}
    //...
  };

class LinkedListIterator
  {
  private:
    LinkedList&       list ;
    LinkedList::Node* current ;
  public:
    LinkedListIterator (LinkedList& l)
      : list (l), current (l.first) {}
    //...
  };
```

It is observed that `LinkedListIterator` requires direct access to `LinkedList`'s **`private`** data member, `first`, and nested **`class`** `Node`. How can this access problem be resolved using **`friend`** classes? Linked lists and their respective iterators will be discussed in more detail in Chapters 12 and 13.

Chapter 12

Pointers

it's good to know the future's Orange

Pointers are a feature that C++ inherited from the C programming language. Although not specifically an object-oriented feature of C++, pointers are nevertheless extremely powerful and form a large part of the language. In several cases, such as the implementation of data structures, pointers are essential for good programming, and give a programmer more control over the manipulation of objects.

In contrast with all of the objects that we have seen so far, a pointer holds the memory address of another object rather than holding the object itself. A pointer is a variable that can point to any object of a specified type. The actual object pointed to by a pointer can be accessed indirectly by a process known as indirection or dereferencing. Pointers can be used in arithmetic expressions, declared as const, passed as arguments to a function and returned by a function. A pointer is not restricted to pointing to a single object, but can point to an array of objects stored contiguously in memory. It is this property of pointers that reveals their similarity to C++ static arrays. Furthermore, C++ allows a pointer to point to another pointer and even to an entire function.

Dynamic memory allocation is made possible with the help of pointers. We will examine both the C malloc()/free() functional and the C++ new and delete operator approaches to dynamic memory allocation. The new and delete operators allocate and deallocate, respectively, a requested block of memory from the operating system and correspondingly allow a programmer to have more control over the allocation and deallocation of memory for objects and arrays of objects. C++ allows the new and delete operators to be overloaded for user-defined classes.

String, Vector and Matrix classes are developed in the present chapter and serve as typical applications of pointers in the implementation of C++ data structures.

12.1 A First Look at Pointers

So far, we have managed to introduce 11 chapters without discussing pointers – but they can wait no more! In theory, pointers are a beautifully simple concept, but in practice they are the root of much confusion, particularly with programmers who are new to C++. I have not met anyone yet who naturally took to pointers; hence the reason for waiting 11 chapters before dis-

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
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cussing them. However, without sounding too pessimistic, and bearing in mind that you are probably a very good C++ programmer by now, let's kick off our tour of pointers.

Simply stated:

A pointer is a variable that contains the memory address of another object.

If you remember, at all times, that (1) a pointer is a *variable* and (2) a pointer contains the memory address of *another* object, and not the object itself, the notion of a pointer will be easier to grasp. Note the emphasis placed on 'another object'. A pointer does not contain the memory address of itself.

The general syntax of a pointer definition is:

```
type_specifier* pointer_name ;
```

This statement defines a pointer variable, `pointer_name`, which points to objects of a C++ integral or user-defined type or **`class`**, `type_specifier`. The asterisk, `*`, is the *indirection* operator, which indicates to the compiler that the defined variable is a pointer and not a *normal* object. For instance:

```
int*     i_ptr ;  // pointer to int
double*  d_ptr ;  // pointer to double
Point*   p_ptr ;  // pointer to Point
Complex* c_ptr ;  // pointer to Complex
```

defines variables `i_ptr`, `d_ptr`, `p_ptr` and `c_ptr` to be pointers to the C++ integral types **`int`** and **`double`** and user-defined classes `Point` and `Complex` respectively. Frequently, the type that a pointer points to is referred to as the *base type.*

Let's examine a program to help clarify pointers:

```
// ptr_1st.cpp
// illustrates pointer variables
#include <iostream>  // C++ I/O

void main ()
   {
   int i (0) ;   // integer variables i & j
   int j (9) ;

   int* i_ptr ;  // pointer variables i_ptr & j_ptr
   int* j_ptr ;

   cout << "i: " << i << endl ;
   cout << "j: " << j << endl ;

   cout << "i_ptr: " << i_ptr << endl ;
   cout << "j_ptr: " << j_ptr << endl ;
   }
```

with output:

```
i: 0
j: 9
i_ptr: 001F5F
j_ptr: 00FFFF
```

The *main()* function begins by defining and initialising two **int** variables i and j:

```
int i (0) ;    // integer variables i & j
int j (9) ;
```

which, by now, should be familiar. Next, two pointer variables i_ptr and j_ptr to the base type **int** are defined in *main()*:

```
int* i_ptr ;  // pointer variables i_ptr & j_ptr
int* j_ptr ;
```

The asterisk in the definition indicates to the compiler that i_ptr and j_ptr are not normal variables of type **int** but pointers to type **int**. So what's the difference? Examining the program output we find that the variables i and j are displayed, as initialised, with the values 0 and 9 respectively. The pointers i_ptr and j_ptr are displayed as the memory addresses 001F5F and 00FFFF respectively (these memory addresses will probably be different when you run the above program). Thus the output illustrates that the compiler treats a pointer variable as containing the memory address of a variable and not the value of a variable. To illustrate this important point further we require the use of the *address-of operator*, &.

12.2 The Address-of Operator

The address-of operator, &, is a unary operator that returns the memory address of the operand that it acts upon[1]. Consider the program:

```
// aoo.cpp
// illustrates the address-of operator &
#include <iostream>  // C++ I/O

void main ()
  {
  int i (0) ;    // integer variables i & j
  int j (9) ;

  cout << "i: " << i << endl ;
  cout << "j: " << j << endl ;

  cout << "&i: " << &i << endl ;  // O/P addresses of i & j
  cout << "&j: " << &j << endl ;
  }
```

1 Refer to Appendix C for a complete list of the available operators in the C++ programming language.

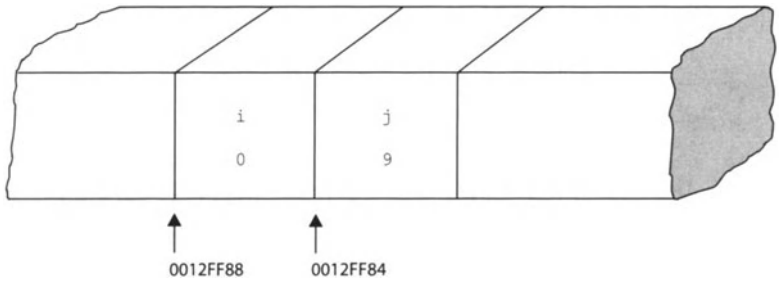


Fig. 12.1 Schematic representation of a segment of the computer's memory which contains the variables `i` and `j`.

with output:

```
i: 0
j: 9
&i: 0012FF88
&j: 0012FF84
```

which illustrates that the address-of operator, `&`, accesses the memory addresses of variables `i` and `j`. Figure 12.1 schematically illustrates the memory addresses of variables `i` and `j`.

Do not confuse the address-of operator with the reference operator, which was introduced in Chapter 6 when we discussed the passing and returning of objects via function arguments by reference rather than by value. The address-of operator precedes a variable or object name, whereas the reference operator follows the variable type:

```
cout << &i ;                  // address-of operator
//...
void Function (int& i) ;    // reference operator
//...
```

We can now combine the above two programs, PTR_1ST.CPP and AOO.CPP, to assign the memory addresses of the variables `i` and `j` to a pointer variable, `ptr`, of base type **`int`**:

```
// ptr.cpp
// illustrates the pointer variables
// and the address-of operator &
#include <iostream>  // C++ I/O

void main ()
  {
  int i (0) ;   // integer variables i & j
  int j (9) ;

  cout << "i: " << i << endl ;
  cout << "j: " << j << endl ;

  int* ptr ;  // pointer to int

  ptr = &i ;  // ptr points to i

  cout << "(ptr=&i); ptr: " << ptr
```

```
        << ", *ptr: " << *ptr << endl ;

    ptr = &j ;  // ptr points to j

    cout << "(ptr=&j); ptr: " << ptr
         << ", *ptr: " << *ptr << endl ;
    }
```

with output:

```
i: 0
j: 9
(ptr=&i); ptr: 0012FF88, *ptr: 0
(ptr=&j); ptr: 0012FF84, *ptr: 9
```

The following statement assigns the memory address of variable `i` to the pointer `ptr`:

```
ptr = &i ;  // ptr points to i
```

This statement does not assign, by copying, the value of variable `i` to the pointer variable `ptr`, but rather assigns the memory address of `i` to `ptr`. Later in the program, the pointer `ptr` points to the memory address of variable `j`:

```
ptr = &j ;  // ptr points to j
```

which illustrates that the pointer `ptr` is indeed a variable and can be assigned the memory address of any variable of type **`int`**.

If the value of a variable of type **`int`** is required rather than its memory address, the unary operator * precedes the pointer:

```
cout << *ptr ;  // access value and not memory address
```

`*ptr` accesses the value held at the memory address pointed to by the operand `ptr`. Alternatively, the **`int`** value held at the memory address pointed to by `ptr` could be assigned to a variable or constant of type **`int`**:

```
int  i ;
int* ptr ;
//...
ptr = &i ;
//...
int value = *ptr ;          // assign value of i to a variable
//...
const int VALUE = *ptr ;    // assign value of i to a constant
```

12.3 Multiple Pointer Definitions in a Single Statement

We have seen with normal variables that C++ allows us to define multiple uninitialised variables in a single statement of the form:

```
int i, j, k ;
```

Similarly, it is possible to define multiple uninitialised pointers in a single statement:

```
int* i_ptr, * j_ptr, * k_ptr ;
```

Note that this is not the same as:

```
int* i_ptr, j_ptr, k_ptr ;
```

which is a common mistake. In this case, `i_ptr` is a pointer to **`int`**, but `j_ptr` and `k_ptr` are normal variables of type **`int`** and not pointers.

12.4 A Note on Pointer Definitions

There are generally two styles of defining a pointer variable:

```
int* ptr1 ;
int *ptr2 ;
```

The first attaches the asterisk to the type specifier, while the second attaches the asterisk to the variable name. Which style you adopt makes no difference to the compiler, but the style that I adopt is to attach the asterisk to the type specifier. This convention helps to clarify that the asterisk is part of the type specification and not the variable name, i.e. **`int`*** is a pointer to **`int`** in the above example.

12.5 Another Look at Pointers

Let's now take another look at pointers from an lvalue/rvalue perspective. Remember that an object's lvalue is its memory address, whereas the rvalue is the actual data held by the object. You may have noticed from the above discussion that the rvalue of a pointer is in fact the lvalue of a different object. Thus we can now extend our definition of a pointer:

A pointer is a variable whose rvalue contains the lvalue or memory address of another object

To help illustrate this definition, let us consider another program example on pointers:

```
// ptr_next.cpp
// another look at pointers
#include <iostream>  // C++ I/O

void main ()
  {
  int  j (9) ; // integer variable j
```

```
int* ptr ;   // pointer to int
cout << "lvalue of j   : " << &j
     << ", rvalue of j   : " << j   << endl ;

cout << "lvalue of ptr: " << &ptr
     << ", rvalue of ptr: " << ptr << endl ;

ptr = &j ;  // ptr points to j

cout << "lvalue of j   : " << &j
     << ", rvalue of j   : " << j   << endl ;
cout << "lvalue of ptr: " << &ptr
     << ", rvalue of ptr: " << ptr << endl ;

*ptr = 10 ;  // assign 10 to rvalue of ptr (j)

cout << "lvalue of j   : " << &j
     << ", rvalue of j   : " << j   << endl ;
cout << "lvalue of ptr: " << &ptr
     << ", rvalue of ptr: " << ptr << endl ;

int k = *ptr ;  // assign rvalue of ptr to k

cout << "lvalue of k   : " << &k
     << ", rvalue of k   : " << k   << endl ;
}
```

with output:

```
lvalue of j   : 0012FF88, rvalue of j   : 9
lvalue of ptr: 0012FF84, rvalue of ptr: 00000100
lvalue of j   : 0012FF88, rvalue of j   : 9
lvalue of ptr: 0012FF84, rvalue of ptr: 0012FF88
lvalue of j   : 0012FF88, rvalue of j   : 10
lvalue of ptr: 0012FF84, rvalue of ptr: 0012FF88
lvalue of k   : 0012FF80, rvalue of k   : 10
```

The program begins by defining and initialising a variable, `j`, of type **`int`** and a pointer variable to **`int`**, `ptr`:

```
int  j = 9 ;  // integer variable j
int* ptr ;    // pointer to int
```

Directly following these definitions the lvalues and rvalues of `j` and `ptr` are displayed on the screen. Both `j` and `ptr` have their own unique lvalues – remember that a pointer is a variable and has to be stored somewhere in memory! The rvalue of `j` is equal to the integer constant 9 while `ptr`'s rvalue is assigned the 0 pointer because at this point the rvalue of `ptr` has not been assigned the memory address of an identifier of type **`int`**; see Fig. 12.2(a).

Next, the following statement assigns the lvalue of `j` to the rvalue of `ptr` by means of the address-of operator, `&`:

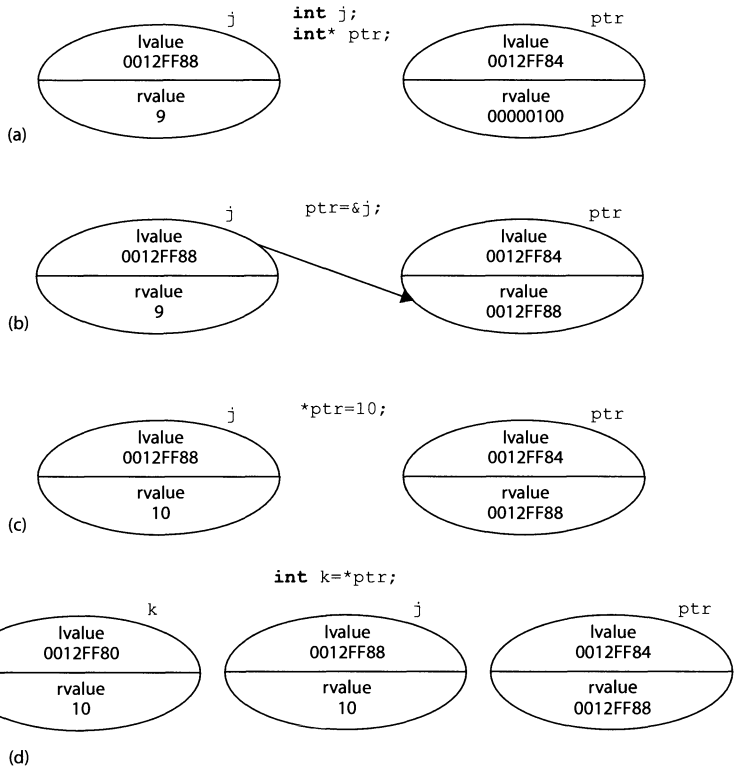


Fig. 12.2 Playing with pointers.

```
ptr = &j ;  // ptr points to j
```

Observe from the program output that the rvalue of `ptr` is now equal to the lvalue of `j` (Fig. 12.2(b)).

The statement:

```
*ptr = 10 ;  // assign 10 to rvalue of ptr (j)
```

uses the process of indirection (i.e. indirect access) to change the rvalue of `j` via the rvalue of the pointer `ptr`. This statement alters the rvalue of `j` because the previous statement, `ptr=&j`, assigned the lvalue or memory address of `j` to the rvalue of `ptr`. Therefore, whatever happens to the rvalue of `ptr` will indirectly happen to `j` (Fig. 12.2(c)). Indirection is frequently referred to as *dereferencing*.

The final assignment statement in program PTR_NEXT.CPP is:

```
int k = *ptr ;  // assign rvalue of ptr to k
```

This statement uses indirection to assign the rvalue of `ptr`, which is the lvalue of `j`, to the rvalue of variable `k` (Fig. 12.2(d)).

Confused? Don't worry - manipulating pointers does take some getting use to.

12.6 Valid Pointers

In an analogous manner to uninitialised variables, similarly C++ assigns the 0 pointer to the memory address held by the rvalue of a pointer variable. However, always ensure that before the rvalue of a pointer variable is accessed that it has been initialised to a valid memory address.

Beware of uninitialised pointers. The following program defines a variable `i` and a pointer `ptr`. The variable `i` is then assigned the constant value of 1. Next, pointer `ptr` is assigned the value of `i` without `ptr` first being initialised:

```
// nonvalid.cpp
// illustrates an uninitialised pointer
#include <iostream>  // C++ I/O

void main ()
  {
  int  i ;
  int* ptr ;

  i = 1 ;

  *ptr = i ;  // error: ptr doesn't point to i
  }
```

The compiler will probably issue a compilation warning of the form 'use of pointer `ptr` before definition' for the above program. The program will compile and link correctly, but will generally generate a run-time application error. Therefore, always ensure that a pointer is defined and initialised to a valid memory address before it is used, e.g. `ptr=&i`.

12.7 The NULL Pointer

C++ inherited from C a `#define` macro called `NULL`[2] which is referred to as 'the `NULL` pointer'. The `NULL` pointer is guaranteed not to point to valid data and will always be interpreted as logical-false (i.e. 0). The `NULL` pointer can be useful for string manipulation since the value of the null terminator character '`\0`' is not fixed and can be machine-dependent. It is still popular to use `NULL` but due to C++'s tighter type checking it is recommended to use the simpler zero (0) pointer instead of the `NULL` macro. Furthermore, try to use either `NULL` or 0 instead of '`\0`' whenever possible in your programs. If you prefer to use an identifier rather than 0 then define `NULL` as a **`const int`**:

```
const int NULL = 0 ;
```

Also, it is good programming practice to initialise a pointer to 0 pointer when a pointer is defined. This convention can eliminate the common programming mistake of accidentally using an uninitialised pointer in a test expression. For example:

---

2 The majority of compilers `#define` the `NULL` pointer in the STDIO.H header file. The Borland C++ (version 5.x) compiler `defines` `NULL` in the _NULL.H header file.

```
// null.cpp
// illustrates the 0 pointer
#include <iostream>  // C++ I/O

void main ()
  {
  int* ptr1 ;       // uninitialised pointer
  int* ptr2 = 0 ;   // pointer initialised to 0

  if (ptr1)
    cout << "uninitialised pointer tested logical-true"
         << endl ;

  if (ptr2)
    cout << "initialised pointer to 0" << endl ;
  }
```

with output:

```
uninitialised pointer tested logical-true
```

Commercial compilers are generally very good at detecting uninitialised pointers and will generally issue a compilation warning for the above program of the form 'possible use of pointer "ptr1" before definition'.

12.8 Don't Mix Types

Consider the program:

```
// mix.cpp
// illustrates the mixing of types with pointers
#include <iostream>  // C++ I/O

void main ()
  {
  double d (1.25) ;
  int*   ptr ;

  ptr = &d ;          // error: can't convert double pointer
                      // to int pointer

  ptr = (int*) &d ;   // cast double pointer to int pointer

  double e = *ptr ;
  cout << "e: " << e << endl ;
  }
```

The statement:

```
ptr = &d ;
```

attempts to assign a **double** pointer to an **int** pointer. Such an assignment will generate a compilation error of the form 'cannot convert a double pointer to an integer pointer'. It is possible to cast the **double** pointer into an **int** pointer:

```
ptr = (int*) &d ;
```

However, such a conversion[3] of one pointer type to another can result in very strange behaviour. Pointer `ptr` is of base type **int** and `d` is of type **double**. Thus, a **double** memory address has been assigned in to an **int** memory address. Hence the statement:

```
double e = *ptr ;
```

assigns only two bytes of data to the **double** eight-byte variable `e`. It is shown in the next section that a pointer to **void** can be defined which allows the mixing of types.

12.9 void Pointers

The previous section illustrated that statements of the form:

```
int* ptr = &d ;
```

are illegal, where `d` is of type **double**. For such cases C++ allows us to define a pointer to **void**:

```
void* ptr ;
```

which at first appears contradictory. A pointer to **void** allows any type of pointer to be assigned to itself. This is illustrated in the following program:

```
// ptr_void.cpp
// illustrates pointers to void

#include <iostream>   // C++ I/O
#include <cstdlib>    // qsort()
#include <string>     // strcmp()

// sorts two arguments
int Sort (const void* a, const void* b)
  {
  return strcmp ((char*)a, (char*)b) ;
  }

void main ()
```

3 It was noted in Chapter 4 that a cast of the form `(int*) &d` is a C-style cast. Unfortunately a C++-style pointer cast `int*(&d)` is not valid.

```
{
int    i, * i_ptr ;  // int variable & pointer
double d, * d_ptr ;  // double variable & pointer
void*  v_ptr ;       // void pointer

i_ptr = &i ;    // ok: int* to int*
//i_ptr = &d ;  // error: double* to int*

d_ptr = &d ;    // ok: double* to double*
//d_ptr = &i ;  // error: int* to double*

v_ptr = &i ;  // ok: int* to void*
v_ptr = &d ;  // ok: double* to void*

// strings:
char names[][5] = { "Jean", "John", "Jane", "Jack" } ;

qsort ((void*) names, 4 , sizeof (names[0]), Sort) ;

for (int j = 0; j < 4; j++)
  cout << names[j] << " " ;
cout<< endl ;

// integers:
int lottery_numbers[] = { 30, 3, 5, 44, 14, 22 } ;

qsort ((void*) lottery_numbers, 6 ,
       sizeof (lottery_numbers[0]), Sort) ;

for (int k = 0; k < 6; k++)
  cout << lottery_numbers[k] << " " ;
}
```

with sorted output:

```
Jack Jane Jean John
3 5 14 22 30 44
```

The statements:

```
v_ptr = &i ;  // ok: int* to void*
v_ptr = &d ;  // ok: double* to void*
```

illustrate that pointers to **int** and **double** can be assigned to a pointer to **void**.

To illustrate the usefulness of pointers to **void**, the above program also sorts an array of strings and an array of integers. The C++ library function *qsort()* is used for performing the sorting of both strings and integers. The signature of the *qsort()* function is:

```
void qsort (void* base, size_t nelem, size_t width,
            int (*fcmp) (const void*, const void*)) ;
```

qsort()[4] is declared in the header file STDLIB.H. Although *qsort()*'s function declarator may appear slightly confusing at the moment, it is an excellent function to illustrate the usefulness of pointers to **void**.

The *qsort()* function sorts an array pointed to by the `base` argument using the *median of three* variant of the quicksort algorithm for sorting. When *qsort()* returns, the sorted array is the array pointed to by `base`. The argument `nelem` is the number of elements in the array, with each element of the array occupying `width` number of bytes. The `fcmp` argument is a pointer to a user-defined comparison function which is called by *qsort()* to sort the elements of the array. In the above program the comparison function is called *Sort()* and defined as:

```
int Sort (const void* a, const void* b)
  {
  return strcmp ((char*)a, (char*)b) ;
  }
```

Note the casting performed on the arguments `a` and `b` of the *Sort()* function from **void*** to **char***, which is necessary because the function declaration of the string compare function, *strcmp()*, is:

```
int strcmp (const char* s1, const char* s2) ;
```

Both function arguments, `a` and `b`, of *Sort()* are of type **const void***, which allows us to define a single *Sort()* function which will deal with a variety of argument types. This eliminates the necessity of defining a comparison function for each possible type:

```
int SortChar (const char* a, const char* b)
//...
int SortInt (const int* a, const int* b)
//...
int SortDouble (const double* a, const double* b)
//...
```

In addition, the first argument to the *qsort()* function is a pointer to **void** which similarly allows the function to sort a variety of types. The function call:

```
qsort ((void*) names, 4 , sizeof (names[0]), Sort) ;
```

sorts an array, `names`, of characters, whereas the function call:

```
qsort ((void*) lottery_numbers, 6 ,
        sizeof (lottery_numbers[0]), Sort) ;
```

sorts an array, `lottery_numbers`, of integers.

In conclusion, the primary use of **void*** is for low-level programming and for passing pointers to functions, as illustrated above. The use of pointers as function arguments is discussed in more detail shortly.

---

4 Refer to your compiler's documentation for a more complete description of the *qsort()* function. See also the related functions declared in STDLIB.H of *bsearch()*, *lfind()* and *lsearch()*.

12.10 const Pointers

The previous section illustrated the use of the **const** modifier with pointers. When dealing with pointers and the **const** modifier it is possible to specify whether the pointer is **const**, the object that it points to is **const** or both. The following program illustrates the use of **const** with pointers:

```
// const_p.cpp
// illustrates the use of const with pointers
#include <iostream>  // C++ I/O

void main ()
  {
  int i (1), j (2), k (3), l (4) ;

  int* ncp_nco ;        // non-const pointer, non-const object

  const int* ncp_co ;       // non-const pointer, const object

  int* const cp_nco = &i ; // const pointer, non-const object

  const int* const cp_co = &j ; // const pointer, const object

  // initialise uninitialised non-const pointers
  ncp_nco = &k ;
  ncp_co  = &l ;

  *ncp_nco = 10 ;
  *ncp_co  = 20 ;  // error: const object
  *cp_nco  = 30 ;
  *cp_co   = 40 ;  // error: const object

  ncp_nco++ ;
  ncp_co++ ;
  cp_nco++ ;        // error: const pointer
  cp_co++ ;         // error: const pointer
  }
```

The following statement defines a **const** object but not a **const** pointer:

```
const int* ncp_co ;       // pointer to const object
```

whereas:

```
int* const cp_nco = &i ; // const pointer to object
```

defines a **const** pointer and not a **const** object. Thus, the rule is that if the **const** modifier appears on the left-hand side of the asterisk, *, then the object is constant; else if **const** appears on the right-hand side of * then the pointer is constant. Note that a **const** pointer must be initialised when defined.

12.11 Pointers and Function Arguments

When functions were discussed in Chapter 6 it was noted that there are three ways in which an argument can be passed to a function in C++: by value, by reference or by pointer. This section will now discuss the passing of an argument to a function by pointer.

With the exception of arrays, objects are (by default) passed as arguments to a function by value. This guarantees that the function cannot alter in any way the object in the calling function, because when an object is passed by value the function automatically makes its own copy of the object. However, there are occasions when it is required that the original object in the calling function is modified. In such situations Chapter 6 showed that a function argument can be passed by reference and consequently acts as an alias for the calling function object. In principle, the passing of function arguments by pointer or by reference are similar (with some minor syntactical differences) in that they both allow data with local scope to a function to be altered. The following program (based on Kernighan and Ritchie (1978, pp. 91–3)) compares the three methods of passing function arguments:

```
// func_arg.cpp
// illustrates passing function arguments by pointer
#include <iostream>  // C++ I/O

// pass by value
void ValueSwap (int x, int y)
  {
  int temp ;  // temporary variable to hold x

  temp = x ;
  x    = y ;
  y    = temp ;
  }

// pass by reference
void ReferenceSwap (int& x, int& y)
  {
  int temp ;  // temporary variable to hold x

  temp = x ;
  x    = y ;
  y    = temp ;
  }

// pass by pointer
void PointerSwap (int* ptr_x, int* ptr_y)
  {
  int temp ;  // temporary variable to hold x

  temp   = *ptr_x ;
  *ptr_x = *ptr_y ;
  *ptr_y = temp ;
  }
```

```
void main ()
  {
  int a (0), b (1) ;
  cout << "a: " << a << ", b: " << b << endl ;

  ValueSwap (a, b) ;
  cout << "ValueSwap ():     a: " << a << ", b: " << b << endl ;

  ReferenceSwap (a, b) ;
  cout << "ReferenceSwap (): a: " << a << ", b: " << b << endl ;

  PointerSwap (&a, &b) ;
  cout << "PointerSwap ():   a: " << a << ", b: " << b << endl ;
  }
```

with output:

```
a: 0, b: 1
ValueSwap ():     a: 0, b: 1
ReferenceSwap (): a: 1, b: 0
PointerSwap ():   a: 0, b: 1
```

The first of the three functions passes two arguments by value:

```
void ValueSwap (int x, int y) ;
```

and the program output illustrates that this function does not alter the integers `a` and `b` from the *`main()`* calling function. The next function passes two arguments by reference:

```
void ReferenceSwap (int& x, int& y) ;
```

The ampersand indicates to the compiler that `x` and `y` are aliases for `a` and `b` in the *`main()`* calling function:

```
ReferenceSwap (a, b) ;
```

The last of the three functions passes pointers to the two integers to be swapped:

```
void PointerSwap (int* ptr_x, int* ptr_y) ;
```

Since the *`PointerSwap()`* function is passed the memory address of two integers it must dereference the arguments to access their values. The statement:

```
temp = *ptr_x ;
```

assigns the integer value held at the memory address pointed to by *`ptr_x`* to the temporary variable `temp`, and similarly for the other two function body assignment statements.

`PointerSwap()` is called in *`main()`* as:

```
PointerSwap (&a, &b) ;
```

Notice the use of the address-of operator, &, since *PointerSwap()* requires that a variable's memory address is passed, not its value.

The key difference between passing by reference and passing by pointer is that a reference is an alias for the calling function variable, whereas a pointer is the actual memory address of the calling function variable. Comparing the two definitions and calls of the functions *ReferenceSwap()* and *PointerSwap()* illustrates two main advantages of passing function arguments by reference rather than by pointer: (1) the function body statements do not require the use of the indirection operator and (2) a user of the function is relieved of the burden of having to provide the function with the memory address of a variable explicitly.

12.12 Pointer Arithmetic

Just as with normal variables, pointers can be used in expressions, provided that certain restrictions are observed. Pointer arithmetic is restricted to the addition (+), subtraction (-), increment (++) and decrement (--) operators. Let us first consider the addition operator.

12.12.1 Pointer Addition

The following program illustrates pointer addition:

```
// p_add.cpp
// illustrates pointer addition
#include <iostream>  // C++ I/O

const int STR_LEN (80) ;

void main ()
  {
  char string[STR_LEN] ;

  cout << "enter a string: " ;
  cin.get (string, STR_LEN) ;

  int   position (0) ;
  char* p_str    = string ;

  cout << "enter a position from the start of the string: " ;
  cin  >> position ;

  if (position >= 0 && position < 80)
    cout << "the character at position " << position
         << " is: " << *(p_str + position) ;
  else
    cout << "position " << position
         << " is out of range-try again!" ;
  }
```

Some user interaction is:

```
enter a string: while you were out
enter a position from the start of the string: 6
the character at position 6 is: y
```

The statement:

```
char* p_str = string ;
```

defines a pointer, `p_str`, to **char** and simultaneously assigns the memory address of a user-entered string to the rvalue of the pointer `p_str`. The statement:

```
*(p_str + position) ;
```

accesses the character held at the memory location (`p_str+position`).

Beware of making the common mistake:

```
*p_str + position ;
```

which would return the value 125 for the string 'while you were out' with `position` equal to 6 because the ASCII decimal equivalent of the letter 'w' of the word 'while' is the integer 119.

The above program illustrates adding an integer, or an *offset*, to a pointer. It is not possible to add pointers in C++. For example, try adding the following two pointers;

```
*(p_str + string) ;
```

This statement will not compile and will issue a compilation warning of the form 'invalid pointer addition'. It makes no sense to add two pointers together, and only an offset can be added to a pointer.

12.12.2 Pointer Subtraction

We saw in the last subsection that pointers cannot be added. Provided that two pointers are of the same base type, one pointer can be subtracted from another pointer. The difference is simply the scalar number of elements of the pointer's base type. To help illustrate this, consider the program:

```
// p_sub.cpp
// illustrates pointer subtraction
#include <iostream>  // C++ I/O

// returns the length of a string
int StrLength (char* string)
  {
  char* p_str = string ;

  while (*p_str)
    p_str++ ;
  return p_str - string ;
  }

void main ()
```

```
{
char string[] = "while you were out" ;

cout << "length of \"" << string << "\" is: "
     << StrLength (string) ;
}
```

Within the body of the *StrLength()* function the pointer p_str is initialised to the start of the string array. While p_str does not point to the 0 pointer, the **while**-loop continues to increment the pointer p_str to the next character in the string. When the end of the string is reached and the **while**-loop terminates, the length of the string is obtained by subtracting the memory address of the first character of the string from the memory address of the last character in the string. The use of the increment operator with the p_str pointer will become clearer from the discussion in the next section.

12.12.3 Increment and Decrement Operators and Pointers

The program STR_ARG.CPP of Chapter 7 examined two functions: *StrCompare()*, which compares two strings, and *StrCopy()* which copies one string into another string. Both functions were passed two arrays of characters using array notation. Let's now modify these functions in terms of pointer notation and simultaneously illustrate the use of the postfix increment operator with pointers:

```
// p_inc.cpp
// illustrates the increment operator and pointers
#include <iostream>  // C++ I/O

// compares two strings
// returns 1 if str1==str2, 0 if str1!=str2
int StrCompare (const char* str1, const char* str2)
  {
  while (*str1++ == *str2++)
    {
    if (*str1 == 0)
      return 1 ;
    }
  return 0 ;
  }

// copies str2 into str1
void StrCopy (char* str1, const char* str2)
  {
  while (*str1++ = *str2++) ;
  }
void main ()
  {
  char string1[] = "this is a string" ;
  char string2[] = "this is a string" ;
  char string3[] = "another string" ;
  // compare string1 and string2
  if (StrCompare (string1, string2) == 1)
```

```
    cout << "string1 == string2" << endl ;
else
    cout << "string1 != string2" << endl ;

// compare string2 and string3
if (StrCompare (string2, string3) == 1)
    cout << "string2 == string3" << endl ;
else
    cout << "string2 != string3" << endl ;

// copy string3 to string1
StrCopy (string1, string3) ;

cout << endl << "string1: " << string1 << endl ;
}
```

StrCompare() performs the comparison of two strings by comparing each character of the two strings that have the same array index. Since the function argument `str1` is a pointer to **char**, `str1++` advances `str1` to the next character. `*str1++` is the actual character pointed to before `str1` is incremented, since the postfix operator is used. The test condition of the **while**-loop tests each character with the same array index of both strings and then proceeds to increment to the next character. Within the body of the **while**-loop each character of the string pointed to by `str1` is tested against the 0 pointer.

How is the expression `*str1++` to be interpreted? It could be read as `*(str1++)` or `(*str1)++`. As always, to resolve this issue we need to refer to the precedence and associativity of the operators * and ++. Appendix C informs us that the binary operators * and ++ both have the same high precedence level of 15 and are right-associative. Hence the expression `*str1++` is interpreted as `*(str1++)` with `str1` incremented first, followed by pointer dereferencing.

StrCopy()[5] copies one string, character by character and including the null terminator, into another string. Note that *StrCopy()* could equally have been written as follows:

```
while ((*str1++ = str2++) != 0)
```

The **while**-loop tests whether the condition within the inner parentheses is not equal to 0. However, since 0 is logical-false in C++ then the explicit test using the relational ! = operator is unnecessary.

It is worth mentioning that the `str1` pointer to **char** of function *StrCopy()* cannot be **const** modified because the string that `str1` points to is altered, whereas the string that `str2` points to is not altered.

The increment and decrement operators, when applied to a pointer, increase or decrease the pointer by a scalar according to what the pointer points to. For instance, the following illustrates incrementing **int**, **float** and **double** pointers and decrementing an **int** pointer:

```
int* i_ptr ; float* f_ptr ; double* d_ptr ;
//...
i_ptr++ ;  // point to next int
f_ptr++ ;  // point to next float
```

5 This function definition may issue a compilation warning 'possible incorrect assignment'.

```
d_ptr++ ;  // point to next double
//...
i_ptr-- ;  // point to previous int
```

Each statement increments or decrements the rvalue (memory address) of its respective pointer by the scalar value of the pointer's base type.

It is also possible to use the prefix increment and decrement operators with pointers. For example:

```
int* ptr ;
//...
++ptr ;   // increment ptr then access memory address
*++ptr ;  // increment ptr then access value
--ptr ;   // decrement ptr then access memory address
*--ptr ;  // decrement ptr then access value
```

12.13 Pointers and Relational Operators

The previous section illustrated that relational operators and pointers can be used in conjunction in expressions. Pointers can be compared using the <, <=, >, >=, == and ! = relational operators, but be sure that you know what you mean! If pointers point to different unrelated arrays or objects, then a pointer comparison will generally be meaningless. If, however, pointers are related somehow pointer comparisons can be useful. For example, we saw in the *StrCompare()* function of the P_INC.CPP program the combined use of the == equality operator with pointers to **char**:

```
int StrCompare (const char* str1, const char* str2)
  {
  while (*str1++ == *str2++)
    {
    if (*str1 == 0)
      return 1 ;
    }
  return 0 ;
  }
```

12.14 Pointers are Quicker

Aside from the obvious benefits, pointers can offer a more efficient way of manipulating data. For example, compare the *StrCompare()* function discussed above for both array indexing and pointer notation. Examining the array notation version first:

```
int StrCompare (char str1[], char str2[])
  {
  int i = 0 ;
  while (str1[i] == str2[i])
```

```
    {
    if (str1[i++] == 0)
//...
}
```

we observe that a variable `i` is required to index through the two string arrays. Considering only the `str1` character array, the lvalue of `str1[0]` must be loaded, then the value of the variable `i` must be loaded, then the value of `i` must be added to the lvalue of `str1[0]` followed by the equality comparison of `str1[i]` and `str2[i]`, and so on.

The relevant code of the pointer version of *`StrCompare()`* is:

```
int StrCompare (const char* str1, const char* str2)
  {
  while (*str1++ == *str2++)
  //...
  }
```

which illustrates that the use of the index variable, `i`, and hence the subscript operator, `[]`, are eliminated. However, the majority of commercial compilers will generate identical code for both cases, so choose the style you feel most confortable with.

12.15 Pointers and Arrays

When arrays were discussed in Chapter 7, the similarity between arrays and pointers was highlighted. This section now discusses in detail the close relationship between arrays and pointers by first examining one-dimensional arrays.

12.15.1 One-Dimensional Arrays

Consider the following program:

```
// parray.cpp
// illustrates pointers and arrays
#include <iostream>  // C++ I/O

void main ()
  {
  const int SIZE (5) ;

  int array[SIZE] = { 1, 2, 3, 4, 5 } ;

  // array notation
  cout << "array[i]  : " ;
  for (int i=0; i<SIZE; i++)
    cout << array[i] << " " ;
  cout << endl ;

  // pointer notation
```

```
    cout << "*(array+j): " ;
    for (int j=0; j<SIZE; j++)
      cout << *(array + j) << " " ;
    }
```

with output:

```
array[i]   : 1 2 3 4 5
*(array+i): 1 2 3 4 5
```

Within *main()* an array, `array`, of `SIZE` elements, each of type **int**, is defined and initialised. The elements of the array are stored consecutively in memory and are indexed using array notation via the subscript operator, `[]`, as `array[0]`, `array[1]`, ..., `array[SIZE-1]`.

Alternatively, and equivalently, the elements of the array can be accessed using pointer notation with the expression:

```
*(array + i)
```

which illustrates that the expression `array[i]` of an arbitrary array `array` is directly equivalent to `*(array+i)`, where `i` is an array index variable. The expression `(array+i)` is the memory address of the `i`th integer element in the array from the start of the array, whereas `*(array+i)` is the value held at that memory address. Note that it is not possible to use the increment operator in conjunction with the array name, `*array++`, although the elements of the array are accessed consecutively. To see why such an expression is not possible, let us first of all remind ourselves that an array name is in fact a synonym for the memory address of the array's zeroth element.

If `ptr` is a pointer to type **int** (i.e. **int*** `ptr`), then the statement:

```
ptr = &array[0] ;
```

assigns the memory address, using the address-of operator, of the first element of `array` (`array[0]`) to the rvalue of pointer `ptr`. However, since the name of an array is a synonym for the memory address of the first or zeroth element of the array, the previous assignment is equivalent to:

```
ptr = array ;
```

Therefore, returning to our expression `*array++`, we now see that such an expression is illegal because the name of an array is a constant, and it is not possible to increment a constant. The name of an array is constant and not a variable simply because the memory address of the array has to be placed somewhere in memory, and if this memory position were to change continually we would be unable to access the array elements[6]. Also, the assignments `array=ptr` and `ptr=&array` are illegal.

However, it is possible to increment a pointer: `ptr++`. The following program illustrates accessing the elements of an array using the increment operator in conjunction with a pointer:

```
// p_inc1.cpp
```

6 A multitasking operating system, such as Microsoft Windows or IBM's OS/2, allows variable or moveable memory addresses of arrays and objects, but such a process is generally beyond our control.

```
// illustrates incrementing a pointer
#include <iostream>  // C++ I/O

void main ()
  {
  const int SIZE (5) ;

  int array[SIZE] = { 1, 2, 3, 4, 5 } ;

  // define & initialise a pointer, ptr, to point to array
  int* ptr = array ;

  for (int i=0; i<SIZE; i++)
    cout << *(ptr++) << " " ;
  }
```

12.15.2 Two-Dimensional Arrays

Let us now examine two-dimensional arrays:

```
// p_a_2d.cpp
// illustrates pointers and two-dimensional arrays
#include <iostream>  // C++ I/O

void main ()
  {
  const int ROW (2), COL (3) ;

  int array[ROW][COL] = { {1, 2, 3}, {4, 5, 6} } ;

  // array notation
  cout << "array[i][j]: " << endl ;
  for (int i=0; i<ROW; i++)
    {
    for (int j=0; j<COL; j++)
      cout << array[i][j] << " " ;
    cout << endl ;
    }

  // pointer notation
  cout << "*(*(array+i)+j): " << endl ;
  for (int k=0; k<ROW; k++)
    {
    for (int l=0; l<COL; l++)
      cout << *(*(array + k) + l) << " " ;
    cout << endl ;
    }
  }
```

with output:

```
array[i][j]:
1 2 3
4 5 6
*(*(array+i)+j):
1 2 3
4 5 6
```

We saw in Chapter 7 that an element of a two-dimensional array at row position `i` and column position `j` can be accessed using the compound subscript operators:

```
array[i][j]
```

The equivalent pointer notation is:

```
*(*(array + i) + j)
```

12.16 Pointers, Arrays and Function Arguments

The topic of passing arrays as function arguments was covered in Chapter 7, but without reference to pointers. Consider the following program:

```
// paafa.cpp
// illustrates pointers, arrays and function arguments
#include <iostream>  // C++ I/O

// returns the sum of an array of integers
// access array elements by subscript operator []
double SubscriptSum (const double array[], int size)
  {
  double sum = 0 ;
  for (int i=0; i<size; i++)
    sum += array[i] ;
  return sum ;
  }

// returns the sum of an array of integers
// access array elements by pointer and increment operator ++
double PointerSum (const double* array, int size)
  {
  double sum = 0 ;
  for (int i=0; i<size; i++)
    sum += *array++ ;
  return sum ;
  }

void main ()
  {
  double a_main[5] = { 1.1, 2.2, 3.3, 4.4, 5.5 } ;
```

```
cout << "SubscriptSum (a_main): "
     << SubscriptSum (a_main, 5) << endl ;
cout << "PointerSum (a_main)  : "
     << PointerSum (a_main, 5)   << endl ;
}
```

First of all, let's remind ourselves of the technique used in Chapter 7 of passing arrays to functions and indexing an array's elements:

```
double SubscriptSum (const double array[], int size)
  {
  //...
  sum += array[i] ;
  //...
  }
```

Both the array name and subscript operator, [ ], are used within the function parentheses to indicate to the compiler that an array is being passed to the function. The elements of the array are accessed using the array subscript operator.

However, it is common practice when passing arrays as function arguments to use pointers rather than the name of an array and the subscript operator. For example, the C++ library function *strcmp()* is passed two constant arrays of characters as pointers:

```
int strcmp (const char* s1, const char* s2) ;
```

The function *PointerSum()* in program PAAFA.CPP uses pointer notation:

```
double PointerSum (const double* array, int size)
  {
  //...
  sum += *array++ ;
  //...
  }
```

The values of the elements of an array are accessed by dereferencing the `array` pointer function argument and using the postfix increment operator, as outlined in the previous sections.

It was mentioned in Chapter 7, with reference to program ARG.CPP and function *Display3()*, that it is not possible to pass a two-dimensional array to a function in the general form:

```
void Display3 (int array[][], int rows, int cols)
  {
  //...
  }
```

This function declarator is illegal because the compiler must know the column dimension of an array so that it can find the memory address of the array. Pointers offer a neat solution to the problem of passing a two-dimensional array to a function whose size is determined by the `row` and `col` arguments:

```
// f_p_a_2d.cpp
// illustrates passing a general
// two-dimensional array to a function
#include <iostream>  // C++ I/O

void Display (int* array, int rows, int cols)
  {
  for (int i=0; i<rows; i++)
    {
    for (int j=0; j<cols; j++)
      cout << array[i*cols + j] << " " ;
    cout << endl ;
    }
  }

void main ()
  {
  const int ROW (2), COL (3) ;

  int array[ROW][COL] = { {1, 2, 3}, {4, 5, 6} } ;

  Display (&array[0][0], ROW, COL) ;
  }
```

with output:

```
1 2 3
4 5 6
```

The function `Display()` is passed a pointer to **int** which points to the start of a two-dimensional array. Note the use of the address-of operator in the `Display()` function call. Within the function body of `Display()` the elements of an array are consecutively accessed using the expression:

```
array[i*col + j]
```

demonstrating that the elements of a two-dimensional array are in fact contiguously stored in memory as a one-dimensional array.

12.17 Pointers to Pointers

We have seen that a pointer can point to the memory address of an another object:

```
int* ptr ;  // rvalue of ptr holds lvalue of object of type int
```

C++ allows us to define a pointer to a pointer. The first pointer points to the memory address of the second pointer which, as we have seen previously, points to the memory address of an object; see Fig. 12.3. The following program helps illustrate a pointer to a pointer:

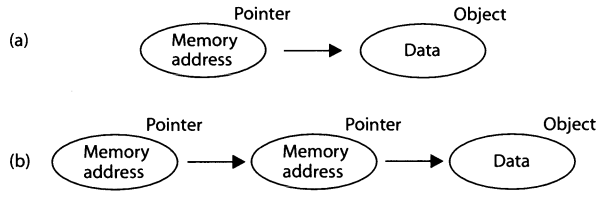


Fig. 12.3 (a) Pointer to an object. (b) Pointer to a pointer, which in turn points to an object.

```
// ptr_ptr.cpp
// illustrates a pointer to a pointer
#include <iostream>  // C++ I/O

void main ()
  {
  int a (1) ;

  int*  ptr1 ;  // pointer to type int
  int** ptr2 ;  // pointer to an int pointer

  ptr1 = &a ;   // assign memory addresses
  ptr2 = &ptr1 ;

  cout << "&a    : " << &a     << endl ;
  cout << "a     : " << a      << endl ;

  cout << "&ptr1 : " << &ptr1  << endl ;
  cout << "ptr1  : " << ptr1   << endl ;
  cout << "*ptr1 : " << *ptr1  << endl ;

  cout << "&ptr2 : " << &ptr2  << endl ;
  cout << "ptr2  : " << ptr2   << endl ;
  cout << "*ptr2 : " << *ptr2  << endl ;
  cout << "**ptr2: " << **ptr2 << endl ;
  }
```

with output:

```
&a    : 0012FF88 // lvalue of a
a     : 1        // rvalue of a
&ptr1 : 0012FF84 // lvalue of ptr1
ptr1  : 0012FF88 // rvalue of ptr1 (loc. of a)
*ptr1 : 1        // rvalue of a
&ptr2 : 0012FF80 // lvalue of ptr2
ptr2  : 0012FF84 // lvalue of ptr1
*ptr2 : 0012FF88 // rvalue of ptr1 (loc. of a)
**ptr2: 1        // rvalue of a
```

The output (with added comments) illustrates that `ptr1` points to the memory address of **`int`** variable `a` and `*ptr1` is the actual value of `a`. Pointer `ptr2` points to the memory address of

`ptr1`, `*ptr2` is the rvalue of `ptr1`, which in turn is the memory address of `a`, and `**ptr2` is the value of `a`. Note that `ptr2` is a pointer to an **`int`** pointer and not a pointer to **`int`**.

12.18 Arrays of Pointers and Pointers to Arrays

C++ allows arrays of pointers:

```
// array_p.cpp
// illustrates an array of pointers
#include <iostream>  // C++ I/O

const int N_ELEMENTS (3) ;
const int N_ARRAYS   (2) ;

// sums elements of a 1D array
int SumArray (int* p, int n_elements)
  {
  int sum (0) ;
  for (int i=0; i<n_elements; i++)
    sum += *p++ ;
  return sum ;
  }

// sum elements of a list of arrays
int Sum (int** pp, int n_list)
  {
  int sum (0) ;
  for (int i=0; i<n_list; i++)
    sum += SumArray (*(pp+i), N_ELEMENTS) ;
  return sum ;
  }

void main ()
  {
  int* array_p[N_ARRAYS] ;  // array of pointers to
                            // type int

  int array1[N_ELEMENTS] = {1, 2, 3} ;
  int array2[N_ELEMENTS] = {4, 5, 6} ;

  array_p[0] = array1 ;
  array_p[1] = array2 ;

  cout << "sum: " << Sum (array_p, N_ARRAYS) ;
  }
```

Within *`main()`*, an array of pointers to **`int`** is defined:

```
int* array_p[N_ARRAYS] ;
```

array_p is an array of N_ARRAYS integer pointers. The above program defines two arrays, array1 and array2, of type **int** and then proceeds to assign the address of each array to a pointer array:

```
array_p[0] = array1 ;
array_p[1] = array2 ;
```

Alternatively, we could have used the address-of and subscript operators explicitly:

```
array_p[0] = &array1[0] ;
array_p[1] = &array2[0] ;
```

The first argument of the *Sum()* function declarator is a pointer to an **int** pointer:

```
int Sum (int** pp, int n_list)
```

The pointer to an **int** pointer allows us to pass to the *Sum()* function an array of pointers, which in the present case is a list of one-dimensional **int** arrays.

A popular use of arrays of pointers is with an array of strings. We saw in Chapter 7 (program INIT_STR.CPP and Fig. 7.6) that a two-dimensional array of constant strings wastes memory space if all the strings are not of the same length. An array of pointers to strings wastes no memory space:

```
// a_p_str.cpp
// illustrates arrays of pointers to strings
#include <iostream>  // C++ I/O

void main ()
  {
  const int MAX_STR_LEN (9) ;
  const int NUM (5) ;

  // array notation (wastes 16 of the 45 bytes allocated)
  char names[NUM][MAX_STR_LEN] =
    { "Graham", "Ian", "Jim", "Jonathan", "Tony" } ;

  // pointer notation
  char* p_names[] =
    { "Graham", "Ian", "Jim", "Jonathan", "Tony" } ;
  //...
  }
```

p_names defines an array of pointers to strings noting that arrays of pointers can be initialised when defined, just like *normal* arrays. Since the strings do not form part of an array the compiler will store them contiguously in memory, thus wasting no space. Refer to Fig. 12.4 for a comparison between a two-dimensional array of strings and an array of pointers to strings.

The above program illustrated an array of pointers. C++ also allows us to define a pointer to an array. The following program illustrates a pointer to an array:

```
// p_array.cpp
// illustrates pointers to arrays
```

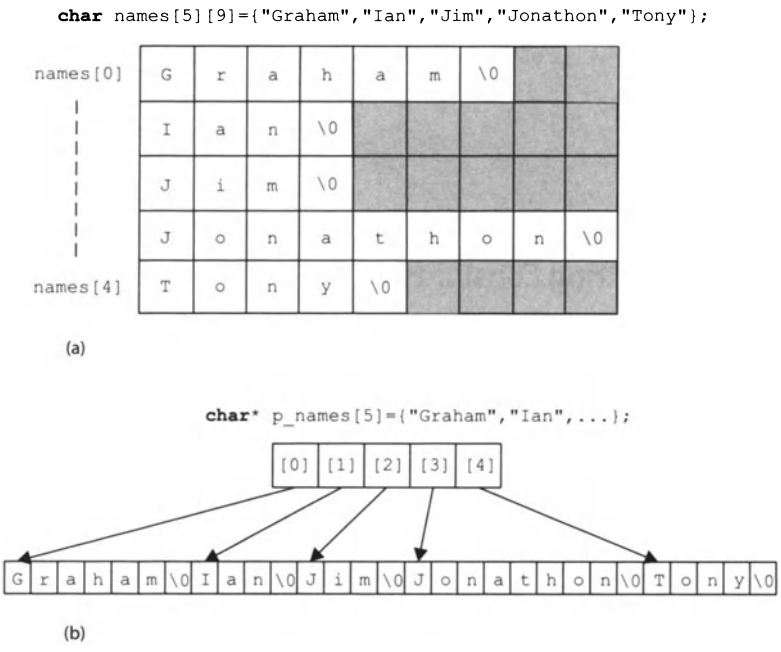


Fig. 12.4 (a) A two-dimensional array of strings which wastes 16 of the 45 bytes allocated. (b) An array of pointers to strings.

```
#include <iostream>  // C++ I/O

const int SIZE (2) ;

void main ()
  {
  int (*p_array)[SIZE] ;       // pointer to an array

  int array[SIZE] = {1, 2} ;  // an array

  // assign memory address of array to pointer
  p_array = &array ;

  cout << "&array[0]: " << &array[0] << endl ;
  cout << "p_array  : " << p_array   << endl ;
  }
```

with output:

```
&array[0]: 0012FF84
p_array  : 0012FF84
```

which illustrates that the `p_array` pointer holds the memory address of `array`.

The statement:

```
int (*p_array)[SIZE] ;
```

defines a pointer, `p_array`, to an array with `SIZE` elements. The memory address of an array, `array`, is assigned to the rvalue of the pointer to an array with the help of the address-of operator:

```
p_array = &array ;
```

12.19 Pointers to String Constants

The previous section illustrated the use of pointers with string constants. Consider the following program:

```
// ptr_str.cpp
// illustrates pointers to string constants
#include <iostream>  // C++ I/O
void main ()
  {
  char string[] = "array string constant" ;

  char* str = "pointer string constant" ;
  //...
  }
```

This program is perfectly legal, although memory has not explicitly been requested in the above program for the string constant 'pointer string constant'. This is because the string constant is placed in a *string table* which is generated by the compiler. The pointer `str` is a pointer to the string in the string table. However, I do not recommend using the pointer-to-string-constant approach if the string is to be modified, because commercial C++ compilers generally generate optimised string tables which allow a string constant to be accessed from different parts of a program. When the **new** and **delete** operators are introduced later we shall see a better method of defining a pointer to a string, allocating sufficient memory for the string and finally deleting the memory allocated to hold the string when we are finished with it.

12.20 Pointers to Functions

This section examines the use of pointers to point to the memory address of a function. This is a powerful use of pointers in C++ which allows a general function to be invoked simply by an application of a pointer.

To date we have seen functions that have been passed objects by value, reference or pointer:

```
double             sqrt   (double x) ;       // value (MATH.H)
inline complex<T> sqrt (const complex<T>& c) ;// reference
                                               // (COMPLEX.H)
size_t             strlen (const char* s) ;    // pointer
                                               // (STRING.H)
```

In program PTR_VOID.CPP the use of the C++ library function *qsort()* illustrated that a pointer to a function can also be passed as a function argument:

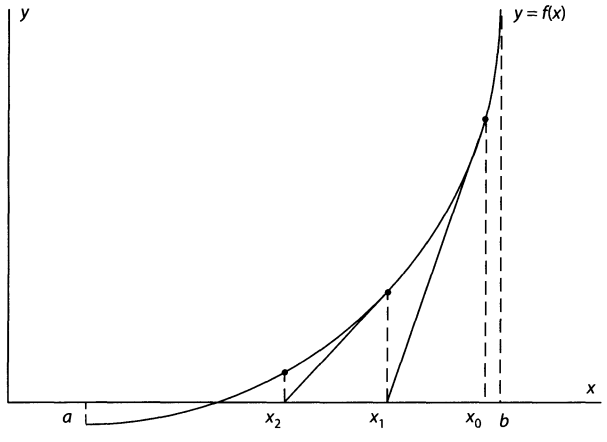


Fig. 12.5 The Newton–Raphson method.

```
void qsort (void* base, size_t nelem, size_t width,
            int (*fcmp) (const void*, const void*)) ;
```

`fcmp` is a pointer to a comparison function.

To illustrate pointers to functions, let's examine the Newton–Raphson iterative method for solving equations of the form $f(x) = 0$, where f is an arbitrary function with a continuous derivative f'. The Newton–Raphson method is a popular technique for determining the roots of an equation because it is simple, efficient and has a graphical interpretation. Figure 12.5 illustrates the graph of $y = f(x)$ for the range $a \leq x \leq b$, where a and b are constants. Given an initial approximate value for x, x_0, we let x_1 be the point of intersection of the tangent to the curve of f at x_0 with the x-axis:

$$\tan\alpha = f'(x_0) = \frac{f(x_0)}{x_0 - x_1}, \text{ or } x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

and similarly for point x_1 and so on until a required degree of convergence has been achieved. Thus, the general iterative Newton–Raphson formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n = 0, 1, 2, \ldots$$

In the program below the termination criteria adopted is that convergence has been achieved when the following condition is satisfied:

$$|x_{n+1} - x_n| \leq \varepsilon$$

where ε is a small non-zero constant.

To help emphasise the generality of pointers to functions, the function which implements the Newton–Raphson method, *`NewtonRaphson()`*, is placed in the file NR.CPP. The *`NewtonRaphson()`* function is developed with no prior knowledge of the functions that it will operate on, only that they will be of the form:

```
double arbitrary_function (double) ;
```

The function declaration of *NewtonRaphson()* is given in the NR.H header file:

```
// nr.h
// header file for Newton-Raphson method
#ifndef _NR_H  // prevent multiple includes
#define _NR_H

#include <cmath>  // fabs()

void NewtonRaphson (double (*f_ptr)(double),
                    double (*df_ptr)(double),
                    int n_iterations, double tolerance,
                    double x0, double& x_new, int& count,
                    bool& converged) ;

#endif // _NR_H
```

The implementation file is NR.CPP:

```
// nr.cpp
// implementation of Newton_Raphson method

#include "nr.h"  // header

// performs Newton-Raphson iteration
void NewtonRaphson (double (*f_ptr)(double),
                    double (*df_ptr)(double),
                    int n_iterations, double tolerance,
                    double x0, double& x_new,
                    int& count, bool& converged)
  {
  double x_old (x0) ;  // initialise x_old to initial guess

  count = 0 ;   // no. of iterations required for convergence
  converged = false ;

  for (int i=0; i<n_iterations; i++)
    {
    count++ ;
    // Newton-Raphson
    x_new = x_old - (f_ptr (x_old) / df_ptr (x_old)) ;

    // if converged
    if (fabs(x_new - x_old) < tolerance)
      {
      converged = true ;
      break ;
      }
    // else continue using new estimate
    else
      x_old = x_new ;
```

```
    }
  }
```

The program which tests the *NewtonRaphson()* function is P_FUNC.CPP:

```
// p_func.cpp
// illustrates pointers to functions
#include <iostream>   // C++ I/O
#include <cmath>      // fabs()
#include <iomanip>    // setprecision()

#include "nr.h"       // NewtonRaphson()

const int    MAX_ITERATIONS (100) ;
const double TOLERANCE      (1e-06) ;

// function prototypes
double Function  (double x) ;
double DFunction (double x) ;

void main ()
  {
  double  x_initial (0.0), x_solution (0.0) ;
  int     iter_to_converge (0) ;
  bool    converge_or_not (false) ;

  cout << endl << "enter an initial value of x
                       for the solution of f(x)=x^2-2=0: " ;
  cin  >> x_initial ;

  NewtonRaphson (Function, DFunction, MAX_ITERATIONS,
                 TOLERANCE, x_initial, x_solution,
                 iter_to_converge, converge_or_not) ;

  if (converge_or_not)
    {
    cout << endl
         << "converged after "
         << iter_to_converge  << " iterations" << endl
         << "solution of f(x)=x^2-2=0: "
         << setprecision (7) << x_solution ;
    }
  else
    cout << "sorry-no convergence" ;
  }

// function
double Function (double x)
  {
  // f(x)=x^2-2
  double fx = x * x - 2.0 ;
```

```
    return fx ;
    }

// derivative of function
double DFunction (double x)
    {
    // f'(x)=2x
    double deriv = 2.0 * x ;
    return deriv ;
    }
```

with some user interaction:

```
enter an initial value of x for the solution of f(x)=x^2-2=0:
10

converged after 7 iterations
solution of f(x)=x^2-2=0: 1.414214
```

As an example, the function $f(x) = x^2 - 2 = 0$, $f'(x) = 2x$ is chosen to test the Newton–Raphson method for computing the square root of 2. The output illustrates that the *NewtonRaphson()* function computes $\sqrt{2}$ to an accuracy of six decimal places in only seven iterations given an initial estimate of $x_0 = 10$.

The function declarator of *NewtonRaphson()* is:

```
void NewtonRaphson (double (*f_ptr)(double),
                    double (*df_ptr)(double), int
                    n_iterations, double tolerance,
                    double x0, double& x_new,
                    int& count, bool& converged) ;
```

The first two arguments are pointers to functions, which we shall examine in more detail shortly. The third and fourth arguments are the maximum number of iterations of the Newton–Raphson method and the specified level of accuracy (i.e. ε), respectively. An upper limit of iterations must be specified so that the function does not iterate indefinitely. Argument number five is the initial guess, x_0, and the sixth argument is the solution to the specified tolerance if convergence is achieved. The last two arguments specify the number of iterations required to reach the specified level of solution accuracy and whether or not convergence was achieved.

It was noted in Chapter 7 that an array name is in fact a synonym for the memory address of the zeroth element of the array. Similarly, C++ adopts the rule that a function's name is the memory address of the function. In C++, a pointer to a function, `f_ptr`, is defined as:

```
return_type_specifier (*f_ptr) (type_specifier1, ...) ;
```

For example, given an arbitrary function of the form:

```
double Function (double) ;
```

a pointer to this function is defined as:

```
double (*f_ptr) (double) ;
```

The memory address of function *Function()* can then be assigned to the pointer f_ptr:

```
f_ptr = Function ;
```

Note that only the function name is used in the assignment.

Since the rvalue of f_ptr contains the memory address of *Function()*, dereferencing f_ptr using the indirection operator * will simply call Function():

```
double d ;
//...
(*f_ptr) (d) ;  // calls the function pointed to by f_ptr
```

The parentheses encompassing *f_ptr are required because the precedence level of the indirection operator is less than that of the function call operator (). Actually, the indirection operator is not required, since attaching the list of arguments to f_ptr dereferences f_ptr by default:

```
double d ;
//...
f_ptr (d) ;  // calls the function pointed to by f_ptr
```

A pointer to a function can also be initialised when it is defined:

```
double (*f_ptr) (double) = Function ;
```

These features of pointers to functions are illustrated in the implementation of the *NewtonRaphson()* function:

```
void NewtonRaphson (double (*f_ptr)(double),
                    double (*df_ptr)(double), /*...*/)
  //...
  {
  for (int i=0; i<n_iterations; i++)
    {
    //...
    x_new = x_old - (f_ptr (x_old) / df_ptr (x_old)) ;
    //...
    }
  }
```

which defines two pointers to functions f_ptr and df_ptr that point to $f(x)$ and $f'(x)$ respectively.

You may think that the use of pointers to functions makes the *NewtonRaphson()* function declarator look a bit messy. If so, then a FunctionPointer **typedef** can be defined as a pointer to a function whose return type is **double** and has one **double** function argument:

```
typedef double (*FunctionPointer) (double) ;
```

typedef FunctionPointer allows us to rewrite the *NewtonRaphson()* declarator as:

```
void NewtonRaphson (FunctionPointer f_ptr,
                    FunctionPointer df_ptr, /* ... */) ;
```

Finally, the equality (==) and inequality (!=) relational operators can be applied to pointers to functions. This can be useful to test whether two or more function pointers point to the same function:

```
void NewtonRaphson (FunctionPointer f_ptr,
                    FunctionPointer df_ptr, /*...*/)
//...
if (f_ptr == df_ptr)  // pointers point to same function
  {
  cout << "pointers to same function" << endl ;
  exit (EXIT_SUCCESS) ;
  }
else  // ok
  {
  //...
  }
```

12.21 Returning a Pointer From a Function

We have seen functions that return an object and a reference to an object. Functions can also return a pointer. Clearly, to return a pointer a function must declare the return type specifier to be a pointer. As an example, the C++ *strcpy()* library function returns a pointer to **char**:

```
char* strcpy (char* dest, const char* src) ;
```

The pointer returned from *strcpy()* is in fact a pointer to the destination string, `dest`. The following program defines a function *StrCopy()* which mimics *strcpy()*:

```
// f_ret_p.cpp
// illustrates functions that return a pointer
#include <iostream>  // C++ I/O

// copies a string, src, and returns a pointer to dest
char* StrCopy (char* dest, const char* src)
  {
  char* start_dest = dest ;  // record start of dest

  while (*dest++ = *src++) ;   // copy src to dest

  return start_dest ;         // return pointer to char
  }

void main ()
  {
  char string1[] = "string1" ;
  char string2[] = "string2" ;
```

```
char* dest = StrCopy (string1, string2) ;
cout << dest ;
}
```

The **return** statement returns a pointer to the destination string.

A function can also return a pointer to **void**. For instance, the C++ library function *memcpy()* returns a **void*** and has the following signature:

```
void* memcpy (void* dest, const void* src, size_t n) ; // MEM.H
```

memcpy() copies n bytes from the source block, `src`, to the destination block, `dest`, and returns a pointer to **void** to the destination block. Receiving and returning pointers to **void** allows *memcpy()* to deal with a variety of different data types.

12.22 Casting

Pointers can be cast from one base type to another. Consider the following program, which prompts a user to enter the size of a square array and then proceeds to display an array of pseudo-random numbers of the requested size:

```
// cast_p.cpp
// illustrates casting pointers
#include <iostream>   // C++ I/O
#include <cstdlib>    // malloc(), exit(), rand(), srand()
#include <ctime>      // time(), time_t
#include <iomanip>    // setw()

// allocate sufficient memory & return
// pointer to memory block
int* Allocate (int n)
  {
  int* i_ptr = (int*) malloc ((size_t)(n*n*sizeof(int))) ;

  if (i_ptr == 0)  // unsuccessful request
    {
    cout << "unsuccessful memory allocation" ;
    exit (EXIT_SUCCESS) ;
    }
  return i_ptr ;
  }

// frees allocated memory
void DeAllocate (int* array)
  {
  if (array != 0)  // prevent freeing 0 pointer
    free ((void*) array) ;
  }

// gets a size
```

```
int GetSize ()
  {
  int size = 0 ;

  do
    {
    cout << "enter size of a square array: " ;
    cin  >> size ;
    } while (size <= 0) ;
  return size ;
  }

// fills an array with random numbers (0:99)
void FillArray (int* array, int size)
  {
  // initialise random number generator
  time_t t ;
  srand (unsigned (time(&t))) ;

  for (int i=0; i<size; i++)
    for (int j=0; j<size; j++, array++)
      *array = rand () % 100 ;
  }

// displays an array
void DisplayArray (int* array, int size)
  {
  for (int i=0; i<size; i++)
    {
    for (int j=0; j<size; j++, array++)
      cout << setw (2) << *array << " " ;
    cout << endl ;
    }
  }

void main ()
  {
  int* array (0), size (0) ;

  size  = GetSize () ;          // get size of array
  array = Allocate (size) ;     // allocate sufficient memory

  FillArray    (array, size) ; // fill array with numbers
  DisplayArray (array, size) ; // display array

  DeAllocate (array) ;          // free memory
  }
```

Some user interaction is:

```
enter size of a square array: 5
```

```
27 59 40 24 59
66 83 64 73 89
81 20 79  4 14
65 52 79 56 71
71 57 89 85 84
```

C++ includes a standard library function called *malloc()* which allocates memory from the heap. The signature of *malloc()* is:

```
void* malloc (size_t size) ;
```

and a program using *malloc()* should include either of the header files STDLIB.H or ALLOC.H. `size` is the requested size of memory allocation in bytes. The return type of *malloc()* is a pointer to **void** and the argument type is `size_t`, which is declared as:

```
typedef unsigned size_t ;
```

Returning a pointer to **void** greatly increases the generality of the *malloc()* function. Within the function body of *Allocate()* the *malloc()* function is used to allocate sufficient memory for an array of integers of size n×n:

```
int* Allocate (int n)
  {
  int* i_ptr = (int*) malloc ((size_t)(n*n*sizeof(int))) ;
  //...
  }
```

Since the return type of *malloc()* is a pointer to **void** we must cast the pointer returned from *malloc()* to be a pointer to **int**. The type cast takes the form of:

```
(int*)
```

If *malloc()* successfully allocates the amount of memory requested, a pointer is returned to the start of the memory block allocated from the heap. If there is insufficient memory available to satisfy the memory request, *malloc()* returns the 0 pointer. In the above program, if the memory request is successful the memory address of the block of memory is assigned to the rvalue of `i_ptr`.

If we specifically request a block of memory from the heap, it is our responsibility to ensure that this memory is deleted when we are finished with it. If we do not delete the block of allocated memory when we are finished using it we are clearly relying on the operating system to delete the memory for us. What the operating system does with memory is generally beyond our control – so always ensure that a memory allocation is accompanied by a deallocation. A C++ library function that frees an allocated block of memory is *free()*:

```
void free (void* block) ;
```

The single function argument of *free()* is a pointer to **void**. The arrays that program CAST_P.CPP deals with are of type **int**, and it is therefore necessary to type cast an **int** array, `array`, to a **void** pointer before *free()* is called:

```
void DeAllocate (int* array)
```

```
{
if (array != 0)  // prevent freeing 0 pointer
  free ((void*) array) ;
}
```

Although the **if**-statement is not essential, it does remove the possibility of deleting a pointer that was not allocated memory. Freeing a 0 pointer using *free()* can cause some serious problems.

The above program is an example of *dynamic memory allocation*. It was mentioned, in connection with program STATIC.CPP of Chapter 7, that the size of an array has to be a constant fixed at the time of compilation. The following code is an extract from STATIC.CPP:

```
void main ()
  {
  int array_size ;

  cout << "enter an array size: " ;
  cin  >> array_size ;

  // define a 1D array of a size determined by a user
  int array[array_size] ;  // error: const. expression
                           // required
  //...
  }
```

The program CAST_P.CPP, through the use of pointers, memory allocation and deallocation functions allows us to allocate memory dynamically at run-time according to the requirements of a program. Use of the functions *malloc()*, *calloc()*, *realloc()* and *free()* to allocate and deallocate memory dynamically is the generally adopted style when programming in C. We shall see shortly that although C++ still supports this C style of dynamic memory allocation, the generally adopted C++ approach to dynamic allocation is through the use of the **new** and **delete** operators.

12.23 Recap

To date we have seen a number of applications of pointers, so before introducing pointers to objects and the **new** and **delete** operators, Table 12.1 summarises several declarations that we've seen so far.

12.24 Pointers to Objects

Pointers would be of limited use if they could not point to objects of user-defined structures and classes. To illustrate pointers to objects we shall examine the `Point` **class** introduced in Chapter 9. An object of **class** `Point` is defined with the statement:

```
Point p ;  // object p
```

Table 12.1 Declarations.

Declaration	*Example*	*Description*
`type name ;`	`int var ;`	Variable
`type name[] ;`	`int array[] ;`	Array
`type* name ;`	`int* i_ptr ;`	Pointer variable
`type* name[] ;`	`int* array_ptr[] ;`	Array of pointers
`type (*name)[] ;`	`int (*ptr_array)[] ;`	Pointer to an array
`type& name ;`	`int& i_ref ;`	Reference
`type name () ;`	`int Func () ;`	Function
`type& name () ;`	`int& Func () ;`	Function returning a reference
`type* name () ;`	`int* Func () ;`	Function returning a pointer
`type (*name) () ;`	`int (*f_ptr) () ;`	Pointer to a function

The syntax for defining a pointer to **class** `Point` is identical to that of defining a pointer variable to an integral type:

```
Point* p_ptr ;  // pointer to class Point
```

The following program examines the above two approaches:

```
// p_obj.cpp
// illustrates pointers to objects
#include <iostream>  // C++ I/O

class Point
  {
  private:
    // private data members
    double x, y, z ;
  public:
    //...
    double& X () { return x ; }
    //...
  };
//...
void main ()
  {
  Point  p (1.0, 2.0, 3.0) ;  // object
  Point* p_ptr (0);           // pointer to Point
  p_ptr = &p ;                // p_ptr points to p

  cout << "p      : " << p      << endl
       << "&p     : " << &p     << endl
       << "p_ptr  : " << p_ptr  << endl
       << "*p_ptr: " << *p_ptr << endl ;

  cout << endl
       << "(*p_ptr).X(): " << (*p_ptr).X () << endl
```

```
        << "p_ptr->X()   : " << p_ptr->X ()    << endl ;
}
```

with output:

```
p      : (1, 2, 3)
&p     : 0012FF6C
p_ptr  : 0012FF6C
*p_ptr: (1, 2, 3)

(*p_ptr).X(): 1
p_ptr->X()   : 1
```

The statement:

```
Point* p_ptr (0) ;
```

defines a pointer to `Point` and simultaneously initialises the pointer to the 0 pointer. Then the rvalue of `p_ptr` is assigned the memory address of object `p`. The output illustrates that `p_ptr` points to the memory address of object `p` and that dereferencing `p_ptr` (i.e. `*p_ptr`) accesses the `Point` object pointed to by `p_ptr`.

The final statement within the *`main()`* function is:

```
cout << endl
     << "(*p_ptr).X(): " << (*p_ptr).X () << endl
     << "p_ptr->X()   : " << p_ptr->X ()    << endl ;
```

and illustrates calling the member function `Point::X()` using the pointer `p_ptr` rather than the object `p`. The *direct member access* operator (`.`) requires that the left-hand operand acted on by the operator is an object of a **`class`** in which the right-hand operand is a member. Hence the following is legal:

```
p.X () ;  // ok: p is an object
```

However, the following statement is illegal:

```
p_ptr.X () ;  // error: p_ptr is not an object
```

Since the direct member access operator must act on an object, we must deference the pointer `p_ptr` to enable us to call the member function `Point::X()`:

```
(*p_ptr).X () ;  // ok: *p_ptr refers to the
                 //     object pointed to by p_ptr
```

This syntax works fine, but C++ offers a more attractive syntax, via the *indirect member access* operator (`->`), which specifically acts on pointers to objects:

```
p_ptr->X () ;  // ok: -> acts on pointers to objects
```

All of the above observations equally apply to structures.

12.25 Pointers to Data Members and Member Functions

Previous sections have illustrated pointers to functions and pointers to objects. The C++ programming language also allows a programmer to define pointers to the data members and member functions of a **class**. There exist in the C++ programming language two binary operators that we have not discussed yet, namely the *pointer to member* operators `.*` and `->*`. The pointer to member operators provide a means of accessing a **class**'s data member or member function via a pointer. To illustrate **class** member pointers, consider the program:

```
// ptr_mem.cpp
// illustrates the pointer-to-member
// operators .* and ->*
#include <iostream>  // C++ I/O

class X
  {
  public:
    int data ;
    X () { data = 0 ; }
    void Set (int i) { data = i ; }
    void Display () { cout << "X::data: " << data << endl ;
}
  };

void main ()
  {
  // define & initialise pointer to X int data member
  int X::*pidm          = &X::data ;
  // define & initialise pointers to X member functions
  void (X::*pimf)(int) = &X::Set ;
  void (X::*pvmf)()    = &X::Display ;

  X x ;  // define object x

  (x.*pvmf)() ;   // call X::Display()
  (x.*pimf)(1) ;  // call X::Set()
  (x.*pvmf)() ;   // call X::Display()

  x.*pidm = 2 ;
  cout << "X::data: " << x.*pidm << endl ;

  X* y = &x ;  // define and initialise pointer to an object

  (y->*pvmf)() ;   // call X::Display()
  (y->*pimf)(1) ;  // call X::Set()
  (y->*pvmf)() ;   // call X::Display()

  y->*pidm = 0 ;
  cout << "X::data: " << y->*pidm << endl ;
  }
```

Within *main()*, three pointers (pidm, pimf and pvmf) are defined and simultaneously initialised. Data member pidm is a pointer to the **int** data data member of **class** X, while pimf and pvmf are pointers to member functions of **class** X. The scope resolution operator (::) associates each pointer with **class** X. An object, x, is then defined of **class** X, the X::*Set()* member function is called and the X::data member is assigned the constant 2 through the use of the pointers and the .* pointer to member operator:

```
X x ;
//...
(x.*pimf)(1) ;   // call X::Set()
//...
x.*pidm = 2 ;    // assign 2 to X::data
```

Clearly, the X::data data member must be accessible in order to perform the above assignment.

Next, a pointer to **class** X is defined and initialised to the object x. The ->* operator is then used to call X::*Set()* and assign a constant to X::data:

```
X* y = &x ;

(y->*pimf)(1) ;   // call X::Set()
//...
y->*pidm = 0 ;    // assign 0 to X::data
```

Why do we need the .* and ->* operators? It turns out that the above pointers, pidm, pimf and pvmf, are in fact offsets and do not define the actual memory addresses of data members or member functions. The offset defines a position in the object that identifies a member of the object.

The .* operator cannot be overloaded, whereas the ->* operator can be overloaded.

12.26 The new and delete Operators

The CAST_P.CPP program illustrated the method generally adopted in C for allocating and deallocating memory dynamically via the *malloc()* and *free()* functions. C++ contains two operators, **new** and **delete**, which perform similar operations to *malloc()* and *free()* through the use of pointers. There exist small syntactical differences between the application of **new** and **delete** for objects and arrays of objects, so we shall begin by examining objects.

12.26.1 Objects

The general syntax of the **new** and **delete** unary operators for objects is:

```
pointer_variable = new class_name (optional_parameter_list) ;

delete pointer_variable ;
```

pointer_variable is a pointer to **class** or type class_name. When the **new** operator is invoked a block of memory is requested from the operating system which is sufficient in size

to hold an object of **class** class_name. If the memory request is successful the **new** operator returns a pointer to the allocated block of memory. If the memory request is unsuccessful **new** returns the 0 pointer. Depending on a **class**'s overloaded constructors (assuming they are defined), an optional list of parameters may accompany class_name.

The **delete** operator frees or deallocates memory allocated using the **new** operator and pointed to by pointer_variable. Do not attempt to use the **delete** operator to free memory allocated by a means other than by the **new** operator.

Let's examine a program which uses the **new** and **delete** operators with objects of **class** Point:

```
// nd_obj.cpp
// illustrates the new and delete operators applied to
// objects
#include <iostream>   // C++ I/O
#include <cstdlib>    // exit()

class Point
  { /* ... */ };
//...
void main ()
  {
  // pointers p_ptr and q_ptr to Point
  Point* p_ptr (0), * q_ptr (0) ;

  // allocate memory
  p_ptr = new Point ;              // default constructor
  q_ptr = new Point (1.0, 2.0) ;   // 3 arg. constructor

  if (!p_ptr || !q_ptr)
    {
    cout << "unsuccessful memory allocation" ;
    exit (EXIT_SUCCESS) ;
    }

  cout << "*p_ptr: " << *p_ptr ;
  cout << "*q_ptr: " << *q_ptr ;

  // free allocated memory
  delete p_ptr ;
  delete q_ptr ;
  }
```

The two statements:

```
p_ptr = new Point ;
q_ptr = new Point (1.0, 2.0) ;
```

request sufficient memory from the heap to hold two objects of **class** Point and assign to p_ptr and q_ptr the memory addresses where the objects will be held in memory. The above two statements make use of Point's default and three-argument constructors, respec-

tively, to initialise the memory area pointed to by `p_ptr` and `q_ptr`. Remember that an object's constructor is called automatically when an object is created. Similarly, constructors are called when memory is allocated dynamically using the **new** operator. When we are finished with the pointers `p_ptr` and `q_ptr` the **delete** operator is used to free the memory pointed to by `p_ptr` and `q_ptr`. Note that although the memory pointed to by `p_ptr` and `q_ptr` is deleted, the pointers themselves are not deleted. Beware of using a pointer that points to a block of memory that has been deleted.

The use of the **new** operator is similar in principle to the *malloc()* function, but does not require casting the pointer to **void** returned from *malloc()* to the appropriate **class**:

```
Point* p_ptr = (Point*) malloc((size_t) sizeof(Point)) ;
```

and similarly for the *free()* function.

When using the **delete** operator you have to know what you are doing. The following program examines three uses of **delete**:

```
// delete.cpp
// illustrates the delete operator
#include <iostream>   // C++ I/O
#include <cstdlib>    // exit()

void main ()
  {
  // 1: define, allocate, test, delete
  int* i_ptr = new int ;  // allocate

  if (!i_ptr)
    {
    cout << "unsuccessful memory allocation" ;
    exit (EXIT_SUCCESS) ;
    }
  delete i_ptr ;      // deallocate

  // 2: define initialised pointer to 0, delete
  int* j_ptr (0) ;
  delete j_ptr ;

  // 3: define uninitialised pointer, delete
  int* k_ptr ;
  delete k_ptr ;  // runtime error
  }
```

The first case defines a pointer to the base type **int**, allocates sufficient memory for an **int** and initialises the pointer to the memory address of the block of allocated memory. If the operating system is unable to allocate the requested memory the program terminates. Finally, the `i_ptr` pointer is deleted. The second case uses the **delete** operator to delete a pointer that is initialised to the 0 pointer. C++ allows us to **delete** a 0 pointer. The third case deletes an uninitialised pointer. This is a very severe error and will generally generate a run-time application error, because an unknown random block of memory will be deleted.

Just as the use of the **new** operator causes a **class**'s constructor to be called automatically, the **delete** operator automatically executes a **class**'s destructor. The `Point` **class** above did not define a destructor, so let's now examine a `String` **class** which incorporates both constructor and destructor member functions. The following `String` **class** also introduces the use of the **new** operator with arrays, which we shall examine in more detail in the next subsection.

A `String` **class** was introduced in Chapter 10, program STRING.CPP, with reference to operator overloading:

```
// string.cpp
// illustrates a String class
//...
const int STR_LEN = 80 ;  // maximum length of String

class String
  {
  private:
    char string[STR_LEN] ;
  public:
    // constructor
    String (char s[]="\0")
      { strcpy (string, s) ; }
    //...
  };
```

The **class** encapsulates a string as an array of 80 characters independent of the actual string length. The string length cannot exceed 80 characters, and if a user of the **class** defines a `String` object of length less than 80 characters memory space is wasted. The **new** and **delete** operators can make the `String` **class** more dynamic:

```
// nd_str.cpp
// illustrates the use of the new and delete
// operators with a String class
#include <iostream>  // C++ I/O
#include <string>    // strlen(), strcpy()
#include <cassert>   // assert()

class String
  {
  private:
    char* string ;  // pointer to char
    int   length ;  // length of string
  public:
    // constructor
    String (const char str[]=0) ;
    // destructor
    ~String () ;
    // friend
    friend ostream& operator << (ostream& s,
                                 const String& str) ;
  }; // String class
```

```
// 1 arg. constructor
String::String (const char str[])
  {
  if (str == 0)  // null string
    {
    length = 0 ;
    string = 0 ;
    }
  else              // get length, allocate, test, copy
    {
    length = strlen (str) ;
    string = new char[length+1] ;
    assert (string != 0) ;
    strcpy (string, str) ;
    }
  cout << "constructor called!" << endl ;
  }

// destructor
String::~String ()
  {
  delete [] string ;  // deallocate memory

  string = 0 ;        // set string pointer to 0
  length = 0 ;        // set string length to zero

  cout << endl << "destructor called!" ;
  }

// friend:

// overloaded insertion operator
inline ostream& operator << (ostream& s, const String& str)
  {
  s << str.string ;
  return s ;
  }

void main ()
  {
  String s = "the time is now" ;

  cout << s ;
  }
```

with output:

```
constructor called!
the time is now
destructor called!
```

class String has two **private** data members, string and length. The data member string is a pointer to **char** and will point to an array of characters held by an object of **class** String. If a user-specified string is not of zero length then String's one-argument constructor first determines the length of the string, requests sufficient memory to hold the string plus the null terminator, verifies that the pointer returned from the **new** operator is not the 0 pointer and finally copies the user-specified string to the string data member. Note the declaration of the String::*String()* constructor in the **class** declaration of String:

```
String (const char str[]=0) ;
```

The argument is passed using array notation, rather than pointer notation, so that uninitialised String objects (null strings) can be defined without having to define a zero argument default String constructor explicitly.

String's destructor simply deallocates the memory allocated from String's constructor and sets the string and length data members to the 0 pointer and zero respectively.

Note the use of the *assert()* macro in the constructor:

```
assert (string != 0) ;
```

Programs CAST_P.CPP, ND_OBJ.CPP and DELETE.CPP tested, using an **if**-statement, the pointer returned from the **new** operator against the 0 pointer. If the memory request was unsuccessful then the programs would first display an error message and then abort. Alternatively, C++ provides a library macro called *assert()* which performs a similar task:

```
void assert (int test) ;   // ASSERT.H
```

The *assert()* macro tests the specified condition in the test argument, and if the test expression evaluates to zero the macro displays the error message 'Assertion failed: test, file filename, line linenum' on stderr[7] and aborts the program by calling the *abort()* function, where filename and linenum are the filename of the source code and line number where the *assert()* macro appears, respectively. If you place the #define NDEBUG directive (i.e. 'no debugging') in the source code before the #include <cassert> preprocessor directive, the *assert()* macro statement is effectively commented out.

The *abort()* function signature is:

```
void abort () ;   // STDLIB.H
```

abort() terminates a program in an abnormal way and displays the run-time message 'Program Aborted' on stderr.

Earlier we saw arrays of pointers to C++ integral types. We can also define arrays of pointers to objects:

```
// a_p_str1.cpp
// illustrates arrays of pointers to String objects
//...
class String
  {
  private:
```

7 stderr is the standard error output device and is #defined in the STDIO.H header file.

```
    char* string ;  // pointer to char
    int   length ;  // length of string
  public:
    //...
    friend istream& operator >> (istream& s, String& str) ;
  };
//...
// overloaded extraction operator
istream& operator >> (istream& s, String& str)
  {
  char buffer[80] ;  // temporary buffer

  s >> buffer ;      // get string

  str.length = strlen (buffer) ;
  delete [] str.string ;
  str.string = new char[str.length+1] ;
  assert (str.string != 0) ;
  strcpy (str.string, buffer) ;
  return s ;
  }

void main ()
  {
  const int SIZE (5) ;

  String* p_str[SIZE] ;  // array of pointers to String

  // I/P
  for (int i=0; i<SIZE; i++)
    {
    // allocate memory for new object
    p_str[i] = new String ;
    assert (p_str[i] != 0) ;
    cout << "enter a string: " ;
    cin  >> *p_str[i] ;
    }
  // O/P
  for (int j=0; j<SIZE; j++)
    cout << *p_str[j] << endl ;
  }
```

Note the addition of the overloaded extraction operator (>>) to the `String` **class**. The statement:

```
String* p_str[SIZE] ;
```

defines an array of pointers, `p_str`, to **class** `String` with `SIZE` elements. The statement:

```
p_str[i] = new String ;
```

requests sufficient memory for a `String` object and assigns the address of the object to the `i`th element of array `p_str`.

12.26.2 Arrays

The previous subsection demonstrated that C++ allows an array of objects to be allocated using the **new** operator. The general syntax is of the form:

```
pointer_variable = new class_name[ARRAY_SIZE] ;
```

This statement requests a block of memory sufficient in size to hold an array of `ARRAY_SIZE` elements of **class** or type `class_name` and assigns the memory address of the block of memory to the pointer `pointer_variable`. Unlike the object form of the **new** operator, the array form does not allow initialisation arguments to be supplied when an array of objects is created using **new**. As a consequence, arrays of objects can only be created using the **new** operator with a default constructor.

To free an array that was allocated dynamically the **delete** operator adopts the syntax:

```
delete [] pointer_variable ;
```

The subscript operator, `[]`, informs the compiler that a block of memory associated with an array is to be deallocated.

To illustrate the use of the **new** and **delete** operators with arrays let's design a `Vector` **class**. Vectors, consisting of three components, are frequently used in computer graphics for modelling objects in a three-dimensional space. However, the term vector can be applied more generally to describe a list of elements of arbitrary size. For instance, the following illustrates an array or matrix of elements a_{ij} ($i = 1,2,...,m; j = 1,2,...,n$):

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The horizontal lines of the matrix are referred to as rows, or *row vectors*, and the vertical lines are columns, or *column vectors*, and are conventionally written:

$$[a_1, a_2, .., a_n], \quad \begin{Bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{Bmatrix}$$

Thus, a `Vector` **class** must be sufficiently dynamic to cater for an arbitrary size of row or column vectors. The following program presents a first implementation of a `Vector` **class**:

```
// vect.cpp
// a Vector class that illustrates the use of the
// new and delete operators for dynamically allocating
// and deallocating memory for arrays
```

```
#include <iostream>   // C++ I/O
#include <cassert>    // assert()
#include <memory>     // memset()
#include <cstdlib>    // exit()

class Vector
  {
  private:
    // data members
    double* array ;
    int     n_elements ;
    // member function
    void    Allocate (int n) ;
  public:
    // constructors
    Vector ()
      : array (0), n_elements (0) {}
    Vector (int n) { Allocate (n) ; }
    // destructor
    ~Vector () { delete [] array ; array = 0 ;
                 n_elements = 0 ; }
    // overloaded operators
    double&       operator [] (int index) ;
    const double& operator [] (int index) const ;
    // friends
    friend ostream& operator << (ostream& s,
                                 const Vector& v) ;
  };

// private member function:

// allocates memory for elements of vector
void Vector::Allocate (int n)
  {
  if (n <= 0)
    {
    cout << "zero or negative number of elements" ;
    exit (EXIT_SUCCESS) ;
    }
  array = new double[n] ;
  assert (array != 0) ;
  n_elements = n ;
  // zero array
  memset ((void*)array, 0, (size_t)n*sizeof(double)) ;
  }

// overloaded operators:

// subscript operator [] (non-const objects)
double& Vector::operator [] (int index)
  {
```

```
    if (index < 0 || index >= n_elements)
      {
      cout << "indexing Vector element out of range" ;
      exit (EXIT_SUCCESS) ;
      }
    return array[index] ;
    }
  //...
  // friend:

  // overloaded insertion operator <<
  ostream& operator << (ostream& s, const Vector& v)
    {
    s << "[" ;
    for (int i=0; i<v.n_elements; i++)
      {
      s << v.array[i] ;
      i!=v.n_elements-1 ? s << ", " : s << "]" ;
      }
    return s ;
    }

  void main ()
    {
    Vector a (3), b (5) ;

    cout << "a: " << a << endl ;
    cout << "b: " << b << endl ;

    a[0] = 1.0 ; a[1] = 2.0 ; a[2] = 3.0 ;

    cout << "a: " << a << endl ;
    }
```

with output:

```
a: [0, 0, 0]
b: [0, 0, 0, 0, 0]
a: [1, 2, 3]
```

The `Vector` **`class`** encapsulates two **`private`** data members:

```
class Vector
  {
  private:
    double* array ;
    int     n_elements ;
    //...
  };
```

`array` is a pointer to **`double`** which will be assigned the memory address of an array of elements of type **`double`** that refer to the `n_elements` elements of a vector. `Vector`'s one-argument constructor calls the *`Allocate()`* **`private`** member function:

```
Vector (int n) { Allocate (n) ; }
```

The default no-argument constructor simply sets `array` to the 0 pointer and `n_elements` to zero. The one-argument constructor allocates sufficient memory for `n` elements. Memory is allocated dynamically by calling the **`private`** *`Allocate()`* function. Note that *`Allocate()`* is declared a **`private`** member function of **`class`** `Vector` to prevent a user from accidentally calling the function. Memory is allocated for an array of `n` **`double`** elements. The pointer returned from **`new`** is tested against the 0 pointer, and if the memory request is successful the number of elements, `n_elements`, in the array is set.

Since C++ does not guarantee that the elements of an array are set to zero (or any other value or object) it is necessary to *zero-out* the array. The obvious way to zero-out an array is to simply use a **`for`**-loop:

```
for (int i=0; i<n_elements; i++)
  array[i] = 0.0 ;
```

However, this is not very efficient. A faster way is to call the *`memset()`* function:

```
void* memset (void* s, int c, int n) ;  // MEM.H
```

`memset()` sets the first `n` bytes of memory pointed to by `s` to the character `c`. The return value of *`memset()`* is `s`. The `Vector::`*`Allocate()`* function calls *`memset()`* to initialise the elements of `array` to zero.

Pay particular attention to the above declaration of the `Vector::array` data member and memory allocation using the **`new`** operator:

```
class Vector
  {
  private:
    double* array ;
    //...
  };
//...
void Allocate (int n)
  {
  //...
  array = new double[n] ;
  //...
  }
```

Alternatively, the following is a common mistake for which the compiler will not issue a compilation or linker warning or error message:

```
//...
void Allocate (int n)
  {
  //...
```

```
    double* array = new double[n] ;
    //...
    }
```

which redefines `array`. This will result in a disastrous run-time error when an attempt is made to access the unallocated elements of a `Vector` object.

`Vector`'s destructor frees the memory allocated to an object, assigns the 0 pointer to the `array` data member and sets the `n_elements` data member to zero. The subscript, `[]`, and insertion, `<<`, operators are overloaded for **`class`** `Vector` so as to enable natural indexing and output of a `Vector`'s elements. The return type of both of the overloaded **`operator`**`[] ()` member functions is a reference to **`double`**. Returning a reference allows the subscript operator to appear on either the left- or right-hand side of the assignment operator:

```
Vector a ;
//...
double d = a[0] ;
a[1] = 2.0 ;
```

Now suppose that we would like to increase or decrease the number of elements of a `Vector` object once it has been defined. The following program adds a *`New()`* member function to **`class`** `Vector` which performs just such a task:

```
// vect1.cpp
// a Vector class that caters for an
// increasing/decreasing number of elements
//...
class Vector
  {
  public:
    //...
    // member function
    void New (int new_n) ;
    //...
  };

// change number of elements
void Vector::New (int new_n)
  {
  delete [] array ;
  array      = 0 ;
  n_elements = 0 ;

  Allocate (new_n) ;
  }
//...
void main ()
  {
  Vector a (3);
  cout << "original a: " << a << endl ;
  a.New (4) ;  // add an element
```

```
//...
}
```

Firstly, *New()* **delete**s the memory allocation associated with an object's original array of elements and then sets the object's `array` and `n_elements` data members to 0. Then *New()* simply calls *Allocate()* to allocate sufficient memory for the new size of a `Vector` object. At present, the *New()* member function is not very smart, since it deletes the existing array of elements of the `Vector` object that calls the function. However, we shall refrain from continuing to extend **class** `Vector` until we discuss templates in Chapter 13.

VECT1.CPP contains some additional features, not shown above, which will be discussed later in connection with returning 'local' objects from member functions.

12.26.3 Objects, Arrays and `delete`

The previous two subsections have illustrated separately the use of the **delete** operator for both objects and arrays of objects, but under general circumstances how does a compiler or programmer know whether a pointer points to an object or an array of objects? Consider the following program code:

```
//...
void Function (X* obj) // obj may point to an object or array
  {
  //...
  delete obj ;      // ?
  delete [] obj ;   // ?
  }
```

It is not possible to determine from a pointer whether it points to an object or an array of objects. It is the programmer's responsibility to ensure that the correct syntax is used when using the **delete** operator – getting it wrong can have disastrous results!

12.26.4 Data Member Initialisation Lists and `new`

Previous sections have demonstrated the use of the **new** operator for allocating memory for data member objects and arrays of objects:

```
class X
  {
  private:
    int* data ;
    int* array ;
  public:
    X ()
      { data = new int ; array = new int[5] ; }
  //...
  };
```

Alternatively, the **new** operator can be placed in a **class**'s constructor data member initialisation list:

```
// new_init.cpp
// illustrates the combined use of the new operator
// with a class's constructor data member initialisation
// list
#include <iostream>  // C++ I/O

class X
  {
  private:
    int* data ;
    int* array ;
  public:
    X ()
      : data (new int), array (new int[5]) {}
    X (int n)
      : data (new int(n)), array (new int[n]) {}
    ~X ()
      { delete data ; delete [] array ; }
  };

void main ()
  {}
```

12.26.5 Overloaded `new` and `delete` Operators

Since **new** and **delete** are operators it is possible to overload them specifically for a given **class**. The general syntax of overloading[8] the **new** and **delete** operators for objects and arrays of objects is:

```
void* operator new (size_t size)
  { /* ... */ }
void* operator new[] (size_t size)
  { /* ... */ }
void operator delete (void* ptr)
  { /* ... */ }
void operator delete[] (void* ptr)
  { /* ... */ }
```

The overloaded **new** operator function must return a pointer to the memory address of the block of memory that it allocates if allocation is successful or the 0 pointer if the memory allocation is unsuccessful. The overloaded **delete** operator function is passed a pointer to the memory address of the block of memory to be deallocated.

To see overloading of the **new** and **delete** operators in action, let's revisit the `Vector` **class** discussed above:

```
// over_nd.cpp
// illustrates overloading the new and delete operators
//...
```

8 You may want to refer to Chapter 10 to remind yourself about operator overloading.

```
class Vector
  {
  //...
  public:
    // constructors
    //...
    // destructor
    ~Vector () { delete [] array ;
                 array = 0 ;
                 n_elements = 0 ;
                 cout << "destructor called" << endl ; }
    //...
    // overloaded operators
    //...
    void* operator new (size_t size) ;
    void  operator delete (void* ptr) ;
    void* operator new[] (size_t size) ;
    void  operator delete[] (void* ptr) ;
    //...
  };

// private member function:

// allocates memory for elements of vector
void Vector::Allocate (int n)
  {
  cout << "constructor called" << endl ;
  //...
  }
//...
// overloaded new operator for object
void* Vector::operator new (size_t size)
  {
  cout << "object overloaded new operator called"  << endl ;
  return malloc (size) ;
  }

// overloaded delete operator for object
void Vector::operator delete (void* ptr)
  {
  cout << "object overloaded delete operator called"  << endl ;
  free (ptr) ;
  }

// overloaded new operator for an array of objects
void* Vector::operator new[] (size_t size)
  {
  cout << "array overloaded new operator called"  << endl ;
  return malloc (size) ;
  }
```

```
// overloaded delete operator for an array of objects
void Vector::operator delete[] (void* ptr)
  {
  cout << "array overloaded delete operator called"  << endl ;
  free (ptr) ;
  }
//...
void main ()
  {
  // allocate object and array of objects
  Vector* v_ptr   = new Vector ;
  Vector* array_v = new Vector[2] ;

  cout << *v_ptr << endl ;

  // deallocate object and array of objects
  delete v_ptr ;
  delete [] array_v ;

  cout << endl ;

  Vector* v1_ptr = ::new Vector (2) ; // global allocate
  cout << *v1_ptr << endl ;
  ::delete v1_ptr ;                   // global deallocate
  }
```

with output:

```
object overloaded new operator called
constructor called
array overloaded new operator called
constructor called
constructor called
[0, 0, 0]
destructor called
object overloaded delete operator called
destructor called
destructor called
array overloaded delete operator called

constructor called
[0, 0]
destructor called
```

For illustration purposes the **new** and **delete** operators, for both objects and an array of objects, simply call the C++ memory manipulation library functions *malloc()* and *free()*, respectively. Note that if a memory allocation request is unsuccessful or the `size` argument is zero the *malloc()* function returns the 0 pointer.

The program output illustrates that the statement:

```
v_ptr = new Vector ;
```

first calls the object overloaded **new** operator and then calls `Vector`'s default constructor. It is important to note that use of the **new** operator will result in an object's appropriate constructor being called automatically. Similarly, but in reverse order to **new**, the statement:

```
delete v_ptr ;
```

will first cause `Vector`'s destructor to be called automatically and then the object's overloaded **delete** operator. Also, the program output illustrates that a **class**'s constructor and destructor are called automatically for each object of an array of objects.

The final three statements within the *main()* function of program OVER_ND.CPP are:

```
Vector* v1_ptr = ::new Vector (2) ; // global allocate
cout << *v1_ptr << endl ;
::delete v1_ptr ;                   // global deallocate
```

If the **new** and **delete** operators are specifically overloaded for a given **class**, use of **new** and **delete** will call the overloaded member functions **operator new** *()* and **operator delete** *()*, respectively. If you require that the global ::**operator new** *()* and ::**operator delete** *()* operators be used then precede the operator by the scope resolution operator (::). The program output illustrates that ::**new** and ::**delete** automatically call an object's constructor and destructor (as usual), but do not call the overloaded **operator new** *()* and **operator delete** *()* operators.

An important point to note when overloading the **new** and **delete** operators is that, if they are overloaded for a given **class**, use of **new** and **delete** on any other type or **class** will result in the default or global **new** and **delete** operators being called. Thus, in the above program, OVER_ND.CPP, the following statement in *main()*:

```
v_ptr = new Vector ;
```

calls `Vector::`**operator new** *()*, but the call to **new** in the *Allocate()* function:

```
void Vector::Allocate (int n)
  {
  //...
  array = new double[n] ;
  //...
  }
```

calls the global **new** operator!

For those readers interested in programming for Windows, the following overloaded **new** and **delete** operators allocate and deallocate, respectively, virtual memory from a disk file:

```
void FAR* Vector::operator new (size_t size)
  {
  // amount of global memory available
  DWORD dwAvailable = GlobalCompact (0) ;

  // request greater than available
  if (size > dwAvailable)
    {
```

```
    MessageBox (GetActiveWindow (),
        (LPCSTR) "insufficient memory",
        (LPCSTR) "Vector Warning",
        MB_ICONEXCLAMATION | MB_OK) ;
    return 0 ;
    }

  // handle to global memory
  HGLOBAL hGlobalMemory =
      GlobalAlloc (GMEM_MOVEABLE | GMEM_ZEROINIT, size) ;

   // if unsuccessful allocation
  if (!hGlobalMemory)
    {
    MessageBox (GetActiveWindow (),
        (LPCSTR) "insufficient memory",
        (LPCSTR) "Vector Warning",
        MB_ICONEXCLAMATION | MB_OK) ;
    return 0 ;
    }

  // void far pointer to global memory
  void FAR* lpvGlobalMemory = GlobalLock (hGlobalMemory) ;

  // return pointer to allocated memory
  return lpvGlobalMemory ;
  }

void Vector::operator delete (void* obj)
  {
  GlobalUnlock (GlobalHandle ((HGLOBAL)obj)) ;  // unlock
  GlobalFree   (GlobalHandle ((HGLOBAL)obj)) ;  // free
  }
```

The Windows Application Programming Interface (API) function *GlobalCompact()* returns the size of the largest block of memory available:

```
DWORD GlobalCompact (DWORD dwMinFree) ;
```

dwMinFree is the size, in bytes, of the requested memory block. If dwMinFree is zero the return value of *GlobalCompact()* is the largest block of free memory available. *GlobalCompact()* determines the size of the largest block of memory by moving moveable blocks and memory that could be obtained if discardable blocks were discarded. The doubleword identifier DWORD is defined as (WINDEF.H):

```
typedef unsigned long DWORD ; // WINDEF.H
```

The *MessageBox()* function declaration is:

```
int MessageBox (HWND hwndParent, LPCSTR lpszText,
                LPCSTR lpszTitle, UINT fuStyle) ;
```

where hwndParent is a handle to the parent window, lpszText is the memory address of the text to be displayed in the message box, lpszTitle is the memory address of the text to be displayed in the title of the message box and fuStyle is the style of the message box.

The *GlobalAlloc()* function allocates the specified number of bytes, dwAlloc, from the global heap:

```
HGLOBAL GlobalAlloc (UINT fuAlloc, DWORD dwAlloc) ;

typedef const void NEAR* HANDLE ;  // WINDEF.H
//...
typedef HANDLE HGLOBAL ;
```

fuAlloc is an identifier that specifies how the memory is to be allocated. GMEM_MOVEABLE allocates moveable memory and GMEM_ZEROINT initialises the contents of the memory allocated to zero. *GlobalAlloc()* returns a handle to the allocated memory block if the request is successful; if not, it returns the NULL pointer. After receiving a handle to a global block of memory it is then necessary to lock the block of memory:

```
void FAR* GlobalLock (HGLOBAL hglb) ;
```

Locking a block of memory is essential so that it will not be moved or discarded unless the memory is explicitly unlocked or reallocated by the Windows API *GlobalReAlloc()* function. The return value of *GlobalLock()* is **void** FAR*. A far pointer to **void** is necessary because memory is being allocated in segments other than the default segment. A far pointer occupies four bytes of memory. The low-order two bytes are the *offset*, while the high-order two bytes are the *selector*.

The overloaded **delete** operator first unlocks the block of memory allocated by **operator new** *()* and then frees the block of memory, assuming that the block of memory is not locked. The function signatures of *GlobalUnlock()* and *GlobalLock()* are:

```
BOOL  GlobalUnlock (HGLOBAL hglb) ;

HGLOBAL GlobalFree (HGLOBAL hglb) ;
```

GlobalFree() invalidates the handle hlgb.

An alternative to overloading the **new** and **delete** operators specifically for each user-defined **class** is to implement a memory **class**. Staying with global memory in a Windows environment, the following is a **class** which encapsulates global memory allocation:

```
// win_mem.cpp
// illustrates a global memory class
// for programming in a Windows environment
#include <iostream>  // C++ I/O
#include <windows>   // Windows header file

class GlobalMemory
  {
  private:
    HGLOBAL   hGlobalMemory ; // handle to memory
    void FAR* mem_address ;   // address of memory
```

```
  public:
    // constructor
    GlobalMemory ()
      : hGlobalMemory (0), mem_address (0) {} ;
    // destructor
    ~GlobalMemory () ;
    // member functions
    DWORD     SizeOfBlock () ;
    void FAR* MemoryAddress () const { return mem_address ; }
    void FAR* Allocate   (UINT size,
        UINT mem_flags=GMEM_MOVEABLE|GMEM_ZEROINIT) ;
    void FAR* ReAllocate (UINT size,
        UINT mem_flags=GMEM_MOVEABLE|GMEM_ZEROINIT) ;
  };

// destructor
GlobalMemory::~GlobalMemory ()
  {
  GlobalUnlock (hGlobalMemory) ;    // unlock
  if (hGlobalMemory)                // free
    GlobalFree (hGlobalMemory) ;
  }

// return size of memory block
DWORD GlobalMemory::SizeOfBlock ()
  {
  if (!hGlobalMemory)
    return 0 ;
  else
    return GlobalSize (hGlobalMemory) ;
  }
// allocate memory
void FAR* GlobalMemory::Allocate (UINT size, UINT mem_flags)
  {
  hGlobalMemory = GlobalAlloc (mem_flags, size) ;

  // if unsuccessful allocation
  if (!hGlobalMemory)
    {
    MessageBox (GetActiveWindow (),
        (LPCSTR) "insufficient memory",
        (LPCSTR) "GlobalMemory Warning",
        MB_ICONEXCLAMATION | MB_OK) ;
    hGlobalMemory = 0 ;
    return 0 ;
    }
  mem_address = GlobalLock (hGlobalMemory) ;
  return mem_address ;
  }

// re-allocate memory
```

```
void FAR* GlobalMemory::ReAllocate (UINT size,
                                    UINT mem_flags)
  {
  HGLOBAL h = GlobalReAlloc (hGlobalMemory, size,
                             mem_flags) ;

   // if unsuccessful allocation
  if (!h)
    {
    MessageBox (GetActiveWindow (),
        (LPCSTR) "insufficient memory",
        (LPCSTR) "GlobalMemory Warning",
        MB_ICONEXCLAMATION | MB_OK) ;
    return 0 ;
    }
  else
    {
    hGlobalMemory = h ;
    mem_address = GlobalLock (hGlobalMemory) ;
    }
  return mem_address ;
  }

void main ()
  {
  const int SIZE (5) ;

  GlobalMemory g_mem ; // global memory object

  // allocate memory
  int* px = (int*) g_mem.Allocate (SIZE*sizeof(int)) ;
  // set
  for (int i=0; i<SIZE; i++)
    *(px+i)=i ;

  // get
  for (int j=0; j<SIZE; j++)
    cout << *(px+j)  << " " ;
  cout << endl ;
  cout << "size: " << g_mem.SizeOfBlock () << endl ;

  // re-allocate memory
  px = (int*) g_mem.ReAllocate (2*SIZE*sizeof(int)) ;

  // get
  for (int k=0; k<2*SIZE; k++)
    cout << *(px+k)  << " " ;
  cout << endl ;
  cout << "size: " << g_mem.SizeOfBlock () << endl ;
  }
```

with output:

```
0 1 2 3 4
size: 20
0 1 2 3 4 0 0 0 0 0
size: 40
```

The `GlobalMemory` **class** declares two **private** data members, `hGlobalMemory` and `mem_address`, which define the handle and memory address of a block of global memory. `GlobalMemory`'s constructor initialises its two data members to the 0 pointer, while its destructor unlocks and frees an allocated global block of memory. `GlobalMemory` contains four member functions. The access member function *`MemoryAddress()`* simply returns the memory address of the block of memory. *`SizeOfBlock()`* returns the size, in bytes, of a global memory object and makes use of the API function *`GlobalSize()`*:

```
DWORD GlobalSize (HGLOBAL hglb) ;
```

The first call to *`SizeOfBlock()`* in *`main()`* displays a size of 20 bytes, as requested for a 32-bit system, for the global memory object `g_mem`. If the size of the memory object returned by *`GlobalSize()`* is greater than requested by *`GlobalAlloc()`* then Windows guarantees that the object has access to the allocated memory. The *`Allocate()`* member function requests a block of memory from the operating system and returns a far pointer to the memory. The *`ReAllocate()`* member function makes use of the API function *`GlobalReAlloc()`* to alter the size and attributes of the memory object:

```
HGLOBAL GlobalReAlloc (HGLOBAL hglb, DWORD cbNewSize,
                       UINT fuAlloc) ;
```

If *`GlobalReAlloc()`* is successful at reallocating the memory associated with the handle to the memory object, `hglb`, the function returns a handle to the object, else 0 is returned. The data held in the block of memory pointed to by the memory object are preserved in the reallocation. This is illustrated in the program output of WIN_MEM.CPP above. Although the memory object `g_mem` is increased from a requested 20 to 40 bytes, the original 5 integers (0:4) are preserved. Note that the default argument `mem_flags` in *`ReAllocate()`* initialises the data of the newly allocated memory object to zero.

You have probably gathered from the above brief discussion that memory management in Windows is a big topic and beyond the scope of this text. However, the above discussion does illustrate that the **new** and **delete** operators can be overloaded to suit the exact needs of your own classes and operating system, or alternatively that a memory **class** can be implemented which is tailored exactly to different types of memory. We shall revisit the subject of memory management when we discuss templates, inheritance and the standard library `allocator` class.

For the reader interested in C programming for Windows I recommend the two introductory books of Lafore (1993) and Petzold (1992).

12.26.6 Placement

C++ allows an object to be placed at a memory address specified by the programmer using the *placement syntax* of the **new** operator:

```
// placemnt.cpp
// illustrates placement
#include <iostream>  // C++ I/O

class X
  {
  private:
    int data ;
  public:
    X ()
      : data (0) {}
    X (int i)
      : data (i) {}
    void* operator new (size_t size, void* ptr)
      {
      return ptr ;  // place object at specified location
      }
  };

void main ()
  {
  int i (0), * i_ptr = &i ;
  void* location = (void*)i_ptr ;  // set to valid memory
                                   // address
  X* x_ptr = new (location) X (1) ;// place

  x_ptr->~X () ;                   // remove

  // int* j_ptr = new (location) int (1) ; // error
  }
```

The overloaded **operator new** *()* function of **class** X places an object of **class** X at a memory position pointed to by the pointer argument `ptr`. Placing an object of **class** X at a specified `location` memory address now requires the **new** `(location) X` syntax with the `size_t` argument `size` passed implicitly.

To remove an object of **class** X pointed to by `x_ptr` from a specified memory address (which was allocated via the **new** placement syntax) the following syntax is used:

```
x_ptr->~X () ;
```

12.27 Returning Local Objects from Member Functions

When an object is returned by a function a temporary object is automatically created by the compiler to hold the object. It is this temporary object which is returned by a function. When the object has been returned by the function the temporary object is destroyed. This can cause some very serious run-time errors:

```
// ret_prob.cpp
// illustrates a problem with returning
```

```
// an object from a function
#include <iostream>  // C++ I/O

class Object
  {
  public:
    int* ptr ;
    Object (int i=0) ;
    ~Object () ;
    Object& operator = (const Object& obj) ;
  }; // class Object

Object::Object (int i)
  {
  ptr = new int (i) ;
  cout << "constructor called, ptr: " << ptr
       << ", ptr.rvalue: " << *ptr << endl ;
  }

Object::~Object ()
  {
  cout << "destructor called, ptr: " << ptr
       << ", ptr.rvalue: " << *ptr << endl ;
  delete ptr ;
  ptr = 0 ;
  }
//...
// return a copy
Object ReturnCopy (const Object& obj)
  {
  Object return_obj (*obj.ptr) ;
  return return_obj ;
  }

void main ()
  {
  Object object1 (1), object2 (2) ;

  object2 = ReturnCopy (object1) ;
  }
```

with output:

```
constructor called, ptr: 007D306C, ptr.rvalue: 1
constructor called, ptr: 007D31CC, ptr.rvalue: 2
constructor called, ptr: 007D31DC, ptr.rvalue: 1
destructor called, ptr: 007D31DC, ptr.rvalue: 1
destructor called, ptr: 007D31DC, ptr.rvalue: 4302204
destructor called, ptr: 007D31DC, ptr.rvalue: 4302204
destructor called, ptr: 007D306C, ptr.rvalue: 1
```

The output illustrates that there are three constructor calls but four destructor calls! As a result, the program will generally result in a run-time application error. `Object`'s constructor is called for each of the two objects, `object1` and `object2`, in *`main()`* and for the 'local' object, `return_obj`, in *`ReturnCopy()`*. `Object`'s destructor is first called when `return_obj` goes out of scope (i.e. when *`ReturnCopy()`* returns) and called a second time when the temporary object returned by *`ReturnCopy()`* is destroyed. Finally, `Object`'s destructor is called twice when objects `object1` and `object2` are destroyed. Following the first destructor call for `return_obj` the output illustrates that the rvalue of `ptr` is garbage.

Note that `Object`'s constructor is not called when a temporary object is created in function *`ReturnCopy()`* since this would reinitialise the object being copied. One solution to this problem is to return an object by reference or pointer. However, this is not always possible. For instance, objects 'local' to a function cannot be returned by reference:

```
Object& ReturnCopy (const Object& obj)
  {
  //...
  return return_obj ;
  }
```

An alternative solution involves the use of a *copy constructor*. It was noted in Chapter 9 that a copy constructor is called for creating a temporary that is to be returned from a function. Try adding the following copy constructor to program RET_PROB.CPP:

```
Object::Object (const Object& obj)
  {
  ptr = new int (*obj.ptr) ;
  cout << "copy constructor called, ptr: "
       << ptr << ", ptr.rvalue: " << *ptr << endl ;
  }
```

The program output will now be:

```
constructor called, ptr: 007D306C, ptr.rvalue: 1
constructor called, ptr: 007D31CC, ptr.rvalue: 2
constructor called, ptr: 007D31DC, ptr.rvalue: 1
copy constructor called, ptr: 007D31EC, ptr.rvalue: 1
destructor called, ptr: 007D31DC, ptr.rvalue: 1
destructor called, ptr: 007D31EC, ptr.rvalue: 1
destructor called, ptr: 007D31DC, ptr.rvalue: 1
destructor called, ptr: 007D306C, ptr.rvalue: 1
```

which illustrates that the temporary object returned by *`ReturnCopy()`* invokes `Object`'s copy constructor, which allocates memory accordingly and makes an exact copy of the `Object::ptr` data member. In addition, class `Object` overloads the assignment operator, the details of which will be clearer when we examine the **`this`** pointer in Section 12.29.

Let's now revisit the VECT1.CPP program, which illustrates a practical application of the copy constructor:

```
// vect1.cpp
//...
class Vector
```

```
  {
  private:
    double* array ;
    int     n_elements ;
    // member function
    void    Allocate (int n) ;
  public:
    // constructors
    Vector ()       { Allocate (3) ; }
    Vector (int n) { Allocate (n) ; }
    // copy constructor
    Vector (const Vector& v) ;
    //...
    Vector&       operator = (const Vector& v) ;
  };
//...
// copy constructor
Vector::Vector (const Vector& v)
  {
  Allocate (v.n_elements) ;
  for (int i=0; i<v.n_elements; i++)
    array[i] = v.Value (i) ;
  }
//...
// assignment operator =
Vector& Vector::operator = (const Vector& v)
  {
  // check not self
  if (this != &v)
    {
    // delete old objects
    delete [] array ;

    // empty vector object
    if (v.n_elements == 0)
      {
      array      = 0 ;
      n_elements = 0 ;
      }
    // something to copy
    else
      {
      double* new_array = new double[v.n_elements] ;
      assert (new_array != 0) ;
      array = new_array ;
      n_elements = v.n_elements ;
      // copy data
      for (int i=0; i<n_elements; i++)
        array[i] = v.Value(i) ;
      }
    }
```

```
    return *this ;
    }
//...
// addition operator +
Vector Vector::operator + (const Vector& v)
    {
    // verify that the two vectors are of equivalent dimensions
    assert(n_elements == v.n_elements) ;

    Vector add (v.n_elements) ;

    for (int i=0; i<v.n_elements; i++)
      add.Value (i) = Value (i) + v.Value (i) ;
    return add ;
    }
//...
void main ()
    {
    //...
    Vector v1, v2, v3 ;
    v3 = v1 + v2 ;  // run-time error if copy-constructor removed
    }
```

The copy constructor calls `Vector::Allocate()` to allocate sufficient memory for the `Vector` object, `v`, being copied and then proceeds to make an exact copy of the array of `Vector` elements. If the copy constructor is removed from VECT1.CPP, the overloaded `Vector::operator +()` function will generate a run-time error.

`Vector` also overloads the assignment operator, the details of which will become clearer when we examine overloading the assignment operator for class `String` in Section 12.29.

12.28 Passing Objects to Functions

The VECT1.CPP program, described in the previous section, illustrated problems that can arise when returning a 'local' object from a member function. Similarly, when constructors and a destructor are defined for a given **class**, passing objects by value to a function can cause unexpected results. To illustrate problems that occur when objects are passed by value to a function, consider the program:

```
// pas_prob.cpp
// illustrates a problem with passing
// an object by value to a function
#include <iostream>  // C++ I/O

class Object
    {
    public:
      int* ptr ;
      Object (int i) ;
```

```
    ~Object () ;
    Object& operator = (const Object& obj) ;
  }; // class Object

Object::Object (int i)
  {
  ptr = new int (i);
  cout << "constructor called, ptr: " << ptr
       << ", ptr.rvalue: " << *ptr << endl ;
  }

Object::~Object ()
  {
  cout << "destructor called, ptr: " << ptr
       << ", ptr.rvalue: " << *ptr << endl ;
  delete ptr ;
  }
//...
// pass by value
void Display (Object obj)
  {
  cout << *obj.ptr << endl ;
  }

void main ()
  {
  Object object (1) ;

  Display (object) ;
  }
```

with output:

```
constructor called, ptr: 007D306C, ptr.rvalue: 1
1
destructor called, ptr: 007D306C, ptr.rvalue: 1
destructor called, ptr: 007D306C, ptr.rvalue: 8192168
```

and will result in a run-time application error. The program output illustrates that `Object`'s constructor is called once but `Object`'s destructor is called twice. When `object` is defined in *`main()`*, `Object`'s constructor is automatically called and allocates memory for the data member `object.ptr`. When `object` is passed by value to *`Display()`* a copy is made and assigned to `obj`. Note that although a copy of an object is made when passed by value to a function, the object's constructor is not called. Calling the appropriate constructor when an object is passed to a function would effectively reinitialise the object – which is not what we want. Thus, when an object is passed by value to a function, the object's constructor is not called. However (and here's the problem), the function argument object, `obj`, destructor is called when the function returns, and `obj` goes out of scope. The destructor will therefore deallocate the memory associated with `obj.ptr`, which in turn is the same memory associated with `object.ptr`. Thus, when `object` is destroyed in *`main()`* an unknown block of memory is deleted, which will generally cause unpredictable run-time errors.

One solution to this problem is simply to pass an object to a function by reference or pointer instead of by value. When an object is passed either by reference or by pointer the original object from the calling function is accessed and no copy is made, thus eliminating a destructor call. Try modifying the *Display()* function in program PAS_PROB.CPP so that `obj` is passed by reference:

```
void Display (const Object& obj)
  { /* ... */ }
```

As illustrated in the previous section, a copy constructor allows a copy of an object to allocate its own memory:

```
Object::Object (const Object& obj)
  {
  ptr = new int (*obj.ptr) ;
  }
```

12.29 The `this` Pointer

When an object calls a member function, a pointer to the object is implicitly returned to the member function. The pointer that is returned to the member function is called the **this** pointer. The **this** pointer enables a member function to know the memory address of the object that invoked the member function. The following program illustrates a use of the **this** pointer with the `Point` **class**:

```
// this.cpp
// illustrates the this pointer
//...
class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    Point* GetObjectAddress () ;
    void   ChangeObject (double xa, double ya, double za) ;
    //...
  };

// return address of object invoking member function
Point* Point::GetObjectAddress ()
  {
  return this ;
  }

// change object's data members via the this pointer
void Point::ChangeObject (double xa, double ya, double za)
  {
  this->x = xa ; this->y = ya ; this->z = za ;
```

```
  }
//...
void main ()
  {
  Point  p (1.0, 2.0, 3.0) ;  // object

  cout << "p: " << p << endl ;
  cout << "address of p (&p)  : "
       << &p << endl ;
  cout << "address of p (this): "
       << p.GetObjectAddress () << endl ;

  p.ChangeObject (4.0, 5.0, 6.0) ;  // change object

  cout << "p: " << p << endl ;
  }
```

with output:

```
p: (1, 2, 3)
address of p (&p)  : 0012FF74
address of p (this): 0012FF74
p: (4, 5, 6)
```

With the help of the two member functions *GetObjectAddress()* and *ChangeObject()*, the above program illustrates the **this** pointer. *GetObjectAddress()* simply returns the memory address of the object that invokes the function by returning the **this** pointer. The program illustrates that the memory address pointed to by **this** is indeed identical to the memory address of the object that invokes an object's member function. The **class** or type of **this** in the above *GetObjectAddress()* function is `Point*` **const**. In general, the type of the **this** pointer in a non-**static** member function of a **class** `X` is `X*` **const**. If a member function is declared as **const** or **volatile**, the type of **this** is **const** `X*` **const** or **volatile** `X*` **const** respectively. The function *ChangeObject()* illustrates, although somewhat hypothetically, that the **this** pointer can be used to alter an object's data members. Note the use of the indirect member access operator (->), since **this** is a pointer.

To date, the overloaded assignment operator for **class** `Point` has been defined as:

```
inline Point Point::operator = (const Point& p)
  {
  x = p.x ; y = p.y ; z = p.z ;
  return Point (x, y, z) ;
  }
```

which copies the data members of a `Point` object on the right-hand side of the = operator exactly to the respective data members of a `Point` object on the left-hand side of the = operator. The **operator**= *()* member function returns a temporary `Point` object by value by invoking `Point`'s three-argument constructor. Returning a `Point` object from **operator**= *()* enables the = operator to be chained in expressions.

The **this** pointer allows us not only to eliminate invoking `Point`'s three-argument constructor but also to return an object by reference:

```
inline Point& Point::operator = (const Point& p)
  {
  x = p.x ; y = p.y ; z = p.z ;
  return *this ;
  }
```

Using the dereferencing operator (*) as a prefix operator to the **this** pointer accesses the actual object at the memory address pointed to by **this**. The following program demonstrates the use of the **this** pointer for returning Point objects for the overloaded = and the prefix and postfix ++ and -- operators:

```
// this_pt.cpp
// illustrates the this pointer applied to class Point
//...
class Point
  {
  private:
    double x, y, z ;
  public:
    //...
    // overloaded operators
    Point& operator = (const Point& p) ;
    Point& operator ++ () ;      // prefix
    Point  operator ++ (int) ;   // postfix
    Point& operator -- () ;      // prefix
    Point  operator -- (int) ;   // postfix
    //...
  };

// assignment operator
inline Point& Point::operator = (const Point& p)
  {
  x = p.x ; y = p.y ; z = p.z ;
  return *this ;
  }

// prefix increment operator
inline Point& Point::operator ++ ()
  {
  x++ ; y++ ; z++ ;
  return *this ;
  }

// postfix increment operator
inline Point Point::operator ++ (int)
  {
  Point temp = *this ;
  x++ ; y++ ; z++ ;
  return temp ;
  }
//...
```

```
void main ()
  {
  Point  p, q (1.0, 2.0, 3.0) ;

  cout << "p     : " << p << endl
       << "q     : " << q << endl ;

  p = q ;

  cout << "(p=q): " << p << endl ;

  cout << "p++   : " << p++ << endl
       << "++q   : " << ++q << endl ;
  }
```

with output:

```
p     : (0, 0, 0)
q     : (1, 2, 3)
(p=q): (1, 2, 3)
p++   : (1, 2, 3)
++q   : (2, 3, 4)
```

In line with the C++ definition of prefix and postfix increment and decrement operators, the overloaded postfix operators for **class** Point first copy the object that the operator acts on to a temporary object, increment the object's data members and then return the temporary object. The overloaded postfix operators return a Point object rather than a reference because a function cannot return a reference to a local object.

One of the key applications of the **this** pointer is in returning an object from an overloaded **operator**= *()* function. This was illustrated above for the Point **class** in THIS_PT.CPP and for the Vector class in VECT1.CPP, where the data members x, y and z of **class** Point and array of Vector are of type **double**. Let's now examine the use of the **this** pointer and the **operator**= *()* function for the String **class** presented in ND_STR.CPP, which encapsulates a pointer to a **char** data member:

```
// this_str.cpp
// illustrates the use of the this pointer with the
// operator=() function for class String
#include <iostream>   // C++ I/O
#include <string>     // strlen(), strcpy()
#include <cassert>    // assert()

class String
  {
  private:
    char* string ;  // pointer to char
    int   length ;  // length of string
  public:
    // constructor
    String (const char str[]=0) ;
    // destructor
```

```
    ~String () ;
    // member function
    int Length () const { return length ; }
    // overloaded operator
    String& operator = (const String& str) ;
    // friend
    friend ostream& operator << (ostream& s,
                                 const String& str) ;
  };
//...
// assignment operator =
String& String::operator = (const String& str)
  {
  if (string != str.string)
    {
    delete [] string ;
    length = str.length ;

    if (str.string == 0)  // null string
      string = 0 ;
    else
      {
      string = new char[str.length+1] ;
      assert (string != 0) ;
      strcpy (string, str.string) ;
      }
    }
  return *this ;
  }

// friend:

// overloaded insertion operator (object)
inline ostream& operator << (ostream& s, const String& str)
  {
  return s << str.string ;
  }

void main ()
  {
  String s1 = "a string" ;
  String s2 = "a longer string" ;

  cout << "before =: " << endl ;
  cout << "s1.string: " << s1 << ", s1.length: "
       << s1.Length () << endl ;

  s1 = s2 ;

  cout << "after =: " << endl ;
  cout << "s1.string: " << s1 << ", s1.length: "
```

```
        << s1.Length () << endl ;
    }
```

with output:

```
before =:
s1.string: a string, s1.length: 8
after =:
s1.string: a longer string, s1.length: 15
```

The overloaded `String::operator=()` function first checks to see whether a `String` object is being assigned to itself. The left-hand side operand of the assignment operator pointed to by the `string` data member is then deleted, a new length assigned to the `length` data member and sufficient memory requested for the new string if it is not a null string. Finally, the `String` object operated on by the **`operator=()`** function is returned with the help of the **`this`** pointer.

12.29.1 Further Illustration of `this`

As a further illustration of the **`this`** pointer, and to help illustrate the explicit use of the indirection and address-of operators, consider the following program:

```
// this_exp.cpp
// experiments with the this pointer
#include <iostream>  // C++ I/O

class X
  {
  private:
    char* charray ;
  public:
    X ()
      { charray = "string" ; }
    void This () ;
    friend ostream& operator << (ostream& s, const X& x)
      { s << x.charray ; return s ; }
  };

void X::This ()
  {
  cout << "this : " << this  << endl
       << "*this: " << *this << endl
       << endl ;

  cout << "this->charray     : " << this->charray      << endl
       << "(*this).charray   : " << (*this).charray    << endl
       << "this->charray[0]  : " << this->charray[0]   << endl
       << "(*this).charray[0]: " << (*this).charray[0] << endl
       << endl ;

  cout << "&this->charray     : " << &this->charray     << endl
```

```
        << "&(*this).charray    : " << &(*this).charray    << endl
        << "&this->charray[0]   : " << &this->charray[0]   << endl
        << "&(*this).charray[0]: " << &(*this).charray[0]<< endl
        << endl ;
    }

  void main ()
    {
    X ().This () ;
    }
```

with output:

```
this : 0012FF88
*this: string

this->charray      : string
(*this).charray    : string
this->charray[0]   : s
(*this).charray[0]: s

&this->charray      : 0012FF88
&(*this).charray    : 0012FF88
&this->charray[0]   : string
&(*this).charray[0]: string
```

12.29.2 Restrictions on `this`

The **`this`** pointer cannot be used in either **`static`** member functions or **`friend`** functions. **`static`** member functions do not have an implicit **`this`** pointer because they are not related to a particular object. **`friend`** functions do not have a **`this`** pointer because **`friend`** functions are not members of a **`class`**.

12.30 A `Matrix` **`class`**

Let's conclude this chapter by developing a `Matrix` **`class`**. A first attempt is given in the program MTRX.CPP:

```
// mtrx.cpp
// a Matrix class
#include <iostream>   // C++ I/O
#include <iomanip>    // setprecision(),...
#include <cassert>    // assert()

class Matrix
  {
  private:
    class Attributes  // nested class
```

```
      {
      public:
        double** m ;  // pointer to matrix
        int      rows, cols ;
      };
    Attributes* pm ;
  public:
    // constructor
    Matrix (int r=1, int c=1, double value=0.0) ;
    // destructor
    ~Matrix () ;
    // member functions
    int Rows () const { return pm->rows ; }
    int Cols () const { return pm->cols ; }
    // overloaded operators
    double&       operator ()(int r, int c) ;
    const double& operator ()(int r, int c) const ;
    // friend
    friend ostream& operator << (ostream& s,
                                 const Matrix& m) ;
  };

// 3 arg. constructor
Matrix::Matrix (int r, int c, double value)
  {
  pm = new Attributes ;
  assert (pm != 0) ;
  pm->m = new double*[r] ;  // array of pointers to double
  assert (pm->m != 0) ;

  for (int i=0; i<r; i++)
    {
    // array of double's for each row
    pm->m[i] = new double[c] ;
    assert (pm->m[i] != 0) ;
    }

  pm->rows = r ;
  pm->cols = c ;

  // initialise matrix
  for (int k=0; k<r; k++)
    for (int l=0; l<c; l++)
      pm->m[k][l] = value ;
  }

// destructor
Matrix::~Matrix ()
  {
  for (int i=0; i<pm->rows; i++)
    delete pm->m[i] ;
```

```
    delete pm->m ;
    delete pm ;
    }

  // overloaded operators:

  // subscript operator () (non-const object)
  double& Matrix::operator () (int r, int c)
    {
    // verify that index values are in range
    assert (r >= 0 && r < pm->rows && c >= 0 && c < pm->cols) ;
    return pm->m[r][c] ;
    }
  //...
  // friend:

  // insertion operator <<
  ostream& operator << (ostream& s, const Matrix& m)
    {
    for (int i=0; i<m.pm->rows; i++)
      {
      for (int j=0; j<m.pm->cols; j++)
        s << setprecision (2)
          << setiosflags (ios::showpoint | ios::fixed)
          << setw (10)
          << m.pm->m[i][j] ;
      s << endl ;
      }
    return s ;
    }

  void main ()
    {
    Matrix m (3, 3, 4.5), n (3, 3) ;

    cout << "Matrix m: " << endl << m ;
    cout << "Matrix n: " << endl << n ;

    cout << "Matrix n (n=m): " << endl << n ;

    // Matrix subscript operator
    n(2, 2) = 222.22 ;

    cout << "Matrix n (n(2,2)): " << endl << n ;
    cout << "n.rows: "   << n.Rows ()
         << ", n.cols: " << n.Cols () << endl ;
    }
```

with output:

```
Matrix m:
```

```
      4.50       4.50       4.50
      4.50       4.50       4.50
      4.50       4.50       4.50
Matrix n:
      0.00       0.00       0.00
      0.00       0.00       0.00
      0.00       0.00       0.00
Matrix n (n(2,2)):
      0.00       0.00       0.00
      0.00       0.00       0.00
      0.00       0.00     222.22
n.rows: 3, n.cols: 3
```

A nested **class**, Attributes, is declared within **class** Matrix:

```
class Attributes  // nested class
  {
  public:
    double** m ;  // pointer to matrix
    int      rows, cols ;
  };
```

class Attributes encapsulates a pointer to a matrix as a pointer to a **double** pointer, m, and the number of rows and columns of a two-dimensional matrix. Note that the data members of Attributes are declared **public** so that the pointer, pm, has direct access to Attributes' data members. Declaring Attributes' data members as **public** is feasible in this instance since Attributes is declared a **private** nested **class** of Matrix. Two **const** access member functions, *Rows()* and *Cols()*, are defined, which return the number of rows and columns of a Matrix object.

The three-argument Matrix constructor first requests sufficient memory for the pointer pm:

```
pm = new Attributes ;
```

Then, sufficient memory is requested for an array of pointers to **double** for each row of the matrix:

```
pm->m = new double*[r] ;
```

For each row of the matrix, sufficient memory is then requested to hold an array of elements of type **double** of size equal to the number of columns of the matrix:

```
for (int i=0; i<r; i++)
  {
  pm->m[i] = new double[c] ;
  //...
  }
```

Each pointer returned by the **new** operator is tested, via the *assert()* macro, against the 0 pointer to verify that memory allocation is successful. The constructor concludes by setting the number of rows and columns of the matrix and initialising the value of all of the elements of the matrix. A **for**-loop is used for the initialisation of the matrix elements since the C++ library function *memset()* cannot be used to set a block of memory to a **double** value.

The `Matrix` destructor first uses a **`for`**-loop to deallocate the memory associated with each row of the matrix, then deallocates the memory that holds the array of pointers to **`double`** and finally deallocates the memory pointed to by the `pm` pointer.

The definitions of the overloaded << operator is fairly straightforward. The `()` operator is overloaded as a matrix subscript operator for indexing the elements of a `Matrix` object:

```
n(2, 2) = 222.22 ;
```

`Matrix::`**`operator`**`()` is overloaded for both non-constant and constant objects of **`class`** `Matrix`:

```
double&       Matrix::operator () (int r, int c) ;
const double& Matrix::operator () const (int r, int c) ;
```

The return type of the two overloaded member functions is a reference to **`double`** rather than a **`double`**. Returning a reference causes the `()` subscript operator to act on an object's data members directly without returning a copy. Returning by reference allows the `Matrix::`**`operator`**`()` function to appear on either the left-hand or right-hand side of the assignment operator.

As it stands, the `Matrix` **`class`** appears to work well. However, if the **`class`** is to be developed further a copy constructor and overloaded assignment operator are essential, for the reasons outlined above of passing objects to and returning objects from functions. The assigned subscript operator for **`class`** `Matrix` is the nasty `()` operator (e.g. `m(1,2)`), which is a cross between something out of Fortran and a C++ function call. But most importantly, the `Matrix` **`class`** has been developed quite independently of the previously developed `Vector` **`class`**. Since a matrix is a system of equal-sized vectors, it makes sense to build our `Matrix` **`class`** using the `Vector` **`class`**. Another very good reason for using `Vector` in our design of `Matrix` is that it solves the problem of subscripting the elements of a `Matrix` object in a C++ array style, e.g. `m[1][2]`.

To increase the modularity of the `Vector` and `Matrix` classes the **`class`** declarations are placed in .H header files and the member function definitions are placed in .CPP implementation files. First, the `Vector` **`class`**:

```
// vector.h
// header file for Vector class

#ifndef _VECTOR_H  // prevent multiple includes
#define _VECTOR_H

#include <iostream>  // C++ I/O
#include <iomanip>   // setprecision(),...
#include <cassert>   // assert()

class Vector
  {
  private:
    double* array ;
    int     n_elements ;
    // member function
    void    Allocate (int n) ;
  public:
```

```
    //...
  };

#endif // _VECTOR_H
```

with corresponding implementation file VECTOR.CPP. And now, the `Matrix` **class**:

```
// matrix.h
// header file for Matrix class

#ifndef _MATRIX_H  // prevent multiple includes
#define _MATRIX_H

#include <iostream>  // C++ I/O
#include <iomanip>   // setprecision(),...
#include <cassert>   // assert()
#include <cmath>     // fabs()

#include "vector.h"

class Matrix
  {
  private:
    Vector*  array ;
    int      rows, cols ;
  public:
    // constructor
    Matrix (int r=1, int c=1, double value=0.0) ;
    // copy constructor
    Matrix (const Matrix& m) ;
    // destructor
    ~Matrix () ;
    // member functions
    double&       Value (int row, int col) ;
    //...
    int           Rows () const { return rows ; }
    int           Cols () const { return cols ; }
    Vector        GaussElimination (Vector& b,
      bool& solved, const bool& operate_on_matrix_rhs=false) ;
    // overloaded operators
    Matrix&       operator = (const Matrix& m) ;
    Vector&       operator [] (int index) ;
    //...
    Matrix        operator + (const Matrix& m) ;
    Matrix        operator - (const Matrix& m) ;
    Matrix        operator * (const Matrix& m) ;
    // friend
    friend ostream& operator << (ostream& s, const Matrix& m) ;
   };

#endif // _MATRIX_H
```

```
// matrix.cpp
// implementation file for Matrix class
#include "matrix.h"

const double TOLERANCE = 1e-06 ;

// swap two doubles
void Swap (double& a, double& b)
  {
  double temp = a ;
  a = b ;
  b = temp ;
  }

// 3 arg. constructor
Matrix::Matrix (int r, int c, double value)
  {
  array = new Vector[r] ; // Vector array
  assert (array != 0) ;

  for (int i=0; i<r; i++)
    array[i] = Vector (c) ; // Vector

  rows = r ;  // set row & column dimensions
  cols = c ;

  // initialise matrix
  for (int k=0; k<r; k++)
    for (int l=0; l<c; l++)
      Value (k, l) = value ;
  }

// copy constructor
Matrix::Matrix (const Matrix& m)
  {
  array = new Vector[m.rows] ; // Vector array
  assert (array != 0) ;

  for (int i=0; i<m.rows; i++)
    array[i] = Vector (m.cols) ; // Vector

  rows = m.rows ;  // set row & column dimensions
  cols = m.cols ;

  // initialise matrix
  for (int k=0; k<m.rows; k++)
    for (int l=0; l<m.cols; l++)
      Value (k, l) = m.Value (k, l) ;
  }

// destructor
```

```
Matrix::~Matrix ()
  {
   if (array != 0)
    delete [] array ;
  rows = cols = 0 ;
  }

// public member functions:

// returns element value (non-const object)
double& Matrix::Value (int row, int col)
  {
  assert (row >= 0 && row < rows && col >=0 && col < cols) ;
  return array[row].Value(col) ;
  }
//...
// Gauss elimination
Vector Matrix::GaussElimination (Vector& b,
  bool& solved, const bool& operate_on_matrix_rhs)
  {
  // operate/leave alone matrix/rhs
  Matrix* matrix ;
  Vector* rhs ;
  if (operate_on_matrix_rhs)
    {
    matrix = this ;
    rhs    = &b ;
    }
  else
    {
    matrix = new Matrix (*this) ;
    rhs    = new Vector (b) ;
    }

  solved = false ;

  int n = rows ;  // assign number of rows to n

  Vector x (n) ;  // solution Vector

  // Gaussian elimination:

  for (int i=0; i<n; i++)
    {
    double max_pivot = fabs (matrix->Value(i, i)) ;
    int    pivot_row = i ;

    // find maximum pivot & row in which it occurs
    for (int k=i+1; k<n; k++)
      {
      if (fabs(matrix->Value(k,i)) > max_pivot)
```

```
        {
        max_pivot = fabs (matrix->Value(k, i)) ;
        pivot_row = k ;
        }
      }

    if (max_pivot > TOLERANCE) // if pivot ok
      {
      solved = true ;

      // if pivot found, swap rows
      if (i != pivot_row)
        {
        for (int j=0; j<n; j++)
          Swap (matrix->Value(i, j),
            matrix->Value(pivot_row, j)) ;
        Swap (rhs->Value(i), rhs->Value(pivot_row)) ;
        }
      for (int j=i+1; j<n; j++)
        {
        double multiplier = matrix->Value(j, i) /
          matrix->Value(i, i) ;
        for (int k=i; k<n; k++)
          matrix->Value(j, k) -=
            multiplier * matrix->Value(i, k) ;
        rhs->Value(j) -= multiplier * rhs->Value(i) ;
        }
      }
    else  // if pivot not ok
      {
      solved = false ;
      return x ;  // return zero solution Vector
      }
    }

  // back substitution:

  // (n-1) element
  x[n-1] = rhs->Value(n-1) / matrix->Value(n-1, n-1) ;

  // for i=n-2,n-3,...0
  for (int s=n-2; s>=0; s--)
    {
    double sum = 0.0 ;
    for (int t=s+1; t<=n-1; t++)
      sum += matrix->Value(s, t) * x[t] ;
    x[s] = (rhs->Value(s) - sum) / matrix->Value(s, s);
    }
  // clean-up
  if (!operate_on_matrix_rhs)
    {
```

```
      delete matrix ;
      delete rhs ;
      }

    return x ;
    } // GaussElimination()

  // overloaded operators:
  //....
  // multiplication operator *
  Matrix Matrix ::operator * (const Matrix& m)
    {
    assert (cols == m.Rows()) ;

    Matrix result (rows, m.cols) ;

    for (int i=0; i<rows; i++)
      {
      for (int j=0; j<m.Cols(); j++)
        {
        double sum = 0.0 ;
        for (int k=0; k<cols; k++)
          sum += Value (i, k) * m.Value(k, j) ;
        result.Value(i, j) = sum ;
        }
      }
    return result ;
    }

  // friend:

  // insertion operator <<
  ostream& operator << (ostream& s, const Matrix& m)
    {
    for (int i=0; i<m.rows; i++)
      {
      for (int j=0; j<m.cols; j++)
        s << setprecision (2)
          << setiosflags (ios::showpoint | ios::fixed)
          << setw (10)
          << m.Value (i, j) ;
      s << endl ;
      }
    return s ;
    }
```

Program MTRX_TST.CPP tests the `Vector` and `Matrix` classes:

```
  // mtrx_tst.cpp
  // tests the modified Matrix class which constructs an array
  // of Vector objects and allows C++-style subscripting of
```

```
// Matrix elements
#include <iostream>  // C++ I/O

#include "vector.h"  // class Vector
#include "matrix.h"  // class Matrix

void main ()
  {
  Vector v (3) ;
  Matrix m (3, 3, 2.5), n (3, 3) ;

  cout << "Vector v: " << endl << v << endl ;
  cout << "Matrix m: " << endl << m ;

  // Matrix assignment
  n = m ;

  // Matrix subscript operator
  n[1][2] = 222.22 ;

  cout << "Matrix n: " << endl << n ;

  // Matrix multiplication
  Matrix m1 (2, 3) ;
  Matrix m2 (3, 3) ;
  m1[0][0] = 3 ; m1[0][1] = 2 ; m1[0][2] = -1 ;
  m1[1][0] = 0 ; m1[1][1] = 4 ; m1[1][2] =  6 ;

  m2[0][0] = 1 ; m2[0][1] = 0 ; m2[0][2] = 2 ;
  m2[1][0] = 5 ; m2[1][1] = 3 ; m2[1][2] = 1 ;
  m2[2][0] = 6 ; m2[2][1] = 4 ; m2[2][2] = 2 ;

  Matrix m3 (2, 3) ;
  m3 = m1 * m2 ;
  cout << "Matrix m3: " << endl << m3 ;

  // Gaussian elimination
  Matrix a (3, 3) ;
  Vector b (3), x (3) ;

  a[0][0] =  1.0 ; a[0][1] =  2.0 ; a[0][2] =  3.0 ;
  a[1][0] =  2.0 ; a[1][1] =  3.0 ; a[1][2] =  4.0 ;
  a[2][0] =  3.0 ; a[2][1] =  4.0 ; a[2][2] =  1.0 ;
  b[0]    = 14.0 ; b[1]    = 20.0 ; b[2]    = 14.0 ;

  cout << "Matrix a: " << endl << a ;
  cout << "Vector b: " << endl << b << endl ;

  bool solved (false) ;

  x = a.GaussElimination (b, solved) ;
```

```
  if (solved)
    cout << "solution Vector x of system [a]{x}={b}: "
         << endl << x << endl ;
  else
    cout << "solution by Gauss elimination failed" << endl ;
  }
```

and generates the following output:

```
Vector v:
[0.00, 0.00, 0.00]
Matrix m:
      2.50        2.50        2.50
      2.50        2.50        2.50
      2.50        2.50        2.50
Matrix n:
      2.50        2.50        2.50
      2.50        2.50      222.22
      2.50        2.50        2.50
Matrix m3:
      7.00        2.00        6.00
     56.00       36.00       16.00
Matrix a:
      1.00        2.00        3.00
      2.00        3.00        4.00
      3.00        4.00        1.00
Vector b:
[14.00, 20.00, 14.00]
solution Vector x of system [a]{x}={b}:
[1.00, 2.00, 3.00]
```

The above declaration of **class** `Vector` is similar to that of program VECT1.CPP.

The **private** data members of **class** `Matrix` are declared as:

```
class Matrix
  {
  private:
    Vector* array ;
    int     rows, cols ;
  //...
  };
```

The **private** data member `array` is a pointer to `Vector`. Within the body of the three-argument constructor of `Matrix` sufficient memory is requested for a matrix by defining an array of `Vectors` for each row of the matrix:

```
array = new Vector[r] ;
```

and then assigning a `Vector` to each row of the matrix:

```
for (int i=0; i<r; i++)
  {
  array[i] = Vector (c) ;
  //...
  }
```

The `Matrix` destructor simply deletes the memory allocated to array and then sets the number of rows and columns to zero:

```
Matrix::~Matrix ()
  {
  if (array != 0)
    delete [] array ;
  rows = cols = 0 ;
  }
```

class `Matrix` defines a copy constructor and two overloaded member functions, *`Value()`*, that access the value of a particular element for both non-constant and constant `Matrix` objects. The insertion operator (<<) is overloaded to enable C++-style output of `Matrix` objects.

The arithmetic binary operators +, - and * are overloaded for `Matrix` objects. If two ($m \times n$) matrices $\mathbf{A} = [a_{jk}]$ and $\mathbf{B} = [b_{jk}]$ are added, the sum matrix $\mathbf{C}[c_{jk}] = \mathbf{A} + \mathbf{B}$ is size ($m \times n$) with elements:

$$c_{jk} = a_{jk} + b_{jk}, \quad j = 1, 2, \ldots, m; \quad k = 1, 2, \ldots, n$$

and similarly for the subtraction of two matrices.

Suppose now that $\mathbf{A} = [a_{jk}]$ is an ($m \times n$) matrix and $\mathbf{B} = [b_{jk}]$ is an ($r \times p$) matrix; then the product $\mathbf{C}[c_{jk}]=\mathbf{AB}$ is size ($m \times p$) with elements:

$$c_{jk} = a_{j1}b_{1k} + a_{j2}b_{2k} + \ldots + a_{jn}b_{nk} = \sum_{i=1}^{n} a_{ji}b_{ik}$$

and is defined only when $r = n$ (i.e. the number of rows in the second matrix is equal to the number of columns of the first matrix). Note that matrices are associative and distributive with respect to addition and subtraction, but are non-commutative with respect to multiplication; that is $\mathbf{AB} \neq \mathbf{BA}$. An example of the multiplication of two `Matrix` objects, `m1` and `m2`, is given in the *`main()`* function of MTRX_TST.CPP.

Let's now examine the indexing of elements of a `Matrix` object.

12.30.1 A Subscript Operator for `class Matrix`

It was noted in Chapter 10 and the first implementation of **class** `Matrix` in program MTRX.CPP that it is not a straightforward operation to overload the subscript operator for a two-dimensional `Matrix` **class**. Consider the use of the subscript operator with an object of **class** `Matrix`:

```
Matrix m (10, 10) ;
//...
m[row][col] = x ;
```

To understand exactly how indexing the elements of a `Matrix` object occurs we must remember that the subscript operator, `[]`, is a binary left-associative operator. Therefore, the above expression is equivalent to:

```
(m[row])[col] = x ;
```

where `m[row]` indicates the `Vector` row of the matrix and `[col]` indicates the element at position `col` of the respective row. Thus, to index an element in a two-dimensional matrix will require the overloading of two individual `[]` operators for both `Matrix` and `Vector` objects. The left-hand overloaded `[]` operator implicitly acts on a `Matrix` object and is passed the row number of the matrix:

```
Vector& Matrix::operator [] (int index) // non-const objects
   {
   //...
   return array[index] ;
   }
```

whereas the right-hand overloaded `[]` operator implicitly acts on a `Vector` object and is passed the element number of a given row:

```
double& Vector::operator [] (int index) // non-const objects
   {
   //...
   return array[index] ;
   }
```

Notice that subscripting the elements of a `Matrix` object in a C++ style is made possible by constructing a `Matrix` object as an array of `Vector` objects.

12.30.2 Gaussian Elimination of a System of Linear Equations

This section examines the development of a member function `GaussElimination()` that solves a system of n linear equations in n unknowns $x_1, x_2, ..., x_n$:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{Bmatrix}$$

where the coefficient matrix $\mathbf{A} = [a_{ij}]$ is an $(n \times n)$ matrix and $\mathbf{x} = [x_i]$ and $\mathbf{b} = [b_i]$ are column vectors.

Let's begin by considering a diagonal matrix (i.e. $a_{ij} = 0$ for $i \neq j$; $a_{ii} \neq 0$ for $i = 1, 2, ..., n$). In this case, the solution vector is known immediately:

$$x_i = \frac{b_i}{a_{ii}}, \quad i = 1, 2, \ldots, n$$

If matrix **A** is upper-triangular with no diagonal element, a_{ii}, equal to zero, then **x** is easily found:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22} & \cdots & a_{2n} \\ & & \ddots & \vdots \\ & & & a_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{Bmatrix}$$

The unknown x_n is found immediately:

$$x_n = \frac{b_n}{a_{nn}}$$

Given x_n, x_{n-1} is found from the $(n-1)$th equation:

$$x_{n-1} = \frac{b_{n-1} - a_{n-1} x_n}{a_{n-1,n-1}}$$

and in general:

$$x_{n-1} = \frac{b_i - a_{i,i+1} x_{i+1} - \ldots - a_{in} x_n}{a_{ii}}, \quad i = n-1, n-2, \ldots, 1$$

This process of determining x_i is referred to as *back-substitution*, an algorithm of which is outlined below:

algorithm: back-substitution

given: upper-triangular form of A, b, n
$x_n = b_n / a_{nn}$
for $i = n - 1, n - 2, ..., 1$

$$x_i = \left(b_i - \sum_{j=i+1}^{n} a_{ij} x_j \right) a_{ii}^{-1}$$

For example, if we apply the back-substitution algorithm to the system:

$$\begin{bmatrix} 3 & 4 & 1 \\ & 1 & 2 \\ & & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 14 \\ 8 \\ 6 \end{Bmatrix}$$

we find:

$$x_3 = b_3 / a_{33} = 6 / 2 = 3$$
$$x_2 = (b_2 - a_{23} x_3) / a_{22} = (8 - 2 \cdot 3) / 1 = 2$$
$$x_1 = (b_1 - a_{12} x_2 - a_{13} x_3) / a_{11} = (14 - 4 \cdot 2 - 1 \cdot 3) / 3 = 1$$

Once a set of linear equations has been reduced to an upper-triangular matrix, the solution vector **x** is easily found by performing back-substitution. The key to solving a set of linear equations lies in the process of reducing a set of simultaneous equations to triangular form. One of the more well-known methods is Gaussian elimination, which is best illustrated by using an example. Consider the system of equations for which a solution is required:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 14 \\ 20 \\ 14 \end{Bmatrix}$$

The first part of the elimination process eliminates x_1 from the second and third equations. This is achieved by subtracting multiples of the first equation from the second and third equations. To eliminate x_1 from the second and third equations *multipliers* of 2/1 = 2 and 3/1 = 3, respectively, are required:

$$\begin{bmatrix} 1 & 2 & 3 \\ & -1 & -2 \\ & -2 & -8 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 14 \\ -8 \\ -28 \end{Bmatrix}$$

During the elimination of x_1 from equations two and three, the first equation is called the *pivotal equation* and its diagonal coefficient the *pivot*. Eliminating x_2 from the third equation now requires that the second equation becomes the pivotal equation, with element a_{12} the pivot and a multiplier of 3/1 = 2:

$$\begin{bmatrix} 1 & 2 & 3 \\ & -1 & -2 \\ & & 4 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 14 \\ -8 \\ 12 \end{Bmatrix}$$

Back-substitution can now be performed to determine vector **x**:

$$\mathbf{x} = \begin{Bmatrix} 1 \\ 2 \\ 3 \end{Bmatrix}$$

An algorithm for Gaussian elimination is:

algorithm: Gaussian elimination

given: A, b, n
for i = 1, 2, ..., n-1
  for k = i+1, ..., n
    $multiplier_{ki} = a_{ki}/a_{ii}$
    for j=i+1, ..., n
      $a_{kj} = a_{kj} - multiplier_{ki} * a_{ij}$
    $b_k = b_k - multiplier_{ki} * b_i$

What if a small or zero pivot element is encountered? One solution would be simply to exit the routine, but a more popular approach is to perform *partial pivoting*. To illustrate the technique of partial pivoting, consider once more the system of equations:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 14 \\ 20 \\ 14 \end{Bmatrix}$$

At present, the element coefficients of x_1 are 1, 2 and 3 for equations 1, 2 and 3, respectively, with multipliers 2 and 3. To make the value of the pivot as large as possible and hence reduce the magnitude of the multipliers, interchange rows 1 and 3:

$$\begin{bmatrix} 3 & 4 & 1 \\ 2 & 3 & 4 \\ 1 & 2 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 14 \\ 20 \\ 14 \end{Bmatrix}$$

This system now has multipliers of 2/3 and 1/3 for rows 2 and 3, respectively. This process is repeated for each part of the triangularisation.

The procedure described above is implemented in the member function `Matrix::GaussElimination()`, remembering that the indexing of an array of size n in C++ is in the range (0:n–1) rather than (1:n). The third argument of `GaussElimination()` allows a user to specify whether the operated-on matrix and right hand side vector `b` are permitted to be altered by the underlying algorithm. In some applications it is important not to corrupt the operated-on and function argument objects (default case) while for other applications we can manipulate the operated-on and function argument objects and therefore save copying objects. This facility is conveniently implemented by working with the `matrix` and `rhs` pointers to `Matrix` and `Vector` respectively:

```
Vector Matrix::GaussElimination (Vector& b, /* ... */)
  {
   Matrix* matrix ;
   Vector* rhs ;
   if (operate_on_matrix_rhs)
    {
    matrix = this ;
    rhs    = &b ;
    }
   else
    {
    matrix = new Matrix (*this) ;
    rhs    = new Vector (b) ;
    }
  //...
  // clean-up
  if (!operate_on_matrix_rhs)
    {
    delete matrix ;
    delete rhs ;
```

```
    }

  return x ;
  }
```

A typical application of *GaussElimination()* is demonstrated in MTRX_TST.CPP:

```
Matrix a (3, 3) ;
Vector b (3), x (3) ;
bool solved ;
//...
x = a.GaussElimination (b, solved) ;
```

It is worth highlighting that swapping all elements of two rows for a large system of equations can be computationally very expensive. Thus, in practice an integer array of 'pointers' is defined which records the order of the rows without actually performing element swapping. To define such an array in *GaussElimination()* would require the use of a dynamic one-dimensional array or vector of integers – which at present we have not discussed.

Note that *complete* pivoting is possible where the element of largest magnitude in the submatrix ($\mathbf{A}'=[a_{jk}]$; $k,j \geq i$) is located and moved to the pivot position by row and column interchanges if necessary. In practice, such a process is generally considered to be computationally too expensive, and partial pivoting is the generally adopted technique.

A measure of the efficiency (or inefficiency) of the Gaussian elimination algorithm is the number of arithmetic operations required to obtain a solution vector. Excluding back-substitution there are $n(n^2-1)/3$ additions, $n(n^2-1)/3$ multiplications and $n(n-1)/2$ divisions. The back-substitution part of the Gaussian elimination algorithm consists of $n(n-1)/2$ additions, $n(n-1)$ multiplications and n divisions.

For a more detailed examination of matrix computations the interested reader is referred to Jennings and McKeown (1992) and Golub and Van Loan (1989).

12.30.3 Return Value of *GaussElimination()*

The return value of *GaussElimination()* is an object of **class** `Vector`. This is feasible for small `Vector` objects, but very inefficient for larger objects because of the time and memory required for the compiler to generate a temporary object to hold the return object of *GaussElimination()*. A better approach is to return a pointer to the solution `Vector` object returned by *GaussElimination()*. Adopting this approach requires a few small alterations to the above MATRIX.H/.CPP and MTX_TST.CPP programs, which are summarised below:

```
// matrix.h
//...
class Matrix
  {
  public:
    Vector* GaussElimination (/* ... */) ;
    //...
  };

// matrix.cpp
//...
```

```
Vector* Matrix::GaussElimination (/* ... */)
  {
  //...
  Vector* x = new Vector (n) ;  // solution Vector
  //...
  return x ;
  }

// mtrx_tst.cpp
//...
void main ()
  {
  Vector* x = a.GaussElimination (b, solved) ;

  //...
  delete x ;
  }
```

12.31 Summary

Hopefully the size of the present chapter has helped to emphasise the level of importance attached to pointers when programming in C++. If there are any topics in this chapter that you feel uncomfortable with, then I seriously recommend referring back to them now before proceeding further.

A pointer is simply a variable that holds the memory address of another variable or object. Pointers are defined using an asterisk (*) after the data type or **class** that is to be pointed to. The rvalue of a pointer is in fact the lvalue or memory address of another variable or object. The lvalue of the pointer is the memory address of the pointer itself. The memory address of an object can be assigned to a pointer by means of the address-of operator (&), and the object itself can be accessed indirectly by means of the indirection operator (*) preceding the pointer.

The data members of an object or member function of the object's **class** are accessed by the direct member access operator (.), whereas the members of an object can be accessed indirectly via a pointer to the object by the indirect member access operator (->).

As with objects, pointers can be **const** modified so that a programmer has complete flexibility of defining either a pointer to a constant object or a constant pointer to an object or both. C++ supports a pointer to **void** which allows any type of pointer to be assigned to itself. Pointers to **void** can greatly increase the generality of a function.

Pointers enable the memory address of an object to be passed to or returned by a function. Passing or returning an object by pointer eliminates any copying of the object, since the actual memory address of the object is being referenced and not the object itself.

Pointers are variables and as such can be used in arithmetical expressions. Pointer arithmetic is, however, restricted to the addition, subtraction, increment and decrement operators.

There is a close relationship between arrays and pointers in C++, since the name of an array is a synonym for the memory address of the array's zeroth element. This property of arrays enables the memory address of an array to be assigned to a pointer and then the elements of the array accessed by pointer arithmetic rather than by the array subscript operator. C++ also adopts the convention that a function name is a synonym for its memory address, which enables the memory address of a function to be assigned to a pointer.

C++ offers a programmer two operators for performing dynamic memory allocation, namely **new** and **delete**. The **new** operator allocates a block of memory from the operating system and returns a pointer to that block. The **delete** operator deallocates a block of memory allocated by the **new** operator. The **new** and **delete** operators can be used either to allocate or deallocate memory for objects or arrays of objects. Since **new** and **delete** are operators they can be overloaded specifically for a given user-defined **class**.

When an object of a **class** calls a member function of that **class**, a pointer to that object is implicitly returned to the member function and is referred to as the **this** pointer.

Exercises

12.1 By dereferencing the pointer to **int** `p_int`, assign the value of 11 to `i`:

```
void main ()
  {
  int i (0) ;

  int* p_int (0) ;
  //...
  }
```

12.2 The trapezoidal rule of numerical integration approximates a continuously differentiable function $f(x)$ within the interval $a \le x \le b$ by a piecewise linear function which is the polygon of n chords of the curve f:

$$I = \int_a^b f(x)\mathrm{d}x = h\left(\frac{1}{2}f(a) + f(x_1) + \ldots + f(x_{n-1}) + \frac{1}{2}f(b)\right)$$

where $h=(b-a)/n$ and x_j $(=a+jh; j=1{:}n-1)$ are the equal interval points. Develop a function which performs trapezoidal integration and is passed an arbitrary function with the following signature:

```
double Function (double x) ;
```

12.3 Rather than implementing a `Matrix` **class** as an array of pointers to `Vector` objects, as shown in the present chapter, declare a `Matrix` **class** which encapsulates a single `Vector` object and consequently stores a two-dimensional matrix as a one-dimensional vector.

12.4 Develop three triangle classes which encapsulate object, reference and pointer `Point` data members.

12.5 The C++ library function *`bsearch()`* has the following signature:

```
void* bsearch (const void* key, const void* base,
               size_t n_elem, size_t width,
               int (*fcmp)(const void*,
               const void*)) ;   // STDLIB.H
```

`bsearch()` performs a binary search of an array (pointed to by `base`) of `n_elem` elements (each of size `width` in bytes) for the search key, `key`. The comparison function, `fcmp`, compares the elements of an array and returns an integer value of <0, ==0 or

>0 if the first element is less than, equal to or greater than the second element respectively. If a match for `key` is found in the array then *`bsearch()`* returns the first array index for which a match is found, else *`bsearch()`* returns 0. The **`typedef`** `size_t` (**`unsigned`**) is defined in STDDEF.H. For *`bsearch()`* to work the array elements must be in order.

Write a program which searches, using *`bsearch()`*, a sorted array of integers for a given integer.

12.6 For the `String` **`class`** of ND_STR.CPP, overload the **`new`** and **`delete`** operators.

12.7 Develop a simple linked list **`class`**, `LinkedList`, to hold a list of integers. A linked list consists of an arbitrary number of nodes which encapsulate both a data value and a pointer to the next node in the list. Implement a node as a `Node` **`class`** nested within `LinkedList`. Provide a `LinkedList::`*`Add()`* member function which allows integers to be added to a `LinkedList` object.

Chapter 13

Templates

Your quote here.
– B. Stroustrup

Templates allow a programmer to create generic functions and parametrised types or generics. The type of data or objects that a template function or class operates on remains unspecified in the function definition and class declaration until a function is called or an object is defined. Such type-independent functions and classes frequently arise when the operation of a function or data of an object is independent of type. The real power of templates is that a function can be defined or a class declared just the once, but can be operated on by all C++ integral or user-defined types.

To see templates in action, a linked list class, LinkedList, is implemented along with an associated iterator class, LinkedListIterator. Iterators are functions or classes whose sole purpose is to iterate through the encapsulated data elements of an object. Iterators provide a safe way of ensuring that out-of-bounds object element indexing does not occur. Vector and Matrix classes are also developed and help to illustrate composed types and the common pitfall when designing template classes of inserting type dependencies.

A Polygon class is developed which makes good use of LinkedList by encapsulating a linked list of Point vertices. Although a relatively simple implementation, class Polygon allows an object to be defined with an arbitrary number of vertices for both convex and concave polygons, and supports operations for determining whether a point is inside a polygon and returning the convex hull of a polygon. Class polygon illustrates that by encapsulating a polygon's vertices as a LinkedList object, rather than a pointer to Point, all the memory management issues behind defining a Polygon object are performed by LinkedList. The use of powerful dynamic data structures such as Vector and LinkedList allows the programmer to focus on the application at hand rather than repeatedly getting involved in the memory management of objects.

13.1 Function Overloading

Let us first examine defining a function which returns the maximum of its two arguments. We could simply define functions `CharMax()`, `IntMax()`, `FloatMax()` etc., which operate on **`char`**, **`int`**, **`float`** data, respectively, or we could alternatively define a set of overloaded functions with the same function name:

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

```
// fun_over.cpp
// implements a Max() function with
// different argument types by function overloading
#include <iostream>  // C++ I/O

// returns the maximum of two chars
char Max (char a, char b)
  {
  return a > b ? a : b ;
  }

// returns the maximum of two ints
int Max (int a, int b)
  {
  return a > b ? a : b ;
  }

// returns the maximum of two floats
float Max (float a, float b)
  {
  return a > b ? a : b ;
  }

// returns the maximum of two doubles
double Max (double a, double b)
  {
  return a > b ? a : b ;
  }

void main ()
  {
  char   ca ('a'), cb ('b') ;
  int    ia (1)  , ib (2)   ;
  float  fa (3.0), fb (4.0) ;
  double da (6.0), db (5.0) ;

  cout << "Max(char, char)    : " << Max (ca, cb) << endl ;
  cout << "Max(int, int)      : " << Max (ia, ib) << endl ;
  cout << "Max(float, float)  : " << Max (fa, fb) << endl ;
  cout << "Max(double, double): " << Max (da, db) << endl ;
  }
```

with output:

```
Max(char, char)    : b
Max(int, int)      : 2
Max(float, float)  : 4
Max(double, double): 6
```

This approach works fine, but we observe from the above definitions of the four overloaded *Max()* functions that each function has exactly the same setup. Each function has two argu-

ments of the same type, returns a type the same as the type of its arguments and has an identical single **return** statement:

```
return a > b ? a : b ;
```

If we now require that the *Max()* function operates on other types or classes, such as Complex or Point, we would have to overload *Max()* specifically for these classes.

13.2 Template Functions

Alternatively, C++ allows a programmer to define *Max()* as a *template* or *generic* function that is defined independent of a particular type or **class**. The type or **class** of an object that a template function operates on is implicitly passed as an argument. Let's modify the above program, FUN_OVER.CPP, defining *Max()* as a **template** function:

```
// tmp_fun.cpp
// implements a Max() function as a template function
#include <iostream>  // C++ I/O

// returns the maximum of two objects
template <class T>
T Max (T a, T b)
  {
  return a > b ? a : b ;
  }

void main ()
  {
  // note: identical to main() in fun_over.cpp
  char   ca ('a'), cb ('b') ;
  int    ia (1)  , ib (2)   ;
  float  fa (3.0), fb (4.0) ;
  double da (6.0), db (5.0) ;

  cout << "Max(char, char)     : " << Max (ca, cb) << endl ;
  cout << "Max(int, int)       : " << Max (ia, ib) << endl ;
  cout << "Max(float, float)   : " << Max (fa, fb) << endl ;
  cout << "Max(double, double): " << Max (da, db) << endl ;
  }
```

Immediately we note the reduced program size of TMP_FUN.CPP compared to FUN_OVER.CPP. Also, it is worth noting that the *main()* functions in both of the above two programs are identical (except for the comment), which illustrates that there is no syntactical difference between a **template** function call and a *normal* function call. In contrast with **template** classes, you do not specify the type of the **template** arguments when calling a **template** function. For example, in the call:

```
int i, j ;
//...
```

```
int k = Max (i, j) ;
```

the compiler knows that the call is of the form *Max(***int**,**int***)*, since the type of `i` and `j` is **int**.

The *Max()* **template** function is defined as:

```
template <class T>
T Max (T a, T b)
  {
  return a > b ? a : b ;
  }
```

and is of the general single type argument or parameter **template** function definition syntax:

```
template <class type>
return_type FunctionName (parameter_list)
  {
  //...
  }
```

template is a C++ keyword. `type`[1] is a type argument which indicates the type on which the function operates and is placed between the set of angle brackets (< and >). The scope of `type` extends from the function declarator to the end of the **template** function definition. Although `type` is preceded by the keyword **class** the type argument can in fact be any valid type, and not necessarily a **class** (e.g. a **double**). Templates are automatically instantiated by the compiler replacing `type` with the actual data type specified in the function call. Note that type `T` is used throughout the function declarator of *Max()*, including the return type. Template functions cannot assign a default argument to `type` whereas template classes accept default arguments.

In program TMP_FUN.CPP the *Max()* function is defined before it is called in *main()*. If the definition of *Max()* appears after the definition of *main()*, we require a **template** declaration of *Max()*:

```
// tmp_dec.cpp
// illustrates a template function declaration
#include <iostream>  // C++ I/O

// global template function declaration
template <class T> T Max (T a, T b) ;

void main ()
  {
  // local-to main() declaration of Abs()
  int Abs (int n) ;
  // local-to-main() declaration of Swap()
```

1 The generally adopted letter and name for a single type argument are `T` and `Type`, respectively. Perhaps the letter `C` should be adopted for **class** or `TC` for type or **class**, since subtle differences do exist between a type and a **class** in C++.

```
    template <class T> void Swap (T& a, T& b) ; // error!

    int i (1), j (2) ;

    int max = Max (i, j) ;
    cout << "max: " << max ;

    int abs = Abs (-1) ;
    cout << "abs: " << abs ;
    }

// returns the maximum of two objects
template <class T>
T Max (T a, T b)
    {
    return a > b ? a : b ;
    }

// returns the absolute value of an int
int Abs (int n)
    {
    return n > 0 ? n : -n ;
    }

// swaps two objects
template <class T>
void Swap (T& a, T& b)
    {
    T temp ;

    temp = a ;
    a = b ;
    b = temp ;
    }
```

The **template** declaration of *Max()* is a global declaration. Template declarations must be global declarations, whereas non-**template** functions can be declared local to a function. This is illustrated by the **template** declaration of *Swap()* within *main()*, which will generate a compilation error and the non-**template** declaration of *Abs()* within *main()*, which is legal.

13.2.1 A Note on Style

The **template** function declarator of *Max()* was written on two lines:

```
template <class T>
T Max (T a, T b)
```

rather than on a single line:

```
template <class T> T Max (T a, T b)
```

As with all style issues, the compiler couldn't care less. However, from a programmer's point of view the first version greatly increases the clarity of the function definition. This issue of style is particularly relevant to defining out-of-line **class** member functions, e.g.:

```
template <class T>
inline Point<T>& Point<T>::operator = (const Point<T>& p)
  { */ ... */ }
```

13.2.2 Templates or Macros?

You may be thinking that a **template** function is a kind of macro:

```
#define Max(a, b) ((a > b) ? a : b)
```

However, using the `#define` pre-processor directive circumvents the static type-checking mechanism of C++ and allows *`Max()`* to perform some strange comparisons. Errors generated by macros can be very difficult to detect, whereas C++ compilers are generally very good at detecting **template** errors. Macros are generally difficult to implement and require extra syntax to avoid precedence problems and to keep the pre-processor happy.

13.2.3 Template Function Arguments

To ensure that the compiler can generate the correct function code from a **template** function, at least one **template** argument type must influence the type of the function. For instance, all of the following are legal **template** function declarations:

```
template <class T> void Function1 (T) ;
template <class T> void Function2 (T, int) ;
template <class T, class C> int Function3 (C) ;
```

whereas the following are illegal:

```
template <class T> void Function3 (int) ;
template <class T> T    Function4 (int) ;
template <class T> void* operator new (size_t size) ;
```

13.2.4 Multiple Type Arguments

Template functions can also be defined with multiple type arguments, but non-type arguments in a function **template** are not allowed. Template classes, as we shall see later, can be declared with both multiple type and non-type arguments. To illustrate multiple type arguments, consider the following program:

```
// mult_arg.cpp
// illustrates multiple type arguments for template functions
#include <iostream>  // C++ I/O

// displays two objects of types T and S
template <class T, class S>
void Display (T a, S b)
```

```
  {
  cout << a << " " << b << endl ;
  }

void main ()
  {
  Display (0, 10.0) ;
  Display ("hello", 0) ;
  }
```

Display() is defined for two generic types, T and S, separated by the comma operator in the **template** parameter type list.

13.2.5 Template Functions do not Perform Implicit Casting

It was noted in Chapter 4 that C++ performs implicit casting from one C++ integral type to another:

```
int    i ;
double d ;
//...
i = d ;  // implicit cast of double to int
```

This can cause problems when passing arguments to a function. The following program illustrates calling a function *IntMax(***int**, **int***)* with two **double** arguments. However, implicit casting is not performed on **template** function arguments:

```
// dif_type.cpp
// illustrates that template functions
// do not perform implicit casting
#include <iostream>  // C++ I/O

// returns the maximum of two int's
int IntMax (int a, int b)
  {
  return a > b ? a : b ;
  }

// returns the maximum of two objects
template <class T>
T Max (T a, T b)
  {
  return a > b ? a : b ;
  }

void main ()
  {
  int    ia (1)  , ib (2)   ;
  double da (3.3), db (4.4) ;

  // IntMax(double,double)
```

```
cout << IntMax (da, db) << endl ; // o.k.

// Max()
cout << Max (ia, ib) << endl ; // o.k. : Max(int,int)
cout << Max (ia, da) << endl ; // error: Max(int,double)
cout << Max (ia, int(da)) << endl ; // o.k.: Max(int,int)
}
```

Within *main()* an attempt is made to call the **template** function *Max()* with two different types. This will result in a compilation error of the form 'no match found for *Max(**int**,**double**)*', since *Max()* is defined with both arguments to be of the same type.

13.2.6 Overloading Template Functions

Template functions can be overloaded:

```
// tmp_over.cpp
// illustrates that template functions can be overloaded
#include <iostream>  // C++ I/O

// returns the maximum of two objects
template <class T>
T Max (T a, T b)
  {
  return a > b ? a : b ;
  }

// returns the maximum of three objects
template <class T>
T Max (T a, T b, T c)
  {
  return c > (a > b ? a : b) ? c : a > b ? a : b ;
  }

void main ()
  {
  int i (1), j (2), k (3) ;
  cout << Max (i, j)    << endl ;
  cout << Max (i, j, k) << endl ;
  }
```

The above program overloads the *Max()* **template** function for two and three arguments.

In addition, **template** functions can be overloaded or *specialised* for specific versions of the **template**. The following program explicitly overloads the *Max()* function for arguments of type **int**:

```
// tmp_o1.cpp
// illustrates overloading a template function
#include <iostream>  // C++ I/O

// returns the maximum of two objects
```

```
template <class T>
T Max (T a, T b)
  {
  return a > b ? a : b ;
  }

// returns the maximum of two int's
int Max (int a, int b)
  {
  cout << "Max(int,int) called" << endl ;
  return a > b ? a : b ;
  }

void main ()
  {
  int ia    (1)  , ib (2)   ;
  double da (3.0), db (4.0) ;

  double max = Max (da, db) ; // calls Max()
  Max (ia, ib) ;              // calls overloaded Max()
  }
```

If a **template** function is explicitly overloaded it is the overloaded function that takes priority in a function call. The ability to overload a **template** function explicitly gives a programmer a greater degree of flexibility in designing **template** functions.

13.3 Template Classes

You've probably already guessed that C++ supports **template** classes. The philosophy is similar to that of **template** functions in that a **template class** declaration is a mould for a set of classes. To illustrate **template** classes, consider the `Point` **template class**:

```
// pt_tmp.cpp
// illustrates a Point template class
#include <iostream>  // C++ I/O

template <class T=double>
class Point
  {
  private:
    T x, y, z ;  // private data members
  public:
    // constructors
    Point ()
      : x (0.0), y (0.0), z (0.0) {}
    Point (T x_arg, T y_arg, T z_arg=0.0)
      : x (x_arg), y (y_arg), z (z_arg) {}
    Point (T t)
      : x (t), y(t), z (t) {}
```

```
    // copy constructor
    Point (const Point& p)
      : x (p.x), y (p.y), z (p.z) {}
    // public member functions
    T& X () { return x ; }  // non-const objects
    //...
    // overloaded operators
    Point& operator = (const Point& p) ;
    Vector3D<T> operator - (const Point& p) ;
  };

// overloaded operators:

// assignment operator =
template <class T>
inline Point<T>& Point<T>::operator = (const Point& p)
  {
  x = p.x ; y = p.y ; z = p.z ;
  return *this ;
  }

// subtraction operator -
template <class T>
inline Vector3D<T> Point<T>::operator - (const Point& p)
  {
  return Point<T> (x-p.x, y-p.y, z-p.z) ;
  }

// overloaded operators/friends:

// overloaded insertion operator <<
template <class T>
inline ostream& operator << (ostream& s, const Point<T>& p)
  {
  return s << "(" << p.X() << ", " << p.Y()
                          << ", " << p.Z() << ")" ;
  }

// overloaded extraction operator >>
template <class T>
inline istream& operator >> (istream& s, Point<T>& p)
  {
  return s >> p.X() >> p.Y() >> p.Z() ;
  }

void main ()
  {
  Point<double> p (1, 2, 3), q (p), r ; // double data members

  cout << "p: " << p << endl ;
  cout << "q: " << q << endl ;
```

```
    r = q ;

    cout << "r: " << r << endl ;

    Point<int>   s ;  //   int data members
    Point<float> t ;  // float data members

    cout << "s: " << s << endl ;
    cout << "t: " << t << endl ;

    cout << "sizeof(Point<int>)   : "
         << sizeof (Point<int>)    << endl ;
    cout << "sizeof(Point<float>) : "
         << sizeof (Point<float>)  << endl ;
    cout << "sizeof(Point<double>): "
         << sizeof (Point<double>) << endl ;
    cout << "sizeof s: " << sizeof s << endl ;
    cout << "sizeof t: " << sizeof t << endl ;
    cout << "sizeof r: " << sizeof r << endl ;

    int   i (0) ;
    float f (0.0) ;

    f = i ;  // ok: implicit conversion

    Point<int>   u ;
    Point<int>   v ;
    Point<float> w ;

    u = v ;  // ok: u & v are of same type
    //w = v ;  // error: no implicit conversion
    }
```

with output:

```
p: (1, 2, 3)
q: (1, 2, 3)
r: (1, 2, 3)
s: (0, 0, 0)
t: (0, 0, 0)
sizeof(Point<int>)   : 12
sizeof(Point<float>) : 12
sizeof(Point<double>): 24
sizeof s: 12
sizeof t: 12
sizeof r: 24
```

The declaration of **template class** `Point` takes the form:

```
template <class T=double>
class Point
```

```
{
private:
  // private data members
  T x, y, z ;
//...
};
```

and is of the general single type argument syntax:

```
template <class type=default_argument>
class ClassName
  { /* ... */ };
```

The prefix **template<class** T> indicates to the compiler that a **template class** is being declared. As with **template** functions, type is a type argument that indicates the type or **class** on which the **class** operates, and is to be specified when the **template class** is instantiated. type is placed between the set of angle brackets (< and >) and can be assigned a default type default_argument. Next, the keyword **class** precedes the name of the **template class**, ClassName.

Following a valid **template class** declaration, an instance or object of the **class** is similar to non-**template** classes except for the inclusion of the type or **class** that the **template class** operates on:

```
ClassName<type> object ;
```

where type is the type or **class** name of the object, object, that the **template class** operates on. For example:

```
Point<int>    p ;  // int data members
Point<double> q ;  // double data members
```

Point<**double**> operates just as if it had been declared as the non-**template class**:

```
class Point
  {
  private:
    double x, y, z ; // private data members
  //...
  };
```

The Point **template class** of program PT_TMP.CPP defines both inline and out-of-line constructors and member functions. Let's examine the inline constructors and member functions first:

```
template <class T=double>
class Point
  {
  //...
  public:
    Point ()
      : x (0.0), y (0.0), z (0.0) {}
```

```
    Point (T x_arg, T y_arg, T z_arg=0.0)
      : x (x_arg), y (y_arg), z (z_arg) {}
    Point (T t)
      : x (t), y(t), z (t) {}
    //...
    T& X () { return x ; }
    //...
};
```

The layout of the `Point` **`template class`** is similar to the non-**`template`** `Point` **`class`** declarations presented in Chapter 9, except that the **`double`** type has been replaced by the argument `T`. Do ensure that argument `T` is correctly used. When you are learning template classes for the first time one of the best techniques for developing a **`template class`** is to copy an existing (working) non-**`template class`**, since it can be difficult to develop a **`template class`** from scratch. The above `Point` **`template class`** was originally copied from the program THIS_PT.CPP of Chapter 12.

The copy constructor of **`template class`** `Point` is defined inline as:

```
Point (const Point& p)
  : x (p.x), y (p.y), z (p.z) {}
```

and not:

```
Point (const Point<T>& p)
  : x (p.x), y (p.y), z (p.z) {}
```

The scope of type argument `T` extends from the **`class`** declaration prefix **`template`** `<`**`class`** `T>` to the end of the declaration and within this scope. Qualification with the argument type `<T>` can be included, but is not necessary for defining inline member functions. This is not the case for **`friend`** functions (as we shall see later) because **`friend`** functions are not implicitly **`template`** functions, whereas member functions of a **`template class`** are implicitly **`template`** functions.

Program PT_TMP.CPP defines a constructor, two overloaded **`operator`** functions and two **`friend`** functions outside the template class declaration. The no-argument constructor is defined as:

```
template <class T>
inline Point<T>::Point ()
  : x (0.0), y (0.0), z (0.0) {}
```

When defining constructors and member functions outside a **`template class`** declaration the keyword **`template`** and the type arguments must precede each function definition. Thus, such functions are defined by function templates. When defining a **`template class`** constructor the **`template`** arguments are specified only the once: `Point<T>::Point()` as opposed to `Point<T>::Point<T>()`.

The `x`, `y` and `z` data members of `Point` are initialised to a floating-point zero, which is OK for **`float`** and **`double`** data types but far from ideal for the **`int`** data type and user-defined classes. An alternative `Point` data member initialisation is to rely on argument `T`'s default constructor:

```
template <class T>
inline Point<T>::Point ()
  : x (T()), y (T()), z (T()) {}
```

which will initialise the x, y and z data members to zero since C++ guarantees that the default value for integral types is set to zero.

In contrast with defining constructors outside a **template class** declaration, member functions and overloaded operator functions must include the **template** argument(s) (but not the keywords **template** or **class**) prior to the scope resolution operator (: :):

```
template <class T>
inline Point<T>& Point<T>::operator = (const Point& p)
  {
  x = p.x ; y = p.y ; z = p.z ;
  return *this ;
  }
```

noting that the return type of the function is Point<T>.

Within the function body of *main()* the three objects p, q and r are defined with **double** data members, while s and t are defined with **int** and **float**, respectively, data members:

```
Point<double> p (1, 2, 3), q (p), r ;
//...
Point<int>   s ;
Point<float> t ;
```

Each of the three data types **int**, **float** and **double** is passed as an argument inside the angle brackets (< and >) when an object is defined.

Also, the *main()* function of PT_TMP.CPP illustrates the use of the **sizeof** operator with both **template class** Point and objects of **template class** Point for **int**, **float** and **double** data members:

```
cout << "sizeof(Point<int>)    : "
     << sizeof (Point<int>)       << endl ;
//...
cout << "sizeof s: " << sizeof s << endl ;
```

The program output illustrates that the size of **template class** Point depends on the type or **class** of argument T.

Finally, the following statements in *main()* illustrate an important point concerning **template** classes:

```
//...
int   i (0) ;
float f (0.0) ;

f = i ;  // ok: implicit conversion

Point<int>   u ;
Point<int>   v ;
Point<float> w ;
```

```
u = v ;  // ok: u & v are of same type
//w = v ;  // error: no implicit conversion
```

We know that C++ implicitly performs conversions from one data type to another (e.g. **int** to **float**), but this does not mean that there is an implicit conversion from Point<**int**> to Point<**float**>. This makes sense when you consider that template **class** Point<T> could have alternatively declared a pointer to the T data member, T*, for which there is no implicit conversion from **int*** to **float***.

13.3.1 Declaring Template Classes in a Header File

At this point it would be a natural step to transfer the declaration and definition of template **class** Point to a .H header file and a .CPP implementation file, respectively, to enable programs other than PT_TMP.CPP to make use of **template class** Point. Unfortunately, unlike other non-**template** functions, a **template** function definition must accompany its declaration within the same file scope. This is because **template** functions do not have external linkage as do non-**template** functions. This is a far from ideal situation, since the generally adopted convention is not to place C++ code in header files but reserve header files for function and **class** declarations. Also, many commercial compilers will not perform compilation on a header file with a .H, .HXX filename extension, restricting compilation to .C, .CPP, .CXX etc. filename extensions.

The following header file, PT_TMP.H, thus contains both Point **template class** declaration and function definitions:

```
// pt_tmp.h

#ifndef _PT_TMP_H  // prevent multiple includes
#define _PT_TMP_H

template <class T>
class Point
  { /* ... */ };
//...
#endif // _PT_TMP_H
```

The following program #includes PT_TMP.H:

```
// tmp_tst.cpp
// ...
#include "pt_tmp.h"  // template class Point
void main ()
  {
  Point<int> p (1, 2) ;
  }
```

An alternative solution is to add **export** to the definition or to a preceding declaration. The keyword **export** makes a template declaration accessible from another translation unit.

```
// pt_tmp.h
export template <class T> class Point ;
```

13.3.2 `friend` Functions

A **`friend`** function of a **`template class`** is not implicitly a **`template`** function, and as a consequence the type argument must be specified in both the declaration and definition of the **`friend`** function. The following program is a modification of program FRI_FUNC.CPP presented in Chapter 11:

```
// fri_func.cpp
// illustrates a friend function of a template class
#include <iostream>   // C++ I/O
#include <cmath>      // atan()

template <class T>
class Circle
  {
  private:
    T radius ;
  public:
    // constructors
    Circle (T r)
      : radius (r) {}
    // friend
    friend T Area (const Circle<T>& c) ;
  };

// friend function Area() of template class Circle
template <class T>
T Area (const Circle<T>& c)
  {
  // area=pi*r^2
  T pi = 4.0 * atan (1.0) ;
  return pi * c.radius * c.radius ;
  }

void main ()
  {
  Circle<double> circle (5.0) ;

  double area = Area (circle) ;

  cout << "area of circle: " << area << endl ;
  }
```

Area() is declared a **`friend`** of **`template class`** `Circle`:

```
friend T Area (const Circle<T>& c) ;
```

and incorporates type `T` in the declaration.

13.3.3 Multiple Type and Non-Type Arguments

When we discussed **template** functions it was noted that all **template** arguments for a function **template** must be type or **class** arguments. Template classes, however, can be declared with both multiple type and non-type arguments. Non-type arguments are rarely used because the same functionality can generally be provided by a constructor. Nevertheless, they do offer a programmer more flexibility in designing **template** classes. Consider the following **template class** `Vector`, which is a modification of VECTOR.H/.CPP from Chapter 12:

```
// mtca.cpp
// multiple template class arguments
#include <iostream>   // C++ I/O
#include <iomanip>    // setprecision(),...
#include <cassert>    // assert()

template <class T, int size>
class Vector
  {
  private:
    T* array ;  // note: no n_elements data member!
  public:
    // constructor
    Vector () ;
    //...
  };

// constructor
template <class T, int size>
Vector<T,size>::Vector ()
  {
  assert (size) ;

  array = new T[size] ;
  assert (array != 0) ;
  // zero array
  memset ((void*)array, 0, (size_t)size*sizeof(T)) ;
  }

// O/P a Vector object to stream s
// which by default is set to cout
template <class T, int size>
void Vector<T,size>::Print (ostream& s)
  {
  if (size > 0)
    {
    // get current precision
    int old_precision = s.precision () ;
    s << "[" ;
    for (int i=0; i<size; i++)
      {
```

```
        s << setprecision (2)
          << setiosflags (ios::showpoint|ios::fixed)
          << array[i] ;
        i!=size-1 ? s << ", " : s << "]" ;
        }
        s << endl ;
      // reset precision and ios flags
      s.precision (old_precision) ;
      resetiosflags (ios::showpoint|ios::fixed) ;
      }
    }

  void main ()
    {
    Vector<int,5> v ;

    v.Value (0) = 1 ;
    v.Value (4) = 4 ;

    v.Display () ;

    Vector<int*,5> vp ;
    int  i (0) ;
    int* i_ptr = &i ;

    vp.Value (0) = i_ptr ;

    cout << endl ;
    cout << "lvalue i    :" << &i
         << ", rvalue i    : " << i      << endl ;
    cout << "lvalue i_ptr:" << &i_ptr
         << ", rvalue i_ptr: " << i_ptr  << endl ;
    vp.Display () ;
    }
```

with output:

```
[1, 0, 0, 0, 4]
lvalue i    : 0012FF5C, rvalue i    : 0
lvalue i_ptr: 0012FF58, rvalue i_ptr: 0012FF5C
[0012FF5C, 00000000, 00000000, 00000000, 00000000]
```

Since the size of a `Vector`, `size`, is explicitly passed as an argument, the `n_elements` data member is no longer required. The **`private`** member function *`Allocate()`* of the `Vector` **`class`** of program VECTOR.CPP (Chapter 12) is no longer required, since a single no-argument constructor suffices in the above design of the **`template class`** `Vector`.

A `Vector` object, `v`, of five elements each of type **`int`** is defined in *`main()`* as:

```
Vector<int,5> v ;
```

Note that the non-type arguments in a **template class** declaration must be constant expressions.

The *main()* function of MTCA.CPP also illustrates defining a Vector<**int***> object whose elements are pointers to type **int**:

```
Vector<int*,5> vp ;
```

13.3.4 Specialisations

Unlike **template** functions, **template** classes cannot be overloaded. As a result, the following program code is illegal:

```
template <class T>
class X
  { /* ... */ };
//...
// error: multiple declaration of template class X
template <class T, class S>
class X
  { /* ... */ };
```

However, a **template class** can have an explicit **class** declaration or complete *specialisation*. For instance:

```
template <class T>
class Vector
  { /* .. */ };
//...
template <>
class Vector<char*>
  { /* ... */ };
```

The specialised **class** declaration will be used for strings and the **template class** declaration will be used for all other types and classes. Note that the **class** name is accompanied by the specific type of **class** in angle brackets for the specialised **template class** declaration. The no argument **template**<> prefix on the specialised **class** informs the compiler that the specialisation can be declared without a **template** argument. All specialisations of a **template class** must be declared after the general **template class** and must be in the same namespace as the **template class** itself.

The following program illustrates in more detail the above explicit **class** declaration for **template class** Vector:

```
// exp_tmp.cpp
// illustrates that a template class can have
// an explicit template class declaration
#include <iostream>   // C++ I/O
#include <iomanip>    // setprecision(),...
#include <cassert>    // assert()
#include <string>     // strlen(), strcpy()

class String
```

```
  { /* ... */ };

// template class Vector
template <class T>
class Vector
  { /* ... */ };

// class Vector specialised for type char*
// (complete specialisation)
template <>
class Vector<char*>
  {
  private:
    Vector<String> array ;
  public:
    // constructor
    Vector (int n) ;
    // member functions
    int           NumberOfElements () const
      { return array.NumberOfElements() ; }
    String&       Value (int index) ;
    const String& Value (int index) const ;
    // overloaded operators
    String&       operator [] (int index) ;
    const String& operator [] (int index) const ;
  };
//...
// 1 arg. constructor
Vector<char*>::Vector (int n)
  {
  assert (n) ;
  array = Vector<String> (n) ;
  }

String& Vector<char*>::Value (int index)
  {
  assert (index >= 0 || index < array.NumberOfElements()) ;
  return array[index] ;
  }

String& Vector<char*>::operator [] (int index)
  {
  assert (index >= 0 || index < array.NumberOfElements()) ;
  return array[index] ;
  }
//...
void main ()
  {
  Vector<int> v (5, 0) ; // Vector<T>

  v.Value (0) = 1 ;
```

```
    v.Value (4) = 4 ;

    cout << v << endl ;

    Vector<char*> vs (3) ; // Vector<char*>

    vs[0] = "a string" ;
    vs[1] = "another string" ;
    vs[2] = "and another string" ;

    cout << vs << endl ;

    cout << vs[0][0] << endl ;
    }
```

with output:

```
[1, 0, 0, 0, 4]
[a string, another string, and another string]
a
```

The above `String` **`class`** is based on the `String` **`class`** declared and defined in ND_STR.CPP of Chapter 12, whereas the above `Vector` **`template class`** is based on the `Vector` **`class`** declaration and definitions in files VECTOR.H/.CPP of Chapter 12. The **`template`** `Vector` **`class`** listed above is discussed in more detail later in the chapter, when both **`template`** `Vector` and `Matrix` classes are discussed.

Although the type of the explicit **`class`** declaration for **`template class`** `Vector` is **`char`***, the underlying implementation of **`class`** `Vector<`**`char`***`>` is a `Vector` of `String` objects. This is used in preference to either a double pointer to **`char`**, **`char`****, or pointer to `String`, `String`*, simply because the `Vector` and `String` classes are available and because it greatly simplifies the implementation of class `Vector<`**`char`***`>`, particularly with respect to allocation and deallocation. However, since **`class`** `String` supports dynamic memory allocation, the elements of an object of **`class`** `Vector<`**`char`***`>` can be of varying lengths, as illustrated in the program output of EXP_TMP.CPP.

Consider the following complete specialisation of **`class`** `Vector` for objects of type **`void`***:

```
template <>
class Vector<void*>  // EXP_TMP.CPP
  {
  private:
    void** array ;
    int    n_elements ;
  public:
    Vector () ;
    void*& operator [] (int index) ;
  };
```

In order to define a `Vector` of pointers for any type of pointer requires a *partial specialisation* as follows:

```
template <class T>
class Vector<T*> : private Vector<void*>  // EXP_TMP.CPP
  {
  public:
    // constructors
    Vector ()
      : Vector<void*> () {}
    Vector (int n)
      : Vector<void*> {}
    // member functions
    T*& Value (int index)
      { return static_cast<T*&>
        (Vector<void*>::Value(index)) ; }
    T*& operator [] (int index)
      { return static_cast<Y*&>
        (Vector<void*>::operator [] (index)) ; }
   };
```

where, for example, `T` could be either `Point` or `Point*`.

13.3.5 Combined Use of Template Functions and Classes

Clearly, if both **`template`** functions and classes are correctly implemented the two can be combined. The following program revisits the *`Max()`* **`template`** function and the `Point` **`template class`**:

```
// tmp_c_f.cpp
// illustrates the use of a template class
// as an argument to a template function
#include <iostream>  // C++ I/O

template <class T>
class Point
  {
  private:
    T x, y, z ;// private data members
  public:
    //...
    //...
    bool operator > (const Point& p) ;
    //...
  };
//...
// greater than operator
template <class T>
inline bool Point<T>::operator > (const Point& p)
  {
  return x>p.x && y>p.y && z>p.z ? true : false ;
  }
//...
// template function Max()
```

```
// returns the maximum of two objects
template <class T>
T Max (T a, T b)
  {
  return a > b ? a : b ;
  }

void main ()
  {
  Point<double> p (1, 2), q (2, 3) ;

  Point<double> max = Max (p, q) ;

  cout << "Max (p, q): " << max << endl ;
  }
```

with output:

```
Max (p, q): (2, 3, 0)
```

Provided the **template class**, in this case `Point`, overloads the > operator, the *`Max()`* **template** function works fine. An overloaded `Point::operator>()` function has been added to the **template class** `Point` in the above program, which performs a comparison on each respective `x`, `y` and `z` data member of two `Point` objects. Try commenting out the `Point::operator>()` function declaration and definition in the above program. This will generate a compilation error of the form 'illegal operation' occurring at the **return** statement of the *`Max()`* function. Neglecting to provide the necessary overloaded operators for **template** classes used in conjunction with **template** functions can result in having to find some very confusing compilation errors.

13.3.6 Nested Classes

A **template class** can contain the declaration of another **class** within its own declaration. For instance, a linked list **class** (`LinkedList`), which is examined in more detail later, contains a nested `Node` **class**:

```
// nested.cpp
// illustrates nested template classes
#include <iostream>  // C++ I/O

template <class T>
class LinkedList
  {
  private:
    // nested Node class
    class Node
      {
      public:
        T     data ;
        Node* next ;
        //...
```

```
    } * first, * last ;
  public:
    LinkedList () { /*...*/ }
    //...
  };

void main ()
  {
  LinkedList<int> ll_obj ;
  }
```

The nested **class** Node is declared **private** to LinkedList<> because it is an implementation detail of LinkedList<>, and an object of LinkedList<> should not have direct access to Node's data members or member functions. Also, it makes no sense to define an object of **class** Node except in the context of a LinkedList<> object. In the present case it is quite legitimate to make Node a nested **class**, since it is a relatively simple **class**. More complex classes are generally, however, best declared separately rather than nested.

Note that the nested **class** Node encapsulates a pointer called next, also of **class** Node! This is an example of *recursive class declaration*, and is a frequently used technique in the design of linked lists.

13.3.7 The typename Keyword

In the previous subsection we saw that a **class** declaration can be nested within another **template class** declaration. If, however, a name is used within a **template class** declaration that is not explicitly declared to be a type or **class**, it is assumed that the name does not name a type unless the name is qualified by the **typename** keyword:

```
// typename.cpp
// illustrates typename
#include <iostream>  // C++ I/O

template <class T>
class A
  {
  private:
    T data ;
  public:
    A ()
      : data () {}
    class X  // local class X
      {};
    X x ;                  // ok: T::X is a type name
    Y y ;                  // error: Y is not a type name
    T::Z z ;               // error: T::Z is not a type name
    typename T::I i ;      // ok: T::I is a type name
    typename T::I* ip ;    // ok: pointer to T::I
  };

class B
```

```
  {
  public:  // B::I publicly accessible
    class I
      {};
  };

template <typename T> // typename replacing class
T Max (T a, T b)
  {
  return a > b ? a : b ;
  }

void main ()
  {
  A<B> a ;

  int x (1), y (2) ;
  cout << "max (x, y): " << Max (x, y) << endl ;
  }
```

Use of the type name `X` within **`template class`** `A` is legal because `X` is explicitly declared within `A`, whereas `Y` and `T::Z` are assumed not to be type names. The use of **`typename`** within `A` indicates that `T::I` is a type name. The above program also illustrates that the **`typename`** keyword can be used to replace the **`class`** keyword in a **`template`** declaration.

13.4 Smart Pointers

Let us develop a **`class`** called `XPtr` which operates like a C++ pointer to another **`class`** called `X`:

```
// smrt_ptr.cpp
// illustrates smart pointers
#include <iostream>  // C++ I/O

class X
  {
  private:
    int x_data ;
  public:
    X ()
      : x_data (0) {}
    int Get ()
      { return x_data ; }
  };

// X pointer class
class XPtr
  {
```

```
  private:
    X* ptr ;
  public:
    // constructor
    XPtr (X* x)
      : ptr (x) {}
    // overloaded operator
    X* operator -> () const
      { return ptr ; }
    // conversion function
    operator X* () const
      { return ptr ; }
  };

// smart pointer template class
template <class T>
class Ptr
  {
  private:
    T* ptr ;
  public:
    // constructor
    Ptr (T* p)
      : ptr (p) {}
    // overloaded operator
    T* operator -> () const
      { return ptr ; }
    // conversion function
    operator T* () const
      { return ptr ; }
  };

void main ()
  {
  X x ;

  XPtr xp = &x ;
  cout << xp->Get () << endl ;
  X* x_ptr0 = xp ;

  Ptr<X> xtp = &x ;
  cout << xtp->Get () << endl ;
  X* x_ptr1 = xtp ;
  }
```

The *main()* function illustrates that an object of **class** XPtr, xp, acts as if it is a C++ pointer to **class** X – often referred to as a *smart pointer*. This is made possible by overloading the **operator**-> *()* function for **class** XPtr, which encapsulates a **private** data member pointer to **class** X. XPtr also supports a conversion function **operator**: :X* *()*, which enables a pointer to **class** X to be defined and initialised to a XPtr object.

The smart pointer **class** XPtr is a perfect candidate to make a **template class**, since in general smart pointers are independent of the **class** of an object that they point to. Such a smart pointer **template class**, Ptr, is declared in the above program and demonstrated in *main()*.

13.5 A GlobalMemory Template class for Windows

Before discussing Vector and Matrix **template** classes let's briefly revisit the GlobalMemory **class** discussed in Chapter 12:

```
// win_tmp.cpp
// illustrates a global memory template class
// for programming in a Windows environment
//...
template <class T>
class GlobalMemory
  {
  private:
    HGLOBAL   hGlobalMemory ; // handle to memory
    void FAR* mem_address ;   // address of memory
  public:
    // constructor
    GlobalMemory ()
      : hGlobalMemory (0), mem_address (0) {} ;
    //...
  };
//...
// allocate memory
template <class T>
void FAR* GlobalMemory<T>::Allocate (UINT size, UINT
                                     mem_flags)
  {
  hGlobalMemory = GlobalAlloc (mem_flags,
                               size*sizeof(T)) ;
  //...
  return mem_address ;
  }

// re-allocate memory
template <class T>
void FAR* GlobalMemory<T>::ReAllocate (UINT size, UINT
                                       mem_flags)
  {
  HGLOBAL h = GlobalReAlloc (hGlobalMemory,
                             size*sizeof(T), mem_flags) ;
  //...
  return mem_address ;
  }
```

```
void main ()
  {
  const int SIZE = 10 ;

  GlobalMemory<int> g_mem ; // global memory object

  // allocate memory
  int* px = (int*) g_mem.Allocate (SIZE) ;
  // re-allocate memory
  px = (int*) g_mem.ReAllocate (2*SIZE) ;
  //...
  }
```

The `GlobalMemory` **`template class`** relieves a user of the **`class`** of having to incorporate the size of the data type or **`class`** to which the memory object refers:

```
int* px = (int*) g_mem.Allocate (SIZE) ;
```

instead of having to indicate the size of the data type or **`class`** explicitly in the *`Allocate()`* or *`ReAllocate()`* member function calls, as was the case for the non-**`template`** `GlobalMemory` **`class`** of Chapter 12:

```
int* px = (int*) g_mem.Allocate (SIZE*sizeof(int)) ;
```

In addition, we could have performed the casting of the far pointer, **`void`** `FAR*`, returned by the Windows API function *`GlobalLock()`* to `T*` within the *`Allocate()`* and *`ReAllocate()`* member functions:

```
template <class T>
T* GlobalMemory<T>::Allocate (UINT size, UINT mem_flags)
  {
  hGlobalMemory = GlobalAlloc (mem_flags, size*sizeof(T)) ;
  //...
  mem_address = GlobalLock (hGlobalMemory) ;
  return (T*)mem_address ;
  }
```

thus relieving a user of explicitly having to perform the casting:

```
int* px = g_mem.Allocate (SIZE) ;
```

This procedure was not adopted in WIN_TMP.CPP because one of three pointers (near, far or huge) can be used to point to a block of memory allocated by Windows, depending on the size of the block. In the case of 16 bits, when a single object is stored in global memory and is less than the 65 536 bytes of a segment, far pointers can be used. If a single object is greater than 65 536 bytes, huge pointers are required. Letting the *`Allocate()`* and *`ReAllocate()`* member functions return a far pointer to **`void`** greatly increases the generality of the functions by allowing a user of the **`class`** to perform the correct casting.

13.6 `Vector` and `Matrix` Template Classes

This section discusses `Vector` and `Matrix` **`template`** classes. These classes are modifications of the non-**`template`** `Vector` and `Matrix` classes discussed towards the end of Chapter 12 (see VECTOR.H/.CPP, MATRIX.H/.CPP and MTX_TST.CPP) The **`template`** `Vector` and `Matrix` classes are tested below in program MTMP_TST.CPP, which also makes use of **`template class`** `Point`, which is declared and defined in PT_TMP.H and is identical to the PT_TMP.H header file discussed earlier:

```
// pt_tmp.h
// template class Point

#ifndef _PT_TMP_H  // prevent multiple includes
#define _PT_TMP_H

#include <iostream>  // C++ I/O

template <class T>
class Point
  {
  private:
    T x, y, z ; // private data members
  };
//...
#endif // _PT_TMP_H
```

Template classes `Vector` and `Matrix` are declared and defined in VEC_TMP.H and MTX_TMP.H, respectively, and are both listed in full below because several modifications have been made to the non-**`template`** classes presented in Chapter 12:

```
// vec_tmp.h
// template class Vector

#ifndef _VEC_TMP_H  // prevent multiple includes
#define _VEC_TMP_H

#include <iostream>  // C++ I/O
#include <iomanip>   // setprecision(),...
#include <cassert>   // assert()

template <class T>
class Vector
  {
  private:
    T*  array ;              // data members
    int n_elements ;
    void Allocate (int n) ; // member function
  public:
    // constructors
    Vector ()
```

```
      { Allocate (3) ; }
    Vector (int n, T obj=T()) ;
    //...
  };
//...
// allocates memory for elements of Vector<T>
template <class T>
void Vector<T>::Allocate (int n)
  {
  assert (n) ;

  array = new T[n] ;
  assert (array != 0) ;
  n_elements = n ;
  for (int i=0; i<n; i++)
    array[i] = T () ;
  }

// 2 arg. constructor
template <class T>
Vector<T>::Vector (int n, T obj)
  {
  Allocate (n) ;
  for (int i=0; i<n; i++)
    array[i] = obj ;
  }
```

The template `Matrix` class is declared and defined in header file MTX_TMP.H:

```
// mtx_tmp.h
// template class Matrix

#ifndef _MTX_TMP_H  // prevent multiple includes
#define _MTX_TMP_H

#include <iostream>   // C++ I/O
#include <iomanip>    // setprecision(),...
#include <cassert>    // assert()
#include "vec_tmp.h"  // template class Vector

template <class T>
class Matrix
  {
  private:
    Vector<T>* array ;              // data members
    int        rows, cols ;
    void Allocate (int r, int c) ; // member function
  public:
    // constructors
    Matrix () { array = 0 ; rows = cols = 0 ; }
    Matrix (int r, int c, T init_obj=T()) ;
```

```
    //...
  };
//...
// allocate sufficient memory for Matrix
template <class T>
void Matrix<T>::Allocate (int r, int c)
  {
  array = new Vector<T>[r] ; // Vector array
  assert (array != 0) ;

  for (int i=0; i<r; i++)
    array[i] = Vector<T> (c) ; // Vector

  rows = r ;  // set row & column dimensions
  cols = c ;
  }
//...
```

The majority of the **template** features of **template class** `Vector` and `Matrix` are similar to those discussed with reference to the `Point` **template class**. The *`Allocate()`* private member function of `Vector` and `Matrix` allocate sufficient memory for objects of type `T` and are called by the constructors of each class.

Remember to include the argument type `T` when requesting memory with the **new** operator (**new** `Vector<T>`). Template **class** `Matrix` has a three-argument constructor. The first two arguments define the row and column dimensions of a `Matrix` object and the third argument is an object of type `T` which is used to initialise the elements of the newly allocated `Matrix` object.

Note that both `Vector` and `Matrix` only overload the `[]` and = operators. If operators such as the arithmetic operators +, - and * had been overloaded then these operators would have to be supported for each class that is used as the template argument type. Constructors of both `Vector` and `Matrix` make use of the default constructor of type `T` via `T()`. As a result a default consructor must be supplied for all template argument types.

So, let's see the above **template** classes in action:

```
// mtmp_tst.cpp
// tests the Point, Vector and Matrix template classes
#include <iostream>  // C++ I/O

#include "pt_tmp.h"     // template class Point
#include "vec_tmp.h"    // template class Vector
#include "mtx_tmp.h"    // template class Matrix

void main ()
  {
  Vector<double> v (3, 0.0) ;
  Matrix<double> m (3, 3, 0.0) ;

  Point<Point<int> > ppi ;                    // composed type
  cout << "ppi: " << ppi << endl ;

  Vector<double> v_init ;
```

```
Vector<Vector<double> > vvd (2, v_init) ; // composed type
cout << "vvd: " << vvd << endl ;

Matrix<int> m_init (2, 2, 1) ;
cout << "m_init: " << endl << m_init << endl ;
Matrix<Matrix<int> > mmi (2, 2, m_init) ; // composed type
cout << "mmi: " << std::endl << mmi << endl ;

Point<double> p1_init ;
Matrix<Point<double> > mpd (2, 2, p1_init) ;
cout << "mpd: " << endl << mpd << endl ;
}
```

The above program illustrates defining both `Vector` and `Matrix` objects with **`double`** elements. Also illustrated is the *composed type* or *nested template* feature of templates, where a **`template class`** is itself a **`template`** argument:

```
Vector<Vector<double> > vvd (2, v_init) ;
```

which defines an object `vvd` with elements of type `Vector<double>`. The two `Vector<double>` elements of object `vvd` are templates also - amazing or what?

Pay particular attention to the space between the two > operators, which is required so that the token > > is not confused with the right-shift or extraction operators. The Borland C++ (version 5.5) compiler issues a compilation warning (not an error) of the form 'use > > instead of >> for nested templates' if the token >> is used.

When designing **`template`** classes and functions, ensure that there is no inherent type or **`class`** dependency in the implementation of the **`template`**, because a user will undoubtedly subject the **`template`** to a whole range of different types, classes and contexts.

13.6.1 Associative Arrays

Associative arrays or *hashes* are vectors which allow the elements to be accessed other than by integer indexing. An associative array stores a (key, object) pair in which retreival of an object is made based on its associated key. It is assumed that only one object is held for each key, meaning that mapping from key to object is unique.

The following program implements an associative array called `AssociativeArray` and uses a `Vector` of `Pair` objects:

```
// assoc_a.cpp
// illustrates associative arrays
#include <iostream>    // C++ I/O
#include "vec_tmp.h"   // class Vector
#include "string.h"    // class String

// pair of objects of type P and Q
template <class P, class Q>
class Pair
  {
  private:
    P p ;
    Q q ;
```

```
  public:
    Pair (const P& _p=P(), const Q& _q=Q())
      : p (_p), q (_q) {}
      P& ObjectP () { return p ; }
      Q& ObjectQ () { return q ; }
   };

template <class KeyType, class ObjectType>
class AssociativeArray
  {
  private:
    Vector<Pair<KeyType,ObjectType> > array ;
  public:
    AssociativeArray ()
      : array () {}
    ObjectType& operator [] (const KeyType& key) ;
   };

template <class KeyType, class ObjectType>
ObjectType& AssociativeArray<KeyType,ObjectType>::
  operator [] (const KeyType& key)
  {
  for (int i=0; i<array.NumberOfElements(); i++)
    if (key == array[i].ObjectP())
      return array[i].ObjectQ() ;

  // not found: add new object at end of array
  Pair<KeyType,ObjectType> obj (key, 0) ;
  array.Add (obj) ;
  return array[array.NumberOfElements()-1].ObjectQ() ;
  }

void main ()
  {
  AssociativeArray<String,int> av ;
  av["john"] = 8 ;
  av["jack"] = 6 ;
  av["jill"] = 7 ;

  cout << "age of jack is: " << av["jack"] << endl ;
  }
```

Class `Pair` associates two objects `p` and `q` which can be either of the same or different types. `AssociativeArray` encapsulates a `Vector` of `Pairs` with the types of `Pair` being the key type, `KeyType`, and object tpye, `ObjectType`. The overloaded access operator `[]` first linearly traverses the elements of the `Vector array` testing `key` against the key objects of `array`. If `key` is not found then a new `Pair` object is defined, with null object, and appended to `array` and finally returned. In practice associative arrays typically use a heap data structure to encapsulate the `Pair` objects rather then the `Vector` used in the above example.

The *`main()`* function illustrates a typical use of an associative array in which the ages of people are accessed directly through their names.

13.7 Iterators

Frequently, when dealing with objects that encapsulate a collection of other objects or elements, we need a mechanism of traversing some or all of the object elements. Such an iteration process is generally implemented as an iterative block within a program. Alternatively, we can implement the iterative process as a member function of a **class** or specifically define an iterator **class** whose sole purpose is to iterate through the elements of an object. Let us first examine the design of iterator functions for **template class** `Vector`.

13.7.1 Iterator Functions

In practice, we are frequently required to perform a given operation on all or some of the elements of an array. For instance, consider truncating the **int** elements of a `Vector` object which are greater than a certain value:

```
const int MAX = BIG ;
//...
void main ()
  {
  Vector v ;
  //...
  for (int i=0; i<v.NumberOfElements(); i++)
    if (v[i] > MAX)
      v[i] = MAX ;
  //...
  }
```

This works fine, but it would be nice to hide away the **for**-loop that iterates over the object elements and make the `Vector` object responsible for performing the iteration. One approach is first to define an iterator function which performs the object element operation:

```
void IteratorFunction (int& i)
  {
  if (i > MAX)
    i = MAX ;
  }
```

and then simply to pass this function to a member function, *`ForEachElement()`* of **class** `Vector` which will iterate through the elements (however many or few there may be) of a `Vector` object:

```
void main ()
  {
  Vector v ;
  //...
  v.ForEachElement (IteratorFunction) ;
  //...
  }
```

Also, we frequently need to test all or some of the elements of an object against a given condition to find the first or last element which satisfies the condition. Similarly, we can pass a user-supplied test condition function to *FirstElement()* or *LastElement()* member functions which perform the necessary iteration:

```
//...
int TestConditionFunction (const int& i)
  {
  //...
  }
//...
void main ()
  {
  Vector v ;
  //...
  int v_first = v.FirstElement (TestConditionFunction) ;
  //...
  int v_last  = v.LastElement  (TestConditionFunction) ;
  //...
  }
```

The above form of iterator is illustrated below for **template class** Vector:

```
// v_it_f.h
// template class Vector with iterators

#ifndef _V_IT_F_H  // prevent multiple includes
#define _V_IT_F_H

#include <iostream>  // C++ I/O
#include <iomanip>   // setprecision(),...
#include <cassert>   // assert()
//...
template <class T>
class Vector
  {
  //...
  public:
    //...
    void     New (int new_n, T init_obj=T(),
                  const bool& keep_old_data=true) ;
    void     ForEachElement (void (*fp)(T&), int first,
                                              int last) ;
    int      FirstElement (bool (*fp)(const T&),
                           int first, int last) const ;
    int      LastElement  (bool (*fp)(const T&),
                           int first, int last) const ;
    //...
  };
//...
// iterate through elements of a Vector object
```

```
template <class T>
void Vector<T>::ForEachElement (void (*fp)(T&), int first,
                                int last)
  {
  assert (first < last) ;
  for (int i=first; i<=last; i++)
    fp (*(array+i)) ;
  }

// returns the index to the first element of a Vector object
// that satisfies a condition, else -1
template <class T>
int Vector<T>::FirstElement (bool (*fp)(const T&),
                             int first, int last) const
  {
  assert (first < last) ;
  assert (first >= 0 && first < n_elements-1) ;
  for (int i=first; i<=last; i++)
    if (fp (*(array+i)) != false)
      return i ;
  return -1 ;
  }

// returns the index of the first element of a Vector object
// that satisfies a condition, else -1
template <class T>
int Vector<T>::LastElement (bool (*fp)(const T&),
                            int first, int last) const
  {
  assert (first < last) ;
  assert (last > 0 && last <= n_elements-1) ;
  for (int i=last; i>=first; i--)
    if (fp (*(array+i)) != false)
      return i ;
  return -1 ;
  }
//...
#endif // _V_IT_F_H
```

The definitions of the member functions `ForEachElement()`, `FirstElement()` and `LastElement()` have been extended so that a user can specify a range of object elements in which the iterator and test condition functions will operate[2]. Let us first examine the `ForEachElement()` iterator member function definition. `ForEachElement()` is passed a pointer to a user-supplied iterator function with the following signature for type `T`:

2 The operation of functions `ForEachElement()`, `FirstElement()` and `LastElement()` is similar in principle to the functions `ForEach()`, `FirstThat()` and `LastThat()`, respectively, supported by the Borland C++ (version 5.5) `Array` and associated container classes. Refer also to the `std::for_each()` function, which is discussed in Chapter 20.

```
void IteratorFunction (T&) ;
```

ForEachElement() simply loops through the specified range of elements, passing each object element to the iterator function.

The *FirstElement()* function loops through the specified range of elements, passing each object element to the test condition function. If the test condition function returns logical-true, then *FirstElement()* returns the index of the first element in the specified range to pass the test. If no element within the specified range passes the test then a value of -1 is returned to indicate that the test was unsuccessful, and similarly for *LastElement()*.

Aside from iterators, the `Vector::New()` member function has been modified in V_IT_F.H to allow the size of a `Vector` object to be increased while maintaining the object's original elements, if desired. A program which tests the above iterators and the `Vector::New()` member function is given in ITER_TST.CPP:

```
// it_f_tst.cpp
// tests the template class Vector with iterators

#include "vec_iter.h"  // template class Vector

// iterator function
void ITFunction (int& i)
  { i = 5 ; }

// test condition function
bool TCFunction (const int& i)
  { return i == 5 ? true : false ; }

void main ()
  {
  Vector<int> v (5, 0) ;
  cout << "v: " << v << endl ;

  v.ForEachElement (ITFunction, 0, 4) ;
  cout << "v: " << v << endl ;

  Vector<int> u (5, 0) ;
  u[2] = 5 ;

  cout << "first: " << u.FirstElement
    (TCFunction, 0, u.NumberOfElements()-1) << endl ;
  cout << "last : " << u.LastElement
    (TCFunction, 0, 1) << endl ;

  u.New (10, 6) ;
  cout << "u: " << u << endl ;
  }
```

with output:

```
v: [0, 0, 0, 0, 0]
v: [5, 5, 5, 5, 5]
first: 2
last : -1
u: [0, 0, 5, 0, 0, 6, 6, 6, 6, 6]
```

13.7.2 Iterator Classes

An alternative approach to iterating through the elements of a `Vector` object is to develop a `VectorIterator` **class** which provides access to a `Vector` object's elements. An object of an iterator **class** hides the details of a `Vector` object and how its elements are accessed from a user, thus eliminating the common user error of stepping out of bounds when indexing the elements of a `Vector` object in a loop construct. The following header file declares a `VectorIterator` **class**:

```
// v_it_c.h
// template class VectorIterator

#ifndef _V_IT_C_H  // prevent multiple includes
#define _V_IT_C_H

#include <iostream>  // C++ I/O
#include <iomanip>   // setprecision(),...
#include <cassert>   // assert()

// forward declaration
template <class T> class VectorIterator ;

template <class T>
class Vector
  {
  public:
    //...
    friend class VectorIterator<T> ;
  };

template <class T>
class VectorIterator
  {
  private:
    int              c_elem ;  // current element
    const Vector<T>* vec ;     // current Vector
  public:
    // constructor
    VectorIterator (const Vector<T>& v)
      : c_elem (0), vec (&v) {}
    // member functions
    bool More () const
      { return c_elem < vec->n_elements ? true : false ; }
    T&   Current () const { return vec->array[c_elem]; }
```

```
    void Next ()     { c_elem++ ; }
    void ReStart () { c_elem = 0 ; }
  };
//...
#endif // _V_IT_C_H
```

`VectorIterator` is made a **friend** of **class** `Vector` so that the declaration of `VectorIterator` has direct access to `Vector`'s **private** data members, `array` and `n_elements`. Note the forward declaration of `VectorIterator` before the declaration of `Vector`. This is necessary because `Vector` makes reference to `VectorIterator` before it is declared.

`VectorIterator` has two **private** data members, `c_elem` and `vec`. The **int** data member `c_elem` acts as an index to an element of a `Vector` object which is currently being pointed to. The `vec` data member is a pointer to a `Vector` object which a `VectorIterator` iterator object operates on. Both `c_elem` and `vec` are initialised when an iterator object is defined via the one-argument constructor, which associates a `Vector` object with a `VectorIterator` iterator object.

Class `VectorIterator` supports four member functions. The function *`More()`* returns logical-true if more elements remain, from the current position, in a `Vector` object. *`Current()`* returns the element object at the current position, *`Next()`* advances the current position to the next element, and *`ReStart()`* returns the current position to the start of the element list.

In the implementation of `VectorIterator`, above, the `vec` data member was declared a pointer to a `Vector` object. Alternatively, `vec` can be declared a reference to `Vector`:

```
template <class T>
class VectorIterator
  {
  private:
    int               c_elem ;  // current element
    const Vector<T>& vec ;      // current Vector
  public:
    // constructor
    VectorIterator (const Vector<T>& v)
      : c_elem (0), vec (v) {}
    //...
  };
```

A program which tests `VectorIterator` is:

```
// it_c_tst.cpp
// tests the template class VectorIterator

#include "v_it_c.h"  // template classes Vector and
                     // VectorIterator

void main ()
  {
  Vector<int> v (5, 0) ;

  // iterate thro' Vector object using integer i
  for (int i=0; i<v.NumberOfElements(); i++)
```

```
    cout << v.Value (i) << " " ;

  cout << endl ;

  // iterate thro' Vector object using iterator object
  for (VectorIterator<int> vi (v); vi.More(); vi.Next())
    cout << vi.Current () << " " ;
  }
```

with output:

```
0 0 0 0 0
0 0 0 0 0
```

The *main()* function illustrates two different methods of iterating through the elements of a `Vector` object `v`. The similarity between the integer variable `i` and the iterator object `vi` is self-evident. We shall revisit iterator classes when we discuss inheritance in Chapter 15 and the STL approach to iterators in Chapter 20.

13.8 A `LinkedList` class

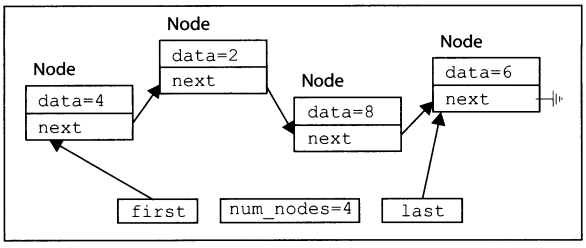
To date we have seen the use of arrays, vectors and matrices, whose size can be either fixed at compile-time or determined at run-time. The dynamic array is clearly more versatile than a compile-time array, but it still suffers from a major flaw. If the size of an array is continually changing as a result of elements being inserted or deleted from the array, then the array mechanism of storing a list of elements contiguously in memory becomes an inefficient structure because existing elements must be repositioned. An alternative structure for storing a list of elements is to use a *linked list*, which is a more responsive structure to the insertion of new elements or deletion of existing elements. A linked list contains a set of *nodes* or *links* in which each node not only stores an object but also a link (implemented as a pointer) to the next node, allowing the nodes not to be stored contiguously in memory. If the linked list maintains pointers to the first and last nodes of the list, then the entire list of nodes can be accessed and new nodes can easily be added or deleted.

The **`template class`** `LinkedList` examined below is based on the non-**`template`** `LinkedList` **`class`** presented in Adams *et al.* (1995, pp. 926–60).

First, let us examine the **`class`** declaration of `LinkedList` and then examine each of `LinkedList`'s member functions more closely:

```
// ll.h
// template classes LinkedList and LinkedListIterator

#ifndef _LL_H  // prevent multiple includes
#define _LL_H

#include <iostream>  // C++ I/O
#include <cassert>   // assert()

// swaps two objects
template <class T>
```

```
void Swap (T& a, T& b)
  {
  T temp ;

  temp = a ;
  a = b ;
  b = temp ;
  }

// forward declaration
template <class T> class LinkedListIterator ;

template <class T>
class LinkedList
  {
  private:
    int num_nodes ;
    // nested class
    class Node
      {
      public:
        T     data ;
        Node* next ;
        // constructors
        Node (const T& d=T(), Node* ptr=0)
          : data (d), next (ptr) {}
        // destructor
        ~Node () { delete next ; }
      } * first, * last ; // class Node
  public:
    enum { START=-1, END=-2 };
    // constructors
    LinkedList ()
      : num_nodes (0), first (0), last (0) {}
    LinkedList (const LinkedList& l) ;
    // destructor
    ~LinkedList ()
      { delete first ; num_nodes = 0 ;
                first = last = 0 ; }
    // member functions
    int  NumberOfNodes () const
      { return num_nodes ; }
    bool IsEmpty ()
      { return num_nodes==0 ? true : false ; }
    int  Search (const T& obj) ;
    void Insert (const T& obj, int loc) ;
    void Head   (const T& obj) ;
    void Append (const T& obj) ;
    void Delete (int loc) ;
    void Sort () ;
    void Print (ostream& s=cout) ;
```



Linked list

Fig. 13.1 A linked list consisting of four `Node` objects.

```
    // overloaded operators
    LinkedList& operator = (const LinkedList& l) ;
    T&          operator [] (int index) ;
    bool        operator == (const LinkedList& l) ;
    bool        operator != (const LinkedList& l) ;
    // friend class
    friend class LinkedListIterator<T> ;
  };
//...
// member function definitions
//...
#endif // _LL_H
```

The **class** declaration of `LinkedList` encapsulates two pointer data members, `first` and `last`, of the nested **class** `Node`, which respectively point to the first and last nodes in a linked list, and an **int** data member called `num_nodes`, which represents the total number of nodes in a linked list; an example of a linked list is shown in Fig. 13.1[3]. The **class** `Node` is a nested **class** because it is an implementation detail specific to **class** `LinkedList`. `Node` encapsulates a `data` data member of **class** or type `T` which represents the object held by a given node. The **class** `Node` is in fact a recursive **class** declaration, since it contains a `next` data member of **class** `Node` which points to a node's next node in a linked list. An object of **class** `Node` can be defined in one of thtree ways using the two-argument constructor. Both `data` and `next` can be assigned `T()` and 0, `next` can be assigned `ptr` or both `data` and `next` can be assigned `d` and `ptr`. Node's destructor deallocates the memory associated with the `next` pointer.

The copy constructor of `LinkedList` first copies the number of nodes of the list being copied. Two pointers to `Node` are then defined for traversing the list being copied and the new list. If the number of nodes being copied is zero then an empty list is simply generated. Otherwise, each node is copied by using the two-argument constructor of **class** `Node` and the **new** operator:

```
template <class T>
LinkedList<T>::LinkedList (const LinkedList& l)
```

3 This type of linked list is frequently referred to as a *double-ended* linked list, since both `first` and `last` pointers are maintained, as opposed to a *single-ended* linked list, which maintains a single `first` pointer. A double-ended linked list is not to be confused with a doubly-linked list in which each node maintains both previous and next pointers.

```
{
num_nodes = l.num_nodes ;

Node* old_node (l.first) ;  // old list
Node* new_node (0) ;        // new list

if (num_nodes == 0)  // if empty list
  first = 0 ;
else  // copy first node
  {
  new_node = new Node (old_node->data) ;  // copy old node
  assert (new_node != 0) ;
  old_node = old_node->next ;
  first = new_node ;
  }
// copy remaining nodes
for (int i=1; i<num_nodes; i++, old_node=old_node->next,
                               new_node=new_node->next)
  {
  // copy old node
  new_node->next = new Node (old_node->data) ;
  assert (new_node->next != 0) ;
  }
last = new_node ;
}
```

The *Search()* member function sequentially searches the nodes of a linked list from the start of the list for a supplied object. If the object is found in the list then its index is returned, else a value of –1 is returned, indicating failure:

```
template <class T>
int LinkedList<T>::Search (const T& obj)
  {
  Node* ptr (first) ;

  for (int i=0; i<num_nodes; i++, ptr=ptr->next)
    if (ptr->data == obj)
      return i ;
  return -1 ;
  }
```

noting that `T` requires the equality operator == to be overloaded.

An object can be inserted at the start or head of a list, the end or tail of a list or at any intermediate position. These three cases are supported by the *Insert()* member function. It is a fairly straightforward process to insert an object either at the start or the end of a list, whereas inserting an object at an intermediate position first requires the previous node to be located from the point of insertion. Then it is a simple matter of exchanging `next` pointers between the previous node and the insertion node:

```
template <class T>
void LinkedList<T>::Insert (const T& obj, int loc)
```

```
{
if (loc < -2 || loc >= num_nodes)
  {
  cout << "invalid argument in LinkedList::Insert()" ;
  exit (EXIT_FAILURE) ;
  }

Node* ptr = new Node (obj) ; // pointer to inserted node
assert (ptr != 0) ;

// insert at start of list
if (loc == 0 || loc == LinkedList<T>::START)
  {
  ptr->next = first ;
  first = ptr ;
  if (num_nodes == 0)  // empty list
    last = ptr ;
  }
// insert at end of list
else if (loc == num_nodes-1 || loc == LinkedList<T>::END)
  {
  if (num_nodes == 0)
    first = ptr ;
  else
    last->next = ptr ;
  last = ptr ;
  }
else  // insert at specified location
  {
  Node* prev_ptr (first) ;

  // locate previous node in list
  for (int j=0; j<loc-1; j++)
    prev_ptr = prev_ptr->next ;

  ptr->next = prev_ptr->next ;
  prev_ptr->next = ptr ;
  }
num_nodes++ ;  // adjust no. of nodes
}
```

With the general *Insert()* function defined, two functions, *Head()* and *Append()*, can be defined which place an object at the start and end of a list respectively:

```
template <class T>
void LinkedList<T>::Head (const T& obj)
  {
  LinkedList::Insert (obj, LinkedList<T>::START) ;
  }

template <class T>
```

```
void LinkedList<T>::Append (const T& obj)
  {
  LinkedList::Insert (obj, LinkedList<T>::END) ;
  }
```

The *Delete()* member function removes a node from a linked list at a specified position. If the node to be removed is at the start of a list then the `first` pointer points to the second node in the list. If the node to be removed is a node other than the first in a list then it is first necessary to locate the node's previous node. Then it is a simple matter of exchanging `next` pointers between the previous node and the node to be removed from the list:

```
template <class T>
void LinkedList<T>::Delete (int loc)
  {
  if (loc < -2 || loc >= num_nodes)
    {
    cout << "invalid argument in LinkedList::Delete()" ;
    exit (EXIT_FAILURE) ;
    }

  Node* ptr (0) ; // pointer to deleted node

  if (loc == LinkedList<T>::START)  // delete first node in
                                    // list
    {
    ptr = first ;
    first = first->next ;
    ptr->next = 0 ;
    }
  else  // delete at specified location
    {
    Node* prev_ptr (first) ;

    // locate previous node in list
    for (int j=1; j<loc; j++)
      prev_ptr = prev_ptr->next ;

    ptr = prev_ptr->next ;
    prev_ptr->next = ptr->next ;
    ptr->next = 0 ;

    if (loc == num_nodes-1)  // last node in list
      last = prev_ptr ;
    }
    delete ptr ;
    num_nodes-- ;  // adjust no. of nodes
  }
```

The *Sort()* member function of **template class** `LinkedList` sorts the elements of a linked list by value using the bubblesort algorithm:

```
template <class T>
void LinkedList<T>::Sort ()
  {
  for (int i=0; i<num_nodes-1; i++)
    for (int j=num_nodes-1; i<j; j--)
      if ((*this)[j] < (*this)[j-1])
        ::Swap ((*this)[j], (*this)[j-1]) ;
  }
```

Bubblesort is not an efficient sorting alogirthm and if efficiency is important then an alternative algorithm should be used, such as quicksort.

The form of the overloaded assignment operator is similar to that of the copy constructor, except that we first need to delete the existing nodes:

```
template <class T>
LinkedList<T>& LinkedList<T>::operator = (const LinkedList& l)
  {
  Node* ptr (first) ;

  // delete existing nodes
  for (int i=0; i<num_nodes; i++, first=first->next, ptr=first)
    delete ptr ;

  num_nodes = l.num_nodes ;

  Node* old_node (l.first) ;  // old list
  Node* new_node (0) ;        // new list

  if (num_nodes == 0)  // if empty list
    first = 0 ;
  else  // copy first node
    {
    new_node = new Node (old_node->data) ;  // copy old node
    assert (new_node != 0) ;
    old_node = old_node->next ;
    first = new_node ;
    }
  // copy remaining nodes
  for (int j=1; j<num_nodes; j++, old_node=old_node->next,
                                  new_node = new_node->next)
    {
    // copy old node
    new_node->next = new Node (old_node->data) ;
    assert (new_node->next != 0) ;
    }
  last = new_node ;
  return *this ;
  }
```

The overloaded random access subscript operator is defined as:

```
template <class T>
T& LinkedList<T>::operator [] (int index)
  {
  assert (index >= 0 || index < num_nodes) ;

  Node* loc (first) ;

  for (int i=0; i<index; i++)
    loc = loc->next ;
  return loc->data ;
  }
```

The overloaded equality relational operator performs a node-wise comparison of two linked lists:

```
template <class T>
bool LinkedList<T>::operator == (const LinkedList& l)
  {
  if (num_nodes != l.num_nodes)
    return false ;

  Node* ptr (first) ;       // initialise pointers to 2 lists
  Node* l_ptr (l.first) ;

  for (int i=0; i<num_nodes; i++, ptr=ptr->next,
                                  l_ptr=l_ptr->next)
    if (ptr->data != l_ptr->data)
      return false ;
  return true ;
  }
```

The overloaded inequality relational operator is implemented as a logical negation of the equality operator:

```
template <class T>
bool LinkedList<T>::operator != (const LinkedList& l)
  {
  if (!(*this == l))  // logical negation of ==
    return true ;
  return false ;
  }
```

The overloaded insertion operator simply passes the work onto the *Print()* member function which displays the object held by each node in a linked list:

```
template <class T>
ostream& operator << (ostream& s, const LinkedList<T>& l)
  {
  l.Print (s) ;
  return s ;
  }
```

A program which tests `LinkedList` is:

```
// ll_tst.cpp
// tests the LinkedList template class

#include <iostream>  // C++ I/O

#include "ll.h"      // template class LinkedList

void main ()
  {
  LinkedList<int> li0 ;
  cout << "li0: " << li0 << endl ;

  // append, head, insert:
  li0.Append (2) ;
  cout << "li0: " << li0 << ", num_nodes: "
       << li0.NumberOfNodes () << endl ;
  li0.Head (4) ;
  cout << "li0: " << li0 << ", num_nodes: "
       << li0.NumberOfNodes () << endl ;
  li0.Insert (8, 1) ;
  cout << "li0: " << li0 << ", num_nodes: "
       << li0.NumberOfNodes () << endl ;

  // assignment
  LinkedList<int> li1 ;
  li1 = li0 ;
  // copy constructor
  LinkedList<int> li2 (li1) ;

  cout << "li2: " << li2 << endl ;
  cout << "li1: " << li1 << endl ;
  cout << "li0: " << li0 << endl ;

  // delete:
  li2.Delete (2) ;
  cout << "li2.Delete(2): " << li2 << endl ;
  // search:
  cout << "li1.Search(8): " << li1.Search (8) << endl ;

  // composed type:
  LinkedList<LinkedList<int> > lli ;
  lli.Append (li0) ;
  lli.Append (li1) ;
  lli.Append (li2) ;

  cout << "lli: " << lli << endl ;

  LinkedList<int> ll_s ;
  ll_s.Append (5) ; ll_s.Append (9) ; ll_s.Append (1) ;
  ll_s.Append (3) ; ll_s.Append (8) ; ll_s.Append (2) ;
  ll_s.Append (7) ; ll_s.Append (0) ; ll_s.Append (4) ;
```

```
ll_s.Append (6) ;

ll_s.Sort () ;
cout << "sorted ll_s: " <<  ll_s << endl ;

// LinkedListIterator:

cout << "LinkedListIterator i (li1): " ;
for (LinkedListIterator<int> i(li1); i.More(); i.Next())
  cout << i.Current () << " " ;
}
```

with output:

```
li0:
li0: 2, num_nodes: 1
li0: 4 2, num_nodes: 2
li0: 4 8 2, num_nodes: 3
li2: 4 8 2
li1: 4 8 2
li0: 4 8 2
li2.Delete(2): 4 8
li1.Search(8): 1
lli: 4 8 2  4 8 2  4 8
sorted ll_s: 0 1 2 3 4 5 6 7 8 9
LinkedListIterator i (li1): 4 8 2
```

`LinkedListIterator` objects are discussed in the next section.

You may have noticed from the above function definitions that a characteristic feature of linked lists is the sequential access of the nodes in a list. A linked list object only holds pointers to the first and last nodes of the list and the total number of nodes in the list. Thus, if a node other than the first or last is to be accessed we must always start at the first node in the list and then sequentially move through the remaining nodes in the list, via the `next` pointer of each node, until we arrive at the appropriate node. This sequential accessing of nodes is the key disadvantage of linked lists, and if a given operation requires a high degree of accessing of interior nodes an array will generally be a more efficient mechanism for storing a list of objects.

There are several variations on the original linked list theme. The nodes of a linked list can be ordered. A linked list can be circular, in which the last node points to the first node of the list. Large linked lists can be broken down into an array of several smaller linked lists. Instead of an array of linked lists, a composed type can be used in which the nodes of a linked list are other linked lists. Another popular form of linked list is the *doubly linked* list, in which each node contains a pointer to its `previous` node as well as its `next` node. A new form of linked list to emerge recently is the *skip list*, which is a linked list with additional pointers that skip over intermediate nodes (Pugh, 1989).

13.8.1 A `LinkedListIterator` **class**

The previous section examined the implementation of a `LinkedList` **class**. Let's now briefly examine a `LinkedListIterator` **class**, which provides access to the elements held by a `LinkedList` object. The general outline of the `LinkedListIterator` **class** is similar to that of the `VectorIterator` iterator **class** discussed previously in that a

`LinkedListIterator` object encapsulates two pointer data members, `c_node` and `c_list`. The `c_node` data member acts as a current position pointer to a given element of a `LinkedList` object pointed to by the `c_list` data member. The `c_node` and `c_list` data members are initialised when an iterator object is defined, thus associating a given `LinkedList` object with a `LinkedListIterator` object. Like `VectorIterator`, `LinkedListIterator` also supports the four member functions *`More()`*, *`Current()`*, *`Next()`* and *`ReStart()`* for determining whether or not more nodes are in a list, returning the current node, advancing to the next node and repositioning the current position to the start of a list, respectively. In addition, `LinkedListIterator` overloads the prefix and postfix increment operators to allow a more natural manipulation of iterator objects:

```
template <class T>  // LL.H
class LinkedListIterator
  {
  private:
    LinkedList<T>::Node* c_node ;  // current node
    const LinkedList<T>* c_list ;  // current list
  public:
    // constructor
    LinkedListIterator (const LinkedList<T>& l)
      : c_node (l.first), c_list (&l) {}
    // member functions
    bool More ()    const
      { return c_node != 0 ? true : false ; }
    T    Current () const { return c_node->data ; }
    void Next ()    { c_node = c_node->next ; }
    void ReStart () { c_node = c_list->first ; }
    // overloaded operators
    bool operator ++ ()     // prefix
      { c_node=c_node->next ; return More() ; }
    bool operator ++ (int)  // postfix
      { c_node=c_node->next ; return More() ; }
  };
```

A typical use of a `LinkedListIterator` **`class`** object was demonstrated in the *`main()`* function of LL_TMP.CPP, listed in the previous subsection.

13.9 A `Polygon` `class`

As a further illustration of the `LinkedList` **`template class`**, let us examine a `Polygon` **`class`** and associated member functions which make good use of linked lists.

13.9.1 A `class` Representation of a Polygon

A polygon is a region consisting of a number (minimum three, i.e. a triangle) of vertices and edges which lie on a planar or curved surface. The edges of a polygon can also be either straight or curved. It is possible to model polygons as a collection of vertices, edges or surfaces, although in the present instance we shall restrict ourselves to defining a *closed* polygon purely in

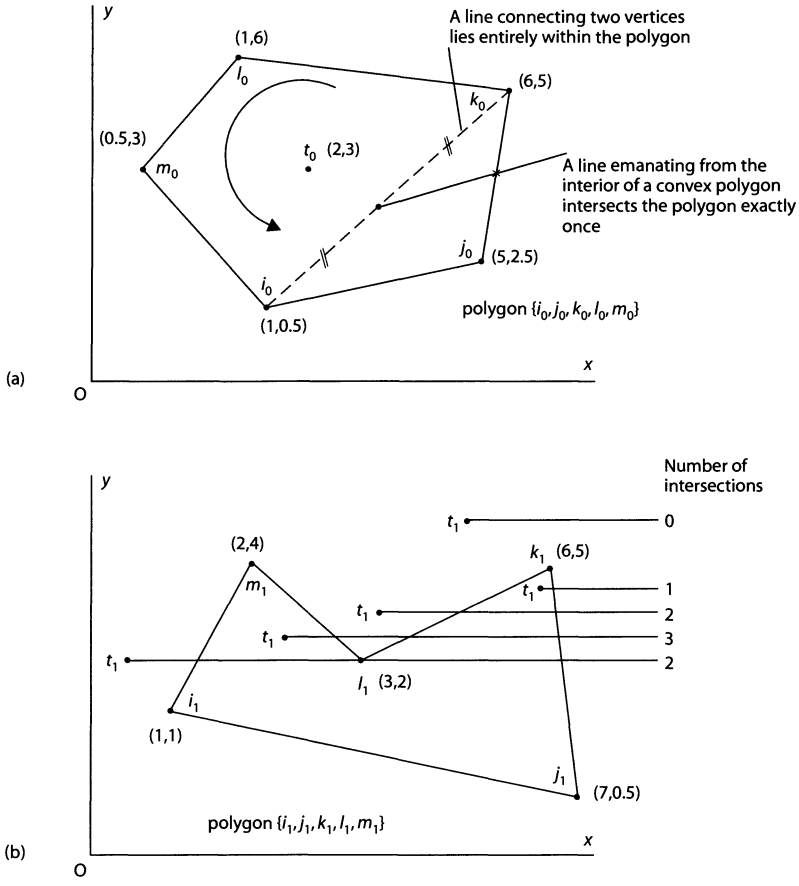


Fig. 13.2 Convex and concave polygons with vertices (a) $\{i_0, j_0, k_0, l_0, m_0\}$ and (b) $\{i_1, j_1, k_1, l_1, m_1\}$.

terms of a set of vertices existing entirely in a two-dimensional x–y plane ($z = 0$). Further, it will be assumed that the edges joining polygon vertices are straight line segments.

In general, polygons are categorised as either *convex* or *concave*; see Fig. 13.2. A convex polygon has the property that a straight line segment joining any two vertices lies entirely within the polygon. Alternatively, a convex polygon can be viewed as a polygon in which all vertices are on the left-hand side of any polygon edge when traversing the polygon in an anticlockwise manner. Apart from simply stating that a concave polygon is one that is not convex, a concave polygon has the property that one or more polygon vertices lie on the right-hand side of the current polygon edge when traversing the polygon edges in an anticlockwise manner.

The **`class`** declaration of `Polygon` can be found in POLY.H and is of the form:

```
// poly.h
// class Polygon

#ifndef _POLY_H  // prevent multiple includes
#define _POLY_H

#include <limits>      // INT_MAX
```

```
#include "pt_line.h"  // classes Point and Line
#include "ll.h"       // template class LinkedList
#include "stack.h"    // template class Stack

const double PI = 4.0 * atan (1.0) ;

class Polygon
  {
  private:
    LinkedList<Point> verts ;  // vertices
    // nested class SortPoint
    class SortPoint
      {
      //...
      }; // nested class SortPoint
  public:
    enum Order { SORTED, UNSORTED_CONVEX, UNSORTED };
    // constructor
    Polygon (const LinkedList<Point>& p_list,
             Order order=SORTED) ;
    // member functions
    int  NumberVertices () const
      { return verts.NumberNodes () ; }
    bool PointInConvexPolygon (const Point& p) ;
    bool PointInPolygon       (const Point& p) ;
    Polygon ThetaRSorted () ;
    Polygon ConvexHull   () ;
  };

#endif // _POLY_H
```

with the implementation file given in POLY.CPP. `Polygon` encapsulates a **`private`** data member called `verts`, which is an object of **`class`** `LinkedList<Point>` and thus represents the vertices of either a convex or concave polygon as a linked list of `Point` objects. The **`class`** declarations of both `Point` and `Line` can be found in PT_LINE.H:

```
// pt_line.h
// classes Point and Line

#ifndef _PT_LINE_H  // prevent multiple includes
#define _PT_LINE_H

#include <iostream>  // C++ I/O
#include <cmath>     // fabs()

const double TOLERANCE = 1e-06 ;

class Point
  {
  private:
    double x, y, z ;
```

```
  public:
    // constructors
    Point ()
      : x (0.0), y (0.0), z (0.0) {}
    Point (double x_arg, double y_arg, double z_arg=0.0)
      : x (x_arg), y (y_arg), z (z_arg) {}
    // copy constructor
    Point (const Point& p)
      : x (p.x), y (p.y), z (p.z) {}
    // member functions
    double& X () { return x ; }
    //...
    int    CAC (const Point& p1, const Point& p2) ;
    // overloaded relational operators
    bool operator == (const Point& p) ;
    bool operator != (const Point& p) ;
    bool operator <  (const Point& p) ;
  };

class Line
  {
  private:
    Point p0, p1 ;
  public:
    // constructors
    Line ()
      : p0 (), p1 () {}
    Line (const Point& p, const Point& q)
      : p0 (p), p1 (q) {}
    // copy constructor
    Line (const Line& l)
      : p0 (l.p0), p1 (l.p1) {}
    // member functions
    Point& P0 () { return p0 ; }
    Point& P1 () { return p1 ; }
    //...
    bool Intersection (const Line& l) ;
  };

#endif // _PT_LINE_H
```

with the **class** member function definitions given in PT_LINE.CPP. The `Point` and `Line` classes presented above are based on those presented in LI_CLASS.CPP of Chapter 9.

The nested **class** `SortPoint`, constructor of **class** `Polygon` and member functions will be discussed in more detail later, but it is sufficient to say at this point that the default behaviour of `Polygon`'s constructor assumes that the list of vertices is in order as we traverse the `Polygon` boundary. For instance, the linked list of points for the convex polygon of Fig. 13.2(a) supplied to the `Polygon` constructor could be in the order i_0, j_0, k_0, l_0, m_0 or i_0, m_0, l_0, k_0, j_0 if we assume that vertex i_0 is the *start* or *anchor* vertex. Clearly, the ordering of the vertices is important to ensure that the polygon edge connectivity is correct.

13.9.2 A Point Inside or Outside a Polygon

In determining whether a point lies inside or outside a polygon it is instructive first to consider the simpler case of when the polygon is convex. When a polygon is convex, as in Fig. 13.2(a), then in developing a member function which informs us whether or not a test point is inside or outside a `Polygon` object we can take advantage of the property that for a point to be inside a polygon the point must lie on the left-hand side of all polygon edges as we traverse the polygon in an anticlockwise manner. The advantage of defining convexity in this manner is that it lends itself to the *`CAC()`* and `Point::`*`CAC()`* functions defined in Chapters 6 and 9. The `Point::`*`CAC()`* member function declarator is:

```
int CAC (const Point& p1, const Point& p2) ;
```

and returns either -1, 1 or 0, depending on whether three `Point` objects are clockwise, anticlockwise or collinear. For example, if we traverse the polygon shown in Fig. 13.2(a) in an anticlockwise manner, using vertex i_0 as the start vertex, then test point t_0 is inside the polygon if the sets of points (t_0, i_0, j_0), (t_0, j_0, k_0), ... all form an anticlockwise rotation. The following `Point::`*`PointInConvexPolygon()`* member function uses this method to test whether a given point is inside or outside a `Polygon` object:

```
bool Polygon::PointInConvexPolygon (const Point& p)
  {
  // direction of test point and first two vertices
  int dir = p.CAC (verts[0], verts[1]) ;

  // remaining vertices
  for (int i=1; i<verts.NumberOfNodes(); i++)
    {
    if (i == verts.NumberOfNodes()-1) // if last vertex
      {
      if (dir != p.CAC (verts[i], verts[0]))
        return false ;
      }
    else
      {
      if (dir != p.CAC (verts[i], verts[i+1]))
        return false ;
      }
    }
  return true ;
  }
```

and returns logical-false if a point is not inside a `Polygon` object, else logical-true. The definition of *`PointInConvexPolygon()`* begins by determining the direction of rotation of the test point and the first two `Polygon` vertices in the `verts` linked list. The direction of rotation is first determined because the vertices of a `Polygon` object could have been defined in either a clockwise or an anticlockwise manner. *`PointInConvexPolygon()`* then proceeds to visit the remaining vertices of a `Polygon` object, being careful to ensure that the last polygon edge consists of the last vertex in the `verts` linked list and the start vertex.

Rather than characterising a convex polygon in terms of the positioning of polygon vertices when traversing the boundary of a polygon in a given direction, we observe that a straight line

emanating (in any direction) from a point in the interior of a convex polygon to a point outside a convex polygon intersects the boundary of the polygon exactly once; see Fig. 13.2(a). This property of convex polygons is a consequence of the so-called *Jordan curve theorem*[4], which informs us whether an arbitrary point is inside or outside either a convex or concave polygon based on the number of intersections of a straight line, emanating from the test point, with the polygon edges. If the number of intersections is odd the point is inside the polygon; else if the number of intersections is even the point is outside the polygon. Fig. 13.2(b), based on Bowyer and Woodwark (1993, Fig. 2(iii)), illustrates horizontal test lines from various points inside and outside a concave polygon. Pay particular attention to the case in which the test line from a point outside the concave polygon intersects a polygon vertex. If we ignore intersections between the test line and a polygon vertex then the number-of-intersections rule still applies. The `Polygon::PointInPolygon()` member function determines whether a point is inside or outside either a convex or concave `Polygon` object using the number-of-intersections rule and is based on the algorithm presented in Sedgewick (1988, pp. 353–5):

```
bool Polygon::PointInPolygon (const Point& p)
  {
  int count (0) ;

  // construct horizontal line from test point
  Line test_line (p, Point (INT_MAX, p.Y())), l ;

  for (int i=0; i<verts.NumberOfNodes(); i++)
    {
    // test if polygon vertex on test line
    l.P0 () = verts[i] ; l.P1 () = verts[i] ;
    if (!test_line.Intersection (l))
      {
      if (i == verts.NumberOfNodes()-1) // if last vertex
        {
        l.P0 () = verts[i] ; l.P1 () = verts[0] ;
        }
      else
        {
        l.P0 () = verts[i] ; l.P1 () = verts[i+1] ;
        }
      if (test_line.Intersection (l))
        count++ ;
      }
    }
  return (count % 2) == 1 ? true : false ;
  }
```

`PointInPolygon()` begins by defining a `count` variable to record the number of intersections between a test line, `test_line`, and each edge, `l`, of the `Polygon` object that it operates on. The test line is a line segment which extends horizontally from the test point, `p`, to a point which has a x-axis component of `INT_MAX` (=32 767 and 2 147 483 647 for 16 and 32 bits respectively) to ensure that the test line extends outside the `Polygon` object it is tested against.

4 Refer to, for example, Preparata and Shamos (1985, Section 3.3).

INT_MAX is defined in the C++ library header file LIMITS.H. *PointInPolygon()* then proceeds to test for a line intersection between each polygon edge formed by adjacent polygon vertices and the test line, ensuring that the last polygon edge is dealt with correctly. As was discussed above, if a polygon vertex falls on the test line then it is not counted as an intersection. If a line intersection occurs then the count variable is incremented and, depending on whether count is odd or even after the test line is tested for intersection with all polygon edges, *PointInPolygon()* returns a value of logical-true or logical-false respectively.

13.9.3 Convex and Concave Polygons, Sorting Vertices and Convex Hulls

Previous subsections considered the cases when all points from the original point set were sorted and formed vertices of either a convex or concave polygon. This does not have to be the case. The constructor of **class** Polygon allows the vertices (in the form of a linked list) of a polygon to be defined in one of three ways: (1) as a sorted list of vertices describing either a convex or concave polygon; (2) as an unsorted list of vertices describing a convex polygon; or (3) as a completely unsorted list of vertices, the *convex hull* of which describes a convex polygon:

```
Polygon::Polygon (const LinkedList<Point>& p_list,
                  Order order)
  {
  // test type of order
  assert (order == Polygon::SORTED ||
          order == Polygon::UNSORTED_CONVEX ||
          order == Polygon::UNSORTED) ;

  // sorted convex or concave polygon
  if (order == Polygon::SORTED)
    verts = p_list ;
  // unsorted convex polygon
  else if (order == Polygon::UNSORTED_CONVEX)
    {
    verts = p_list ;
    verts = ThetaRSorted ().verts ;
    }
  // unsorted convex or concave polygon
  else
    {
    verts = p_list ;
    verts = ConvexHull ().verts ;
  . }
  }
```

The type of ordering of the vertices is characterised by an enumerated data type called Order, which is **public**ly declared in **class** Polygon. The constructor argument order indicates the type of ordering, is set to SORTED by default and refers to a sorted list of vertices for either a convex or concave polygon. When the list of vertices is sorted it is a simple matter of assigning the linked list of vertices, p_list, to the Polygon::verts data member. If the list of vertices is unsorted but known to form a convex polygon, then again the linked list of vertices, p_list, is assigned to verts, but the Polygon::*ThetaRSorted()* member function is subsequently called. *ThetaRSorted()* sorts the vertices of a polygon according to angle

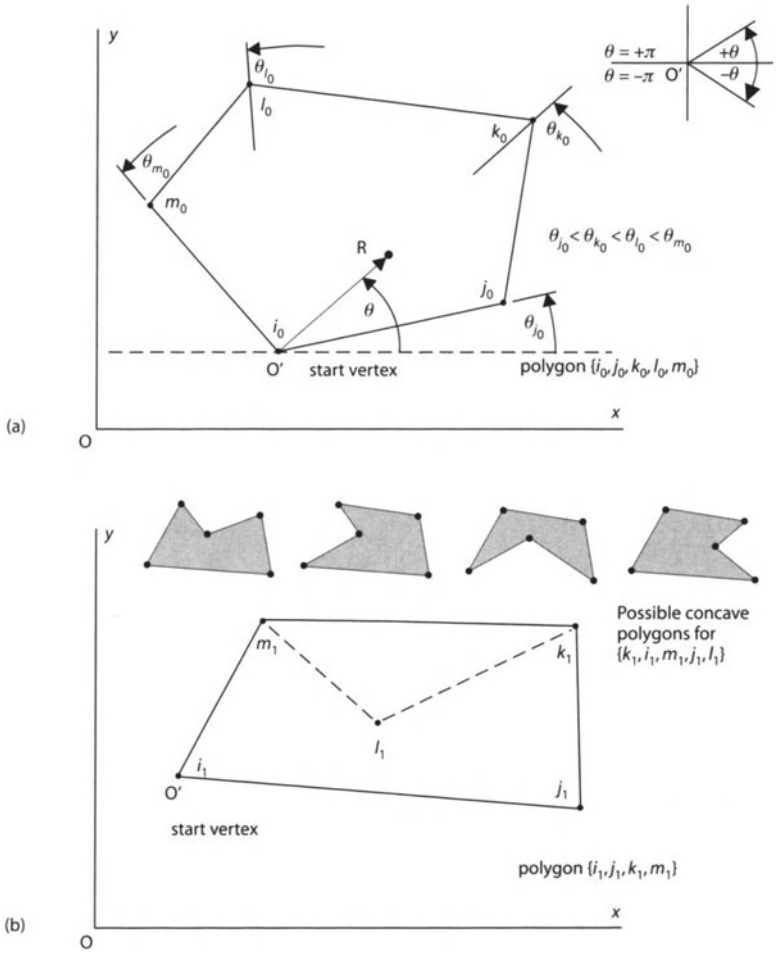


Fig. 13.3 (a) Convex polygon generated from an initial unsorted set of vertices $\{k_0, i_0, m_0, j_0, l_0\}$ to the sorted set of vertices $\{i_0, j_0, k_0, l_0, m_0\}$. (b) Convex polygon generated from an initial unsorted set of vertices $\{k_1, i_1, m_1, j_1, l_1\}$ to the convex hull set of vertices $\{i_1, j_1, k_1, m_1\}$. Also shown are the four concave polygons which could have been formed from the set of vertices $\{k_1, i_1, m_1, j_1, l_1\}$.

and radial distance from the polygon start vertex, but before we describe the exact details of `ThetaRSorted()`, let us examine how the polygon vertices are sorted and the need for the `Polygon::SortPoint` nested **`class`**.

Consider sorting the unsorted set of vertices $\{k_0, i_0, m_0, j_0, l_0\}$ for the convex polygon of Fig. 13.3(a). One method of sorting the vertices is to sort the vertices relative to an additional point known to be internal to the set of vertices. However, in practice it suffices to sort the vertices relative to the start point, which is chosen as the leftmost ordinate (i.e. minimum y–minimum x) vertex of the original set of vertices. With a start vertex established the remaining polygon vertices are sorted according to polar angle and radial distance from the start vertex. The radial distance comparison is only required if two vertices have the same polar angle. The polar angle, θ ($-\pi \leq \theta \leq +\pi$), is measured from the positive horizontal axis passing through the start vertex. Note the necessity of sorting the vertices relative to the start vertex (or local origin O′) rather than the global origin O.

It would have been straightforward to develop a function which is passed a linked list of unsorted points and returns a sorted linked list of points in accordance with the polar angle–radial distance sort discussed above. However, since the polar angle–radial distance sorting of points is unique to a polygon, it makes sense to declare a `SortPoint` **private**ly nested **class** within `Polygon` whose sole purpose is to perform a polar angle–radial distance sort of polygon vertices. The declaration of `Polygon::SortPoint` is:

```
class Polygon
  {
  private:
    LinkedList<Point> verts ;  // vertices
    // nested class SortPoint
    class SortPoint
      {
      private:
        double x, y, z ;
      public:
        SortPoint ()
          : x (0.0), y (0.0), z (0.0) {}
        SortPoint (double x_arg, double y_arg,
                                 double z_arg=0.0)
          : x (x_arg), y (y_arg), z (z_arg) {}
        SortPoint (const Point& p)
          : x (p.X()), y (p.Y()), z (p.Z()) {}
        double& X () { return x ; }
        double& Y () { return y ; }
        double& Z () { return z ; }
        double R () { return sqrt(x*x+y*y+z*z) ; }
        double Theta () ;
        bool operator <  (const SortPoint& sp) ;
        bool operator >  (const SortPoint& sp) ;
        bool operator == (const SortPoint& sp) ;
        bool operator != (const SortPoint& sp) ;
      };
  public:
    //...
};
```

The declaration of **class** `SortPoint` is similar to that of **class** `Point` in that it encapsulates `x`, `y` and `z` data members, but in addition `SortPoint` defines *`R()`*, *`Theta()`* and alternative overloaded relational operator member functions to those of `Point`. The *`R()`* member function returns the radial distance of a `SortPoint` object from the origin, whereas *`Theta()`* returns the polar angle of a `SortPoint` object and is defined as:

```
double Polygon::SortPoint::Theta ()
  {
  double theta (0.0) ;
  if (fabs(x) < TOLERANCE && y > 0.0)       // upper vertical
    theta  = 90.0 ;
  else if (fabs(x) < TOLERANCE && y < 0.0) // lower vertical
    theta = -90.0 ;
```

```
  else if (fabs(y) < TOLERANCE && x > 0.0) // right horizontal
    theta = 0.0 ;
  else if (fabs(y) < TOLERANCE && x < 0.0) // left horizontal
    theta = 180.0 ;
  else  // non-vertical/non-horizontal
    {
    if (x > 0.0 && y > 0.0)
      theta = atan (fabs(y)/fabs(x))*180/PI ;
    else if (x < 0.0 && y > 0.0)
      theta = 90.0 + atan (fabs(x)/fabs(y))*180/PI ;
    else if (x < 0.0 && y < 0.0)
      theta = -90.0 - atan (fabs(x)/fabs(y))*180/PI ;
    else  // dx>0.0 && dy<0.0
      theta = - atan (fabs(y)/fabs(x))*180/PI ;
    }
  return theta ;
  }
```

Member function *Theta()* first tests whether the `SortPoint` object lies in a horizontal or vertical plane which passes through the start vertex. *Theta()* then proceeds to determine which quadrant the `SortPoint` falls in relative to the start vertex; see Fig. 13.3(a).

The overloaded relational (<, >, ==, !=) operator member functions of **class** `SortPoint` are defined as:

```
bool Polygon::SortPoint::operator < (const SortPoint& sp)
  {
  if (Theta() < sp.Theta())
    return true ;
  else if ((fabs(Theta()-sp.Theta())<TOLERANCE) &&
           (R() < sp.R()))
    return true ;
  else
    return false ;
  }

bool Polygon::SortPoint::operator > (const SortPoint& sp)
  {
  if (Theta() > sp.Theta())
    return true ;
  else if ((fabs(Theta()-sp.Theta())<TOLERANCE) &&
           (R() > sp.R()))
    return true ;
  else
    return false ;
  }

bool Polygon::SortPoint::operator == (const SortPoint& sp)
  {
  return (fabs(x-sp.x)<TOLERANCE && fabs(y-sp.y)<TOLERANCE &&
```

```
            fabs(z-sp.z)<TOLERANCE) ? true : false ;
    }

bool Polygon::SortPoint::operator != (const SortPoint& sp)
    {
    if (!(*this == sp))  // logical negation of ==
      return true ;
    return false ;
    }
```

The overloaded < and > operators compare two `SortPoint` objects with respect to the radial distance and angle of the objects from the origin, with priority given to the angular comparison. The equality and inequality operators == and != compare two `SortPoint` objects simply by comparing the `x`, `y` and `z` data members of each object.

Note that the declaration of **class** `SortPoint` would have been greatly simplified if the C++ inheritance mechanism (refer to Chapter 15) had been used and `SortPoint` derived from `Point`.

Let's now return to the examination of the `Polygon::ThetaRSorted()` member function, which sorts the vertices of a polygon according to the angle and radial distance from the polygon start vertex:

```
Polygon Polygon::ThetaRSorted ()
    {
    // sorted list of Polygon vertices
    LinkedList<Point> sorted_poly ;

    // find pt. with min y and min x (i.e. leftmost ordinate pt.)
    Point min_pt (verts[0]) ;
    int   min_pt_index (0) ;
    for (int i=1; i<verts.NumberOfNodes(); i++)
      if ((verts[i].Y() <  min_pt.Y() &&
           verts[i].X() <= min_pt.X()) ||
          (verts[i].Y() <= min_pt.Y() &&
           verts[i].X() <  min_pt.X()))
        {
        min_pt = verts[i] ;
        min_pt_index = i ;
        }

    // use min_pt as 'local' origin point and
    // first point in list of sorted vertices
    sorted_poly.Append (min_pt) ;

    // translate points to 'local' point and add to
    // separate linked list excluding 'local' origin point
    LinkedList<SortPoint> sp ;
    for (int j=0; j<verts.NumberOfNodes(); j++)
      if (j != min_pt_index)
        sp.Append (SortPoint (
                   Point (verts[j].X()-min_pt.X(),
```

```
                        verts[j].Y()-min_pt.Y())) ;

  // sort translated points
  sp.Sort () ;

  // add sorted points to polygon vertices,
  // remembering to translate back
  for (int k=0; k<sp.NumberOfNodes(); k++)
    sorted_poly.Append (Point (sp[k].X()+min_pt.X(),
                        sp[k].Y()+min_pt.Y())) ;

  return Polygon (sorted_poly) ;
  }
```

The object `sorted_poly` is of **class** `LinkedList<Point>` and encapsulates a linked list of the θ–R sorted vertices. Next, the polygon start vertex (leftmost ordinate point) is found by performing a linear search through the entire list of original polygon vertices, `verts`. Also, the position of the start vertex within the `verts` linked list is recorded. The start vertex is then placed at the head of the `sorted_poly` object. Next, an object `sp` of **class** `LinkedList<SortPoint>` is defined, to which all of the polygon vertices (excluding the start vertex) are appended and then sorted. Note that a `Point` vertex is first translated to the start vertex and then cast to a `SortPoint` vertex. Since object `sp` is of **class** `LinkedList<SortPoint>`, the `LinkedList<T>::Sort()` member function will call the `SortPoint::operator<()` overloaded member function. Finally, the linked list of sorted vertices is appended to `sorted_poly` and a `Polygon` object returned.

Consider now the problem of defining a `Polygon` object as the convex hull of a set of unsorted arbitrary points. Figure 13.3(b) illustrates the convex polygon $\{i_1, j_1, k_1, m_1\}$ of the initially unsorted set of points $\{k_1, i_1, m_1, j_1, l_1\}$. So how do we find computationally the convex hull of a set of arbitrary points in a two-dimensional plane? At first glance the problem appears fairly straightforward, but in fact it is deceptively difficult. Several established algorithms exist, such as *gift-wrapping* or *Jarvis's march*, *Graham's scan*, and the *quickhull* or *divide and conquer* algorithm. Incidentally, an excellent treatment of convex hulls for both two and three dimensions can be found in O'Rourke (1994). The algorithm that we will examine is Graham's scan (Graham, 1972), because it is a reasonably efficient algorithm and makes good use of the previously developed `Polygon::ThetaRSorted()` and `Point::CAC()` member functions.

Suppose we are given a θ–R sorted set of points relative to a start point, as discussed above. With reference to Fig. 13.4, the Graham's scan algorithm commences at the start point by traversing the entire point set in an anticlockwise manner, examining triples of consecutive points and testing whether each triple of points forms an anticlockwise or clockwise rotation until we return to the start point. If points $p_0p_1p_2$ form an anticlockwise rotation then p_1 is a point on the convex hull, else if $p_0p_1p_2$ form a clockwise rotation then p_1 is not a hull point. `Polygon::ConvexHull()` implements the Graham's scan algorithm:

```
Polygon Polygon::ConvexHull ()
  {
  // sort list and convex hull of polygon vertices
  LinkedList<Point> sorted_poly = ThetaRSorted().verts ;

  // convex hull vertices (local origin and
  // next point are hull vertices)
  Stack<Point> hull_vertices ;
```

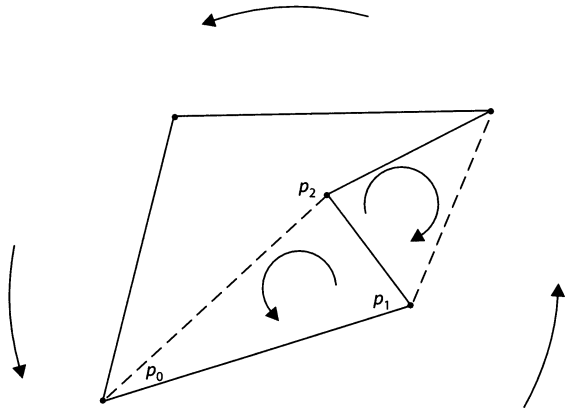


Fig. 13.4 The Graham's scan algorithm for determining the convex hull of a set of points. Point p_1 is not a hull point if points p_0, p_1 and p_2 form a clockwise rotation.

```
hull_vertices.Push (sorted_poly[0]) ;
hull_vertices.Push (sorted_poly[1]) ;

int dir ;
for (int i=2; i<sorted_poly.NumberOfNodes(); i++)
  {
  dir = sorted_poly[i].CAC(hull_vertices.NextToTop(),
                           hull_vertices.Top()) ;
  // push points to stack if anticlockwise
  // or collinear orientation, else pop stack
  while (dir == -1)
    {
    hull_vertices.Pop () ;
    dir = sorted_poly[i].CAC(hull_vertices.NextToTop(),
                             hull_vertices.Top()) ;
    }
  hull_vertices.Push (sorted_poly[i]) ;
  }

// convex hull polygon (move Points from stack to list)
LinkedList<Point> hull_poly ;
while (!hull_vertices.IsEmpty())
  {
  hull_poly.Append(hull_vertices.Top()) ;
  hull_vertices.Pop () ;
  }
return Polygon (hull_poly) ;
}
```

First, a `LinkedList<Point>` object, `sorted_poly`, is defined which encapsulates a θ–R sorted linked list of the vertices of the `Polygon` object operated on by *`ConvexHull()`*. The object `hull_poly` is the convex hull of the polygon vertices. By definition, the start vertex and the next sorted point must be hull vertices and are therefore appended to `hull_poly` before the remaining vertices are examined in triples. As discussed above, if an anticlockwise ro-

tation is described by a triple vertex set, then the intermediate vertex of the triple vertex set is a hull vertex and is therefore appended to the linked list of hull vertices, `hull_poly`. Note the use of class `Stack` which is ideally suited for either popping or pushing a point onto the stack depending on whether the point forms either a clockwise or anticlockwise/collinear rotation with the two previously established hull points. Class `Stack` will be discussed in Chapter 15 when we discuss inheritance.

Since *`ConvexHull()`* returns a `Polygon` object, `Polygon`'s constructor uses *`ConvexHull()`* to define a `Polygon` object which is the convex hull of the `Point` linked list `p_list` if the `Polygon` constructor argument `order` is set to `UNSORTED`:

```
Polygon::Polygon (const LinkedList<Point>& p_list, Order order)
  {
  //...
  // sorted convex or concave polygon
  if (order == Polygon::SORTED)
  //...
  // unsorted convex or concave polygon
  else
    {
    verts = p_list ;
    verts = ConvexHull ().verts ;
    }
  }
```

Why form a convex hull polygon from an unsorted list of vertices? Simply because a user could supply a whole variety of point sets defining a `Polygon` object, and the `Polygon` constructor has no way of determining exactly which concave or convex polygon the user intended. Consequently, the `Polygon` constructor constructs the convex hull of an unsorted list of points. Figure 13.3(b) illustrates the four possible combinations of concave polygons from the point set $\{k_1, i_1, m_1, j_1, l_1\}$.

The following program tests **`class`** `Polygon`:

```
// poly_tst.cpp
// tests class Polygon

#include <iostream>    // C++ I/O

#include "pt_line.h"   // classes Point and Line
#include "poly.h"      // class Polygon

void main ()
  {
  // define points of convex polygon and test point
  Point i0 (1.0, 0.5), j0 (5.0, 2.5), k0 (6.0, 5.0),
          l0 (1.0, 6.0), m0 (0.5, 3.0) ;
  Point test_point0 (2.0, 3.0) ;
  // define points of concave polygon and test point
  Point i1 (1.0, 1.0), j1 (7.0, 0.5), k1 (6.0, 5.0),
          l1 (3.0, 2.0), m1 (2.0, 4.0) ;
  Point test_point1 (3.0, 3.0) ;
```

```
// sorted convex polygon:

LinkedList<Point> ll_p0 ;
ll_p0.Append (i0) ; ll_p0.Append (j0) ; ll_p0.Append (k0) ;
ll_p0.Append (l0) ; ll_p0.Append (m0) ;

Polygon convex_poly (ll_p0) ;
cout << "sorted convex polygon:" << endl
     << convex_poly << endl ;

// test if test point is inside or outside convex polygon
if (convex_poly.PointInConvexPolygon (test_point0))
  cout << "point " << test_point0
       << " in convex polygon" << endl ;
else
  cout << "point " << test_point0
       << " outside convex polygon " << endl ;

// sorted concave polygon:

LinkedList<Point> ll_p1 ;
ll_p1.Append (i1) ; ll_p1.Append (j1) ; ll_p1.Append (k1) ;
ll_p1.Append (l1) ; ll_p1.Append (m1) ;

Polygon concave_poly (ll_p1) ;
cout << "sorted concave polygon:" << endl
     << concave_poly << endl ;

// test if test point is inside or outside
// convex polygon
if (concave_poly.PointInPolygon (test_point1))
  cout << "point " << test_point1
       << " in concave polygon" << endl ;
else
  cout << "point " << test_point1
       << " outside concave polygon " << endl ;

// unsorted convex polygon:

LinkedList<Point> ll_p2 ;
ll_p2.Append (k0) ; ll_p2.Append (i0) ; ll_p2.Append (m0) ;
ll_p2.Append (j0) ; ll_p2.Append (l0) ;

Polygon un_convex_poly (ll_p2, Polygon::UNSORTED_CONVEX) ;
cout << "unsorted convex polygon:" << endl
     << un_convex_poly << endl ;

// unsorted concave polygon:

LinkedList<Point> ll_p3 ;
ll_p3.Append (k1) ; ll_p3.Append (i1) ; ll_p3.Append (m1) ;
```

```
    ll_p3.Append (j1) ; ll_p3.Append (l1) ;

    Polygon un_concave_poly (ll_p3, Polygon::UNSORTED) ;
    cout << "unsorted concave polygon:" << endl
         << un_concave_poly << endl ;
    }
```

with output:

```
sorted convex polygon:
(1, 0.5, 0) (5, 2.5, 0) (6, 5, 0) (1, 6, 0) (0.5, 3, 0)
point (2, 3, 0) in convex polygon
sorted concave polygon:
(1, 1, 0) (7, 0.5, 0) (6, 5, 0) (3, 2, 0) (2, 4, 0)
unsorted convex polygon:
(1, 0.5, 0) (5, 2.5, 0) (6, 5, 0) (1, 6, 0) (0.5, 3, 0)
unsorted concave polygon:
(2, 4, 0) (6, 5, 0) (7, 0.5, 0) (1, 1, 0)
```

The above `Point` and `Polygon` objects correspond exactly to those shown in Figs. 13.2 and 13.3.

13.10 Summary

This chapter has illustrated **`template`** functions and classes. By defining a **`template`** function or **`class`** you define a mould for a family of related overloaded functions or classes by letting the data types or objects that the function or **`class`** operates on be arguments.

Template functions and classes are very powerful because they allow a programmer to design a generic function or **`class`** that can operate on a variety of different types and classes without requiring a separate implementation for each function or type of **`class`**. The use of templates is ideally suited to functions and classes that have the same logic and perform similar operations irrespective of the type or **`class`** of the data or objects that they operate on.

A **`template`** or generic function can be thought of as a function that has the ability to overload itself an infinite number of times according to the objects that it operates on. Template functions can have multiple type arguments but not non-type arguments. A **`template class`** can have both multiple type and non-type arguments. A **`template class`** can itself be used as a **`template`** argument, thus defining a nested **`template`** or a composed type.

Beware of inserting type, **`class`** or context dependencies into a **`template`** function or **`class`**. A developer of a **`template`** function or **`class`** generally has no knowledge of the variety of types, classes and applications to which the **`template`** will be applied. If the **`template`** function or **`class`** is not completely generic, compilation errors will result which can be very difficult to trace.

This chapter has examined several popular **`template`** data structures, such as `Vector`, `Matrix`, `LinkedList` and associated iterator **`class`** `LinkedListIterator`. In addition to linked list, vector and matrix data structures, other popular **`template`** data structures are the `Stack` and `Queue`, `Trees` (`BinarySearchTree`, `AVL` etc.), `Polynomial`, `Graph` and `Hash` (Budd, 1994; Ford and Topp, 1996). We shall examine `Stack` and `Queue` **`template`** classes in Chapter 15 when we discuss templates in connection with inheritance.

In conclusion, it is worth noting that the ANSI C++ standard adds the Standard Template Library (STL) container template classes `vector`, `list`, `deque`, `set`, `map`, `stack`, `queue`, `valarray` and many more. All of these classes are examined in detail in chapter 20.

Exercises

13.1 Overload the *Max()* **template** function specifically for type **char***:

```
template <class T>
T Max (T a, T b)
  {
  return a > b ? a : b ;
  }
```

13.2 For the following `Complex` **class**, define an *Abs()* **friend** function which returns the absolute value of a complex number:

```
template <class T>
class Complex
  {
  private:
    T re, im ;
  public:
    Complex ()
      : re (0.0), im (0.0) {}
    Complex (T real, T imag)
      : re (real), im (imag) {}
  };
```

13.3 Implement simple `Polygon` classes which encapsulate a polygon's `Point` vertices as a pointer to `Point`, vector and linked list using **templates** throughout. Consider the advantages and disadvantages of each implementation.

13.4 Change the `LinkedList` **class** presented in this chapter from a double-ended to a single-ended linked list so that `LinkedList` encapsulates only a `first` pointer.

13.5 The `VectorIterator` iterator **class** for `Vector` objects declared in V_IT_C.H is of the form:

```
template <class T>
class VectorIterator
  {
  private:
    int             c_elem ;  // current element
    const Vector<T>* vec ;    // current Vector
  //...
  };
```

It was mentioned in the text that the `vec` data member could equally be declared a reference instead of a pointer. Modify `VectorIterator` so that `vec` is now a reference to `Vector`.

13.6 Implement a `Polynomial` **class** which models a polynomial such as:

$$p(x) = 7x^4 + 6x^2 + 4x + 1$$

To assist in the implementation of `Polynomial`, declare a **`class`** called `Term` which encapsulates coefficient and exponent data members. Encapsulate a linked list of `Terms` within `Polynomial`. Use templates throughout.

For further details of polynomial representations in C++ refer to Budd (1994), Wilt (1994) and Horowitz *et al.* (1995).

13.7 Develop an undirected graph **`class`**, `Graph`, which encapsulates lists of vertices and undirected edges. Provide a default constructor and a two-argument constructor which allows a graph object to be defined in terms of graph vertices and edges. Provide operations for adding and deleting either a vertex or an edge from the graph. Use **`template`** classes throughout.

For an excellent discussion of graph data structures in C++ refer to Horowitz *et al.* (1995), Weiss (1996) and Ford and Topp (1996).

Chapter 14

Exception Handling

We must keep looking
Bauby (1998)

Errors are the burden of all programming. Run-time errors are particularly troublesome because of their associated difficulty in tracing. C++ supports a language-based mechanism called exception handling for handling run-time errors. When an abnormal situation arises at run-time, a program should terminate. If, however, before program termination occurs an exception or error is thrown to a section of the program which deals with exceptions, information can be gathered to assist in diagnosing the source of the problem which resulted in program termination.

The C++ exception handling mechanism is based on three keywords: try, catch and throw. If an exception occurs within a block of program statements that are tested for errors then the exception is thrown and caught by a program's exception handler. Functions and class member functions can have an associated exception specification which indicates a function's exception-handling capabilities.

C++ supports various exception handling functions such as abort(), exit(), terminate() and unexpected() and the standard library exception class hierarchy and the message class xmsg. Alternatively, user-defined exception handling classes can be declared, and we will examine Index and Range classes, which are useful in the implementation of Vector and Matrix classes. The chapter finishes by taking a look at exception handling and the new operator. It will be shown that by default the new operator throws a bad_alloc exception in the case of unsuccessful memory allocations.

14.1 `try`, `catch` and `throw`

Exception handling in C++ revolves around three keywords: **`try`**, **`catch`** and **`throw`**. A block of program code in which an exception may occur is prefixed by the **`try`** keyword:

```
try
   {
   // block of statements monitored for exceptions
   //...
   }
```


G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

The block of statements within the opening and closing braces of the **try** statement is monitored for exceptions. If an exception occurs within this block, program flow is interrupted and the exception is thrown, using the **throw** keyword:

```
throw exception ; // throw point
```

and caught, using the **catch** keyword, to a part of the program which processes exceptions:

```
catch (T obj)
  {
  // process exception
  //...
  }
```

The scope of `obj` is local to the **catch** statement. The section of a program which deals with handling exceptions is frequently referred to as the *exception handler.*

If an exception occurs and an exception is thrown, the program searches for an associated exception handler. If a handler is found, program control is transferred to the handler. If no handler is found, the program will be terminated by calling the *terminate()* function; see later. Although several variations exist, the above discussion outlines the general procedure of exception handling in C++.

Let's see **try**, **catch** and **throw** in action:

```
// tct.cpp
// introduces the try, catch and throw keywords
#include <iostream>  // C++ I/O

void main ()
  {
  int number (0), i ; // note: definition of i!

  cout << "enter a positive integer: " ;
  cin  >> number ;

  // code may throw an exception
  try
    {
    if (number < 0)
      throw number ;  // throw exception to handler
    else
      cout << "number entered: " << number << endl ;
    }
  // process exception
  catch (int i)
    {
    cout << "negative number entered!" << endl ;
    }
  }
```

with some user interaction:

```
enter a positive integer: -7
negative number entered!
```

Note that the variable `i`, defined between the parentheses of the **`catch`** statement, is local to the **`catch`** statement block. This program prompts a user to enter a positive integer. If a positive integer is entered by the user the program executes normally. If the user enters a negative integer an exception is thrown from within the **`try`** statement and caught by the **`catch`** statement. The **`try`** statement block encompasses an **`if-else`** statement which tests whether the number entered is greater than zero or not. If the number entered is less than zero an exception is thrown by the unary **`throw`** expression:

```
throw number ;
```

The exception will be caught by its corresponding **`catch`** statement (assuming one exists). The type of the exception, in this case integer, determines which **`catch`** statement is used for processing the exception. The **`catch`** statement can only be used directly after a **`try`** statement or after another **`catch`** statement exception handler. Thus the following will generate a compilation error:

```
try
   {
   //...
   }
int j (1) ;
//...
catch (int i)
   {
   //...
   }
```

There can be any number of **`catch`** statements following a **`try`** statement:

```
try
   {
   //...
   }
catch (T1 obj)
   {
   //...
   }
catch (T2 obj)
   {
   //...
   }

//...

catch (Tn obj)
   {
   //...
   }
```

If an exception is thrown to which there is no corresponding **catch** statement, the program will abnormally terminate by calling the *terminate()* function. To illustrate abnormal termination, let us modify the program TCT.CPP as follows:

```
// term.cpp
// illustrates abnormal program termination
// by calling the terminate() function
#include <iostream>  // C++ I/O

void main ()
  {
  int number (0) ;

  cout << "enter a positive integer: " ;
  cin  >> number ;

  // code may throw an exception
  try
    {
    if (number < 0)
      throw number ;  // throw exception to handler
    else
      cout << "number entered: " << number << endl ;
    }
  // process exception
  catch (double d)
    {
    cout << "negative number entered!" << endl ;
    }
  }
```

with some user interaction:

```
enter a positive integer: -7
program aborted
```

If you are programming in a Windows (generic sense) environment you may alternatively see a dialog box displayed with the message 'Program Aborted'.

The key difference between programs TCT.CPP and TERM.CPP is that the **int** type in the **catch** statement has been changed to type **double**. When the program executes and a user enters a negative integer this exception will not be caught by the **catch** (**double** d) statement block. Each **catch** statement exception handler will only evaluate an exception that matches, or can be converted to, the data type or **class** specified in the **catch** statement argument list. If no match is found in the set of **catch** statements the *terminate()* function is called. In general, a **throw** expression of type T is a match for an exception handler of type T, **const** T, T& and **const** T&. Also, the same match sequence applies to a **throw** expression that is a pointer to a type that can be converted to the handler type by standard pointer conversion. This is illustrated in the following program:

```
// types.cpp
// illustrates different handler types
```

```
#include <iostream>  // C++ I/O

void main ()
  {
  int number (0) ;
  //int* pi = new int (0) ;

  cout << "enter a positive integer: " ;
  cin  >> number ;
  //cin >> *pi ;

  // code may throw an exception
  try
    {
    if (number < 0)
    //if (*pi < 0)
      throw number ;  // throw exception to handler
      //throw pi ;
    else
      cout << "number entered: " << number << endl ;
      //cout << "number entered: " << *pi << endl ;
    }
  // process exception
  //catch (void* pv)
  //catch (int i)
  //catch (const int i)
  //catch (int& i)
  catch (const int& i)
    {
    cout << "negative number entered!" << endl ;
    }
  }
```

Incidentally, if your program contains multiple **catch** statements that clash a compilation error will be generated:

```
try
  {
  //...
  }
catch (int i)
  {
  //...
  }
catch (const int& i)  // error: duplicate exception
                      // handler
  {
  //...
  }
```

14.2 Multiple `catch` Statements

The previous section illustrated that multiple **`catch`** statements are allowed provided they do not duplicate one another:

```
// mult.cpp
// illustrates multiple catch statements
#include <iostream>  // C++ I/O

char NEG_NO[25] = "negative number entered!" ;

void main ()
  {
  int number (0) ;

  cout << "enter a positive integer: " ;
  cin  >> number ;

  // code throws an exception
  try
    {
    if (number < 0)
      throw NEG_NO ;  // throw char* exception to handler
    else
      throw number ;  // throw int exception to handler
    }
  // process exceptions
  catch (int i)
    {
    cout << "number entered: " << i << endl ;
    }
  catch (char* string)
    {
    cout << string << endl ;
    }
  }
```

If a user enters a positive integer then the type **`int`** expression is thrown, else a **`char`*** expression is thrown. Each **`catch`** statement deals with a specific type of **`throw`** expression. Note that for each **`catch`** statement either an integer or a string is passed to the **`catch`** statement block as a temporary variable or object. The operand of the **`throw`** expression is identical in principle to an argument in a function call or the operand of a **`return`** statement. If the argument to the **`catch`** statement is not used it can be neglected:

```
//...
catch (int) // note: no variable
  {
  cout << "negative number entered!" << endl ;
  }
```

14.3 Handling all Exceptions

C++ supports a certain kind of **catch** statement which catches all exceptions regardless of type or **class**. This **catch** statement uses an ellipsis (. . .) in place of an exception declaration and is of the general syntax:

```
catch (...)  // catch all exceptions
  {
  //...
  }
```

This catch-all exception is illustrated in the following program:

```
// catch_al.cpp
// illustrates the catch-all statement
#include <iostream>  // C++ I/O

char NEG_NO[25] = "negative number entered!" ;

void main ()
  {
  int number (0) ;

  cout << "enter a positive integer: " ;
  cin  >> number ;

  // code throws an exception
  try
    {
    if (number < 0)
      throw NEG_NO ;  // throw char* exception to handler
    else
      throw number ;  // throw int exception to handler
    }
  // process all exceptions
  catch (...)
    {
    cout << "I catch all exceptions" ;
    }
  }
```

The catch-all statement is particularly useful for handling exceptions that are not explicitly dealt with in a set of **catch** statements. The catch-all statement is akin to the **default** case in a **switch** statement. The following program illustrates the combined use of an **int catch** statement and a catch-all statement:

```
// int_all.cpp
// illustrates both catch-int and catch-all statements
#include <iostream>  // C++ I/O
```

```
void main ()
  {
  int    number (0) ;
  double d (0.0) ;

  cout << "enter a positive integer: " ;
  cin  >> number ;

  // code throws an exception
  try
    {
    if (number < 0)
      // throw double exception to handler
      throw d ;
    else
      // throw int exception to handler
      throw number ;
    }
  // process exceptions
  catch (int i) // int exceptions
    {
    cout << "number entered: " << i << endl ;
    }
  catch (...)   // all other exceptions
    {
    cout << "negative number entered!" ;
    }
  }
```

Note that the **catch**(. . .) handler must be placed after all other handlers for a given **try** block. Thus, the following is illegal:

```
//...
catch (...)   // all other exceptions
  {
  cout << "negative number entered!" ;
  }
catch (int i) // int exceptions
  {
  cout << "number entered: " << i << endl ;
  }
```

14.4 Functions and Exception Handling

Up to now we have seen the use of **try**, **catch** and **throw** restricted to a single function (namely the *main()* function), and all exceptions have been thrown from within the **try** statement block. Let's now examine throwing an exception from a function other than *main()* that is outside a **try** block:

```
// out_try.cpp
// illustrates throwing an exception from outside a
// try block
#include <iostream>  // C++ I/O

// throws an int exception
void Function (int i)
  {
  throw i ;
  }

void main ()
  {
  try
    {
    Function (0) ;
    Function (1) ;
    }
  // process int exceptions
  catch (int i)
    {
    cout << "caught: " << i << endl ;
    }
  }
```

with output:

```
caught: 0
```

Note that only one exception is thrown although there are two calls to *Function()*.

The **int** type exception thrown from *Function()* is outside the **try** block within *main()*. This is possible because the **throw** expression is in the body of a function that is called from within the **try** block.

The **try-catch-throw** setup can be transferred entirely to a function other than *main()*:

```
// tct_func.cpp
// illustrates the try-catch-throw
// setup from within a function
#include <iostream>  // C++ I/O

// try-catch-throw within function
void Function (int i)
  {
  try
    {
    throw i ;
    }
  catch (int i)
    {
    cout << "caught number: " << i << endl ;
```

```
    }
  }

void main ()
  {
  Function (0) ;
  Function (1) ;
  }
```

with output:

```
caught number: 0
caught number: 1
```

The output illustrates that each time *Function()* is called the exception handling statements of the function are reset.

14.4.1 Exception Specification

C++ allows a function declaration to specify exactly what set of exceptions the function can **throw** via an *exception specification*. The exception specification is used as a suffix to a function declarator and is of the general syntax:

```
return_type FunctionName (argument_list) throw (type_list)
  { /* ... */ }
```

The exception specification is not part of a function's type. Both `argument_list` and `type_list` are optional, and an empty exception specification (i.e. **throw** *()*) indicates that a function cannot **throw** any exception. A function with no exception specification is capable of throwing any exception and is the default function declarator. Each of the different types or classes that can be thrown by a function is separated by the comma operator in the `type_list` argument. A function that attempts to **throw** an exception that is not listed in its exception specification will result in a call to the *unexpected()* function. The default behaviour of *unexpected()* is to call *abort()*, which in turn causes abnormal program termination. A description of *unexpected()* is given later.

The following program illustrates exception specifications:

```
// ex_spec.cpp
// illustrates exception specifications
#include <iostream>  // C++ I/O

// throw only int & double
void FThrowIntDouble (int i) throw (int, double)
  {
  if (i==0)
    throw 0 ;     // throw int
  if (i==1)
    throw 0.1 ;   // throw double
  }
```

```
// throw no exceptions
void FThrowNone (int i) throw ()
  {
  // these statements can't be thrown!
  if (i==0)
    throw 0 ;     // throw int
  if (i==1)
    throw 0.1 ;   // throw double
  }

// throw any exceptions
void FThrowAny (int i)
  {
  if (i==0)
    throw 0 ;     // throw int
  if (i==1)
    throw 0.1 ;   // throw double
  }

void main ()
  {
  try
    {
    //FThrowIntDouble (0) ;  // ok
    //FThrowIntDouble (1) ;  // ok
    //FThrowAny (1) ;        // ok
    FThrowNone (0) ;         // abnormal program termination
    }
  catch (int i)
    {
    cout << "caught int: " << i << endl ;
    }
  catch (double d)
    {
    cout << "caught double: " << d << endl ;
    }
  }
```

The function *FThrowIntDouble()* can throw either **int** or **double** exceptions, but if an attempt is made to throw any other type of exception abnormal program termination will occur and the *unexpected()* function will be called. *FThrowNone()* cannot throw any exceptions and an attempt to do so will result in an abnormal program termination. *FThrowAny()* illustrates that a function without an exception specification can throw any exception.

14.5 Re-Throwing an Exception

An exception can be re-thrown. This is accomplished by using the **throw** keyword without an operand and causes the original exception to be caught by an outer **try-catch** block of state-

ments. The following program illustrates re-throwing an exception within the *Function1()* function, which throws back the original exception that is caught by the **catch** statement within *Function0()*:

```
// rethrow.cpp
// illustrates re-throwing an exception
#include <iostream>  // C++ I/O

// prototype
void Function1 (int i) ;

void Function0 (int i)
  {
  try
    {
    Function1 (i) ;
    }
  catch (int i)
    {
    cout << "Function0(): " << i << endl ;
    }
  }

void Function1 (int i)
  {
  try
    {
    if (i==-1)
      throw i ;
    }
  catch (int i)
    {
    cout << "Function1(): " << i << endl ;
    throw ;  // re-throw exception
    }
  }

void main ()
  {
  Function0 (-1) ;
  }
```

with output:

```
Function1(): -1
Function0(): -1
```

14.6 The EXCEPT.H Header File

The EXCEPT.H header file contains several function and **class** declarations for exception handling: *terminate()*, *set_terminate()*, *unexpected()*, *set_unexpected()* and class xmsg. Each of these functions and class are described separately in the following subsections.

14.6.1 The *terminate()* Function

When an exception is thrown but not caught by an associated handler, the *terminate()* function is called:

```
void terminate () ;  // EXCEPT.H
```

The default action taken by *terminate()* is to call the *abort()* function, which results in immediate abnormal program termination. Alternatively, if you do not want a program to terminate with a call to *abort()* you can define a terminate function which is called instead. The terminate function must be registered using the *set_terminate()* function. If a terminate function has been registered, *terminate()* will execute the most recent function given as an argument to *set_terminate()*.

The *abort()* C++ library function has the following signature:

```
void abort () ; // STDLIB.H
```

and is similar to the *exit()* function:

```
void exit (int status) ;  // STDLIB.H
```

exit() terminates program execution. Before termination commences, all files are closed, buffered output is written and any registered exit functions are called. status can be assigned either of the following identifiers:

```
EXIT_SUCCESS (0) Normal program termination.
EXIT_FAILURE (1) Abnormal program termination.
```

Unlike *exit()*, the *abort()* function does not return any termination information to the operating system.

14.6.2 The *set_terminate()* Function

It was noted above that the default action taken by *terminate()* is to call *abort()*. Alternatively, *terminate()* calls the last function given as an argument to the *set_terminate()* function:

```
typedef void (*terminate_handler) () ;  // EXCEPT.H
terminate_handler set_terminate (terminate_handler t_handle) ;
```

The *set_terminate()* function gives you more control over the actions to be taken when a handler to an exception cannot be found. The sequence of actions to be taken when abnormal

program termination occurs is defined in the function pointed to by `t_handle`. The return value of *`set_terminate()`* is the previous function given as an argument to the function. The definition of the function pointed to by `t_handle` must terminate the program, since it is an error to return the function pointed to by `t_handle` to its caller. Also, it is an error for the function pointed to by `t_handle` to **throw** an exception.

The following program illustrates the *`set_terminate()`* function by installing an abnormal termination function called *`MyTerminate()`*, which displays a 'program aborted' message and then calls *`exit()`*:

```
// set_term.cpp
// illustrates the set_terminate() function
#include <iostream>   // C++ I/O
#include <except>     // exception handling, exit()

// abnormal termination function
void MyTerminate ()
  {
  cout << "program aborted" << endl ;
  exit (EXIT_FAILURE) ;
  }

void Function () throw (int)
  {
  int number (0) ;

  cout << "enter a positive integer: " ;
  cin  >> number ;

  if (number < 0)
    throw number ;
  else
    cout << "number entered: " << number << endl ;
  }

void main ()
  {
  // install MyTerminate() function
  set_terminate (MyTerminate) ;

  try
    {
    Function () ;
    }
  catch (double)
    {
    cout << "negative number entered!" << endl ;
    }
  }
```

and some user interaction:

```
enter a positive integer: -7
program aborted
```

The *MyTerminate()* function is called because an **int** type exception is thrown to which there is no corresponding handler.

14.6.3 The *unexpected()* Function

The signature of the *unexpected()* function is:

```
void unexpected () ;   // EXCEPT.H
```

The *unexpected()* function is called when a function throws an exception which is not declared in its exception specification. The default action taken by *unexpected()* is to call *terminate()*, which in turn calls *abort()*. Alternatively, if you do not want *unexpected()* to call *terminate()* you can define an unexpected function which is called instead. If an unexpected function has been registered, then *unexpected()* executes the last function given as an argument to the *set_unexpected()* function.

14.6.4 The *set_unexpected()* Function

The *set_unexpected()* function allows a programmer to install a function which defines program behaviour when a function throws an exception that is not in its exception specification. The signature of *set_unexpected()* is:

```
typedef void (*unexpected_handler) () ;   // EXCEPT.H
unexpected_handler set_unexpected (unexpected_handler
                                   u_handle) ;
```

The sequence of actions to be taken when a function throws an exception not listed in its exception specification is defined in the function pointed to by u_handle. The return value of *set_unexpected()* is the previous function given as an argument to the function. If a function is registered with *set_unexpected()* the *unexpected()* function is still called automatically and then control is passed to the registered function pointed to by u_handle. Also, the function pointed to by u_handle can call the functions *abort()*, *exit()* or *terminate()*.

The following program illustrates the *set_unexpected()* function:

```
// set_unx.cpp
// illustrates the set_unexpected() function
#include <iostream>   // C++ I/O
#include <except>     // exception handling, exit()

// abnormal unexpected exception specification function
void MyUnexpected ()
  {
  cout << "program aborted" << endl ;
  exit (EXIT_FAILURE) ;
  }
```

```
void Function () throw (double)
  {
  int number (0) ;
  cout << "enter a positive integer: " ;
  cin  >> number ;

  if (number < 0)
    throw number ;
  else
    cout << "number entered: " << number << endl ;
  }

void main ()
  {
  // install MyUnexpected() function
  set_unexpected (MyUnexpected) ;

  try
    {
    Function () ;
    }
  catch (int)
    {
    cout << "negative number entered!" << endl ;
    }
  }
```

and some user interaction:

```
enter a positive integer: -7
program aborted
```

The *MyUnexpected()* function is called because a type **int** exception is thrown from *Function()*, but the exception specification of *Function()* does not support this type.

14.6.5 The xmsg Class

The xmsg **class** is declared in EXCEPT.H and is of the form:

```
class xmsg : public exception
  {
  private:
      string* str ;
  public:
      xmsg (const string& msg) ;
      xmsg (const xmsg& msg) ;
      ~xmsg () throw () ;
      virtual const char* what () const throw () ;
      const string& why() const { return *str ; }
      void          raise() throw (xmsg) ;
```

```
    xmsg&          operator = (const xmsg& src) ;
};
```

As the **class** name suggests, xmsg is used for reporting messages that are related to an exception and is derived from the base class exception. The constructor sets the string data member to a given message. The C++ library string **class** is defined in the STRING.H header file. The *raise()* member function causes an xmsg exception to be thrown, whereas the *what()* and *why()* member functions return **char*** and string strings indicating the type of exception.

The xmsg class will be examined further in the header file VEC_X1.H below and in the next chapter, when we discuss inheritance with reference to the C++ string library **class**.

14.7 The EXCEPTION.H and STDEXCEPT.H Header Files

The C++ standard library supports a number of exception handling classes declared in EXCEPTION.H and STDEXCEPT.H. Figure 14.1 illustrates the class hierarchy with class exception acting as base class. Classes exception and bad_exception are declared in EXCEPT.H with all other classes declared in STDEXCEPT.H. The declaration of exception is of the form:

```
class exception
  {
  public:
    exception () throw () ;
    exception (const exception&) throw () ;
    virtual ~exception () throw () ;
    virtual const char* what () const throw () ;
    exception& operator = (const exception&) throw () ;
  };
```

exception supports default and copy constructors and overloads the assignment operator. In addition, exception declares a virtual destructor and the virtual member function *what()* which returns a string indicating the type of exception. Every class in the exception handling class hierarchy overrides the destructor and *what()* function and does not add to the functions and operators provided by exception. Note that all operations of exception cannot throw exceptions so as to prevent recursive exceptions being thrown.

The hierarchy of Figure 14.1 consists of three key divisions: bad exceptions, logic errors and runtime errors. Bad exceptions are simply ill-formed exceptions. Logic errors are errors that could be caught before a program commences execution or by tests performed on function arguments and constructors. Runtime errors are all other types of errors. The exception class bad_alloc will be discussed in more detail in Section 14.11 in connection with the **new** operator.

The following program illustrates the invalid_argument exception via a square root function, *Sqrt()*, which if passed a negative value for x throws an invalid_argument exception:

```
// std_exc.cpp
// illustrates the C++ standard exceptions
#include <iostream>    // C++ I/O
```

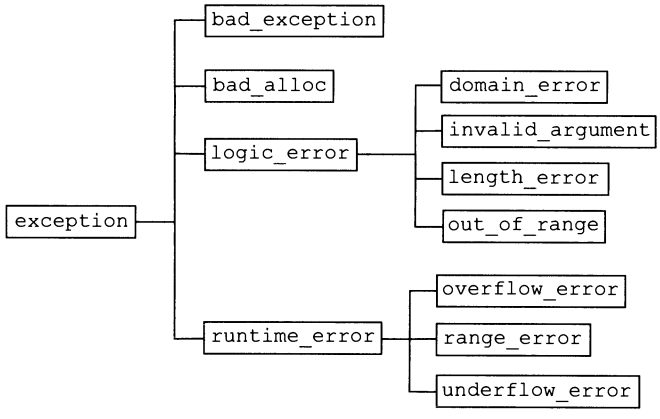


Fig. 14.1 Standard library exception handling class hierarchy.

```
#include <exception>  // C++ exception class
#include <stdexcept>  // C++ standard exceptions

double Sqrt (double x) throw (invalid_argument)
  {
  if (x < 0.0)
    throw invalid_argument ("Invalid argument x in Sqrt()") ;

  double sqrt (0.0) ;
  //...
  return sqrt ;
  }

void main ()
  {
  try
    {
    double sqrt = Sqrt (-1.0) ;
    }
  catch (exception& e)
    {
    cout << "exception: " << e.what () << endl ;
    }
  }
```

14.8 Constructors, Copy Constructors and Destructors

Exception handling generates its fair share of calls to a **class**'s constructors, copy constructor and destructor. Consider the following program:

```
// con_des.cpp
// illustrates calls to constructors,
// copy constructors and destructors with exception handling
#include <iostream>  // C++ I/O

class X
  {
  private:
    int data ;
  public:
    X ()
      { cout << "constructor called" << endl ;
        data = 0 ; }
    X (const X& x)
      { cout << "copy constructor called" << endl ;
        data = x.data ; }
    ~X ()
      { cout << "destructor called" << endl ; }
  };
void Function (const X& x) throw (X)
  {
  throw x ;
  }

void main ()
  {
  try
    {
    X x ;
    Function (x) ;
    }
  catch (X)
    {
    cout << "X exception caught" << endl ;
    }
  }
```

with output:

```
constructor called
copy constructor called
copy constructor called
destructor called
destructor called
copy constructor called
X exception caught
destructor called
destructor called
```

If the **throw** expression, **throw** `x;`, is commented out in the above program, then the program output is as expected:

```
constructor called
destructor called
```

When an exception is thrown the copy constructor of the thrown object is called. The copy constructor call initialises a temporary object at the point where the exception is thrown (the **throw** point). When a program is interrupted by an exception, destructors are called for all objects created since the **try** block was first entered. An object that has only been partially created when an exception is thrown (e.g. an array of objects) will only have destructors called for the objects that have been fully constructed.

14.9 Exception Handling and a `Vector` **class**

Let us now examine the `Vector` **class** with respect to exception handling. On examining the `Vector` **class** declared and defined in the header file VEC_TMP.H of Chapter 13 we observe that there are three key repeatable exceptions to handle: memory, size and range. Allocating memory dynamically using the **new** operator requires us to test the return pointer against the 0 pointer to verify that the memory request was successful. The size (i.e. the number of elements) of a `Vector` object must be verified when a `Vector` object is defined and during assignment and arithmetical operations, such as the overloaded =, +, - and * operators. Each time an element of a `Vector` is indexed (using the *`Value()`* member functions or the overloaded `[]` operator of VEC_TMP.H) a range test must be performed. The `Vector` **class** of VEC_TMP.H relied heavily on the *`assert()`* macro. The `Vector` **class** below is a modification of VEC_TMP.H using the exception handling techniques outlined in the present chapter:

```
// vec_x.h
// template class Vector with exception handling

#ifndef _VEC_X_H  // prevent multiple includes
#define _VEC_X_H

#include <iostream>  // C++ I/O
#include <iomanip>   // setprecision(),...

#define NDEBUG (0)   // debugging:ON

// template/exception handling Assert() function
template <class T, class X>
inline void Assert (T exp, X x)
  {
  if (!NDEBUG) // compatible with assert() macro NDEBUG
    if (!exp)
      throw x ;
  }

template <class T>
class Vector
  {
  public:
    // exception handling classes
```

```
    class Memory {};
    class Size   {};
    class Range  {};
  private:
    // data members
    T*  array ;
    int n_elements ;
    // member function
    void Allocate (int n) throw (Vector<T>::Size,
                                 Vector<T>::Memory) ;
  public:
    // constructors
    Vector () { Allocate (3) ; }
    Vector (int n, T obj) ;
    Vector (const Vector& v) ;
    // destructor
    ~Vector () throw () { delete [] array ; array = 0 ;
                          n_elements = 0 ; }
    // member functions

    int      NumberOfElements () const { return n_elements ; }
    T&       Value (int index) throw (Vector<T>::Range) ;
    const T& Value (int index) const
                               throw (Vector<T>::Range) ;
    void     New (int new_n) ;
    // overloaded operators
    Vector&  operator = (const Vector& v)
                        throw (Vector<T>::Memory) ;
    T&       operator [] (int index)
                         throw (Vector<T>::Range) ;
    //...
    Vector   operator + (const Vector& v)
                        throw (Vector<T>::Size) ;
    //...
    // friend
    friend ostream& operator << (ostream& s,
                                 const Vector<T>& v) ;
  };

// allocates memory for elements of Vector<T>
template <class T>
void Vector<T>::Allocate (int n) throw (Vector<T>::Size,
                                        Vector<T>::Memory)
  {
  if (n <= 0)
    throw Size () ;

  array = new T[n] ;
  Assert (array!=0, Memory ()) ;
  //if (array == 0)
  // throw Memory () ;
```

```
    n_elements = n ;
    for (int i=0; i<n; i++)
      array[i] = T () ;
    }

// 2 arg. constructor
template <class T>
Vector<T>::Vector (int n, T obj)
    {
    Allocate (n) ;
    for (int i=0; i<n; i++)
      array[i] = obj ;
    }

// copy constructor
template <class T>
Vector<T>::Vector (const Vector& v)
    {
    Allocate (v.NumberElements()) ;
    for (int i=0; i<v.NumberElements(); i++)
      *(array+i) = v.Value (i) ;
    }

// public member functions:

// returns element value (non-const object)
template <class T>
T& Vector<T>::Value (int index) throw (Vector<T>::Range)
    {
    if (index < 0 || index >= n_elements)
      throw Range () ;
    return *(array+index) ;
    }
//...
// change number of elements
template <class T>
void Vector<T>::New (int new_n)
    {
    delete [] array ;
    array      = 0 ;
    n_elements = 0 ;

    Allocate (new_n) ;
    }

// overloaded operators:
//...

// subscript operator [] (non-const object)
template <class T>
```

```
T& Vector<T>::operator [] (int index) throw
    (Vector<T>::Range)
  {
  if (index < 0 || index >= n_elements)
    throw Range () ;
  return *(array+index) ;
  }
//...
// addition operator +
template <class T>
Vector<T> Vector<T>::operator + (const Vector& v)
                                  throw (Vector<T>::Size)
  {
  // verify that the two vectors are of equivalent
  // dimensions
  if (n_elements != v.n_elements)
    throw Size () ;

  Vector<T> add (v.n_elements,T()) ;

  for (int i=0; i<v.n_elements; i++)
      add.Value (i) = Value (i) + v.Value (i) ;
  return add ;
  }
//...
#endif // _VEC_X_H
```

The following program tests the above `Vector` **`template class`**:

```
// vec_tst.cpp
// tests the exception handling features
// of the template class Vector
#include <iostream>   // C++ I/O
#include "vec_x.h"    // template class Vector

void main ()
  {
  try
    {
    Vector<int> v (5, 0) ;
    cout << "Vector<int> v: " << v << endl ;

    // run-time range error
    cout << "v[5]: " << v[5] << endl ;
    }
  catch (Vector<int>::Memory)
    {
    cout << "insufficient memory for allocation" << endl ;
    }
  catch (Vector<int>::Size)
    {
```

```
    cout << "incorrect Vector<T> size defined" << endl ;
    }
  catch (Vector<int>::Range)
    {
     cout << "out-of-bounds indexing of Vector<T> elements"
          << endl ;
    }
}
```

with output:

```
Vector<int> v: [0, 0, 0, 0, 0]
out-of-bounds indexing of Vector<T> elements
```

The above program code illustrates several new features which are worth examining separately.

14.9.1 Exception Classes

The **template class** `Vector` incorporates three **public**ly declared empty exception nested classes called `Memory`, `Size` and `Range`:

```
template <class T>
class Vector
  {
  public:
    // exception handling classes
    class Memory {};
    class Size   {};
    class Range  {};
  //...
  };
```

The sole purpose of these three exception classes is to enable **template class** `Vector` to **throw** a memory, size or range exception to appropriate exception handlers. The three exception classes are nested classes, because it is assumed at present that these types of exception are specific to `Vector`. Also, the three exception classes are **public**ly declared to allow the exception handlers access to the nested classes.

14.9.2 Constructors

Since constructors have no return type, it is difficult to report an exception to a caller when an object is defined. The C++ method of exception handling allows a construction error to be conveniently thrown out of the constructor and dealt with accordingly. This is illustrated above in VEC_X.H, although indirectly, through the use of the *`Allocate()`* member function:

```
template <class T>
Vector<T>::Vector (int n, T obj)
  {
```

```
  Allocate (n) ;
  //...
  }
//...
template <class T>
void Vector<T>::Allocate (int n) throw (Vector<T>::Size,
                                        Vector<T>::Memory)
  {
  if (n <= 0)
    throw Size () ;
  //...
  }
```

If a user defines a `Vector` object with a size dimension less than or equal to zero, a `Size` exception is thrown allowing the user to **`catch`** the exception and deal with the error:

```
void main ()
  {
  try
    {
    Vector<int> v(5, 0) ;
    //...
    }
  //...
  catch (Vector<int>::Size)
    {
    cout << "incorrect Vector<T> size defined" << endl ;
    }
  //...
  }
```

14.9.3 *Assert()* `template` Function

The **`template`** `Vector` and `Matrix` classes of Chapter 13 (VEC_TMP.H and MTRX_TMP.H) made good use of the C++ *`assert()`* macro for testing a given assertion:

```
template <class T>
void Vector<T>::Allocate (int n)
  {
  assert (n) ;

  array = new T[n] ;
  assert (array != 0) ;
  //...
  }
```

Alternatively, we can define a **`template`** exception-handling *`Assert()`* function (Stroustrup, 1994, pp. 397–8):

```
template <class T, class X>
inline void Assert (T exp, X x)
```

```
{
if (!NDEBUG) // compatible with assert() macro NDEBUG
  if (!exp)
    throw x ;
}
```

Assert() throws an exception of **class** X if exp is logical-false. A typical use of *Assert()* is given in the *Allocate()* member function and tests whether the **new** operator cannot allocate the requested block of memory:

```
template <class T>
void Vector<T>::Allocate (int n) throw (Vector<T>::Size,
                                        Vector<T>::Memory)
  {
  //...
  array = new T[n] ;
  Assert (array!=0, Memory ()) ;
  //...
  }
```

If the memory request is unsuccessful a Memory exception is thrown.

14.9.4 Declaring Exceptions

Several member functions and overloaded operator functions of **template class** Vector incorporate exception specifications as part of their function declarator:

```
template <class T>
class Vector
  {
  //...
  void Allocate (int n) throw (Vector<T>::Size,
                               Vector<T>::Memory) ;
  public:
    //..
    ~Vector () throw () ;
    //...
    Vector&  operator = (const Vector& v)
                          throw (Vector<T>::Memory) ;
    //...
  };
```

The declaration of **template class** Vector indicates that *Allocate()* can **throw** exceptions of classes Size and Memory, but no others. The Vector **class** destructor cannot **throw** any exception. Since a function declaration that has no exception specification can **throw** any type of exception, the above exception specifications could be viewed as redundant. However, explicitly declaring the types of exception that a member function can **throw** as part of a **class** declaration makes a **class** declaration much more expressive.

14.10 Passing Information with an Exception

The above use of exception handling with the `Vector` **`template class`** demonstrated throwing exceptions with the help of three exception classes: `Memory`, `Size` and `Range`. However, because these three exception classes are empty classes (i.e. no data members) no information is sent to the appropriate handlers, via the thrown objects, concerning the exact reason for an exception. Therefore, let us extend the `Memory`, `Size` and `Range` exception classes:

```
// vec_x1.h
// template class Vector with exception handling
//...
template <class T>
class Vector
  {
  public:
    // exception handling classes
    class Memory
      {
      public:
        xmsg xm_mem ;
            int  allocation ;
        Memory (const string& msg, int alloc)
          : xm_mem (msg), allocation (alloc) {}
      };
    class Size
      {
      public:
        xmsg xm_size ;
            int  size ;
        Size (const string& msg, int siz)
          : xm_size (msg), size (siz) {}
      };
    class Range
      {
      public:
        xmsg xm_range ;
        int  index ;
        Range (const string& msg, int i)
          : xm_range (msg), index (i) {}
      };
  //...
  };
//...
// public member functions:

// allocates memory for elements of Vector<T>
template <class T>
void Vector<T>::Allocate (int n) throw (Vector<T>::Size,
                                        Vector<T>::Memory)
  {
```

```
    if (n <= 0)
      throw Size (string("Vector::Allocate(); invalid
                          size defined"), n) ;

    array = new T[n] ;
    Assert (array!=0,
      Memory (string("Vector::Allocate(); insufficient memory
                           for allocation"), n*sizeof(T))) ;

    n_elements = n ;
    for (int i=0; i<n; i++)
      array[i] = T () ;
    }

  // returns element value (non-const object)
  template <class T>
  T& Vector<T>::Value (int index) throw (Vector<T>::Range)
    {
    if (index < 0 || index >= n_elements)
      throw Range (string(Vector::Value(); out-of-bounds
                   indexing"), index) ;
    return *(array+index) ;
    }
  //...
  #endif // _VEC_X_H
```

Program VEC_TST1.CPP tests the above **template** `Vector` **class**:

```
  // vec_tst1.cpp
  // tests the exception handling features of the template
  // class Vector having modified the Memory, Size and Range
  // exception classes
  #include <iostream>  // C++ I/O
  #include <string>    // C++ string class
  #include "vec_x1.h"  // template class Vector

  void main ()
    {
    try
      {
      Vector<int> v (5, 0) ;
      cout << "Vector<int> v: " << v << endl ;

      // run-time range error
      cout << "v.Value(5): " << v.Value(5) << endl ;
      }
    catch (Vector<int>::Memory m)
      {
      cout << m.xm_mem.why ()
           << "; bytes requested: " << m.allocation << endl ;
      }
```

```
  catch (Vector<int>::Size s)
    {
    cout << s.xm_size.why ()
         << "; size: " << s.size << endl ;
    }
  catch (Vector<int>::Range r)
    {
    cout << r.xm_range.why ()
         << "; index: " << r.index << endl ;
    }
}
```

with output:

```
Vector<int> v: [0, 0, 0, 0, 0]
Vector::Value(); out-of-bounds indexing; index: 5
```

All three nested classes `Memory`, `Size` and `Range` encapsulate a `xmsg` data member and associated `allocation`, `size` and `index` data members. The exception handling classes simply initialise their respective data members. `Memory`, `Size` or `Range` exceptions can now be thrown with the appropriate information relating to the exception rather than simply the type of exception. A thrown object is examined by a handler by giving a name to the exception object.

14.10.1 Range and Index Classes

Previous sections have illustrated the use of nested exception handling classes `Memory`, `Size` and `Range` for the **`template class`** `Vector`:

```
template <class T>
class Vector
  {
  public:
    // exception handling classes
    class Memory { /* ... */ };
    class Size   { /* ... */ };
    class Range  { /* ... */ };
    //...
};
```

However, classes `Memory`, `Size` and `Range` could equally be used by a variety of classes other than `Vector`, and not only as exception handling classes. As a result we shall now remove the nested `Memory`, `Size` and `Range` classes from the declaration of **`template class`** `Vector` and develop a `Range` **`class`**. Although the `Memory` and `Size` classes could equally be developed, we shall rely on the C++ standard library exception handling classes `bad_alloc` and `invalid_argument` declared in the NEW.H and STDEXCEPT.H header files for handling both memory and size exceptions. In addition to the `Range` **`class`**, an `Index` **`class`** is developed to assist the user in accessing the elements of a `Vector`.

In working with arrays, vectors and matrices we are continually accessing elements within a bounded range. Thus, it would be nice to have a **`class`** which is specifically designed to work with ranges. Such a **`class`**, called `Range`, is declared below:

```
// range.h
// class Range

#ifndef _RANGE_H  // prevent multiple includes
#define _RANGE_H

#include <iostream>  // C++ I/O

class Range
  {
  private:
    int min, max ;
  public:
    // constructors
    Range ()
      : min (0), max (0) {}
    Range (int _max)
      : min (0), max (_max) {}
    Range (int _min, int _max)
      : min (_min), max (_max) {}
    // copy constructor
    Range (const Range& r)
      : min (r.min), max (r.max) {}
    // member functions
    int          Min () const { return min ; }
    int          Max () const { return max ; }
    unsigned int Mag () const { return max - min + 1 ; }
    bool         In  (int i)
      { return i >= min && i <= max ? true : false ; }
    bool         Out (int i)
      { return i <  min || i >  max ? true : false ; }
    // overloaded operators
    Range& operator = (const Range& r)
      { min = r.min ; max = r.max ; return *this ; }
    bool operator == (const Range& r)
      { return min==r.min && max==r.max ? true : false ; }
    bool operator != (const Range& r)
      { return min!=r.min || max!=r.max ? true : false ; }
  };

ostream& operator << (ostream& s, const Range& r)
  { return s << "(" << r.Min() << ", " << r.Max() << ")" ; }

#endif // _RANGE_H
```

`Range` encapsulates two **`int`** data members which represent the lower and upper bounds of a given range. The four constructors of `Range` allow a user to define a `Range` object with ranges 0:0, 0:`max` and `min`:`max`, and to initialise one `Range` object to another. The member functions *`In()`* and *`Out()`* test whether an **`int`** number is within or outside a range, respectively.

To assist the user in indexing elements of an array, vector or matrix within a given range, let us develop an `Index` **`class`**:

```
// index.h
// class Index

#ifndef _INDEX_H  // prevent multiple includes
#define _INDEX_H

#include <iostream>  // C++ I/O
#include "assert.h"  // Assert()
#include "range.h"   // class Range

class Index
  {
  private:
    Range range ;
    int   index ;
  public:
    // constructors
    Index (int _min, int _max)
      : range (_min, _max), index (_min) {}
    Index (const Range& r)
      : range (r), index (r.Min()) {}
    Index (const Range& r, int _index)
      : range (r), index (_index) {}
    // copy constructor
    Index (const Index& i)
      : range (i.range), index (i.index) {}
    // member functions
    Range GetRange () const { return range ; }
    int   GetIndex () const { return index ; }
    int   Min () const { return range.Min () ; }
    int   Max () const { return range.Max () ; }
    int   Mag () const { return range.Mag () ; }
    // overloaded operators
    Index& operator = (const Index& i)
      { range = i.range ; index = i.index ; return *this ; }
    int operator [] (const Index& i)
      { return i.index ; }
    bool operator == (int i)
      { return index == i ? true : false ; }
    bool operator != (int i)
      { return index != i ? true : false ; }
    Index operator ++ () throw (Index) ;    // prefix
    Index operator ++ (int) throw (Index) ; // postfix
    // conversion function
    operator int () const { return index ; }
  };

Index Index::operator ++ () throw (Index) // prefix
  {
  Assert (!(index>range.Max()), *this) ;
  index++ ; return *this ;
```

```
  }
//...
ostream& operator << (ostream& s, const Index& i)
  { return s << i.GetRange() << " " << i.GetIndex() ; }

#endif // _INDEX_H
```

Index encapsulates both Range and **int** objects, range and index. The majority of **class** Index is fairly straightforward, but note the conversion function Index::**operator int** *()*, which is used to cast an Index object to an **int** object. The **class** Index also makes use of the *Assert()* **template** function, which is defined in the ASSERT.H header file.

Note that an Index is *a kind of* Range, and a more suitable implementation of Index would be to derive Index from Range:

```
class Index : public Range  // Index derived from Range
  {
  protected:
    int index ;
  //...
  };
```

In this case, Index would inherit the Range data members min and max and the *Min()*, *Max()* and *Mag()* member functions of Range, thus eliminating the present replication of these member functions in the Index **class** and the Index::range data member. However, since we have not yet covered the topic of inheritance we will encapsulate a Range object within an Index object:

```
class Index  // Index contains a Range object
  {
  private:
    Range range ;
  //...
  };
```

With the Range and Index classes at our disposal we can now redesign the Vector **template class** of VEC_X1.H as follows:

```
// rng_indx.cpp
// illustrates the use of Range and Index classes
// with a Vector template class
//...
#include "range.h"  // class Range
#include "index.h"  // class Index

template <class T>
class Vector
  {
  private:
    // data members
    T*    array ;
    Range range ;
```

```
    // member function
    void  Allocate (const Range& r)
      throw (bad_alloc, invalid_argument) ;
  public:
    //...
    T& operator [] (int i) throw (Index) ;
    //...
  };
//...
template <class T>
void Vector<T>::Allocate (const Range& r)
  throw (bad_alloc, invalid_argument)
  {
  Assert (!(r.Mag()<=0), invalid_argument
  ("Vector::Allocate(); invalid argument r")) ;

  array = new T[r.Mag()] ;

  range = r ;
  for (Index i (r); i<=r.Max(); i++)
    *(array+int(i)-range.Min()) = T () ;
  }
//...
template <class T>
T& Vector<T>::operator [] (int i) throw (Index)
  {
  Assert (!range.Out(i), Index (range, i)) ;
  return *(array+i-range.Min()) ;
  }
//...
void main ()
  {
  try
    {
    Vector<int> v (Range (6, 10), 0) ;
    cout << "Vector<int> v: " << v << endl ;

    v[11] = 1 ;  // out-of bounds indexing
    }
  catch (std::bad_alloc ba)
    {
    cout << "bad_alloc exception: "
         << ba.what () << std::endl ;
    }
  catch (Range r)
    {
    cout << "Range exception: " << r << endl ;
    }
  catch (Index i)
    {
    cout << "Index exception: " << i << endl ;
```

```
    }
  }
```

with output:

```
Vector<int> v: [0, 0, 0, 0, 0]
Index exception caught (6, 10) 11
```

The elements of a `Vector` object can now be within a bounded range specified when a `Vector` object is defined, as illustrated above within the *`main()`* function:

```
Vector<int> v (Range (6, 10), 0) ;
```

`Vector` no longer encapsulates an `n_elements` data member as in VEC_X1.H, but instead a `range` data member. However, since the elements of a `Vector` object are no longer indexed relative to zero we have to be careful when defining member functions which access elements of a `Vector` object. For instance, consider the *`Allocate()`* member function:

```
template <class T>
void Vector<T>::Allocate (const Range& r)
  throw (bad_alloc, invalid_argument)
  {
  //...
  for (Index i (r); i<=r.Max(); i++)
    *(array+int(i)-range.Min()) = T () ;
  }
```

The **`for`**-loop defines and initialises (via a `Range` object) a control object `i` of **`class`** `Index` for performing an iteration through the elements of a newly created `Vector` object. The single statement of the **`for`**-loop body initialises each `Vector` object element to the default constructor of type `T` for the specified range `r`.

14.11 Exception Handling and the `new` Operator

Let's conclude the present chapter by examining the **`new`** operator with reference to exception handling. To date, when requesting a block of memory using the **`new`** operator we have explicitly tested the pointer returned by **`new`** to verify whether the memory request was successful or not:

```
//...
X* x = new X ;
if (x == 0)
  {
  // memory allocation unsuccessful
  }
//...
```

or:

```
//...
X* x = new X ;
assert (x != 0) ;
//...
```

Testing every pointer returned by the **new** operator can become laborious and is easily forgotten. Fortunately, the **new** operator, by default, throws a `bad_alloc` exception if a memory allocation is unsuccessful. Class `bad_alloc` is derived from the standard library exception handling class `exception` and is declared in NEW.H:

```
class bad_alloc : public exception
  {
  public:
    bad_alloc () throw () ;
    bad_alloc (const bad_alloc&) throw () ;
    virtual ~bad_alloc () throw () ;
    virtual const char* what () const throw () ;
    bad_alloc& operator = (const bad_alloc&) throw () ;
  };
```

The declaration of `bad_alloc` is derived from and analogous to `exception`, discussed in Section 14.6. The following program helps illustrate exception handling and the **new** operator:

```
// x_new.cpp
// illustrates exception handling and the new operator
#include <iostream>   // C++ I/O
#include <exception>  // exception handling

const unsigned int SIZE (1e06) ;

class X
  {
  private:
    double* data ;
  public:
    X ()
      : data (new double[SIZE]) {}
    ~X ()
      { delete [] data ; }
  };

void main ()
  {
  try
    {
    X* x = new X[SIZE] ;
    }
  catch (bad_alloc ba)
    {
    cout << ba.what () << endl ;
    }
```

with output:

```
bad alloc exception thrown
```

If the **new** operator is unsuccessful in allocating the requested amount of memory it returns a 0 pointer and throws a `bad_alloc` exception.

A closer inspection of NEW.H will reveal that the standard library supports several variants of the **new** and **delete** operators:

```
// standard global new & delete operators
void* operator new (size_t) throw (bad_alloc) ;
void  operator delete (void*) throw () ;
void* operator new[] (size_t) throw (bad_alloc) ;
void  operator delete[] (void*) throw () ;
// placement syntax
void* operator new (size_t, void*) throw () ;
void  operator delete (void*, void) throw () ;
void* operator new[] (size_t, void*) throw () ;
void  operator delete[] (void*, void*) throw () ;
// nothrow
void* operator new (size_t, const nothrow_t&) throw () ;
void  operator delete (void*, const nothrow_t&) throw () ;
void* operator new[] (size_t, const nothrow_t&) throw () ;
void  operator delete[] (void*, const nothrow_t&) throw () ;
```

where `nothrow_t` is a **struct** which indicates that an allocator/deallocator does not throw an exception:

```
struct nothrow_t {};
extern const nothrow_t nothrow ;
```

A **new** operator with an empty exception specification cannot indicate a memory allocation exception by throwing a `bad_alloc` exception and alternatively returns a 0 pointer. Furthermore, the *nothrow* **new** operators cannot throw a `bad_alloc` exception and indicate a memory allocation failure by returning 0. For example:

```
// nothrow.cpp
// illustrates the nothrow new operator
#include <iostream>   // C++ I/O
#include <exception>  // exception handling

const unsigned int SIZE (1e06) ;

void main ()
  {
  try
    {
    double* array = new(nothrow) double[SIZE] ;
    if (array == 0)
      {
       cout << "memory allocation failed" << endl ;
```

```
            exit (EXIT_FAILURE) ;
            }
         else
           { /* ... */ }
         }
      catch (bad_alloc ba)
        {
        cout << ba.what () << endl ;
        }
    }
```

In the same vein as the *set_terminate()* and *set_unexpected()* functions, a user-defined new_handler function can be defined which is called if a memory allocation using the **new** operator is unsuccessful. The new_handler function has no arguments and returns **void** and is registered by passing the new_handler function name to the *set_new_handler()* function. The signature of *set_new_handler()* is:

```
typedef void (*new_handler) () ;
new_handler set_new_handler (new_handler new_p) throw () ;
```

and is declared in the NEW.H header file. *set_new_handler()* returns the previous handler if one was registered.

The new_handler function should specify an appropriate sequence of actions to be taken when the **new** operator cannot successfully allocate a requested block of memory. The new_handler function should free previously allocated redundant memory, **throw** an exception or initiate program termination.

An example of a user-defined new_handler function is given in the following program:

```
// set_hand.cpp
// illustrates the set_new_handler() function
#include <iostream>   // C++ I/O
#include <cstdlib>    // exit()
#include <new>        // set_new_handler()

const unsigned int SIZE (1e06) ;

class X
  {
  private:
    double* data ;
  public:
    X ()
      : data (new double[SIZE]) {}
    ~X ()
      { delete [] data ; }
  };

// abnormal termination function
void MyHandler ()
  {
  cout << "MyHandler () called" << endl ;
```

```
    exit (EXIT_FAILURE) ;
    }

  void main ()
    {
    // install MyHandler() function
    set_new_handler (MyHandler) ;

    X* x = new X[SIZE] ;
    }
```

with output:

```
MyHandler () called
```

14.12 Summary

Exception handling enables communication to take place between the developer and a user of a function, **class** or library: *'The author of a library can detect run-time errors but does not in general have any idea what to do about them. The user of a library may know how to cope with such errors but cannot detect them – or else they would have been handled in the user's code and not left for the library to find. The notion of an exception is provided to help deal with such problems'* (Stroustrup, 1991, p. 293).

There are several different methods that can be adopted for handling exceptions when programming. The C++ *termination* method of exception handling provides a programmer with a structured, language-based way of dealing with exceptions. Exceptions can be passed or thrown to a *non-local* part of a program whose sole purpose is to deal with exceptions and errors. Exception handling in C++ is based around the three keywords: **try**, **catch** and **throw**. A block of program statements that is to be monitored for exceptions is prefixed by the keyword **try**. If, at run-time, an exception occurs within this monitored block of statements, the exception is thrown, using the **throw** keyword, to an appropriate handler which deals with a certain type of exception. The exception is caught, using the **catch** keyword, and is then processed accordingly.

Exercises

14.1 Define a function called `Sqrt()` which simply calls the C++ standard library function `sqrt()`, but which accounts for the possibility of calculating a negative square root by making use of the standard library exception class `invalid_argument` and an appropriate exception specification for `Sqrt()`.

14.2 Exercises 9.2, 10.2 and 11.4 have examined an `Int` **class**. Exercise 10.2 added an overloaded modulus operator `%` to `Int`, but did not cater for the possibility of division by zero. By declaring an exception-handling **class** called `DivisionByZero`, modify the `Int::operator%()` member function so that division by zero exceptions can be caught.

14.3 Use the *set_terminate()* function to install your own function which deals with uncaught exceptions.

14.4 Program ND_STR.CPP of Chapter 12 declared a simple `String` **class**:

```
class String
  {
  private:
    char* string ;  // pointer to char
    int   length ;  // length of string
  public:
    String (const char str[]=0) ;
    ~String () ;
    //...
  };
```

Declare an exception handling **class** called `InsufficientMemory` and modify the one-argument constructor of `String` so that it now has an exception specification of **class** `InsufficientMemory`. Also, ensure that `String`'s destructor can **throw** no exceptions.

14.5 The following is a simpler implementation of the `Vector` **class** declared in VEC_X.H:

```
template <class T>
class Vector
  {
  private:
    T*   array ;
    int  n_elements ;
    void Allocate (int n) ;
  public:
    //...
    T& operator [] (int index) ;
  };
```

Declare an exception handling **class** called `OutOfBounds` to handle out-of-bounds indexing of the elements of a `Vector` object using the overloaded **operator**`[]` *()* member function.

14.6 Previous implementations of the `Vector::`*Allocate()* member function used the *assert()* macro to test whether the requested size of a `Vector` object is greater than zero:

```
template <class T>
void Vector<T>::Allocate (int n)
                          throw (NegativeVectorSize)
  {
  assert (n) ;
  //...
  }
```

By declaring an exception handling **class** called `NegativeVectorSize`, modify the *Allocate()* member function so that *assert()* is no longer required.

14.7 By using the implicit `bad_alloc` exception handling capabilities of the **new** operator, rewrite the following `Vector::`*Allocate()* member function:

```
template <class T>
void Vector::Allocate (int n)
  {
  //...
  array = new T[n] ;
  assert (array != 0) ;
  //...
  n_elements = n ;
  }
```

Chapter 15

Inheritance

Inheritance, in C++, is the ability of a class to inherit the features of one or more other classes. It is exactly this ability of deriving from an existing class or classes which makes inheritance such an important feature in the C++ language, not only from the point of view of object-oriented programming, but also in the design and development of reusable code. Similar classes with inherent relationships can be placed into class hierarchies to assist in the conceptualisation and organisation of program code.

C++ gives us complete control in specifying inheritance relationships between classes, through the use of the private, protected and public access specifiers. Just as an object can always maintain private data while supporting a public interface to other objects, a class can equally control the access of its data members to derived classes. A derived class can also be derived from by other classes, and a class can be derived from multiple base classes – multiple inheritance.

We shall address the issue of containment versus inheritance in the design of a class. Containment exercises the 'has a' relationship, in which a class has or contains an object of another class. Inheritance exercises the 'is a' relationship, in which a class is a kind of another class. These two differing approaches of containment and inheritance will be illustrated by way of developing Triangle and Tetrahedra classes, where a tetrahedron object can be viewed as a kind of triangle or as having four triangle faces.

C++ supports run-time polymorphism using inheritance by declaring base class member functions as virtual. Derived classes can specifically override these base class virtual functions. The use of virtual functions and abstract base classes will be illustrated with the help of a Shapes class hierarchy. Classes such as Polygon, Triangle, Quadrilateral, Circle and Tetrahedra are all derived from an abstract base class called Shape for the modelling of popular primitive geometric objects.

Several powerful classes are presented throughout the present chapter. As mentioned above, several Shapes classes are presented. Vector3D, RVector, PtrVector and SortedVector classes are derived from Vector for modelling a vector in three-dimensional space, an indexed range vector, an array of pointer memory addresses and a sorted vector. The Matrix and Windows memory classes presented in Chapter 13 are revisited, and Stack and Queue classes are derived from class LinkedList.

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
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15.1 Base and Derived Classes

Inheritance in C++ is achieved by creating a new or *derived* **class** from an existing or *base* **class**, and is introduced in the following program:

```
// b_d.cpp
// introduces base and derived classes
#include <iostream>  // C++ I/O

// base class
class Base
  {
  private:
    int bdm ;
  public:
    Base ()
      : bdm (0) {}
    int  Get () { return bdm ; }
    void Set (int i) { bdm = i ; }
  };

// derived class
class Derived : public Base
  {};

void main ()
  {
  cout << "before:" << endl ;

  // base class object
  Base base ;
  cout << "Base::bdm    : " << base.Get () << endl ;

  // derived class object
  Derived derived ;
  cout << "Derived::bdm: " << derived.Get () << endl ;

  cout << "after: " << endl ;

  // alter
  derived.Set (1) ;

  cout << "Base::bdm    : "    << base.Get () << endl ;
  cout << "Derived::bdm: " << derived.Get () << endl ;
  }
```

with output:

```
before:
Base::bdm    : 0
```

```
Derived::bdm: 0
after:
Base::bdm:  : 0
Derived::bdm: 1
```

The above program declares a base **class**, called, appropriately, `Base`, which encapsulates a **private** data member called `bdm`, a no-argument constructor which initialises `bdm` to zero and two **public** member functions which **get and set** the value of `bdm`. Another **class**, called `Derived`, is declared in the above program, and is **public**ly derived from the base **class**:

```
class Derived : public Base
  {};
```

Notice that `Derived` is an empty **class** that has no data members or member functions of its own. Although the **class** declaration of `Derived` encapsulates no data members or member functions, the following statement within *`main()`* illustrates that `Derived` has access to the `Base::`*`Get()`* member function:

```
//...
cout << "Derived::bdm: " << derived.Get () << endl ;
```

Similarly, `Derived` has access to the `Base::`*`Set()`* member function:

```
//...
derived.Set (1) ;
```

which is in fact assigning the constant value of 1 to the `bdm` data member of the object of **class** `Derived` and not **class** `Base`. This important point is emphasised in the program output, which illustrates that objects `base` and `derived` each have their own `bdm` data members and `derived` does not share the `bdm` data member with object `base`. Objects of **class** `Derived` behave as if we had declared **class** `Derived` as:

```
class Derived
  {
  private:
    int bdm ;
  public:
    Derived ()
      : bdm (0) {}
    int  Get () { return bdm ; }
    void Set (int i) { bdm = i ; }
  };
```

Figure 15.1 schematically illustrates **class** `Derived` inheriting the features of **class** `Base`. Note the direction of the inheritance arrow.

It may sound obvious, but a **class** cannot be derived from itself or from an unknown **class**. It is only possible to derive from a **class** which has previously been declared.

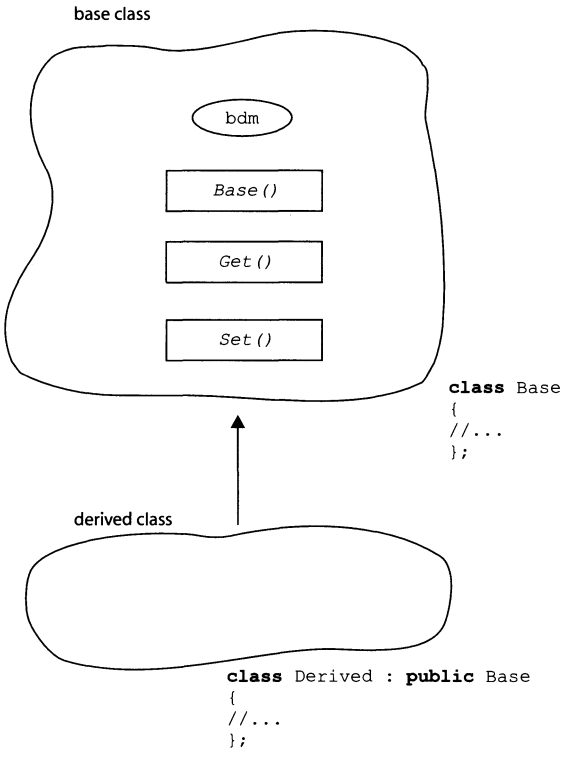


Fig. 15.1 A derived **class** inheriting the features of a base **class**.

15.2 Inheritance by Declaration

Inheritance is specified by incorporating the base **class** in the declaration of the derived **class**. This was demonstrated in the above program for base **class** `Base` and derived **class** `Derived`:

```
class Derived : public Base
  {};
```

The general syntax of **class** inheritance is of the form[1]:

```
class DerivedClass : access_specifier BaseClass
  { /* ... */ };
```

The use of a single colon (`:`) and the optional access specifier in the declaration inform the compiler that `DerivedClass` is derived from `BaseClass`. The next section examines the different access specifiers available in C++.

1 Note that is illegal to derive a **class** from a C++ integral type such as **int**.

15.3 Access Specifiers

Program B_D.CPP illustrated the use of the **public** access specifier for **class** Derived:

```
class Derived : public Base
  {};
```

The keyword **public** allows objects of **class** Derived access to Base's **public** member functions, but not access to Base's **private** data members. Alternatively, we can use the **private** keyword as the access specifier, which prevents objects of Derived from accessing Base's **public** member functions, thus preventing a Derived object from accessing any data member or member function of an object of **class** Base.

Note that an object of a base or derived **class** never has access to another **class**'s **private** data members or **private** member functions.

Consider the following program which examines the use of the **public** and **private** access specifiers for derived classes:

```
// acc_spec.cpp
// examines the public and private
// access specifiers for derived classes
#include <iostream>  // C++ I/O

class Base
  {
  private:
    int pri_bdm ;
  public:
    int pub_bdm ;
  };

// publicly derived class
class PubDerived : public Base
  {
  private:
    int pri_ddm ;
  public:
    void Function ()
      {
      pri_ddm = pri_bdm ;  // error: not accessible
      pri_ddm = pub_bdm ;  // ok: accessible
      }
  };

// privately derived class
class PriDerived : private Base
  {
  private:
    int pri_ddm ;
  public:
    void Function ()
```

```
        {
        pri_ddm = pri_bdm ;   // error: not accessible
        pri_ddm = pub_bdm ;   // ok: accessible
        }
    };

void main ()
    {
    // publicly derived class object
    PubDerived pub_derv ;

    pub_derv.pri_bdm ;   // error: not accessible
    pub_derv.pub_bdm ;   // ok: accessible

    // privately derived class object
    PriDerived pri_derv ;

    pri_derv.pri_bdm ;   // error: not accessible
    pri_derv.pub_bdm ;   // error: not accessible
    }
```

Functions of derived classes cannot access the **private** data members of a base **class**, whereas they can access **public** data or function members of a base **class**. Similarly, objects of a derived **class** cannot access the **private** data or function members of a base **class**. Objects of a **public**ly derived **class** can access the **public** data or function members of a base **class**, whereas objects of a **private**ly derived **class** cannot access the **public** data or function members of a base **class**.

In addition, we shall see shortly that the keyword **protected** can also be used as an access specifier.

15.3.1 Default Specifier

If an access specifier is not used when declaring a derived **class**, then **private** is assumed:

```
class Derived : Base   // privately derived class
    { /* ... */ };
```

This is identical in principle to declaring a **class**'s data members, which we know are **private** by default:

```
class X
    {
    // private data members
    };
```

15.3.2 Access Declaration

There are occasions in C++ when a base **class** is **private**ly inherited but it is required that certain data members retain their **public** specification. For such cases, C++ supports an *access declaration*, which is appropriately placed in the **public** section of a derived **class**.

The following program declares a base **class**, Base, which encapsulates two **public** data members. A **class**, Derived, is **private**ly derived from Base, thus making Base's **public** data members **private** data members of Derived:

```
// chng_acc.cpp
// illustrates changing an access specification
// associated with data members of a derived class
#include <iostream>  // C++ I/O

class Base
  {
  public:
    int pub_bdm0 ;
    int pub_bdm1 ;
  };

// privately derived class
class Derived : private Base
  {
  public:
    Base::pub_bdm1 ; // restore public pub_bdm1
  };

void main ()
  {
  Derived d_obj ;
  d_obj.pub_bdm0 ;  // error: not accessible
  d_obj.pub_bdm1 ;  // ok
  }
```

The **public** section of Derived contains the following access declaration:

```
Base::pub_bdm1 ;
```

which restores the **public** specification of Base **class** data member pub_bdm1. The statements within *main()* illustrate that without the use of the access declaration for Base::pub_bdm1 this data member would not be accessible to an object of **class** Derived.

Similarly, an access declaration can be used to restore a **protected** (to be discussed shortly) access specification for a derived **class** data member. Note that it is not possible to modify the original access specification of a base **class** data member using an access declaration, because this would effectively allow a base **class** declaration to be modified by a derived **class**.

When the C++ **namespace** is discussed in Chapter 19 it will be shown that the so-called *using declaration* offers a programmer a more general form of the access declaration, thus making the access declaration redundant.

15.4 Properties of Base and Derived Classes

Before examining the **protected** keyword, let's examine three obvious, but nevertheless important, properties of inheritance:

- Inheritance is a one-way process. A base **class** knows nothing of its derived class/es.
- A base **class** remains unchanged during inheritance.
- In theory, an infinite number of classes can be derived from a single base **class**.

The following program illustrates these three properties:

```
// props.cpp
// illustrates certain properties of inheritance
#include <iostream>  // C++ I/O

class Base
  {
  private:
    int bdm ;
  public:
    int BaseGet ()
      { return bdm ; }
    void BFunction ()
      { bdm = DerivedGet () ; } // error: undefined function
  };

// declare multiple derived classes from a single base class:

class Derived0 : public Base
  {
  private:
    int ddm ;
  public:
    int DerivedGet ()
      { return ddm ; }
    void DFunction ()
      { ddm = BaseGet () ; } // ok: accessible
  };

class Derived1 : public Base
  {};

class Derived2 : public Base
  {};

void main ()
  {
  // base class remains unaltered by derived classes
  Base b_obj ;
  }
```

15.5 Deriving from a Derived `class`

Once a derived **`class`** has been adequately declared, additional classes can be derived from the derived **`class`**:

```
// der_der.cpp
// illustrates deriving from a derived class
#include <iostream>  // C++ I/O

class Base
  {
  protected:
    int bdm ;
  public:
    int  Get ()
      { return bdm ; }
    void Set (int i)
      { bdm  = i ; }
  };

class Derived0 : public Base
  {};

class Derived1 : public Derived0
  {};

void main ()
  {
  Base b_obj ;
  b_obj.Set (0) ;
  cout << b_obj.Get () << endl ;

  Derived0 d0_obj ;
  d0_obj.Set (1) ;
  cout << d0_obj.Get () << endl ;

  Derived1 d1_obj ;
  d1_obj.Set (2) ;
  cout << d1_obj.Get () << endl ;
  }
```

The above hierarchy is schematically illustrated in Fig. 15.2.

15.6 `protected` Data Members

To date, all of the classes that we have seen have declared their data members and member functions as either **`private`** or **`public`**. C++ supports a third keyword, called **`protected`**, to declare members of a **`class`** when programming for inheritance. Declaring a data or function

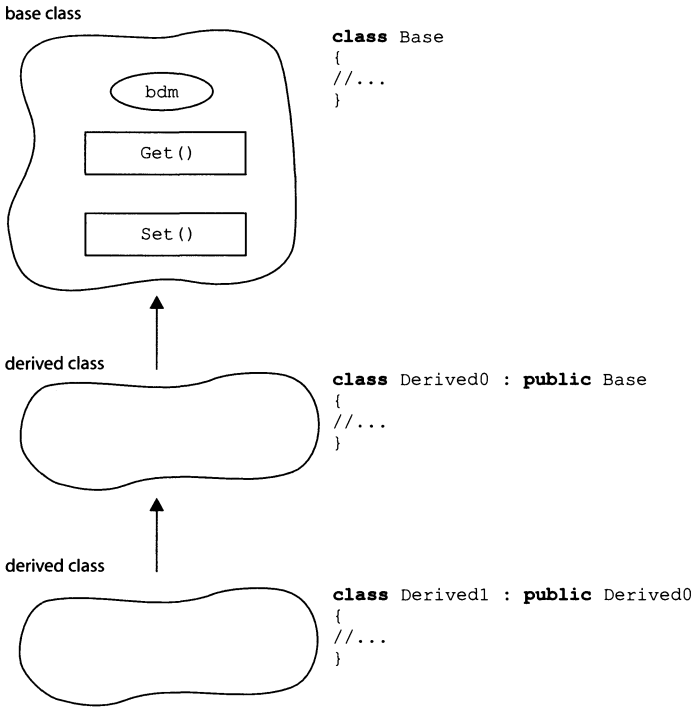


Fig. 15.2 Deriving from a derived **class**.

member of a **class** as **protected** allows the member to be accessed not only by other data or function members of the same **class**, but also by any **class** derived from it. Objects of a **class** do not have access to a **class**'s **protected** members, just as if they were declared as **private**. Consider the program:

```
// prot.cpp
// illustrates the protected keyword
#include <iostream>  // C++ I/O

class Base
  {
  private:
    int pri_bdm ;
  protected:
    int pro_bdm ;
  public:
    int pub_bdm ;
  };

class Derived : public Base
  {
  private:
    int pri_ddm ;
  public:
```

```
    void Function ()
      { pri_ddm = pro_bdm ; } // ok: accessible
  };

void main ()
  {
  Base bobj ;

  bobj.pro_bdm ;  // error: not accessible

  Derived dobj ;

  dobj.pro_bdm ;  // error: not accessible
  }
```

The **protected** data member of **class** Base, pro_bdm, is accessible to **class** Derived, but not accessible to objects of either Base or Derived.

15.7 protected Access Specifier

We saw in the previous section that the **private** and **public** keywords can be used as access specifiers. Also, the **protected** keyword can be used as an access specifier for derived classes. When a **class** is **protected**ly derived from a base **class**, all **protected** and **public** data or function members of the base **class** are **protected** members of the derived **class**:

```
// pro_accs.cpp
// examines the protected access specifier for derived classes
#include <iostream>  // C++ I/O

class Base
  {
  private:
    int pri_bdm ;
  protected:
    int pro_bdm ;
  public:
    int pub_bdm ;
  };

class ProDerived : protected Base
  {
  private:
    int pri_ddm ;
  public:
    void Function ()
      {
      pri_ddm = pri_bdm ;  // error: not accessible
      pri_ddm = pro_bdm ;  // ok: accessible
```

```
        pri_ddm = pub_bdm ;  // ok: accessible
        }
    };

void main ()
    {
    // protectedly derived class object
    ProDerived pro_derv ;

    pro_derv.pri_bdm ;  // error: not accessible
    pro_derv.pro_bdm ;  // error: not accessible
    pro_derv.pub_bdm ;  // error: not accessible
    }
```

Both the **protected** and **public** data members, `pro_bdm` and `pub_bdm`, of the base **class**, `Base`, are **protected** data members of the derived **class**, `ProDerived`, thus preventing objects of `ProDerived` having access to `Base`'s **public** data member.

15.8 Base and Derived `class` Constructors and Destructors

15.8.1 No Derived `class` Constructors or Destructor

Several of the derived classes in the above programs did not explicitly define their own constructor or destructor. To examine exactly what initialisation takes place for an object of a derived **class** without a constructor or destructor explicitly defined, consider the program:

```
// no_d_cd.cpp
// illustrates a derived class
// with no constructor or destructor
#include <iostream>  // C++ I/O

class Base
    {
    public:
      Base ()
        { cout << "Base class constructor called" << endl ; }
      ~Base ()
        { cout << "Base class destructor called"  << endl ; }
    };

class Derived : public Base
    {};

void main ()
    { Derived d_obj ; }
```

with output:

```
Base class constructor called
Base class destructor called
```

The output illustrates that if a constructor or destructor is not defined for **class** Derived the compiler will use the appropriate constructor or destructor of **class** Base.

15.8.2 Derived class Constructors and Destructor

Let's now extend the above program by defining a constructor and destructor for **class** Derived:

```
// d_cd.cpp
// illustrates a derived class
// with constructor and destructor
#include <iostream>  // C++ I/O

class Base
  {
  public:
    Base ()
      { cout << "Base class constructor called" << endl ; }
    ~Base ()
      { cout << "Base class destructor called"  << endl ; }
  };

class Derived : public Base
  {
  public:
    Derived ()
      { cout << "Derived class constructor called" << endl ; }
    ~Derived ()
      { cout << "Derived class destructor called"  << endl ; }
  };

void main ()
  { Derived d_obj ; }
```

with output:

```
Base class constructor called
Derived class constructor called
Derived class destructor called
Base class destructor called
```

The program output illustrates that although a constructor is defined for **class** Derived, the constructor for **class** Base is called, and before Derived's constructor is called, when an object of Derived is defined. It makes sense to call a base **class** constructor before a derived **class** constructor, because generally the correct operation of a derived **class** object will depend on the data members of the base **class** being correctly initialised. As expected,

the destructors are called in reverse order, i.e. the derived **class** destructor is called before the base **class** destructor.

Let's extend the above D_CD.CPP by declaring a data member of a given **class** in the derived **class**:

```
// d_cd_d.cpp
// illustrates a derived class
// with constructor, destructor and data members
#include <iostream>  // C++ I/O

class X
  {
  public:
    X ()
      { cout << "X class constructor called" << endl ; }
    ~X ()
      { cout << "X class destructor called"  << endl ; }
  };

class Base
  {
  public:
    Base ()
      { cout << "Base class constructor called" << endl ; }
    ~Base ()
      { cout << "Base class destructor called"  << endl ; }
  };

class Derived : public Base
  {
  protected:
    X x ;
  public:
    Derived ()
      { cout << "Derived class constructor called" << endl ; }
    ~Derived ()
      { cout << "Derived class destructor called"  << endl ; }
  };

void main ()
  { Derived d_obj ; }
```

with output:

```
Base class constructor called
X class constructor called
Derived class constructor called
Derived class destructor called
X class destructor called
Base class destructor called
```

The program output illustrates that if data members are declared for a derived **class**, the order in which constructors are called when an object of a derived **class** is defined is: base, data member and derived. The order is reversed for **class** destructors.

Even if a derived **class** constructor has a different number of arguments than its base **class** constructor, the above rule of calling base/derived constructors/destructor still applies:

```
// odd_cd.cpp
// further illustrates a derived class
// with constructor and destructor
#include <iostream>  // C++ I/O

class Base
  {
  protected:
    int bdm ;
  public:
    Base ()
      : bdm (0)
      { cout << "Base class constructor called" << endl ; }
    ~Base ()
      { cout << "Base class destructor called"  << endl ; }
  };

class Derived : public Base
  {
  private:
    int ddm ;
  public:
    Derived (int i)
      : ddm (i)
      { cout << "Derived class constructor called" << endl ; }
    ~Derived ()
      { cout << "Derived class destructor called"  << endl ; }
  };

void main ()
  {
  //Derived d_obj ;  // error: no match found

  Derived d_obj (1) ;
  }
```

with output identical to that generated by program D_CD.CPP:

```
Base class constructor called
Derived class constructor called
Derived class destructor called
Base class destructor called
```

Note that an object of `Derived` can only be defined using the one-argument constructor of `Derived`. The compiler will not use the no-argument constructor of `Base`. Also, if the default constructor of **`class`** `Base` is removed a compilation error will result, because objects of the base **`class`** `Base` cannot be initialised correctly when an object of `Derived` is defined.

Alternatively, what if we wanted to define an object of **`class`** `Derived` and at the same time initialise an inherited data member of **`class`** `Base`? For example, let's modify program ODD_CD.CPP to define a `Derived` object and simultaneously initialise the inherited **`protected`** data member, `bdm`, of **`class`** `Base` via the none and one-argument constructors:

```
// d_to_b.cpp
// illustrates passing constructor argument objects
// from a derived class constructor to a base class
// constructor
#include <iostream>  // C++ I/O

class Base
  {
  protected:
    int bdm ;
  public:
    Base ()
      : bdm (0) {}
    Base (int i)
      : bdm (i) {}
    ~Base () {}
    int Get () { return bdm ; }
  };

class Derived : public Base
  {
  protected:
    int ddm ;
  public:
    Derived () : Base () {}
    Derived (int i) : Base (i), ddm (i) {}
    ~Derived () {}
  };

void main ()
  {
  Derived no_arg_obj ;
  Derived one_arg_obj (1) ;

  cout << "no_arg_obj.bdm : " << no_arg_obj.Get () << endl
       << "one_arg_obj.bdm: " << one_arg_obj.Get () ;
  }
```

with output:

```
no_arg_obj.bdm : 0
one_arg_obj.bdm: 1
```

The above program illustrates that C++ allows a derived **class** constructor to call its base **class** constructor and simultaneously to pass derived **class** constructor argument objects to its base **class** constructor using the general syntax:

```
DerivedClass (argument_list)
  : BaseClass (argument_list)
  { /* ... */ };
```

Note that if a non-zero-argument base **class** constructor is called by a derived **class** constructor, the default constructor of the base **class** is no longer called.

This syntax is identical to the data member initialisation list syntax that we have seen in previous chapters:

```
ClassName (type_specifier arg1, ..., type_specifier argn)
  : data_member1 (arg1), ..., data_membern (argn)
  { /* ... */ };
```

Hence a base **class** constructor acts just as if it is a data member of the derived **class** when initialising an object of a derived **class**.

Both of the constructor bodies for **class** `Derived` in program D_TO_B.CPP are empty:

```
class Derived : public Base
  {
  protected:
    int ddm ;
  public:
    Derived () : Base () {}
    Derived (int i) : Base (i), ddm (i) {}
    //...
  };
```

As well as passing constructor arguments to a base **class**, a derived **class** can also include statements, function calls and expressions involving the passed constructor arguments in the constructor function body:

```
// d_to_b1.cpp
// further illustrates passing constructor argument objects
// from a derived class constructor to a base class
// constructor
#include <iostream>  // C++ I/O

class Base
  {
  protected:
    int bdm ;
  public:
    Base ()
      : bdm (0) {}
    Base (int i)
      : bdm (i) {}
    ~Base () {}
```

```
    int Get () { return bdm ; }
  };

class Derived : public Base
  {
  protected:
    int ddm ;
  public:
    Derived () : Base () {}
    Derived (int i) : Base (i), ddm (i)
      { cout << "der. con. arg. sq.: " << i*i << endl ; }
    ~Derived () {}
  };

void main ()
  {
  Derived one_arg_obj (2) ;
  cout << "one_arg_obj.bdm: " << one_arg_obj.Get () << endl ;
  }
```

with output:

```
der. con. arg. sq.: 4
one_arg_obj.bdm: 2
```

15.9 Copy Constructors, Overloaded Assignment Operator and Inheritance

15.9.1 Copy Constructors

If a base **class** defines a copy constructor but a derived **class** does not, then the base **class** copy constructor is invoked when an object of the derived **class** is defined and initialised (either by assignment or copy constructor) to another object of the derived **class**. This is demonstrated in the following program:

```
// cc_inh.cpp
// illustrates a base class copy constructor and inheritance
#include <iostream>  // C++ I/O

class Base
  {
  protected:
    int bdm ;
  public:
    Base ()
      : bdm (0)
    { cout << "Base::Base() called" << endl ; }
    // copy constructor
```

```
    Base (const Base& b)
      : bdm (b.bdm)
    { cout << "Base::Base(const Base&) called" << endl ; }
  };

class Derived : public Base
  {
  protected:
    int ddm ;
  public:
    Derived ()
      : ddm ()
    { cout << "Derived::Derived() called" << endl ; }
  };

void main ()
  {
  Derived d0 ;       // define
  Derived d1 = d0 ; // define and initialise by assignment
  //Derived d1 (d0) ; // define and initialise by copy con.
  }
```

with output:

```
Base::Base() called
Derived::Derived() called
Base::Base(const Base&) called
```

The first two lines displayed are a result of defining object `d0`. The final line displayed is due to defining and initialising object `d1` to object `d0`. The data members of both the base and derived classes are initialised by memberwise assignment.

If a derived **`class`** defines a copy constructor, it will be invoked when a derived **`class`** object is defined and initialised (either by assignment or copy constructor) to another object of the derived **`class`**. This is demonstrated in the program below, which simply adds a copy constructor to **`class`** `Derived`:

```
// cc_inh1.cpp
// illustrates the use of both base and derived class
// copy constructors and inheritance
#include <iostream>  // C++ I/O

class Base
  { /* ... */ };

class Derived : public Base
  {
  protected:
    int ddm ;
  public:
    Derived ()
      : ddm ()
```

```
    { cout << "Derived::Derived() called" << endl ; }
    // copy constructor
    Derived (const Derived& d)
      : Base (d), ddm (d.ddm)
    { cout << "Derived::Derived(const Derived&) called"
           << endl ; }
};

void main ()
  {
  Derived d0 ;       // define
  Derived d1 = d0 ; // define and initialise by assignment
  //Derived d1 (d0) ; // define and initialise by copy con.
  }
```

with output:

```
Base::Base() called
Derived::Derived() called
Base::Base(const Base&) called
Derived::Derived(const Derived&) called
```

Note that the `Derived` copy constructor invokes the `Base` copy constructor, since it is defined and it is the responsibility of a derived **`class`** copy constructor to ensure that data members of a base **`class`** are appropriately initialised:

```
Derived (const Derived& d)
  : Base (d), ddm (d.ddm) { /*...*/ }
```

The program output illustrates that, as with normal constructors, a base **`class`** copy constructor is invoked before a derived **`class`** copy constructor. If the base **`class`** copy constructor is not defined but the derived **`class`** copy constructor is defined, the compiler will perform the necessary default memberwise initialisation of the base **`class`** data members.

15.9.2 Overloaded Assignment Operator

The operation of the overloaded assignment **`operator`**`=()` function is similar to the operation of the copy constructor:

```
// op_inh.cpp
// illustrates the use of a base class
// overloaded assignment operator and inheritance
#include <iostream>  // C++ I/O

class Base
  {
  protected:
    int bdm ;
  public:
    // constructors
```

```
    Base ()
      : bdm (0) {}
    Base (int i)
      : bdm (i) {}
    Base& operator = (const Base& b)
      {
      cout << "Base::operator =() called" << endl ;
      bdm = b.bdm ; return *this ;
      }
};

class Derived : public Base
  {
  protected:
    int ddm ;
  public:
    Derived () : Base () { ddm = 0 ; }
    Derived (int i) : Base (i), ddm (i) {}
    void Display ()
      { cout << "Derived::bdm: "   << bdm
             << ", Derived::ddm: " << ddm << endl ; }
  };

void main ()
  {
  Base b1 (1), b2 (2) ;
  b2 = b1 ;

  Derived d1 (1), d2 (2) ;
  d2 = d1 ;

  d2.Display () ;
  }
```

with output:

```
Base::operator=() called
Base::operator=() called
Derived::bdm: 1, Derived::ddm: 1
```

The output illustrates that if a base **class** defines an **operator**= *()* function, but a derived **class** does not define **operator**= *()*, the base **class operator**= *()* function is invoked when one derived **class** object is assigned to another derived **class** object. The output also illustrates that the default assignment of objects of a derived **class** which does not overload **operator**= *()* is memberwise assignment.

If a derived **class** also defines the **operator**= *()* function, then it is invoked when one derived **class** object is assigned to another. This is illustrated in the following program:

```
// op_inh1.cpp
// illustrates the use of both base and derived class
```

```
// overloaded assignment operator and inheritance
#include <iostream>  // C++ I/O

// base class
class Base
  { /* ... */ };

class Derived : public Base
  {
  protected:
    int ddm ;
  public:
    Derived () : Base () { ddm = 0 ; }
    Derived (int i) : Base (i), ddm (i) {}
    Derived& operator = (const Derived& d)
      {
      cout << "Derived::operator=() called" << endl ;
      Base::operator = (d) ;
      ddm = d.ddm ; return *this ;
      }
  };

void main ()
  {
  Base b1 (1), b2 (2) ;
  b2 = b1 ;

  Derived d1 (1), d2 (2) ;
  d2 = d1 ;
  }
```

with output:

```
Base::operator=() called
Derived::operator=() called
Base::operator=() called
```

Note that the `Derived` **operator**= *()* function invokes the `Base` **operator**= *()* function, since it is defined and it is the responsibility of a derived **class** **operator**= *()* function to ensure that data members of a base **class** are appropriately assigned:

```
Derived& operator = (const Derived& d)
  {
  Base::operator = (d) ;
  ddm = d.ddm ; return *this ;
  }
```

When both a base and a derived **class** define **operator**= *()* functions, and an object of a derived **class** is assigned to an object of a base **class**, the **operator**= *()* function of the **class** on the left-hand side of the assignment operator is called:

```
// op_inh2.cpp
// illustrates base and derived class
// overloaded assignment operator functions and inheritance
#include <iostream>  // C++ I/O

class Base
  {
  public:
    //...
    Base& operator = (const Base& b) { /*...*/ }
  };

class Derived : public Base
  {
  public:
    //...
    Derived& operator = (const Derived& d) { /*...*/ }
  };

void main ()
  {
  Base    b (0) ;
  Derived d (1) ;

  b = d ;  // ok : assign derived object to base object
  //d = b ;  // error: assign base object to derived object
  }
```

with output:

```
Base::operator=() called
```

Assigning a derived **class** object to a base **class** object is often referred to as *slicing*. Note that the reverse situation of assigning a base **class** object to a derived **class** object is illegal.

Let us conclude this section by examining a program in which both base and derived classes define a copy constructor and an **operator**= *()* function:

```
// op_inh3.cpp
// illustrates the use of a copy constructor and overloaded
// assignment operator function for both base and derived
// classes
#include <iostream>  // C++ I/O

class Base
  {
  protected:
    int bdm ;
  public:
    // constructors
    Base ()
```

```
        : bdm (0) {}
      Base (int i)
        : bdm (i) {}
      Base (const Base& b)
        : bdm (b.bdm)
        { cout << "Base::Base(const Base&) called" << endl ; }
      Base& operator = (const Base& b)
        {
        cout << "Base::operator=() called" << endl ;
        bdm = b.bdm ; return *this ;
        }
    };

  class Derived : public Base
    {
    protected:
      int ddm ;
    public:
      Derived () : Base () { ddm = 0 ; }
      Derived (int i) : Base (i), ddm (i) {}
      Derived (const Derived& d)
        : Base (d), ddm (d.ddm)
        { cout << "Derived::Derived(const Derived&) called"
               << endl ; }
      Derived& operator = (const Derived& d)
        {
        cout << "Derived::operator=() called" << endl ;
        Base::operator = (d) ;
        ddm = d.ddm ; return *this ;
        }
    };

  void main ()
    {
    Derived d0 (0) ;

    cout << "Derived d1=d0:" << endl ;
    Derived d1 = d0 ;

    cout << "d1=d0:" << endl ;
    d1 = d0 ;
    }
```

with output:

```
Derived d1=d0:
Base::Base(const Base&) called
Derived::Derived(const Derived&) called
d1=d0:
Derived::operator=() called
Base::operator=() called
```

Pay particular attention to the distinction between the use of the assignment operator for initialisation of an object and assignment of one object to another.

15.10 `new` and `delete` Operators and Inheritance

If the **new** and **delete** operators are overloaded for a base **class**, then a derived **class** inherits the overloaded **new** and **delete** operators. This is illustrated in the following program:

```
// nd_inh.cpp
// illustrates overloading the new and delete operators
// with respect to inheritance
#include <iostream>   // C++ I/O
#include <calloc>     // malloc(), free()

class Base
  {
  protected:
    int bdm ;
  public:
    //...
    void* operator new (size_t size)
      {
      cout << "Base::operator new() called, size: "
           << size << endl ;
      return malloc (size) ;
      }
    void operator delete (void* ptr)
      {
      cout << "Base::operator delete() called" << endl ;
      free (ptr) ;
      }
  };

class Derived : public Base
  {
  protected:
    int ddm ;
  public:
    Derived () : Base () { ddm = 0 ; }
    Derived (int i) : Base (i), ddm (i) {}
  };

void main ()
  {
  Base*    pb (new Base) ;
  Derived* pd (new Derived) ;

  delete pb ;
```

```
    delete pd ;
    }
```

with output:

```
Base::operator new() called, size: 4
Base::operator new() called, size: 8
Base::operator delete() called
Base::operator delete() called
```

The interesting point to note is that the size of a `Derived` object is larger than the size of a `Base` object, and yet the **`new`** and **`delete`** operators allocate and deallocate the memory appropriately. If **`new`** and **`delete`** are overloaded for a base **`class`**, correct allocation and deallocation is inherited for all derived classes.

15.11 Overriding `class` Member Functions

Let's now examine base and derived classes that have member functions with identical function names:

```
// override.cpp
// illustrates derived class member functions
// which override base class member functions
#include <iostream>  // C++ I/O

class Base
   {
   protected:
      int bdm ;
   public:
      //...
      void Display ()
         {
         cout << "Base::bdm: " << bdm << endl ;
         }
   };

class Derived : public Base
   {
   protected:
      int ddm ;
   public:
      //...
      void Display ()
         { cout << "Derived::bdm: " << bdm << endl
                << "Derived::ddm: " << ddm << endl ; }
   };
```

```
void main ()
  {
  Base    b_obj ;
  Derived d_obj ;

  b_obj.Display () ;
  d_obj.Display () ;
  }
```

with output:

```
Base::bdm: 0
Derived::bdm: 0
Derived::ddm: 0
```

Both `Base` and `Derived` classes define a member function called *`Display()`*, which simply displays each **`class`**'s **`protected`** data members. The output illustrates that for objects of a derived **`class`** the compiler will choose the member function of the derived **`class`** if the same function name exists in both base and derived classes. For objects of a base **`class`**, the compiler has no other option than to choose the member defined in the base **`class`**, since the base **`class`** knows nothing of the derived **`class`**. The *`Display()`* member function in **`class`** `Derived` is said to *override* the *`Display()`* member function in **`class`** `Base`. This is a convenient and frequently used property of derived classes, because it allows the same function name to be used for objects of base and derived classes.

What about overriding base **`class`** overloaded member functions? Consider the following program, which overloads the *`Display()`* member function in **`class`** `Base`:

```
// oo.cpp
// illustrates overriding base class overloaded member
// functions
#include <iostream>  // C++ I/O

class Base
  {
  protected:
    int bdm ;
  public:
    //...
    void Display ()
      { cout << "Base::bdm: " << bdm << endl ; }
    void Display (char* str)
      { cout << str << bdm << endl ; }
  };

class Derived : public Base
  {
  protected:
    int ddm ;
  public:
    //...
    void Display ()
```

```
        { cout << "Derived::bdm: " << bdm << endl
               << "Derived::ddm: " << ddm << endl ; }
    };

  void main ()
    {
    Base    b_obj ;
    Derived d_obj ;

    b_obj.Display () ;                // Base::Display()
    b_obj.Display ("Base::bdm: ") ; // Base::Display(char*)
    d_obj.Display () ;
    d_obj.Display ("Derived::bdm: ") ; // error: undefined
                                       // function
    }
```

Objects of **class** Base have access to both of the overloaded *Display()* member functions. However, objects of **class** Derived do not have access to Base::*Display(**char***)*, and the compiler will issue an 'undefined function' compilation error if an attempt is made to call this member function via a Derived **class** object.

The next program illustrates calling a base **class** overridden member function using the scope resolution operator (::):

```
// scpe_res.cpp
// illustrates the use of the scope resolution operator to
// access an overridden base class member function from a
// derived class
#include <iostream>  // C++ I/O

class Base
  {
  protected:
    int bdm ;
  public:
    //...
    void Display ()
      { cout << "Base::bdm: " << bdm << endl ; }
  };

class Derived : public Base
  {
  protected:
    int ddm ;
  public:
    //...
    void Display ()
      {
      Base::Display () ;
      cout << "Derived::ddm: " << ddm << endl ;
      }
  };
```

```
void main ()
  {
  Derived d_obj ;

  d_obj.Display () ;
  }
```

with output:

```
Base::bdm: 0
Derived::ddm: 0
```

Rather than defining `Derived::Display()` as:

```
void Display ()
  {
  cout << "Derived::bdm: " << bdm << endl
       << "Derived::ddm: " << ddm << endl ;
  }
```

as was the case in program OVERRIDE.CPP, SCPE_RES.CPP makes use of the scope resolution operator to call `Base::Display()` from within `Derived::Display()`:

```
void Display ()
  {
  Base::Display () ;
  cout << "Derived::ddm: " << ddm << endl ;
  }
```

If the `Base` **`class`** name and the scope resolution operator were not used, the compiler would view the call to `Display()` within `Derived::Display()` as a call to itself, and thus generate an infinite recursive loop.

15.12 Overloaded Operator Functions and Inheritance

Except for the **`operator`**`=()` function, all other overloaded operator functions of a base **`class`** are inherited by a derived **`class`**.

15.13 Friendship and Inheritance

Class friendship is not inherited:

```
// fr_inh.cpp
// illustrates friendship and inheritance
```

```
#include <iostream>  // C++ I/O

class X
  {
  private:
    int x_data ;
  public:
    // friend
    friend class Y ;
  };

// friend of X
class Y
  {
  public:
    void YFunction (X x)
      { x.x_data++ ; }  // ok
  };

// Z derived from Y
class Z : public Y
  {
  public:
    void ZFunction (X x)
      { x.x_data++ ; }  // error: X::x_data not accessible
  };

void main ()
  {}
```

The **class** Y is a **friend** of **class** X and therefore has access to the **private** data member X::x_data. However, just because **class** Z is derived from **class** Y it does not similarly allow members of **class** Z access to the **private** data member of **class** X. Class friendship is not inherited.

A **friend** function of a derived **class** has access to a **protected static** data member of a base **class**. In the case of non-**static** data members, a derived **class friend** function only has access to a non-**static** data member of a base **class** via an object, pointer or reference to the derived **class**:

```
// fr_inh1.cpp
// further illustrates friendship and inheritance
#include <iostream>  // C++ I/O

class Base
  {
  protected:
    int        ns_bdm ;
    static int s_bdm ;
  };

class Derived : public Base
```

```
  {
  public:
    friend void Function0 (Base vb, Base* pb, Base& rb)
      {
      vb.s_bdm++ ;    // ok

      vb.ns_bdm++ ;  // error: not accessible
      pb->ns_bdm++ ; // error: not accessible
      rb.ns_bdm++ ;  // error: not accessible
      }
    friend void Function1 (Derived vd, Derived* pd,
                           Derived& rd)
      {
      vd.ns_bdm++ ;     // ok
      pd->ns_bdm++ ;    // ok
      rd.ns_bdm++ ;     // ok
      }
  };

void main ()
  {}
```

15.14 Increased Functionality and Code Reusability

Let's now examine how an existing **class** can be derived from to create a related **class** which has increased functionality. The base **class** that we will use is the `Point` **class** discussed in previous chapters, and is basically identical to previous versions of `Point` except that the `x`, `y` and `z` data members are now declared **protected**:

```
// point.h
// class Point

#ifndef _POINT_H  // prevent multiple includes
#define _POINT_H

#include <iostream>  // C++ I/O

class Point
  {
  protected:
    double x, y, z ; // protected data members
  public:
    // constructors
    Point ()
      : x (0.0), y (0.0), z (0.0) {}
    Point (double x_arg, double y_arg, double z_arg=0.0)
      : x (x_arg), y (y_arg), z (z_arg) {}
    Point (double d)
      : x (d), y(d), z (d) {}
```

```
    // copy constructor
    Point (const Point& p)
      : x (p.x), y (p.y), z (p.z) {}
    // public member functions
    double& X () { return x ; }
    //...
    Point& operator = (const Point& p) ;
  };
//...
#endif // _POINT_H
```

The implementation file for **class** Point is:

```
// point.cpp
// implementation file for class Point
#include "point.h"  // header file

// assignment operator =
inline Point& Point::operator = (const Point& p)
  {
  x = p.x ; y = p.y ; z = p.z ;
  return *this ;
  }

// insertion operator <<
ostream& operator << (ostream& s, const Point& p)
  { return s << "(" << p.X() << ", " << p.Y() << ", "
             << p.Z() << ")" ; }

// extraction operator >>
istream& operator >> (istream& s, Point& p)
  { return s >> p.X() >> p.Y() >> p.Z() ; }
```

The above Point **class** is a non-**template** and non-exception-handling **class** in order to help focus our attention on design considerations relevant to inheritance. The following program extends the Point **class** by **public**ly deriving from it a NumberedPoint **class**:

```
// num_pt.cpp
// a numbered Point class
#include <iostream>  // C++ I/O
#include "point.h"   // Point class

class NumberedPoint : public Point
  {
  protected:
    int number ; // protected data member
  public:
    // constructors
    NumberedPoint ()
      : Point (), number (0) {}
    NumberedPoint (double x_arg, double y_arg,
```

```
                    double z_arg=0.0, int n=0)
      : Point (x_arg, y_arg, z_arg), number (n) {}
    NumberedPoint (double d, int n=0)
      : Point (d), number (n) {}
    // copy constructor
    NumberedPoint (const NumberedPoint& p)
      : Point (p), number (p.number) {}
    // member functions
    int& Number () { return number ; }
    //...
    NumberedPoint& operator = (const NumberedPoint& p) ;
  };

// overloaded operator:

// assignment operator =
inline NumberedPoint& NumberedPoint::operator =
                                    (const NumberedPoint& p)
  {
  Point::operator = (p) ;
  number = p.number ;
  return *this ;
  }

// insertion operator <<
inline ostream& operator << (ostream& s,
                             const NumberedPoint& p)
  {
  return s << "(" << p.X() << ", " << p.Y() << ", "
           << p.Z() << ", " << p.Number() << ")" ;
  }

// extraction operator >>
inline istream& operator >> (istream& s, NumberedPoint& p)
  {
  return s >> p.x >> p.y >> p.z >> p.number ;
  }

void main ()
  {
  Point p (1.0, 2.0) ;
  cout << "p: " << p << endl ;

  NumberedPoint np ;
  cout << "np: " << np << endl ;

  NumberedPoint nq (3.0, 4.0, 0.0, 1) ;
  cout << "nq: " << nq << endl ;
  cout << "nq.X(): " << nq.X () << endl ;
  cout << "nq.Number (): " << nq.Number () << endl ;
  }
```

with output:

```
p: (1, 2, 0)
np: (0, 0, 0, 0)
nq: (3, 4, 0, 1)
nq.X (): 3
nq.Number (): 1
```

NumberedPoint adds further functionality to Point by an additional **protected** data member called number, which assigns an integer number to a given Point. The number data member is declared **protected** to allow further classes to be derived from NumberedPoint while maintaining data protection. NumberedPoint defines similar constructors, including a copy constructor, to Point. For instance:

```
//...
NumberedPoint ()
  : Point (), number (0) {}
```

and the copy constructor of NumberedPoint invokes the Point copy constructor and initialises the number data member:

```
//...
NumberedPoint (const NumberedPoint& p)
  : Point (p), number (p.number) {}
```

Note that, when initialising the data members of a derived **class**, a base **class** constructor acts just as if it is a data member of the derived **class**. NumberedPoint adds a single access member function called *Number ()* for both constant and non-constant objects which simply returns the value of the number data member. The three overloaded operator functions =, << and >> of **class** Point must be redefined for **class** NumberedPoint. The **operator**= *()* function is overloaded for **class** NumberedPoint first calls the function Point::**operator**= *()*, passing a NumberedPoint object argument, then assigns object p's number data member to number, and finally returns a reference to the object operated on by the **operator**= *()* function. Also, the overloaded functions **operator**<< *()* and **operator**>> *()* are **friend** functions, and as a result must be redefined for **class** NumberedPoint, since they are not inherited.

You may be thinking that developing the derived **class** NumberedPoint has generated a great deal of extra unnecessary work when we could have simply added the number data member to **class** Point, plus the additional modifications to Point's constructors, member functions and overloaded **operator** *()* functions:

```
class Point
  {
  private:
    double x, y, z ;
    int    number ;
  public:
    Point ()
      : x (0.0), y (0.0), z (0.0), number (0) {}
    //...
  };
```

This is not always possible, and in certain cases is inadvisable. The `Point` **`class`** may have been supplied as part of a **`class`** library for which we do not have access to the implementation files. In modifying the `Point` **`class`** we lose the original structure of the **`class`** and may in the process of modification damage the original **`class`**. Also, there is a knock-on affect associated with modifying **`class`** `Point`, in that any previously developed classes which use the original `Point` **`class`** would have to be modified. These issues help to emphasise the key advantages of inheritance and of developing derived classes rather than modifying existing classes. Inheritance permits *code reusability* by allowing a programmer to add increased functionality to a **`class`** by reusing existing classes and at the same time guaranteeing that existing code remains unaltered. Aside from code reusability, inheritance assists a programmer in the conceptualisation of a problem and in structuring the design of a program. General low-level base classes can be developed which allow derived classes to add more and more functionality, through specialisation, to base classes.

15.15 A `Vector` `class` for Three-dimensional Space

Let us continue our examination of inheritance by developing a `Vector3D` **`class`** which is derived from **`class`** `Vector`. A `Vector` **`class`** has been discussed in some detail in previous chapters, with reference to the topics of pointers, templates and exception handling. However, the `Vector` **`class`** listed in the present section is a non-**`template`** and non-exception-handling **`class`** based on the **`class`** declared in VECTOR.H in Chapter 12. A non-**`template`**/exception-handling approach is adopted to help focus our attention on inheritance. Template classes `RVector` (range vectors) and `SortedVector` (sorted vectors) are presented towards the end of the chapter, when templates and inheritance are discussed.

Class `Vector` encapsulates an array of `n_elements` **`double`** floating-point numbers (`array[0],array[1],...,array[n_elements-1]`):

```
// vec.h
// header file for Vector class

#ifndef _VEC_H  // prevent multiple includes
#define _VEC_H

#include <iostream>  // C++ I/O
#include <iomanip>   // setprecision(),...
#include <cassert>   // assert()

class Vector
  {
  protected:
    double* array ;
    int     n_elements ;
    // member function
    void    Allocate (int n) ;
  public:
    // constructors
    Vector () ;
    Vector (int n, double initial_value=0.0) ;
```

```
    // copy constructor
    Vector (const Vector& v) ;
    // destructor
    ~Vector () { delete [] array ; array = 0 ;
                 n_elements = 0 ; }
    // member functions
    int           NumberOfElements () const
      { return n_elements ; }
    double&       Value (int index) ;
    const double& Value (int index) const ;
    void          New (int new_n) ;
    // overloaded operators
    Vector&       operator = (const Vector& v) ;
    double&       operator [] (int index) ;
    //...
  };

#endif // _VEC_H
```

with member function definitions given in VEC.CPP. Note that `Vector`'s data members are declared **`protected`** so as to make `Vector` *ready for* inheritance.

Although **`class`** `Vector` allows us to work in `n_elements`-dimensional space, we are frequently interested in using pairs or triples of numbers to locate points in a plane or three-dimensional space when performing geometric modelling. Hence a **`class`** `Vector3D` is **`public`**ly derived from **`class`** `Vector` which specifically caters for points in two- and three-dimensional space:

```
// vec3d.h
// header file class Vector3D

#ifndef _VEC3D_H  // prevent multiple includes
#define _VEC3D_H

#include <cmath>  // sqrt()
#include "vec.h"  // class Vector

class Vector3D : public Vector
  {
  public:
    // constructors
    Vector3D ()
      : Vector (3, 0.0) {}
    Vector3D (double arg1, double arg2, double arg3=0.0)
      : Vector (3)
      { array[0]=arg1; array[1]=arg2; array[2]=arg3;}
    // copy constructor
    Vector3D (const Vector3D& v)
      : Vector (v) {}
    // member functions
    double   Norm         () ;
    void     Normalise    () ;
```

```
    double   DotProduct   (const Vector3D& v) ;
    Vector3D CrossProduct (const Vector3D& v) ;
  };

#endif // _VEC3D_H
```

with member function definitions given in VEC3D.CPP:

```
// vec3d.cpp
// implementation file for Vector3D class

#include "vec3d.h"

const double TOLERANCE = 1e-06 ;

// returns the norm or magnitude (i.e. length) of a vector
double Vector3D::Norm ()
  {
  return sqrt ( array[0]*array[0] +
                array[1]*array[1] +
                array[2]*array[2] ) ;
  }

// returns a normalised vector
void Vector3D::Normalise ()
  {
  double n = this->Vector3D::Norm () ;
  assert (n > TOLERANCE) ;
  array[0]/=n ; array[1]/=n ; array[2]/=n ;
  }

// returns the dot product of two vectors
double Vector3D::DotProduct (const Vector3D& v)
  {
  return ( array[0]*v.array[0] +
           array[1]*v.array[1] +
           array[2]*v.array[2] ) ;
  }

// returns the cross product of two vectors
// (right-hand screw rule)
Vector3D Vector3D::CrossProduct (const Vector3D& v)
  {
  return Vector3D ( array[1]*v.array[2] - array[2]*v.array[1],
                    array[2]*v.array[0] - array[0]*v.array[2],
                    array[0]*v.array[1] -
                    array[1]*v.array[0] ) ;
  }
```

`Vector3D` adds no new data members to `Vector`, only member functions. Both of `Vector3D`'s constructors call the default `Vector` constructor, which encapsulates a vector

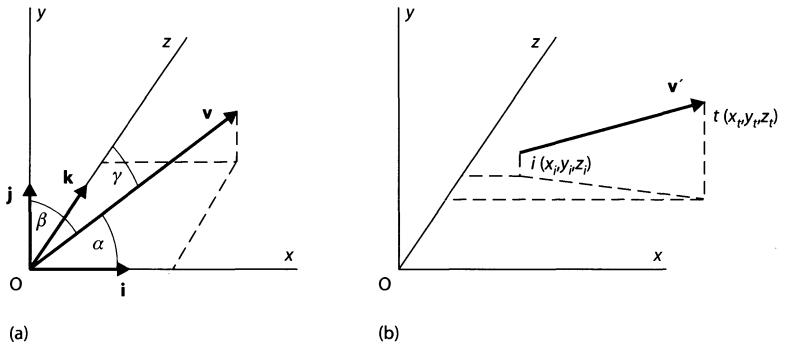


Fig. 15.3 (a) A vector **v** (a,b,c) in a three-dimensional space (x,y,z) with initial point at the global origin O. (b) A vector **v**′ $(x_t-x_i, y_t-y_i, z_t-z_i)$ in a three-dimensional space (x,y,z) with initial point i at (x_i,y_i,z_i) and terminal point t at (x_t,y_t,z_t).

with three components. The `Vector3D` copy constructor simply calls the `Vector` copy constructor, since `Vector3D` does not add additional data members to `Vector`. In addition, `Vector3D` does not define a destructor but calls, by default, the `Vector` destructor.

Note that the overloaded functions **`operator`**`= ()`, **`operator`**`+ ()`, **`operator`**`- ()` and **`operator`**`<< ()` are not specifically overridden for **`class`** `Vector3D`, since `Vector3D` adds no new data members to **`class`** `Vector`.

`Vector3D` defines four additional member functions, called `Norm()`, `Normalise()`, `DotProduct()` and `CrossProduct()`, for manipulating `Vector3D` objects. To understand the workings of these functions let's first examine an arbitrary vector **v**(a, b, c) in a three-dimensional space shown in Fig. 15.3(a). Constants a, b and c are the scalar components of **v**. Figure 15.3(a) shows the initial point of **v** to be at the origin O of the Cartesian coordinate system (x,y,z), although the initial point of a vector can, in general, be positioned arbitrarily: Fig. 15.3(b).

The length or *norm* of a vector **v**(a,b,c) follows from Pythagoras' theorem and is given by:

$$\|\mathbf{v}\| = (a^2 + b^2 + c^2)^{1/2}$$

Note the distinction between the norm of a vector, **v**, which is denoted by $\|\mathbf{v}\|$, and the absolute value of a scalar, s, which is denoted by $|s|$. A vector of norm 1 is called a **unit** vector, and Fig. 15.3(a) illustrates the three unit vectors **i**, **j** and **k** directed along the positive x-, y- and z-axes, respectively. Member function `Vector3D::Norm()` returns the norm of a `Vector3D` object.

Normalising a vector is performed by multiplying a non-zero norm vector, **v**, by the reciprocal of its norm to obtain a resultant vector of norm 1 or a unit vector:

$$\frac{1}{\|\mathbf{v}\|}\mathbf{v}$$

which is implemented by `Normalise()` and uses `assert()` to confirm that the `Vector3D` object operated on by the member function is of non-zero norm to eliminate division by zero.

Two important properties of vectors are the dot and cross product. The dot product of two non-zero norm vectors **u**(u_1, u_2, u_3) and **v**(v_1, v_2, v_3) is denoted by **u** · **v** and given by:

$$\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta \quad \text{for} \quad \|\mathbf{u}\| \neq 0 \quad \text{and} \quad \|\mathbf{v}\| \neq 0$$

where θ is the angle between **u** and **v**. In terms of components it is easily found that:

$$\mathbf{u}\cdot\mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3$$

The dot product of two `Vector3D` objects is implemented by *`DotProduct()`*. Above, the norm of a vector was determined in terms of the vector components. Alternatively, we could have defined the norm of a vector **v** in terms of a dot product $\|\mathbf{v}\| = (\mathbf{v}\cdot\mathbf{v})^{1/2}$, since $\mathbf{v}\cdot\mathbf{v} = \|\mathbf{v}\|^2$. The dot product of two non-zero norm vectors, **u** and **v**, is particularly useful for determining the angle, θ ($0 \le \theta \le \pi$), between **u** and **v**. Angle θ is acute if $\mathbf{u}\cdot\mathbf{v} > 0$, obtuse if $\mathbf{u}\cdot\mathbf{v} < 0$ and equal to $\pi/2$ if $\mathbf{u}\cdot\mathbf{v} = 0$. Thus, two vectors **u** and **v** are orthogonal (i.e. perpendicular) if and only if $\mathbf{u}\cdot\mathbf{v} = 0$.

The cross product of two vectors $\mathbf{u}(u_1,u_2,u_3)$ and $\mathbf{v}(v_1,v_2,v_3)$ is denoted by $\mathbf{u}\times\mathbf{v}$ and defined by:

$$\mathbf{u}\times\mathbf{v} = \left(\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}, -\begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix}, \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}\right) = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1)$$

Note that the cross product of two vectors **u** and **v** is a vector that is orthogonal to **u** and **v**. Also, it is worth noting that the cross product of two vectors is anticommutative; i.e. $\mathbf{u}\times\mathbf{v} = -\mathbf{v}\times\mathbf{u}$. The cross product of two `Vector3D` objects is defined by member function *`CrossProduct()`*.

VEC_TST.CPP illustrates typical uses of classes `Vector` and `Vector3D` and member functions `Vector3D::`*`DotProduct()`* and `Vector3D::`*`CrossProduct()`*:

```
// vec_tst.cpp
// tests base and derived classes Vector and Vector3D
#include <iostream>   // C++ I/O
#include "vec.h"      // Vector class
#include "vec3d.h"    // Vector3D class

void main ()
  {
  Vector v (5) ;
  cout << "Vector v: " << v << endl ;

  Vector3D v3d (1.0, 2.0), u3d (4.0, 3.0) ;
  cout << "Vector3D v3d: " << v3d << "; "
       << "Vector3D u3d: " << u3d << endl ;

  double   vu_dot   = v3d.DotProduct (u3d) ;
  Vector3D vu_cross = v3d.CrossProduct (u3d) ;

  cout << "v3d.u3d: " << vu_dot   << endl
       << "v3dxu3d: " << vu_cross << endl ;
  }
```

with output:

```
Vector v: [0.00, 0.00, 0.00, 0.00, 0.00]
Vector3D v3d: [1.00, 2.00, 0.00]; Vector3D u3d: [4.00, 3.00, 0.00]
uv3d.u3d: 10.00
v3dxu3d: [0.00, 0.00, -5.00]
```

Before examining `Triangle` and `Tetrahedra` classes, which make good use of `Vector3D` objects, let's revisit Fig. 15.3. The angles α, β and γ between vector $\mathbf{v}(a, b, c)$ and the unit vectors **i**, **j** and **k** are called the direction angles of **v**, and $\cos\alpha, \cos\beta$ and $\cos\gamma$ are the direction cosines of **v**, given by:

$$\cos\alpha = \frac{a}{\|\mathbf{v}\|}, \quad \cos\beta = \frac{b}{\|\mathbf{v}\|}, \quad \cos\gamma \frac{c}{\|\mathbf{v}\|}$$

and have the following property: $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$. In addition, the normalised unit vector $\mathbf{v}/\|\mathbf{v}\|$ is equal to:

$$\frac{\mathbf{v}}{\|\mathbf{v}\|} = (\cos\alpha, \cos\beta, \cos\gamma)$$

which proves useful when evaluating the unit normal vector to a plane or surface at a given point, since the components of the vector are in fact the direction cosines of the vector.

Although the functions *`Norm()`*, *`Normalise()`* and *`DotProduct()`* are specific to **`class`** `Vector3D`, they could equally be member functions of **`class`** `Vector`, since each function can be generalised for n-dimensional Euclidean space.

15.16 Containment Versus Inheritance

This section examines the development of `Triangle`, `Tetrahedra` and associated classes such as `Point` and `Line`. The `Tetrahedra` **`class`** is developed using the two different approaches of containment and inheritance. However, before we examine containment versus inheritance, let's first discuss the `Vector3D`, `Point` and `Line` classes used in the design of the `Triangle` and `Tetrahedra` classes. The `Vector3D` **`class`** used below is identical to the `Vector3D` **`class`** declared and implemented in VEC3D.H/.CPP. The `Point` **`class`** is similar to that of POINT.H/.CPP, but does add some extra operations:

```
// pt.h
// class Point

#ifndef _PT_H  // prevent multiple includes
#define _PT_H

#include <iostream>  // C++ I/O
#include <cmath>     // fabs(), sqrt()
#include "v3d.h"     // Vector3D class

class Point
  {
  protected:
    double x, y, z ; // protected data members
  public:
    // constructors
    Point ()
      : x (0.0), y (0.0), z (0.0) {}
```

```
    //...
    // public member functions
    double& X () { return x ; }
    //...
    double DistanceToOrigin () ;
    // overloaded operator
    Point&  operator = (const Point& p) ;
    double& operator [] (int n) ;
    //...
  };

#endif // _PT_H

// pt.cpp
// implementation file for Point class
#include "pt.h"  header file

const double TOLERANCE = 1e-06 ;

// returns distance from Point to origin
double Point::DistanceToOrigin ()
  { return sqrt (x*x + y*y + z*z) ; }

// assignment operator =
Point& Point::operator = (const Point& p)
  {
  x = p.x ; y = p.y ; z = p.z ;
  return *this ;
  }

// subscript operator [] (non-const objects)
double& Point::operator [] (int n)
  {
  assert (n >= 0 && n < 3) ;
  if (n == 0)
    return x ;
  else if (n == 1)
    return y ;
  else
    return z ;
  }

// subtraction operator -
Vector3D Point::operator - (const Point& p)
  {
  return Vector3D (x-p.x, y-p.y, z-p.z) ;
  }

// compares two Points
bool Point::operator == (const Point& p)
  {
```

```
    if (fabs(x-p.x)<TOLERANCE &&
        fabs(y-p.y)<TOLERANCE &&
        fabs(z-p.z)<TOLERANCE)
      return true ;
    return false ;
    }
//...
```

The `Point` **class** listed above overloads the subscript operator for both constant and non-constant `Point` objects, the subtraction operator and the == and != relational operators. Note that the subtraction of two `Point` objects is a `Vector3D` object. A `Line` **class** is as follows:

```
// line.h
// header file for Line class

#ifndef _LINE_H  // prevent multiple includes
#define _LINE_H

#include "pt.h"  // Point class

class Line
  {
  protected:
    Point p0, p1 ;
  public:
    // constructors
    Line ()
      : p0 (), p1 () {}
    //...
    // member functions
    Point&       P0 () { return p0 ; }
    Point&       P1 () { return p1 ; }
    double       Length () ;
    //...
    // overloaded operators
    Line&        operator = (const Line& l) ;
    Point&       operator [] (int n) ;
    //...
  };

#endif // _LINE_H

// line.cpp
// implementation file for class Line
#include "line.h"  // header file

// returns the length of a Line
double Line::Length ()
  {
  double x = p1.X () - p0.X () ;
```

```
    double y = p1.Y () - p0.Y () ;
    double z = p1.Z () - p0.Z () ;

    return sqrt (x*x + y*y + z*z) ;
    }

// subscript operator [] (non-const objects)
Point& Line::operator [] (int n)
    {
    assert (n >= 0 && n < 2) ;
    return n == 0 ? p0 : p1 ;
    }
```

The **class** Line encapsulates two Point data members which represent the end-points of a three-dimensional straight line segment.

15.16.1 Containment

A **class** declaration can encapsulate a data member which is an object of another **class**:

```
class X
    { /* ... */ };

class Y
    {
    //...
    X x ;  // object x of class X
    };
```

class Y *has an* X object. Such a relationship is frequently referred to as a *hasa* relationship. The **class** Line *hasa* relationship with **class** Point, since Line has two Point data members:

```
class Line
    {
    protected:
      Point p0, p1 ;
    //...
    };
```

Let's now examine the slightly more complicated Triangle **class**, which models an arbitrary planar triangle with straight-line edges in three-dimensional space:

```
// tri_con.h
// header file for Triangle class to illustrate containment

#ifndef _TRI_CON_H  // prevent multiple includes
#define _TRI_CON_H

#include "vec_tmp.h"  // class Vector
#include "pt.h"       // class Point
```

```
#include "line.h"      // class Line

class Triangle
  {
  private:
    Vector<Point> verts ;
    Vector<Line>  edges ;
    int           number ;
  public:
    // constructors
    Triangle ()
      : verts (), edges (), number (0) {}
    Triangle (const Point& p0, const Point& p1,
              const Point& p2, int n=0) ;
    // copy constructor
    Triangle (const Triangle& t)
      : verts(t.verts), edges(t.edges), number(t.number) {}
    // destructor
    ~Triangle () { number = 0 ; }
    // member functions
    int&      Number () { return number ; }
    Point&    Vertex (int n) ;
    Line&     Edge (int n) ;
    //...
    Point     Centroid () ;
    double    Perimeter () ;
    double    SurfaceArea () ;
    Vector3D Normal () ;
    // overloaded operator
    Triangle& operator = (const Triangle& t) ;
  };

#endif // _TRI_CON_H
```

`Triangle` declares three **private** data members: `verts`, `edges` and `number`. The `verts` data member is a `Vector` of three `Point` objects which represent the three vertices of a triangle. `edges` is a `Vector` of three `Line` objects which represent the three edges of a triangle. `number` is an integer constant assigned to a `Triangle` object and is zero by default. Note that it is not necessary to encapsulate both `Point` and `Line` data members within a `Triangle` **class**, and significant savings in memory can be made if we simply declare a `Vector` of `Point` objects. However, declaring also a `Vector` of `Line` objects greatly simplifies the operations performed on a `Triangle` object.

The member functions are defined as:

```
// tri_con.cpp
// implementation file for Triangle class
#include "tri_con.h"  // header file

Triangle::Triangle (const Point& p0, const Point& p1,
                    const Point& p2, int n)
  : verst (3), edges (3), number (n)
```

```
  {
  verts[0] = p0 ; verts[1] = p1 ; verts[2] = p2 ;
  edges[0].P0() = p0 ; edges[0].P1() = p1 ;
  edges[1].P0() = p1 ; edges[1].P1() = p2 ;
  edges[2].P0() = p2 ; edges[2].P1() = p0 ;
  }

// member functions:

Point& Triangle::Vertex (int n)
  {
  assert (n >= 0 && n < 3) ;
  return verts[n] ;
  }

Line& Triangle::Edge (int n)
  {
  assert (n >= 0 && n < 3) ;
  return edges[n] ;
  }
//...
// returns the centroid of a Triangle
Point Triangle::Centroid ()
  {
  Point cent ;

  cent.X () = (verts[0].X()+verts[1].X()+verts[2].X()) / 3.0 ;
  cent.Y () = (verts[0].Y()+verts[1].Y()+verts[2].Y()) / 3.0 ;
  cent.Z () = (verts[0].Z()+verts[1].Z()+verts[2].Z()) / 3.0 ;
  return cent ;
  }

// returns the perimeter of a Triangle
double Triangle::Perimeter ()
  {
  return (edges[0].Length () + edges[1].Length () +
          edges[2].Length ()) ;
  }

// returns the surface area of a Triangle
double Triangle::SurfaceArea ()
  {
  Vector3D u, v, cross ;
  // use vertex v0 as 'local' origin
  u = verts[1] - verts[0] ;
  v = verts[2] - verts[0] ;
  cross = u.CrossProduct (v) ;
  return cross.Norm () / 2.0 ;
  }

// returns the normalised normal of a Triangle
```

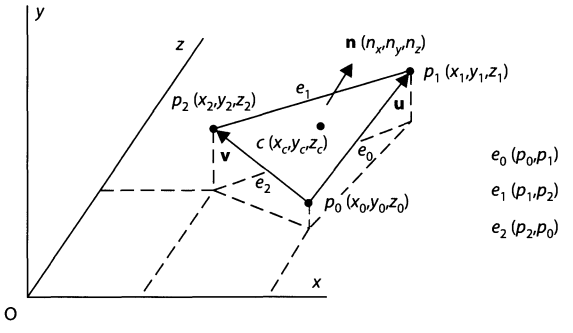


Fig. 15.4 An arbitrary planar triangle in a three-dimensional space.

```
Vector3D Triangle::Normal ()
  {
  Vector3D u, v, normal ;
  // use vertex v0 as 'local' origin
  u = verts[1] - verts[0] ;
  v = verts[2] - verts[0] ;
  normal = u.CrossProduct (v) ;
  normal.Normalise () ;
  return normal ;
  }
```

Member functions *Number()*, *Vertex()* and *Edge()* allow a user of **class** Triangle to get and set the number, vertices and edges of a Triangle object. Triangle also defines four geometric member functions called *Centroid()*, *Perimeter()*, *SurfaceArea()* and *Normal()* which allow basic operations to be performed on a Triangle object. Let's begin examining these member functions by considering the centroid of a triangle. The centroid $c(x_c,y_c,z_c)$ of a triangle defined by the three points $p_0(x_0,y_0,z_0)$, $p_1(x_1,y_1,z_1)$ and $p_2(x_2,y_2,z_2)$ is given by (see Fig. 15.4):

$$(x_c, y_c, z_c) = \left(\frac{x_0 + x_1 + x_2}{3}, \frac{y_0 + y_1 + y_2}{3}, \frac{z_0 + z_1 + z_2}{3} \right)$$

The perimeter of a triangle (Fig. 15.4), is simply the sum of the lengths of the three triangle edges e_0, e_1 and e_2. The area, A, of a triangle is one half the area of a parallelogram determined by two vectors **u** and **v** (Fig. 15.5):

$$A = \frac{1}{2}||\mathbf{u}||\,||\mathbf{v}||\sin\theta = \frac{1}{2}||\mathbf{u}\times\mathbf{v}||$$

In *SurfaceArea()* the first vertex of the Triangle object is used as a *local* origin to form two Vector3D objects u and v.

It was noted earlier that the cross product of two vectors **u** and **v** is another vector and is orthogonal to both **u** and **v**. Thus, the normal **n** (n_x, n_y, n_z) vector of a triangle is easily determined from the cross product of two vectors **u** $(p_1 - p_0)$ and **v** $(p_2 - p_0)$; see Fig. 15.4. Note that $-\mathbf{n} = \mathbf{v} \times \mathbf{u}$.

Before a normal vector is returned by *Normal()* it is first normalised ($\mathbf{n}/||\mathbf{n}||$) so that the components of the normal vector are the direction cosines of the vector. The normal vector to a

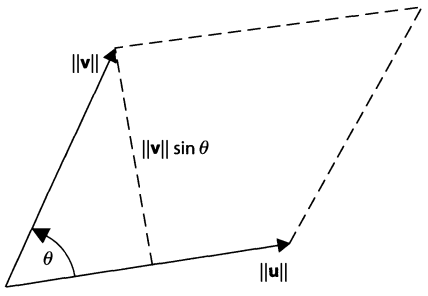


Fig. 15.5 A parallelogram formed by the two vectors **u** and **v**.

plane or surface at a point is conventionally written as a unit vector, since the magnitude of **n** is generally not required, only its direction.

It is worth noting in the above declaration and member function definitions just how much **class** `Triangle` relies on the classes `Point`, `Line`, `Vector3D` and how naturally they are used. Also, the above declaration/definition of `Triangle` helps to emphasise the importance of good robust design of classes (particularly *primitive* classes), because classes can quickly become highly integrated.

The `Tetrahedra` **class** declaration is:

```
// tet_con.h
// header file for Tetrahedra class to illustrate containment

#ifndef _TET_CON_H  // prevent multiple includes
#define _TET_CON_H

#include "tri_con.h"   // class Triangle

class Tetrahedra
  {
  private:
    Vector<Triangle> faces ;
    int              number ;
  public:
    // constructors
    Tetrahedra ()
      : faces (), number (0) {}
    Tetrahedra (const Triangle& t0, const Triangle& t1,
                const Triangle& t2, const Triangle& t3,
                int n=0) ;
    // copy constructor
    Tetrahedra (const Tetrahedra& t)
      : faces (t.faces), number (t.number) {}
    // destructor
    ~Tetrahedra () { number = 0 ; }
    // member functions
    int&       Number () { return number ; }
    Triangle& Face (int n) ;
    //...
```

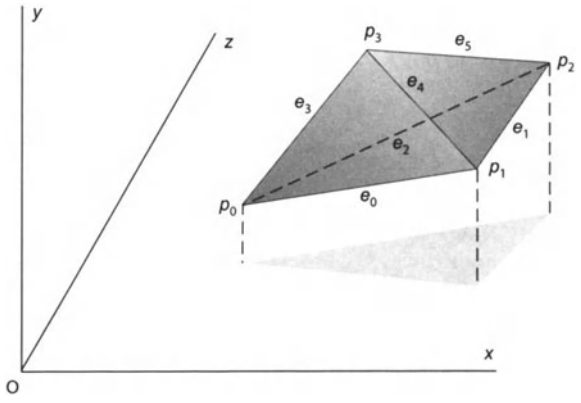


Fig. 15.6 An arbitrary tetrahedron in three-dimensional space.

```
    Vector<Point>     Vertices () ;
    Point             Centroid () ;
    double            Perimeter () ;
    double            SurfaceArea () ;
    double            Volume () ;
    // overloaded operator
    Tetrahedra& operator = (const Tetrahedra& t) ;
  };

#endif // _TET_CON_H
```

Figure 15.6 illustrates an arbitrary tetrahedron in three-dimensional space consisting of four vertices, six straight-line edges and four planar triangle faces. `Tetrahedra` models a three-dimensional tetrahedron by using containment to encapsulate a pointer data member, `faces`, to `Triangle`. The member function definitions of `Tetrahedra` are:

```
// tet_con.cpp
// implementation file for Tetrahedra class
#include "tet_con.h"  // header file

Tetrahedra::Tetrahedra (const Triangle& t0, const Triangle& t1,
                        const Triangle& t2, const Triangle& t3,
                        int n)
  : faces (4), number (n)
  {
  faces[0] = t0 ; faces[1] = t1 ;
  faces[2] = t2 ; faces[3] = t3 ;
  }

// member functions:

Triangle& Tetrahedra::Face (int n)
  {
  assert (n >= 0 && n < 4) ;
```

```
  return faces[n] ;
  }

// return a pointer to the four vertices of a Tetrahedra
Vector<Point> Tetrahedra::Vertices ()
  {
  Vector<Point> vertices (4) ;

  // use faces[0] as base Triangle
  vertices[0] = faces[0].Vertex (0) ;
  vertices[1] = faces[0].Vertex (1) ;
  vertices[2] = faces[0].Vertex (2) ;

  // find fourth vertex
  Point pt4 ;

  for (int i=0; i<3; i++)
    for (int j=0; j<3; j++)
    if ( (pt4 = faces[1].Vertex(i))!=(faces[0].Vertex(j)) )
      break ; // found 4th. vertex
  vertices[3] = pt4 ;

  return vertices ;
  }

// returns the centroid of a Tetrahedra
Point Tetrahedra::Centroid ()
  {
  Point         cent ;  // centroid
  Vector<Point> verts = Vertices () ; // get 4 vertices

  cent.X() = (verts[0].X()+verts[1].X()+verts[2].X()+
              verts[3].X())/4.0 ;
  cent.Y() = (verts[0].Y()+verts[1].Y()+verts[2].Y()+
              verts[3].Y())/4.0 ;
  cent.Z() = (verts[0].Z()+verts[1].Z()+verts[2].Z()+
              verts[3].Z())/4.0 ;
  return cent ;
  }

// returns the perimeter of a Tetrahedra
// uses vertices of a Tetrahedra to find the perimeter
// length since a Tetrahedra edge is common to 2 Triangles
double Tetrahedra::Perimeter ()
  {
  Vector<Point> verts = Vertices () ; // get 4 vertices

  double perm = (verts[1]-verts[0]).Norm () +
                (verts[2]-verts[0]).Norm () +
                (verts[3]-verts[0]).Norm () +
                (verts[3]-verts[1]).Norm () +
```

```
                (verts[2]-verts[1]).Norm () +
                (verts[3]-verts[2]).Norm () ;
  return perm ;
  }

// returns the surface area of a Tetrahedra
double Tetrahedra::SurfaceArea ()
  {
  return (faces[0].SurfaceArea () + faces[1].SurfaceArea () +
          faces[2].SurfaceArea () + faces[3].SurfaceArea ()) ;
  }

// returns the volume of a Tetrahedra
double Tetrahedra::Volume ()
  {
  Vector<Point> verts = Vertices () ; // get 4 vertices

  // vol=(1/6)|a.(bxc)|
  // use vertex v0 as 'local' origin
  return fabs ( (verts[1]-verts[0]).DotProduct ( (verts[2]-
                 verts[0]).CrossProduct (verts[3]-verts[0]) )
              ) / 6.0 ;
  }
```

Because `Tetrahedra` uses containment to model a tetrahedron as four `Triangle` objects, and each `Triangle` face shares its three `Point` vertices with the other three `Triangle` faces, a *`Vertices()`* member function is required which returns a `Vector` of the four `Point` vertices. *`Vertices()`* loops through the vertices of the first two faces of a `Tetrahedra` object to find the four unique vertices of a `Tetrahedra` object.

The centroid $c\,(x_c, y_c, z_c)$ of a tetrahedron defined by the four points $p_0\,(x_0, y_0, z_0)$, $p_1\,(x_1, y_1, z_1)$, $p_2\,(x_2, y_2, z_2)$ and $p_3\,(x_3, y_3, z_3)$ is given by:

$$(x_c, y_c, z_c) = \left(\frac{x_0 + x_1 + x_2 + x_3}{4}, \frac{y_0 + y_1 + y_2 + y_3}{4}, \frac{z_0 + z_1 + z_2 + z_3}{4} \right)$$

The perimeter length of a tetrahedron is the sum of its six edges. The surface area of a tetrahedron is easily found from the sum of the area its four triangular faces. `Tetrahedra` does not define a *`Normal()`* member function, since the normal vector is undefined for a tetrahedron. However, the normal of face `n` of a `Tetrahedra` object `tet_obj` is easily found:

```
tet_obj.Face (n).Normal () ;
```

A tetrahedron does have volume and this is determined from considering the volume of a parallelepiped determined by the three vectors **u**, **v** and **w**, shown in Fig. 15.7. The volume, *V*, of a tetrahedron determined by **u**, **v** and **w** is one-sixth the volume of the parallelepiped:

$$V = \frac{1}{6}\|\mathbf{v}\times\mathbf{w}\|\frac{|\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})|}{\|\mathbf{v}\times\mathbf{w}\|} = \frac{1}{6}|\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})|$$

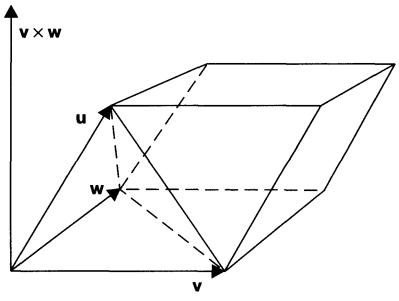


Fig. 15.7 A parallelepiped formed by the three vectors **u**, **v** and **w**.

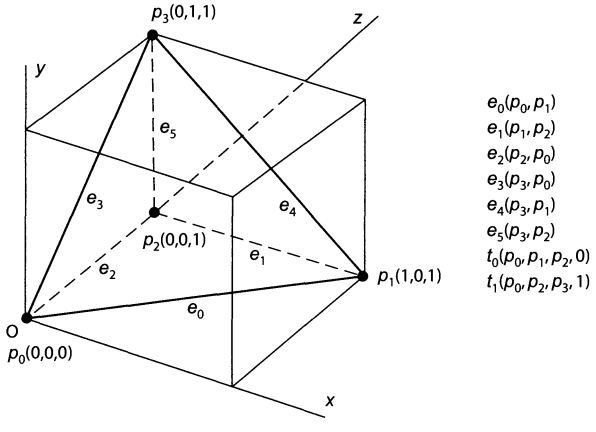


Fig. 15.8 A tetrahedron inside a unit cube.

`Point` and `Vector3D` classes allow a very concise definition of `Tetrahedra::Volume()`.

File TTT_CON.CPP tests the above `Triangle` and `Tetrahedra` classes for the tetrahedron described by the four points p_0, p_1, p_2 and p_3 shown in Fig. 15.8:

```
// ttt_con.cpp
// tests the Triangle and Tetrahedra classes
#include <iostream>    // C++ I/O
#include "pt.h"        // Point class
#include "tri_con.h"   // Triangle class
#include "tet_con.h"   // Tetrahedra class

void main ()
  {
  Triangle tri_def ;
  cout << "tri_def: " << endl << tri_def << endl ;

  Point p0 (0.0, 0.0, 0.0), p1 (1.0, 0.0, 1.0) ;
  Point p2 (0.0, 0.0, 1.0), p3 (0.0, 1.0, 1.0) ;
```

```
    Triangle t0 (p0, p1, p2, 0), t1 (p0, p2, p3, 1) ;
    Triangle t2 (p1, p2, p3, 2), t3 (p0, p1, p3, 3) ;


    Tetrahedra tet (t0, t1, t2, t3, 0) ;
    cout << "tet: " << endl << tet << endl
       << "tet centroid    : "  << tet.Centroid()      << endl
       << "tet.perimeter   : "  << tet.Perimeter ()    << endl
       << "tet.surface_area: "  << tet.SurfaceArea () << endl
       << "tet.volume      : "  << tet.Volume ()       << endl ;
    }
```

with output:

```
tri_def:
number   : 0
vertices: (0, 0, 0), (0, 0, 0), (0, 0, 0)
edges    : [(0, 0, 0), (0, 0, 0)], [(0, 0, 0), (0, 0, 0)],
           [(0, 0, 0), (0, 0, 0)]

tet:
number: 0
faces:
number   : 0
vertices: (0, 0, 0), (1, 0, 1), (0, 0, 1)
edges    : [(0, 0, 0), (1, 0, 1)], [(1, 0, 1), (0, 0, 1)],
           [(0, 0, 1), (0, 0, 0)]
number   : 1
vertices: (0, 0, 0), (0, 0, 1), (0, 1, 1)
edges    : [(0, 0, 0), (0, 0, 1)], [(0, 0, 1), (0, 1, 1)],
           [(0, 1, 1), (0, 0, 0)]
number   : 2
vertices: (1, 0, 1), (0, 0, 1), (0, 1, 1)
edges    : [(1, 0, 1), (0, 0, 1)], [(0, 0, 1), (0, 1, 1)],
           [(0, 1, 1), (1, 0, 1)]
number   : 3
vertices: (0, 0, 0), (1, 0, 1), (0, 1, 1)
edges    : [(0, 0, 0), (1, 0, 1)], [(1, 0, 1), (0, 1, 1)],
           [(0, 1, 1), (0, 0, 0)]
tet.centroid    : (0.25, 0.25, 0.75)
tet.perimeter   : 7.242264
tet.surface_area: 2.36603
tet.volume      : 0.166667
```

15.16.2 Inheritance

When a **class** Y is derived from a **class** X, Y is a kind of X, and such a relationship is frequently referred to as a *isa* relationship:

```
class X
  { /* ... */ };
```

```
class Y : public X
  { /* ... */ };
```

class Y *isa* kind of X. The **class** Vector3D *isa* relationship with **class** Vector since Vector3D is a kind of Vector:

```
class Vector3D : public Vector
  { /* ... */};
```

Let us now derive **class** Tetrahedra from **class** Triangle, thus making Tetrahedra a kind of Triangle. First, we need to modify the data members of Triangle from **private** to **protected** to enable them to be inherited:

```
class Triangle
  {
  protected:
    Vector<Point> verts ;
    Vector<Line>  edges ;
    int           number ;
  public:
    //...
  };
```

The declaration of Tetrahedra is now of the form:

```
class Tetrahedra : public Triangle
  {
  public:
    Tetrahedra (const Point& p0, const Point& p1,
                const Point& p2, const Point& p3, int n=0) ;
    //...
  };
//...
Tetrahedra::Tetrahedra (const Point& p0, const Point& p1,
                        const Point& p2, const Point& p3,
                        int n)
  : verts (4), edges (6), number (n)
  {
  //...
  }
```

Tetrahedra does not add any additional data members to those inherited from Triangle, and consequently does not need to define its own destructor but simply calls, by default, the destructor of Triangle.

15.16.3 Containment or Inheritance?

In view of the above, how do we choose between containment and inheritance? Unfortunately, there is no simple answer to this question, and generally the choice between containment and inheritance depends on the application. From a user's perspective, both approaches operate in

a similar manner and can generally be designed to operate identically. In the present case, it is tempting to assume that `Tetrahedra` is derived from `Triangle`. However, on closer examination we see that `Tetrahedra` is not a kind of `Triangle` but instead has four `Triangle` objects. In other words, `Tetrahedra` *hasa* relationship to `Triangle`.

A frequently adopted method of choosing between containment and inheritance is to consider whether or not a derived **`class`** can have two or more base **`class`** objects – the so-called 'can it have two?' question. If the answer is yes, then use containment rather than inheritance. In the case of a `Tetrahedra`, it is much easier to visualise a tetrahedron as having four triangular faces rather than a tetrahedron being a kind of triangle. Similarly, a `Line` was modelled as having two `Point` data members and a `Triangle` as having three `Points` and three `Lines`.

In addition, when choosing between containment and inheritance bear in mind the implicit conversion from a derived **`class`** to a base **`class`**. Clearly, a conversion from a `Tetrahedra` to a `Triangle` is meaningless.

15.17 Multiple Inheritance

So far, all of the derived classes discussed have been derived from a single base **`class`**. C++ also allows us to declare a **`class`** which is derived from multiple base classes. Deriving a **`class`** from more than one base **`class`** is referred to as *multiple inheritance*. Figure 15.9 schematically illustrates multiple inheritance in which **`class`** `Derived` is derived from classes `Base0` and `Base1`. The following program demonstrates multiple inheritance:

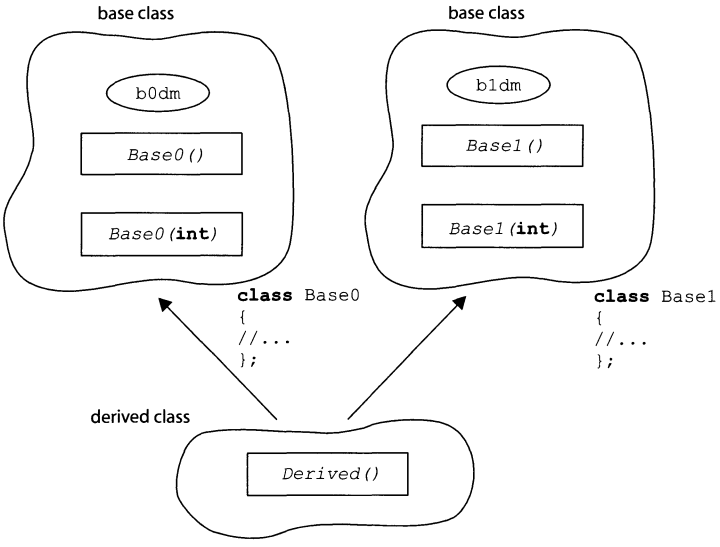


Fig. 15.9 A derived **`class`** inheriting the features of two base classes.

```
// mult_inh.cpp
// illustrates multiple inheritance
#include <iostream>  // C++ I/O

class Base0
  {
  protected:
    int b0dm ;
  public:
    Base0 ()
      : b0dm (0) {}
    Base0 (int i)
      : b0dm (i) {}
  };

class Base1
  {
  protected:
    int b1dm ;
  public:
    Base1 ()
      : b1dm (0) {}
    Base1 (int i)
      : b1dm (i) {}
  };

class Derived : public Base0, public Base1
  {
  public:
    Derived (int i)
      : Base0 (i), Base1 (i) {}
    friend ostream& operator << (ostream& s, const Derived& d)
      { return s << d.b0dm << " " << d.b1dm ; }
  };

void main ()
  {
  Derived d (1) ;
  cout << d ;
  }
```

with output:

```
1 1
```

The declaration of **class** Derived is:

```
class Derived : public Base0, public Base1
  { /* ... */ };
```

and is of the general form:

```
class DerivedClass : access_specifier BaseClass0,
                     ..., access_specifier BaseClassn
  { /* ... */ };
```

The multiple base classes are separated by the comma operator. Note that each base **class** inherited is accompanied by its own access specifier and not, for instance:

```
class Derived : public Base0, Base1
  { /* ... */ };
```

15.17.1 Constructors, Destructors and Multiple Inheritance

Let's now examine the execution of constructors and destructors for multiple inheritance more closely. First of all we'll declare an empty derived **class** with no constructor(s) or destructor explicitly defined:

```
// m_no_cd.cpp
// illustrates a class with no constructor and destructor
// derived from two base classes
#include <iostream>  // C++ I/O

class Base0
  {
  public:
    Base0 ()
      { cout << "Base0 class constructor called" << endl ; }
    ~Base0 ()
      { cout << "Base0 class destructor called"  << endl ; }
  };

class Base1
  {
  public:
    Base1 ()
      { cout << "Base1 class constructor called" << endl ; }
    ~Base1 ()
      { cout << "Base1 class destructor called"  << endl ; }
  };

class Derived : public Base0, public Base1
  {};

void main ()
  { Derived d_obj ; }
```

with output:

```
Base0 class constructor called
Base1 class constructor called
Base1 class destructor called
Base0 class destructor called
```

The program output illustrates that if a constructor and destructor are not defined for `Derived` then the appropriate constructor or destructor of the base classes `Base0` and `Base1` is called. The order in which the base **`class`** constructors are called matches the order in which they are specified in the derived **`class`**'s inheritance list. As with single inheritance, base **`class`** destructors are called in reverse order.

Let's now extend the above program by simply defining a constructor and destructor for **`class`** `Derived`:

```
// m_cd.cpp
// illustrates a class with a constructor and destructor
// derived from two base classes
#include <iostream>  // C++ I/O

class Base0
  {
  public:
    Base0 ()
      { cout << "Base0 class constructor called" << endl ; }
    ~Base0 ()
      { cout << "Base0 class destructor called"  << endl ; }
  };

class Base1
  {
  public:
    Base1 ()
      { cout << "Base1 class constructor called" << endl ; }
    ~Base1 ()
      { cout << "Base1 class destructor called"  << endl ; }
  };

class Derived : public Base0, public Base1
  {
  public:
    Derived ()
      { cout << "Derived class constructor called" << endl ; }
    ~Derived ()
      { cout << "Derived class destructor called"  << endl ; }
  };

void main ()
  { Derived d_obj ; }
```

with output:

```
Base0 class constructor called
Base1 class constructor called
Derived class constructor called
Derived class destructor called
Base1 class destructor called
Base0 class destructor called
```

The output illustrates that base **class** constructors are called in the order that they are specified in the derived **class** inheritance list prior to the derived **class** constructor, and vice versa for **class** destructors.

As with single inheritance, derived **class** constructor arguments can be passed to base **class** constructors using the following general syntax:

```
DerivedClass (argument_list)
   : BaseClass0 (argument_list), ..., BaseClassn (argument_list)
   { /* ... */ }
```

This is demonstrated in the following program:

```
// d_to_bs.cpp
// illustrates passing constructor argument objects
// from a derived class constructor to base class constructors
#include <iostream>  // C++ I/O

class Base0
   {
   protected:
      int b0dm ;
   public:
      Base0 ()
         : b0dm (0) {}
      Base0 (int i)
         : b0dm (i) {}
   };

class Base1
   {
   protected:
      int b1dm ;
   public:
      Base1 ()
         : b1dm (0) {}
      Base1 (int i)
         : b1dm (i) {}
   };

class Derived : public Base0, public Base1
   {
   public:
      Derived (int i, int j)
         : Base0 (i), Base1 (j) {}
   };

void main ()
   { Derived d_obj (0, 1) ; }
```

15.18 `virtual` Base Classes

When a **`class`** is derived from two or more base classes ambiguities can arise when accessing data members and member functions. Consider the following program:

```
// m_prob.cpp
// illustrates ambiguous accessing of data members by a
// derived class
#include <iostream>  // C++ I/O

class Base0
  {
  protected:
    int bdm ;
  public:
    void Display () { cout << bdm << endl ; }
  };

class Base1
  {
  protected:
    int bdm ;
  public:
    void Display () { cout << bdm << endl ; }
  };

class Derived : public Base0, public Base1
  {
  public:
    void Function (int i)
      {
      bdm = i ;          // error: bdm is ambiguous
      Base1::bdm = i ; // ok: ambiguity resolved
      }
  };

void main ()
  {
  Derived d_obj ;

  d_obj.Display () ;          // error: Display() is ambiguous

  d_obj.Base0::Display () ; // ok: ambiguity resolved
  d_obj.Base1::Display () ; // ok: ambiguity resolved
  }
```

Base classes `Base0` and `Base1` have a data member called `bdm` and a member function called *`Display()`*. The **`class`** `Derived` is derived from both base classes, `Base0` and `Base1`, and `Derived` does not override any of its base **`class`** members. The function `Derived::`*`Function()`* illustrates an ambiguous reference to the `bdm` data member, since

the compiler has no way of knowing which of the bdm data members of classes Base0 and Base1 is being referred to. Similarly, the object d_obj of **class** Derived calls the *Display()* member function – but which one? The solution to these access problems is through the use of the scope resolution operator. If ambiguity exists, simply prefix the data member or member function by the name of its associated **class**.

Let us consider another slightly different program in which access ambiguity can occur:

```
// m_prob1.cpp
// further illustrates ambiguous accessing
// of data members by a derived class
#include <iostream>  // C++ I/O

class Base
  {
  protected:
    int bdm ;
  public:
    void Display () { cout << bdm << endl ; }
  };

class Derived0 : public Base
  {};

class Derived1 : public Base
  {};

class Derived2 : public Derived0, public Derived1
  {};

void main ()
  {
  Derived2 d_obj ;

  d_obj.Display () ;     // error: Display() is ambiguous

  d_obj.Derived0::Display () ;  // ok: ambiguity resolved
  d_obj.Derived1::Display () ;  // ok: ambiguity resolved
  }
```

In this program, **class** Derived2 is derived from classes Derived0 and Derived1 which are in turn both derived from **class** Base. As a result, there are two copies of **class** Base present in an object of **class** Derived2, and unless the problem of scope is resolved the compiler will not know which copy of Base to use.

What if we required that **class** Derived2 did not inherit two copies of Base but only one? In this instance C++ allows a derived **class** to be derived from **virtual** base classes. Let's modify the above program M_PROB1.CPP by using **virtual** base classes to ensure that Derived2 contains only a single copy of Base:

```
// v_b_clss.cpp
// illustrates virtual base classes
#include <iostream>  // C++ I/O
```

```
class Base
  {
  protected:
    int bdm ;
  public:
    void Display () { cout << bdm << endl ; }
  };

class Derived0 : virtual public Base
  {};

class Derived1 : virtual public Base
  {};

class Derived2 : public Derived0, public Derived1
  {};

void main ()
  {
  Derived2 d_obj ;
  d_obj.Display () ;  // ok: unambiguous
  }
```

With the help of the keyword **virtual**, classes Derived0 and Derived1 declare **class** Base as **virtual** in each of their inheritance lists. Any multiple inheritance involving classes Derived0 and Derived1 will result in a single copy of Base being made in the derived **class**es Derived0 and Derived1.

15.19 virtual Functions

Previous chapters have examined function and operator overloading. These are examples of compile-time polymorphism in which a function name can operate in a variety of different ways. The term *polymorphism* refers to the general concept that a single name may denote a variety of different objects of different derived classes which are related by a common base **class**. Each object denoted by a common name will respond differently to a given set of instructions.

Run-time polymorphism exists when dynamic binding and inheritance combine. C++ supports run-time polymorphism through the use of pointers to derived classes and **virtual** functions[2]. Virtual functions allow a function of a base **class** to be redefined in a derived **class**.

Polymorphism is one of the most powerful features of an object-oriented programming language and is one of the key features that distinguishes an object-oriented language from other languages. Polymorphism allows a programmer to design a highly generalised **class** with operations that are common to all classes derived from the **class** and at the same time allow derived classes to add additional operations specific to the derived classes. Thus, polymorphism gives a programmer the opportunity to design a common interface based on a base

2 A **class** containing one or more **virtual** functions is often referred to as a *polymorphic* **class**.

class which dictates the general operation of the interface, while derived classes define multiple specialised operations. This is why polymorphism is frequently referred to as 'one interface, multiple methods'. An interface can be developed relatively independently of derived classes, and new derived classes can be easily added without affecting the overall design of the interface.

The use of **virtual** functions relies heavily on pointers, so let's begin by examining a program which accesses non-**virtual** member functions using pointers:

```
// non_virt.cpp
// illustrates the use of pointers to non-virtual member
// functions
#include <iostream>  // C++ I/O

class Base
  {
  public:
    void Display ()
      { cout << "Base::Display() called" << endl ; }
  };

class Derived0 : public Base
  {
  public:
    void Display ()
      { cout << "Derived0::Display() called" << endl ; }
  };

class Derived1 : public Base
  {
  public:
    void Display ()
      { cout << "Derived1::Display() called" << endl ; }
  };

void main ()
  {
  Base* b_ptr (0) ;

  Derived0 d0_obj ;
  Derived1 d1_obj ;

  b_ptr = &d0_obj ;  // b_ptr pointer points to d0_obj object
  b_ptr->Display () ;

  b_ptr = &d1_obj ;  // b_ptr pointer points to d1_obj object
  b_ptr->Display () ;
  }
```

with output:

```
Base::Display() called
Base::Display() called
```

The *main()* function of NON_VIRT.CPP defines a pointer to **class** Base, b_ptr, and two objects of classes Derived0 and Derived1, d0_obj and d1_obj. The memory address of object d0_obj is then assigned to the Base **class** pointer:

```
b_ptr = &d0_obj ;
```

This mixing of classes is a valid assignment in the present case because C++ allows a base **class** pointer to point to an object of a **class** derived from the base **class**. However, the output illustrates that although a base **class** pointer may point to objects of a derived **class**, the overridden member functions are still called according to the **class** or type of the pointer, and not what it points to.

A base **class** pointer can be used to access data members or member functions of a derived **class** by explicitly casting the base **class** pointer to the appropriate derived **class**. For instance, the Derived0::*Display()* member function could be accessed by b_ptr as:

```
((Derived0*)b_ptr)->Display () ;
```

An important point to note when using a pointer to a base **class** is that the pointer should generally not be incremented or decremented. This is because the pointer will be incremented or decremented relative to the base **class**, but the base **class** pointer could be pointing to a derived **class** object!

Let us now modify the above program by declaring the Base::*Display()* member function as a **virtual** member function by using the **virtual** keyword:

```
// virt.cpp
// illustrates the use of virtual member functions
#include <iostream>  // C++ I/O

class Base
  {
  public:
    virtual void Display ()
      { cout << "Base::Display() called" << endl ; }
  };

class Derived0 : public Base
  {
  public:
    void Display ()
      { cout << "Derived0::Display() called" << endl ; }
  };

class Derived1 : public Base
  {
  public:
    void Display ()
      { cout << "Derived1::Display() called" << endl ; }
```

```
};

void main ()
  {
  Base* b_ptr (0) ;

  Derived0 d0_obj ;
  Derived1 d1_obj ;

  b_ptr = &d0_obj ;  // b_ptr pointer points to d0_obj object
  b_ptr->Display () ;

  b_ptr = &d1_obj ;  // b_ptr pointer points to d1_obj object
  b_ptr->Display () ;
  }
```

with output:

```
Derived0::Display() called
Derived1::Display() called
```

The output now illustrates that the member function of the derived **class** pointed to by the base **class** pointer is called. The particular member function called is determined by the **class** or type of the object pointed to by the base **class** pointer. Thus, the following function-call statement:

```
b_ptr->Display () ;
```

is now capable of executing different functions with the same name, with each function performing its own set of operations. This is an example of run-time polymorphism. The type of polymorphism is run-time rather than compile-time because at the time of compilation the **class** or type operated on by the above function-call statement is undetermined.

A member function is declared as **virtual** in a base **class** using the **virtual** keyword:

```
class BaseClass
  {
  //...
  virtual return_type_specifier MemberFunction (/*...*/) ;
  //...
  };
```

When *MemberFunction()* is redefined in a **class** or **class**es derived from BaseClass, the **virtual** keyword is not repeated. Repeating the **virtual** keyword in an overriding derived **class** member function declaration is legal but redundant.

There are a number of important properties of **virtual** functions which are worth highlighting:

- **virtual** functions are not simply overloaded functions with the keyword **virtual** attached. An overloaded function must differ in either or both of its **class** or type and number of arguments. A redefined **virtual** function must have exactly the same **class** or type and number of arguments. If two member functions have different

class, type or number of arguments they are considered different functions and the **virtual** function mechanism is ignored. Generally, an overriding member function declaration cannot differ only by return type from an overridden member function. The exception to this rule is when the overridden **virtual** member function returns a pointer or reference to the base **class** and the overriding member function returns a pointer or a reference to the derived **class**.

- **virtual** functions are inherited and hierarchical irrespective of the depth of inheritance.
- A **virtual** function must be defined for a **class** in which it is first declared. The exception to this is when the function is declared as a *pure* virtual function. Pure **virtual** functions are discussed later.
- If a derived **class** does not override a **virtual** function then the **virtual** function defined in the base **class** is used.
- A **virtual** function must be a member function of a given **class** and not a **friend**.
- A **virtual** function can be declared a **friend** of another **class**.
- **virtual** functions cannot be **static** member functions.
- **virtual** functions can be declared as **inline**.
- Destructor functions can be **virtual** functions.
- Constructor functions cannot be **virtual** functions.

If we consider the first of the above list of points, the following program code illustrates declaring a function called *Display()* in a derived **class** which has a different number of arguments from the base **class** **virtual** function *Display()*:

```
class Base
  {
  //...
  virtual void Display () ;
  //...
  };

class Derived : public Base
  {
  //...
  // warning: Derived::Display() hides Base::Display()
  void Display (int i) ;
  };
```

Most compilers will issue a compilation warning message indicating that `Derived::`*Display() hides* the **virtual** function `Base::`*Display()*. However, the following is illegal because the overriding **virtual** function `Derived::`*Display()* conflicts with base **class** `Base`, owing to the two functions having different return types:

```
class Base
  {
  //...
  virtual void Display () ;
  //...
  };
```

```
class Derived : public Base
  {
  //...
  // error: Derived::Display() conflicts with
  // Base::Display()
  int Display () ;
  };
```

whereas the following program code is legal because `Base::Function()` is returning a reference to `Base` and `Derived::Function()` is returning a reference to `Derived`:

```
class Base
  {
  //...
  virtual Base& Display () ;
  //...
  };

class Derived : public Base
  {
  //...
  Derived& Display () ;  // ok
  };
```

To illustrate that **`virtual`** functions are hierarchical, consider the following program:

```
// v_hier.cpp
// illustrates that virtual functions are hierarchical
#include <iostream>  // C++ I/O

class Base
  {
  public:
    virtual void Display ()
      { cout << "Base::Display() called" << endl ; }
  };

class Derived0 : public Base
  {
  public:
    void Display ()
      { cout << "Derived0::Display() called" << endl ; }
  };

class Derived1 : public Derived0
  {};

void main ()
  {
  Base* b_ptr (0) ;
```

```
    Derived0 d0_obj ;
    Derived1 d1_obj ;

    b_ptr = &d0_obj ;  // b_ptr pointer points to d0_obj object
    b_ptr->Display () ;

    b_ptr = &d1_obj ;  // b_ptr pointer points to d1_obj object
    b_ptr->Display () ;
    }
```

with output:

```
Derived0::Display() called
Derived0::Display() called
```

The **class** `Derived1` is now derived from `Derived0` and not `Base`. The output illustrates that when object `d1_obj` is assigned to the `Base` **class** pointer and *`Display()`* is called, it is the function `Derived0::`*`Display()`* that is called and not `Base::`*`Display()`*. This is because `Derived0` is hierarchically closer to `Derived1` than is `Base`.

Generally, it is necessary to ensure that destructors are **virtual**. Consider the following program:

```
// v_dest.cpp
// illustrates the need for virtual destructors
#include <iostream>  // C++ I/O

class Base
  {
  private:
    int* int_p ;
  public:
    Base ()
      : int_p (new int[10]) {}
    ~Base ()
      {
      cout << "~Base()" << endl ;
      delete [] int_p ;
      }
  };

class Derived : public Base
  {
  private:
    int* int_q ;
  public:
    Derived ()
      : int_q (new int[10]) {}
    ~Derived ()
      {
      cout << "~Derived()" << endl ;
```

```
        delete [] int_q ;
        }
    };

void main ()
    {
    Base*  b_ptr (new Derived) ;
    delete b_ptr ;
    }
```

with output:

```
~Base ()
```

Both `Base` and `Derived` encapsulate pointers to **`int`** which allocate memory upon object definition. Although the base **`class`** pointer, `b_ptr`, is pointing to `Derived`, the call to **`delete`** will only invoke the `Base` destructor at run-time. As a consequence, the memory allocated by the `Derived` constructor will not be deallocated. If the `Base` destructor is made **`virtual`**, then at run-time the appropriate destructor will be called for the object pointed to by the base **`class`** pointer:

```
~Derived()
~Base()
```

15.19.1 Pure `virtual` Functions and Abstract Classes

If a **`virtual`** function is not defined in the base **`class`** in which it is first declared, then it is referred to as a *pure* **`virtual`** function. The declaration syntax of a pure **`virtual`** function is:

```
virtual return_type_specifier MemberFunction (/*...*/) = 0 ;
```

A **`virtual`** function is made pure by the initialiser `=0`. If a **`class`** declares one or more pure **`virtual`** functions it is referred to as an *abstract* **`class`**. It is illegal to define an object of an abstract **`class`**, although pointers of an abstract **`class`** can still be defined:

```
// pure_abs.cpp
// illustrates pure virtual functions and abstract classes
#include <iostream>  // C++ I/O

// abstract base class
class Base
    {
    public:
      virtual void Display () = 0 ; // pure virtual function
    };

class Derived0 : public Base
    {
    public:
```

```
    void Display ()
      { cout << "Derived0::Display() called" << endl ; }
  };

class Derived1 : public Base
  {
  public:
    void Display ()
      { cout << "Derived1::Display() called" << endl ; }
  };

void main ()
  {
  Base* b_ptr (0) ;

  Derived0 d0_obj ;
  Derived1 d1_obj ;

  Base b_obj ;    // error: abstract class objects not
                  // allowed

  b_ptr = &d0_obj ;  // b_ptr pointer points to d0_obj
                     // object
  b_ptr->Display () ;

  b_ptr = &d1_obj ;  // b_ptr pointer points to d1_obj
                     // object
  b_ptr->Display () ;
  }
```

The **class** Base is an abstract **class**, since Base::*Display()* is declared as a pure **virtual** function:

```
class Base
  {
  public:
    virtual void Display () = 0 ; // pure virtual function
  };
```

Note that a pure **virtual** function in a base **class** which is not defined by a derived **class** remains a pure **virtual** function in the derived **class** and therefore results in the derived **class** being an abstract **class**.

Pure **virtual** functions are useful in preventing a user from defining an object of a generic base **class**, since it is frequently meaningless to define an object of such a **class**.

To help discuss inheritance, **virtual** functions, pure **virtual** functions and abstract classes further, let's revisit an example discussed in Chapter 1 for developing a series of shape classes using inheritance. First, we require some primitive classes, such as Position, Colour and Size, to model the position, colour and size of a given shape:

```
class Position
  {
```

```
  protected:
    int x, y ;  // (x, y) coordinates in space
  public:
    Position () ;
    //...
  };

class Colour
  {
  protected:
    int red, green, blue ; // 3 primary components of colour
  public:
    Colour () ;
    //..
  };
class Size
  {
  protected:
    int bottom_left, bottom_right, top_left, top_right ;
  public:
    Size () ;
    //...
  };
```

`Position` encapsulates integer coordinates of a point in two-dimensional space. `Colour` encapsulates the red, green and blue primary components of light. `Size` encapsulates the size of an object as a two-dimensional, planar bounding box.

With classes such as `Position`, `Colour` and `Size` available we are able to develop a **`class`** which encapsulates the general properties of all shapes:

```
class FuzzyShape
  {
  protected:
    Colour   colour ;
    Size     size ;
    Position position ;
    //...
  public:
    // constructors & destructor
    FuzzyShape () ;
    //...
    virtual ~FuzzyShape () ;
    // member functions
    Position GetPosition () const ;
    void     SetPosition (const Position& p) ;
    //...
    // pure virtual member functions
    virtual void Draw () = 0 ;
    //...
  };
```

The FuzzyShape **class** captures the position, colour, size etc. features of all objects by declaring appropriate **protected** data members. Note that FuzzyShape is an abstract **class**, since member functions such as FuzzyShape : : *Draw( )* are declared as pure **virtual** member functions. Thus it is illegal and meaningless to define an object of **class** FuzzyShape because it defines pure **virtual** functions, and an object of FuzzyShape has no physical equivalent.

We are now in a position to derive specific shape classes from FuzzyShape using inheritance:

```
class Triangle : public FuzzyShape
  {
  protected:
    Position p0, p1, p2 ;
  public:
    Triangle () ;
    //...
    ~Triangle () ;
    void Draw () ;
    //...
  };

class Rectangle : public FuzzyShape
  {
  protected:
    double height, width ;
  public:
    Rectangle () ;
    //...
    ~Rectangle () ;
    void Draw () ;
    //...
  };

class Circle : public FuzzyShape
  {
  protected:
    double radius ;
  public:
    Circle () ;
    //...
    ~Circle () ;
    void Draw () ;
    //...
  };
//...
```

A typical interface could be:

```
//...
void Function ()
  {
```

```
FuzzyShape* fs_array[N] ;
//...
Triangle  t_obj ;
Rectangle r_obj ;
Circle    c_obj ;
//...
fs_array[0] = &t_obj ;
fs_array[1] = &r_obj ;
//...
fs_array[N-1] = &s_obj ;
//...
// draw all shape objects using a single function call
for (int i=0; i<N; i++)
  fs_array[i]->Draw () ;
//...
}
```

15.19.2 An Application

To further illustrate typical applications of single and multiple inheritance, non-pure and pure **virtual** functions, and abstract classes let's revisit the `Triangle`, `Tetrahedra` and related classes previously discussed. To begin with let's declare a `Shape` **class** which plays the role of base **class** to all shapes:

```
// shape_v.h
// header file for Shape class

#ifndef _SHAPE_V_H  // prevent multiple includes
#define _SHAPE_V_H

#include "pt.h"     // class Point

class Shape
  {
  protected:
    int number ;
  public:
    Shape ()
      : number (0) {}
    Shape (int n)
      : number (n) {}
    // destructor
    virtual ~Shape () { number = 0 ; }
    // member functions
    int& Number () { return number ; }
    //...
    // pure virtual member functions
    virtual Point  Centroid ()    = 0 ;
    virtual double Perimeter ()   = 0 ;
    virtual double SurfaceArea () = 0 ;
    virtual double Volume ()      = 0 ;
```

```
};

#endif // _SHAPE_V_H
```

Rather than simply encapsulating a number, **class** Shape could encapsulate a whole list of properties, such as colour, size, position and surface, as in the FuzzyShape example discussed in the previous subsection. The abstract **class** Shape declares four pure **virtual** functions, *Centroid()*, *Perimeter()*, *SurfaceArea()* and *Volume()*, since it is assumed that all shapes have these properties but that they are specific to each shape. Note the **virtual** destructor for Shape.

Another **class** which will act as a base **class** for numerous derived classes is **class** Plane:

```
// plane_v.h
// header file for Plane class

#ifndef _PLANE_V_H  // prevent multiple includes
#define _PLANE_V_H

#include "vec_tmp.h"  // class Vector
#include "pt.h"       // class Point

class Plane
  {
  protected:
    Vector3D normal ;     // A, B, C of plane
    double   d ;          // D of plane
  public:
    // constructors
    Plane ()
      : d (0.0) {}
    Plane (double a, double b, double c, double d) ;
    Plane (Vector<Point>& p_array) ;
    // copy constructor
    Plane (const Plane& p) ;
    // destructor
    ~Plane () ;
    // member functions
    Vector3D& Normal () { return normal ; }
    double&   D () { return d ; }
    // overloaded operator
    Plane& operator = (const Plane& p) ;
  };

#endif // _PLANE_V_H
```

with definitions residing in PLANE_V.CPP:

```
// plane_v.cpp
// implementation file for Plane class
```

```
#include "plane_v.h"  // header file

// Ax+By+Cz+D=0
Plane::Plane (double a, double b, double c, double d_arg)
  {
  normal[0] = a ; normal[1] = b ; normal[2] = c ;
  d = d_arg ;
  normal.Normalise () ;
  }

// plane through three points
Plane::Plane (Vector<Point>& p_array)
  {
  Vector3D u, v ;
  // use vertex v0 as 'local' origin
  u = p_array[1] - p_array[0] ;
  v = p_array[2] - p_array[0] ;

  normal = u.CrossProduct (v) ;
  normal.Normalise () ;
  d = - (p_array[0].X()*normal[0] +
         p_array[0].Y()*normal[1] +
         p_array[0].Z()*normal[2]) ;
  }

// overloaded assignment operator =
Plane& Plane::operator = (const Plane& p)
  {
  normal   = p.normal ;
  d        = p.d ;
  return *this ;
  }

// overloaded insertion operator <<
ostream& operator << (ostream& s, const Plane& p)
  {
  return s << "normal: " << p.Normal() << ", d: " << p.D() ;
  }
```

`Plane` encapsulates the characteristics of a general first-degree planar surface which is given by the following linear equation in x, y and z:

$$Ax + By + Cz + D = 0$$

where the constant coefficients A, B, C and D are specific to a particular plane in three-dimensional space. The normal vector, $\mathbf{n}$, to the plane is conveniently given in terms of the coefficients A, B and C ($\mathbf{n}(A, B, C)$), and if the normal is normalised then D is the distance from the origin to the nearest point on the plane; see Fig. 15.10. Since the coefficients A, B and C of a plane are conveniently stored in terms of the unit normal vector to the plane, the **`protected`** data members of **`class`** `Plane` are:

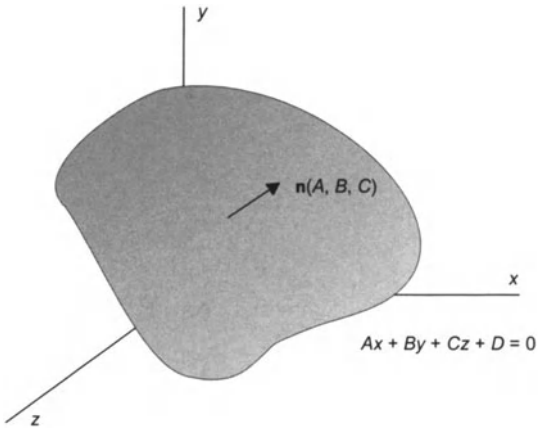


Fig. 15.10 A general planar surface in a three-dimensional Cartesian coordinate system.

```
class Plane
  {
  protected:
    Vector3D normal ;     // A, B, C of plane
    double   d ;          // D of plane
  //...
  };
```

and not:

```
class Plane
  {
  protected:
    double a, b, c, d ;     // A, B, C and D of plane
  //...
  };
```

The four- and one-argument constructors of `Plane` allow a plane to be defined explicitly in terms of the coefficients A, B, C and D or in terms of three points. It is worth highlighting that `Plane` is not derived from `Shape`, because an unbounded plane having an undefined centroid and infinite perimeter and surface area is not considered to be a *real* shape.

Now, let's declare a `Polygon` **`class`** to model a straight-edged plane polygon of arbitrary number of vertices in a three-dimensional space:

```
// poly_v.h
// header file for Polygon class

#ifndef _POLY_V_H  // prevent multiple includes
#define _POLY_V_H

#include "pt.h"        // class Point
#include "line.h"      // class Line
#include "shape_v.h"   // class Shape
```

```
#include "plane_v.h"  // class Plane

class Polygon : public Shape, public Plane
  {
  protected:
    Vector<Point> verts ;
    Vector<Line>  edges ;
  public:
    // constructors
    Polygon (int number_of_vertices=0)
      : Shape (0), Plane (),
        verts (number_of_vertices),
        edges (number_of_vertices) {}
    Polygon (Vector<Point>& p_array, int n=0) ;
    // copy constructor
    Polygon (const Polygon& p)
      : Shape (p), Plane (p),
        verts (p.verts), edges (p.edges) {}
    // destructor
    ~Polygon () {}
    // member functions
    int    NumberOfVertices () const
      { return verts.NumberOfElements() ; }
    Point& Vertex (int n) ;
    Line&  Edge (int n) ;
    // overridden member functions
    Point  Centroid () ;
    double Perimeter () ;
    double SurfaceArea () ;
    double Volume () ;
    //...
  };

#endif // _POLY_V_H
```

and implementation file:

```
// poly_v.cpp
// implementation file for Polygon class
#include "poly_v.h"  // header file

// constructor
Polygon::Polygon (Vector<Point>& p_array, int n)
  : Shape (n), Plane (p_array),
    verts (p_array.NumberOfElements()),
    edges (p_array.NumberOfElements())
  {
  int nverts = p_array.NumberOfElements() ;
  for (int i=0; i<nverts; i++)
    {
    verts[i]      = p_array[i] ;
```

```
    edges[i].P0() = p_array[i] ;
    edges[i].P1() = p_array[i+1] ;
    if (i == nverts-1) // final edge
      {
      verts[nverts-1]      = p_array[nverts-1] ;
      edges[nverts-1].P0() = p_array[nverts-1] ;
      edges[nverts-1].P1() = p_array[0] ;
      }
    }
  }

Point& Polygon::Vertex (int n)
  {
  assert (n >= 0 && n < verts.NumberOfElements()) ;
  return verts[n] ;
  }

Line& Polygon::Edge (int n)
  {
  assert (n >= 0 && n < verts.NumberOfElements()) ;
  return edges[n] ;
  }
//...
// returns the centroid of a Polygon
Point Polygon::Centroid ()
  {
  Point cent ;
  int nverts = verts.NumberOfElements() ;
  for (int i=0; i<nverts; i++)
    {
    cent.X () += verts[i].X () ;
    cent.Y () += verts[i].Y () ;
    cent.Z () += verts[i].Z () ;
    }
  return Point (cent.X()/nverts, cent.Y()/nverts,
                cent.Z()/nverts) ;
  }

// returns the perimeter of a Polygon
double Polygon::Perimeter ()
  {
  double perm (0.0) ;
  for (int i=0; i<verts.NumberOfElements(); i++)
    perm += edges[i].Length () ;
  return perm ;
  }

// returns the surface area of a convex or concave Polygon
double Polygon::SurfaceArea ()
  {
  Vector3D vi1v0, vi2v0, sum_vector ;
```

```
    for (int i=0; i<verts.NumberOfElements(); i++)
      {
      // use vertex v0 as 'local' origin
      vi1v0 = verts[i+1] - verts[0] ;
      vi2v0 = verts[i+2] - verts[0] ;
      sum_vector += vi1v0.CrossProduct (vi2v0) ;
      }
    return sum_vector.Norm () / 2.0 ;
    }

  // returns the volume of a Polygon
  double Polygon::Volume ()
    { return 0.0 ; }
  //...
```

`Polygon` steals many of its features from the `Triangle` **class**, discussed in previous sections, which was developed to model a straight-edge planar triangle in three-dimensional space. As we shall see shortly, a `Polygon` **class** greatly simplifies the development of a whole range of planar shapes, such as triangles and quadrilaterals. Also, by 'taking a step backwards' and developing a `Polygon` **class** we shall eliminate a great deal of repetition in program code, and more importantly help to categorise similar characteristics of different shapes. For instance, the *`Vertex()`* and *`Edge()`* member functions are not overloaded for classes `Triangle` and `Quadrilateral` but are inherited from `Polygon`.

Multiple inheritance is used to derive **class** `Polygon` from `Shape` and `Plane`:

```
  class Polygon : public Shape, public Plane
    {
    protected:
      Vector<Point> verts ;
      Vector<Line>  edges ;
    //...
    };
```

The **protected** data members `verts` and `edges` are `Vectors` of vertices and edges of a polygon. The default `Polygon` object has no vertices or edges and is described by the one-argument constructor, whereas the three-argument constructor describes a polygon assigned `p_array` vertices and integer number `n`.

The overridden member functions *`Centroid()`*[3], *`Perimeter()`* and *`Volume()`* are fairly straightforward and similar to those developed for **class** Triangle, but *`SurfaceArea()`* deserves a closer examination. With reference to Fig. 15.11(a) and `Triangle::`*`SurfaceArea()`* we know that the area, A, of a triangle described by the three non-collinear points p_0, p_1 and p_2 is given by:

$$A = \tfrac{1}{2}(p_1 - p_0) \times (p_2 - p_0)$$

where point p_0 is taken as a reference vertex. Similarly, the area of a general convex or concave polygon of n vertices is given by:

3 Beware of using the present algorithm (i.e. taking the mean of the x-, y- and z-coordinates of the polygon vertices) for determining the centroid of a polygon when clustering of vertices exists (Bashein and Detmer, 1994).

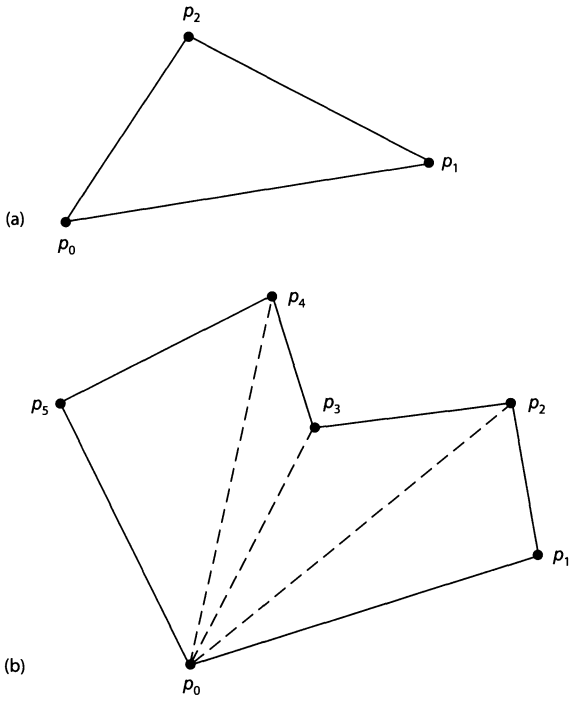


Fig. 15.11 (a) A three-vertex polygon. (b) A six-vertex concave polygon consisting of four triangles.

$$A = \frac{1}{2}\sum_{i=0}^{n-1}(p_{i+1} - p_0) \times (p_{i+2} - p_0)$$

since there are $n - 2$ signed triangular areas for an n-vertex polygon; see Fig. 15.11(b).

Deriving **class** `Triangle` from `Polygon` is now easily implemented as:

```
// tri_v.h
// header file for Triangle class

#ifndef _TRI_V_H  // prevent multiple includes
#define _TRI_V_H

#include "pt.h"       // class Point
#include "poly_v.h"   // class Plane

class Triangle : public Polygon
  {
  public:
    // constructors
    Triangle ()
      : Polygon (3) {}
    Triangle (Vector<Point>& p_array, int n=0)
      : Polygon (p_array, n) {}
```

```
    // copy constructor
    Triangle (const Triangle& t)
      : Polygon (t) {}
    // destructor
    ~Triangle () { Polygon::~Polygon () ; }
    //...
  };

#endif // _TRI_V_H
```

Note that the extraction operator is overloaded specifically for **class** `Triangle`, whereas **operator**<<`(ostream, Polygon)` is used by `Triangle`. A similar procedure is adopted for **class** `Quadrilateral`:

```
// quad_v.h
// header file for Quadrilateral class

#ifndef _QUAD_V_H  // prevent multiple includes
#define _QUAD_V_H

#include "pt.h"       // class Point
#include "poly_v.h"   // class Polygon

class Quadrilateral : public Polygon
  {
  public:
    // constructors
    Quadrilateral ()
      : Polygon (4) {}
    Quadrilateral (Vector<Point>& p_array, int n=0)
      : Polygon (p_array, n) {}
    // copy constructor
    Quadrilateral (const Quadrilateral& q)
      : Polygon (q) {}
    // destructor
    ~Quadrilateral ()  { Polygon::~Polygon () ; }
    //...
  };

#endif // _QUAD_V_H
```

A `Circle` **class** which encapsulates two data members `centre` and `radius` and derived from `Shape` and `Plane` is declared as:

```
// circ_v.h
// header file for Circle class

#ifndef _CIRC_V_H  // prevent multiple includes
#define _CIRC_V_H

#include <cmath>      // atan()
```

```
#include "pt.h"        // class Point
#include "shape_v.h"   // abstract class Shape
#include "plane_v.h"   // class Plane

class Circle : public Shape, public Plane
  {
  protected:
    Point  centre ;
    double radius ;
  public:
    Circle ()
      : Shape (), Plane (), centre (), radius (0.0) {}
    Circle (double a, double b, double c, double d,
            const Point& cent, double r, int n=0)
      : Shape (n), Plane (a, b, c, d),
        centre (cent), radius (r) {}
    // copy constructor
    Circle (const Circle& c)
      : Shape (c.number),
        Plane (c.normal[0], c.normal[1],
                            c.normal[2], c.d),
        centre (c.centre), radius (c.radius) {} ;
    // member functions
    double&       Radius () { return radius ; }
    Point&        Centre () { return centre ; }
    // overridden member functions
    Point  Centroid () ;
    double Perimeter () ;
    double SurfaceArea () ;
    double Volume () ;
    //...
  };

#endif // _CIRC_V_H
```

The four-argument constructor of `Circle` defines a `Circle` object via the A, B, C and D coefficients of a plane, a centre point, a radius and an assigned number. The definition file for **class** `Circle` is:

```
// circ_v.cpp
// implementation file for Circle class
#include "circ_v.h"  // header file

Point Circle::Centroid ()
  { return centre ; }

double Circle::Perimeter ()
  { return 2.0 * 4.0 * atan (1.0) * radius ; }

double Circle::SurfaceArea ()
  { return 4.0 * atan (1.0) * radius * radius ; }
```

```
double Circle::Volume ()
  { return 0.0 ; }
//...
```

The perimeter (circumference) and area of a circle are given by $2\pi r$ and πr^2 respectively, with $\pi = 4 \tan^{-1} 1$.

Finally, a `Tetrahedra` **class** is derived from `Shape` and uses containment to represent its four `Triangle` faces:

```
// tet_v.h
// header file for Tetrahedra class

#ifndef _TET_V_H  // prevent multiple includes
#define _TET_V_H

#include "shape_v.h"  // abstract class Shape
#include "tri_v.h"    // class Triangle

class Tetrahedra : public Shape
  {
  protected:
    Vector<Triangle> faces ;
  public:
    // constructors
    Tetrahedra ()
      : Shape (0), faces (4) {}
    Tetrahedra (const Triangle& t0, const Triangle& t1,
                const Triangle& t2, const Triangle& t3,
                int n=0) ;
    // copy constructor
    Tetrahedra (const Tetrahedra& t)
      : Shape (t), faces (t.faces) {}
    // destructor
    ~Tetrahedra () {}
    // member functions
    Triangle&       Face (int n) ;
    Vector<Point>   Vertices () ;
    // overridden member functions
    Point    Centroid () ;
    double   Perimeter () ;
    double   SurfaceArea () ;
    double   Volume () ;
    //...
  };

#endif // _TET_V_H
```

The implementation file for **class** `Tetrahedra` is similar to that of TTT_CON.CPP:

```
// tet_v.cpp
// implementation file for Tetrahedra class
```

```
#include "tet_v.h"  // header file

Tetrahedra::Tetrahedra (const Triangle& t0, const Triangle& t1,
                        const Triangle& t2, const Triangle& t3,
                        int n)
  : Shape (n), faces (4)
  {
  faces[0] = t0 ; faces[1] = t1 ;
  faces[2] = t2 ; faces[3] = t3 ;
  }

Triangle& Tetrahedra::Face (int n)
  {
  assert (n >= 0 && n < 4) ;
  return faces[n] ;
  }

// return a pointer to the four vertices of a Tetrahedra
Vector<Point> Tetrahedra::Vertices ()
  {
  Vector<Point> vertices (4) ;

  // use faces[0] as base Triangle
  vertices[0] = faces[0].Vertex (0) ;
  vertices[1] = faces[0].Vertex (1) ;
  vertices[2] = faces[0].Vertex (2) ;

  // find fourth vertex
  Point pt4 ;

  for (int i=0; i<3; i++)
    for (int j=0; j<3; j++)
    if ( (pt4 = faces[1].Vertex(i))!=(faces[0].Vertex(j)) )
      break ; // found 4th. vertex
  vertices[3] = pt4 ;

  return vertices ;
  }

// returns the centroid of a Tetrahedra
Point Tetrahedra::Centroid ()
  {
  Point         cent ;  // centroid
  Vector<Point> verts = Vertices () ; // get 4 vertices

  cent.X() = (verts[0].X()+verts[1].X()+verts[2].X()+
              verts[3].X())/4.0 ;
  cent.Y() = (verts[0].Y()+verts[1].Y()+verts[2].Y()+
              verts[3].Y())/4.0 ;
  cent.Z() = (verts[0].Z()+verts[1].Z()+verts[2].Z()+
              verts[3].Z())/4.0 ;
```

```
    return cent ;
    }

  // returns the perimeter of a Tetrahedra
  // uses vertices of a Tetrahedra to find the perimeter
  // length since a Tetrahedra edge is common to 2 Triangles
  double Tetrahedra::Perimeter ()
    {
    Vector<Point> verts = Vertices () ; // get 4 vertices
    double perm = (verts[1]-verts[0]).Norm () +
                  (verts[2]-verts[0]).Norm () +
                  (verts[3]-verts[0]).Norm () +
                  (verts[3]-verts[1]).Norm () +
                  (verts[2]-verts[1]).Norm () +
                  (verts[3]-verts[2]).Norm () ;
    return perm ;
    }

  // returns the surface area of a Tetrahedra
  double Tetrahedra::SurfaceArea ()
    {
    return (faces[0].SurfaceArea () + faces[1].SurfaceArea () +
            faces[2].SurfaceArea () + faces[3].SurfaceArea ()) ;
    }

  // returns the volume of a Tetrahedra
  double Tetrahedra::Volume ()
    {
    Vector<Point> verts = Vertices () ; // get 4 vertices
    // vol=(1/6)|a.(bxc)|
    // use vertex v0 as 'local' origin
    return fabs ( (verts[1]-verts[0]).DotProduct ( (verts[2]-
                   verts[0]).CrossProduct (verts[3]-verts[0]) )
                  ) / 6.0 ;
    }
  //...
```

Note that `Tetrahedra` could have been derived from a more general `Polyhedra` **class**.

The classes presented above form the **class** hierarchy shown in Fig. 15.12. A program which tests the run-time polymorphic properties of the above classes is given in SHAPES.CPP:

```
// shapes.cpp
// tests abstract base class Shape and derived classes

#include <iostream>    // C++ I/O
#include <cstdlib>     // exit()
#include "shape_v.h"   // abstract class Shape
#include "circ_v.h"    // class Circle
#include "quad_v.h"    // class Quadrilateral
#include "tri_v.h"     // class Triangle
```

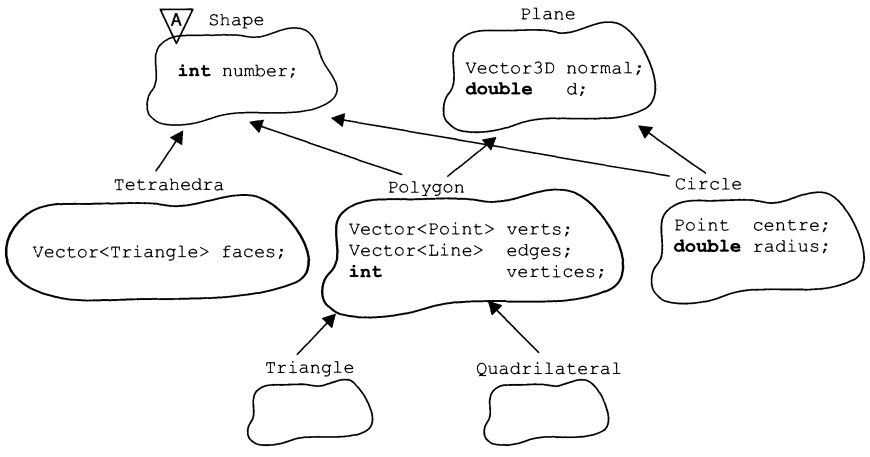


Fig. 15.12 Class hierarchy of the *Shapes* program.

```
void main ()
  {
  Shape*  s (0) ;
  Polygon* p (0) ;        Triangle* t (0) ;
  Quadrilateral* q (0) ; Circle* c (0) ;

  char ch ('x') ;

  cout << "enter c, p, q or t for circle, polygon,
           quadrilateral or triangle: " ;
  cin  >> ch ;

  switch (ch)
    {
    case 'c':
      c = new Circle ;
      cout << "enter centre properties: " ;
      cin  >> *c ;
      s = c ;
      break ;
    case 'p':
      p = new Polygon ;
      cout << "enter polygon properties: " ;
      cin  >> *p ;
      s = p ;
      break ;
    case 'q':
      q = new Quadrilateral ;
      cout << "enter quadrilateral properties: " ;
      cin  >> *q ;
      s = q ;
      break ;
    case 't':
```

```
        t = new Triangle ;
        cout << "enter triangle properties: " ;
        cin  >> *t ;
        s = t ;
        break ;
      default:
        cout << "enter c, p, q or t" << endl ;
        exit (EXIT_SUCCESS) ;
      }

    cout << "number    : " << s->Number ()       << endl
         << "centroid  : " << s->Centroid ()     << endl
         << "perimeter: " << s->Perimeter ()     << endl
         << "area      : " << s->SurfaceArea () << endl
         << "volume    : " << s->Volume ()       << endl ;
    //...
    }
```

A user is prompted to enter the properties of a circle, polygon, quadrilateral or triangle. The `cout` statement which displays the characteristics of a given shape makes use of a pointer to the abstract **`class`** `Shape`, and is thus independent of any specific shape. Some user interaction is:

```
enter c, p, q or t for circle, polygon, quadrilateral or
triangle: c
enter circle properties: 0 -1 0 1 1 0 0 1 5
number    : 5
centroid  : (1, 0, 0)
perimeter: 6.28319
area      : 3.14159
volume    : 0
```

which illustrates various properties for a circle lying in a plane characterised by the plane coefficients $A = 0, B = -1, C = 0$ and $D = 1$ with centre at the point $(1,0,0)$, radius 1 and number 5.

The classes presented above are far from ideal, but should have demonstrated just how powerful inheritance is in the C++ language and object-oriented programming, and how naturally the topic of computer graphics lends itself to inheritance. A possible modification and extension of the Shapes **`class`** hierarchy is shown in Fig. 15.13. By recognising that a polygon can have straight or curved edges and lie on either a planar or complex three-dimensional surface, **`class`** `Polygon` is now derived from `Shape`, with `PlanarPolygon` derived from `Polygon`. In addition to defining a polygon in terms of vertices and edges, you may need to associate a surface with the polygon. The surface could be defined either in terms of a surface constructed of objects of the Shapes **`class`** hierarchy or a surface defined by a given function. A `Mesh` **`class`** hierarchy is also shown in Fig. 15.13 to enable a Shapes surface to be constructed. `Mesh` encapsulates a pointer to abstract base **`class`** `Shape` so that an array of similar or different Shapes objects can be used to construct a mesh. A `Polyhedra` **`class`** has been inserted into the Shapes **`class`** hierarchy to model a solid body consisting of polygon faces. `Tetrahedra` is now derived from `Polyhedra`. A `SuperQuadrics` **`class`** has been added to the Shapes **`class`** hierarchy. Quadric surfaces include spheres, ellipsoids and tori. A superquadric is a generalisation of a quadric which allows complex surfaces and solids to be

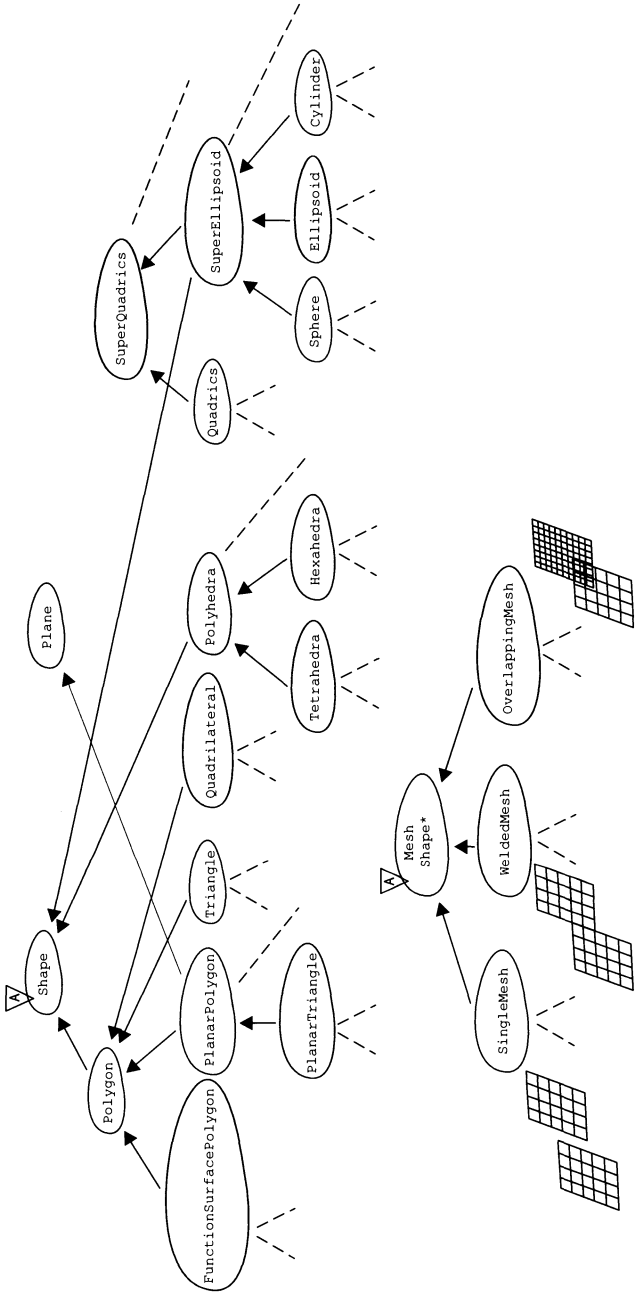


Fig. 15.13 Possible extension of the *Shapes* **class** hierarchy.

defined by incorporating additional parameters in a quadric equation. For instance, the equation of a superellipse is given by:

$$\left(\frac{x}{r_x}\right)^{2/s} + \left(\frac{y}{r_y}\right)^{2/s} = 1$$

where s is real and non-zero. When $s = 1$ and $rx \neq ry$ we have an ellipse, and when $s = 1$ and $rx = ry$ we have a circle.

I hope Fig. 15.13 has demonstrated just how complex **class** hierarchies can become. Even in its simplest implementation, modifying the **class** hierarchy shown in Fig. 15.12 to that of Fig. 15.13 requires a lot of work in restructuring **class** declarations, data members and member functions. Therefore – and this is very important – when designing and implementing a **class** hierarchy ensure that you spend a significant amount of time and effort in the design stage to make your **class** hierarchy as general and accommodating to change as possible.

15.20 Templates and Inheritance

The use of **template** classes and inheritance is similar to that of non-**template** classes and inheritance:

```
// tmp_inh.cpp
// illustrates the use of template classes and inheritance
#include <iostream>  // C++ I/O

class X
  {};

// template class derived from non-template base class
template <class T>
class A : public X
  {
  protected:
    T a_data ;
  public:
    A ()
      : a_data (T()) {}
  };

// non-template class derived from a template base class
class B : public A<int>
  {
  protected:
    int b_data ;
  public:
    B ()
      : A<int> (), b_data (0) {}
  };
```

```
// template class derived from template base class
template <class T>
class C : public A<T>
  {
  protected:
    T c_data ;
  public:
    C ()
      : A<T> (), c_data (T()) {}
  };

void main ()
  {}
```

Let's develop a `RVector` **`template class`** which is derived from **`template class`** `Vector`. `RVector` allows a user to specify an index range for the elements of an `RVector` object rather than relying on the 0:n – 1 rule:

```
// rv_tmp.h
// template class RVector
#ifndef _RV_TMP_H  // prevent multiple includes
#define _RV_TMP_H

#include <iostream>    // C++ I/O
#include "vec_tmp.h"   // class Vector

template <class T>
class RVector : public Vector<T>
  {
  private:
    int lowerbound ;
  public:
    RVector ()
      : Vector<T> (), lowerbound (0) {}
    RVector (int upper, int lower, T obj)
      : Vector<T> (upper-lower+1, obj),
                  lowerbound (lower) {}
    // member functions
    int Lower () const { return lowerbound ; }
    int Upper () const { return lowerbound +
                          NumberElements () - 1 ; }
    int& operator [] (int index)
      { return Vector<T>::operator [] (index - lowerbound) ; }
  };

#endif // _RV_TMP_H
```

A program which tests the above **`template class`** `RVector` is:

```
// rv_tst.cpp
// tests the template class RVector which
// is inherited from template class Vector
#include <iostream>  // C++ I/O
#include "rv_tmp.h"  // template class RVector

void main ()
  {
  RVector<int> rv (5, 1, 0) ; // indexing from 1!

  cout << "rv.first: " << rv[1] << ", rv.last: " << rv[5] ;
  }
```

with output:

```
rv.first: 0, rv.last: 0
```

Object `rv` of **`template class`** `RVector` has five elements of type **`int`**, whose indexes range from 1 to 5 rather than 0 to 4.

It was noted when we discussed non-**`template`** base and derived classes that C++ implicitly converts from a derived **`class`** object to a base **`class`** object, but not vice versa. A similar rule applies in the case of **`template`** base and derived classes providing both base and derived **`template`** classes operate on the same type or **`class`**:

```
// convert.cpp
// illustrates inheritance, template classes and conversion
#include <iostream>  // C++ I/O

class B
  {};

class D : public B
  {};

template <class T>
class TB
  {};

template <class T>
class TD : public TB<T>
  {};

void main ()
  {
  B b_obj ;
  D d_obj ;

  b_obj = d_obj ;      // ok: slice Derived to Base
  d_obj = b_obj ;      // error: can't convert Base to
                       // Derived
```

```
TB<int> tb_iobj ;
TD<int> td_iobj ;

tb_iobj = td_iobj ; // ok: slice TD<int> to TB<int>
td_obj  = tb_obj ;  // error: can't convert TB<int> to TD<int>

TD<float> td_fobj ;

tb_iobj = td_fobj ; // error: can't convert TD<float> to TB<int>
}
```

Let's now derive a **class** PtrVector<T> from Vector<**void***> which will enable the elements of an object of **class** PtrVector<T> to hold the memory addresses of pointers to type T:

```
// vec_ptr.h
// Vector of pointers template class

#ifndef _VEC_PTR_H  // prevent multiple includes
#define _VEC_PTR_H

#include <iostream>   // C++ I/O
#include "vec_tmp.h"  // class Vector

template <class T>
class PtrVector : public Vector<void*>
  {
  public:
    // constructor
    PtrVector (int i, T obj)
      : Vector<void*> (i, obj) {}
    // member function
    T*& Value (int index)
      {return (T*&)Vector<void*>::Value (index) ;}
    // overloaded operator
    T*& operator [] (int index)
      {return (T*&)Vector<void*>::operator [] (index) ;}
  }; // template class PtrVector

#endif // _VEC_PTR_H
```

Note the necessary casting in the member functions PtrVector<T>::*Value()* and PtrVector<T>::**operator**[] *()* from **void*** to T*.

A program which tests the **template class** VectorPtr is:

```
// ptrv_tst.cpp
// tests the template class PtrVector which
// is inherited from template class Vector
#include <iostream>   // C++ I/O
#include "vec_ptr.h"  // template class Vector
```

```
void  main ()
  {
  int  i (0) ;
  int* i_ptr = &i ;

  cout << "lvalue i     :" << &i
       << ", rvalue i     : " << i       << endl ;
  cout << "lvalue i_ptr:" << &i_ptr
       << ", rvalue i_ptr: " << i_ptr  << endl ;

  Vector<int*> piv (3, 0) ;
  cout << "piv : " << piv << endl ;
  piv[0] = i_ptr ;
  cout << "piv : " << piv << endl ;

  PtrVector<int*> pipv (3, 0) ;
  cout << "pivp: " << pipv << endl ;
  pipv[0] = &i_ptr ;
  cout << "pivp: " << pipv << endl ;
  }
```

The output of PTRV_TST.CPP is:

```
lvalue i     : 0012FF54, rvalue i     : 0
lvalue i_ptr: 0012FF50, rvalue i_ptr: 0012FF54
piv : [00000000, 00000000, 00000000]
piv : [0012FF54, 00000000, 00000000]
pivp: [00000000, 00000000, 00000000]
pivp: [0012FF50, 00000000, 00000000]
```

As demonstrated in Chapter 13, the elements of a `Vector<T*>` object are the rvalues of pointers to type `T`. The elements of a `PtrVector<T*>` object are the lvalues of pointers to type `T`.

15.20.1 A `SortedVector` class

As a further illustration of the combined use of templates and inheritance, let's examine a `SortedVector` **template class** which is **public**ly derived from **template class** `Vector` and maintains a sorted array of elements:

```
// vec_sort.h
// template class SortedVector

#ifndef _VEC_SORT_H  // prevent multiple includes
#define _VEC_SORT_H

#include <iostream>  // C++ I/O
#include <iomanip>   // setprecision(),...
#include <cassert>   // assert()

// swaps two objects
```

```
template <class T>
void Swap (T& a, T& b)
  {
  T temp ;
  temp = a ;
  a = b ;
  b = temp ;
  }

template <class T>
class Vector
  {
  //...
  public:
    //...
    void Sort () const ;
    int  Find (const T& obj) const ;
    //...
  };
//...
// sorts a Vector object (performs a bubble sort)
template <class T>
void Vector<T>::Sort () const
  {
  for (int i=0; i<n_elements-1; i++)
    for (int j=n_elements-1; i<j; j--)
      if (array[j] < array[j-1])
        Swap (array[j], array[j-1]) ;
  }

// returns the first index of a given Vector object, else -1
// linear search
template <class T>
int Vector<T>::Find (const T& obj) const
  {
  for (int i=0; i<n_elements; i++)
    if (array[i] == obj)
      return i ;
  return -1 ;
  }
//...
// template class SortedVector
template <class T>
class SortedVector : public Vector<T>
  {
  public:
    // constructors
    SortedVector ()
      : Vector<T> () {}
    SortedVector (int n, T obj)
      : Vector<T> (n, obj) {}
```

```
    SortedVector (const Vector<T>& v)
      : Vector<T> (v) { Sort () ; }
    // copy constructor
    SortedVector (const SortedVector& sv)
      : Vector<T> (sv) {}
    // member functions
    void Add  (const T& obj) ;
    int  Find (const T& obj) const ;
  };

// adds an element to SortedVector<T>
template <class T>
void SortedVector<T>::Add (const T& obj)
  {
  // increase no. of elements by 1
  New (n_elements+1, T(), true) ;

  int insert (0) ;
  while (obj > array[insert] && insert < n_elements-1)
    insert++ ;
  if (insert < n_elements-1)
    for (int i=n_elements-1; i>insert; i--)
      array[i] = array[i-1] ;
  array[insert] = obj ;
  }

// returns the first index of a given Vector object, else -1
// binary search
template <class T>
int SortedVector<T>::Find (const T& obj) const
  {
  int first (0), last (n_elements - 1), middle (0) ;

  while (first <= last)
    {
    middle = (first + last) / 2 ;

    if (obj == array[middle])  // found & return index
      return middle ;
    else if (obj < array[middle])
      last = middle - 1 ;
    else
      last = middle + 1 ;
    }
  return -1 ;  // not found
  }

#endif // _VEC_SORT_H
```

Two new member functions, *Sort()* and *Find()*, have been added to **template class** Vector. The *Sort()* member function uses the bubblesort algorithm to sort the elements of

a `Vector`, while *`Find()`* performs a linear search to return the first index of a `Vector` object element which matches a supplied object.

`SortedVector` supports a default constructor, a two-argument constructor to initialise all the elements of a `SortedVector` object to a given object and a one-argument constructor which enables a `SortedVector` object to be initialised using the elements of a `Vector` object:

```
SortedVector (const Vector<T>& v)
  : Vector<T> (v) { Sort () ; }
```

The constructor calls the inherited `Vector<T>::`*`Sort()`* function to ensure that upon definition of a `SortedVector` object that the object's elements are sorted.

The `SortedVector<T>::`*`Add()`* member function is slightly more tricky in that it allows an element to be added to a `SortedVector` object while simultaneously maintaining a sorted array of object elements. First, the size of a `SortedVector` object is increased to accommodate the addition of a new element. Then the point of insertion of the new element is determined, and if this insertion point is not the last element position the existing element objects are transferred to their respective new positions.

The `SortedVector<T>::`*`Find()`* member function overrides the `Vector<T>::`*`Find()`* member function so that a binary search rather than a linear search algorithm can be used when searching for a given object in a sorted list of elements. The binary search algorithm is a popular algorithm for searching a sorted list and begins by comparing the search object with the element in the middle of the list of elements. If the search object is less than or greater than the middle list element the procedure is repeated for the first or second half sublists, respectively, of the list until the search object is either found or we establish that the search object is not present in the list. The binary search algorithm is much faster than a linear search algorithm, since the number of remaining elements to be searched in the list is approximately halved for each iteration[4]. For a more detailed discussion of the bubblesort and binary search algorithms in C++, refer to Budd (1994) and Horowitz *et al.* (1995).

The **`template class`** `SortedVector` is tested in the following program:

```
// sort_tst.cpp
// tests the template class SortedVector which is
// inherited from template class Vector

#include "vec_sort.h"  // template class SortedVector

void main ()
  {
  SortedVector<int> sv (1, 0) ;
  cout << "sv: " << sv << endl ;

  sv.Add (5) ;
  cout << "sv: " << sv << endl ;
  sv.Add (10) ;
  cout << "sv: " << sv << endl ;
  sv.Add (2) ;
```

4 It can be shown that the worst-case running time of the binary search algorithm is of the order $\log_2 n$ for n elements in the list to be searched.

```
    cout << "sv: " << sv << endl ;
    sv.Add (-1) ;
    cout << "sv: " << sv << endl ;

    Vector<int> v (5, 0) ;
    v[0] = 3 ; v[1] = 4 ; v[2] = 2 ; v[3] = 0 ; v[4] = 1 ;
    cout << "v: " << v << endl ;

    SortedVector<int> sv1 (v) ;
    cout << "sv1: " << sv1 << endl ;

    cout << "sv1.Find (2): " << sv1.Find (2) << endl ;
    }
```

with output:

```
sv: [0]
sv: [0, 5]
sv: [0, 5, 10]
sv: [0, 2, 5, 10]
sv: [-1, 0, 2, 5, 10]
v: [3, 4, 2, 0, 1]
sv1: [0, 1, 2, 3, 4]
sv1.Find (2): 2
```

The function body of *main()* first defines an object, sv, of **class** SortedVector which consists of one element of type **int**. Next, four additional **int** elements are added to sv using the SortedVector<T>::*Add()* member function. An object, sv1, of **class** SortedVector is then defined and initialised to the Vector object v. The output of sv1 illustrates that directly after definition the elements of object sv1 are sorted in increasing order. Finally, *main()* accesses the third element of sv1 using the SortedVector<T>::*Find()* member function.

15.20.2 Memory, LocalMemory and GlobalMemory Classes for Windows

In Chapters 12 (WIN_MEM.CPP) and 13 (WIN_TMP.CPP) we examined a GlobalMemory **class** for allocating global memory when programming in a Windows environment. For allocating global memory we saw that the Windows Application Programming Interface (API) supports functions such as *GlobalSize()*, *GlobalAlloc()*, *GlobalReAlloc()*, *GlobalLock()*, *GlobalUnlock()* and *GlobalFree()*. The Windows API similarly supports functions for managing local memory blocks, such as *LocalSize()* and *LocalAlloc()*. Therefore it would be nice to declare a LocalMemory **class** with similar functionality to that of GlobalMemory, but for local memory.

Recognising the similarity between the Windows API functions for local and global memory and that objects of LocalMemory and GlobalMemory both require encapsulated memory handle and address data members then it seems natural to develop a base **class** Memory from which LocalMemory and GlobalMemory are derived:

```
// win_lgm.cpp
// illustrates local and global memory template classes
// for programming in a Windows environment
```

```
#include <iostream>  // C++ I/O
#include <windows>   // Windows header file

template <class T>
class Memory
  {
  protected:
    HANDLE    hMemory ;        // handle to memory
    void FAR* mem_address ;    // address of memory
  public:
    // constructor
    Memory ()
      : hMemory (0), mem_address (0) {}
    // destructor
    ~Memory () {}
    // non-virtual member function
    void FAR* MemoryAddress () const { return mem_address ; }
    // virtual member functions
    virtual DWORD     SizeOfBlock () = 0 ;
    virtual void FAR* Allocate   (UINT size,
            UINT mem_flags=LMEM_MOVEABLE|LMEM_ZEROINIT) = 0 ;
    virtual void FAR* ReAllocate (UINT size,
            UINT mem_flags=LMEM_MOVEABLE|LMEM_ZEROINIT) = 0 ;
  };

template <class T>
class LocalMemory : public Memory<T>
  {
  public:
    // constructor
    LocalMemory ()
      : Memory<T> () {} ;
    // destructor
    ~LocalMemory () ;
    // overridden member functions
    DWORD     SizeOfBlock () ;
    void FAR* Allocate   (UINT size,
        UINT mem_flags=LMEM_MOVEABLE|LMEM_ZEROINIT) ;
    void FAR* ReAllocate (UINT size,
        UINT mem_flags=LMEM_MOVEABLE|LMEM_ZEROINIT) ;
  };

template <class T>
class GlobalMemory : public Memory<T>
  {
  public:
    // constructor
    GlobalMemory ()
      : Memory<T> () {} ;
```

```
    // destructor
    ~GlobalMemory () ;
    // overridden member functions
    DWORD     SizeOfBlock () ;
    void FAR* Allocate   (UINT size,
        UINT mem_flags=GMEM_MOVEABLE|GMEM_ZEROINIT) ;
    void FAR* ReAllocate (UINT size,
        UINT mem_flags=GMEM_MOVEABLE|GMEM_ZEROINIT) ;
  };

// LocalMemory:

// destructor
template <class T>
LocalMemory<T>::~LocalMemory ()
  {
  LocalUnlock (hMemory) ;    // unlock
  if (hMemory)               // free
    LocalFree (hMemory) ;
  }

// return size of memory block
template <class T>
DWORD LocalMemory<T>::SizeOfBlock ()
  {
  if (!hMemory)
    return 0 ;
  else
    return LocalSize (hMemory) ;
  }

// allocate memory
template <class T>
void FAR* LocalMemory<T>::Allocate (UINT size, UINT mem_flags)
  {
  hMemory = LocalAlloc (mem_flags, size*sizeof(T)) ;

  // if unsuccessful allocation
  if (!hMemory)
    {
    MessageBox (GetActiveWindow (),
        (LPCSTR) "insufficient memory",
        (LPCSTR) "LocalMemory Warning",
        MB_ICONEXCLAMATION | MB_OK) ;
    hMemory = 0 ;
    return 0 ;
    }
  mem_address = LocalLock (hMemory) ;
  return mem_address ;
  }
```

```
// re-allocate memory
template <class T>
void FAR* LocalMemory<T>::ReAllocate (UINT size, UINT mem_flags)
  {
  HANDLE h = LocalReAlloc (hMemory, size*sizeof(T), mem_flags) ;

  // if unsuccessful allocation
  if (!h)
    {
    MessageBox (GetActiveWindow (),
        (LPCSTR) "insufficient memory",
        (LPCSTR) "LocalMemory Warning",
        MB_ICONEXCLAMATION | MB_OK) ;
    return 0 ;
    }
  else
    {
    hMemory = h ;
    mem_address = LocalLock (hMemory) ;
    }
  return mem_address ;
  }

// GlobalMemory:
//...
void main ()
  {
  const int SIZE = 10 ;

  // pointer to abstract base class Memory<T>
  Memory<int>* mem ;

  LocalMemory<int>  l_mem ; // local memory object
  GlobalMemory<int> g_mem ; // global memory object

  //mem = &g_mem ;
  mem = &l_mem ;  // point to local memory object

  // allocate memory
  int* px = (int*) mem->Allocate (SIZE) ;

  // set
  for (int i=0; i<SIZE; i++)
    *(px+i)=i ;

  // get
  for (int j=0; j<SIZE; j++)
    cout << *(px+j)  << " " ;
  cout << endl ;
  cout << "size: " << mem->SizeOfBlock () << endl ;
```

```
// re-allocate memory
px = (int*) mem->ReAllocate (2*SIZE) ;

// get
for (int k=0; k<2*SIZE; k++)
  cout << *(px+k)  << " " ;
cout << endl ;
cout << "size: " << mem->SizeOfBlock () << endl ;
}
```

with output:

```
0 1 2 3 4 5 6 7 8 9
size: 40
0 1 2 3 4 5 6 7 8 9 0 0 0 0 0 0 0 0 0 0
size: 80
```

The memory handle and address data members, `hMemory` and `mem_address`, have been transferred from the `GlobalMemory` classes presented in previous chapters to the `Memory` **`template class`**. The member function `Memory<T>::MemoryAddress()` is declared non-**`virtual`**, since its operation is common to both derived classes. The member functions `SizeOfBlock()`, `Allocate()` and `ReAllocate()` of `Memory<T>` are, however, declared **`virtual`** because their use of the API memory functions is different for `LocalMemory<T>` and `GlobalMemory<T>`. Further, `Memory<T>` is an abstract base **`template class`**, since a memory object in Windows can be either local or global.

The function body of `main()` in the above program WIN_LGM.CPP is similar to that presented WIN_MEM.CPP and WIN_TMP.CPP of Chapters 12 and 13, except that the member functions now operate on a pointer to the abstract base **`template class`** `Memory` rather than operating directly on objects of classes `LocalMemory<T>` and `GlobalMemory<T>`.

15.20.3 The `Matrix` `class` Revisited

Chapter 12 presented a non-template (type **`double`**) `Matrix` class in MATRIX.H/.CPP while a template implementation of `Matrix` was presented in Chapter 13 in MTX_TMP.H. The template class removed the floating-point type-dependent implementations from the non-template implementation such as the function `GaussElimination()` and the overloaded **`operator`**`*()` function. In the present section we will re-introduce these floating-point routines via specialised non-template floating-point classes:

```
// mtx_tmp.h
// template class Matrix and floating-point classes
// DoubleMatrix and FloatMatrix
//...
template <class T>
class Matrix
  {
  protected:
    Vector<T>* array ;
    //...
  public:
    Matrix () ;
```

```
    //...
  };
//...
class DoubleMatrix : public Matrix<double>
  {
  public:
    // constructors
    DoubleMatrix ()
      : Matrix<double> () {}
    DoubleMatrix (int r, int c, double init_value)
      : Matrix<double> (r, c, init_value) {}
    // member function
    Vector<double> GaussElimination (Vector<double>& b,
                                     bool& solved) ;
    // overloaded operator
    DoubleMatrix operator * (const DoubleMatrix& m) ;
  };

class FloatMatrix : public Matrix<float>
  {
  public:
    // constructors
    FloatMatrix ()
      : Matrix<float> () {}
    FloatMatrix (int r, int c, float init_value)
      : Matrix<float> (r, c, init_value) {}
    // member function
    Vector<float> GaussElimination (Vector<float>& b,
                                    bool& solved) ;
    // overloaded operator
    FloatMatrix operator * (const FloatMatrix& m) ;
  };
//...
// DoubleMatrix:

// Gauss elimination
Vector<double> DoubleMatrix::GaussElimination
               (Vector<double>& b, bool& solved)
  {
  solved = false ;

  int n = rows ;  // assign number of rows to n

  Vector<double> x (n, 0.0) ;  // solution Vector
  //...
  }
```

Floating-point classes `DoubleMatrix` and `FloatMatrix` are now derived from `Matrix<T>` for **double** and **float** types respectively. This solution is not ideal, since the function definitions of *GaussElimination()* and **operator*** *()* have been repeated for

both **float** and **double** types, which conflicts greatly with the object-oriented programming philosophy.

The following program tests the `DoubleMatrix`, `FloatMatrix` and `Matrix` classes:

```
// mtxt_tst.cpp
// tests the Point, Vector and Matrix template classes
// and the DoubleMatrix and FloatMatrix classes
#include <iostream>  // C++ I/O

#include "pt_tmp.h"   // template class Point
#include "vec_tmp.h"  // template class Vector
#include "mtx_tmp.h"  // template class Matrix, ...

void main ()
  {
  Matrix<int> m_init (3, 4, 0.0) ;
  Matrix<Matrix<int> > mm (2,2,m_init) ;  // composed type

  //FloatMatrix a (3, 3, 0.0) ;
  //Vector<float> b (3, 0.0), x (3, 0.0) ;
  DoubleMatrix   a (3, 3, 0.0) ;
  Vector<double> b (3, 0.0), x (3, 0.0) ;

  a[0][0] =  1.0 ; a[0][1] =  2.0 ; a[0][2] =  3.0 ;
  a[1][0] =  2.0 ; a[1][1] =  3.0 ; a[1][2] =  4.0 ;
  a[2][0] =  3.0 ; a[2][1] =  4.0 ; a[2][2] =  1.0 ;
  b[0]    = 14.0 ; b[1]    = 20.0 ; b[2]    = 14.0 ;

  cout << "DoubleMatrix   a: " << endl << a ;
  cout << "Vector<double> b: " << endl << b << endl ;

  bool solved (false) ;

  x = a.GaussElimination (b, solved) ;

  if (solved)
    cout << "soln. Vector<double> x of system [a]{x}={b}: "
         << endl << x << endl ;
  else
    cout << "soln. by Gauss elimination failed" << endl ;
  }
```

with output:

```
Matrix<double> a:
      1.00       2.00       3.00
      2.00       3.00       4.00
      3.00       4.00       1.00
Vector<double> b:
[14.00, 20.00, 14.00]
```

```
soln. Vector<double> x of system [a]{x}={b}:
[1.00, 2.00, 3.00]
```

15.20.4 `Stack` and `Queue` Classes

Let's now examine `Stack` and `Queue` **`template`** classes, which are both derived from the `LinkedList` **`template class`** presented in Chapter 13. The following `Stack` and `Queue` classes are based on the non-**`template`** `Stack` and `Queue` classes presented in Adams *et al.* (1995, pp. 972–90).

A *stack* is a special form of linked list in which an object is added to and removed from the same end of the list. The most recently added object to a stack list is the first object removed: the so-called *last in, first out* order. Since a stack is *a kind of* linked list it makes sense to derive `Stack` from `LinkedList` and add operations specific to **`class`** `Stack`.

The only difference between the present **`class`** declaration of `LinkedList` and the declaration of presented in Chapter 13 is that the **`private`** data members have been changed to **`protected`**. The declaration of `Stack` is as follows:

```
// stack.h
// template class Stack

#ifndef _STACK_H  // prevent multiple includes
#define _STACK_H

#include "ll.h"   // template class LinkedList

template <class T>
class Stack : protected LinkedList<T>
  {
  public:
    // constructors
    Stack ()
      : LinkedList<T> () {}
    Stack (const Stack& s)
      : LinkedList<T> (s) {}
    // member functions
    bool IsEmpty ()
      { return LinkedList<T>::IsEmpty () ; }
    void Push (T obj)
      { Insert (obj, Stack<T>::START) ; }
    T Pop ()
      { assert(num_nodes>0) ; T obj=first->data ;
        Delete(Stack<T>::START) ; return obj ; }
  };

#endif // _STACK_H
```

Generally, there are two key operations performed on a stack: insert or *push* an object to the start of the stack, and delete or *pop* an object from the start of the stack. `Stack::` *`Push()`* and `Stack::` *`Pop()`* simply call `LinkedList::` *`Insert()`* and `LinkedList::` *`Delete()`* respectively to perform the push and pop operations.

The access specifier in the **class** declaration of Stack is **protected** rather than **public**, indicating that the **public** member functions of LinkedList are not **public** to Stack. This is because the majority of member functions of LinkedList are undefined for Stack. The exception is the member function *Empty()*, which is also required by Stack and has therefore been specifically overridden for Stack.

Another special form of linked list is a *queue* in which objects are added to the start of a list and removed from the end of a list. Contrary to a stack, the most recently added object to a queue list is the last object removed: the so-called *first in, first out* order. Since a queue is *a kind of* linked list, **class** Queue is derived from LinkedList with operations specific to Queue added. The declaration of Queue is:

```
// queue.h
// template class Queue

#ifndef _QUEUE_H  // prevent multiple includes
#define _QUEUE_H

#include "ll.h"  // template class LinkedList

template <class T>
class Queue : protected LinkedList<T>
  {
  public:
    // constructors
    Queue ()
      : LinkedList<T> () {}
    Queue (const Queue& q)
      : LinkedList<T> (q) {}
    // member functions
    bool IsEmpty ()
      { return LinkedList<T>::IsEmpty () ; }
    void Add (T obj)
      { Insert (obj, Queue<T>::END) ; }
    T Remove ()
      { assert(num_nodes>0) ; T obj=first->data ;
        Delete(Queue<T>::START) ; return obj ; }
  };

#endif // _QUEUE_H
```

Instead of member functions *Push()* and *Pop()* for **class** Stack we define the functions *Add()* and *Remove()*, respectively, for **class** Queue. The member function *Add()* inserts an object to the end of a queue and *Remove()* deletes an object from the start of a queue. Like **class** Stack, the access specifier in the **class** declaration of Queue is **protected** so that the **public** member functions of LinkedList are not **public** to Queue.

A program which tests the above Stack and Queue classes is given below:

```
// s_q_tst.cpp
// tests the LinkedList template class

#include "stack.h"  // template class Stack
```

```
#include "queue.h"  // template class Queue

void main ()
  {
  // Stack:
  Stack<int> s ;

  s.Push (1) ; s.Push (2) ; s.Push (3) ;

  while (!s.IsEmpty ())
    cout << s.Pop () ;

  cout << endl ;

  // Queue:
  Queue<int> q ;

  q.Add (1) ; q.Add (2) ; q.Add (3) ;

  while (!q.IsEmpty ())
    cout << q.Remove () ;
  }
```

with output:

```
321
123
```

15.20.5 `VectorIterator` and `LinkedListIterator` Classes

In Chapter 13 (V_IT_C.H and LLI.H) we examined iterator classes for classes `Vector` and `LinkedList`, namely `VectorIterator` and `LinkedListIterator`. It is noted that the four member functions *`More()`*,*`Current()`*,*`Next()`* and *`ReStart()`* are common to both `VectorIterator` and `LinkedListIterator`. Thus, let us now declare an abstract **`class`**, `Iterator`, from which `VectorIterator` and `LinkedListIterator` will be derived:

```
// iter.h
// template class Iterator

#ifndef _ITER_H  // prevent multiple includes
#define _ITER_H

template <class T>
class Iterator
  {
  public:
    // virtual member functions
    virtual bool More ()    const = 0 ;
    virtual T&   Current () const = 0 ;
    virtual void Next ()    = 0 ;
```

```
    virtual void    ReStart () = 0 ;

    // virtual overloaded operators
    virtual bool operator ++ () = 0 ;      // prefix
    virtual bool operator ++ (int) = 0 ;   // postfix
  };

#endif // _ITER_H
```

The **virtual** member functions *More()*, *Current()*, *Next()* and *ReStart()* determine whether or not more elements or nodes are within an object, return the current element or node, advance to the next element or node and reposition the current position to the start of a vector or list respectively. Virtual overloaded prefix and postfix increment operators are also declared for iterator object manipulation similar to that of C++ integral types. Note that `Iterator` has no data members.

It is now a simple matter of deriving `VectorIterator` from `Iterator`:

```
// vec_iter.h
//...
// forward declaraion
template <class T> class VectorIterator ;

template <class T>
class Vector
  {
  //...
  friend class VectorIterator<T> ;
  };

template <class T>
class VectorIterator : public Iterator<T>
  {
  private:
    int    c_elem ;          // current element
    const Vector<T>& vec ;   // current Vector
  public:
    // constructor
    VectorIterator (const Vector<T>& v)
      : c_elem (0), vec (v) {}
    // member functions
    bool More ()    const
      { return c_elem < vec.n_elements ? true : false ; }
    T&   Current () const { return vec.array[c_elem] ; }
    void Next ()    { c_elem++ ; }
    void ReStart () { c_elem = 0 ; }
    // overloaded operators
    bool operator ++ ()     // prefix
      { c_elem++ ; return More() ; }
    bool operator ++ (int)  // postfix
      { c_elem++ ; return More() ; }
  };
```

An object of `VectorIterator` holds a reference to the `Vector` object being manipulated and the current element index. Similarly, `LinkedListIterator` is given by:

```
// ll_iter.h
// forward declaraion
template <class T> class LinkedListIterator ;

template <class T>
class LinkedList
  {
  //...
  friend class LinkedListIterator<T> ;
  };

template <class T>
class LinkedListIterator : public Iterator<T>
  {
  private:
    LinkedList<T>::Node* c_node ;  // current node
    const LinkedList<T>& c_list ;  // current list
  public:
    // constructor
    LinkedListIterator (const LinkedList<T>& l)
      : c_node (l.first), c_list (l) {}
    // member functions
    bool More ()    const
      { return c_node != 0 ? true : false ; }
    T&   Current () const { return c_node->data ; }
    void Next ()    { c_node = c_node->next ; }
    void ReStart () { c_node = c_list.first ; }
    // overloaded operators
    bool operator ++ ()     // prefix
      { c_node=c_node->next ; return More() ; }
    bool operator ++ (int)  // postfix
      { c_node=c_node->next ; return More() ; }
  };
```

The following program illustrates the above `VectorIterator` and `LinkedList-Iterator` classes:

```
// iter_tst.cpp
// tests the iterator classes VectorIterator and
// LinkedListIterator
//...
void main ()
  {
  // LinkedList & LinkedListIterator
  LinkedList<int> ll ;
  ll.Append (2) ;
  ll.Append (4) ;
  ll.Append (6) ;
```

```
    cout << "LinkedListIterator lli (ll): " ;
    for (LinkedListIterator<int> lli(ll); lli.More(); ++lli)
      cout << lli.Current () << " " ;
    cout << endl ;

    // Vector & VectorIterator
    Vector<int> v (3, 0) ;
    v[0] = 2 ;
    v[1] = 4 ;
    v[2] = 6 ;

    cout << "VectorIterator vi (v)       : " ;
    for (VectorIterator<int> vi(v); vi.More(); vi++)
      cout << vi.Current () << " " ;
    cout << endl ;
    }
```

with output:

```
LinkedListIterator lli(ll): 2 4 6
VectorIterator vi(v)      : 2 4 6
```

15.21 Exception Handling and Inheritance

This section examines exception handling and inheritance. To remind ourselves of the **try-catch-throw** exception handling mechanism of C++, consider the following program, which simply throws and catches an exception of a derived **class**:

```
// exp_inh.cpp
// illustrates exception handling and inheritance
#include <iostream>  // C++ I/O

class Base
  {};

class Derived : public Base
  {};

void main ()
  {
  try
    {
    Derived derived ;

    // throw exception to handler
    throw derived ;
    }
  // process exception
  catch (Derived)
```

```
    {
    cout << "Derived exception caught" << endl ;
    }
  //catch (const Derived)  // duplicate handlers
  //catch (Derived&)
  //catch (const Derived&)
  }
```

with output:

```
Derived exception caught
```

Remember from Chapter 14 that a **throw** expression of type T is a match for an exception handler of type T, **const** T, T& or **const** T&.

Let us now examine more closely the interaction of base and derived **class** exception handlers:

```
// bd_x_inh.cpp
// illustrates base and derived classes,
// exception handling and inheritance
#include <iostream>  // C++ I/O

class Base
  {};

class PubDerived : public Base
  {};

void main ()
  {
  try
    {
    PubDerived pub_derived ;

    // throw exception to handler
    throw pub_derived ;
    }
  // process exception
  catch (Base)
    {
    cout << "Base exception caught" << endl ;
    }
  catch (PubDerived)
    {
    cout << "PubDerived exception caught" << endl ;
    }
  }
```

with output:

```
Base exception caught
```

The output of the above program illustrates that a **throw** expression of a derived **class** is a match for an exception handler of a base **class**, provided the base **class** is accessible to a derived **class**. Compilers will issue a compilation warning message of the form 'PubDerived handler is hidden by Base handler' regarding the hiding of the PubDerived handler by the Base handler.

Consider now the above program, but declaring a **private**ly derived **class**:

```
// b_pri_d.cpp
// illustrates base and privately derived classes,
// exception handling and inheritance
#include <iostream>  // C++ I/O

class Base
  {};

class PriDerived : private Base
  {};

void main ()
  {
  try
    {
    PriDerived pri_derived ;

    // throw exception to handler
    throw pri_derived ;
    }
  // process exception
  catch (Base)
    {
    cout << "Base exception caught" << endl ;
    }
  catch (PriDerived)
    {
    cout << "PriDerived exception caught" << endl ;
    }
  }
```

with output:

```
PriDerived exception caught
```

which illustrates that a base **class** handler is not a match for a **private**ly derived **class** exception, and similarly for a **protected**ly derived **class** exception.

Generally, a derived **class** exception is caught by a base **class** handler rather than by a handler of the derived **class**. This can result in the derived **class** exception thrown being *sliced* if the exception object is thrown by value. Alternatively, if a thrown exception object is caught by reference the thrown exception is not sliced:

```
// bd_catch.cpp
// illustrates catching a derived class exception
```

```
#include <iostream>  // C++ I/O

class Base
  {
  public:
    virtual void Function ()
      { cout << "Base::Function() called" << endl ; }
  };

class Derived : public Base
  {
  public:
    void Function ()
      { cout << "Derived::Function() called" << endl ; }
  };

void main ()
  {
  try
    {
    Derived derived ;

    // throw exception to handler
    throw derived ;
    }
  // process exception
  catch (Base& b)
    {
    b.Function () ;
    }
  }
```

with output:

```
Derived::Function() called
```

The above program illustrates that even though a derived **class** exception object is thrown and the exception handler catches objects of the base **class** the `Derived::Function()` is called because the thrown exception object is caught by reference and not by value.

15.21.1 Exception Specification

Chapter 14 illustrated that the *exception specification* of a function declaration enables a function to specify exactly what exceptions it is allowed to **throw**. With respect to inheritance, a function that is able to **throw** an exception of a given **class** can also **throw** an exception of any **class** **public**ly derived from that **class**:

```
// x_sp_inh.cpp
// illustrates function exception specifications and
// inheritance
#include <iostream>  // C++ I/O
```

```
class Base
  {};

class Derived : public Base
  {};

// throw Base exception specification
void FThrowBase (int i) throw (Base)
  {
  Base    b ;
  Derived d ;

  if (i == 0)
    throw b ;  // throw Base
  if (i == 1)
    throw d ;  // throw Derived
  }

void main ()
  {
  try
    {
    //FThrowBase (0) ;  // ok
    FThrowBase (1) ;  // ok
    }
  // process exception
  catch (Base)
    {
    cout << "exception caught" << endl ;
    }
  }
```

with output:

```
exception caught
```

15.21.2 Re-Throwing an Exception

It was noted above that if a derived **class** exception is caught by a base **class** handler that this may result in the thrown object being sliced. However, if such an object is then re-thrown it is the original object that is re-thrown, not the sliced object.

15.22 Summary

Inheritance allows a *derived* **class** to inherit the features of one or more other **base** classes. Inheritance relationships are specified during the declaration of a derived **class**. The three access specifiers **private**, **protected** and **public** give a programmer complete flexibility in controlling data member and member function accessibility during inheritance.

Generally, two alternative approaches can be adopted when developing a new **class**: containment or inheritance. Containment exploits the **has a** relationship, in which case a given **class** has objects of another **class**. On the other hand, inheritance exploits the **is a** relationship, in which case one **class** is a kind of another **class**.

Combining the features of inheritance and **virtual** functions enables a set of base and derived **class** member functions to exhibit run-time polymorphic behaviour.

Exercises

15.1 A given **class** is declared as:

```
class Base
  {
  protected:
    int bdm ;
  public:
    // constructors
    Base ()
      : bdm (0) {}
    Base (int i)
      : bdm (i) {}
    // copy constructor
    Base (const Base& b)
      : bdm (b.bdm) {}
    // overloaded operator
    Base& operator = (const Base& b)
      {
      bdm = b.bdm ;
      return *this ;
      }
  };
```

Publicly derive a **class**, `Derived`, from `Base` which adds its own **protected** data member, `ddm`, and defines both default and one-argument constructors, a copy constructor and an overloaded = operator member function.

15.2 What is wrong with the following assignment expression in *main()*?

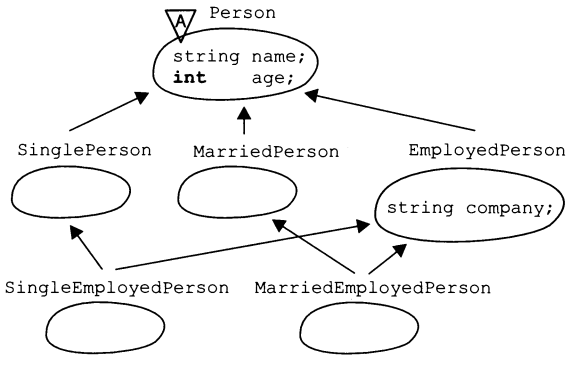
```
class Base
  {};

class Derived : public Base
  {};

void main ()
  {
  Base    b ;
  Derived d ;
  //...
  d = b ;
  }
```

15.3 Declare an abstract **class** called `Person` which encapsulates `name` and `age` data members.

15.4 Declare classes for the following **class** hierarchy:



15.5 The following program will not compile. Why?

```
template <class T>
class Point
  { /* ... */ };

void main ()
  {
  Point<int>    p (1, 2) ;
  Point<double> q (3.0, 4.0) ;
  //...
  q = p ;
  }
```

15.6 In Exercise 9.7 a `CartCoorSys` **class** was developed to represent a Cartesian coordinate system. Consider the development of classes for cylindrical, polar and spherical coordinate systems. Should containment or inheritance be used?

15.7 In the following **catch** statement, which *`Message()`* member function is called?

```
class Exception
  {
  //...
  public:
    virtual string& Message () ;
    //...
  };

class InsufficientMemory : public Exception
  {
  public:
    string& Message () ;
    //...
  };
```

```
void main ()
  {
  try
    {
    InsufficientMemory memory_exp ;
    //...
    throw memory_exp ;
    }
  catch (Exception& exp)
    {
    exp.Message () ;
    }
  }
```

Chapter 16

Run-Time Type Information and Casting

We saw in the previous chapter that similar classes can be placed in a class hierarchy, with numerous classes derived from a single base class. Working in terms of a pointer to the base class, rather than objects of the derived classes, we are able to operate on functions common to all derived classes via a single pointer object that can point to any derived class object. Such a run-time capability is essential to object-oriented programming, but unfortunately moves us away from the strongly typed compile-time mechanism so characteristic of the C++ language.

Run-time support is given through the use of virtual member functions, and C++ complements this with language-based Run-Time Type Information (RTTI) features. The RTTI approach, as the name suggests, is concerned with determining the type of an object at run-time and can be summarised in terms of two key operators: dynamic_cast<>() and typeid(). The dynamic_cast<>() operator allows a pointer or reference of a base class to be cast to a derived class provided the base class is polymorphic. Using the dynamic_cast<>() operator to perform such down-casting is, importantly, checked at run-time, whereas the non-RTTI casting approach goes unchecked at run-time. The typeid() operator allows the exact type of an object to be determined for both polymorphic and non-polymorphic classes.

Casting is the process of converting an object's type. This chapter will examine three cast operators (static_cast<>(), reinterpret_cast<>() and const_cast<>()) which replace the (T)e C-style and T(e) C++-style casts that we have seen used in previous chapters. Three different cast operators are defined because the single (T)e cast style is, in certain circumstances, not sufficiently explicit in disclosing the exact intentions behind a cast operation.

16.1 Recap

When we discussed **virtual** functions in the previous chapter, an abstract base **class** Shape, a base **class** Plane and associated derived classes were developed:

```
class Shape  // abstract base class of all shapes
  {
  //...
```


G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

```
  virtual Point Centroid () = 0 ;  // pure virtual function
  //...
  };
//...
class Plane  // base class
  {};
//...
class Polygon : public Shape, public Plane
  {
  //...
  Point Centroid () ;  // overridden member function
  };
//...
class Triangle : public Polygon
  { /* ... */ };
//...
class Rectangle : public Polygon
  { /* ... */ };
//...
class Circle : public Shape, public Plane
  {
  Point Centroid () ;  // overridden member function
  };
//...
```

A simple application of the above classes was demonstrated in SHAPES.CPP, which is of the form:

```
void main ()
  {
  Shape* s (0) ;  // pointer to abstract base class Shape
  Circle c ; Rectangle r ; Triangle t ;  // derived class
                                         // objects
  //...
  // if Circle:
  s = &c ;  // point to Circle
  //...
  // if Rectangle:
  s = &r ;  // point to Rectangle
  //...
  // if Triangle:
  s = &t ;  // point to Triangle
  //...
  }
```

The problem with working with derived **class** objects via a pointer to a base **class**, which can point to any number of derived **class** objects, is that we have no way of determining from the pointer alone which type of object the pointer is in fact pointing to. Run-time polymorphism, via the use of **virtual** functions, focuses our attention on the common characteristics of different objects and allows us to use these common characteristics in designing a general interface with complete disregard for the type information of each of the objects. There are occasions when this is exactly what we require. However, there are other occasions when we

require the run-time polymorphic behaviour of a set of objects while at the same time maintaining the type information of each of the objects.

16.2 Use of `virtual` Functions

16.2.1 A Direct Method

One solution to the problem of recovering type information about an object from a pointer is simply to define a member function for each **`class`** which returns a `string` object indicating an object's **`class`** and then call this function when required:

```
class Shape  // abstract base class of all shapes
  {
  //...
  string WhoAmI () const { return string ("Shape") ; }
  //...
  };
//...
class Plane  // base class
  {
  //...
  string WhoAmI () const { return string ("Plane") ; }
  };
//...
class Polygon : public Shape, public Plane
  {
  //...
  string WhoAmI () const { return string ("Polygon") ; }
  //...
  }
//...
```

Alternatively, we can define the *`WhoAmI()`* member functions for the derived classes so that they return a specific code which indicates a **`class`**'s type and which is defined in the `Shape` abstract base **`class`**:

```
// who.cpp
// illustrates detecting the type of an object
// through the use of an abstract base class type code
// and type casting an abstract base class pointer
#include <iostream>  // C++ I/O

// abstract base class of all shapes
class Shape
  {
  public:
    enum Who { POLYGON, TRIANGLE, RECTANGLE };
    virtual Who  WhoAmI () const = 0 ;
```

```
    virtual void Display ()       = 0 ;
  };

class Plane
  {};

class Polygon : public Shape, public Plane
  {
  public:
    Who  WhoAmI () const { return POLYGON ; }
    void Display () { cout << "Polygon" << endl ; }
  };

class Triangle : public Polygon
  {
  public:
    Who  WhoAmI () const { return TRIANGLE ; }
    void Display () { cout << "Triangle" << endl ; }
  };

class Rectangle : public Polygon
  {
  public:
    Who  WhoAmI () const { return RECTANGLE ; }
    void Display () { cout << "Rectangle" << endl ; }
  };

void main ()
  {
  void Function (Shape* s) ;  // prototype

  Shape*   s (0) ;
  Triangle t ;

  s = &t ;  // Shape pointer points to a Triangle object

  Function (s) ;
  }

void Function (Shape* s)
  {
  Shape::Who who_are_you = s->WhoAmI () ;

  if (who_are_you == Shape::POLYGON)
    {
    Polygon* p = (Polygon*)s ;
    p->Display () ;
    }
  else if (who_are_you  == Shape::TRIANGLE)
    {
    Triangle* t = (Triangle*)s ;
```

```
    t->Display () ;
    }
  else if (who_are_you == Shape::RECTANGLE)
    {
    Rectangle* r = (Rectangle*)s ;
    r->Display () ;
    }
}
```

with output:

```
Triangle
```

The type code for each derived **class** is defined using an **enum** Who, which is **public**ly declared in Shape. The **virtual** function *WhoAmI()* is defined for each derived **class** and simply returns an integer corresponding to the **class** of an object which the function operates on.

Function() is passed a pointer, s, to abstract base **class** Shape which could be pointing to any type. The exact type that s is pointing to is determined from the *WhoAmI()* function, and by using a block of **if-else- if** statements the appropriate *Display()* function is called. On considering the first **if** statement, we observe that the pointer, s, to Shape is *down-cast*[1] to **class** Polygon:

```
//...
Polygon* p = (Polygon*)s ;
```

It was discussed in Chapter 15 that it is natural to cast from a derived **class** object to a base **class** object, or *slice* a derived **class** object. However, the reverse process actually adds information to a base **class** object. The compiler knows nothing of such type casts and can therefore offer no support, warning or error messages to a programmer.

16.2.2 The Double-Dispatch Method

An alternative solution to the problem of recovering the type information of an object at run-time is a technique called *double-dispatch*. The technique used above for determining run-time type information involved defining an overridden *WhoAmI()* **virtual** function which returns a unique type code for each derived **class**, explicitly testing the type code returned by *WhoAmI()* and then performing the appropriate operation. The double-dispatch approach is still based on the use of **virtual** functions, but now involves a *Dispatch()* function which is declared a pure **virtual** function in the abstract base **class** Shape and overridden for each derived **class**. The single argument to *Dispatch()* is a reference to a **class** called Accept. The *Dispatch()* function for each derived **class** simply calls the appropriate accept function, which is to be defined in a **class** derived from **class** Accept. The abstract **class** Accept declares a pure **virtual** function for each **class** derived from Shape, such as *AcceptPolygon*(**const** Polygon& p), and has a single function argument which is a reference to the **class** or type of object for which the function is to provide an operation:

---

1 Type casting a base **class** object to a derived **class** object is referred to as *down-casting* because we are going down the **class** inheritance diagram rather than up; see Fig. 15.1.

```
// dbl_dis.cpp
// illustrates double dispatch
#include <iostream>  // C++ I/O

class Accept ;

class Polygon ; class Triangle ; class Rectangle ;

// abstract base class of all shapes
class Shape
  {
  public:
    virtual void Dispatch (Accept& a) const = 0 ;
  };

class Accept
  {
  public:
    virtual void AcceptPolygon (const Polygon& p) = 0 ;
    virtual void AcceptTriangle (const Triangle& t) = 0 ;
    virtual void AcceptRectangle (const Rectangle& r) = 0 ;
  };

class Plane
  {};

class Polygon : public Shape, public Plane
  {
  public:
    void Dispatch (Accept& a) const
      { a.AcceptPolygon (*this) ; }
  };

class Triangle : public Polygon
  {
  public:
    void Dispatch (Accept& a) const
      { a.AcceptTriangle (*this) ; }
  };

class Rectangle : public Polygon
  {
  public:
    void Dispatch (Accept& a) const
      { a.AcceptRectangle (*this) ; }
  };

class DisplayShape : public Accept
  {
  public:
    void AcceptPolygon (const Polygon& p)
```

```
    { cout << "Polygon" << endl ; }
  void AcceptTriangle (const Triangle& t)
    { cout << "Triangle" << endl ; }
  void AcceptRectangle (const Rectangle& r)
    { cout << "Rectangle" << endl ; }
};

void main ()
  {
  Shape*   s (0) ;
  Triangle t ;

  s = &t ;  // Shape pointer points to a Triangle object

  DisplayShape dis_shape ;

  s->Dispatch (dis_shape) ;
  }
```

with output:

```
Triangle
```

Since we are currently interested in simply displaying the **class** name of an object, we derive a **class** called `DisplayShape` from `Accept` and override the required accept function(s) in `DisplayShape`. A `DisplayShape` object is then passed to *`Dispatch()`*, indicating that we want to display an object pointed to by the `Shape` abstract base **class** pointer `s`.

The double-dispatch approach completely bypasses the need to determine the type of a given object at run-time in order to call the appropriate function – it figures out the details for you. The price to pay is that there are now two calls to **virtual** functions instead of the one **virtual** function call used in the more direct method presented in WHO.CPP. The first **virtual** function call is to the *`Dispatch()`* function, while the second call is to the appropriate accept function. Once the *`Dispatch()`* function is overridden for each derived **class** we can add new functions to the existing classes via the `Accept` **class** and classes derived from `Accept` without modification to the classes derived from `Shape`.

16.3 The RTTI Approach

Before C++ offered direct support for run-time type information, a programmer had to rely solely on the use of **virtual** functions to resolve the issue of an object's type at run-time. The C++ RTTI approach focuses around two operators called `dynamic_cast<>()` and `typeid()` and several support classes. Let's first examine the `dynamic_cast<>()` operator, which is probably the most important feature of RTTI.

16.3.1 The `dynamic_cast<>()` Operator

The `dynamic_cast<>()` operator can be used to cast to either a pointer or a reference type. Let's first examine the cast to a pointer version:

```
dynamic_cast<T*>(p)
```

This will convert and return the operand `p` to type `T*` if, and only if, `*p` is of type `T` or of a **`class`** or type derived from `T`; else the value of the returned pointer is 0. Note that `p` must be a pointer or an expression which resolves to a pointer. To help illustrate `dynamic_cast<T*>()` let's examine a program:

```
// dyn_cstp.cpp
// illustrates the dynamic_cast<T*>() operator
#include <iostream>  // C++ I/O

// non-polymorphic class
class X
  {};

// polymorphic class
class Base
  {
  public:
    virtual void Function () {} ;
  };

// derived from Base
class Derived : public Base
  {};

void main ()
  {
  Derived d ;
  Base*   bp = &d ;
  X*      xp (0) ;

  // ok: downcast from Base to Derived-unchecked by compiler
  Derived* dp0 = (Derived*)bp ;

  // ok: downcast from Base to Derived-checked by compiler
  // at run-time
  Derived* dp1 = dynamic_cast<Derived*>(bp) ;

  // error: X is a non-polymorphic class
  Derived* dp2 = dynamic_cast<Derived*>(xp) ;
  // ok: operand expression
  Derived* dp3 = dynamic_cast<Derived*>(bp++) ;

  // ok: cast from Base to void
  void* vp = dynamic_cast<void*>(bp) ;
  }
```

The following static down-cast from `Base` to `Derived`:

```
Derived* dp0 = (Derived*)bp ;
```

is legal but unchecked at compilation. However, the same cast operation using the `dynamic_cast<T*>()` operator:

```
Derived* dp1 = dynamic_cast<Derived*>(bp) ;
```

is checked at run-time.

Type `T` in the `dynamic_cast<T*>(p)` operator can be of type **`void`** provided operand `p` is a pointer:

```
void* vp = dynamic_cast<void*>(bp) ;
```

The above program illustrates that a cast from a base **`class`** to a derived **`class`** can only be performed if the base **`class`** is a polymorphic **`class`**, i.e. a **`class`** with one or more **`virtual`** functions. RTTI only supports polymorphic types for which objects can be manipulated via a base **`class`**. Thus, there is no RTTI support for integral types such as **`int`** and **`double`**.

Let's now examine the use of the `dynamic_cast<>()` operator for casting to a reference type:

```
dynamic_cast<T&>(r)
```

The reference version of `dynamic_cast<>()` is similar to the pointer version except that if a cast to a reference type is unsuccessful a `bad_cast` exception is thrown. The exception-handling **`class`** `bad_cast` is declared in the TYPEINFO.H header file, which is discussed later. For example:

```
// dyn_cstr.cpp
// illustrates the dynamic_cast<T&>() operator
#include <iostream>  // C++ I/O
#include <typeinfo>  // RTTI

// non-polymorphic class
class X
  {};

// polymorphic class
class Base
  {
  public:
    virtual void Function () {} ;
  };

// derived from Base
class Derived : public Base
  {};

void main ()
  {
  try
    {
    Derived d ;
    Base&   br = d ;
```

```
    Derived& dr = dynamic_cast<Derived&>(br) ;
    }
  catch (bad_cast)
    {
    cout << "dynamic_cast<T&>() failed" << endl ;
    }
}
```

Using the conventional C-style cast syntax it is illegal to cast from a **virtual** base **class** to a derived **class**. The `dynamic_cast<>()` operator, however, allows such a cast to be performed:

```
// v_b_c.cpp
// illustrates the use of the dynamic_cast<T*>() operator
// with virtual base classes
#include <iostream>  // C++ I/O

// polymorphic class
class Base
  {
  public:
    virtual void Function () {} ;
  };

// derived from virtual base class, Base
class Derived : virtual public Base
  {};

void main ()
  {
  Derived d ;
  Base*   bp = &d ;

  // error: can't cast from virtual base class to derived class
  Derived* dp0 = (Derived*)bp ;

  // ok: checked at run-time
  Derived* dp1 = dynamic_cast<Derived*>(bp) ;
  }
```

16.3.2 The `typeid()` Operator

The `dynamic_cast<>()` operator informs us of whether or not an object pointed to is of a given **class** or any **class** derived from that **class**, but does not allow us to determine the exact type of an object. The `typeid()`[2] operator, however, does allow the exact type of an object to be determined, and is of the following general syntax:

2 Type identity

```
typeid (type_name)
typeid (expression)
```

The `typeid()` operator returns a **`const`** reference to an object of **`class`** `type_info`; see below. The type of the object returned by `typeid()` represents the type of the `type_name` or `expression` operand. Unlike `dynamic_cast<>()`, the `typeid()` operator can have an integral non-polymorphic data type operand. If the operand is non-polymorphic `typeid()` returns an object of the static type, whereas if the operand is polymorphic `typeid()` returns the dynamic type of the object pointed to. If the operand to `typeid()` is a dereferenced null pointer a `bad_typeid` exception is thrown. The following program illustrates `typeid()`:

```
// typeid.cpp
// illustrates the use of the typeid() operator
#include <iostream>  // C++ I/O
#include <typeinfo>  // RTTI

// polymorphic class
class Base
  {
  public:
    virtual void Function () {} ;
  };

// derived from Base
class Derived : public Base
  {};

void main ()
  {
  Derived d ;
  Base*   bp = &d ;

  if (typeid (*bp) == typeid (Derived))
    cout << "Base pointer pointing to Derived object"
         << endl ;
  }
```

with output:

```
Base pointer pointing to Derived object
```

Note the inclusion of the TYPEINFO.H header file, which is necessary if the object returned by `typeid()` is to be used.

16.3.3 The TYPEINFO.H Header File and the `type_info`, `bad_cast` and `bad_typeid` Classes

The TYPEINFO.H header file includes **`class`** declarations for the run-time type information classes `type_info`, `bad_cast` and `bad_typeid`. The implementation-independent part

of type_info[3] and the empty bad_cast and bad_typeid class declarations are as follows:

```
class type_info
  {
  private:
    // private copy constructor & assignment operator
    type_info (const type_info& rhs) ;
    type_info& operator = (const type_info& rhs) ;
  public:
    // virtual destructor
    virtual ~type_info () ;
    // member functions
    bool        before (const type_info& rhs ) const ;
    const char* name () const ;
    // overloaded operators
    bool operator == (const type_info& rhs) const ;
    bool operator != (const type_info& rhs) const ;
  };
//...
class bad_cast
  {};

class bad_typeid
  {};
//...
```

The copy constructor and overloaded assignment operator of **class** type_info are declared **private** so that type_info objects cannot be defined and copied. type_info supports the two comparison operators == and !=. The *before()* member function compares the lexical order of different types and returns either logically true or false, e.g.:

```
bool i = typeid (int).before (typeid (double)) ;  // false
```

The *name()* member function returns the type name of the operand to the typeid() operator, e.g.:

```
cout << typeid (*p).name () ;
```

If a cast to a reference type is performed using the dynamic_cast<T&>() operator and is unsuccessful then a bad_cast exception is thrown. If the typeid() operator is unsuccessful in identifying the type of its operand it throws a bad_typeid exception. This is best illustrated by the following pseudo declarations of typeid() had it been implemented as a function rather than an operator:

```
const type_info& typeid (type_name) throw () ;
const type_info& typeid (expression) throw (bad_typeid) ;
```

3 The exact definition of the type_info polymorphic **class** is implementation dependent.

The following program helps illustrate the `type_info` **class** via the `typeid()` operator:

```
// typeinfo.cpp
// illustrates the type_info class
#include <iostream>  // C++ I/O
#include <typeinfo>  // RTTI

// polymorphic class
class Base
  {
  public:
    virtual void Function () {} ;
  };

// derived from Base
class Derived : public Base
  {};

void main ()
  {
  try
    {
    Derived d ;
    Base*   bp = &d ;

    cout << "typeid(Derived).name(): "
         << typeid (Derived).name () << endl ;
    cout << "typeid(int).name()    : "
         << typeid (int).name ()     << endl ;

    typeid (Base).before (typeid (Derived))
      ? cout << "Base before Derived" << endl
      : cout << "Base after Derived"  << endl ;

    typeid (int).before (typeid (double))
      ? cout << "int before double" << endl
      : cout << "int after double"  << endl ;

    // throw Bad_typeid exception
    bp = 0 ;
    cout << typeid (*bp).name () << endl ;
    }
  catch (bad_typeid)
    {
    cout << "bad_typeid exception caught" << endl ;
    }
  }
```

with output:

```
typeid(Derived).name(): Derived
typeid(int).name()    : int
Base before Derived
int after double
bad_typeid exception caught
```

16.4 Casting

The C-style syntax of casting that we have seen to date is of the form `(T)e` and casts an expression `e` to type `T`. The C++-style of casting which we have seen in previous chapters uses the constructor notation `T(e)` as an alternative to `(T)e`. However, the `dynamic_cast<>()` operator introduces a new style of performing conversions from one type to another. In addition to the `dynamic_cast<>()` operator previously discussed, C++ also supports three additional cast operators, `static_cast<>()`, `reinterpret_cast<>()` and `const_cast<>()`, each having its own specific application. These three cast operators help separate and clarify the different forms of conversion performed by the traditional `(T)e` cast notation. This is necessary because the simple `(T)e` cast notation can have several different interpretations, making it difficult for a programmer to establish from a cast expression alone what the original programmer had in mind.

16.4.1 The `static_cast<>()` Operator

The syntax of the `static_cast<>()` operator is:

```
static_cast<T>(e)
```

where type `T` can be either an integral or user-defined type, pointer, reference or **enum**. No runtime check is performed when using the `static_cast<>()` operator; only static type checking.

Integral types, such as **int** and **double**, can be cast to **enum** types using `static_cast<>()`. Both pointer and reference types must be complete at the time of compilation for `static_cast<>()` to perform the conversion from one type to another. To cast a pointer of a **class** `X` to a pointer of a **class** `Y`, `X` must be a non-**virtual** base **class** of `Y` and an unambiguous conversion must exist. A given object can be cast to an object of another **class** provided that the appropriate constructor or conversion function exists.

One of the serious faults of the C-style cast notation, `(T)e`, is that it allows a programmer to cast-away the **const** attribute attached to an object. The `static_cast<>()` operator rectifies this fault by respecting **const**ness.

The following program illustrates some of the above-mentioned properties of `static_cast<>()`:

```
// stat_cst.cpp
// illustrates the static_cast<>() operator

enum Integers { ZERO, ONE, TWO, THREE, FOUR,
                FIVE, SIX, SEVEN, EIGHT, NINE };

class Base
```

```
  {
  };

// derived from Base
class Derived : public Base
  {
  public:
    // conversion function - Derived to int
    operator int () { return int () ; } ;
  };

// derived from virtual base class, Base
class VDerived : virtual public Base
  {};

// class forward-declarations - incomplete classes
class X ;
class Y ;

void main ()
  {
  Derived d ;
  Base*   bp = &d ;

  // cast Base* to Derived* using (T)e notation
  Derived* dp0 = (Derived*)bp ;

  // cast Base* to Derived* using static_cast<>() operator
  Derived* dp1 = static_cast<Derived*>(bp) ;

  int i (0) ;
  // cast int to enum
  Integers integer = static_cast<Integers>(i) ;

  // cast from Derived to int using Derived conversion
  // function
  int i_obj0 = (Derived)d ;
  int i_obj1 = static_cast<Derived>(d) ;
  // cast using undefined types X and Y
  X* xp ;
  Y* yp0 = (Y*)xp ;  // ok
  Y* yp1 = static_cast<Y*>(xp) ;  // error

  // cast from virtual base class
  VDerived* vdp0 = (VDerived*)bp ;  // error
  VDerived* vdp1 = static_cast<VDerived*>(bp) ;  // error

  // cast an object to a reference type
  Base b ;
  Derived& dp4 = (Derived&)b ;  // o.k.
  Derived& dp5 = static_cast<Derived&>(b) ;  // ok
```

```
// cast-away const
const Base* cbp ;
Derived* dp2 = (Derived*) cbp ;  // ok
Derived* dp3 = static_cast<Derived*>(cbp) ;  // error
}
```

16.4.2 The `reinterpret_cast<>()` Operator

The `reinterpret_cast<T>(e)` operator allows a programmer to perform casts which are generally unsafe and implementation-dependent, with `reinterpret_cast<>()` returning a reinterpretation of its operand. An integral type can be converted to a pointer type (e.g. **int** to **char***) and vice versa using `reinterpret_cast<>()`. Unlike `static_cast<>()` it is possible with `reinterpret_cast<>()` to perform both pointer and reference type conversions with types that are not complete at the time of compilation. As with `static_cast<>()`,`reinterpret_cast<>()` does not cast-away the **const** attribute of an object. A pointer of a given type to a function can be cast to a function pointer of another type using `reinterpret_cast<>()`. The program below illustrates a few properties of `reinterpret_cast<>()`:

```
// rein_cst.cpp
// illustrates the reinterpret_cast<>() operator
#include <iostream>  // C++ I/O

void Function (int i)
  {
  cout << "Function () called" << endl ;
  }

class Base
  {};

// derived from Base
class Derived : public Base
  {};

// class forward-declarations - incomplete classes
class X ;
class Y ;

void main ()
  {
  int* ip ;

  // cast int* to char* using (T)e notation
  char* cp0 = (char*)ip ;

  // cast int* to char* using reinterpret_cast<>() operator
  char* cp1 = reinterpret_cast<char*>(ip) ;

  Derived* dp0 = (Derived*)ip ;
  Derived* dp1 = reinterpret_cast<Derived*>(ip) ;
```

```
// cast using undefined types X and Y
X* xp ;
Y* yp0 = (Y*)xp ;  // ok
Y* yp1 = reinterpret_cast<Y*>(xp) ;  // ok

// cast-away const
const Base* cbp ;
Derived* dp2 = (Derived*) cbp ;  // ok
Derived* dp3 = reinterpret_cast<Derived*>(cbp) ;  // error

void Function (int) ;
typedef void (*pfv) () ;

// cast pointer to function of one type to another type
pfv p_function0 = (pfv)Function ;  // ok
pfv p_function1 = reinterpret_cast<pfv>(Function) ;  // ok

p_function0 () ;
p_function1 () ;
}
```

16.4.3 The `const_cast<>()` Operator

The `const_cast<T>(e)` operator allows access to an object with the **`const`** or **`volatile`** attribute attached. Type `T` must be the same type as operand `e` except for **`const`** and **`volatile`** modifiers, and the result returned by `const_cast<>()` is the same as `e` but of type `T`. The `const_cast<>()` operator also respects the **`const`**ness of an object's **`const`** attribute. The following program helps illustrate `const_cast<>()`:

```
// cnst_cst.cpp
// illustrates the const_cast<>() operator
class Base
  {};

// derived from Base
class Derived : public Base
  {};

void Function (char* cp)
  {}

void main ()
  {
  // cast-away const
  const Base* cbp ;
  Derived* dp2 = (Derived*) cbp ;  // ok
  Derived* dp3 = const_cast<Derived*>(cbp) ;  // error

  const char* ccp = "string" ;
  Function (ccp) ;  // error: can't cast const char* to char*
  Function ((char*)ccp) ;  // ok
```

```
Function (const_cast<char*>(ccp)) ;  // ok

volatile char* vc ;
Function ((char*)vc) ;
Function (const_cast<char*>(vc)) ;

const volatile char* cvcp ;
Function ((char*)cvcp) ;
Function (const_cast<char*>(cvcp)) ;
}
```

16.5 Summary

The C++ run-time type information mechanism provides support for resolving an object's type at run-time. The `dynamic_cast<>()` operator can have one of two forms: pointer or reference. `dynamic_cast<T*>()` returns a pointer to an object of a derived **`class`** given a pointer to a base **`class`** if, and only if, the type of the object pointed to matches that of the derived **`class`**; else 0 is returned. The `dynamic_cast<T&>()` operator performs a similar operation for reference types, except that if a cast to a reference is unsuccessful a `Bad_cast` exception is thrown. The `typeid()` operator allows the exact type of an object to be extracted at run-time. If the `typeid()` operator is unsuccessful in identifying the type of its operand it throws a `Bad_typeid` exception. The `typeid()` operator returns a reference to an object of the polymorphic **`class`** `type_info`, which supports equality and inequality comparisons and *`before()`* and *`name()`* member functions.

C++ also supports three operators for performing various types of casting: `static_cast<>()`, `reinterpret_cast<>()` and `const_cast<>()`. These three cast operators act as a replacement for the C-style `(T)e` cast notation, each performing its own specific operation.

Regarding both run-time type information and casting, C++'s long-term philosophy is to eliminate both and rely exclusively on the static type-checking mechanism because it is inherently less error-prone and generally results in better program style. At present it is not always possible to develop programs without reference to RTTI and casting. If this is the case, try to adopt the `dynamic_cast<>()` and `typeid()` operator approaches rather than **`virtual`** functions for extracting run-time type information and the `static_cast<>()`, `reinterpret<>()` and `const_cast<>()` style of casting rather than the traditional `(T)e` cast style.

Exercises

16.1 Use the `dynamic_cast<T*>()` operator to cast a pointer to `Base` to a pointer to `Derived`:

```
class Base
  {
  public:
    virtual void Function () ;
  };
```

```
class Derived : public Base
  {};
```

16.2 For the `Base` and `Derived` classes of Exercise 16.1 use the `dynamic_cast<T&>()` operator to cast a reference to `Base` to a reference to `Derived`.

16.3 Use the `typeid()` operator to determine exactly which **`class`** derived from `Base` the `bp` pointer is pointing to:

```
void main ()
  {
  Base* bp ;
  //...
  }
```

16.4 Use the `(T)e` style cast to cast the `Base` **`class`** pointer, `bp`, to a `Derived` **`class`** pointer:

```
class Base
  {};

class Derived : public Base
  {};

void main ()
  {
  Base* bp ;
  //...
  }
```

16.5 Rather than using the `(T)e` style cast of Exercise 16.4 use the `static_cast<>()` operator.

16.6 Use the `reinterpret_cast<>()` operator to cast the following pointer to a function, `pfv`, to point to the *`Sqrt()`* function:

```
double Sqrt(double) ;

void main ()
  {
  typedef void (*pfv) () ;
  //...
  }
```

16.7 The following program results in a compilation error because it is not possible to convert from a **`const char*`** to a **`char*`**. Use the `const_cast<>()` operator to remove this compilation error and hence determine the length of `string`:

```
// returns the length of a string
int StrLength (char* string)
  {
  //...
  }

void main ()
  {
```

```
const char* string = "MEMORY_ERROR" ;
//...
cout << "string length: " << StrLength (string) ;
}
```

Chapter 17

Input and Output, Files and Streams

from atoms to bits
Negroponte (1995)

The vast majority of computer programs involve some degree of input or output or both. This chapter examines the C++ approach to the input and output of objects.

To begin with, the notion of a stream is discussed. The standard input and output streams are the keyboard and the display screen, respectively. The C++ language offers a programmer an extensive input and output class library, with high-level classes such as basic_istream and basic_ostream for the manipulation of input and output.

Ever since the simplest of programs in Chapter 3 we have seen the repeated use of the input and output extraction and insertion operators >> and <<. The use of >> and >> for both integral and user-defined types is examined more closely in the present chapter. The formatting of input and output is essential for achieving good program–user interaction. In C++, input and output can be formatted either by the use of formatting functions or manipulators which can be both non-parametrised or parametrised and specifically defined for user-defined classes.

The reading from and writing to disk of objects of integral and user-defined types is covered, along with the associated C++ file stream classes and their operations. User-defined file stream classes are developed which are derived from the C++ file stream classes. This is then followed by spending some time discussing the development of a slightly larger class called World, which reads and writes a set of geometric objects from disk in a purely free-format manner.

C++ supports the redirection of standard input and output to alternative devices. The main() function is revisited from the point of view of command-line arguments. For console-based applications, C++ provides a library class called constream with several corresponding console stream manipulators.

Although not object-oriented, the latter half of the chapter presents an overview of the C function-based stream mechanism because it is so well supported and still widely used in C/C++ programming. The chapter concludes by developing a device-independent bitmap class, DIBitmap, for Windows. The class is capable of reading from and writing to file bitmaps of 1, 4, 8 or 24 bits per pixel.

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

17.1 Why Wait Until Now?

Until now we have made good use of the standard input and output stream objects `cin` and `cout`, respectively, with little explanation of the input and output **`class`** libraries from which they are defined. Also, the topics of writing and reading objects to and from a disk file have somehow been bypassed except for a brief discussion in Chapter 4. This is because input and output are not central to the object-oriented and **`class`** concepts of the C++ language, and to appreciate the C++ input and output mechanism requires a reasonably good understanding of *the object* and inheritance. Furthermore, the C++ stream classes form a comprehensive **`class`** library, so don't worry if it takes you a great deal of time and effort before you feel comfortable with the library. I certainly recommend referring to your compiler's **`class`** library reference manual for detailed information regarding the data members and member functions of each of the stream classes.

17.2 Streams and Stream Classes

A frequently used term when discussing input and output is the *stream*. A stream is an abstraction which refers to a common interface to the various input and output devices of a computer. There are two forms of stream: text and binary. Text streams are used with ASCII characters, whereas binary streams can be used with any type of data. The synonyms *extracting* or *getting* are generally used when referring to the input of data from a device and *inserting* or *putting* when referring to the output of data to a device.

The C++ stream classes form a **`class`** hierarchy shown in Fig. 17.1. It is seen that the stream **`class`** hierarchy is divided into two main areas: classes derived from `ios_base` and classes derived from `basic_streambuf`. Note that access from `basic_ios`-based classes to `basic_streambuf`-based classes is obtained via a pointer declared in `basic_ios`. The following sections briefly describe the `ios_base` and `basic_streambuf` base classes.

17.2.1 Character Traits

In the following sub-sections we will observe the template argument names `CharT` and `Traits`. `CharT` denotes the character type such as **`char`** or **`wchar_t`** whereas type `Traits` specifies the character traits used by a **`class`**. The properties of a character type are defined by a specialisation of the `char_type` **`struct`** declared in STRING.H:

```
template <class CharT>
struct char_traits
  { /* ... */ };
```

To define a valid character type for a stream then a specialisation must provide all of the types and member functions declared by `char_traits`. For the two most frequently used types of **`char`** and **`wchar`** the C++ standard library provides appropriate specialisations:

```
template<> struct char_traits<char>    { /* ... */ };
template<> struct char_traits<wchar_t> { /* ... */ };
```

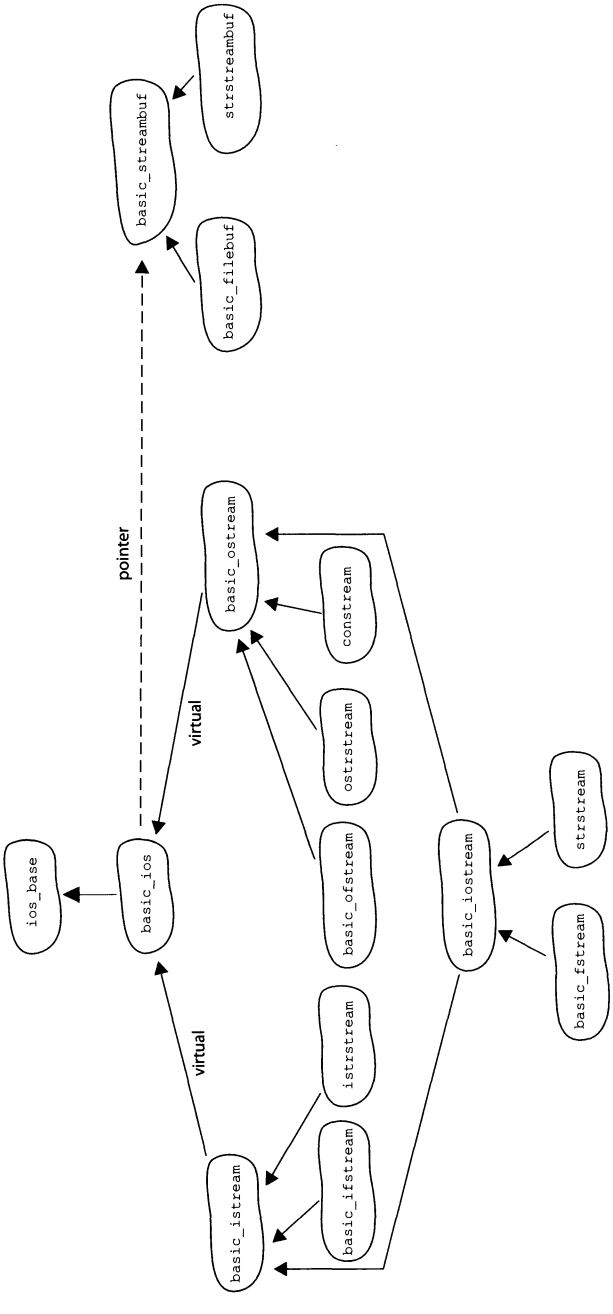


Fig. 17.1 C++ stream **class** hierarchy illustrating base classes `ios_base` and `basic_streambuf` and their derived classes.

17.2.2 The ios_base Base class

The ios_base **class** provides methods common to both formatted input and output with error checking. The classes basic_istream and basic_ostream are both derived from basic_ios which in turn is derived from ios_base. Class ios_base is declared in header IOSBASE.H.

Input and output streams implemented using either **char** or **wchar_t** are supported via the **typedef**s ios and wios:

```
typedef basic_ios<char,char_traits<char> >        ios ;
typedef basic_ios<wchar_t,char_traits<wchar_t> > wios ;
```

For convenience, throughout this chapter the stream classes will be generally referred to as their **char typedef** equivalent. It is worth noting that the **typedef** names are compatible with the pre-standardisation names for the stream classes.

17.2.3 The basic_streambuf Base class

The basic_streambuf **class** provides a buffer interface between data and storage areas, such as memory and physical devices, and supports low-level methods for handling streams and data buffering when negligible formatting is required. The buffering classes basic_filebufand strstreambuf are both derived from basic_streambuf which supports the following **char** and **wchar_t typedef**s:

```
typedef basic_streambuf<char,char_traits<char> >
  streambuf ;
typedef basic_streambuf<wchar_t,char_traits<wchar_t> >
  wstreambuf ;
```

17.2.4 Stream Classes Derived from basic_ios and basic_streambuf

Figure 17.1 illustrates that the derivation of classes from ios_base and basic_streambuf is comprehensive, so the following sections briefly describe the most frequently used classes, basic_istream, basic_ostream, basic_iostream, basic_ifstream, basic_ofstream and basic_fstream, which are derived from the ios_base base **class**.

The basic_istream **class**

Class basic_istream allows you to define an input stream and supports methods for both formatted and unformatted input. The extraction operator, >>, is overloaded for all C++ integral data types, thus making high-level input operations possible. Classes basic_ifstream, istrstream and basic_iostream are all derived from basic_istream.

It is worth noting that **class** basic_istream is derived from **class** basic_ios using **virtual** inheritance to avoid multiple **class** member declarations when **class** basic_iostream is derived from both basic_istream and basic_ostream using multiple inheritance:

```
template <class CharT, class Traits> // ISTREAM.H/IOSTREAM.H
class basic_istream : virtual public basic_ios<CharT,Traits>
  { /* ... */ };
```

For convenience input streams implemented using either **char** or **wchar_t** are supported via the **typedef**s istream and wistream:

```
typedef basic_istream<char,char_traits<char> >    istream ;
typedef basic_istream<wchar_t,char_traits<char> > wostream ;
```

The basic_ostream **class**

Similarly, the basic_ostream **class** allows a user to define an output stream and supports methods for both formatted and unformatted output. The insertion operator, <<, is overloaded for all C++ integral data types. Classes basic_ofstream, ostrstream and basic_iostream are all derived from basic_ostream.

As with basic_istream, **class** basic_ostream is **virtual**ly derived from **class** basic_ios to avoid multiple declarations when declaring basic_iostream:

```
template <class CharT, class Traits>  // OSTREAM.H/IOSTREAM.H
class basic_ostream : virtual public basic_ios<CharT,Traits>
  { /* ... */ };
```

In an analogous manner to **class** basic_istream there are **char** and **wchar_t** **typedef**s for basic_ostream:

```
typedef basic_ostream<char,char_traits<char> >    ostream ;
typedef basic_ostream<wchar_t,char_traits<char> > wostream ;
```

The basic_iostream **class**

The basic_iostream **class** is derived from basic_istream and basic_ostream, and as a result caters for both input and output of a stream. Classes basic_fstream and strstream are both derived from basic_iostream.

Since **class** basic_iostream is derived from both basic_istream and basic_ostream, basic_iostream is declared using multiple inheritance:

```
template <class CharT, class Traits>// ISTREAM.H/IOSTREAM.H
class basic_iostream : public basic_istream<CharT,Traits>,
                       public basic_ostream<CharT,Traits>
  { /* ... */ };
```

The **typedef**s iostream and wiostream are declared for types **char** and **wchar_t**:

```
typedef basic_iostream<char,char_traits<char> >
  iostream ;
typedef basic_iostream<wchar_t,char_traits<wchar_t> >
  wiostream ;
```

The basic_ifstream **class**

The basic_ifstream **class** is derived from basic_istream, and therefore allows file input of a stream. The **class** declaration of basic_ifstream can be found in FSTREAM.H and is of the form:

```
template <class charT, class traits> // FSTREAM.H
class basic_ifstream : public basic_istream<CharT,Traits>
  { /* ... */ };
```

The `basic_ifstream` **typedef**s are:

```
typedef basic_ifstream<char,char_traits<char> >
  ifstream ;
typedef basic_ifstream<wchar_t,char_traits<wchar_t> >
  wifstream ;
```

The `basic_ofstream` **class**

The `basic_ofstream` **class** is derived from `basic_ostream`, and therefore allows file output of a stream:

```
template <class charT, class traits> // FSTREAM.H
class basic_ofstream : public basic_ostream<CharT,Traits>
  { /* ... */ };
```

The `basic_ofstream` **typedef**s are:

```
typedef basic_ofstream<char,char_traits<char> >
  ofstream ;
typedef basic_ofstream<wchar_t,char_traits<wchar_t> >
  wofstream ;
```

The `basic_fstream` **class**

The `basic_fstream` **class** is derived from `basic_iostream` and therefore caters for both file input and output of a stream:

```
template <class CharT, class Traits> // FSTREAM.H
class basic_fstream : public basic_iostream<CharT,Traits>
  { /* ... */ };
```

The `basic_fstream` **typedef**s are:

```
typedef basic_fstream<char,char_traits<char> >
  fstream ;
typedef basic_fstream<wchar_t,char_traits<wchar_t> >
  wfstream ;
```

17.2.5 Header Files

Table 17.1 associates each stream **class** with its appropriate header file. IOSBASE.H and IOS.H provide general support for all input and output streams, STREAMBUF.H stream buffers, IOSTREAM.H memory buffers, FSTREAM.H files and STRSTREA.H for strings.

Header file IOSFWD.H contains declarations of all stream classes which are sufficient to refer to them but not to apply operations to them. The header file also includes all standard

Table 17.1 C++ stream header files and associated stream classes.

Header file	*Stream classes*
IOSBASE.H/IOS.H/STREAMBUF.H	`ios_base / basic_ios / basic_streambuf`
IOSTREAM.H	`basic_istream, basic_ostream, basic_iostream`
FSTREAM.H	`basic_filebuf, basic_ifstream, basic_ofstream, basic_fstream`
STRSTREA.H	`istrstream, ostrstream, strstream, strstreambase, strstreambuf`

typedefs of the stream classes. Header file IOMANIP.H declares classes `smanip` and `smanip_fill` and defines manipulators such as *`setw()`* and *`setprecision()`*.

It is worth emphasing that each of the stream header files also declares iterator classes such as `istream_iterator` and `ostream_iterator`. These classes are used for traversing the elements of a stream and are discussed in more detail in Chapter 20 when we discuss the C++ Standard Template Library.

I recommend examining the stream **class** declarations of the six header files listed in Table 17.1 in more detail. Since the manipulation of stream objects in C++ relies heavily on standard C++ stream libraries I also recommend that you consult the ANSI/ISO standard, Standard (1998).

17.2.6 Predefined Streams

C++ defines eight predefined *open* streams in the IOSTREAM.H header file for types **char** and **wchar_t**:

```
extern istream cin ;
extern ostream cout ;
extern ostream cerr ;
extern ostream clog ;
extern wistream wcin ;
extern wostream wcout ;
extern wostream wcerr ;
extern wostream wclog ;
```

Each of the **char** stream objects `cin`, `cout`, `cerr` and `clog` is defined as an object of one of the `istream` and `ostream` classes. As we have seen throughout previous chapters, the `cin` and `cout` stream objects are associated with standard input and standard output respectively. The `cerr` and `clog` stream objects are generally used for error handling in which an object is sent to the standard output device. The only difference between `cerr` and `clog` is that `cerr` outputs an object immediately whereas `clog` buffers its output.

As an example the following generates a domain error by attempting to calculate the square of a negative number:

```
//...
errno = EZERO ;  // flush system error variable
sqrt (-1.0) ;
if (errno == EDOM)
  cerr << "sqrt() not defined for negative argument" ;
//...
```

The domain error is detected using the system error variable `errno` which is an integer and defined in ERRNO.H. `EDOM` is an identifier which indicates a domain error and note that numerous other identifiers exist such as `ENOFILE` (file not found), `ENOPATH` (path not found) and `EINVDRV` (invalid drive).

17.3 Standard Stream Output

Consider the following program, which outputs various objects of C++ integral types using the `cout` stream object:

```
// cout.cpp
// illustrates the cout stream object and insertion operator <<
#include <iostream>  // C++ O/P

void main ()
  {
  char   string[] = "hi there!" ;
  int    i (1) ;
  double d (1.25) ;

  cout << string << endl ;
  cout << i      << endl ;
  cout << d      << endl ;

  cout << "string: " << string << "; i: " << i
                                << "; d: " << d << endl ;

  cerr << d << endl ;
  clog << i << endl ;

  // void* data type
  cout << &d << endl ;
  }
```

with output to the display screen:

```
hi there!
1
1.25
string: hi there!; i: 0; d:1.25
1.25
1
0012FF78
```

As we have seen on numerous occasions in previous chapters, stream output is achieved by a combination of the `cout` stream object and the insertion, <<, operator. Operator << is the left shift binary operator, which has been overloaded for all C++ integral data types and can be overloaded for user-defined types. The left-hand operand of << is an object of **`class`** `ostream` while the right-hand operand is an object for which stream output has been defined.

The << operator is left-associative and returns a reference to the `ostream` object which invoked the operator. Returning a reference allows stream objects to be cascaded, as illustrated above in COUT.CPP.

By default, the standard output stream is directed to the console. However, we shall see later that output streams can be redirected to alternative devices, such as disk files and printers.

Program COUT.CPP also illustrates the use of output stream objects `cerr` and `clog`. Note that the **`void`*** overloaded insertion operator is called for displaying the memory address, in hexadecimal, of object `d`.

17.4 Standard Stream Input

To illustrate basic stream input of integral data types consider the program:

```
// cin.cpp
// illustrates the cin stream object and extraction operator >>
#include <iostream>  // C++ I/O

void main ()
  {
  char*  string (new char(80)) ;
  int    i (0) ;
  double d (0.0) ;

  cout << "enter a string: " ;
  cin  >> string ;

  // skip 80 char. in input stream; stop if newline encountered
  //cin.ignore (80, '\n') ;

  cout << "enter an int and a double: " ;
  cin  >> i >> d ;

  cout << "string: " << string << "; i: " << i
                                << "; d: " << d << endl ;
  }
```

with some user interaction:

```
enter a string: hi there!
enter an int and a double: 1 1.25
string: hi; i: 1; d: 1.25
```

As with stream output, stream input is a combination of the `cin` stream object and the extraction, >>, operator. The left-hand operand of >> is an object of **`class`** `istream`, while the right-hand operand is an object for which the stream input has been defined. The >> operator is left-associative and returns a reference to the `istream` object which invoked the operator, thus allowing input stream objects to be cascaded.

By default, the standard input stream is from the keyboard. As with output streams, an input stream can be redirected so that input arrives from a device other than the keyboard.

By default the >> operator extracts input from the `istream` and then transfers the input to the right-hand object until a white space character[1] (such as a space, tab, vertical tab, carriage return, new line or form feed) is encountered. This is why the string object displayed in CIN.CPP consists only of the first two letters 'hi' of the total 'hi there!' string entered, since a space exists between 'hi' and 'there!'. Any entered characters will remain in the `istream` and will thus be read by subsequent input streams. In CIN.CPP this is not what was intended, and hence the reason for the commented out statement:

```
cin.ignore (80, '\n') ;
```

Function *ignore()* is a member function of **class** `basic_istream` and has the following signature:

```
istream_type& ignore (streamsize n=1,
  int_type delim=traits::eof()) ;  // IOSTREAM.H
```

It skips over `n` characters in the input stream and stops if `delim` is encountered before the `n` characters. The call to `basic_istream::`*ignore()* in CIN.CPP assumes that at most 80 characters will be entered on a single line and the delimiter is a new line.

Try commenting out the call to *ignore()*. The output will now be:

```
enter a string: hi there!
enter an int and a double: string: hi; i: 0; d: 0
```

A user is not given the opportunity to enter integer and floating-point numbers. After the string 'hi there!' is entered, followed by a carriage return, the second line shown above is immediately displayed.

In addition, note that by default the >> operator skips any leading white space. As an illustration, execute CIN.CPP once more, but this time press carriage return several times before entering the string 'hi there!' and press the spacebar several times before entering integer and floating-point numbers:

```
enter a string:


hi there!
enter an int and a double:           1 1.25
string: hi; i: 1; d: 1.25
```

White space can be controlled by the `ios_base::skipws` flag, which skips white space on input. The `skipws` flag can be set using the `ios_base::`*setf()* member function and unset with the `ios_base::`*unsetf()* member function. The `ws` manipulator can also be used to skip leading white space.

The member functions of classes `basic_istream` and `basic_ios`, `ios_base` flags and manipulators will be discussed in more detail later in the chapter.

---

1 To test whether a given character is a white space character use the *isspace()* (**int** *isspace*(**int** `c`) ;) C++ library function, which returns non-zero if `c` is a white space character. Similar functions are *isdigit()*, *isalpha()* and *isalnum()* to test whether a character is a digit, a letter and a digit or a letter. Refer to the standard library header file CTYPE.H

17.5 Overloading the Insertion and Extraction Operators

In Chapters 10 and 11 we examined the overloading of the C++ insertion, <<, and extraction, >>, operators for a given **class**. Both the insertion and extraction operators are overloaded for C++'s integral data types in the IOSTREAM.H header file, so that the appropriate input or output function is called automatically:

```
template <class CharT, class Traits> // IOSTREAM.H
class basic_istream : virtual public basic_ios<CharT,Traits>
  {
  //...
  public:
    typedef basic_istream<CharT,Traits> istream_type ;
    //...
    istream_type& operator >>
      (istream_type& (*pf)(istream_type&)) ;
    //...
    //...
    istream_type& operator >> (int&) ;
    istream_type& operator >> (float&) ;
    istream_type& operator >> (double&) ;
    //...
  };
//...
template <class CharT, class Traits> // IOSTREAM.H
class basic_ostream : virtual public basic_ios<CharT,Traits>
  {
  //...
  public:
    typedef basic_ostream<CharT,Traits> ostream_type ;
    //...
    ostream_type& operator <<
      (ostream_type& (*pf)(ostream_type&)) ;
    ostream_type& operator << (int) ;
    ostream_type& operator << (float) ;
    ostream_type& operator << (double) ;
    //...
  };
//...
```

The C++ integral data types supported are **short**, **int**, **long**, **float**, **double**, **long double** and the pointer to **void**: **void***.

Similarly, << and >> can be overloaded for a user-defined **class**. As an illustration let's examine overloading << and >> for a basic `Point` **class**:

```
// usr_type.cpp
// illustrates overloading the insertion and extraction
// operators for a user-defined class
#include <iostream>  // C++ I/O
```

```
class Point
  {
  private:
    double x, y, z ;
  public:
    // constructors
    Point ()
      : x (0.0), y (0.0), z (0.0) {}
    Point (double x_arg, double y_arg, double z_arg=0.0)
      : x (x_arg), y (y_arg), z (z_arg) {}
    // member functions
    double& X () { return x ; }
    //...
  };

ostream& operator << (ostream& s, const Point& p)
  {
  return s << "(" << p.X() << ", "
                  << p.Y() << ", " << p.Z() << ")" ;
  }

istream& operator >> (istream& s, Point& p)
  { return s >> p.X() >> p.Y() >> p.Z() ; }

void main ()
  {
  Point p, q (4.0, 5.0, 6.0) ;

  cout << "enter a Point object p: " ;
  cin  >> p ;
  cout << "Point objects p and q: " <<  p << " " << q ;
  }
```

with some user interaction:

```
enter a Point object p: 1.0 2.0 3.0
Point objects p and q: (1, 2, 3) (4, 5, 6)
```

Provided the << and >> operators are appropriately overloaded for **class** `Point`, the use of `cin` and >> and `cout` and << for objects of **class** `Point` is identical to that for integral data types.

The **operator**<< *()* function returns a reference to `ostream`, which is necessary to enable cascading of multiple objects. Since << is a binary operator and **operator**<< *()* is a non-member function it has two arguments. The first is a reference to an `ostream` object and is the left-hand operand of <<. The second is a **const** reference to a `Point` object which is the right-hand operand of <<. The single statement of **operator**<< *()* outputs the `x`, `y` and `z` data members of a `Point` object, each separated by a comma, and with all three data members enclosed within a pair of parentheses.

The declaration and definition of **operator**>> *()* for **class** `Point` is similar to **operator**<< *()* except that it uses **class** `istream` instead of `ostream` and a non-**const** `Point` **class** object as second argument.

It is tempting when defining **operator**<< *()* to use the `cout` output stream object instead of the argument output stream object `s`:

```
ostream& operator << (ostream& s, const Point& p)
  {
  cout << "(" << p.X() << ", "
               << p.Y() << ", " << p.Z() << ")" ;
  return s ;
  }
```

This works fine provided you only intend to use **operator**<< *()* for standard stream output. The whole idea behind passing a reference to an `ostream` object as first argument is that it allows **operator**<< *()* to operate on any `ostream` object and not just `cout`.

Similarly, it is equally tempting to place user prompts within the definition of **operator**>> *()* which are hard-wired to `cout`:

```
istream& operator >> (istream& s, Point& p)
  {
  cout << "enter a Point object; x y z: " ;
  return s >> p.X() >> p.Y() >> p.Z() ;
  }
```

You may have noticed from the program USR_TYPE.CPP that there is an incompatibility between the overloaded insertion and extraction operators for **class** `Point`. The extraction operator expects the `x`, `y` and `z` data members of **class** `Point` to be in the format 1 2 3, whereas the insertion operator outputs the data members in the format (1, 2, 3). For the standard input and output streams this may not be a problem, but in the case of file streams, which we shall examine later, this incompatibility in format will result in problems when writing and reading `Point` objects to and from a disk file. Therefore, let's modify the overloaded extraction operator of **class** `Point` presented in USR_TYPE.CPP:

```
// in_out.cpp
// further illustrates overloading the insertion
// and extraction operators for class Point
#include <iostream>  // C++ I/O

class Point
  { /* ... */ };
//...
istream& operator >> (istream& s, Point& p)
  {
  char left ('('), right (')'), comma (',') ;
  return s >> left
           >> p.X() >> comma >> p.Y() >> comma >> p.Z()
           >> right ;
  }

void main ()
  {
  Point p ;
```

```
cout << "enter a Point object p: " ;
cin  >> p ;
cout << "Point object p: " <<  p ;
}
```

With some user interaction:

```
enter a Point object p: (1,2,3)
Point object p: (1, 2, 3)
```

The overloaded extraction operator of **class** `Point` now defines three temporary objects, `left`,`right` and `comma`, to enable the input format to consist of the `x`,`y` and `z` data members each separated by a comma and enclosed within a pair of parentheses.

17.6 Formatted Input and Output

C++ allows a programmer complete control over the formatting of input and output. In general there are two ways in which formatting can be specified by a programmer: manipulators and member functions of **class** `ios_base`. Let's first examine the manipulator approach.

17.6.1 Formatting via Manipulators

Throughout previous chapters we have frequently seen the use of *manipulators*, such as `endl`, to alter the format of a stream. Manipulators are function-like operators and take a stream reference as argument and return a reference to the same stream and can therefore be used in cascading several insertions or extractions. For instance, the declaration of the `endl` manipulator can be found in the IOSTREAM.H header file and is of the form:

```
template <class CharT, class Traits> // OSTREAM.H/IOSTREAM.H
basic_ostream<CharT,Traits>&
endl (basic_ostream<CharT,Traits>& os)
  {
  os.put(CharT('\n')) ;
  os.flush () ;
  return os ;
  }
```

It is seen that the `endl` manipulator is implemented as a global template function and puts a new-line into the output stream, flushes the stream and then returns a reference to `basic_ostream`. Table 17.2 lists the stream manipulators available in C++.

Note that the non-parametrised manipulators (`dec`, `endl`, `ends`, `flush`, `hex`, `oct` and `ws`) are declared in the IOSTREAM.H header file, whereas the other parametrised manipulators (*`resetiosflags()`*, *`setbase()`*, *`setfill()`*, *`setiosflags()`*, *`setprecision()`* and *`setw()`*) are declared in IOMANIP.H.

The following program illustrates typical applications of several manipulators:

```
// manip.cpp
// illustrates the C++ stream manipulators
```

Table 17.2 C++ stream manipulators.

Manipulator	*Action*
`dec`	Decimal conversion base
`endl`	Insert new line and flush an `ostream`
`ends`	Insert `NULL` terminator in string
`flush`	Flush an `ostream`
`hex`	Hexadecimal conversion base
`oct`	Octal conversion base
`ws`	Skip leading white space
`resetiosflags (long f)`	Turn off format flags specified by `f`
`setbase (int n)`	Set conversion base format flag to `n`
`setfill (int c)`	Set fill character to `c`
`setiosflags (long f)`	Turn on format flags specified by `f`
`setprecision (int p)`	Set floating-point precision to `p`
`setw (int n)`	Set field width to `n`

```
#include <iostream>   // C++ I/O
#include <iomanip>    // C++ manipulators

void main ()
  {
  int    i (1), j (11) ;
  double d1 (1.0), d125 (1.25), d12345 (1.2345) ;

  cout << i << endl ;
  cout << d1 << " " << d125 << " " << d12345 << endl ;

  cout << setprecision (3)
       << d1 << " " << d125 << " " << d12345 << endl ;

  cout << setprecision (4)
       << setw (7) << d1
       << setw (7) << d125
       << setw (7) << d12345
       << endl ;

  cout << setfill ('?')
       << setw (7) << d1
       << setw (7) << d125
       << setw (7) << d12345
       << endl ;

  cout << dec << j << " "
       << hex << j << " "
       << oct << j << " "
       << endl ;

  cout << setbase (16)
       << j
```

```
        << endl ;

    cout << dec << j << endl ;  // reset conversion base

    cout << flush ;
    }
```

with output:

```
1
1 1.25 1.2345
1 1.25 1.23
      1   1.25  1.234
??????1???1.25??1.234
11 b 13
b
11
```

In addition to the C++ manipulators it is possible to define your own manipulators. The user-defined manipulators can be either non-parametrised or parametrised. Let's first examine the slightly easier case of non-parametrised manipulators by considering the following program, which illustrates two user-defined manipulators `PointPrompt` and `PointFormat` for the input and output, respectively, of objects of user-defined **`class`** `Point`:

```
// my_manip.cpp
// illustrates user-defined non-parametrised stream manipulators
#include <iostream>   // C++ I/O
#include <iomanip>    // C++ manipulators

class Point
  {
  private:
    double x, y, z ;
  public:
    //...
  };

ostream& operator << (ostream& s, const Point& p)
  {
  return s << "(" << p.X() << ", "
           << p.Y() << ", " << p.Z() << ")" ;
  }

istream& operator >> (istream& s, Point& p)
  {
  return s >> p.X() >> p.Y() >> p.Z() ;
  }

// user-defined output manipulator
ostream& PointFormat (ostream& s)
```

```
    {
    return s << setiosflags (ios::fixed)
             << setprecision (2)
             << setw (2) ;
    }

  // user-defined input manipulator
  istream& PointPrompt (istream& s)
    {
    cout << "enter a Point object; x y z: " ;
    return s ;
    }

  void main ()
    {
    Point p, q (4.0, 5.0, 6.0) ;

    cin >> PointPrompt >> p ;

    cout << PointFormat << p << PointFormat << q << endl ;
    }
```

with some user interaction:

```
enter a Point object; x y z: 1 2 3
 (1.00, 2.00, 3.00) (4.00, 5.00, 6.00)
```

The `PointPrompt` manipulator simply prompts a user to enter the `x`, `y` and `z` data members of a `Point` object. The form of the manipulator is characteristic of a user-defined non-parametrised manipulator function in that it has a single argument which is a reference to the stream object that it is operating on, and that the manipulator function returns a reference to the same stream object.

The definition of the output manipulator `PointFormat` is similar in outline to `PointPrompt`, but illustrates that user-defined manipulators are useful for compressing several other manipulators into a single manipulator.

It is also possible in C++ to define parametrised manipulators. As an example let's revisit the `endl` manipulator mentioned above so that a variable number of new lines can be inserted into a stream instead of a single new line. The key problem in parametrising the manipulator is the limitation of the overloaded insertion, `<<`, operator syntax. The overloaded insertion operator is a binary operator with an `ostream` object as the left-hand operand and an object to be output as the right-hand operand:

```
ostream& operator << (ostream& s, const Point& p) ;
```

with a typical use:

```
Point p ;
//...
cout << p << endl ;
```

Hence there is no way of attaching an additional argument to **operator**<< *()* which specifies the number of new lines to be inserted. One solution is to declare an additional **class**, say X, which encapsulates both an additional argument, data, a data member and a pointer to a function, pf, data member:

```
// p_manip.cpp
// illustrates user-defined parametrised stream manipulators
#include <iostream>  // C++ I/O

class X
  {
  private:
    int data ;
    ostream& (*pf) (ostream&, int) ;
  public:
    X (ostream& (*pf_arg)(ostream&, int), int data_arg)
      : pf (pf_arg), data (data_arg) {}
    friend ostream& operator << (ostream& s, const X& x)
      { return (*x.pf)(s, x.data) ; }
  }; // class X

ostream& insert_lines (ostream& s, int n)
  {
  for (int i=0; i<n; i++)
    s << endl ;
  return s ;
  }

X Endl (int n)
  { return X (insert_lines, n) ; }

class Point
  { /* ... */ };
//...
void main ()
  {
  Point p, q (1.0, 1.0, 1.0), r (2.0, 2.0, 2.0) ;

  cout << p << endl ;           // non-parametrised
  cout << q << Endl (3) << r ;  // parametrised
  }
```

with output:

```
(0, 0, 0)
(1, 1, 1)


(2, 2, 2)
```

The overloaded insertion operator for **class** X calls a function specified in the definition of an object of **class** X:

```
friend ostream& operator << (ostream& s, const X& x)
  { return (*x.pf)(s, x.data) ; }
```

To define a parametrised version of the endl manipulator it is a simple matter of now defining a function Endl so that it is passed a variable argument which indicates the number of new lines to be inserted into the stream and returns an object of **class** X, which in turn calls (via the function pointer) a hidden function, *insert_lines()*, which actually inserts the specified number of new lines into the stream.

An alternative approach to user-defined parametrised manipulators is to make use of the **template** smanip and smanip_fill classes declared in IOMANIP.H.

```
template <class T>
class smanip
  {
  public:
    ios_base& (* pf)(ios_base&, T) ;
    T data ;
    smanip (ios_base& (*pf_arg)(ios_base&, T), T data_arg)
        : pf (pf_arg), data (data_arg) {}
  };

template <class T, class Traits>
class smanip_fill
  {
  public:
    basic_ios<T,Traits>& (*pf)(basic_ios<T,Traits>&,T) ;
    T data ;
    smanip_fill (basic_ios<T,Traits>&
      (*pf_arg)(basic_ios<T,traits>&, T), T data_arg)
        : pf (pf_arg), data (data_arg) {}
  };
```

Class smanip_fill is analogous to that of smanip except that it caters for character traits, Traits. As a result we will focus on smanip for which the << and >> operators are overloaded:

```
template <class CharT, class Traits, class T>
basic_ostream<CharT,Traits>& operator << (basic_
ostream<CharT,Traits>& os, const smanip<T>& m)
  { (*m.pf)(os, m.data) ; return os ; }

template <class T, class Traits>
basic_istream<T,Traits>& operator >>
(basic_istream<T,Traits>& is, const smanip_fill<T,Traits>& m)
  { (*m.pf)(is, m.data) ; return is ; }
```

On comparing the **class** declarations of smanip and X in P_MANIP.CPP we observe that the declarations are of identical format except that X has been replaced by smanip and **int** re-

placed by the parametrised type T. The smanip and smanip_fill **template** classes declared in IOMANIP.H can be used to create user-defined parametrised input, output, and input and output stream manipulators respectively. Their operation is identical to that outlined for **class** X in P_MANIP.CPP in that a parametrised manipulator calls (via a pointer to a function) a user-defined function which actually does the work. The following program again parametrises the endl manipulator, as in P_MANIP.CPP, but using smanip instead of X:

```
// p_manip1.cpp
// an alternative approach to user-defined
// parametrised stream manipulators
#include <iostream>   // C++ I/O
#include <iomanip>    // C++ manipulators

ios_base& insert_lines (ios_base& s, int n)
  {
  // cast from ios_base to ostream
  ostream& sc = dynamic_cast<ostream&>(s) ;
  for (int i=0; i<n; i++)
      sc << std::endl ;
  return sc ;
  }

smanip<int> endl (int n)
  {
  return smanip<int> (insert_lines, n) ;
  }

class Point
  { /* ... */ };
//...
void main ()
  {
  Point p, q (1.0, 1.0, 1.0), r (2.0, 2.0, 2.0) ;

  cout << p << endl ;            // non-parametrised
  cout << q << endl (3) << r ;   // parametrised
  }
```

with output identical to that generated by P_MANIP.CPP. Note that unlike program P_MANIP.CPP the endl manipulator has now been overloaded but has required a cast from ios_base to ostream in *insert_lines()* since ios_base does not overload <<.

17.6.2 Formatting via Member Functions of **class** ios_base

The previous subsection illustrated the use of an ios_base format flag, ios_base::fixed, with the C++ *setiosflags()* manipulator in the definition of the user-defined manipulator PointFormat. The ios_base format flags determine how both input and output are to be formatted and are generally specified as an enumerated **long** integer within the fmt_flags **enum** in IOSBASE.H:

```
class ios_base
  {
  public:
    //...
    enum fmt_flags
      {
      boolalpha   = 0x0001, // symbolic rep. of true and
                            // false
      dec         = 0x0002, // deicmal conversion
      fixed       = 0x0004, // use floating-point notation
      hex         = 0x0008, // hexadecimal conversion
      internal    = 0x0010, // padding
      left        = 0x0020, // left-justified
      oct         = 0x0040, // octal conversion
      right       = 0x0080, // right-justified
      scientific  = 0x0100, // use scientific notation
      showbase    = 0x0200, // show base
      showpoint   = 0x0400, // show decimal point
      showpos     = 0x0800, // show positive sign
      skipws      = 0x1000, // skip whitespace
      unitbuf     = 0x2000, // flush O/P streams
      uppercase   = 0x4000, // use uppercase for e and x
      adjustfield = left|right|internal, // field adj. flags
      basefield   = dec|oct|hex, // int base flags
      floatfield  = scientific|fixed // floating-point flags
      };
  };
```

When a format flag is set, the respective formatting feature is turned on; otherwise the formatting feature is turned off and default formatting is performed.

Let's describe each of the above `ios_base` format flags briefly. The `boolalpha` flag uses the symbolic representation of true and false rather than 0 and 1. The flags `dec`, `hex` and `oct` are the decimal, hexadecimal and octal conversion bases respectively. The `internal` flag pads the field of a numeric number after the sign or base indicator of the number. The `left` and `right` flags left- and right-justify stream output respectively. The `showbase` flag ensures that the base of a numeric number is displayed, whereas `showpoint` displays the decimal point for floating-point numbers. The `skipws` flag skips leading white space for stream input and is set by default. When the `uppercase` flag is set, the lowercase letters `e` and `x` for scientific notation and hexadecimal numbers will be displayed in uppercase. The `showpos` flag causes a positive sign to prefix positive numeric numbers. The flag `scientific` causes a numeric number to be displayed using scientific notation, whereas the `fixed` flag causes floating-point numbers to be displayed to six decimal places with a decimal point. When the `unitbuf` flag is set output streams are flushed following insertion. Finally, `adjustfield`, `basefield` and `floatfield` are flags relating to field adjustment, integer base and floating-point output.

The `ios_base` format flags are generally set, unset and read using the *`setf()`*, *`unsetf()`* and *`flags()`* member functions of **`class`** `ios_base`:

```
fmtflags setf (fmtflags fmtfl) ;
fmtflags setf (fmtflags fmtfl, fmtflags mask) ;
void     unsetf (fmtflags mask) ;
```

```
fmtflags flags () const ;
fmtflags flags (fmtflags fmtfl) ;
```

The first version of *setf()* is the most frequently used and sets the format flag(s) specified by the argument `fmtfl` and returns the previous settings of the format flags. The second form of *setf()* clears the flags specified in `mask` and sets them to those specified in `fmtfl`. The three flags `adjustfield`, `basefiled` and `floatfield` are intended to be used with `mask`. Member function *unsetf()* clears the flag(s) specified by `mask`. The first version of *flags()* simply returns the current format flag settings, while the second version sets the flags specified by `fmtfl` and returns the previous settings. The call *flags(0)* sets the default values of the formatting flags.

The following program helps to illustrate the `ios_base` format flags and the *setf()*, *unsetf()* and *flags()* member functions:

```
// ios_f_mf.cpp
// illustrates the format flags and the setf(),
// unsetf() and flags() member functions of class ios_base
#include <iostream>  // C++ I/O

// display the state of the fifteen ios_base format flags
void DisplayFlags (long f)
  {
  for (long i=0x4000; i; i=i>>1)
    i & f ? cout << "1" : cout << "0" ;
  cout << endl ;
  }

void main ()
  {
  long f = cout.flags () ;

  cout << "default format flags: " ;
  DisplayFlags (f) ;

  cout.setf (ios::showpos | ios::fixed) ;
  f = cout.flags () ;
  DisplayFlags (f) ;
  cout << 10.01 << endl ;

  cout.unsetf (ios::showpos) ;
  f = cout.flags () ;
  DisplayFlags (f) ;

  // set the default format flags
  cout.flags (0) ;
  f = cout.flags () ;
  DisplayFlags (f) ;
  }
```

with output:

```
default format flags: 001000000000010
001100000000110
+10.010000
001000000000110
000000000000000
```

The program output illustrates that in the present case the default settings for the 15 `ios_base` format flags is that `skipws` and `unitbuf` are turned on while the other 13 flags are turned off. Note that the use of the bitwise inclusive operator | allows two or more format flags to be specified in a single function call.

Formatting can also be specified via the *`fill()`*, *`precision()`* and *`width()`* member functions of **`class`**es `ios_base` and `basic_ios`. The two versions of `basic_ios::`*`fill()`* are:

```
char_type fill () const ;
char_type fill (char_type c) ;
```

The first version simply returns the current fill character, which is a space by default. The second version sets the current character to `c` and returns the previous fill character.

There are two versions of `ios_base::`*`precision()`*:

```
streamsize precision () const ;
streamsize precision (streamsize n) ;
```

The first version returns the current precision setting whereas the second version sets the floating-point precision to `n`, returning the previous precision setting. The default number of decimal places is six.

The `ios_base::`*`width()`* member function takes one of two forms:

```
streamsize width () const ;
streamsize width (streamsize n) ;
```

You have probably guessed that the first version of *`width()`* simply returns the current field width setting, whereas the second version sets the field width to `n` and returns the previous field width.

The operations of the member functions `basic_ios::`*`fill()`*,`ios_base::`*`precision()`* and `ios_base::`*`width()`* are similar to the *`setfill()`*, *`setprecision()`* and *`setw()`* manipulators, respectively, discussed in the previous subsection.

An interesting application of `ios::`*`width()`* is the prevention of overflowing when entering the characters of a fixed-sized string:

```
// width.cpp
// illustrates the ios::width() member function
#include <iostream>  // C++ I/O

void main ()
  {
  char string[10] ;

  // prevent overflow
  cin.width (sizeof (string)) ;
```

```
    cin  >> string ;
    cout << string << endl ;
    }
```

with some user interaction:

```
abcdefghijklmnop
abcdefghi
```

17.7 Locale

The standard library supports a class called `locale` that controls the appearance of numeric value on both input and output, classification of characters into letters, digits etc and the collation order of strings. Locale objects are used by the stream library to ensure that culture conventions are adhered to. Class `locale` is declared in LOCALE.H:

```
class locale
  {
  public:
    locale () throw () ;
    explicit locale (const char* name) ;
    locale (const locale& l) throw () ;
    ~locale () ;
    string name () const ;
    const locale& operator = (const locale& l) throw () ;
    bool          operator == (const locale& l) const ;
    bool          operator != (const locale& l) const ;
    static locale global (const locale&) ;
    static const  locale& classic () ;
  };
```

The *global()* member function sets the global locale, *classic()* returns the classic "C" locale and *name()* returns the name of a locale and returns "*" if the locale is unnamed. Also, class `ios_base` supports the *getloc()* and *imbue()* member functions which return the current locale and set the locale:

```
class ios_base
  {
  public:
    locale getloc () const ;
    locale imbue (const locale& loc) ;
  };
```

The following program illustrates class `locale`:

```
// locale.cpp
// illustrates class locale
#include <iostream>  // C++ I/O
#include <locale>    // class locale
```

```
void main ()
  {
   cout << "current locale name: "
        << cin.getloc().name() << endl ;

   locale new_loc ("POSIX") ;

   cin.imbue (new_loc) ; // let cin use POSIX locale

   cin.imbue (locale::classic()) ; // reset locale to classic
   }
```

17.8 Working with Disk Files

Let's now examine file handling. The C++ stream **class** library contains the three classes `basic_ifstream`, `basic_ofstream` and `basic_fstream`, and associated methods for creating files and handling file input and output. The stream **class** hierarchy shown in Fig. 17.1 illustrates that the input file stream **class** `basic_ifstream` inherits the stream and extraction operations of `basic_istream`, whereas the output file stream **class** `basic_ofstream` inherits the stream and insertion operations of `basic_ostream`. The `basic_fstream` **class** is derived from `basic_iostream` and consequently allows for both input and output.

All three classes `basic_ifstream`, `basic_ofstream` and `basic_fstream` are declared in the header file FSTREAM.H, which, incidentally, includes the IOSTREAM.H header file so that you do not need to `#include` IOSTREAM.H explicitly if header file FSTREAM.H is included. For convenience the header file FSTREAM.H defines the **typedef**s `ifstream`, `ofstream` and `fstream` when using type **char**.

17.8.1 File Input Using **class** `basic_ifstream`

To see file handling in action, let us first examine a program called IFSTR.CPP, which reads a file and then proceeds to display the contents of the file to the screen. The program makes use of the `basic_ifstream` **class**, so first let's take a look at the **class** declaration of `basic_ifstream`:

```
template <class CharT, class Traits> // FSTREAM.H
class basic_ifstream : public basic_istream<CharT,Traits>
  {
  public:
    basic_ifstream () ;
    explicit basic_ifstream
      (const char* s, ios_base::openmode mode=ios_base::in) ;
    virtual ~basic_ifstream() ;
    basic_filebuf<CharT,Traits>* rdbuf () const ;
    bool is_open () ;
    void open (const char* s,
               ios_base::openmode mode=ios_base::in) ;
    void close () ;
  };
```

The no-argument constructor defines a `basic_ifstream` object which is not attached to a file. The two-argument constructor defines a `basic_ifstream` object, opens the file of filename `s` in open for reading mode and then attaches the object to the file. Note that, by default, the file is not created if it does not exist. The `mode` argument will be discussed with reference to the *`open()`* member function later, but note at this point that the default value for `mode` is `ios_base::in`, which indicates input.

Member function *`rdbuf()`* returns a pointer to the `basic_filebuf` used. The *`open()`* member function will be examined in more detail later in the present chapter for classes `basic_ifstream`, `basic_ofstream` and `basic_fstream`. Member function *`close()`* closes an opened file.

Here is the program IFSTR.CPP:

```
// ifstr.cpp
// illustrates the ifstream stream class
// for handling file input
#include <fstream>  // C++ file I/O

void main ()
  {
  char file_name[] = "IFSTR.CPP" ;

  // define an ifstream object
  ifstream in_file (file_name) ;

  if (!in_file)
    cerr << "sorry-can't open file \""
         << file_name << "\" " << endl ;

  char ch ;

  while (in_file)
    {
    in_file.get (ch) ; // get a character
    cout << ch ;
    }
  }
```

which displays itself on the screen.

An object `in_file` is defined of **`class`** `ifstream` as follows:

```
ifstream in_file (file_name) ;
```

in which `file_name` is a **`char`*** string. The filename passed to the constructor of `ifstream` can also include a path name; for instance:

```
ifstream in_file ("C:\\BCC55\\INCLUDE\\FSTREAM.H") ;
```

Note the use of the double backslash characters (`\\`). If a path name is not explicitly given it is assumed that the file resides in the current directory. If the file does not exist or cannot be opened the associated stream object is set to zero, which is useful for confirming that the requested file has indeed been opened before you begin to operate on it:

```
if (!in_file)
  cerr << "sorry-can't open file \""
       << file_name << "\" " << endl ;
//...
```

Assuming that the requested file is successfully opened, IFSTR.CPP proceeds to read the contents of the file character by character using the `basic_istream::get()` member function (which `basic_ifstream` inherits from `basic_istream`) and immediately outputs each character to the screen. There are a few versions of `basic_istream::get()`:

```
// extract single character
int_type      get () ;
istream_type& get (char_type& c) ;
// extract single character in to buffer
istream_type& get (char_type* s, streamsize n) ;
istream_type& get (char_type* s, streamsize n,
                   char_type delim) ;
// extract character in to a streambuf
istream_type& get (streambuf_type& sb) ;
istream_type& get (streambuf_type& sb, char_type delim) ;
```

The no-argument version extracts a character or end-of-file. The second version extracts a character from the input stream, places it into `c` and returns a reference to a `basic_istream` object (that is, **typdef** `basic_istream<CharT,Traits> istream_type`). The third and fourth versions extract a character and place it in buffer `s` until a delimiter is encountered. The fifth and sixth versions extract a character into the supplied `streambuf` object until a delimiter is encountered. The default delimiter is a new line character or until the end-of-file is encountered. Alternatively, a delimiter can be supplied. The array pointed to by buffer `s` is null terminated.

Rather than using the `get()` member function, which extracts a single character, the `basic_istream::getline()` member function could have been used, which is similar to `get()` but extracts characters and places them in an array pointed to by buffer `s` until either the default or user-supplied delimiter `delim` is encountered or until the end-of-file is encountered:

```
istream_type& getline (char_type* s, streamsize n) ;
istream_type& getline (char_type* s, streamsize n,
                       char_type delim) ;
```

The `getline()` member function is useful for reading a string that contains white space. Note that `getline()` extracts the delimiter, whereas `get()` does not. The following program modifies IFSTR.CPP for `getline()`:

```
// ifstr1.cpp
// illustrates the ifstream stream class for handling file
// input and the istream::getline() member function
#include <fstream>  // C++ file I/O

void main ()
  {
  char file_name[] = "IFSTR1.CPP" ;
```

```
    // define an ifstream object
    ifstream in_file (file_name) ;

    if (!in_file)
      cerr << "sorry-can't open file \""
           << file_name << "\" " << endl ;

    const int BUF_SIZ (80) ;
    char      buffer[BUF_SIZ] ;

    while (in_file)
      {
      // get a line of characters
      in_file.getline (buffer, BUF_SIZ-1) ;
      cout << buffer << endl ;
      }
    }
```

Let's consider another application of the `basic_istream::getline()` member function to read the number of lines of a given file requested by a user:

```
// no_lines.cpp
// illustrates the use of the istream::getline() member
// function to determine the number of lines of a file
#include <fstream>  // C++ file I/O
#include <ctype>    // toupper()

void main ()
  {
  char file_name[50] ;

  cout << "enter a filename: " ;
  cin  >> file_name ;

  // convert file_name entered to uppercase
  for (int i=0; i<50; i++)
    file_name[i] = toupper (file_name[i]) ;

  // define an ifstream object
  ifstream in_file (file_name) ;

  if (!in_file)
    cerr << "sorry-can't open file \""
         << file_name << "\" " << endl ;

  const int BUF_SIZ (120) ;
  char      buffer[BUF_SIZ] ;
  int       line_count (0) ;  // line count

  while (in_file)
```

```
    {
    in_file.getline (buffer, BUF_SIZ-1) ;
    line_count++ ;  // increment line count
    }
  cout << "no. of lines of \"" << file_name << "\: "
       << line_count << endl ;
  }
```

with some user interaction:

```
enter a filename: no_lines.cpp
no. of lines of "NO_LINES.CPP": 35
```

The above program simply increments a counter, `line_count`, after each line of the input file is extracted by *`getline()`*. For the purposes of illustration, NO_LINES.CPP also makes use of the C++ library function *`toupper()`*[2]:

```
int toupper (int ch) ;  // CTYPE.H
```

which translates character `ch` to uppercase if it is not already. Function *`toupper()`* is used in NO_LINES.CPP to convert the filename entered by a user to uppercase before defining the `ifstream` object `in_file`.

Note that the number of lines returned by the above program is only approximate, since several new line characters are usually floating around at the end of a file.

Before discussing **`class`** `basic_ofstream` there are two additional member functions of **`class`** `basic_istream` which deserve a mention: *`peek()`* and *`putback()`*, with declarations:

```
int_type      peek () ;
istream_type& putback (char_type c) ;
```

`peek()` returns the next character in an input stream without actually extracting the character from the stream, while *`putback()`* places a character back into the input stream.

17.8.2 File Output Using `class basic_ofstream`

The `ofstream` **`class`** is designed for handling file output, and has the following declaration:

```
template <class CharT, class Traits> // FSTREAM.H
class basic_ofstream : public basic_ostream<CharT,Traits>
  {
  public:
    basic_ofstream () ;
    explicit basic_ofstream
      (const char* s, ios_base::openmode mode=ios_base::out) ;
    virtual ~basic_ofstream () ;
    basic_filebuf<CharT,Traits>* rdbuf () const ;
```

2 The complement to *`toupper()`* is the function **`int`** *`tolower(`***`int`** `ch`*`)`*, which translates character `ch` to lowercase if it is not already.

```
    bool is_open () ;
    void open (const char* s,
               ios_base::openmode mode=ios_base::out) ;
    void close () ;
};
```

It is of a similar form to `basic_ifstream`. The default value for `mode` is `ios_base::out`, indicating output. If a given file pointed to by `s` does not exist then, by default, it is created. The following program illustrates output to a file called OUT.TXT using an `ofstream` object:

```
// ofstr.cpp
// illustrates the ofstream stream class
// for handling file output
#include <fstream>  // C++ file I/O
#include <string>   // strlen()

void main ()
  {
  char file_name[] = "OUT.TXT" ;

  // define an ofstream object
  ofstream out_file (file_name) ;

  if (!out_file)
    cerr << "sorry-can't open file \""
         << file_name << "\" " << endl ;

  char   string[] = "I am a string" ;
  int    i (0) ;
  double d (1.25) ;

  // <<
  for (int j=0; j<strlen (string); j++)
    out_file << *(string+j) ;
  out_file << endl ;
  out_file << i << endl ;
  out_file << d << endl ;

  char* an_string (new char[20]) ;
  an_string = "I am another string" ;

  // ostream::put()
  while (*an_string)
    out_file.put (*an_string++) ;
  out_file.put ('\n') ;
  }
```

After execution of the above program the contents of file OUT.TXT should read:

```
I am a string
0
1.25
I am another string
```

OFSTR.CPP illustrates two forms of writing to a disk file. The first uses the << insertion operator, while the second approach uses the `basic_ostream::put()` member function, which **class** `basic_ofstream` inherits from **class** `basic_ostream`:

```
ostream_type& put (char_type c) ;
```

The member function *put()* inserts a character `c` into the output stream and returns a reference to a `basic_ostream` object.

17.8.3 File Input and Output Using **class** `basic_fstream`

The stream **class** `basic_fstream` is derived from `basic_iostream`, and correspondingly caters for both file input and output:

```
template <class CharT, class Traits>  // FSTREAM.H
class basic_fstream : public basic_iostream<CharT,Traits>
  {
  public:
    basic_fstream () ;
    explicit basic_fstream (const char* s,
      ios_base::openmode mode=ios_base::in | ios_base::out) ;
    virtual ~basic_fstream () ;
    basic_filebuf<CharT,Traits>* rdbuf () const ;
    bool is_open () ;
    void open (const char* s,
      ios_base::openmode mode=ios_base::in | ios_base::out) ;
    void close () ;
  };
```

The following program illustrates simple file input and output using an `fstream` object:

```
// fstr.cpp
// illustrates the fstream stream class
// for handling file input and output
#include <fstream>  // C++ file I/O

void main ()
  {
  char file_name[] = "IN_OUT.TXT" ;

  // define an ofstream object
  fstream io_file (file_name) ;

  if (!io_file)
    cerr << "sorry-can't open file \""
```

```
        << file_name << "\" " << endl ;

  // read from file and echo to screen
  char ch ;

  while (io_file)
    {
    io_file.get (ch) ; // get a character
    cout << ch ;
    }

  // write to file
  char* string (new char[46]) ;
  string = "I am probably a different string than the one read" ;
  while (*string)
    io_file.put (*string++) ;
  io_file.put ('\n') ;
  }
```

The program uses an object, `io_file`, of **`class`** `fstream` to first read the contents of file IN_OUT.TXT and then, using the same object, append an additional string to IN_OUT.TXT.

17.8.4 Stream Opening `mode` and `access`

In the previous subsection we saw the use of the `ios_base::in` and `ios_base::out` stream operation and opening modes for the second argument of the `fstream` constructor. The stream opening modes determine how a file is to be opened and are generally specified as an enumerated **`int`** declared in **`class`** `ios_base` in the IOSBASE.H header file:

```
enum open_mode
  {
  app       = 0x01, // append to end-of-file
  binary    = 0x02, // binary mode
  in        = 0x04, // open for reading (default for
                    // ifstream)
  out       = 0x08, // open for writing (default for
                    // ofstream)
  trunc     = 0x10, // truncate file to zero if exists
  ate       = 0x20, // seek to end-of-file
  nocreate  = 0x40, // open() fails if file does not exist
  noreplace = 0x80  // open() fails if file exists
  };
typedef int openmode ;
```

The flags `in` and `out` indicate that a file is capable of input (open for reading) and output (open for writing) respectively. We saw in the above **`class`** declarations that `ios_base::in` and `ios_base::out` are the default opening modes for the two-argument constructors and *`open()`* member functions of `basic_ifstream` and `basic_ofstream`, respectively. The two-argument constructor of **`class`** `basic_fstream` specifies both `ios_base::in` and `ios_base::out`, since `basic_fstream` is capable of both input and output.

Table 17.3 MS-DOS and Windows access permission values.

access attribute	*Permission*
0	Read and write file (default)
1	Read only file
2	Hidden file
4	System file
8	Archive bit set

The `ios_base::ate` flag results in a seek to the end-of-file, while `ios_base::app` appends output to any existing file contents. The `ios_base::trunc` deletes the existing contents of a file (if any), thus truncating the file length to zero.

The `ios_base::nocreate` and `ios_base::noreplace` flags cause the *open()* member function to fail if a file does not exist and to fail if a file does exist, respectively.

Finally, the `ios_base::binary` flag ensures that a file is opened in binary mode rather than text mode, text mode being the default. For instance, when a file is opened in text mode the carriage return/line feed sequence is converted to the new line '`\n`' character and vice versa for file output. Such conversions are not performed when in binary mode.

The `access` attribute specifies the access permission of a file. For Microsoft MS-DOS and Windows, Table 17.3 lists the five available attributes.

17.8.5 Stream Status Flags

The base stream **`class`** `ios_base` declares an enumerated type called `io_state` with appropriate members that indicate the state of a stream:

```
enum io_state
  {
  goodbit = 0x00, // no bit set- no errors
  badbit  = 0x01, // invalid operation attempted
  eofbit  = 0x02, // end-of-file
  failbit = 0x04  // last operation failed
  };
typedef int iostate ;
```

When handling files it is important to know if any error(s) have occurred and if so to know specifically which error(s) so that appropriate action can be taken. For instance, the `basic_istream::`*get()* member function fails if it encounters an end-of-file before a character is extracted and then proceeds to set the `ios_basic::failbit` flag to inform the stream of failure.

Class `basic_ios` supports a number of member functions to help diagnose the stream status flags and stream errors:

```
iostate rdstate () const ; // IOS.H
bool    eof ()  const ;
bool    fail () const ;
bool    bad ()  const ;
bool    good () const ;
void    clear (iostate state=goodbit) ;
void    setstate (iostate state) ;
```

rdstate() returns the current state of a stream. The member function *eof()* returns non-zero if at the end of a file, *fail()* returns non-zero if an operation failed and *bad()* returns non-zero if an error occurred. The *good()* member function returns non-zero if no stream status flags have been set (i.e. no stream errors have occurred). *clear()* and *setstate()* set the state of a stream to a given value.

The following program illustrates the ios_base stream status member functions:

```
// ios_stat.cpp
// illustrates the ios stream status flags
// and associated member functions
#include <fstream>  // C++ file I/O
#include <iomanip>  // setw(), ...

void main ()
  {
  // file stream object
  ifstream file ;

  // get file to open
  char file_name[50] ;
  cout << "enter a filename: " ;
  cin  >> file_name ;

  // open
  file.open (file_name, ios::in | ios::nocreate) ;

  // test
  if (!file)
    cerr << "file \"" << file_name
         << "\" not opened " << endl ;
  else
    cout << "file \"" << file_name
         << "\" opened " << endl ;

  cout << setw (15) << "file.rdstate():"
       << setw (2) << file.rdstate () << endl ;
  cout << setw (15) << "file.good()   :"
       << setw (2) << file.good ()    << endl ;
  cout << setw (15) << "file.eof()    :"
       << setw (2) << file.eof ()     << endl ;
  cout << setw (15) << "file.fail()   :"
       << setw (2) << file.fail ()    << endl ;
  cout << setw (15) << "file.bad()    :"
       << setw (2) << file.bad ()     << endl ;

  // close
  file.close () ;
  }
```

with some user interaction for a non-existent file:

```
enter a filename: xxx.xxx
file "xxx.xxx" not opened
file.rdstate(): 4
file.good()   : 0
file.eof()    : 0
file.fail()   : 1
file.bad()    : 4
```

Incidentally, the `basic_ios::eof()` member function is useful for detecting when the end of a file has been reached. The following program illustrates `eof()` via modification of IFSTR.CPP:

```
// eof.cpp
// illustrates the ios::eof() member function
// for detecting the end of a file
#include <fstream>  // C++ file I/O

void main ()
  {
  char file_name[] = "EOF.CPP" ;

  // define an ifstream object
  ifstream in_file (file_name) ;

  if (!in_file)
    cerr << "sorry-can't open file \""
         << file_name << "\" " << endl ;

  char ch ;

  while (!in_file.eof ())
    {
    in_file.get (ch) ; // get a character
      cout << ch ;
    }
  }
```

17.8.6 The `open()` and `close()` Member Functions

Each of the three classes discussed above, `basic_ifstream`, `basic_ofstream` and `basic_fstream`, define both `open()` and `close()` member functions.

The `open()` member function for the three classes has the following declaration:

```
void open (const char* s, ios_base::openmode mode) ;
```

The argument `s` is the name of the file to be opened, whereas `mode` is the opening mode attribute discussed above. If `open()` fails the value of the object which invoked the member function is set to zero.

The declaration of `close()` is:

```
void close () ;
```

which closes a file associated with a given stream.

Since classes `basic_ifstream`, `basic_ofstream` and `basic_fstream` allow file stream objects to be defined which are not attached to a file at the time of definition, clearly the *`open()`* and *`close()`* member functions are useful for associating or disassociating, respectively, a file stream object with or from a file after object definition:

```
fstream file ;
//...
file.open (/*...*/) ;
//...
file.close () ;
```

As an illustration of the *`open()`* and *`close()`* functions, consider the following program:

```
// op_cl.cpp
// illustrates the open() and close() stream functions
#include <fstream.h>  // C++ file I/O

void main ()
  {
  // file stream object
  ifstream file ;

  // get file to open
  char file_name[50] ;
  cout << "enter a filename you wish to open and display: " ;
  cin  >> file_name ;

  // open
  file.open (file_name, ios::in | ios::nocreate) ;

  // test
  if (!file)
    cerr << "sorry-can't open file \""
         << file_name << "\" " << endl ;

  // display
  char ch ;

  while (file)
    {
    file.get (ch) ; // get a character
    cout << ch ;
    }

  // close
  file.close () ;
  }
```

What happens if an attempt is made to open a file that is already open and similarly close a file that has previously been closed? Try duplicating the call to *open()* in the above program:

```
//...
// open
file.open (file_name, ios::in | ios::nocreate) ;
file.open (file_name, ios::in | ios::nocreate) ;
```

If the above program is now recompiled and executed, then upon entering a valid filename the 'sorry can't open file' message will be displayed. A valid stream object is made invalid by further calls to *open()*. The *close()* function is not as fussy, in that a closed file remains closed following repeated *close()* calls.

You may be thinking that you will always know whether a file is open or not, or that it is simply a matter of referring to the program code to determine whether a file is open. In a large and complex program with numerous function calls it is not always a straightforward matter to establish whether a file has been previously opened or whether an open file has been inadvertently closed.

Each of three file stream classes `basic_ifstream`, `basic_ofstream` and `basic_fstream` declare a member function called *is_open()* which returns a non-zero integer value if a file is open:

```
bool is_open() const ;// FSTREAM.H
```

The use of *is_open()* is demonstrated in the following program:

```
// op_or_cl.cpp
// illustrates the is_open() function for
// determining whether a file is open or closed
#include <fstream>  // C++ file I/O

void main ()
  {
  // file stream object
  ifstream file ;

  // get file to open
  char file_name[50] ;
  cout << "enter a filename: " ;
  cin  >> file_name ;

  // open
  file.open (file_name, ios::in | ios::nocreate) ;

  // test
  if (!file)
    cerr << "sorry-can't open file \""
         << file_name << "\" " << endl ;

  if (file.is_open())
    cout << "file is open" << endl ;
```

```
// close
file.close () ;

if (!file.is_open())
  cout << "file is closed" << endl ;
}
```

with some user interaction:

```
enter a filename: op_or_cl.cpp
file is open
file is closed
```

It is worth noting that operations such as opening and closing files can also be performed via a `basic_filebuf` object which is encapsulated in each of the three file stream classes. For instance, **`class`** `basic_filebuf` declares an *`is_open()`* member function:

```
template <class CharT, class Traits>
class basic_filebuf : public basic_streambuf<CharT,Traits>
  {
  public:
    bool is_open() const ;
    //...
  };
```

which could then be called by an `ifstream` object as follows:

```
ifstream file ;
//...
if (file.rdbuf()->is_open())
//...
```

17.8.7 The *`read()`* and *`write()`* Member Functions

The `basic_istream::`*`read()`* member function extracts `n` characters or bytes from an input stream and places them in an array pointed to by `s`:

```
istream_type& read (char_type* s, streamsize n) ;
```

If an end-of-file is reached before the specified number of characters have been read then *`read()`* stops reading. In such cases the `basic_istream::`*`gcount()`* function is useful for determining the number of characters extracted from the last operation:

```
streamsize gcount() const ;
```

The `basic_ostream::`*`write()`* member function inserts `n` characters or bytes from the array pointed to by `s` and places them into an output stream:

```
ostream_type& write (const char_type* s, streamsize n) ;
```

The following program illustrates *read(), write()* and *gcount()*:

```
// rw.cpp
// illustrates the read(), write() and
// gcount() member functions
#include <fstream>  // C++ file I/O

void main ()
  {
  const int BUF_SIZ = 1e03 ;
  char in_file_name[]  = "RW.CPP" ;
  char out_file_name[] = "RW.TXT" ;
  char buffer[BUF_SIZ] ;

  // read & display
  ifstream in_file (in_file_name) ;

  if (!in_file)
    cerr << "sorry-can't open file \""
         << in_file_name << "\" " << endl ;
  in_file.read (buffer, sizeof buffer) ;

  cout << in_file.gcount () << endl ;
  for (int i=0; i<in_file.gcount(); i++)
    cout << buffer[i] ;

  // write
  ofstream out_file (out_file_name) ;

  if (!out_file)
    cerr << "sorry-can't open file \""
         << out_file_name << "\" " << endl ;

  out_file.write (buffer, in_file.gcount ()) ;

  // close
  out_file.close () ;
  in_file.close () ;
  }
```

This program uses the *read()* member function to fill an array of characters, `buffer`, of fixed size (1000). Since an end-of-file could occur before 1000 characters are read the *gcount()* member function is used to determine the actual number of characters read. The contents of `buffer` are then written to an output file `out_file`.

The entire contents of the input file will be read provided the length of the input file is less than or equal to the size of `buffer`. In general this cannot be guaranteed, and an alternative mechanism would have to be devised to read the contents of an arbitrary sized file. In the next subsection we shall revisit this problem.

17.8.8 Random File Access

The file access that we have seen so far has been purely sequential, in that we have either read or written a file from beginning to end. However, there are occasions in which it is required to access an arbitrary position of a file. Random file access is generally described in terms of a *file pointer* (not to be confused with C++ pointers). A file pointer can be viewed as a finger which points to a given position or the *current position* of a file between the beginning and the end of a file. The current position is the point at which file access next commences. Since file access can be generally categorised in terms of input and output, the current position can more specifically be referred to as a *current get position* and a *current put position*. The terms current *get* and *put* positions will help introduce the associated member functions of classes `basic_istream` and `basic_ostream` for performing random or non-sequential file access.

Let's first examine the two random file access functions, `seekg()` and `tellg()`, of **class** `basic_istream`:

```
istream_type& seekg(pos_type pos) ;
istream_type& seekg(off_type offset, ios_base::seekdir dir) ;

pos_type      tellg () ;
```

The one-argument version of `seekg()` moves the current get position to an absolute position, `pos`, in the input stream. The argument `pos` is of type `pos_type`, which is a **typedef** of `Traits::pos_type` (position type):

```
template <class CharT, class Traits>
class basic_istream : virtual public basic_ios<CharT,Traits>
  {
  public:
    typedef typename Traits::pos_type pos_type ;
    typedef typename Traits::off_type off_type ;
    //...
  };
```

The two-argument version of `seekg()` moves `offset` bytes relative to the current *get* position in the direction specified by `dir`. As with `pos_type`, `off_type` is similarly a **typedef** of of `Traits::off_type` (offset type).

The type of `dir` is an enumerated type `seek_dir` and is declared a member of **class** `ios_base`:

```
enum seek_dir  // IOSBASE.H
  {
  beg = 0x0, // beginning
  cur = 0x1, // current position
  end = 0x2  // end
  };
```

illustrating that there are three possibilities: an offset from the beginning (0); an offset from the current position (1); and an offset from the end (2): refer to Fig. 17.2.

The `basic_istream::tellg()` member function returns the current position of the input stream and returns a value of –1 if an error occurs.

The following program helps illustrate the `seekg()` and `tellg()` member functions:

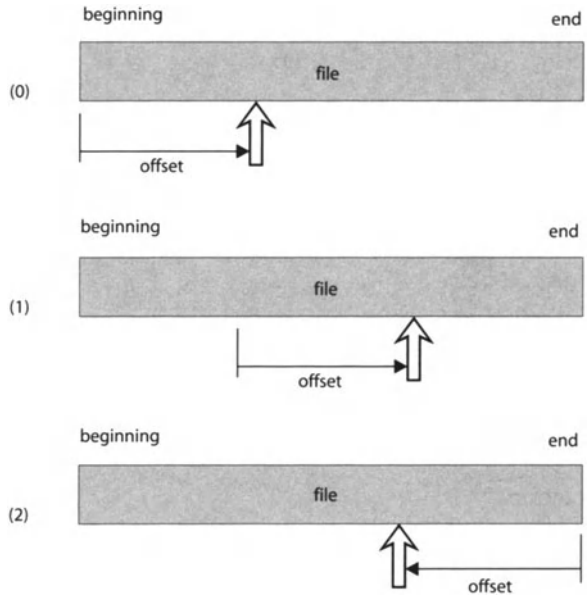


Fig. 17.2 The three possibilities for defining a relative offset in a given direction from (0) the beginning; (1) the current position; and (2) the end of a stream.

```
// cgp.cpp
// illustrates the seekg() and tellg() member functions
// of class istream for extracting the current get position
#include <fstream>  // C++ file I/O

void main ()
  {
  char file_name[] = "CGP.CPP" ;

  ifstream in_file (file_name) ;

  if (!in_file)
    cerr << "sorry-can't open file \""
         << file_name << "\" " << endl ;

  // determine file length
  streampos start = in_file.seekg (0, ios::beg).tellg () ;
  streampos end   = in_file.seekg (0, ios::end).tellg () ;

  // get an offset
  streamoff offset (0) ;

  cout << "enter an offset(+) from the beginning of file \""
       << file_name << "\" " << endl
       << "which has a file length of "
       << (end-start) << " characters: " ;
  cin  >> offset ;
```

```
// go to offset
in_file.seekg (offset, ios::beg) ;

// display file to screen
char ch ;

while (in_file)
  {
  in_file.get (ch) ;
    cout << ch ;
  }
// close
in_file.close () ;
}
```

with some user interaction:

```
enter an offset(+) from the beginning of file "CGP.CPP"
which has a file length of 910 characters: 840
  while (in_file)
    {
    in_file.get (ch) ;
    cout << ch ;
    }
  // close
  infile.close () ;
  }
```

Similarly, **class** `basic_ostream` has two member functions, *`seekp()`* and *`tellp()`*:

```
ostream_type& seekp(pos_type pos) ;
ostream_type& seekp(off_type offset, ios_base::seekdir dir) ;

pos_type      tellp () ;
```

The one-argument version of *`seekp()`* moves the current *put* position to an absolute position, `pos`, in the output stream. The two-argument version of *`seekp()`* moves `offset` bytes relative to the current put position in the direction specified by `dir`. The *`tellp()`* member function returns the current position of the output stream and returns a value of –1 if an error occurs.

When we examined program RW.CPP it was mentioned that we would readdress the problem of reading an arbitrary sized file using a fixed sized buffer and the `basic_istream::`*`read()`* member function. One solution is to view the entire input file as a series of adjacent segments, each of size equal to the buffer size; see Fig. 17.3. We can use the `basic_istream::`*`seekg()`* member function to determine the length of the input file and therefore determine the number of segments to read, and then simply use the *`read()`* and *`write()`* member functions to transfer the input file contents to the output file:

```
// rw1.cpp
// further illustrates the read(), write() and
// gcount() member functions
#include <fstream>  // C++ file I/O
```

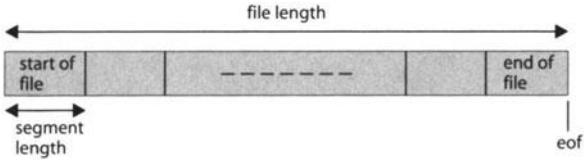


Fig. 17.3 Fragmentation of a file.

```
#include <cmath>     // ceil()

void main ()
  {
  ifstream in_file  ("RW1.CPP") ;
  ofstream out_file ("RW1.TXT") ;

  // determine file length
  streampos start = in_file.seekg (0, ios::beg).tellg () ;
  streampos end   = in_file.seekg (0, ios::end).tellg () ;
  double file_length = end - start ;

  const int BUF_SIZ (100) ;
  char buffer[BUF_SIZ] ;

  // return to start of in-file
  in_file.seekg (0, ios::beg) ;

  // for each segment of the in-file
  for (int i=0; i<ceil(file_length/BUF_SIZ); i++)
    {
    in_file.read (buffer, BUF_SIZ) ; // fill buffer
    int siz = in_file.gcount () ;
    if (siz < BUF_SIZ) // if chars read < buffer size
      out_file.write (buffer, siz) ;
    else               // a full buffer
      out_file.write (buffer, BUF_SIZ) ;
    }
  // close
  out_file.close () ;
  in_file.close () ;
  }
```

Care is required when reading the last segment, since the actual number of characters read could be less than the buffer size, `BUF_SIZ`, if an end-of-file occurred before `BUF_SIZ` characters were read. Rather than flushing the buffer after each call to *`write()`* to ensure that no characters remain in the buffer from the last segment, we can alternatively use the `basic_istream::`*`gcount()`* member function to return the exact number of characters extracted from the last *`read()`* operation. Note the use of the *`ceil()`*[3] C++ library function in RW1.CPP, which rounds up a floating-point argument to the smallest integer not less than the argument value:

3 The complement to *`ceil()`* is the function *`floor()`*, which rounds down.

```
double ceil (double x) ;   // MATH.H
```

17.8.9 Objects and Files

We saw earlier with reference to USR_TYPE.CPP the use of the overloaded insertion and extraction operators for the user-defined **class** `Point` with the standard stream objects `cout` and `cin`. Let us now modify USR_TYPE.CPP to read a number of `NumberedPoint` objects from the disk file POINTS.TXT and display them to the screen:

```
// ob_file.cpp
// illustrates the use of user-defined objects of
// class NumberedPoint and an input file stream
#include <fstream>     // C++ file I/O
#include "pt_tmp.h"    // template class Point
#include "npt_tmp.h"   // template class NumberedPoint
#include "vec_tmp.h"   // template class Vector

void main ()
  {
  char file_name[] = "POINTS.TXT" ;
  int  n_points (0) ;

  ifstream in_file ;

  in_file.open (file_name) ;

  in_file >> n_points ;

  Vector<NumberedPoint<double> > points (n_points, 0.0) ;

  for (int i=0; i<n_points; i++)
    in_file >> points[i] ;

  cout << "points: " << endl << points ;

  in_file.close () ;
  }
```

with output:

```
points:
[(1, 2, 3, 0), (4, 5, 6, 1), (7, 8, 9, 2)]
```

The data file POINTS.TXT, from which the three `NumberedPoint` objects are read, is:

```
3
1.0 2.0 3.0 0
4.0 5.0 6.0 1
7.0 8.0 9.0 2
```

OB_FILE.CPP defines an `ifstream` object called `in_file` from which the file POINTS.TXT is opened and the number of `NumberedPoint` objects, `n_points`, in the file read. The **`template class`** `NumberedPoint` is derived from **`template class`** `Point`; they are declared in NPT_TMP.H and PT_TMP.H respectively:

```
template <class T>  // PT_TMP.H
class Point
  {
  protected:
    T x, y, z ;
  public:
    //...
  };
//...
template <class T>  // NPT_TMP.H
class NumberedPoint : public Point<T>
  {
  protected:
    int number ;
  public:
    // constructors
    NumberedPoint ()
      : Point<T> (), number (0) {}
    NumberedPoint (T x_arg, T y_arg, T z_arg=T(), int n=0)
      : Point<T> (x_arg, y_arg, z_arg), number (n) {}
    //...
  };
```

`NumberedPoint` associates an integer number with a `Point` object. The **`template`** classes `Point`, `NumberedPoint` and `Vector` are described in more detail in Chapters 13 and 15.

OB_FILE.CPP then proceeds to read each `NumberedPoint` object in the POINTS.TXT file into the `points` object of **`class`** `Vector<NumberedPoint<`**`double`**`> >`:

```
Vector< NumberedPoint<double> > points (n_points, 0.0) ;

for (int i=0; i<n_points; i++)
  in_file >> points[i] ;
```

which illustrates that the use of the `ifstream` object `in_file`, overloaded extraction operator and object `points` of a user-defined **`class`** is identical in syntax to the use of the `cin` predefined stream object with objects of integral data types.

The above program, OB_FILE.CPP, read a series of `NumberedPoint` objects from a file called POINTS.TXT by first reading the number of objects stored in the file, then defining an object of **`class`** `Vector` to store the `NumberedPoint` objects to be read, and finally reading each `NumberedPoint` object using a **`for`**-loop. The operation of reading the number of objects stored in the file may seem unnecessary, and could be a difficult task to specify exactly if the number of objects is very large. Therefore, let us now modify OB_FILE.CPP so that it determines the number of objects stored in a file rather than requiring a user to specify the number of objects explicitly. The file POINTS.TXT is replaced by POINTS1.TXT:

```
1.0 2.0 3.0 0
4.0 5.0 6.0 1
7.0 8.0 9.0 2
```

and OB_FILE.CPP is replaced by OB_FILE1.CPP:

```
// ob_file1.cpp
// further illustrates the use of user-defined objects of
// class NumberedPoint and an input file stream
#include <fstream>     // C++ file I/O
#include "pt_tmp.h"    // template class Point
#include "npt_tmp.h"   // template class NumberedPoint
#include "ll.h"        // template class LinkedList

void main ()
  {
  char file_name[] = "POINTS1.TXT" ;

  ifstream in_file ;
  in_file.open (file_name) ;

  LinkedList <NumberedPoint<double> > points ;

  while (!in_file.eof ())
    {
    NumberedPoint<double> pt ;
    in_file >> pt ;
    if (!in_file.eof ())
      points.Append (pt) ;
    }

  cout << points.NumberOfNodes () << " points: " << endl
       << points ;

  in_file.close () ;
  }
```

with output:

```
3 points:
(1, 2, 3, 0) (4, 5, 6, 1) (7, 8, 9, 2)
```

The `NumberedPoint` objects read from the file are stored in a linked list object, `points`, of the **`template class`** `LinkedList` (see Chapter 13) because a linked list is more responsive to the addition and deletion of elements or nodes than an array-based `Vector` **`class`** object. Since the number of objects to be read is no longer explicitly specified, a **`while`**-loop is used instead of a **`for`**-loop. The **`while`**-loop will continue to read objects until the end-of-file is reached. Each object read is first placed in the temporary object `pt` and then appended to the linked list object `points` via the `LinkedList<T>::`*`Append()`* member function. Note the necessity of having to perform a second test for end-

of-file within the body of the **while**-loop so as not to append the last object read twice to the linked list.

You may be wondering why the `LinkedList<T>::Append()` member function was used in preference to an overloaded extraction operator of **template class** `LinkedList`:

```
//...
while (!in_file.eof ())
  {
  //...
  if (!in_file.eof ())
    in_file >> points[i] ;
  }
```

The extraction operator is not overloaded for **template class** `LinkedList` because a node object can be added to a linked list at an arbitrary location, inserted at the head or appended to the tail of the linked list. Thus, it is simpler to use the *`Insert()`*, *`Head()`* or *`Append()`* member functions rather than imposing conditions on an overloaded extraction operator for **template class** `LinkedList`.

17.9 `ReadFile`, `WriteFile` and `ReadAndWriteFile` Classes

This section discusses the development of three classes called `ReadFile`, `WriteFile` and `ReadAndWriteFile` for the manipulation of input, output and input and output file streams respectively. The key reason for examining these classes is to demonstrate that user-defined classes can be derived from C++ library classes. Although it sounds obvious that classes can be derived from C++ library classes, it is a property that is frequently neglected. The C++ library classes have taken professional programmers years to develop and as a result are extremely powerful and robust; this is particularly applicable to the stream classes because they have a complex hierarchy.

The `ReadFile`, `WriteFile` and `ReadAndWriteFile` classes are derived from the C++ library file stream classes `basic_ifstream`, `basic_ofstream` and `basic_fstream`, and therefore inherit the features of these classes, but in addition are derived from a **class** called `PathAndFile`, which encapsulates a filename. The filename will in fact incorporate the drive, directory path, filename and extension (assuming we are using MS-DOS) if each is specified by a user.

The `ReadFile` **class** is, in part, derived from `basic_ifstream`, so before we examine `ReadFile` let's remind ourselves of the **class** declaration of `basic_ifstream`:

```
template <class CharT, class Traits> // FSTREAM.H
class basic_ifstream : public basic_istream<CharT,Traits>
  {
  public:
    basic_ifstream () ;
    explicit basic_ifstream
      (const char* s, ios_base::openmode mode=ios_base::in) ;
    virtual ~basic_ifstream() ;
    basic_filebuf<CharT,Traits>* rdbuf () const ;
    bool is_open () ;
```

```
    void open (const char* s,
               ios_base::openmode mode=ios_base::in) ;
    void close () ;
};
```

In the declaration of **class** ReadFile we shall focus our attention on replicating the constructors and the *open()* and *close()* member functions of basic_ifstream. The two-argument constructor and *open()* member function of basic_ifstream each specify a filename argument, s, of type **char***. However, when defining a file stream object it is nice to be able to work with filenames which are of the C++ library **class** string (STRING.H) rather than **char***. Therefore, ReadFile will declare a constructor and an *Open()* member function with a filename argument of **class** string. When the C++ library member functions *open()* and *close()* were discussed previously it was noted that *open()* can be called to open a file which is already open, and similarly for *close()*. Therefore, the ReadFile *Open()* and *Close()* member functions will detect whether a file is open or not and act accordingly.

In declaring the ReadFile **class** we could encapsulate either a **private** or **protected** data member of **class** string, which attaches a path and filename to a stream object. Alternatively, we can declare another **class** called PathAndFile which encapsulates a path and file object because the path and filename of a file are common features to all three classes, ReadFile, WriteFile and ReadAndWriteFile. It is then a simple matter for each stream **class** to derive from PathAndFile. The following header file RW_FILE.H includes the **class** declarations of PathAndFile and the user-defined stream classes:

```
// rw_file.h
// ReadFile, WriteFile and ReadAndWriteFile class
declarations

#ifndef _RW_FILE_H  // prevent multiple includes
#define _RW_FILE_H

#include <fstream>  // C++ file I/O
#include <string>   // C++ class string
#include <ctype>    // isalnum()

class PathAndFile
  {
  protected:
    string pathfilename ;
  public:
    // constructors
    PathAndFile () ;
    PathAndFile (const string& s) ;
    // member functions
    void   SetPathAndFileName (const string& s) ;
    string PathAndFileName () const ;
    string FileName () ;
    string FileExtension () ;
    string Path () ;
    char   Drive () ;
  };
```

```
class ReadFile : public ifstream, public PathAndFile
  {
  public:
    // constructors
    ReadFile () ;
    ReadFile (const char* name,
              ios_base::openmode mode=ios_base::in) ;

    ReadFile (const string& s,
              ios_base::openmode mode=ios::in) ;
    // member functions
    int  Open (const string& s,
               ios_base::openmode mode=ios::in) ;
    int  Open (ios_base::openmode mode=ios::in) ;
    void Close () ;
  };

class WriteFile : public ofstream, public PathAndFile
  {
  public:
    // constructors
    WriteFile () ;
    WriteFile (const char* name,
               ios_base::openmode mode=ios::out) ;
    WriteFile (const string& s,
               ios_base::openmode mode=ios::out) ;
    // member functions
    int  Open (const string& s,
               ios_base::openmode mode=ios::out) ;
    int  Open (ios_base::openmode mode=ios::out) ;
    void Close () ;
  };

class ReadAndWriteFile : public fstream, public PathAndFile
  {
  public:
    // constructors
    ReadAndWriteFile () ;
    ReadAndWriteFile (const char* name,
                      ios_base::openmode mode) ;
    ReadAndWriteFile (const string& s,
                      ios_base::openmode mode) ;
    // member functions
    int  Open (const string& s, ios_base::openmode mode) ;
    int  Open (ios_base::openmode mode) ;
    void Close () ;
  };

#endif // _RW_FILE_H
```

PathAndFile declares one **protected** data member, two constructors and six member functions. The pathfilename data member is of **class** string and holds the path and filename of a given file. The no-argument constructor initialises pathfilename to a zero length string, whereas the one-argument constructor simply sets pathfilename to the constructor argument. Before discussing the member functions of PathAndFile, let's list the key elements of RW_FILE.CPP, which is the corresponding implementation file of RW_FILE.H:

```
// rw_file.cpp
// implementation file for classes
// ReadFile, WriteFile and ReadAndWriteFile
#include "rw_file.h"

// PathAndFile:

PathAndFile::PathAndFile ()
  : pathfilename ()
  {}

PathAndFile::PathAndFile (const string& s)
  : pathfilename (s)
  {}

// sets the path and filename
void PathAndFile::SetPathAndFileName (const string& s)
  { pathfilename = s ; }

// returns the path and filename
string PathAndFile::PathAndFileName () const
  { return pathfilename ; }

// extracts the filename from a path and filename string
string PathAndFile::FileName ()
  {
  string filename ;
  for (int i=pathfilename.rfind('\\')+1;
           i<pathfilename.length(); i++)
    filename.append (1, pathfilename[i]) ;
  return filename ;
  }

// extracts the filename extension from a path and filename
// string
string PathAndFile::FileExtension ()
  {
  string file_ext ;
  for (int i=pathfilename.rfind('.')+1;
           i<pathfilename.length(); i++)
    file_ext.append (1, pathfilename[i]) ;
  return file_ext ;
  }
```

```
// extracts the path from a path and filename string
string PathAndFile::Path ()
  {
  string path ;
  if (pathfilename.rfind ('\\') != string::npos) // if path
    for (int i=0; i<pathfilename.rfind('\\'); i++)
      path.append (1, pathfilename[i]) ;
  return path ;
  }

// extracts the drive from a path and filename string
char PathAndFile::Drive ()
  {
  char drive (0) ;
  if (pathfilename.find (':') != string::npos) // if drive
    drive = pathfilename[0] ;
  return drive ;
  }

// ReadFile:

ReadFile::ReadFile ()
  : ifstream (), PathAndFile ()
  {}

ReadFile::ReadFile (const char* name,
                    ios_base::openmode mode)
  : ifstream (name, mode), PathAndFile (name)
  {}

ReadFile::ReadFile (const string& s,
                    ios_base::openmode mode)
  : ifstream (s.c_str (), mode), PathAndFile (s)
  {}

// opens a specified file
// returns 1 if opened successfully and
// -1 if the file is already open
int ReadFile::Open (const string& s,
                    ios_base::openmode mode)
  {
  pathfilename = s ;    // set path and filename
  if (is_open())        // already open
    return -1 ;
  else  // open
    {
    ifstream::open (pathfilename.c_str (), mode) ;
    return 1 ;
    }
  }
```

```
// opens a file when filename has previously been defined
// returns 1 if opened successfully, 0 if a filename has
// not been set and -1 if the file is already open
int ReadFile::Open (ios_base::openmode mode)
  {
  if (pathfilename.length () == 0)
    return 0 ;
  else if (is_open())
    return -1 ;
  else  // open
    {
    ifstream::open (pathfilename.c_str (), mode) ;
    return 1 ;
    }
  }

// close an open file
void ReadFile::Close ()
  {
  if (is_open())
    ifstream::close () ;
  }

// WriteFile:
//...
// ReadAndWriteFile:
//...
```

The *SetPathAndFileName()* member function allows a path and filename to be set to a specified string and, as we shall see later, is useful for defining an object of **class** ReadFile without simultaneously associating a filename with the object at the time of object definition. The *PathAndFileName()* access member function returns the pathfilename string data member.

FileName() returns the filename (including extension) of pathfilename. The temporary object filename holds the filename. The **for**-loop uses the string::*rfind()* member function to find the *last* occurrence of the backslash character '\' in the pathfilename string and initialise the control variable i of the **for**-loop to this position. The **for**-loop then proceeds to append (using string::*append()*) each character of pathfilename to the filename string until the end of the string is reached (using string::*length()*). The declarations of member functions *rfind()*, *append()* and *length()* of **class** string (actually basic_string) are, STRING.H:

```
template <class CharT, class Traits, class Allocator>
class basic_string
  {
  public:
    typedef typename Allocator::size_type size_type ;
    //...
    size_t rfind (const basic_string& s,
                  size_type i=0) const ;
    //...
```

```
    basic_string& append (const basic_string& s) ;
    string&       append (const string& s,
                          size_type pos, size_type n)  ;
    basic_string& append (const CharT* s) ;
    basic_string& append (const CharT* p, size_type);
    basic_string& append (size_type n, CharT c) ;
    //...
    size_type length () const ;
    //...
  };
//...
typedef basic_string<char,char_traits<char>,
                     allocator<char> > string ;
```

where `size_type` is a **typedef** of `Allocator::size_type` (`size_t` or equivalently **unsigned int**).

rfind() finds the last occurrence of string `s` in the target string that is not beyond position `pos` in the target string if specified. If the string `s` is not found then *rfind()* returns `basic_string::npos`:

```
#define npos (size_type)-1 ;
```

`npos` is equivalent to the largest value of `size_type`. *append()* appends the string `s` to the target string or the first `n` characters of `s` to the target string at the specified location `pos`. The string `s` can be specified either as a `string` object, a pointer to an array of characters or as a number `n` of a single character `c`. The number of characters in a `string` object is returned by *length()*. It is worth mentioning that indexing the characters of a `string` object starts at zero and that a `string` object can hold null characters.

The member functions *FileExtension()* and *Path()* are similar in principle to *FileName()* except that *FileExtension()* uses *rfind()* to locate the last occurrence of the '.' character instead of '\', and *Path()* confirms that a path is specified before performing a search using *rfind()*. The *Drive()* member function is slightly different in that it returns a single character rather than a `string` object to indicate the drive and uses the `basic_string::` *find()* member function to confirm that a drive is specified before performing the assignment.

The `basic_string::` *find()* member function has the following declaration:

```
size_type find (const basic_string& s, size_type i=0) const ;
size_type find (const CharT* p, size_type i,
                                size_type n) const ;
size_type find (const CharT* p, size_type i=0) const ;
size_type find (CharT c, size_type i=0) const ;
```

and finds the *first* occurrence of `basic_string s`, **char*** `p` and character `c`.

With `PathAndFile` declared we can now declare **class** `ReadFile` using multiple inheritance:

```
class ReadFile : public ifstream, public PathAndFile
  { /* ... */ };
```

The first two constructors of `ReadFile` replicate those of `basic_ifstream`, whereas the third constructor allows a stream object to be defined using a `string` filename argument:

```
ReadFile::ReadFile (const string& s, ios_base::openmode mode)
  : ifstream (s.c_str (), mode), PathAndFile (s)
  {}
```

Note the use of the `basic_string::c_str()` member function:

```
const CharT* c_str () const ;
```

which returns a pointer to a null-terminated character array which contains the same characters as the string object operated on by the member function.

Similarly, the two-argument `ReadFile::Open()` member function allows a stream object to be opened in terms of a `string` filename:

```
int ReadFile::Open (const string& s, ios_base::openmode mode)
  { /* ... */ }
```

The `ReadFile::Open()` member function could equally have been named `open()` to overload `ifstream::open()`, but was named `Open()` purely on the grounds of notational convention. `Open()` first confirms, using `is_open()`, that the function is not called to open an already open file. If the associated file is already open a value of –1 is returned. If the file is being opened for the first time the `pathfilename` data member is set, the file is opened by simply calling `basic_ifstream::open()` and a value of 1 is returned to indicate a successful file open call.

Alternatively, the one-argument `ReadFile::Open()` member function allows a file to be opened for which the path and filename have been previously set. The member function first confirms that a path and filename have indeed been defined, and if not a value of 0 is returned. Next, the function tests if the specified file is already open and if so a value of –1 is returned. Finally, if a path and filename have been set and the file is not already open, the file is opened and a value of 1 is returned.

The `ReadFile::Close()` member function will close a file only if it is open.

The `WriteFile` and `ReadAndWriteFile` **class** declarations are of a similar form to that of `ReadFile`. Use of the `ReadFile` **class** presented above is illustrated in the following program:

```
// rw_tst.cpp
// tests the ReadFile, WriteFile and ReadAndWriteFile classes
#include <iostream>    // C++ I/O
#include <iomanip>     // setw()
#include "rw_file.h"   // classes ReadFile, ...

void main ()
  {
  string filename ("c:\\bcc55\\include\\fstream.h") ;

  //ReadFile in_file ;
  //in_file.Open (filename) ;

  ReadFile in_file (filename) ;
```

```
cout << setw (21) << "path and filename: "
     << in_file.PathAndFileName () << endl ;
cout << setw (21) << "filename: "
     << in_file.FileName () << endl ;
cout << setw (21) << "filename extension: "
     << in_file.FileExtension () << endl ;
cout << setw (21) << "path: "  << in_file.Path ()  << endl ;
cout << setw (21) << "drive: " << in_file.Drive () << endl ;

in_file.close () ;

// open or not open?
ReadFile in_file1 ;       // define object-no file associated
if (in_file1.Open () == 0)      // try and open
  cout << "filename not set" << endl ;
in_file1.SetPathAndFileName("obj_in.dat") ; // set filename
if (in_file1.Open ())  // now open
  cout << "file opened" << endl ;
if (in_file1.Open () == -1)     // try to open an open file
  cout << "file already open" << endl ;
in_file1.Close () ;
}
```

with output:

```
path and filename : c:\bcc55\include\fstream.h
         filename : fstream.h
filename extension: h
              path: c:\bc55\include
             drive: c
filename not set
file opened
file already open
```

The standard library **class** `basic_string` is discussed in greater detail in Chapter 20.

17.10 Reading a Collection of Objects from a Disk File

Continuing our examination of reading from and writing to a disk, let's now address the problem of reading a disk file which contains a collection of user-defined geometrical objects.

Frequently when performing geometric and solid modelling it is desirable to be able to read objects from an *object data file* rather than defining them explicitly in either a header or implementation file. The type of objects that we shall consider are those discussed in the Shapes program of Chapter 15; see SHAPE_V.H and SHAPES.CPP and Fig. 15.12 for an overview of the **class** hierarchy of the Shapes program. Excluding the abstract base **class** `Shape` and base

class Plane, the actual shapes or objects considered are a circle, sphere, polygon, quadrilateral, triangle and tetrahedron.

Shown below is the disk file OBJ_IN.DAT, which represents a typical object data file:

```
CIRCLE
  {
  [0,-1,0,1,(1,0,0),1,1] // normal=(0,-1,0),c=(1,0,0),
  r=1 & no.=1
  }
// CIRCLE { [0,-1,0,1,(1,0,0),1,5] }
SPHERE
  {
  [(5,5,5),1,2]
  }
Triangle // 2 Triangle objects
  {
  [(1,1,0),(4,2,0), (3,  4,2),   3]
  [(1,1,1),(2,2,1),(3,3,3),4]
  }
QUADRILATERAL
  {
  [(0,0,0),(1,0,0),(1,1,0),(0,1,0),5] [/*...*/,6]
  }
//TRIANGLE { [/*...*/,7] } POLYGON { [/*...*/,8] }
  POLYGON
  {
  // a Polygon object
  [5,(1,0,2),(3,0,1),(4,0,3),(3,0,6),(0.5,0,5),7] // 5
vertices
  }
TRIANGLE { [/*...*/,8] } POLYGON { [/*...*/,9] }
TETRAHEDRA { [(1,1,0),(3,2,0),(4,5,2),(2,4,0),10] }
```

The start of an object is indicated by a *keyword* or *object name descriptor*, which for the sake of convenience, as we shall see later, is in fact equivalent to the object's **class** name. The keyword can be specified entirely in lowercase, entirely in uppercase or a mixture of both lowercase and uppercase characters. The data of an object (such as vertices or number) are enclosed by two square brackets ([and]), which are similarly enclosed by the braces ({ and }), which are similar to the C++ syntax. Encompassing an individual object between square brackets allows multiple objects to be defined between a single pair of braces as illustrated above for the first of the two triangle objects.

The object description language is fully free-format and allows C++-style comments (// . . .) to be placed anywhere within a data file. The above object data file illustrates that any number of objects can be placed in any order, at any position within a line or with multiple objects on a single line. Similarly, comments can be positioned at the start of a line, after a keyword, within the braces or directly following the square brackets of an object's data. Also, the free-format nature of the object description language applies within the square brackets as illustrated above for the first of the two triangle objects.

Note that the C-style comments (/ * . . . * /) are not part of the object description language, but are used above to illustrate the variety of ways in which objects can be defined.

A first solution in the implementation of a **class** which reads a collection of objects from a file would be simply to extend the `ReadFile` **class** presented in the previous section to read `Shape` objects from a file. However, such an implementation would inappropriately specialise `ReadFile` because there is no relationship between reading a file and a list of arbitrary objects. Therefore, the following implementation will introduce a new **class** called `World` which encapsulates a list of *world* objects. Thus, the `PathAndFile`, `ReadFile`, `WriteFile` and `ReadAndWriteFile` **class** declarations and definitions remain the same as those in RW_FILE.H and RW_FILE.CPP.

The following lists the declaration of **class** `World`:

```
// world.h
// a World class

#ifndef _WORLD_H  // prevent multiple includes
#define _WORLD_H

#include <typeinfo>    // RTTI
#include <string>      // C++ class string

#include "rw_file.h"   // classes ReadFile, ...
#include "ll.h"        // template class LinkedList

#include "shape_v.h"   // class Shape
#include "plane_v.h"   // class Plane
#include "poly_v.h"    // class Polygon
#include "tri_v.h"     // class Triangle
#include "quad_v.h"    // class Quadrilateral
#include "circ_v.h"    // class Circle
#include "sphere_v.h"  // class Sphere
#include "tet_v.h"     // class Tetrahedra

class World
  {
  private:
    // nested class
    class SortKeyword
      {
      public:
        // data members
        string kw ;
        size_t pos ;
        // constructors
        SortKeyword ()
          : kw (), pos (0) {}
        SortKeyword (const string& s,
                     const size_t& p)
          : kw (s), pos (p) {}
        // overloaded relational operators
        bool operator <  (const SortKeyword& rhs)
          { return pos < rhs.pos ? true :
                                   false ; }
```

```
        bool operator == (const SortKeyword& rhs)
          { return (pos == rhs.pos &&
                           kw == rhs.kw) ?
                           true : false ; }
        bool operator != (const SortKeyword& rhs)
          { return (pos != rhs.pos &&
                           kw != rhs.kw) ?
                           true : false ; }
      }; // nested class SortKeyword

    // data members
    ReadFile            obj_infile ;  // obj in file
    WriteFile           obj_outfile ; // obj out file
    LinkedList<Shape*> l_obj ;        // list of world objects
    // member function
    LinkedList<SortKeyword>
              KeywordPositions (const string& str,
                                LinkedList<string>& l_kw) ;
    bool                CheckBracesNoNesting
                               (const streampos& fp) ;
    Shape*              ReadParticularObject
                               (const string& obj_type) ;
    LinkedList<Shape*> ReadListOfObjects
                               (const string& obj_type,
                                const streampos& fp_arg) ;
  public:
    // constructor
    World (const string& o_infile,
           const string& o_outfile) ;
    // member functions
    LinkedList<string> ReadKeywords
                      (const string& kw_fname="OBJ_KW.DAT") ;
    void ReadObjects (const string& kw_fname="OBJ_KW.DAT") ;
    void WriteObjects () ;
    LinkedList<Shape*> ObjectList () const { return l_obj ; }
  }; // class World

// string utilities:
bool IsLowerCase (char c) ;
char ToUpperCase (char c) ;
void StringToUpperCase (string& s) ;

#endif // _WORLD_H
```

It is worth noting that in addition to the header files included in RW_FILE.H the above file WORLD.H also includes the C++ RTTI header file TYPEINFO.H and the `Shape` and `LinkedList` associated header files.

The two-argument constructor of **class** `World` sets the path and filenames of the in and out object data files without opening the files:

```
World::World (const string& o_infile, const string& o_outfile)
```

```
{
obj_infile.SetPathAndFileName (o_infile) ;
obj_outfile.SetPathAndFileName (o_outfile) ;
}
```

The in and out object data files will be opened and closed, as required, by `World` member functions.

Eight member functions are defined for **class** `World`. *`KeywordPositions()`* is a **private** member function and returns a linked list of keywords (with their respective positions via the nested **class** `SortKeyword`) found in a given string. *`CheckBracesNoNesting()`* returns logical-true if an open brace is followed by a close brace from a specified file pointer position, else logical-false is returned. Thus, as the name suggests, *`CheckBracesNoNesting()`* does not allow the nesting of braces. *`ReadKeywords()`* returns a linked list of keywords listed in a specified file. The default filename is OBJ_KW.DAT, which in the present case, is:

```
Triangle
QUADRILATERAL
CIRCLE
SPHERE
TETRAHEDRA
POLYGON
```

Listing the type of objects considered in a separate data file removes them from a .H or .CPP file and therefore makes it easier for a user to alter the object type list or specify an alternative list. The listed object types can be specified entirely in lowercase, entirely in uppercase or a mixture of both lowercase and uppercase characters. *`ReadParticularObject()`* returns a pointer to a given object via the `Shape` abstract base **class**. *`ReadListOfObjects()`* returns a linked list of objects between the next pair of open and close braces from a specified file pointer position. *`ReadObjects()`* is the most important member function to be defined for **class** `World` and associates a linked list of all the objects read from a file to the `World::l_obj` data member. *`WriteObjects()`* writes a linked list of objects to a file and *`ObjectList()`* returns the `l_obj` data member. `World` also declares a **private** nested **class** called `SortKeyword` which will be discussed later in connection with *`ReadObjects()`*.

So, let's now examine the eight member functions of `World` discussed above in more detail. The member functions quickly become very involved, and I am not suggesting that you attempt to understand every detail of their operation. As always, try to focus on the overall design of the functions and their integration with the `World` and `Shape` classes. Further, the following functions will be far from ideal, so try to look for areas which could be improved or simplified. Here we go:

```
// world.cpp
// implementation file for class World
#include "world.h"

// private:

// returns a linked list of the
// keywords (with their positions) in a string
// accounts for '//' comments in the string
```

```
LinkedList<World::SortKeyword>
  World::KeywordPositions (const string& str,
                           LinkedList<string>& l_kw)
  {
  LinkedList<SortKeyword> l_skw ;

  // if string is worth examining
  if (str.length() > 0)
    {
    // find position of '//' in string-if present
    bool    comment_found (false) ;
    size_t  comment_pos (0) ;
    for (int i=0;i<str.length()-1; i++)
      if (str[i] == '/' && str[i+1] == '/')
        {
        comment_found = true ;
        comment_pos = i ;
        break ;
        }
    // find positions of keywords in string-if any
    for (int j=0; j<l_kw.NumberNodes(); j++)
      {
      // find first occurrence of keyword
      size_t kw_pos = str.find (l_kw[j]) ;
      if (kw_pos != string::npos &&
        comment_found && kw_pos < comment_pos)
        {
        SortKeyword skw (l_kw[j], kw_pos) ;
        l_skw.Append (skw) ;
        }
      if (kw_pos != string::npos && !comment_found)
        {
        SortKeyword skw (l_kw[j], kw_pos) ;
        l_skw.Append (skw) ;
        }
      // find other occurrences-if any
      while (kw_pos != string::npos)
        {
        // commence search from end of last kw. found
        kw_pos =
          str.find (l_kw[j],
          kw_pos+l_kw[j].length()-1) ;
        if (kw_pos != string::npos &&
           comment_found && kw_pos < comment_pos)
          {
          SortKeyword skw (l_kw[j], kw_pos) ;
          l_skw.Append (skw) ;
          }
        if (kw_pos != string::npos && !comment_found)
          {
          SortKeyword skw (l_kw[j], kw_pos) ;
```

```
            l_skw.Append (skw) ;
            }
          }
        }
      }
    // sort and return
    l_skw.Sort () ;
    return l_skw ;
    } // World::KeywordPositions()

// returns logical-true if the sequence {...} is found
// from the specified file pointer position (fp),
// else logical-false is returned
bool World::CheckBracesNoNesting (const streampos& fp)
    {
    bool check (false) ;

    // first get current file position
    streampos orig_fp = obj_infile.tellg () ;

    // go to specified file position
    obj_infile.seekg (fp, ios::beg) ;

    char  ch ;
    int   number (0) ;
    char  bra[2] = { 'x', 'x' } ; // ensure != '{' or '}'

    while (!obj_infile.eof ()) // till eof
      {
      obj_infile.get (ch) ;
      // test for open and close braces
      if (ch == '{' || ch == '}')
        {
        number++ ;
        bra[number] = ch ;
        }
      if (number == 2) // found first two braces
        break ;
      }
    (bra[1] == '{' && bra[2] == '}') ? check = true
                                     : check = false ;

    // return file position to original value
    obj_infile.seekg (orig_fp, ios::beg) ;

    return check ;
    } // World::CheckBracesNoNesting()

// returns a pointer to a particular object type
Shape* World::ReadParticularObject (const string& obj_type)
    {
```

```
  Shape* obj (0) ;

  if (obj_type == "CIRCLE")
    {
    Circle* c (new Circle) ;
    obj_infile >> *c ;
    obj = c ;
    }
  if (obj_type == "SPHERE")
    {
    Sphere* s (new Sphere) ;
    obj_infile >> *s ;
    obj = s ;
    }
  if (obj_type == "POLYGON")
    {
    Polygon* p (new Polygon) ;
    obj_infile >> *p ;
    obj = p ;
    }
  if (obj_type == "TRIANGLE")
    {
    Triangle* t (new Triangle) ;
    obj_infile >> *t ;
    obj = t ;
    }
  if (obj_type == "QUADRILATERAL")
    {
    Quadrilateral* q (new Quadrilateral) ;
    obj_infile >> *q ;
    obj = q ;
    }
  if (obj_type == "TETRAHEDRA")
    {
    Tetrahedra* t (new Tetrahedra) ;
    obj_infile >> *t ;
    obj = t ;
    }
  return obj ;
  } // World::ReadParticularObject()

// returns a linked list of objects read
// of a particular object type between braces { and }
LinkedList<Shape*> World::ReadListOfObjects
    (const string& obj_type,
     const streampos& fp_arg)
  {
  // first get current file position
  streampos orig_fp = obj_infile.tellg () ;

  // go to specified file position
```

```
obj_infile.seekg (fp_arg, ios::beg) ;

// pointer to abstract base class Shape
Shape* obj (0) ;
LinkedList<Shape*> l_obj ; // linked list of objects

streampos fp = fp_arg ;  // file pointer to start of line
string    str, r_str ;   // original and reduced strings

// read file
while (r_str.find ('}') == string::npos)
  {
  r_str.erase () ;      // flush r_str

  // a line at a time
  ::getline (obj_infile, str, '\n') ;

  // get current file pointer
  streampos cp = obj_infile.tellg () ;

  // convert string to uppercase-if not already
  StringToUpperCase (str) ;

  // check for comments and develop reduced string
  for (int i=0; i<str.length(); i++)
    {
    // '//' comment: quit reading characters
    if (i<str.length()-1 &&
      str[i] == '/' && str[i+1] == '/')
      break ; // leave for-loop
    else  // read '[' and ']' characters
      {
      if (str[i] == '[' || str[i] == ']' ||
          str[i] == '}')
        {
        r_str.append (1, str[i]) ;
        if (str[i] == '}') // upto first '}'
          break ; // leave for-loop
        }
      }
    }

  // line is worth examining further
  if (r_str.length () > 0)
    {
    // find first '[' in line-if any
      size_t r_pos = r_str.find ('[') ;
      if (r_pos != string::npos) // '[' found
        {
        // find first '[' in original line
        size_t o_pos = str.find ('[') ;
```

```
            if (o_pos != string::npos)
              {
              streampos fp_pos = fp + streampos(o_pos) ;
              obj_infile.seekg (fp_pos, ios::beg) ;

              // read object and append to list
              obj = ReadParticularObject
                (obj_type) ;
              l_obj.Append (obj) ;
              }
            }

          // find other '['-if any
          int number_of (0) ;
          for (int i=0; i<r_str.length(); i++)
            if (r_str[i] == '[')
              number_of++ ;

          // if more than one '[' per line
          if (number_of >= 2)
            {
            LinkedList<int> pos ;

            // find position/s in line of '['
            for (int i=0; i<str.length(); i++)
              if (str[i] == '[')
                pos.Append (i) ;

            // already read first object above,
            // so start from second
            for (int j=1; j<pos.NumberOfNodes(); j++)
              {
              streampos fp_pos = fp + pos[j] ;
              obj_infile.seekg (fp_pos, ios::beg) ;

              // read object and append to list
              obj = ReadParticularObject
                (obj_type) ;
              l_obj.Append (obj) ;
              }
            }
        } // if r_str.length()>0
      obj_infile.seekg (cp, ios::beg) ;// re-pos'n file
pointer
      fp = obj_infile.tellg () ;        // update file pointer
      } // while ()

    // return file position to original value
    obj_infile.seekg (orig_fp, ios::beg) ;

    return l_obj ;
```

```
  } // World::ReadListOfObjects()

// public:

// set object in and out data files but do not
// open files!
World::World (const string& o_infile,
              const string& o_outfile)
  {
  obj_infile.SetPathAndFileName (o_infile) ;
  obj_outfile.SetPathAndFileName (o_outfile) ;
  }

// returns a linked list of the object keywords in the
// specified file
LinkedList<string> World::ReadKeywords (const string& kw_fname)
  {
  // open keyword file
  ifstream kw_file (kw_fname.c_str ()) ;
  if (!kw_file)
    {
    cerr << "sorry-can't open file \"" << kw_fname
         << "\" " << endl ;
    exit (EXIT_FAILURE) ;
    }

  // place keywords in linked list
  LinkedList<string> keywords ;
  while (!kw_file.eof ())
    {
    string str ;
    kw_file >> str ;           // read string
    StringToUpperCase (str) ;  // convert to uppercase
    if (!kw_file.eof ())
      keywords.Append (str) ;
    }
  // close and return
  kw_file.close () ;
  return keywords ;
  } // World::ReadKeywords()

// reads a series of objects from a file
void World::ReadObjects (const string& kw_fname)
  {
  streampos fp (0) ;          // file pointer to start of
line
  size_t    line_count (0) ;  // line count
  string    str ;             // line string

  // open in object data file
  obj_infile.Open () ;
```

```
// get keywords
LinkedList<string> keywords = ReadKeywords (kw_fname) ;

// read file
while (!obj_infile.eof ())
  {
  // a line at a time
  ::getline (obj_infile, str, '\n') ;

  // convert string to uppercase-if not already
  StringToUpperCase (str) ;

  // get keywords (with their respective positions) in line
  LinkedList<SortKeyword> l_skw =
    KeywordPositions (str, keywords) ;

  line_count++ ;

  // line is worth examining further
  if (l_skw.NumberOfNodes() > 0)
    {
    // for each keyword in line string
    for (int i=0; i<l_skw.NumberOfNodes(); i++)
      {
      // adjust 'exact' fp position
      // and record current fp
      streampos fp_pos = fp + streampos(l_skw[i].pos) ;
      streampos cp     = obj_infile.tellg () ;
      // if braces o.k.
      if (CheckBracesNoNesting (fp_pos))
      {
        obj_infile.seekg (cp, ios::beg) ;
        // read objects and append to list
        LinkedList<Shape*> l_pobj =
        ReadListOfObjects (l_skw[i].kw,
                           fp_pos) ;
        for (int j=0;
          j<l_pobj.NumberOfNodes(); j++)
          l_obj.Append (l_pobj[j]) ;
        }
      else // simply quit!
        {
        cout << "invalid braces on line: "
             << line_count << endl ;
          exit (EXIT_FAILURE) ;
        }
      }
    }
  fp = obj_infile.tellg () ;  // update file pointer
  } // while (!eof())
```

```
  // close in object data file
  obj_infile.close () ;
  } // World::ReadObjects()

// simply write all objects to the object out file
void World::WriteObjects ()
  {
  // open out object data file
  obj_outfile.Open () ;

  if (l_obj.NumberOfNodes() == 0)  // objects not yet read
    this->ReadObjects () ;

  // write objects to out-file
  for (int i=0; i<l_obj.NumberOfNodes(); i++)
    {
    obj_outfile << typeid (*l_obj[i]).name () << endl ;
    obj_outfile << *l_obj[i] << endl ;
    }

  // close out object data file
  obj_outfile.close () ;
  }
```

Let's begin by taking a look at the *ReadObjects()* member function. The first question to ask when designing a function to read a collection of objects is just how the set of objects is to be stored. The storage mechanism has to be dynamic due to the unknown nature, at the time of compilation, of the object data file. The two most popular dynamic storage mechanisms are clearly vectors and linked lists. A linked list approach is chosen owing to its increased flexibility for the addition and deletion of nodes or elements when compared to a vectorial approach. Two approaches were considered for storing a linked list of objects; see Fig. 17.4. The approach in Fig. 17.4(a) is made possible by the fact that `LinkedList<T>` is a **`template class`** which accommodates composed types. The approach shown in Fig. 17.4(b) is the adopted approach

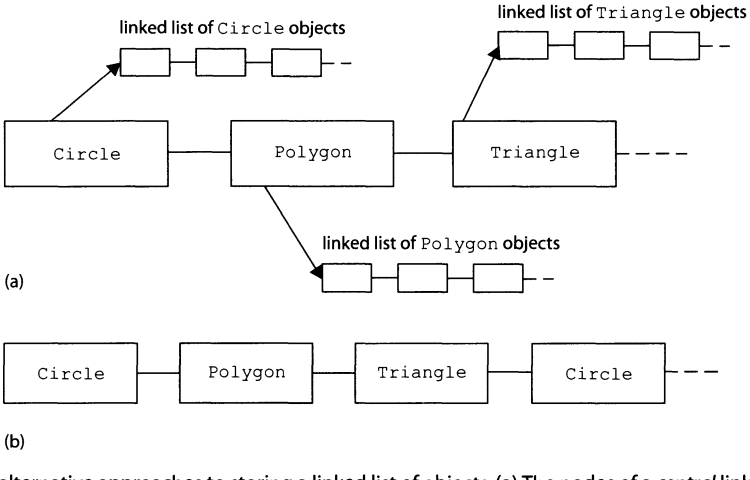


Fig. 17.4 Two alternative approaches to storing a linked list of objects. (a) The nodes of a *central* linked list point to linked lists of objects of a given type or **`class`**. (b) Each node of a *single* linked list is a unique object.

because it allows a simpler implementation than that of Fig. 17.4(a) and because the RTTI features of C++ allow the type or **class** of an object in a single linked list to be determined at run-time. Since we are dealing with objects of a **class** in the Shape inheritance hierarchy, the type or **class** of the linked list of objects is a pointer to Shape.

ReadObjects() first opens the in-object data file and then calls *ReadKeywords()* to obtain a linked list of the keywords which will be searched for in the object data file. The object data file is then read line by line till the end-of-file is reached using the global *getline()* function. Following a call to *getline()* the file pointer position is incremented to the start of the next line. The version of *getline()* used is particularly useful because it places the string of characters read into an object of the C++ library **class** string, and will continue to read characters until the new-line delimiter is encountered rather than specifying a fixed number of characters to be read. For the present application it is more advantageous to read the object data file line by line rather than a character at a time, for reasons which will become evident. Working with a string object rather than a C++ array of characters greatly simplifies the implementation. The line string read from the object data file is then converted to uppercase (if not already) in order to simplify future string comparisons. For the line string just read the *KeywordPositions()* member function is then called to obtain a linked list of keywords found in the string (if any) with their associated positions in the string. An association between a keyword and its position in a string is achieved by declaring an additional **private** nested **class** called World::SortKeyword, which encapsulates both string, kw, and position, pos, data members. Three overloaded relational operators are defined for World::SortKeyword to assist in the sorting of a linked list of SortKeyword objects. The < operator compares two SortKeyword objects in terms of position only, which guarantees that the keywords in a line string containing multiple keywords will be read in the order in which they appear from left to right.

For each keyword found in a line string the exact file position of the keyword is first determined and then passed to the member function *CheckBracesNoNesting()* to confirm that the non-nested syntax of the open and close braces is correct. If the braces pass the test then the *ReadListOfObjects()* member function is called to return a linked list of all the objects read from the object data file between the braces. The linked list of objects read between braces is then appended to the total linked list of objects previously read. *ReadObjects()* concludes by closing the previously opened in-object data file.

The World::*ReadKeywords()* member function returns a linked list of the keywords read from a specified file, which is OBJ_KW.DAT by default. The keywords read from the keyword data file are converted to uppercase if not already.

World::*KeywordPositions()* is passed a string and a linked list of keywords, for which it is to search the string and return a linked list of SortKeyword objects. Comments may exist in the passed string, and if the comment characters (//) are found then the string position of the first forward slash character is recorded. *KeywordPositions()* then proceeds to search for each keyword in the string using the string::*find()* member function, being careful to ensure that commented-out keywords are not appended to the linked list of keywords returned. Note the second form of the string::*find()* member function used in the **while**-loop of *KeywordPositions()*:

```
//...
while (kw_pos != string::npos)
   {
   // commence search from end of last kw. found
   kw_pos = str.find (l_kw[j], kw_pos+l_kw[j].length()-1) ;
   //...
   }
```

which commences searching for a keyword from the end of a previously found keyword.

The `World::CheckBracesNoNesting()` member function returns logical-true if, from a specified file position, first an open brace is encountered followed by a close brace with other characters possibly appearing before the open brace and between the two braces. Any other permutation of braces will result in `CheckBracesNoNesting()` returning a value of logical-false.

`World::ReadListOfObjects()` returns a linked list of objects of a particular type from a specified file pointer position. `ReadListOfObjects()` checks for the comment characters within the line string and works with two `string` objects: `str`, which is the original line string read, and `r_str`, which is a *reduced* string containing only the characters `[,]` and `}`. The reduced string object is then examined to determine the file pointer positions of the start of an object's data indicated by the character `[`. If an object is found then its data are read by calling the `ReadParticularObject()` member function and appended to a linked list of total objects read of a particular type or **`class`**. Locating objects by the character `[` allows multiple objects of a particular type or **`class`** to be defined between a single pair of `{` and `}` braces.

Finally (in connection with the reading of a list of objects) the `ReadParticularObject()` member function creates a new object of a specified type or **`class`**, reads the object from the current file pointer position, assigns the object to a pointer to `Shape` and returns the `Shape` pointer.

The `World::WriteObjects()` member function first opens the out-object data file (OBJ_OUT.DAT by default), confirms that the `World::l_obj` linked list of objects is not empty (an example of anticipation) and then proceeds to write each object encapsulated in the `World::l_obj` linked list data member to a disk file. The linked list of objects is written to a file with the aid of the overloaded insertion operator and the `Print()` overridden member function. Note the use of the `typeid()` operator and the `Type_info::name()` member function to extract, at run-time, the **`class`** name of each object via the pointer to abstract base **`class`** `Shape`. Finally, `WriteObjects()` closes the previously opened out-object data file.

The access member function `World::ObjectList()` simply returns the `l_obj` data member to allow access to the linked list of `World` objects.

The following program reads a collection of objects from a specified in-object data file and then writes the objects read to an out-object data file:

```
// read_obj.cpp
// tests the World class by reading a set of objects from a
// disk file
#include "world.h"  // class World

void main ()
  {
  World world ("obj_in.dat", "obj_out.dat") ;

  // read world objects
  world.ReadObjects () ;

  // write world objects
  world.WriteObjects () ;
  }
```

Following execution of READ_OBJ.EXE the OBJ_OUT.DAT disk file will be of the form:

```
Circle
number: 1, centre: (1, 0, 0), radius: 1
Sphere
number: 2, centre: (5, 5, 5), radius: 1
Triangle
number: 3
vertices: (1, 1, 0), (4, 2, 0), (3, 4, 2)
edges: [(1, 1, 0), (4, 2, 0)], [(4, 2, 0), (3, 4, 2)],
       [(3, 4, 2), (1, 1, 0)]
//...
Tetrahedra
number: 10
faces:
number: 0
vertices: (1, 1, 0), (3, 2, 0), (4, 5, 2)
//...
```

When we discussed the Shape inheritance hierarchy in Chapter 15 the insertion operator was overloaded for classes derived from Shape to assist in the stream output of objects. For the present chapter the approach taken is to declare a pure **virtual** function for abstract **class** Shape called *Print()* and then to override this function for each **class** in the Shape inheritance hierarchy:

```
// shape_v.h
// header file for Shape class
//...

class Shape
  {
  public:
    //...
    virtual void Print (ostream& s=cout) = 0 ;
  }; // class Shape

ostream& operator << (ostream& s, Shape& shp)
  { shp.Print (s) ; return s ; }

// poly_v.h
//...
class Polygon : public Shape, public Plane
  {
  public:
    //...
    void Print (ostream& s) ;
  };
//...
```

Note the use of the predefined ostream object cout as a default argument for s in the member function declaration of Shape::*Print()*. The overloaded insertion operator for abstract base **class** Shape redirects any object of a **class** derived from Shape to a specified output stream by means of Shape::**operator**<<*()* invoking the correct overridden **vir-**

tual *Print()* function. READ_OBJ.CPP called the `World::WriteObjects()` member function and illustrated an application of the overloaded << operator for writing a linked list of objects (pointers to `Shape`) to a disk file pointed to by the `WriteFile` object `out_file`:

```
void World::WriteObjects()
  {
  //...
  for (int i=0; i<l_obj.NumberOfNodes(); i++)
    {
    //...
    out_file << *l_obj[i] ;
    //...
    }
  //...
  }
```

17.11 String Streams

The header file STRSTREAM.H declares classes which support in-memory string formatting. The **class** `strstreambuf` is derived from `basic_streambuf` and provides operations for manipulating in-memory strings. Class `istrstream` is derived from `basic_istream` and provides support for input stream operations on a `strstreambuf` object, whereas **class** `ostrstream` is derived from `basic_ostream` and provides output stream operations on a `strstreambuf` object. The **class** `strstream` is derived from `basic_iostream` and provides both input and output stream operations.

The **class** declarations of `istrstream`, `ostrstream` and `strstream` are of the following form:

```
class istrstream
  : public basic_istream<char,char_traits<char> >
  {
  private:
    strstreambuf sb ;
  public:
    // types
    typedef char              char_type ;
    typedef char_traits<char> traits ;
    typedef traits::int_type  int_type ;
    typedef traits::pos_type  pos_type ;
    typedef traits::off_type  off_type ;
    // constructors and destructor
    explicit istrstream (char* s) ;

    explicit istrstream (const char* s) ;
    istrstream (char* s, streamsize n) ;
    istrstream (const char *s, streamsize n) ;
    virtual ~istrstream () ;
    // member functions
    strstreambuf* rdbuf () const ;
```

```
    char*          str () ;
  };

class ostrstream
  : public basic_ostream<char,char_traits<char> >
  {
  //...
  public:
    // types
    //...
    // constructors and destructor
    ostrstream () ;
    ostrstream (char* s, streamsize n,
                ios_base::openmode=ios_base::out) ;
    virtual ~ostrstream () ;
    strstreambuf* rdbuf () const ;
    void           freeze (bool freezefl=true) ;
    char*          str () ;
    int            pcount () const ;
  };

class strstream
  : public basic_iostream<char,char_traits<char> >
  {
  //...
  public:
    // types
    //...
    // constructors and destructor
    strstream () ;
    strstream (char* s, int n,
               ios_base::openmode=ios_base::out |
                                  ios_base::in) ;
    virtual ~strstream () ;
    // member functions
    void           freeze (bool freezefl=true) ;
    int            pcount () const ;
    strstreambuf* rdbuf () const ;
    char*          str () ;
  };
```

The constructors of the three classes initialise an object with a null terminated string `s`. Both classes `ostrstream` and `strstream` have default constructors whereas `istrstream` does not. The three-argument constructors of `ostrstream` and `strstream` initialise an object with a string array of `n` elements for a given mode; the mode is `ios_base::out` by default for objects of **class** `ostrstream` and `ios_base::in` and `ios_base::out` for **class** `strstream`.

Let's take a look at a program which uses the `istrstream` **class**:

```
// istrstr.cpp
// illustrates the C++ input string stream class istrstream
```

```
#include <strstrea>   // strstream
#include <string>     // strlen()

const int BUF_SIZ = 80 ;

void main ()
  {
  cout << "enter an int, double and string: " ;

  int    i (0) ;
  double d (0.0) ;
  char   s[BUF_SIZ/2], buffer[BUF_SIZ] ;

  cin.getline (buffer, BUF_SIZ);

  istrstream iss (buffer, strlen (buffer)) ;

  iss >> i >> d >> s ;

  cout << "i: " << i << ", d: " << d << ", string: "
       << s << endl ;
  }
```

with some user interaction:

```
enter an int, a double and a string: 9 12.2 c++
i: 9, d: 12.2, string: c++
```

17.12 Console Streams

The header file CONSTREA.H[4] includes the **class** declarations of `conbuf` and `constream` and various console manipulators for console-mode applications. CONSTREA.H provides similar **class** and manipulator based functionality to that of the header file CONIO.H.

The `conbuf` **class** is derived from `streambuf` for console output and supports numerous operations; refer to your compiler's documentation for a more detailed discussion of `conbuf`'s operations. The **class** `constream` is derived from `ostream` and is of the form:

```
class constream : public ostream  // CONSTREA.H
  {
  private:
    conbuf   buf ;
    ostream* oldtie ;
  public:
    constream () ;
    ~constream () ;
```

4 Header file CONSTREA.H and its associated classes are not part of the C++ standard library.

```
    // member functions
    conbuf*     rdbuf () ;
    void        clrscr () ;
    void        window (int l, int t, int r, int b) ;
    void        textmode (int mode) ;
    static int isCon (ostream& s) ;
};
```

Class constream encapsulates conbuf and ostream* data members and has a single default constructor and destructor. The member function *clrscr()* clears the console screen, *rdbuf()* returns a pointer to the conbuf data member of the object operated on by the function, *textmode()* sets the screen mode to text mode, *window()* defines the rectangular dimensions of an active window and *isCon()* indicates whether stream s is a console stream.

The CONSTREA.H header file also declares several manipulators specifically for formatting console streams. Before we examine the manipulators, let's examine the **class** declaration of a two-argument version of the smanip<T> parametrised **template class** declaration declared in IOMANIP.H which was previously discussed in connection with user-defined parametrised manipulators:

```
template <class T, class S>  // CONSTREA.H
class omanip2
  {
  private:
    ostream& (*pf)(ostream&, T, S) ;
    T data1 ;
    S data2 ;
  public:
    omanip2<T,S>
      (ostream& (*pf_arg)(ostream&, T, S),
      T data1_arg, S data2_arg)
      : pf (pf_arg), data1 (data1_arg), data2 (data2_arg)
      {}
    friend ostream& operator << (ostream& s,
                                 omanip2<T,S>& m)
      { return(*m.pf)(s, m.data, m.data2) ; }
};
```

omanip2 is useful when defining two-argument parametrised manipulators such as the *setxy(**int**,**int**)* manipulator discussed below. Note also that CONSTREA,H declares a **class** called omanip1which is analogous to omanip2 except that it caters for one-argument parameterised manipulators.

Let's first examine the non-parametrised console stream manipulators declared in CONSTREA.H:

```
ostream& clreol     (ostream& s) ;  // CONSTREA.H
ostream& delline    (ostream& s) ;
ostream& highvideo  (ostream& s) ;
ostream& insline    (ostream& s) ;
ostream& lowvideo   (ostream& s) ;
ostream& normvideo  (ostream& s) ;
```

clreol() clears to the end of a line in the text window. *delline()* deletes a line in a text window, while *insline()* inserts a line in the text window. *highvideo()* selects the high-intensity characters, whereas *normvideo()* selects the normal intensity characters.

The standard C++ parametrised console stream manipulators are:

```
omanip1<int>      setattr     (int a) ;  // CONSTREA.H
omanip1<int>      setbk       (int cc) ;
omanip1<int>      setclr      (int c) ;
omanip1<int>      setcrsrtype (int t) ;
omanip2<int,int>  setxy       (int x, int y) ;
```

setattr() sets various screen attributes. *setbk()* sets the background character colour, whereas *setclr()* sets the foreground colour. Each allowable colour has an associated symbolic constant and numeric value (e.g. BLACK (0) and WHITE (15)). *setcrsrtype()* sets the cursor type and *setxy()* positions the cursor at the specified position.

The following program illustrates constream:

```
// constr.cpp
// illustrates the C++ console stream class constream
#include <constrea>  // constream

void main ()
  {
  // console objects
  constream win1, win2 ;

  // define windows and clear
  win1.window (2,  2, 39, 15) ;
  win2.window (41, 4, 78, 17) ;
  win1.clrscr () ;
  win2.clrscr () ;

  // O/P to win1
  win1 << setbk (BLUE)
       << setxy (7, 25)
       << "I'm window #1" ;

  // O/P to win2
  win2 << setbk (CYAN)
       << setxy (7, 25)
       << "I'm window #2" ;
  }
```

17.13 Redirection

The standard input and output streams of cin and cout are, respectively, the keyboard and the console. C++ supports a feature called *redirection* in which the standard input and output streams can be redirected to alternative devices. For MS-DOS the syntax for redirecting command-line input and output involves redirection characters and is of the general form:

- The operator > redirects output from the console to an alternative device such as a disk file or printer.
- The operator < redirects input from the keyboard to an alternative device such as a disk file or printer.
- The >> operator redirects output from the console to a disk file and appends the output to existing data (if any) in the file.

To illustrate redirection, consider first the program:

```
// redirect.cpp
// illustrates redirection
#include <iostream>  // C++ I/O

void main ()
  {
  char ch ;

  while (cin.get (ch)) // while I/P, get and echo
    cout << ch ;
  }
```

and some user interaction when REDIRECT.EXE is executed at the command line of the C: root directory:

```
C:\>redirect
hello there!
hello there!
^Z
```

When carriage return is pressed REDIRECT.CPP echoes the user string to the display screen. The program is terminated either by entering Ctrl-Z or by pressing the F6 function key, since both of these are equivalent to end-of-file.

17.13.1 Redirecting Output from the Display Screen to a Disk File

To redirect output from the console to a disk file the > operator is used on the command line. The following user interaction redirects a user-entered string 'hi there!' from the display screen to a disk file called HI_THERE.TXT:

```
C:\>redirect > hi_there.txt
hi there!
^Z
```

If the file HI_THERE.TXT does not already exist MS-DOS creates the file. If the file does exist any existing data in the file is replaced by the output from the REDIRECT command.

The use of the program REDIRECT.CPP for redirecting output is identical to, say, redirecting a directory listing from the display screen to a disk file called DIR_LIST.TXT using the `dir` command:

```
C:\>dir > dir_list.txt
```

Alternatively, if you require that any existing data in a given file is not deleted then the >> redirection character ensures that any new output is appended to existing data:

```
C:\>redirect >> hi_there.txt
hi there again!
^Z
```

The content of file HI_THERE.TXT is now:

```
hi there!
hi there again!
```

17.13.2 Redirecting Input from the Keyboard to a Disk File

Similarly, to redirect input from the keyboard to a file or printer the < redirection character can be used. For example, the following redirects the input for REDIRECT.CPP from the file HI_THERE.TXT and displays the contents to the display screen:

```
C:\>redirect < hi_there.txt
hi there!
hi there again!
```

17.13.3 Redirecting both Input and Output

It is possible to redirect both input and output with a single command using both the < and > redirection characters. For example, the following redirects FILE1.TXT to FILE2.TXT:

```
C:\>redirect < file1.txt > file2.txt
```

17.14 Command Line Arguments

There are occasions when it is required to pass information to a program from the command line. For instance, the MS-DOS COPY command allows one or more source files to be copied to a destination file which is resident in the same directory or another directory or disk:

```
C:\>copy src.txt dst.txt
```

The filenames SRC.TXT and DST.TXT are known as *command line arguments* and are passed to the COPY command before execution commences.

One of the most surprising features of *main()* is that it is a function (just like any other function) which has two optional arguments. All of the *main()* functions that we have seen to date have been of the form:

```
void main ()
  { /* ... */ }
```

but in fact *main()* has the following more general form:

```
void main (int argc, char* argv[])
  { /* ... */ }
```

Although any argument names can be used, it has become convention that the names `argc` and `argv` are used as the first and second arguments to *`main()`*, respectively. The first argument, `argc`, is the argument count and is used to inform the program of the number of command line arguments. Note that `argc` will be at least equal to one, since the program name itself qualifies as a command line argument. The second argument, `argv`, is the argument vector, and is an array of pointers to **`char`** that point to the command line arguments. Since the argument vector is of type **`char`*** the command line arguments are passed to *`main()`* as null-terminated strings, and if an alternative argument type is required then a conversion is necessary.

Let's take a look at a program which is passed a string command line argument:

```
// hello.cpp
// illustrates command line arguments
// and the passing of strings
#include <iostream>   // C++ I/O
#include <cstdlib>    // exit()

void main (int argc, char* argv[])
  {
  // check command-line arguments
  if (argc != 2)
    {
    cout << "format: hello name" << endl ;
    exit (EXIT_FAILURE) ;
    }

  // display message
  cout << "hi there " << argv[1] << "!" << endl ;
  }
```

With some user interaction (assuming that HELLO.EXE is executed at the command line of `C:` root directory):

```
C:\>hello zak
hi there zak!
```

The program name and string argument are separated by a space. The function body of *`main()`* first checks that the number of command line arguments is equal to two (program name plus the string argument). If the user enters the incorrect number of command line arguments a message is displayed, informing the user of the required format of command line arguments for successful program execution and the program terminates. If the correct number of command line arguments are entered the program simply displays a message to the user via the standard output stream.

The string following the program name is accessed as `argv[1]`, not `argv[0]`. Some compilers and operating systems flush the first element of `argv` when program execution commences. Other compilers and operating systems place the disk and directory path and program name in `argv[0]` (e.g. `argv[0]`="C:\HELLO.EXE" for the above example using the Borland C++ (version 5.5) command-line compiler).

The following program passes numeric rather than character data to *`main()`* to evaluate the square root of a number:

```
// sq_root.cpp
// illustrates command-line arguments
// and the passing of numerics
#include <iostream>   // C++ I/O
#include <cstdlib>    // exit(), atof()
#include <cath>       // sqrt()

void main (int argc, char* argv[])
  {
  // check command-line arguments
  if (argc != 2)
    {
    cout << "format: sq_root number" << endl ;
    exit (EXIT_FAILURE) ;
    }

  // display square root
  cout << sqrt (atof (argv[1])) << endl ;
  }
```

with some user interaction:

```
C:/>sq_root 2
1.41421
```

The structure of SQ_ROOT.CPP is identical to that of HELLO.CPP, except that the C++ library function *atof()* is used to convert a string to a floating-point **double** number. The declaration of *atof()* is:

```
double atof (const char* string) ;  // STDLIB.H or MATH.H
```

The C++ library includes a variety of other functions which perform conversions between a string and a numeric, and vice versa. For a complete list refer to your compiler's documentation, but the most popular are *atoi()*, which converts a string to an **int**; *atol()*, which converts a string to a **long**; *strtod()*, which converts a string to a **double**; *strtol()*, which converts a string to a **long**; *strtoul()*, which converts a string to an **unsigned long**; *itoa()*, which converts an **int** to a string; and *ltoa()*, which converts a **long** to a string.

Let's conclude our discussion of command line arguments by developing our own program which mimics the MS-DOS COPY command. It must be mentioned that the COPY command is a comprehensive command and has the general syntax:

```
COPY [/A|/B]src[/A|/B][+src[/A|/B][+...]]
     [dst[/A|/B]][/V][/Y|/-Y]
```

where `src` is the source file(s) to be copied, `dst` is the new destination file, `/A` indicates an ASCII text file, `/B` indicates a binary file, `/V` verifies that the destination file is written correctly, `/Y` and `/-Y` suppress and verify that an existing destination file is to be overwritten and + is used for the appending of source files. However, we shall develop a simplified version of COPY which performs a copy of a source file to a destination file:

```
// cpp_cpy.cpp
// illustrates command-line arguments
#include <fstream>  // C++ file I/O
#include <cstdlib>  // exit()
#include <string>   // strcmp()

void main (int argc, char* argv[])
  {
  // check command-line arguments
  if (argc != 3)
    {
    cerr << "format: cpp_cpy src_file dst_file" << endl ;
    exit (EXIT_FAILURE) ;
    }

  // prevent self copy
  if (strcmp (argv[1], argv[2]) == 0)
    {
    cerr << "source file must be different from
             destination file" << endl ;
    exit (EXIT_FAILURE) ;
    }

  // open source and destination files
  ifstream src_file ;
  ofstream dst_file ;

  src_file.open (argv[1], ios::binary) ;
  if (!src_file)
    {
    cerr << "sorry-can't open file \"" << argv[1]
         << "\" " << endl ;
    exit (EXIT_FAILURE) ;
    }
  dst_file.open (argv[2], ios::binary) ;
  if (!dst_file)
    {
    cerr << "sorry-can't open file \"" << argv[2]
         << "\" " << endl ;
    exit (EXIT_FAILURE) ;
    }

  // copy file
  while (src_file)
    dst_file.put (char(src_file.get ())) ;

  // close files
  src_file.close () ;
  dst_file.close () ;
  }
```

with some user interaction:

```
C:\>cpp_cpy src.txt dst.txt
```

CPP_CPY.CPP first performs a check to confirm that the number of command line arguments (i.e. three) is correct. Then the C++ library function *strcmp()* is used to confirm that the names of the source and destination files are different so as to prevent the source file being copied onto itself. If these two checks are successfully passed the source and destination files are opened with further checks for each file object to confirm that the files are opened for reading and writing respectively. Next, the contents of the source file are copied to the destination file using the `basic_istream::`*get()* and `basic_ostream::`*put()* member functions. Finally, both the opened source and destination files are closed.

17.15 The C Approach to Streams

Let's now examine the C language approach to streams. Although C's mechanism for handling streams is not object-oriented, it is nevertheless extremely powerful and cannot be ignored if you are considering migrating from programming in C to C++. This section begins by introducing the STDIO.H header file and the C predefined streams. Then the functions *printf()* and *scanf()* are discussed in detail, since they are the most frequently used functions for performing output and input operations, respectively, on the standard input and output streams. The section concludes by examining file handling in C.

17.15.1 The STDIO.H Header File

The STDIO.H (standard input and output) header file is the C equivalent to the C++ IOSTREAM.H header file. STDIO.H incorporates the necessary structure and function declarations for the high-level manipulation of streams.

17.15.2 Predefined Streams

In a similar manner to the C++ predefined standard stream objects `cin`, `cout`, `cerr` and `clog`, the C language defines five predefined open streams in the STDIO.H header file:

```
extern FILE _streams[];
/* ... */
#define stdin  (&_streams[0])
#define stdout (&_streams[1])
#define stderr (&_streams[2])
#define stdaux (&_streams[3])
#define stdprn (&_streams[4])
```

The `FILE` structure is discussed later with reference to disk files.

The stream `stdin` is dedicated to the standard input device, which is generally the keyboard. The stream `stdout` refers to the standard output device, which is usually the display screen. The stream `stderr` is the standard error output device, which again is usually the display screen. The streams `stdaux` and `stdprn` are, respectively, the standard auxiliary and printing devices.

17.15.3 Output to the Display Screen Using the *printf()* Function

The C approach to directing a stream to the `stdout` stream (i.e. the display screen) is to use the *printf()* function, which has the following declaration, STDIO.H:

```
int printf (const char* fmt_string[, argument, ...]) ;
```

The `fmt_string` argument, as the name suggests, is called the *format string* and contains text and *format specifiers* or *conversion characters*. The *printf()* function returns the number of bytes or characters output if successful, else the `EOF` (=–1) identifier (STDIO.H) is returned if unsuccessful. Before discussing the available conversion characters let's examine a program which displays a string:

```
/*
printf0.c
illustrates the printf() function
*/

#include <stdio.h>  /* C I/O */

void main ()
  { printf ("hello, world\n") ; }
```

with output:

```
hello, world
```

printf() outputs the string of characters between the double quotation characters (" . . . "). The character '\n' is the new line escape sequence and advances the current position to the first position of the next line. The C++ equivalent to '\n' is the `endl` manipulator. Escape sequences were discussed in Chapter 4; refer to Table 4.1 for a list of the available escape sequences.

Let's now consider an example program which uses a conversion character:

```
/*
printf1.c
further illustrates the printf() function
*/

#include <stdio.h>  /* C I/O  */
#include <math.h>   /* sqrt() */

void main ()
  {
  double number = 2.0 ;

  printf ("the square root of %f is: %f\n",
           number, sqrt (number)) ;
  }
```

with output:

Table 17.4 Conversion characters for `printf()`.

Conversion character	*Input*	*Output*
Character		
`c`	Character	Single character
`s`	String pointer	Character string
`%`	None	The % sign
Numeric		
`d`	Integer	Signed decimal integer
`e`	Floating-point	Scientific notation: e[+/–]number
`e`	Floating-point	Scientific notation: e[+/–]number
`f`	Floating-point	Signed decimal floating-point
`g`	Floating-point	`e` or `f` form, whichever is shorter
`g`	Floating-point	`e` or `f` form, whichever is shorter
`i`	Integer	Signed decimal integer
`o`	Integer	Unsigned octal integer
`u`	Integer	Unsigned decimal integer
`x`	Integer	Unsigned hexadecimal integer (`a`, `b`, `c`, `d`, `e`, `f`)
`x`	Integer	Unsigned hexadecimal integer (`a`, `b`, `c`, `d`, `e`, `f`)
Pointer		
`n`	Pointer to **`int`**	Number of characters output up to that point
`p`	pointer	A pointer

```
the square root of 2.000000 is: 1.414214
```

A conversion character sequence starts with the percentage sign (`%`) and is followed by the format code. The conversion character in the above program is `f`, which performs a conversion to a decimal floating-point number. The `printf()` function now has three arguments, each separated by the comma operator. Note that for each conversion character there is a corresponding function argument and that the order of the conversion characters matches the order of the arguments.

Conversion characters have the following general form:

```
%[flags][width][precision][F|N|h|l|L]type
```

where:

`[flags]`	Flag characters (optional)
`[width]`	Width specifiers (optional)
`[precision]`	Precision specifiers (optional)
`[F|N|h|l|L]`	Input size modifiers (optional)
`type`	Conversion type character (required)

Table 17.4 lists the available conversion characters and the available flag characters are listed in Table 17.5. For instance, `%-10d` left-justifies a signed decimal integer with a total character field width of 10. Note that + takes precedence over a blank space if both are specified.

The available width specifiers are shown in Table 17.6. The width specifiers set the minimum field width for output and can be specified directly through the use of a decimal or indirectly through the use of an asterisk. If a field width is specified directly using the `n` specifier and the

Table 17.5 Flag characters for `printf()`.

Flag	*Meaning*
-	Left-justify, right filled with blanks
+	Signed conversion
blank space	If non-negative, output starts with a blank instead of +
#	Use one of the alternate forms below
Alternate forms	
c, d, i, s, u	No effect
0	0 prepended to non-zero argument
x, X	0x, 0X prepended to argument
e, E, f	Decimal point shown even if no digit follows
g, G	Same as e, E except that trailing zeros are not removed

Table 17.6 Width specifiers for `printf()`.

Width specifier	*Meaning*
n	A minimum of n characters are output (blank padding)
0n	A minimum of n characters are output (zero padding)
*	Argument list contains width specifier

output width is less than n characters the output is padded with blank spaces, whereas if a field width is specified directly using the 0n specifier and the width is less than n characters the output is padded with zeros. If an asterisk is used the next argument in the call to `printf()` specifies the field width. Note that if the result of a conversion is wider than the specified field width the field is expanded accordingly to accommodate the conversion result.

The precision specifiers are shown in Table 17.7. A precision specifier sets the number of characters or digits for output and is preceded by the dot operator (.) so as to distinguish it from a width specifier. Similarly to the width specifier, a precision specifier can be specified directly through the use of a decimal or indirectly through the use of an asterisk. If the output value exceeds n characters the output is truncated or rounded depending on the type of conversion character. For conversion characters d, i, o, u, x and X, if the input value has fewer than n digits the value is left-padded with zeros, whereas if the input value has more than n digits the value is not truncated. For e, E and f the last input value digit is rounded when the n digits are exceeded, while n specifies the maximum number of significant digits to output for g

Table 7.7 Precision specifiers for `printf()`.

Precision specifier	*Meaning*
None	Default precision used: c = no effect d, i, o, u, x, X = 1 e, E, f = 6 g, G = all significant digits s = print to NULL character
.0	e, E, f: no decimal point is output d, u, o, u, x: precision is set to defaults
.n	n characters or decimal places are output
.*	Argument list contains precision specifier

Table 17.8 Input size modifiers for `printf()`.

Input size modifier	*Meaning*
F	Argument interpreted as a far pointer
N	Argument interpreted as a near pointer (memory model-dependent)
h	Argument interpreted as a **short int** for d, i, o, u, x, X
l	Argument interpreted as a **long int** for d, i, o, u, x, X and as a **double** for e, E, f, g, G
L	Argument interpreted as a **long double** for e, E, f, g, G

and G. In the case of the s conversion character, n specifies the maximum number of characters to output.

Table 17.8 illustrates the input size modifiers. The input size modifiers specify how the *printf()* function is to interpret the next input argument.

Consider the following program, which illustrates a few of the character conversion specifications mentioned above:

```
/*
printf2.c
further illustrates the printf() function
*/

#include <stdio.h>   /* C I/O   */
#include <math.h>    /* sqrt()  */

void main ()
  {
  char   ch       = 'c' ;
  char   string[] = "love of wisdom" ;
  int    i        = 1 ;
  float  vat      = 17.5 ;
  double number   = 2.0 ;
  double mass     = 5.974e21 ;

  printf ("character ch: %c\n", ch) ;
  printf ("philosophy: %s\n", string) ;
  printf ("lvalue of i: %p; rvalue of i: %d\n", &i, i) ;
  printf ("V.A.T is currently: %+3.1f %%\n", vat) ;
  printf ("the square root of %g is: %f\n",
           number, sqrt (number)) ;
  printf ("the mass of earth, including atmosphere, is:
           %-10.3E tonnes\n", mass) ;
  }
```

with output:

```
character ch: c
philosophy: love of wisdom
lvalue of i: 22F2; rvalue of i: 1
V.A.T. is currently: +17.5 %
the square root of 2 is: 1.414214
```

```
the mass of earth, including atmosphere, is: 5.974E+21
tonnes
```

It is worth mentioning the function *sprintf()* (also declared in STDIO.H) which is similar in form and name to *printf()* but writes formatted output to a string, buffer, rather than stdout:

```
int sprintf (char* buffer, const char* format[, argument, ...]);
```

An illustration of *sprintf()* is:

```
/*
sprintf.c
illustrates the sprintf() function
*/

#include <stdio.h>  /* C I/O */

void main ()
  {
  char string0[80] ;
  char string1[] = "a string" ;

  sprintf (string0, "this is %s\n", string1) ;

  printf ("%s", string0) ;
  }
```

with output:

```
this is a string
```

From the above discussion it is clear that the *printf()* function is extremely versatile, but at the same very cumbersome when compared with the C++ standard output stream approach.

17.15.4 Input from the Keyboard Using the *scanf()* Function

Let's now examine the *scanf()* function, which scans and formats input from the stdin stream (i.e. the keyboard) and has the following declaration, STDIO.H:

```
int scanf (const char* fmt_string[, address, ...]) ;
```

scanf() returns the number of input fields scanned, converted and stored. If the end-of-file is reached then *scanf()* returns EOF. Conversion characters have the following general form:

```
%[*][width][F|N][h|l|L]type
```

where:

`[*]`	Assignment suppression character (optional)
`[width]`	Width specifiers (optional)
`[F|N]`	Pointer size modifier (optional)
`[h|l|L]`	Argument type modifier (optional)
`type`	Conversion type character (required)

Similarly to *`printf()`*, an input conversion character sequence is preceded with the percent sign (`%`) and is followed by the format code. For each conversion character there is a corresponding function argument, and the order of the conversion characters matches the order of the arguments.

The assignment suppression character (`*`) suppresses assignment of the next input field by allowing the input field to be scanned but not assigned to the next address argument. The width specifier defines the maximum number of characters to be read from the input field. If the input field has fewer characters than the value specified in the width specifier, all characters in the input field are read. If a non-white space or non-convertible character is encountered, all character scanning is terminated and all characters read up to that point are converted and stored. The pointer input size modifiers (`F` and `N`) override the default size of the address argument, whereas the argument type modifiers (`h`, `l` and `L`) override the default type of the address argument. The available conversion characters for *`scanf()`* are listed in Table 17.9.

To input a **`long double`** using the *`scanf()`* function combine the `l` and `f` conversion characters (i.e. `%lf`). The pointer input size and argument type modifiers for *`scanf()`* are listed in Table 17.10.

It is important to note that *`scanf()`* interprets any white space character (blank space, new line, form feed, horizontal tab, vertical tab) as a signal to terminate input. This is illustrated in the following program:

Table 17.9 Conversion characters for *`scanf()`*.

Conversion character	*Type of argument*	*Input*
Character		
`c`	Pointer to **`char`**	Character
`s`	Pointer to string	String
`%`	% character stored	% character
Numeric		
`d`	Pointer to decimal **`int`**	Decimal integer
`D`	Pointer to decimal **`long`**	Decimal integer
`e,E`	Pointer to **`float`**	Floating-point
`f`	Pointer to **`float`**	Floating-point
`g,G`	Pointer to **`float`**	Floating-point
`i`	Pointer to **`int`**	Decimal integer
`l`	Pointer to **`long`**	Decimal integer
`o`	Pointer to **`int`**	Octal integer
`O`	Pointer to **`long`**	Octal integer
`u`	Pointer to **`unsigned int`**	Unsigned decimal
`U`	Pointer to **`unsigned long`**	Unsigned decimal
`x`	Pointer to **`int`**	Hexadecimal integer
`X`	Pointer to **`long`**	Hexadecimal integer
Pointer		
`n`	Pointer to **`int`**	
`p`	Pointer	Hexadecimal integer

Table 17.10 Input size and argument type modifiers for `scanf()`.

Modifier	*Meaning*
Input size modifiers	
F	Overrides default size
N	Overrides default size
Argument type modifiers	
h	Converts input to **short int** for d, i, o, u, x
l	Converts input to **long int** for d, i, o, u, x Converts input to **double** for e, f, g No effect for c, D, I, n, O, p, s, U, X
L	Converts input to **long double** for e, f, g

```
/*
scanf0.c
illustrates the scanf() function
*/

#include <stdio.h>  /* C I/O */

void main ()
  {
  char name[20] ;

  printf ("enter your forename and surname: ") ;
  scanf  ("%s", name) ;
  printf ("hi %s\n", name) ;
  }
```

with some user interaction:

```
enter your forename and surname: eddie edwards
hi eddie
```

It may be evident from the above program that input variables are passed by their memory address by the use of pointers; remember that the name of an array in C/C++ is a synonym for its memory address.

Let's now take a look at a few programs which illustrate several features of the `scanf()` function. The first program illustrates that a non-white space character is discarded by `scanf()`:

```
/*
scanf1.c
further illustrates the scanf() function
*/

#include <stdio.h>  /* C I/O */

// Point data structure
struct Point
```

```
  { double x, y, z ;};

void main ()
  {
  struct Point p ;  /* note: keyword struct in definition */

  printf ("enter x, y and z members of a Point (x,y,z): ") ;
  scanf  ("(%lf,%lf,%lf)", &p.x, &p.y, &p.z) ;
  printf ("point p: (%.2f, %.2f, %.2f)\n", p.x, p.y, p.z) ;
  }
```

with some user interaction:

```
enter x, y and z members of a Point (x,y,z): (1,2,3)
point p: (1.00, 2.00, 3.00)
```

The above program illustrates the use of the "`(%lf,%lf,%lf)`" format specification for inputting the data members of a `Point` data structure object in the form "`(x,y,z)`". The parentheses and commas are read and discarded by *`scanf()`* and the three **`double`** floating-point numbers read. Note the use of the address-of operator (`&`) for inputting the floating-point numbers. If the non-white space characters are neglected then the input data must be separated by white space characters:

```
/*
scanf2.c
further illustrates the scanf() function
*/
/* ... */
void main ()
  {
  /* ... */
  scanf ("%lf%lf%lf", &p.x, &p.y, &p.z) ;
  /* ... */
  }
```

with some user interaction:

```
enter x, y and z members of a Point (x,y,z): 1 2 3
point p: (1.00, 2.00, 3.00)
```

However, if the input values 1, 2 and 3 are separated by a non-white space character such as a comma then *`scanf()`* will fail and enter zero for `y` and `z`:

```
enter x, y and z members of a Point (x,y,z): 1,2,3
point p: (1.00, 0.00, 0.00)
```

The following program illustrates the use of the assignment suppression (`*`) character:

```
/*
scanf3.c
further illustrates the scanf() function
```

```
*/
/* ... */
void main ()
  {
  /* ... */
  scanf ("%*c%lf%*c%lf%*c%lf", &p.x, &p.y, &p.z) ;
  /* ... */
  }
```

with some user interaction:

```
enter x, y and z members of a Point (x,y,z): [1,2;3}
point p: (1.00, 2.00, 3.00)
```

The assignment suppression character (*) suppresses assignment of the next input field. Thus, the x, y and z `Point` data members can be input with any non-white space characters between and enclosing the members.

The following program illustrates the use of a width specifier with *`scanf()`* to prevent a user from entering characters outside the bounds of a string array:

```
/*
scanf4.c
further illustrates the scanf() function
*/

#include <stdio.h>  /* C I/O */

void main ()
  {
  char string[10] ;

  printf ("enter a string: ") ;
  scanf  ("%9s", string) ;  /* leave room for null-terminator */
  printf ("string entered: %s\n", string) ;
  }
```

with some user interaction:

```
enter a string: aspacelessstring
string entered: aspaceles
```

Try removing the width format specifier from *`scanf()`* in the above program and enter a string of ten characters or more.

The *`scanf()`* function also contains a format specifier '`%[...]`' often referred to as the *scanset*. *`scanf()`* will only scan for characters specified within the square brackets:

```
/*
scanf5.c
illustrates the scanset feature of the scanf() function
*/
```

```
#include <stdio.h>  /* C I/O */

void main ()
  {
  char string[80] ;

  printf ("enter a string: ") ;
  scanf  ("%[aeiou]", string) ;
  printf ("string entered: %s\n", string) ;
  }
```

with some user interaction:

```
enter a string: utopia
string: u
```

As soon as a character is input which is not specified within the square brackets (in the present case a vowel) `string` will be null-terminated.

If the character ^ is used as the first character between the square brackets *`scanf()`* scans for all characters except those specified within the square brackets:

```
/*
scanf6.c
further illustrates the scanset feature of the scanf()
function
*/
/* ... */
void main ()
  {
  /* ... */
  scanf  ("%[^aeiou]", string) ;
  /* ... */
  }
```

with some user interaction:

```
enter a string: scanf
string: sc
```

Further, *`scanf()`* can scan for a range of characters by specifying a hyphen (-) format specifier:

```
/*
scanf7.c
further illustrates the scanset feature of the scanf()
function
*/
/* ... */
void main ()
  {
  /* ... */
```

```
    scanf  ("%[a-zA-Z]", string) ;  /* case sensitive */
    /* ... */
    }
```

with some user interaction:

```
enter a string: Non-Whitespace
string: Non
```

In conclusion, the function *sscanf()* is similar to *scanf()* except that it scans and formats input from a string pointed to by `buffer` rather than `stdin` and has the following declaration, STDIO.H:

```
int sscanf (const char* buffer,
            const char* fmt_string [,address, ...]) ;
```

An example of *sscanf()* is:

```
/*
sscanf.c
illustrates the sscanf() function
*/

#include <stdio.h>  /* C I/O */

void main ()
  {
  char* polyhedra[5] = {"tetrahedron",
                        "hexahedron",
                        "octahedron",
                        "dodecahedron",
                        "icosahedron"} ;
  int   faces[5] = {4, 6, 8, 12, 20} ;
  char  buffer[5][50] ;
  char* poly[20] ;
  int   face ;
  int   i ;

  for (i=0; i<5; i++)
    sprintf (buffer[i], "%s %d", polyhedra[i], faces[i]) ;

  for (i=0; i<5; i++)
    {
    sscanf (buffer[i],"%s %d", &poly, &face) ;
    printf ("%12s: %2d faces\n", poly, face) ;
    }
  }
```

with output:

```
  tetrahedron: 4  faces
```

```
   hexahedron: 6  faces
   octahedron: 8  faces
dodecahedron: 12 faces
 icosahedron: 20 faces
```

SSCANF.CPP first copies the array of strings of polyhedra names, `polyhedra`, and number of faces, `faces`, to the temporary buffer, `buffer`, using the *`sprintf()`* function. The *`sscanf()`* function then scans and formats the array of strings buffer and stores the formatted input in the memory addresses specified in the arguments `poly` and `face`. The *`printf()`* function displays the formatted output shown of arguments `poly` and `face`.

17.15.5 The *`vprintf()`*, *`vscanf()`*, *`vsprintf()`* and *`vsscanf()`* Functions

The function declarations of *`vprintf()`*, *`vscanf()`*, *`vsprintf()`* and *`vsscanf()`* are given in STDARG.H and are:

```
int vprintf (const char* format, va_list arg_list) ;
int vscanf  (const char* format, va_list arg_list) ;

int vsprintf (char* buffer,
              const char* format, va_list arg_list) ;
int vsscanf (const char* buffer,
             const char* format, va_list arg_list) ;
```

`vprintf()` and *`vsprintf()`* write formatted output to `stdout` and to a string respectively. *`vscanf()`* and *`vsscanf()`* scan and format input from `stdin` and from a string respectively. The *`v...printf()`* and *`v...scanf()`* functions are similar to *`printf()`* and *`scanf()`* except that they accept a pointer to a list of arguments rather than a list of arguments. The **`typedef`** `va_list` is declared in STDARG.H and is a pointer to **`void`**:

```
typedef void* va_list ;
```

17.15.6 File Input and Output

As with standard input and output in C, the C file-handling mechanism is function-based and revolves around the STDIO.H library header file. The majority of the C file-handling functions make use of a pointer to a `FILE` structure which encapsulates various information about a given file, such as the file's name, mode and so on. We will begin this section by first examining the `FILE` structure and then several of the most popular C functions used for file handling.

17.15.7 The `FILE` Structure

The `FILE` structure encapsulates information relevant to the manipulation of a file stream, is declared in the STDIO.H header file and is of the form:

```
typedef struct
  {
  int             level ;
  unsigned        flags ;
```

```
char            fd ;
unsigned char   hold ;
int             bsize ;
unsigned char*  buffer ;
unsigned char*  curp ;
unsigned        istemp ;
short           token ;
} FILE ;
```

The `level` data member indicates how many characters remain in the file buffer. The `flags` member informs the operating system of the status of a file, such as read or write only, read and write, binary or text, file data is incoming or outgoing, error and end-of-file indicators. The file descriptor is denoted by `fd` and provides an association between a file and an object of the `FILE` structure. The `hold` member is used for holding file access if no buffer exists. The file buffer is denoted by `buffer` and is of size `bsize`. The current or active pointer of the file buffer is denoted by `curp`. The `istemp` member is a temporary file indicator and `token` is for file validity checking.

17.15.8 The *fopen()* Function

The declaration of the *fopen()* C library function is:

```
FILE* fopen (const char* filename, const char* mode) ;
```

fopen() opens a file specified by `filename` (which can also incorporate a path) in a mode specified by the second argument `mode`. If *fopen()* is successful in opening the specified file, a pointer to a `FILE` structure stream object is returned which is your link with the operating system for handling the file, else if *fopen()* is unsuccessful the `NULL` pointer is returned.

The permissible modes in which a file can be opened are listed in Table 17.11. The mode strings listed in Table 17.11 `"r"`, `"w"`, `"a"`, `"+"` and `"b"` represent reading, writing, appending, updating and binary respectively. By default, a file is assumed to be a text file. If a file is opened for reading it is assumed that the file exists, and an error will be returned if it does not exist. If a file is opened for writing and it does not exist then it will be created, and if it already exists then all existing data will be deleted and the file length truncated to zero. Thus, beware of writing to an existing file. If a file is opened for appending it will be created if it does not exist. If a file is opened for reading and writing and it does not exist then it will be created.

Before we examine an example program which illustrates *fopen()*, let's discuss the *fclose()* and *fgetc()* functions.

17.15.9 The *fclose()* Function

Having opened a file using the *fopen()* function it is the responsibility of the programmer to close the file after use. It is important to formally close a file when access to the file is no longer required so that the file is not accidentally damaged, and because the operating system stipulates a limit on the number of files which can be simultaneously open at any given time[5]. The function *fclose()* closes a file whose pointer to the `FILE` structure is given by `f_ptr`:

5 The Borland C++ command-line compiler (version 5.5) specifies the number of simultaneously open files by the `define FOPEN_MAX` in STDIO.H, which is set to the `define _NFILE_` in _NFILE.H and can equal either 20 or 40.

Table 17.11 *fopen()* mode values.

mode	*Meaning*
Text	Open for read only
"r"	Create for write only
"w"	Open to append or if file does not exist then create for write only
"a"	Open for read and write
"r+"	Open for read and write
"w+"	Create for read and write
"a+"	Open or create to append
Binary	
"rb"	Open for read only
"wb"	Create for write only
"ab"	Open to append or if file does not exist then create for write only
"rb+"	Open for read and write
"wb+"	Create for read and write
"ab+"	Open or create to append

```
int fclose (FILE* f_ptr) ;
```

fclose() returns zero if the file was successfully closed, else EOF if an error occurred. If *fclose()* is successful the FILE pointer, f_ptr, is no longer valid and if a file is to re-opened a further call to *fopen()* is required.

17.15.10 The *fgetc()* Function

The *fgetc()* function reads a character at a time from an input file stream given a FILE structure pointer f_ptr returned by *fopen()*:

```
int fgetc (FILE* f_ptr) ;
```

If successful, *fgetc()* returns the next character from the specified file input stream after it is converted to **int**. If *fgetc()* is unsuccessful or end-of-file is reached then EOF is returned.

17.15.11 Reading a Disk File

The following program illustrates the C *fopen()*, *fclose()* and *fgetc()* functions for reading a file and displaying it to the screen:

```
/*
ordc.c
illustrates opening, reading, displaying and closing a file
using the C fopen(), fgetc() and fclose() file handling
functions
*/
#include <stdio.h>    /* C I/O  */
#include <stdlib.h>   /* exit() */

void main ()
```

```
{
char  file_name[50] ;
FILE* f_ptr ;
int   c ;

/* get a filename */
printf ("enter a filename: ") ;
gets (file_name) ;

/* open file */
if ((f_ptr = fopen (file_name, "r")) == 0)
  {
  printf ("sorry-can't open file \"%s\"\n", file_name) ;
    exit (EXIT_FAILURE) ;
  }

/* read and display file */
while ((c = fgetc (f_ptr)) != EOF)
  putchar (c) ;

/* close file */
fclose (f_ptr) ;
}
```

with some user interaction:

```
enter a filename: ordc.c
```

and outputs itself to the display screen.

The *gets()* function gets a string of characters, `string`, from the `stdin` stream until a new-line character is encountered and has the declaration:

```
char* gets (char* string) ;  /* STDIO.H */
```

gets() returns a pointer to the string when successful and the `NULL` pointer when an error or end-of-file is reached.

The *putchar()* function outputs the character `c` to the `stdout` stream:

```
int putchar (int c) ;  /* STDIO.H */
```

putchar() returns the character when successful and `EOF` when unsuccessful.

The above program illustrated a use of *fopen()* to open a file requested by a user. The file is opened for reading only (`"r"`) and the `FILE` structure pointer returned by *fopen()* is tested against the `NULL` pointer within the **if**-statement. If the file-opening procedure is unsuccessful a message is displayed to `stdout` using *printf()* and the program is terminated using *exit()*:

```
/* ... */
if ((f_ptr = fopen (file_name, "r")) == 0)
  {
```

```
    printf ("sorry-can't open file \"%s\"\n", file_name) ;
    exit (EXIT_FAILURE) ;
    }
```

If the file is successfully opened a **while**-loop is used to get each character of the file using *fgetc()*. Each character obtained from the file is tested against EOF to see if the end of file has been reached:

```
  while ((c = fgetc (f_ptr)) != EOF)
    putchar (c) ;
```

Finally, the file opened is closed using the *fclose()* function:

```
  fclose (f_ptr) ;
```

17.15.12 The *fgets()* Function

The previous program example worked fine reading the contents of a file a character at a time, but we can alternatively read the contents of a file line by line using the *fgets()* function:

```
  char* fgets (char* buffer, int n, FILE* f_ptr) ;  /* STDIO.H */
```

fgets() will continue to read a line of data from a file specified by f_ptr until either n-1 characters have been read or a new-line character is read, placing each character in the string array pointed to by buffer. If successful, *fgets()* returns the string pointed to by buffer, else it returns NULL if an end of file or error occurred. Note the similarity between the C *fgetc()* and *fgets()* functions and the C++ member functions basic_istream::*get()* and basic_istream::*getline()*.

Let's now modify ORDC.C to use *fgets()*:

```
  /*
  ordc1.c
  illustrates opening, reading, displaying and closing a file
  using the C fopen(), fgets() and fclose() file handling
  functions
  */
  #include <stdio.h>   /* C I/O */
  #include <stdlib.h>  /* exit() */

  #define MAX_CHAR 80

  void main ()
    {
    char  file_name[50] ;
    FILE* f_ptr ;
    char  buffer[MAX_CHAR] ;

    /* get a filename */
    printf ("enter a filename: ") ;
    gets (file_name) ;
```

```
/* open file */
if ((f_ptr = fopen (file_name, "r")) == 0)
  {
  printf ("sorry-can't open file \"%s\"\n", file_name) ;
  exit (EXIT_FAILURE) ;
  }

/* read and display file */
while (fgets (buffer, MAX_CHAR, f_ptr) != 0)
  printf ("%s", buffer) ;

/* close file */
fclose (f_ptr) ;
}
```

17.15.13 The *fputc()* Function

The *fputc()* function outputs a character c to the stream pointed to by the FILE structure pointer f_ptr:

```
int fputc (int c, FILE* f_ptr) ;
```

If *fputc()* is successful the character output is returned, else EOF is returned.

17.15.14 Writing to a Disk File

To illustrate the *fputc()* function, let's revisit the CPP_CPY.CPP program, which illustrated command line arguments in C++, by developing a program which operates similarly to the MS-DOS COPY command:

```
/*
c_cpy.c
illustrates the fputc() function
*/
#include <stdio.h>   /* C I/O    */
#include <stdlib.h>  /* exit()   */
#include <string.h>  /* strcmp() */

void main (int argc, char* argv[])
  {
  FILE* fin_ptr, * fout_ptr ;
  char  c ;

  /* check command-line arguments */
  if (argc != 3)
    {
    printf ("format: c_cpy src_file dst_file\n") ;
    exit (EXIT_FAILURE) ;
```

```
    }

  /* prevent self copy */
  if (strcmp (argv[1], argv[2]) == 0)
    {
    printf ("source file must be different from
             destination file\n") ;
    exit (EXIT_FAILURE) ;
    }

  /* open source and destination files */
  if ((fin_ptr = fopen (argv[1], "rb")) == 0)
    {
    printf ("sorry-can't open file \"%s\"\n", argv[1]) ;
    exit (EXIT_FAILURE) ;
    }
  if ((fout_ptr = fopen (argv[2], "wb")) == 0)
    {
    printf ("sorry-can't open file \"%s\"\n", argv[2]) ;
    exit (EXIT_FAILURE) ;
    }

  /* copy file */
  while ((c = fgetc (fin_ptr)) != EOF)
    fputc (c, fout_ptr) ;

  /* close files */
  fclose (fin_ptr) ;
  fclose (fout_ptr) ;
  }
```

with some user interaction:

```
C:\>c_cpy autoexec.bat autoexec.old
```

after which a copy of the file AUTOEXEC.BAT is made and written to AUTOEXEC.OLD

C_CPY.CPP first performs a check to confirm that the number of command line arguments is correct, and then uses the C++ library function *strcmp()* to confirm that the names of the source and destination files are different. The source and destination files are then opened for reading and writing, respectively. The contents of the source file are copied to the destination file using the *fputc()* function, after which both the opened source and destination files are closed.

17.15.15 The *fputs()* Function

The *fputs()* function outputs a string pointed to by `string` to the stream pointed to by the `FILE` structure pointer `f_ptr`:

```
int fputs (const char* string, FILE* f_ptr) ;
```

fputs() returns the last character written if successful, else EOF if unsuccessful. Note that *fputs()* does not append the new line character and does not copy the null- terminator. The following program modifies C_CPY.C to use the *fputs()* function instead of *fputc()*:

```
/*
c_cpy1.c
illustrates the fputs() function
*/
/* ... */
#define MAX_CHAR 80

void main (int argc, char* argv[])
  {
  FILE* fin_ptr, * fout_ptr ;
  char  buffer[MAX_CHAR] ;
  /* ... */
  while (fgets (buffer, MAX_CHAR, fin_ptr) != 0)
    fputs (buffer, fout_ptr) ;
  /* ... */
  }
```

17.15.16 The *fread()* and *fwrite()* Functions

The *fread()* and *fwrite()* functions allow a block of data (text or binary) to be read from or written to a file.

The *fread()* function reads n items, each of size size, from a file pointed to by f_ptr and places them in memory at a location pointed to by buffer:

```
size_t fread (void* buffer, size_t size, size_t n,
              FILE* f_ptr) ;  /* STDIO.H */
```

If *fread()* is successful the number of items read is returned, else a short count is returned if an error is encountered or end-of-file is reached.

In contrast to *fread()*, the *fwrite()* function writes n items, each of size size, from a memory location pointed to by buffer to a file pointed to by f_ptr:

```
size_t fwrite (const void* buffer, size_t size, size_t n,
               FILE* f_ptr) ;  /* STDIO.H */
```

If *fwrite()* is successful the number of items written is returned, else a short count is returned.

The following program helps to illustrate the *fread()* and *fwrite()* functions:

```
/*
frfw.c
illustrates the fread() and fwrite() functions
*/
#include <stdio.h>    /* C I/O */
#include <stdlib.h>   /* exit(), rand(), srand() */
#include <time.h>     /* time() */
```

```
#define MAX_NUM 10

void main ()
  {
  FILE*  f_ptr ;
  char   filename[] = "FRFW.TXT" ;
  int    rand_num, i ;
  time_t t ;

  /* fwrite(): */

  /* open file for writing */
  if ((f_ptr = fopen (filename, "wb")) == 0)
    {
    printf ("sorry-can't open file \"%s\"\n", filename) ;
    exit (EXIT_FAILURE) ;
    }

  /* write random numbers (0:99) to file */
  srand((unsigned)time (&t)) ;
  for (i=0; i<MAX_NUM; i++)
    {
    rand_num = rand () % 100 ;
    if ((fwrite (&rand_num, sizeof (int), 1, f_ptr)) == 0)
      {
      printf ("sorry-write error\n") ;
      exit (EXIT_FAILURE) ;
      }
    }

  /* close file */
  fclose (f_ptr) ;

  /* fread(): */

  /* open file for reading */
  if ((f_ptr = fopen (filename, "rb")) == 0)
    {
    printf ("sorry-can't open file \"%s\"\n", filename) ;
    exit (EXIT_FAILURE) ;
    }

  /* read and display random numbers from file */
  for (i=0; i<MAX_NUM; i++)
    {
    if ((fread (&rand_num, sizeof (int), 1, f_ptr)) == 0)
      {
      printf ("sorry-read error\n");
      exit (EXIT_FAILURE) ;
      }
    printf ("%2d\n", rand_num) ;
```

```
    }

  /* close file */
  fclose (f_ptr) ;
  }
```

The above program first opens a file for writing in binary mode called FRFW.TXT. Ten random numbers, between zero and 100, are then written one at a time using a **for**-loop to FRFW.TXT using the *fwrite()* function. FRFW.TXT is then closed for writing and reopened for reading in binary mode, when the random numbers previously written are read using *fread()* and displayed to the screen.

17.15.17 The *fprintf()* and *fscanf()* Functions

In a similar way to *printf()* and *scanf()*, the C file stream library includes the functions *fprintf()* and *fscanf()*, which write formatted output to a file stream and format input from a file stream, respectively. The declarations of *fprintf()* and *fscanf()* are to be found in STDIO.H and are of the form:

```
int fprintf (FILE* f_ptr, const char* format[, argument, ...]) ;

int fscanf  (FILE* f_ptr, const char* format[, address, ...]) ;
```

The operation of *fprintf()* and *fscanf()* is identical to that of *printf()* and *scanf()*, except that they operate on file streams rather than the standard input and output streams.

17.15.18 The *vfprintf()* and *vfscanf()* Functions

The functions *vfprint()* and *vfscanf()* write formatted output to and scan and format input from a stream, respectively, and are declared in the STDARG.H header file:

```
int vfprintf (FILE* f_ptr, const char* format,
                                   va_list arg_list) ;
int vfscanf  (FILE* f_prr, const char* format,
                                   va_list arg_list) ;
```

The *vfprintf()* and *vfscanf()* functions are similar to *fprintf()* and *fscanf()*, except that they accept a pointer to a list of arguments rather than a list of arguments.

17.15.19 Random File Access

When we discussed random file access in C++ we saw the use of objects of classes `basic_ifstream`, `basic_ofstream` and `basic_fstream` for manipulating file input, output, and input and output streams, respectively. For stream input the `basic_istream::`*seekg()* member function was used for moving to a specified position relative to the beginning, end or current pointer position of the stream, whereas the `basic_istream::`*tellg()* member function returns the current position of an input stream. In the case of stream output the *seekp()* and *tellp()* member functions of **class** `basic_`

ostream are used. Recall that basic_ifstream is derived from **class** basic_istream and that ofstream is derived from **class** basic_ostream.

Random file access in C is similar in principle to that discussed previously in C++, in that access is described in terms of a file pointer, except that functions rather than member functions are used to operate on both input and output file streams. The four main functions for performing random file access in C are described in the next subsection.

17.15.20 The *fseek()*, *ftell()*, *fgetpos()* and *fsetpos()* Functions

The key functions for performing random file access in C are *fseek()*, *ftell()*, *fgetpos()* and *fsetpos()*, which are all declared in STDIO.H:

```
int fseek (FILE* f_ptr, long offset, int whence) ;

long int ftell (FILE* f_ptr) ;

int fgetpos (FILE* f_ptr, fpos_t* pos) ;

int fsetpos (FILE* f_ptr, const fpos_t* pos) ;
```

where fpos_t is a **long typedef** which represents the file position type and is declared in STDIO.H.

The *fseek()* function sets the current file pointer in the file pointed to by f_ptr to offset bytes from the position specified in the whence argument. The whence argument can be one of the define identifiers listed in Table 17.12.

To set the file pointer position offset bytes from the start of a file, whence should be set to SEEK_SET, whereas SEEK_END sets the file pointer position offset bytes from the end of a file and SEEK_CUR sets the file pointer position offset bytes from the current position. The *fseek()* function returns zero if successful or EBADF, indicating a bad file pointer, EINVAL, indicating an invalid argument, or ESPIPE, indicating an illegal seek.

The *ftell()* function returns a **long int** value, which is the offset in bytes measured from the start of the file, specified by f_ptr, to the current file position. If *ftell()* is unsuccessful the value -1L is returned and the global variable errno is set to either EBADF, indicating a bad file pointer, or ESPIPE, which indicates an illegal seek.

The *fgetpos()* function retrieves the current file pointer position, pos, which corresponds to the file specified by f_ptr. If *fgetpos()* is successful zero is returned, else the global variable errno is set to either EBADF or EINVAL.

The *fsetpos()* function sets the current file pointer position to pos for the file specified by f_ptr. If *fsetpos()* is successful zero is returned, else errno is set to either EBADF or EINVAL.

The following program illustrates the *fseek()* and *fgetpos()* functions and is a C version of the program CGP.CPP:

Table 17.12 File pointer identifiers for *fseek()*.

whence	*Value*	*Meaning*
SEEK_SET	0	Beginning of file
SEEK_CUR	1	Current file pointer position
SEEK_END	2	End of file

```
/*
c_rnd_ac.c
illustrates random file access in C
using the fseek(), ftell(), fgetpos() and fsetpos() functions
*/
#include <stdio.h>   /* C I/O  */
#include <stdlib.h>  /* exit() */

#define MAX_CHAR 80

void main ()
  {
  FILE*  f_ptr ;
  char   filename[] = "C_RND_AC.C";
  char   buffer[MAX_CHAR] ;
  long   start, end, offset ;
  fpos_t position ;

  /* open file */
  if ((f_ptr = fopen (filename, "rb")) == 0)
    {
    printf ("sorry-can't open file \"%s\"\n", filename) ;
    exit (EXIT_FAILURE) ;
    }

  /* determine file length */
  fseek (f_ptr, 0, SEEK_SET) ; /* go to start */
  fgetpos (f_ptr, &position) ;
  start = position ;

  fseek (f_ptr, 0, SEEK_END) ; /* go to end */
  fgetpos (f_ptr, &position) ;
  end = position ;

  /* get an offset */
  printf ("enter an offset(+) from the beginning of file
         \"%s\"\nwhich has a file length of %d characters: ",
           filename, (end-start)) ;
  offset = atoi (gets (buffer)) ;

  /* go to offset */
  fseek (f_ptr, offset, SEEK_SET) ;

  /* display file to screen */
  while (fgets (buffer, MAX_CHAR, f_ptr) != 0)
    printf ("%s", buffer) ;

  /* close file */
  fclose (f_ptr) ;
  }
```

with some user interaction:

```
enter an offset(+) from the beginning of file "C_RND_AC.C"
which has a file length of 1159 characters: 1040
  hile (fgets (buffer, MAX_CHAR, f_ptr) != 0)
    printf ("%s", buffer) ;

  /* close file */
  fclose (f_ptr) ;
  }
```

17.15.21 The *feof()*, *ferror()*, *fflush()*, *flushall()*, *freopen()*, *clearerr()*, *remove()*, *rename()* and *rewind()* Functions

The C file stream library is very comprehensive, and as a result it is simply not possible to examine every function in detail. It is therefore recommended that you consult your compiler's documentation for a more complete discussion. However, there are a few C file stream functions that are frequently used and warrant a mention. All of the following functions are declared in the STDIO.H header file.

The *feof()* function tests the file stream specified by `f_ptr` for an end-of-file:

```
int feof (FILE* f_ptr) ;
```

If the end-of-file is reached a non-zero value is returned, else zero is returned. By using *feof()* the **while**-loop of C_RND_AC.C could be written as:

```
while (!feof (f_ptr))
    {
    fgets (buffer, MAX_CHAR, f_ptr) ;
    printf ("%s", buffer) ;
    }
```

The *ferror()* function tests the file stream specified by `f_ptr` for read and write errors:

```
int ferror (FILE* f_ptr) ;
```

ferror() returns a non-zero value if an error is detected. Once a stream error has been set it remains set until either *clearerr()* or *rewind()* is called. The functions *clearerr()* and *rewind()* have the following similar declarations:

```
void clearerr (FILE* f_ptr) ;

void rewind (FILE* f_ptr) ;
```

Both *clearerr()* and *rewind()* reset the error and end-of-file indicators to zero for the file stream specified by `f_ptr`, but *rewind()* also moves the file pointer position to the start of the file pointed to by `f_ptr`.

The function *fflush()* has the following declaration:

```
int fflush (FILE* f_ptr) ;
```

and flushes the file stream specified by `f_ptr` when the stream has buffered output. If *`fflush()`* is successful zero is returned, else `EOF` is returned. The similar function *`flushall()`*:

```
int flushall () ;
```

flushes all buffers corresponding to *open* input streams and writes buffered output for *open* output streams. *`flushall()`* returns the number of open streams.

The *`freopen()`* function associates a new file, `filename`, in place of an open stream, `stream`, and has the following declaration:

```
FILE* freopen (const char* filename,
               const char* mode, FILE* stream) ;
```

The new file is opened according to the value of `mode`, which may have the same values as for the *`fopen()`* function. If *`freopen()`* is successful the argument stream is returned, else `NULL` is returned if an error occurs. *`freopen()`* is useful for redirecting the standard streams `stdin`, `stdout` and `stderr` to a specified file.

The *`rename()`* function renames a file specified by `oldname` to the file specified by `newname`:

```
int rename (const char* oldname, const char* newname) ;
```

and returns zero if successful, else -1 with `errno` set to either `EACCES`, indicating that permission is denied, `ENONET`, indicating that the specified file or directory is non-existent, or `ENOTSAM`, indicating that the drives of the two files are different.

Finally, the *`remove()`* function deletes the file specified by `filename`:

```
int remove (const char* filename) ;
```

and returns zero if successful, else -1 with `errno` set to either `EACCES` or `ENOENT`.

17.16 A Device-Independent Bitmap `class` for Windows

Let's conclude this chapter by discussing a device-independent bitmap (DIB) **`class`** for the Windows operating system. The Windows bitmap file format is a relatively simple and general-purpose format for the storing of device-independent bitmaps, which are usually attributed either the BMP or DIB file extension. Owing to the popularity of the Windows operating system and acknowledging that most commercial graphics programs support the Windows bitmap file format, it is a useful format to examine.

There are several different bitmap file formats, of which the Windows DIB is just one. If you are interested in alternative formats or completely confused after my description of the DIB format, then an excellent source of further reading of the major bitmap file formats can be found in Luse (1993).

The following two subsections first examine the Windows bitmap file format and then discuss the implementation of the `DIBitmap` **`class`**.

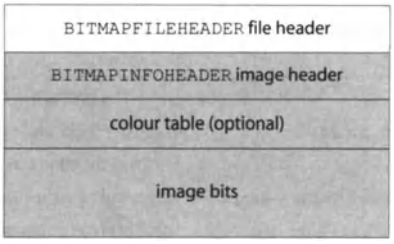


Fig. 17.5 The components of the Windows bitmap file format.

17.16.1 The Windows Device-Independent Bitmap File Format

The Windows bitmap file format generally consists of the following four components: (1) a file header defined by the `BITMAPFILEHEADER` structure; (2) an image header defined by the `BITMAPINFOHEADER` structure; (3) a colour table or a palette array; and (4) an array of bits that define the actual bitmap image. Figure 17.5 schematically illustrates the Windows bitmap file format. Contrary to the majority of bitmap file formats, the Windows bitmap file format stores the image bits from the bottom row to the top row, with each row beginning with the leftmost pixel.

The `BITMAPFILEHEADER` structure is declared in the WINDOWS.H/WINGDI.H header file and has five fields, which are listed in Table 17.13.

The `BITMAPFILEHEADER` structure is present at the start of all DIB files. The `bfType` field indicates whether or not the file format is compatible with that of a Windows bitmap file. The signature 'BM' (4D42h) identifies the file as a Windows bitmap file. The `bfSize` field indicates the total size of the bitmap in bytes. The fields `bfReserved1` and `bfReserved2` are reserved structure members and their values should be set to zero. The `bfOffBits` field holds the offset position, in bytes, from the beginning of the file to the start of the image bits.

The `BITMAPINFOHEADER` structure follows the `BITMAPFILEHEADER` structure and encapsulates the fields listed in Table 17.14, which describe the bitmap image.

The `biSize` field indicates the size of the `BITMAPINFOHEADER` structure and should be set to 40. `biWidth` and `biHeight` designate the width and height of the bitmap in pixels respectively. `biPlanes` is the number of colour planes of the bitmap image, and at present the Windows bitmap file format only supports single-plane bitmaps of either 1, 4, 8 or 24 bits per pixel. `biBitCount` is the number of colour bits per pixel and can be set to either 1, 4, 8 or 24. The `biCompression` field indicates the compression scheme used, if any. The Windows bitmap file format supports only the RLE image compression scheme, with only 4- and 8-bit images allowing compression. A value of 1 for `biCompression` indicates RLE-8 compression, and a value of 2 for `biCompression` indicates RLE-4 compression. Since compression is only supported for 4- and 8-bit images, and because the majority of applications do not compress DIB images, the `DIBitmap` **`class`** described below will ignore image compression.

Table 17.13 Fields of the `BITMAPFILEHEADER` structure.

Field	*Type*	*Description*
`bfType`	`DWORD` (**`unsigned long`**)	Signature 'BM' (bitmap)
`bfSize`	`DWORD`	File size in bytes
`bfReserved1`	`DWORD`	Set to 0
`bfReserved2`	`DWORD`	Set to 0
`bfOffBits`	`DWORD`	Offset to image bits from start of file

Table 17.14 Fields of the BITMAPINFOHEADER structure.

Field	*Type*	*Description*
biSize	DWORD	Size of this structure in bytes (should be 40)
biWidth	LONG (**signed long**)	Width of the bitmap in pixels
biHeight	LONG	Height of the bitmap in pixels
biPlanes	WORD (**unsigned short**)	Number of colour planes (set to 1)
biBitCount	WORD	Number of colour bits per pixel (1, 4, 8 or 24)
biCompression	DWORD	Compression scheme used
biSizeImage	DWORD	Size of the bitmap bits in bytes
biXPelsPerMeter	LONG	Horizontal resolution in pixels per metre
biYPelsPerMeter	LONG	Vertical resolution in pixels per metre
biClrUsed	DWORD	Number of colours in bitmap or set to zero
biClrImportant	DWORD	Number of colours in bitmap that are considered important or set to zero

biSizeImage indicates the size of the bitmap bits in bytes, but should generally only be used if compression is used because there are a number of applications which do not assign a value to biSizeImage. Also, bear in mind that each row of the bitmap bits is padded to the right with an integral number of double words, so that the number of bytes in each bitmap row is evenly divisible by 4. The two fields biXPelsPerMeter and biYPelsPerMeter indicate the horizontal and vertical resolution of the bitmap in pixels per metre. biClrUsed indicates the number of colours used in the bitmap. If biClrUsed is set to zero and the number of colour bits per pixel is either 1, 4 or 8 then the BITMAPINFOHEADER structure is followed by a colour table. Finally, the biClrImportant field is the number of colours considered important in the bitmap image.

An additional structure declared in the WINDOWS.H header file, which will prove useful later, is the BITMAPINFO structure, which encapsulates both a BITMAPINFOHEADER object field, bmiHeader, and a pointer to the RGBQUAD structure field, bmiColors:

```
typedef struct tagBITMAPINFO  // WINDOWS.H
  {
  BITMAPINFOHEADER bmiHeader ;
  RGBQUAD          bmiColors[1] ;
  } BITMAPINFO ;
```

The RGBQUAD structure encapsulates the four fields listed in Table 17.15. The rgbBlue, rgbGreen and rgbRed fields are each of type **unsigned char**, and respectively represent the blue, green and red primary intensities of a single colour as an 8-bit value. The rgbReserved field is an unused reserved data member.

So why the emphasis on device independence, and what is a colour table? Basically, the DIB format is referred to as *device-independent* because it incorporates a colour table or an array of palette entries. The colour table is a concise way of representing an RGB colour by assigning an integer value to a particular colour and then using the value as an index to the colour table rather than the RGB colour itself. If each bitmap image contains its own colour table the bitmap does not have to be compatible with the specification of a particular display device. In the case of a monochrome bitmap consisting of 1 colour bit per pixel, the most significant bit of each image byte is used as an index to the two-entry colour table. For a 16 colour bitmap consisting of four colour bits per pixel the four most significant bits (0:15) act as an index to the 16-entry colour table. For a 256 colour bitmap each byte corresponds exactly to a single pixel with a 256-

Table 17.15 Fields of the RGBQUAD structure.

Field	*Type*	*Description*
`rgbBlue`	`BYTE (unsigned char)`	Blue intensity (0:255)
`rgbGreen`	`BYTE`	Green intensity (0:255)
`rgbRed`	`BYTE`	Red intensity (0:255)
`rgbReserved`	`BYTE`	Reserved byte (set to zero)

Table 17.16 A typical colour table for a 16-colour bitmap.

Index	Red	Green	Blue	RGB colour
0	0	0	0	Black
1	128	0	0	Maroon
2	0	128	0	Dark green
3	128	128	0	Mustard
4	0	0	128	Navy blue
5	128	0	128	Purple
6	0	128	128	Olive
7	128	128	128	Dark grey
8	192	192	192	Grey
9	255	0	0	Red
10	0	255	0	Green
11	255	255	0	Yellow
12	0	0	255	Blue
13	255	0	255	Magenta
14	0	255	255	Cyan
15	255	255	255	White

entry colour table. In the case of a 24-bit colour bitmap a colour table is not required because each set of three bytes correspond exactly to an RGB colour, thus giving a total of 2^{24} (16 777 216) different colours. Table 17.16 illustrates a typical non-dithered colour table corresponding to a 16-colour bitmap.

17.16.2 The `DIBitmap` `class`

This section examines an implementation of a **`class`** called `DIBitmap` for the manipulation of bitmapped images in the Windows bitmap file format. Unless you are familiar with programming for Windows much of what follows may appear confusing, particularly the use of data type **`typedef`**s specific to Windows and the use of the Windows API functions. However, try to focus your attention on the design of the `DIBitmap` **`class`** and the reading and writing of data structures and bitmap bits from a disk file using a C approach. The **`class`** declaration of `DIBitmap` is listed below in DIB.H with corresponding implementation file DIB.CPP:

```
// dib.h
// device independent bitmap class for Windows
```

```
#ifndef _DIB_H  // prevent multiple includes
#define _DIB_H

#define STRICT  // Windows strict conformance

#include <windows>  // Windows
#include <mem>      // memcpy()
#include <cmath>    // log()

class DIBitmap
  {
  private:
    HANDLE          handle ;
    BYTE FAR*       bits ;
    BITMAPINFO FAR* bmp_info ;
    int             width ;
    int             height ;
    WORD            pixel_bits ;
    WORD            mode ;
  public:
    enum { ACTUAL, STRETCH } ;
    // constructors & destructor
    DIBitmap (const char* filename) ;
    DIBitmap (int w, int h, WORD bit_count,
              WORD m=DIB_RGB_COLORS) ;
    DIBitmap (const DIBitmap& dib) ;
    ~DIBitmap () ;
    // member functions
    int   Width () const { return width ; }
    int   Height () const { return height ; }
    DWORD NumberOfColours () const ;
    WORD  PixelBits () const { return pixel_bits ; }
    WORD  Mode () const { return mode ; }
    const BITMAPINFO FAR* BMPInfo () const
      { return bmp_info ; }
    BITMAPINFO FAR*       BMPInfo () { return bmp_info ; }
    const BITMAPINFOHEADER FAR* BMPInfoHeader () const
      { return &bmp_info->bmiHeader ; }
    BITMAPINFOHEADER FAR*       BMPInfoHeader ()
      { return &bmp_info->bmiHeader ; }
    const RGBQUAD FAR* BMPColours () const
      { return bmp_info->bmiColors ; }
    RGBQUAD FAR*       BMPColours ()
      { return bmp_info->bmiColors ; }
    RGBQUAD Colour (int index) const ;
    const BYTE FAR* Bits () const { return bits ; }
    BYTE FAR*       Bits () { return bits ; }
    HBITMAP HandleToBitmap (const HDC& hdc) ;
    void    DisplayBitmap (const HDC& hdc,
          int x0=0, int y0=0, int w=0, int h=0,
          WORD display_mode=DIBitmap::ACTUAL) ;
```

```
    BOOL    Write (const char* filename) ;
  };

#endif // _DIB_H
```

DIBitmap encapsulates seven **private** data members. handle indicates the global memory block in which the bitmap BITMAPINFOHEADER structure, colour table (if one is present) and the array of bits will be stored. It is important to note that the allocated global memory block is not just for the array of bits but is of sufficient size to hold the bitmap image header, colour table and array of bits. The bits data member is a far pointer to store a pointer to the bitmap bits. The bits pointer is a far pointer to BYTE because bitmap images are generally greater than 64 kbyte in size and consequently occupy more than one memory segment. bmp_info is a far pointer to the BITMAPINFO structure discussed above. The width and height of the bitmap are designated by the width and height data members, whereas pixel_bits indicates the number of colour bits per pixel (i.e. 1, 4, 8 or 24). The mode data member indicates the type of bitmap colour table. Various access member functions are defined to allow access to DIBitmap's data members, such as *Width()*, *Height()*, *PixelBits()*, *Mode()*, *BMPInfo()* and *Bits()*.

There are two global functions, *ImageColours()* and *ImageBits()*, defined in DIB.CPP which, respectively, return the number of image colours given the number of colour bits per pixel and, conversely, the number of colour bits per pixel given the number of colours:

```
DWORD ImageColours (WORD bit_count)
  {
  if (bit_count == 1 || bit_count == 4 || bit_count == 8)
    return 1 << bit_count ;
  else if (bit_count == 24)
    return 16777216L ;
  return 0L ;
  }

WORD ImageBits (DWORD num_colours)
  {
  if (num_colours == 2 || num_colours == 16 ||
      num_colours == 256 || num_colours == 16777216L)
    return (WORD)(log (num_colours) / log (2.0)) ;
  else
    return 0 ;
  }
```

Before examining the remaining member functions of DIBitmap, let's take a look at the two constructors, copy constructor and destructor of DIBitmap. The one-argument constructor defines a DIBitmap object given a bitmap path and filename:

```
DIBitmap::DIBitmap (const char* filename)
  : handle (0), bits (0), bmp_info (0),
    width (0), height (0), pixel_bits (0), mode (DIB_RGB_
COLORS)
  {
  // open bmp file
  HFILE hfile ;
```

```
if (HFILE_ERROR == (hfile = _lopen(filename, OF_READ)))
  {
  MessageBox (0,
              "can't open file","DIBitmap()",
              MB_ICONEXCLAMATION|MB_OK) ;
  return ;
  }
// read bmp header
BITMAPFILEHEADER bfh ;

if (_lread(hfile, (LPVOID)&bfh, sizeof bfh) != sizeof bfh)
  {
  MessageBox (0,
              "header file problem","DIBitmap()",
              MB_ICONEXCLAMATION|MB_OK) ;
  _lclose (hfile) ;
  return ;
  }
// check that bmp is a Windows bmp
if (bfh.bfType != 'BM')
  {
  MessageBox (0,
              "DIB is not of Windows format","DIBitmap()",
              MB_ICONEXCLAMATION|MB_OK) ;
  _lclose (hfile) ;
  return ;
  }
// check bmp info header size and width
DWORD info_header_size ;
BITMAPINFOHEADER bih ;
if (_lread(hfile, &info_header_size, sizeof info_header_size)
    != sizeof info_header_size
  && info_header_size != sizeof(BITMAPINFOHEADER))
  {
  MessageBox (0,
              "header size problem","DIBitmap()",
              MB_ICONEXCLAMATION|MB_OK) ;
  _lclose (hfile) ;
  return ;
  }
bih.biSize = info_header_size ;
if (_lread(hfile, &bih.biWidth,
  (int)info_header_size-sizeof(DWORD))
  != (int)info_header_size-sizeof(DWORD))
  {
  MessageBox (0,
              "header width problem","DIBitmap()",
              MB_ICONEXCLAMATION|MB_OK) ;
  _lclose (hfile) ;
  return ;
```

```
  }
// colour bits per pixel
pixel_bits = bih.biBitCount ;

// width and height
width  = (int)bih.biWidth ;
height = (int)bih.biHeight ;

// size of bits
int   bytes_per_row = ((width*bih.biBitCount+31)/32)*4 ;
DWORD bmp_size      = (DWORD)bytes_per_row * (DWORD)height ;
// 1, 4 or 8 bits
if (pixel_bits == 1 || pixel_bits == 4 || pixel_bits == 8)
  {
  // if biClrUsed=0 then a colour table is present
  if (!bih.biClrUsed)
    bih.biClrUsed = NumberOfColours () ;

  int colour_table_size = (int)NumberOfColours () *
                          sizeof (RGBQUAD) ;

  // if compression used
  if (bih.biCompression)
    bih.biSizeImage = bmp_size ;

  // handle
  handle = GlobalAlloc (GMEM_ZEROINIT|GMEM_MOVEABLE,
           bih.biSize+colour_table_size+bmp_size) ;
  if (!handle)
    {
    MessageBox (0,
                "handle problem","DIBitmap()",
                MB_ICONEXCLAMATION|MB_OK) ;
    return ;
    }
  bmp_info = (LPBITMAPINFO)GlobalLock (handle) ;
  bmp_info->bmiHeader = bih ;

  // get colour table
  if (_lread(hfile, (BYTE FAR*)bmp_info +
        (int)bih.biSize, colour_table_size)
    != colour_table_size)
    {
    MessageBox (0,
                "colour table problem","DIBitmap()",
                MB_ICONEXCLAMATION|MB_OK) ;
    _lclose (hfile) ;
    return ;
    }

  // get bits
```

```
    bits = (BYTE FAR*)bmp_info + (int)bih.biSize +
           colour_table_size ;
    _llseek (hfile, bfh.bfOffBits, 0) ;
    if (_hread(hfile, bits, bmp_size) != bmp_size)
      {
      MessageBox (0,
                  "reading bits problem","DIBitmap()",
                  MB_ICONEXCLAMATION|MB_OK) ;
      _lclose (hfile) ;
      return ;
      }
    // close file
    _lclose (hfile) ;
    }
  else  // 24 bit (assume no compression)
    {
    // handle
    handle = GlobalAlloc (GMEM_ZEROINIT|GMEM_MOVEABLE,
                          bih.biSize+bmp_size) ;
    if (!handle)
      {
      MessageBox (0,
                  "handle problem","DIBitmap()",
                  MB_ICONEXCLAMATION|MB_OK) ;
      return ;
      }
    bmp_info = (LPBITMAPINFO)GlobalLock (handle) ;
    bmp_info->bmiHeader = bih ;

    // get bits
    bits = (BYTE FAR*)bmp_info + (int)bih.biSize ;
    _llseek (hfile, bfh.bfOffBits, 0) ;
    if (_hread(hfile, bits, bmp_size) != bmp_size)
      {
      MessageBox (0,
                  "reading bits problem","DIBitmap()",
                  MB_ICONEXCLAMATION|MB_OK) ;
      _lclose (hfile) ;
      return ;
      }
    // close file
    _lclose (hfile) ;
    }
  } // DIBitmap()
```

The above constructor makes use of several Windows file-handling functions, declared in WINDOWS.H/WINBASE.H, which are prefixed with an underscore and differ slightly from those discussed in previous sections. The *_lopen()* function opens an existing file, `filename`, in a specified mode, `openmode`, and sets the current file pointer to the start of the file and has the following signature:

```
HFILE _lopen (LPCSTR filename, int openmode) ;
```

The return value of *_lopen()* is a file handle (HFILE (**typedef int**)) if the function is successful, else HFILE_ERROR. The different file modes are specified by #defines such as OF_READ, OF_WRITE, OF_READWRITE, etc.

The *_lread()*, *_lwrite()*, *_lclose()*, *_llseek()*, *_hread()* and *_lcreat()* are similar to their C counterparts *fread()*, *fwrite()*, *fclose()* and so on, but accept far pointers and can therefore be used with global memory blocks:

```
UINT  _lread  (HFILE fh, LPVOID buff, UINT buff_size) ;
UINT  _lwrite (HFILE fh, LPCSTR buff, UINT buff_size) ;
HFILE _lclose (HFILE fh) ;
LONG  _llseek (HFILE fh, LONG offset, int origin) ;
long  _hread  (HFILE fh, LPVOID buff, long buff_size) ;
HFILE _lcreat (LPCSTR buff, int attribute) ;
```

where LPVOID is a **void** FAR* **typedef**.

Returning now to the DIBitmap constructor listed above, the constructor begins by using the *_lopen()* function to open the specified bitmap file for reading. If the file is unsuccessfully opened then the API function *MessageBox()* is called to display an error message, and the constructor returns. A more suitable error-handling mechanism would be to use the C++ exception-handling mechanism, but as a first implementation we shall simply use the *MessageBox()* approach.

If the file is successfully opened it is sequentially read from the beginning of the file, first reading the BITMAPFILEHEADER structure, then the BITMAPINFOHEADER structure, followed by the colour table (if one is present for the cases of 1, 4 or 8 colour bits per pixel) and finally the array of image bits.

Once the BITMAPFILEHEADER structure has been read the bfType field is checked for the 'BM' signature. The BITMAPINFOHEADER biSize and biWidth fields are then read and checked. After the number of colour bits per pixel, bitmap width and height are set, the size of the bitmap is determined, ensuring that the bitmap row size is evenly divisible by 4.

Reading the remainder of the bitmap file depends on whether or not the bitmap has 24 colour bits per pixel. If the bitmap has either 1, 4 or 8 colour bits per pixel a colour table is present. The size of the colour table is the number of image colours multiplied by the size of a single RGBQUAD structure. Having calculated the size of the colour table a global memory block is allocated which is sufficient in size to accommodate the BITMAPINFOHEADER structure, the colour table and the array of image bits. A similar procedure is followed for the 24-bit case except that a colour table is not present.

The four-argument DIBitmap constructor allows a DIBitmap object to be defined which is of a given width, height, number of colour bits per pixel and mode, and has the following definition:

```
DIBitmap::DIBitmap (int w, int h, WORD bit_count, WORD m)
  : handle (0), bits (0), bmp_info (0),
    width (w), height (h), pixel_bits (bit_count), mode (m)
  {
  // DIB info header
  BITMAPINFOHEADER bih ;
  bih.biSize          = sizeof (BITMAPINFOHEADER) ;
  bih.biWidth         = width ;
  bih.biHeight        = height ;
```

```
bih.biPlanes        = 1 ;
bih.biBitCount      = pixel_bits ;
bih.biCompression   = BI_RGB ;
bih.biXPelsPerMeter = 0L ;
bih.biYPelsPerMeter = 0L ;
bih.biClrUsed       = NumberOfColours () ;
bih.biClrImportant  = 0L ;

// 1, 4 or 8 bits and 1 plane
if (pixel_bits == 1 || pixel_bits == 4 || pixel_bits == 8)
  {
  // remainder of DIB info header
  int bytes_per_row   = ((width*bih.biBitCount+31)/32)*4 ;
  bih.biSizeImage     = bytes_per_row * height ;
  // handle
  int   colour_table_size = (int)NumberOfColours () *
                            sizeof (RGBQUAD) ;
  DWORD file_size = bih.biSize + colour_table_size +
                            bih.biSizeImage ;
  handle = GlobalAlloc (GMEM_ZEROINIT|GMEM_MOVEABLE,
                        file_size) ;
  if (!handle)
    {
    MessageBox (0,
                "handle problem","DIBitmap()",
                MB_ICONEXCLAMATION|MB_OK) ;
    return ;
    }
  bmp_info = (LPBITMAPINFO)GlobalLock (handle) ;
  bmp_info->bmiHeader = bih ;

  // get system palette
  if ((GetDeviceCaps(GetDC(0), RASTERCAPS) & RC_PALETTE)
              || (GetSystemPaletteEntries (GetDC(0), 0,
                (UINT)NumberOfColours(),
                (LPPALETTEENTRY)bmp_info->bmiColors) == 0))
    {
    MessageBox (0,
                "palette problem","DIBitmap()",
                MB_ICONEXCLAMATION|MB_OK) ;
    return ;
    }
  for (int i=0; i<NumberOfColours(); i++)
    bmp_info->bmiColors[i].rgbReserved = 0 ;

  // bits
  bits = (BYTE FAR*)bmp_info +
         (int)bmp_info->bmiHeader.biSize +
         colour_table_size ;
  }
else if (pixel_bits == 24)
```

```
    {
    // remainder of DIB info header
    bih.biSizeImage     = 3L * width * height ;

    // handle (no colour table for 24 bit image)
    DWORD file_size = bih.biSize + bih.biSizeImage ;
    handle = GlobalAlloc (GMEM_ZEROINIT|GMEM_MOVEABLE,
                          file_size) ;
    if (!handle)
      {
      MessageBox (0,
                  "handle problem","DIBitmap()",
                  MB_ICONEXCLAMATION|MB_OK) ;
      return ;
      }
    bmp_info = (LPBITMAPINFO)GlobalLock (handle) ;
    bmp_info->bmiHeader = bih ;

    // bits
    bits = (BYTE FAR*)bmp_info +
           (int)bmp_info->bmiHeader.biSize ;
    }
  else
    {
    MessageBox (0,
                "DIB should be 1, 4, 8 or 24 bits",
                "DIBitmap()",
                MB_ICONEXCLAMATION|MB_OK) ;
    return ;
    }
  } // DIBitmap()
```

The above constructor begins by filling in as many fields of the `BITMAPINFOHEADER` as possible. If the bitmap is a 24-bit image a global memory block is allocated and the far pointer to the `BITMAPINFO` structure data member, `bmp_info`, and the array of bits, `bits`, are set. In the case of either a 1-, 4- or 8-bit image the colour table has to be defined. The Windows API function *`GetSystemPaletteEntries()`* is used to retrieve a selection of palette entries from the system palette. The API function *`GetDeviceCaps()`* is also used to confirm that the device is palette based.

The majority of the `DIBitmap` copy constructor is self-explanatory, but note the use of the C-style *`memcpy()`* function which copies from a source buffer to a destination buffer, which was allocated using the *`GlobalAlloc()`* function. The `DIBitmap` destructor first unlocks the global memory block associated with the `handle` data member using the *`GlobalUnlock()`* function, and then proceeds to deallocate the global memory block using the *`GlobalFree()`* function. The `DIBitmap` copy constructor and destructor definitions are listed below:

```
DIBitmap::DIBitmap (const DIBitmap& dib)
  : handle (0), bits (0), bmp_info (0),
    width (dib.width), height (dib.height),
    pixel_bits (dib.pixel_bits), mode (dib.mode)
```

```
    {
    handle = GlobalAlloc (GMEM_ZEROINIT|GMEM_MOVEABLE,
                          GlobalSize(dib.handle)) ;
    if (!handle)
      MessageBox (0,
                  "handle problem","DIBitmap()",
                  MB_ICONEXCLAMATION|MB_OK) ;

    bmp_info = (LPBITMAPINFO)GlobalLock (handle) ;
    memcpy (bmp_info, dib.bmp_info, GlobalSize(dib.handle)) ;

    if (pixel_bits == 1 || pixel_bits == 4 || pixel_bits == 8)
      bits = (BYTE FAR*)bmp_info +
           (int)bmp_info->bmiHeader.biSize +
           (int)NumberOfColours()*sizeof(RGBQUAD) ;
    else  // 24 bit
      bits = (BYTE FAR*)bmp_info +
             (int)bmp_info->bmiHeader.biSize ;
    }

  DIBitmap::~DIBitmap ()
    {
    if (handle)
      {
      GlobalUnlock (handle) ;
      GlobalFree (handle) ;
      }
    }
```

The *NumberOfColours()* member function simply returns the number of image colours using the global function *ImageColours()*:

```
  DWORD DIBitmap::NumberOfColours () const
    {
    return ImageColours (pixel_bits) ;
    }
```

DIBitmap::*Colour()* returns an RGBQUAD colour object corresponding to the specified colour table index:

```
  RGBQUAD DIBitmap::Colour (int index) const
    {
    RGBQUAD rgbq ;
    if (pixel_bits != 24 &&
          index >= 0 && index < NumberOfColours())
      rgbq = BMPColours()[index] ;
    else
      {
      rgbq.rgbRed = 0 ; rgbq.rgbGreen = 0 ; rgbq.rgbBlue = 0 ;
      rgbq.rgbReserved = 0 ;
      }
```

```
    return rgbq ;
    }
```

DIBitmap::*HandleToBitmap()* uses the Windows API function *CreateDIBitmap()* to return a handle to a bitmap, HBITMAP, given a handle to a device context:

```
HBITMAP DIBitmap::HandleToBitmap (const HDC& hdc)
  {
  return CreateDIBitmap (hdc,
                         this->BMPInfoHeader(),
                         CBM_INIT,
                         (const void FAR*)this->Bits(),
                         this->BMPInfo(),
                         this->Mode()) ;
  }
```

The signature of the *CreateDIBitmap()* function is:

```
HBITMAP CreateDIBitmap (HDC hdc,
                        BITMAPINFOHEADER FAR* bih,
                        DWORD initialiser,
                        const void FAR* bits,
                        BITMAPINFO FAR* bi,
                        UINT mode) ;
```

The *DisplayBitmap()* member function displays a DIBitmap object to a specified device context, hdc:

```
void DIBitmap::DisplayBitmap (const HDC& hdc, int x0, int y0,
                              int w, int h, WORD display_mode)
  {
  if (pixel_bits == 1 || pixel_bits == 4 || pixel_bits == 8)
    {
    // stretch or compress
    if (display_mode == DIBitmap::STRETCH)
      {
      SetStretchBltMode (hdc, COLORONCOLOR) ;
      StretchDIBits (hdc,
                     x0, y0, w, h,
                     0, 0, this->width, this->height,
                     (const void FAR*)this->Bits(),
                     this->BMPInfo(),
                     this->Mode(),
                     SRCCOPY) ;
      }
    // display actual size
    else
      {
      HDC     hdcMem ;
      POINT   bmp_size, bmp_origin ;
```

```
        bmp_origin.x = 0 ; bmp_origin.y = 0 ;

        // create memory DC and select bitmap object
        hdcMem = CreateCompatibleDC (hdc) ;
        SelectObject (hdcMem, this->HandleToBitmap(hdc)) ;
        SetMapMode (hdcMem, GetMapMode(hdc)) ;

        bmp_size.x = width ; bmp_size.y = height ;

        // convert device to logical coordinates
        DPtoLP (hdc, &bmp_size, 1) ;
        DPtoLP (hdc, &bmp_origin, 1) ;

        BitBlt (hdc, x0, y0, bmp_size.x, bmp_size.y,
                hdcMem, bmp_origin.x, bmp_origin.y,
                SRCCOPY) ;

        DeleteDC (hdcMem) ;
        }
      }
    else  // 24 bit (display actual size)
      {
      // row-wise scan
      for (int j=0; j<height; j++)
        {
        for (int i=0; i<width; i++)
          {
          // note orientation of screen coor.
          SetPixel (hdc, i, (height-j),
            RGB(bits[3*((DWORD)j*width+i)],
            bits[3*((DWORD)j*width+i)+1],
            bits[3*((DWORD)j*width+i)+2]));
          }
        }
      }
    } // DisplayBitmap()
```

DisplayBitmap() will display a 24-bit image using the original dimensions of the image or display a 1-, 4- or 8-bit image using either the original dimensions of the image or by stretching the image to fit the display window. In the case of 1-, 4- or 8-bit images the origin of the image can also be translated to a point (x0, y0) from the window origin. An image is stretched using the Windows API function *StretchDIBits()*:

```
int StretchDIBits (HDC hdc,
                   int xdest, int ydest,
                   int wdest, int hdest,
                   int xsrc, int ysrc, int wsrc, int hsrc,
                   const void FAR* bits,
                   BITMAPINFO FAR* bi,
                   UINT mode,
                   DWORD rop) ;
```

For the 1-, 4- or 8-bit images an image could have been displayed actual size using the API function *SetDIBitsToDevice()*, but an alternative approach has been taken which is based on the *DrawBitmap()* function presented in Petzold (1992, p. 631). First, a memory device context is created using *CreateCompatibleDC()*, and the bitmap image is then selected into the memory device context using *SelectObject()*. The mapping mode is then set to the same as that of the display device context. Both the origin and size of the bitmap coordinates are then converted from device coordinates to logical coordinates and the *BitBlt()* function is called to copy the bitmap from the memory device context to the display device context.

For a 24-bit image it is a simple matter of using the Windows API function *SetPixel()* to set each pixel in the display window to the specified RGB colour. Care is required in ensuring that the RGB components are accessed correctly from the `DIBitmap::bits` array and that the image is oriented correctly due to the display window origin being centred at the top left corner rather than the bottom left corner. The signature of *SetPixel()* is:

```
COLORREF SetPixel (HDC hdc, int x, int y, COLORREF colour) ;
```

where `COLORREF` is a `DWORD` **typedef**.

The definition of the `DIBitmap::`*Write()* function is:

```
BOOL DIBitmap::Write (const char* filename)
  {
  // open bmp file for writing
  HFILE hfile ;
  if (HFILE_ERROR == (hfile = _lopen(filename, OF_WRITE)))
    {
    if (HFILE_ERROR == (hfile = _lcreat(filename, 0)))
      {
      MessageBox (0,
        "DIBitmap::Write()","can't open/create file",
        MB_ICONEXCLAMATION|MB_OK) ;
      return FALSE ;
      }
    }

  // approx size
  DWORD size = GlobalSize (handle) ;

  BITMAPFILEHEADER bfh ;
  bfh.bfType      = 'BM' ;
  bfh.bfSize      = sizeof bfh + size ;
  bfh.bfReserved1 = 0 ;
  bfh.bfReserved2 = 0 ;
  bfh.bfOffBits   = sizeof bfh + (BYTE FAR*)bits -
                    (BYTE FAR*)bmp_info ;

  // write headers and bits to file
  if ((_hwrite(hfile,
      (LPCSTR)&bfh, sizeof bfh) != sizeof bfh) ||
      (_hwrite(hfile, (LPCSTR)bmp_info, size) != size))
    return FALSE ;
```

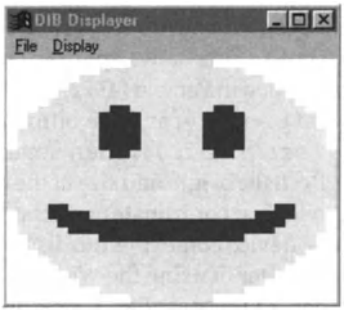


Fig. 17.6 The DIB Displayer main window displaying the SMILE.BMP bitmap.

```
// close file
_lclose (hfile) ;
return TRUE ;
} // Write()
```

Note the use of the Windows API *_hwrite()* function for the writing memory blocks greater than 64 kbyte to a disk file:

```
long _hwrite (HFILE hf, LPCSTR buffer, long buffer_size) ;
```

The `DIBitmap` **class** presented above is tested using a simple Windows DIB file format displayer program called DIB Displayer. A typical example of DIB Displayer displaying, stretched to fit the window, the SMILE.BMP 32 × 32 16-colour bitmap (which was generated using the Borland Resource Workshop bitmap editor) is shown in Fig. 17.6.

The DIB Displayer program is a minimal Windows program consisting of a single-document interface and two pull-down menus, File and Display. File | Open displays the *File Open* dialog box from the Windows common dialog box library to extract the path and filename of a bitmap image. The one-argument constructor of `DIBitmap` is then used to create a pointer, `bmp`, to a `DIBitmap` object. File | Exit simply closes the DIB Displayer application. Display | Actual.size displays the bitmap actual size while Display | Stretch.to.window stretches the bitmap to the size of DIB Displayer's client area.

The Windows main program function, *WinMain()*, and window procedure function, *WndProc()*, are listed in DIB_TST.CPP:

```
// dib_tst.cpp
// a DIB displayer for Windows

#define STRICT  // Windows strict conformance

#include <windows>  // Windows
#include <commdlg>  // Windows common dialog
#include <stdlib>   // _MAX_PATH,...

#include "dib.h"       // DIBitmap class
#include "min_max.h"   // Min() & Max() template functions
#include "dib_tst.rh" // resource script header file
```

```
long FAR PASCAL _export WndProc (HWND, UINT, UINT, LONG) ;

char szAppName [] = "DIBDisplayer" ;

int PASCAL WinMain (HINSTANCE hInstance,
                    HINSTANCE hPrevInstance,
                    LPSTR lpszCmdLine, int nCmdShow)
  {
  WNDCLASS wndclass ;
  if (!hPrevInstance)
    {
    wndclass.style         = CS_HREDRAW|CS_VREDRAW ;
    wndclass.lpfnWndProc   = WndProc ;
    wndclass.cbClsExtra    = 0 ;
    wndclass.cbWndExtra    = 0 ;
    wndclass.hInstance     = hInstance ;
    wndclass.hIcon         = LoadIcon (hInstance, szAppName) ;
    wndclass.hCursor       = LoadCursor (0, IDC_ARROW) ;
    wndclass.hbrBackground = (HBRUSH)GetStockObject
                                          (WHITE_BRUSH) ;
    wndclass.lpszMenuName  = (char*)MAIN_WINDOW_MENU ;
    wndclass.lpszClassName = szAppName ;

    RegisterClass (&wndclass) ;
    }
  HWND hwnd = CreateWindow (szAppName, "DIB Displayer",
                            WS_OVERLAPPEDWINDOW,
                            CW_USEDEFAULT, CW_USEDEFAULT,
                            CW_USEDEFAULT, CW_USEDEFAULT,
                            0, 0, hInstance, 0) ;

  ShowWindow (hwnd, nCmdShow) ;
  UpdateWindow (hwnd) ;

  MSG msg ;
  while (GetMessage (&msg, 0, 0, 0))
    {
    TranslateMessage (&msg) ;
    DispatchMessage (&msg) ;
    }
  return msg.wParam ;
  } // WinMain()

long FAR PASCAL _export WndProc (HWND hwnd, UINT message,
                                 UINT wParam, LONG lParam)
  {
  static char  szFileName  [_MAX_PATH],
               szTitleName [_MAX_FNAME+_MAX_EXT] ;
  static char* szFilter[] = { "Bitmap files (*.bmp)",
                              "*.bmp" } ;
```

```
static DIBitmap* bmp (0) ;
static OPENFILENAME ofn ;
static short        cxClient, cyClient ;
static WORD         wDisplay = CM_DISPLAYACTUAL ;
HDC                 hdc ;
HMENU               hMenu ;
PAINTSTRUCT         ps ;

switch (message)
  {
  case WM_CREATE:
    ofn.lStructSize       = sizeof (OPENFILENAME) ;
    ofn.hwndOwner         = hwnd ;
    ofn.lpstrFilter       = szFilter[0] ;
    ofn.lpstrFile         = szFileName ;
    ofn.nMaxFile          = _MAX_PATH ;
    ofn.lpstrFileTitle    = szTitleName ;
    ofn.nMaxFileTitle     = _MAX_FNAME + _MAX_EXT ;
    ofn.lpstrDefExt       = "bmp" ;
    return 0 ;

  case WM_SIZE:
    cxClient = LOWORD (lParam) ;
    cyClient = HIWORD (lParam) ;
    return 0 ;

  case WM_COMMAND:
    hMenu = GetMenu (hwnd) ;

    switch (wParam)
        {
        case CM_FILEOPEN:
          if (GetOpenFileName(&ofn))
            {
            if (bmp)
              delete bmp ;
            bmp = new DIBitmap
              (szFileName) ;

            if (bmp == 0)
              MessageBox (hwnd,
                          szAppName,
                          "can't create DIB object",
                          MB_ICONEXCLAMATION|MB_OK) ;
            InvalidateRect (hwnd, 0, TRUE) ;
            }
          return 0 ;

        case CM_DISPLAYACTUAL:
        case CM_DISPLAYSTRETCH:
          CheckMenuItem (hMenu, wDisplay,
```

```
                                MF_UNCHECKED) ;
            wDisplay = wParam ;
            CheckMenuItem (hMenu, wDisplay,
                                MF_CHECKED) ;
            InvalidateRect (hwnd, 0, TRUE) ;
            return 0 ;

          case CM_FILEEXIT:
            SendMessage (hwnd, WM_CLOSE, 0, 0L) ;
            return 0 ;
          }
          break ;

        case WM_PAINT:
          hdc = BeginPaint (hwnd, &ps) ;
          // display dib
          if (bmp)
            {
            if (wDisplay == CM_DISPLAYACTUAL)
              bmp->DisplayBitmap (hdc) ;
            else
              bmp->DisplayBitmap (hdc,
                                  0, 0,
                                  cxClient, cyClient,
                                  DIBitmap::STRETCH) ;
             }
          EndPaint (hwnd, &ps) ;
          return 0 ;

        case WM_DESTROY:
          if (bmp)
            delete bmp ;

          PostQuitMessage (0) ;
            return 0 ;
        }
          return DefWindowProc (hwnd, message, wParam, lParam) ;
      } // WndProc()
```

DIB Displayer's main menu is implemented in the resource script file DIB_TST.RC, with identifiers listed in DIB_TST.RH:

```
// dib_tst.rc
// resource script file

#include "dib_tst.rh"

DIBDisplayer ICON "dib_tst.ico"

MAIN_WINDOW_MENU MENU
  {
```

```
  POPUP "&File"
    {
    MENUITEM "&Open...", CM_FILEOPEN
    MENUITEM SEPARATOR
    MENUITEM "E&xit", CM_FILEEXIT
    }
  POPUP "&Display"
    {
    MENUITEM "&Actual size", CM_DISPLAYACTUAL, CHECKED
    MENUITEM "Stretch to window", CM_DISPLAYSTRETCH
    }
  }

// dib_tst.rh
// resource script header file

#define MAIN_WINDOW_MENU  101

#define CM_FILEOPEN       102
#define CM_FILEEXIT       103
#define CM_DISPLAYSTRETCH 104
#define CM_DISPLAYACTUAL  105
```

The Windows module definition file DIB_TST.DEF is:

```
; DIB_TST.DEF
; module definition file

NAME           DIB_TST

DESCRIPTION    'DIB Displayer for Windows by G. M. Seed,
2000'
EXETYPE        WINDOWS
STUB           'C:\BC5\BIN\WINSTUB.EXE'
CODE           PRELOAD MOVEABLE DISCARDABLE
DATA           PRELOAD MOVEABLE MULTIPLE
HEAPSIZE       1000000
STACKSIZE      1000000
```

It is beyond the scope of this book to describe in detail the implementation of DIB_TST.CPP, DIB_TST.RC and DIB_TST.DEF, but the interested reader is referred to Petzold (1992).

17.17 Summary

You may have gathered from this chapter that there are a great deal of C++ stream classes and associated member functions and C stream functions. The C and C++ stream libraries are complex, and it does take time before you will feel reasonably comfortable with them. However, the C++ and C stream libraries are very powerful, and because of the importance and necessity

attached to input and output it is well worth spending some time and effort in developing a good understanding of the libraries.

Of the many classes in the complex C++ standard stream **class** library, the most frequently used are `basic_istream`, `basic_ostream` and `basic_iostream` and the file stream classes `basic_ifstream`, `basic_ofstream` and `basic_fstream`.

C++ predefines four open streams for use with the standard input and output streams, namely `cin`, `cout`, `cerr` and `clog`. Similar open streams are predefined for **`wchar_t`**. Stream input and output can be formatted either by using manipulators or member functions of **class** `ios_base`. Manipulators can be either parametrised or non-parametrised and can be overloaded and defined for user-defined classes.

The insertion and extraction operators are overloaded for all of the C++ integral data types and can be overloaded in a device-independent manner for user-defined classes. Provided the insertion and extraction operators are overloaded in a device-independent manner there is no difference between the manipulation of standard stream objects and file stream objects.

The C++ standard stream library provides a programmer with excellent support for the handling of file streams.

This chapter discussed classes called `ReadFile` , `WriteFile` and `ReadAndWriteFile` for reading and writing a collection of objects from and to an object data file. These classes illustrated just how powerful an object-oriented approach to file handling can be.

We have seen the use of redirection and command line arguments. Redirection allows the standard input and output streams to be redirected to an alternative device. Command line arguments allow a programmer to pass information to a program before the program commences execution.

Although not object-oriented, C's approach to the handling of streams is nevertheless extremely powerful and popular, and this chapter has presented an overview of the C file-handling mechanism.

The chapter concluded with a discussion of a Windows device-independent bitmap file format **class** called `DIBitmap`. The `DIBitmap` **class** and its associated member functions greatly assists the handling of bitmap images stored in a DIB file format.

Exercises

17.1 Write a program that displays the first 20 lines of a user-specified file.

17.2 For the following `Vector` **class**:

```
class Vector
  {
  private:
    double a, b, c ;
  public:
    Vector (double _a, double _b, double _c)
      : a (_a), b (_b), c (_c) {}
    friend ostream& operator << (ostream& s,
                                 const Vector& v) ;
  };
```

define a one-argument parametrised manipulator called `VectorPrecision` using **class** `smanip` so that a `Vector` object can be output to a specified precision:

```
Vector v ;
//...
cout << "v: " << VectorPrecision (3) << v << endl ;
```

17.3 Write a program that reads an arbitrary number of integers from a file that are in the following format:

```
6
5
4
2
4
1
```

and subsequently displays the numbers to the display screen.

17.4 The set of numbers of Exercise 17.3 is assumed to represent the numbers that turned up on a die. Write a program which reads the data file of Exercise 17.3 and then displays the mean and standard deviation of the numbers on the display screen. The mean, $\bar{x}$, and variance, s^2, of a sample $x_1, x_2, ..., x_n$ are defined as:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

The square root of the variance is the standard deviation.

17.5 Write a program which uses command line arguments to display the current time or date, with both the time and date displayed by default. Use the C++ *time()* and *ctime()* library functions:

```
time_t time (time_t* timeptr) ;
char* ctime (const time_t* timeptr) ;
```

both declared in TIME.H. The *ctime()* function returns a pointer to a string which contains 26 characters of the following form:

```
Fri Nov 24 15:59:08 2000
```

assuming a 24-hour clock.

17.6 Write the following to a file called 17_06.DAT using the C *fprintf()* function:

```
5
15.15
stream
```

17.7 Rewrite Exercise 17.3 using C functions.

Chapter 18

The Preprocessor

This chapter describes the preprocessor directives. Although not strictly part of the C++ language, the preprocessor directives, particularly the #include directive, are frequently used when programming in C++.

The preprocessor is increasingly becoming redundant because of equivalent C++ language-based features. The const keyword allows constant identifiers to be defined. Macros are replaced by inline functions, with the template feature defining and declaring type-independent functions and classes. We shall see in the next chapter that the namespace feature associates a name to a given scope. All of const, inline, template and namespace are statically type-checked, whereas the preprocessor bypasses the static type checking mechanism of C++.

18.1 The Preprocessor

This chapter examines the preprocessor, which C++ inherited from the C language. C++ programmers do not like the preprocessor. In fact, use of the preprocessor is strongly discouraged by the original designer of C++ (Stroustrup, 1994). The key problems with the preprocessor are that it is not part of the C++ programming language, it is not object-oriented, it is unstructured and it bypasses the strong type-checking mechanism of C++.

The compilation of a program generally involves several different phases or passes. The first pass on a program is the preprocessor pass. The second pass is generally a syntax and semantic pass, which is followed by a token and a code optimisation pass. This chapter is concerned with the first pass and the preprocessor's associated directives.

18.2 Preprocessor Directives

To date, we have seen various preprocessor directives used, such as `#include`, `#define`, `#ifndef` and `#endif`. A complete list of the preprocessor directives is given in Table 18.1. Each preprocessor directive is prefixed by the number sign, #, which must be the first non-

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Table 18.1 Preprocessor directives.

Preprocessor directive	*Meaning*
#	Null directive
`#define`	Defines a symbolic constant or macro
`#error`	Stop compilation and generate an error message
`#include`	Include files in source code
`#line`	Line numbers
`#pragma`	Implementation-specific directive
`#undef`	Undefine previously defined macro *conditional directives*
Conditional directives	
`#elif`	Else if
`#else`	Else
`#endif`	End of if
`#if`	Conditional if
`#ifdef`	If defined
`#ifndef`	If not defined

whitespace character and must be placed on its own line. Preprocessor directives can be placed anywhere within a program but are usually placed at the start of a program.

We shall examine each of the preprocessor directives in more detail in the following subsections.

18.2.1 The # Null Directive

The operator # on a single line is the null directive and is ignored by the preprocessor.

18.2.2 The `#define` Directive

The `#define` preprocessor directive defines a symbolic constant. For example, the following `#define` has been extracted from the C++ standard header file MATH.H:

```
#define M_PI 3.14159265358979323846  // MATH.H
```

and associates the symbolic constant `M_PI` with π. Note that no semicolon is specified to terminate a preprocessor directive and that no assignment operator is used to assign the approximate value of π to the `#define M_PI`. It was mentioned above that a preprocessor directive must be placed on its own line, and consequently a directive terminates when a new line is encountered.

`#define` can also be used to define a parametrised macro:

```
#define MIN(a, b) (((a) < (b)) ? (a) : (b))
```

which defines a parametrised macro, *`MIN()`*, that determines whether `a` is less than `b`. Note that white space between the macro name and the open parenthesis '`(`' is not allowed.

The general syntax of `#define` is:

```
#define MacroName <token_sequence>
```

Each occurrence of `MacroName` in a program will be identically replaced by the optional (between < and >) *body* of the parametrised macro `token_sequence`. Such a process of replacement is generally referred to as *macro replacement* or *macro expansion*. Because, by definition, a preprocessor directive must be placed on its own line, the backslash character (\) is used for token sequences that extend onto a new line:

```
#define MEM_ERROR "Insufficient memory\
                   available"
```

The `token_sequence` can be null:

```
#define EMPTY
```

Following a suitable macro definition the macro name can be used to define subsequent macros or form part of expressions:

```
#define PI_OVER2 M_PI / 2
```

You may have noticed an excessive amount of parentheses in the `#define` *MIN()* macro above. To illustrate why each macro argument is enclosed in parentheses, consider a macro called *SQUARE()* which simply multiplies x by itself:

```
#define SQUARE(x) ((x) * (x))
```

Consider using *SQUARE()* to find the square of an expression:

```
int sq = SQUARE (1+1) ;
```

This statement will be expanded by the preprocessor to:

```
int sq = ((1+1) * (1+1)) ;
```

which will, as expected, assign the value of four to `sq`.

Consider now the case when the parentheses are neglected from the definition of *SQUARE()*:

```
#define SQUARE(x) x * x
```

Now, *SQUARE(1+1)* will be expanded as:

```
int sq = 1 + 1 * 1 + 1 ;
```

which assigns the value of three, and not four, to `sq` because the * operator has a higher precedence than the + operator. The above example illustrates that a macro is not expanded during its own expansion. Therefore, always remember to use parentheses in the definition of parametrised macros.

Macro names cannot be overloaded and parameterised macros do not support recursive calls:

```
#define MIN(a,b) (((a) < (b)) ? (a) : (b)
#define MIN(a,b,c) /* ... */              // error: overloaded
#define FACT(n) (n>1) ? n*FACT(n-1) : 1 // error: recursive
```

It is important to note that in the above `#defines` the type of `M_PI` or `a` and `b` in *`MIN()`* was not declared. You may be thinking that defining a type-independent `#define` is great, but in fact it is `#define`'s greatest weakness because it bypasses the type-checking mechanism of the compiler – which means that you're on your own when it comes to weird and wonderful compilation and run-time errors.

Since a macro is expanded inline there is no associated function call overhead. Thus an increase in performance can be experienced by using a macro instead of a function call. Remember though, as with **`inline`** functions, that because a macro is expanded inline any increase in performance can quickly be lost if the macro is large in size or is called a large number of times. Also, if a macro is called a large number of times the resultant size of the program's executable file will be significantly greater than if a function call had been used instead.

The following program illustrates the various properties of `#define` discussed above:

```
// define.cpp
// illustrates the #define preprocessor directive
#include <iostream>   // C++ I/O
#include <cmath>      // M_PI

// common errors: '=' and ';'
#define GEGA = 1e09
#define GMS "Geometric Modelling Society" ;

#
#define MIN(a, b) (((a) < (b)) ? (a) : (b))
#define EMPTY
#define PI_OVER2 M_PI / 2
#define SQUARE_WITHOUT(x) x * x
#define SQUARE(x) ((x) * (x))
#define CUBE(x) (SQUARE (x) *\
                 (x))
void main ()
  {
  cout << "MIN(1,2): "      << MIN (1, 2)      << endl
       << "MIN(1.1,2.2): " << MIN (1.1, 2.2) << endl
       << "MIN('a','b'): " << MIN ('a', 'b') << endl
       << "M_PI: "          << M_PI           << endl
       << "PI_OVER2: "      << PI_OVER2       << endl
       << "SQUARE_WITHOUT(1+1): "
                  << SQUARE_WITHOUT (1+1)  << endl
       << "SQUARE(1+1): "   << SQUARE (1+1)   << endl
       << "CUBE(2): "       << CUBE (2)       << endl ;
  }
```

with output:

```
MIN(1,2): 1
MIN(1.1,2.2): 1.1
MIN('a','b'): a
M_PI: 3.14159
PI_OVER2: 1.5708
SQUARE_WITHOUT(1+1): 3
```

```
SQUARE(1+1): 4
CUBE(2): 8
```

C++ provides a language-based alternative for each of `#define`'s main uses. C++ offers the keyword **`const`** for constant identifiers, the keyword **`inline`** for functions expanded inline and the keyword **`template`** for parametrised function classes or types. Also, the C++ **`namespace`** feature discussed in the next chapter allows a name to be associated with a given scope. Such language-based features as **`const`**, **`inline`**, **`template`** and **`namespace`** make `#define` redundant:

```
// cpp_ver.cpp
// a C++ version of DEFINE.CPP
#include <iostream>  // C++ I/O
#include <cmath>     // M_PI

template <class T>
inline T Min (T a, T b)
  {
  return a < b ? a : b ;
  }

const double PI_OVER2 = M_PI / 2 ;

void main ()
  {
  cout << "Min(1,2): "      << Min (1, 2)      << endl
       << "Min(1.1,2.2): " << Min (1.1, 2.2) << endl
       << "Min('a','b'): " << Min ('a', 'b') << endl
       << "PI_OVER2: "      << PI_OVER2        << endl ;
  }
```

with output:

```
Min(1,2): 1
Min(1.1,2.2): 1.1
Min('a','b'): a
PI_OVER2: 1.5708
```

18.2.3 The `#error` Directive

The `#error` directive stops compilation and displays `error_message`:

```
#error error_message
```

Note that `error_message` is not enclosed within double quotation marks. The program below illustrates the `#error` directive:

```
// error.cpp
// illustrates the #error preprocessor directive

#error preprocessor error message
```

```
void main ()
  {}
```

18.2.4 The #include Directive

The #include directive has the following general syntax:

```
#include <file_name>
#include "file_name"
#include macro_identifier
```

The first two versions of #include replace the line containing the #include directive with the contents of the specified file file_name between either the angle brackets (<>) or the double quotation marks (" "). The file file_name is generally a header file with the filename extension .H, but the extension name is optional. Spaces within < > or " " are recognised so be careful of accidentally introducing spaces:

```
#include < iostream >  // IOSTREAM.H not found
```

The file file_name is allowed to contain further #include directives. If file_name is enclosed by angle brackets then file_name refers to a standard include file and is searched for in the include directories of the compiler. Alternatively, if file_name is enclosed between double quotation marks then file_name generally refers to a user-defined file, with file searching commencing in the current working directory. If a file enclosed within double quotation marks is not found in the current working directory then the file search algorithm reverts to that of a file enclosed within angle brackets. A standard include file can also be placed between double quotation marks. If file_name contains an explicit directory path then only that directory will be searched.

The macro version of #include expands macro_identifier to a filename of either the <> or " " formats. Therefore, neither of the < or " characters is allowed as the first character of macro_identifier.

The following program illustrates the three versions of the #include directive:

```
// inc.cpp
// illustrates the #include preprocessor directive

#define IOSTREAM "d:\borland\bcc55\include\iostream.h"

#include <cmath>
#include "inc.h"
#include IOSTREAM

void main ()
  {}
```

18.2.5 The #line Directive

The #line directive allows you to specify the next line number, line_number, and filename, file_name, of the source file and has the following general syntax:

```
#line line_number <"file_name">
```

where `line_number` is a positive integer. The filename token is optional, and if neglected is assumed to be the last filename specified in the `#line` directive. If no previous `#line` directive exists then the filename is assumed to be the source filename. Consider the program:

```
// line.cpp
// illustrates the #line preprocessor directive
#include <iostream>  // C++ I/O

void main ()
  {
  #line 1
  cout << "line: " << __LINE__
       << "; file: " << __FILE__ << endl ;
  #line 9 "file.cpp"
  cout << "line: " << __LINE__
       << "; file: " << __FILE__ << endl ;
  }
```

with output:

```
line: 1; file: LINE.CPP
line: 9; file: file.cpp
```

The `__LINE__` and `__FILE__` predefined macros contain the current line number and filename of the source file and will be described in more detail later.

The `#line` directive is generally used for program debugging, error reporting and line number and filename referencing of code extracted from files other than the source file.

18.2.6 The `#pragma` Directive

The `#pragma` directive allows implementation-specific directives to be defined and has the following syntax:

```
#pragma implementation_specific_name
```

Since the exact details of `#pragma` vary from compiler to compiler it is recommended that you consult your compiler's documentation. For instance, the Borland C++ compiler (version 5.x) defines 14 `#pragma`s, four of which are `argsused`, `hdrfile`, `startup` and `exit`. The `#pragma argsused` disables the warning message 'Parameter `name` is never used in function `FunctionName`'. The `#pragma hrdfile "filename.csm"` sets the name of the precompiled header file to FILE_NAME.CSM. The `startup` and `exit` `#pragma`'s allow a programmer to specify functions to be called upon program startup (before *`main()`* is called) and exit (just before the program terminates).

18.2.7 The `#undef` Directive

The `#undef` directive allows you to undefine a previously defined macro, and has the syntax:

```
#undef macro_identifier
```

#undef is illustrated in the following program:

```
// undef.cpp
// illustrates the #undef preprocessor directive
#include <iostream>  // C++ I/O
#define BUF_SIZ 80   // define directive

void main ()
  {
  cout << "BUF_SIZ: " << BUF_SIZ << endl ;
  #undef BUF_SIZ       // undefine directive
  #define BUF_SIZ 120  // redefine directive
  cout << "BUF_SIZ: " << BUF_SIZ << endl ;
  }
```

with output:

```
BUF_SIZ: 80
BUF_SIZ: 120
```

18.2.8 The #elif, #else, #endif, #if, #ifdef and #ifndef Conditional Directives

The #if, #elif, #else and #endif conditional directives operate in a similar manner to the conditional operators:

```
#if test_expression
<statement_block>
<#elif test_expression statement_block>
<#elif test_expression statement_block>
//...
<#elif test_expression statement_block>
<#else <newline> statement_block>
#endif
```

If the #if test_expression evaluates to logical-true the corresponding statement_block is preprocessed, else if the #if test_expression evaluates to logical-false the #if statement_block is ignored. If the #if test_expression evaluates to logical-true, then after the corresponding statement_block is processed control is directed to the #endif directive, else if the #if test_expression evaluated to logical-false control is directed to the next optional #elif directive. If the #elif test_expression evaluates to logical-true its corresponding statement_block is preprocessed, else control is directed to the next optional #elif directive. This process is repeated until either an #elif or #else directive is encountered. An #else directive is an alternative condition if all previous #if and #elif directives evaluated to logical-false. Note that for each #if there must a corresponding #endif.

The following program illustrates the `#if`, `#elif`, `#else` and `#endif` directives for controlling compilation:

```
// if.cpp
// illustrates the #if, #elif, #else and #endif
// preprocessor directives
#include <iostream>  // C++ I/O

#define STRING_TYPE 2
#if STRING_TYPE == 0
#include <string.h>   // C string
#elif STRING_TYPE == 1
#include <cstring.h>  // C++ string
#elif STRING_TYPE == 2
#include <string>     // C++ string in std namespace
#else
#include "my_str.h"   // user-defined string
#endif

void main ()
  {
  std::string str ("hi!") ;
  }
```

In addition to the above conditional preprocessor directives there is the `defined` operator which has the following general syntax:

```
defined(identifier)
```

or

```
defined identifier
```

The parentheses are optional and `defined` can be preceded by the logical negation operator (`!`). For example:

```
#if !defined(_BOOL_H)
#include "bool.h"
#endif
```

The main advantage of the `defined` operator is that it can be used in conditional expressions:

```
#if defined(BOOL) || !defined(_BOOL_H) && !defined(_BOOLEAN_
H)
//...
```

The `#ifdef` and `#ifndef` directives have the following general syntax:

```
#ifdef identifier
<statement_block>
#endif
```

```
#ifndef identifier
<statement_block>
#endif
```

and allow you to test whether `identifier` is defined or not defined, respectively. Both `#ifdef` and `#ifndef` can use `#else`, but not `#elif`. An illustration of `#ifndef` is given in the program below:

```
// ifndef.cpp
// illustrates the #ifndef preprocessor directive

#ifndef BOOL
typedef int BOOL ;
#define FALSE 0
#define TRUE  1
#endif

void main ()
  {
  BOOL boolean (TRUE) ;
  }
```

18.3 The # and ## Operators

The # and ## preprocessor operators are used in conjunction with the `#define` directive. The # operator converts the macro argument that it operates on to a string enclosed by double quotation marks. The ## operator concatenates two tokens. The following program illustrates # and ##:

```
// #&##.cpp
// illustrates the # and ## preprocessor operators
#include <iostream>  // C++ I/O

#define STRINGISE(s) #s
#define CONCATENATE(x, y) x ## y

void main ()
  {
  cout << STRINGISE (gymnopedies) << endl
       << CONCATENATE ("inter", "national") << endl ;
  }
```

with output:

```
gymnopedies
international
```

Table 18.2 Predefined macros.

Predefined macro name	*Meaning*
`__cplusplus`	C++ program
`__DATE__`	Calendar data (*mmm dd yyyy*)
`__FILE__`	Source code filename
`__LINE__`	Current line number
`__STDC__`	ANSI-C compliant
`__TIME__`	Compilation time (*hh::mm::ss*)

18.4 Predefined Macros

C++ predefines the six implementation-independent macros shown in Table 18.2. Refer to your compiler's documentation regarding additional compiler-dependent predefined macros.

The `__cplusplus` macro is defined as equal to 1 if the program being compiled is a C++ program, else `__cplusplus` is undefined. Note that `__cplusplus` does not end with two underscore characters and contains entirely lowercase alphanumeric characters, unlike the other five predefined macros. With a quick browse through the C++ standard library header files you will find `__cplusplus` used as the test expression of an `#ifdef` conditional directive to ensure that a C++ header file is included in a source file with a C++ extension. The following example has been extracted from the C++ exception-handling header file EXCEPT.H:

```
// except.h
//...
#ifndef __cplusplus
#error Must be a C++ source file to include except.h
#endif
//...
```

The `__DATE__` macro is a string literal that contains the calendar date on which the preprocessor commenced processing the source file. The format of the date string is *mmm dd yyyy*, where *mmm* represents the month (e.g. Jan), *dd* represents the day (e.g. 16) and *yyyy* represents the year (e.g. 2000).

The `__FILE__` macro is a string literal that contains the name of the source file being processed. This macro is updated by the `#include` and `#line` directives.

The `__LINE__` macro is an integer number which represents the current line number being processed. This macro is updated by the `#line` directive.

The `__STDC__` macro is defined as equal to 1 if a compiler is ANSI C-compliant, else `__STDC__` is undefined.

Finally, the `__TIME__` macro is a string literal that contains the time at which the preprocessor commenced processing the source file. The format of the time string is *hh::mm::ss*, where *hh* represents the hour (e.g. 22), *mm* represents the minutes (e.g. 58) and *ss* represents the seconds (e.g. 45).

The following program illustrates the predefined macros:

```
// predef.cpp
// illustrates the predefined macros
```

```
#include <iostream>  // C++ I/O

void main ()
  {
  cout << "__cplusplus: " << __cplusplus << endl
       << "__DATE__: "    << __DATE__    << endl
       << "__FILE__: "    << __FILE__    << endl
       << "__LINE__: "    << __LINE__    << endl ;

  #ifdef __STDC__
    cout << "__STDC__: "    << __STDC__    << endl ;
  #endif

  cout << "__TIME__: "    << __TIME__    << endl ;
  }
```

which generated the following output when I ran the program:

```
__cplusplus: 1
__DATE__: Aug 28 2000
__FILE__: PREDEF.CPP
__LINE__: 12
__TIME__: 13:59:22
```

Let's conclude by mentioning the ever popular *assert()* macro defined in the ASSERT.H header file:

```
assert (expression)
```

If expression evaluates to logical-false a message is directed to stderr (generally the display screen) and program execution is aborted by calling the *abort()* function. The form of the message displayed is:

```
Assertion failed: expression, file file_name, line line_number
```

line_number is the line number in which the *assert()* macro appears in the source file file_name.

If the directive #define NDEBUG is placed in the source file before the #include <assert.h> directive then the *assert()* macro is effectively turned off.

The following program illustrates *assert()*:

```
// assert.cpp
// illustrates the assert() macro
#include <iostream>  // C++ I/O

//#define NDEBUG

#include <assert>    // assert()

void main ()
```

```
{
#define FALSE 0

assert (FALSE) ;
}
```

with output:

```
Assertion failed: FALSE, file ASSERT.CPP, line 13
Abnormal program termination
```

18.5 Summary

The preprocessor allows a programmer to control the compilation process by operating on directives. A number of implementation-independent directives are supplied by the preprocessor, such as `#define` and `#include` and a compiler-dependent directive called `#pragma`. The `#define` directive can be used to define identifiers and parametrised macros and several predefined macros are available to a programmer.

Since its earliest days C++ has gradually introduced language-based features, such as **`const`**, **`inline`**, **`template`** and **`namespace`** that have made the preprocessor redundant. Admittedly there are still facilities offered by the preprocessor for which C++ offers no language-based equivalent, but whenever you have the option of choosing between a preprocessor directive and a language-based feature then choose the latter.

Exercises

18.1 Develop two macros and **`inline`** functions which return a Boolean value depending on whether a given number is zero or infinite (which are to be defined as 1e-20 and 1e+20, respectively).

18.2 Use the `#ifdef`, `#else`, `#endif` and `#include` directives and the `__cplusplus` predefined macro to include either the IOSTREAM.H or STDIO.H header files, depending on whether a user is compiling either a C++ or C program.

18.3 It is desired that the WINDOWS.H header file is included in a program that declares the following `Boolean` type with the `FALSE` and `TRUE` enumerators:

```
#include <windows>  // Windows header file
//...
enum Boolean { FALSE, TRUE };
```

Unfortunately WINDOWS.H `#defines` `FALSE` and `TRUE`:

```
#define FALSE 0
#define TRUE  1
```

so that name clashes are experienced. Use the `#ifdef`, `#undef` and `#endif` conditional directives so that the WINDOWS.H header file and the `Boolean` **`enum`** can both define the `FALSE` and `TRUE` literals.

18.4 The function argument `i` is not used in the definition of *`Function()`*:

```
void Function (int i)
  {
  }

void main ()
  {
  int i (0) ;

  Function (i) ;
  }
```

Use the `#pragma` directive to eliminate the compilation warning 'Parameter `i` is never used in function *`Function()`*'.

18.5 A header file, POINT.H, for a **`class`** `Point` is of the form:

```
// point.h

class Point
  {
  //...
  };
```

Modify the header file so that only C++ source files can include the POINT.H header file.

18.6 Develop an *`Assert()`* function which mimics the C++ standard library macro *`assert()`*.

18.7 Compare the use of the C++ standard library macro *`assert()`* declared in ASSERT.H with the *`Assert()`* **`template`** function defined in Chapter 14:

```
template <class T, class X>
inline void Assert (T exp, X x)
  {
  if (!NDEBUG) // compatible with assert() macro NDEBUG
    if (!exp)
      throw x ;
  }
```

Chapter 19

Namespaces

A namespace declaration defines a scope, and can therefore be used to prevent name clashes. The namespace feature is required because C++ defines a single global scope namespace outside the scope of if-else or switch statement blocks, loops, functions and classes. As programs steadily grow in size they correspondingly become more susceptible to name clashes. A large program will consist of numerous function and class library files, and namespaces allow functions and classes to be associated with a scope other than the global file scope. The namespace feature is particularly useful for eliminating name clashes between applications and function and class libraries.

A namespace name and the scope resolution operator are used to qualify access explicitly. A using declaration allows access to a namespace member without explicit qualification whereas a using directive allows access to all members of a namespace without explicit qualification.

19.1 Namespaces – What are They and Why do we Need Them?

The namespace is one of the newest major features to be added to the C++ programming language. C++ provides a namespace for **`if-else`** and **`switch`** statement blocks, loops, function declarations and definitions, **`union`**s, **`struct`**s, **`class`**es or translational units[1]. Outside these namespaces C++ provides one single global namespace which can result in name clashes when multiple files are linked together. For example, consider including the WINDOWS.H library header file in the following program:

```
// n_clash.cpp
// illustrates name clashes

#include <windows>  // Windows header file
```

1 A translational unit in C++ is a file and all included files corresponding to `#include` directives.

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

```
class Polygon
  { /* ... */ };

void main ()
  {
  Polygon poly ;
  }
```

This is straightforward enough, but this program will generate a number of confusing compilation errors because of the name clash between the **class** `Polygon` declaration in N_CLASH.CPP and the `Polygon` function declaration in WINDOWS.H:

```
// windows.h
#ifndef _WINDOWS_H  // prevent multiple includes
#define _WINDOWS_H
//...
BOOL Polygon (HDC, const POINT*, int) ;
//...
#endif // _WINDOWS_H
```

In addition, note the trouble that we have to go to prevent multiple definitions and `#includes` using a preprocessor macro guard in WINDOWS.H.

One solution to this problem, which you may have seen used in previous chapters, is simply to rename any conflicting names:

```
// n_clash1.cpp
// illustrates name clashes

#include <windows>  // Windows header file

class PlanarPolygon
  { /* ... */ };

void main ()
  {
  PlanarPolygon poly ;
  }
```

Renaming is a tedious solution and not always possible, since you may be working with shipped libraries.

19.2 The `namespace` Declaration

The general form of the **namespace** declaration is:

```
namespace Name
  {
  // declarations & definitions
  }
```

Note that there is no semicolon directly after the closing brace of the **namespace** declaration. Declarations within the braces of **namespace** Name do not clash with those declared in another **namespace** or global names because each **namespace** defines its own unique scope. For example, the following namespaces, X and Y, and global names are completely separate:

```
// nspace.cpp
// illustrates the namespace declaration

void Function () ;  // global

namespace X
  {
  void Function () ;

  void Gunction ()
    {
    //...
    Function () ;  // X::Function()
    }
  //...
  class Point
    { /*...*/ };
  }

namespace Y
  {
  void Function () ;
  //...
  class Point
    { /*...*/ };
  }
//...
void main ()
  {
  X::Function () ;
  X::Point p ;

  Y::Function () ;
  Y::Point q ;
  }
```

The name of a **namespace** and the scope resolution operator (::) are used to resolve access to a given **namespace**'s members explicitly. As with classes, explicit use of the scope resolution operator is not necessary for names enclosed within a **namespace**:

```
namespace X
  {
  void Function () ;
  //...
  void Gunction ()
    {
```

```
    //...
    Function () ;  // X::Function ()
    }
}
```

A **namespace** is said to be *open*, whereas classes are not:

```
namespace X
  {
  void Function () ;
  //...
  }
//...
namespace X
  {
  void Function1 () ;
  //...
  }
```

which inherently leaves a **namespace** open for namespace clashing. Thus, when choosing a name for a **namespace** choose appropriately descriptive names and not simply X as shown above! Namespaces have been designed to be open so as to allow a programmer to place program code in several different, identically named, **namespace** declarations rather than one single **namespace** declaration. This is similar in principle to having multiple access specifiers within a **class** declaration.

It is possible to use an alias for a namespace name and is useful when referring to long namespace names:

```
namespace Library_Mathematical
  {
  }
//...
namespace Math = Library_Mathematical ;
```

Namespaces can be nested:

```
// ns_nest.cpp
// illustrates nested namespaces

namespace X
  {
  void Function () ;
  //...
  namespace Y
    {
    void Function () ;
    //...
    class Point
      {
      //...
      public:
```

```
            //...
            double Z () ;
            //...
          };
        }
    }
//...
```

with the usual use of the scope resolution operator for qualification:

```
void X::Y::Function ()
  { /* ... */ }
//...
double X::Y::Point::Z ()
  { /* ... */ }
```

Let's briefly return to our original problem of a name clash between **class** `Polygon` and the Windows API *`Polygon()`* function. By using namespaces the name clash is easily resolved:

```
// no_clash.cpp
// modified N_CLASH.CPP using namespaces

#include <windows>     // Windows header file
#include "polygon.h"   // class Polygon
#include "sphere.h"    // class Sphere

void main ()
  {
  Shapes::Polygon poly ;
  Shapes::Sphere  sphere ;
  //...
  }
```

With POLYGON.H and SPHERE.H:

```
// polygon.h
// class Polygon

namespace Shapes
  {
  class Polygon
    { /* ... */ };
  }

// sphere.h
// class Sphere

namespace Shapes
  {
  class Sphere
```

```
    { /* ... */ };
}
```

In addition the declaration of **class** Polygon has been placed in the header file POLYGON.H within the scope of Shapes. Advantage is also taken of the property that namespaces are, by definition, *open* by also placing **class** declaration Sphere within Shapes in SPHERE.H. Thus, in addition to providing name clashes, namespaces also provide a mechanism for logical grouping.

19.3 The `using` Declaration

The *using declaration* allows a member from a **namespace** to be accessed without explicit qualification:

```
// use_dec.cpp
// illustrates the using-declaration

namespace X
  {
  int  data = 0 ;
  void Function () ;
  //...
  }
//...
void main ()
  {
  using X::Function ;  // using-declaration

  data = 1 ;      // error: not accessible

  Function () ;  // ok: X::Function()
  }
```

A **using** declaration adds an object to local scope just as a *normal* object declaration:

```
namespace X
  {
  int data = 0 ;
  //...
  }

void Function ()
  {
  int data = 0 ;
  using X::data ;  // error: multiple definition of data!
  //...
  }
```

A **using** declaration brings into scope every declaration of the specified name. For example, the following illustrates a `String` class and two versions of the overloaded concatenation operator in the **namespace** `StringLibrary`:

```
// use_dec1.cpp
// using declaration applied to classes

namespace StringLibrary
{
class String { /* ... */ };

String operator + (const String&, const String&) ;
String operator + (const String&, const char*) ;
}

namespace LinguisticsLibrary
{
using StringLibrary::String ;
using StringLibrary::operator + ;

void Function ()
        {
        String s1, s2 ;
        char*  c ;

        String s3 = s1 + s2 ;
        String s4 = s1 + c ;
        }
}
```

Within the `LinguisticsLibrary` namespace the single using declaration of the overloaded + operator brings into local scope both versions of the overloaded operator, as illustrated in *`Function()`*.

19.4 The `using` Directive

The **using** directive allows *all* members of a **namespace** to be accessed without explicit qualification:

```
// use_dir.cpp
// illustrates the using-directive

namespace X
  {
  int  data = 0 ;
  void Function () ;
  //...
  }
//...
```

```
void main ()
  {
  using namespace X ;  // using-directive

  data = 1 ;       // X::data
  Function () ;    // X::Function()
  }
```

Do note that a **using** directive allows access to all members of a given **namespace**, and if this is not what is intended then use the safer, more explicit, using declaration approach.

A **using** directive can easily introduce ambiguities between similarly named members of different namespaces:

```
namespace X
  {
  int data = 0 ;
  }
//...
namespace Y
  {
  int data = 0 ;
  }

void Function ()
  {
  using namespace X ;
  using namespace Y ;
  //...
  data = 1 ;  // error: X::data or Y::data?
  }
```

In contrast with a **using** declaration, a **using** directive does not add an object to local scope:

```
namespace X
  {
  int data = 0 ;
  //...
  }

void Function ()
  {
  int data = 0 ;
  using namespace X ;  // o.k.: non-declaration
  //...
  }
```

Since a **using** directive does not add to local scope, the above **using** directive within *Function()* does not redefine `data`, but makes `X::data` accessible.

19.5 The Nameless namespace Declaration

A **namespace** can be declared unnamed:

```
namespace
  {
  //...
  }
```

which is a useful feature of the **namespace** declaration when the name of a **namespace** is irrelevant provided that it does not clash with other nameless **namespace** declarations. Thus we observe from the above discussions that the nameless **namespace** is equivalent to:

```
namespace Name
  {
  //...
  }
//...
using namespace Name ;
```

Since **namespace** declarations are open by definition, all nameless **namespace** declarations in the same scope have the same *name*.

19.6 Global Scope

On numerous occasions in previous chapters we have seen the use of the global scope qualification `::Function()`, meaning the global namespace declaration of `Function()`:

```
// global.cpp
// namespaces and global scope

void Function () ;  // global

namespace X
  {
  void Function () ;
  //...
  void Function1 ()
    {
    Function () ;      // X::Function ()
    X::Function () ;   // X::Function ()
    ::Function () ;    // global Function()
    }
  }
//...
```

In view of the previous section, note that all nameless **namespace** declarations in the same scope form part of the global scope.

Be careful of name clashes arising as a result of global objects and the **using** directive:

```
int data = 0 ;  // global

namespace X
  {
  int data = 0 ;
  }

void Function1 ()
  {
  using namespace X ;

  ::data  = 1 ;  // ok: global data
  X::data = 2 ;  // ok: X::data
  data    = 3 ;  // error: global data or X::data?
  }
```

A namespace can be placed into global scope by placing a **using** directive outside the scope of any function and in file scope. For example, the C++ standard library **namespace** `std` can be placed in to global scope so as to eliminate the use of the `std::` prefix:

```
#include <iostream>

using namespace std ;

void main ()
  {
  cout << "hello, object" << endl ;
  }
```

Generallly speaking, placing a namespace in to global scope is not to be recommended since it encourages name clashes.

19.7 Overloading Namespaces

Members of a **namespace** can be overloaded with the aid of either a **using** declaration or directive:

```
// overload.cpp
// overloading namespaces

namespace X
  {
  void Function (int i) ;
  }

using namespace X ;
```

```
namespace Y
  {
  void Function (double d) ;
  }

using namespace Y ;

void main ()
  {
  Function (1.0) ;  // call to Y::Function(double)
  }
```

19.8 Namespaces and Inheritance

Namespaces integrate with inheritance. A **using** declaration that is a member of a derived **class** *must* name a member of a base **class**, while a **using** directive is simply not allowed as a **class** member:

```
// inherit.cpp
// illustrates namespaces and inheritance

class Base
  {
  protected:
    int data ;
  public:
    Base () ;
    //...
  };

class Derived : public Base
  {
  public:
    // o.k.: u-dec & data is a member of Base
    using Base::data ;
    // error: Function() is not a member of Base
    using Base::Function ;
    // error: u-dir not allowed as class member
    using namespace Base ;
    //...
  };
//...
```

The **using** declaration offers a new angle on the problem of hiding a member in a base **class** by a similarly named derived **class** member. In the following program code `Derived::Function(double)` hides `Base::Function(int)`:

```
class Base
  {
```

```
    public:
      void Function (int i) ;
      //...
    };

  class Derived : public Base
    {
    public:
      void Function (double d) ;
      //...
    };
  //...
  void main ()
    {
    derived d ;
    //...
    d.Function (2) ;  // call to Derived::Function(double)
    }
```

A **using** declaration in **class** Derived adds Base::*Function()* to Derived's scope:

```
  // un_hide.cpp
  // a namespaces solution to the problem of a derived class
  // member hiding a base class member

  class Base
    {
    public:
      void Function (int i) ;
      //...
    };
  //...
  class Derived : public Base
    {
    public:
      void Function (double d) ;
      using Base::Function ;    // using declaration
      //...
    };
  //...
  void main ()
    {
    Derived d ;

    d.Function (2) ;  // call to Derived::Function(int)
    }
```

In program CHNG_ACC.CPP of Chapter 15 we saw the use of a **public**ly placed access declaration in a **private**ly derived **class** to recover the **public** access specification of a data member in the base **class**:

```
//...
class Base
  {
  public:
    int pub_bdm0 ;
    int pub_bdm1 ;
  };

// privately derived class
class Derived : private Base
  {
  public:
    Base::pub_bdm1 ; // access-declaration
  };

void main ()
  {
  Derived d_obj ;
  d_obj.pub_bdm0 ;  // error: not accessible
  d_obj.pub_bdm1 ;  // ok
  }
```

A **using** declaration now makes the access declaration redundant:

```
//...
// privately derived class
class Derived : private Base
  {
  public:
    using Base::pub_bdm1 ; // using-declaration
  };
//...
```

Stroustrup (1994) discusses an important implication of the **using** declaration for eliminating ambiguities as a result of multiple inheritance:

```
class Base0
  {
  public:
    void Function (int i) ;
  };
//...
class Base1
  {
  public:
    void Function (double d) ;
  };
//...
class Derived : public Base0, public Base1
  {
```

```
  public:
    void DerivedFunction ()
      {
      // error: Base0::Function() or Base1::Function()?
      Function (1) ;
      }
  };
```

Two **using** declarations resolve the above ambiguity:

```
// multiple.cpp
// namespaces and multiple inheritance

class Base0
  {
  public:
    void Function (int i) ;
  };
//...
class Base1
  {
  public:
    void Function (double d) ;
  };
//...
class Derived : public Base0, public Base1
  {
  public:
    using Base0::Function ;
    using Base1::Function ;

    void DerivedFunction ()
      {
      Function (1) ;  // Base0::Function(int)
      }
  };
//...
```

which is a neat solution to a tricky problem!

19.9 The Standard Library Namespace

All C standard library rountines are placed in a header <X.H> with coresponding C++ standard header file <cX>. The C++ standard library is placed into the **namespace** std and to ensure backward compatibility the routines of the *old* header files are wrapped in the std **namespace**. For example, the MATH.H header file is of the form:

```
#ifdef __cplusplus
namespace std {

extern "C" {
#endif
//...
double sqrt (double x) ;
//...
#ifdef __cplusplus
} // extern "C"
} // std
using namespace std ;
#endif
```

and illustrates the use of a **using** directive to place the C standard library in to global scope. If it is required that the C standard library routines are not placed in to global scope then the <cX> header file should be `included`; e.g. CMATH.H.

19.10 Java and Namespaces

Name clashes are not unique to programming in C++ and are a problem with all programming languages. For instance, Java is a pure object-oriented programming language and as a result does not support global data or functions and therefore eliminates entirely global name clashes. Java's solution to name clashes is via the name spaces: *packages*, *classes* and *fields*. Every public class is part of a package (.java file) while every data member/operation (field) is declared as a member of a class. Thus, every Java data member or method is accessed by its package-class-field fully qualified name. For example, to access the *`substring()`* method of class `String` of package `java.lang` we would write:

```
java.lang.String.substring () ;
```

19.11 Summary

The C++ **namespace** declaration is a solution to the problem of conflicting names in a translational unit. A **namespace** has its own scope and can be either named or nameless, nested, and overloaded. Access to a **namespace**'s members is via the scope resolution operator. A **using** declaration allows access to a specified **namespace** member without explicit qualification, whereas a **using** directive allows access to *all* members of a specified **namespace** without explicit qualification.

Namespaces also provide a mechanism for logically grouping the components of a large program or library. In fact, the larger a program or library then the more useful namespaces become.

Try not to draw too many comparisons between **class** declarations and **namespace** declarations. Classes and namespaces have their similarities, but a **namespace** is not a **class**.

Stroustrup (1993b) and (1994) highlight that the **using** directive should be viewed as a transitional tool and when writing new program code a **using** declaration approach should be adopted rather than a **using** directive.

Exercises

19.1 A program uses two different library `Point` classes placed in the header files POINT1.H and POINT2.H:

```
// point1.h                    // point2.h

template <class T>             template <class T>
class Point                    class Point
  {                            {
  //...                        //...
  };                           };
```

Use namespaces so that both header files can be included in a program file and objects defined for each **class**.

19.2 It is required that a number of geometric classes are nested within a single **namespace** called `Geometry`. The classes can be categorised as points and vectors, lines and curves, surfaces and geometric shapes each of which is to be placed within its own appropriate namespace. Design the outline of the namespaces.

19.3 For Exercise 19.2 define concise alias names for each nested **namespace**.

19.4 Use the property that namespaces are open by placing the namespaces of Exercise 19.2 into separate header files.

19.5 With the help of a **using** declaration rewrite the *main()* function for the following program code so that explicit qualification is not required to define object `c`:

```
namespace Mathematical
  {
  class Complex
    {
    private:
      double re, im ;
    public:
      Complex ()
                  : re (0.0), im (0.0) {}
      Complex (double _re, double _im=0.0)
                  : re (_re), im (_im) {}
      //...
    };
  }
//...
void main ()
  {
  Mathematical::Complex c (1.0, 2.0) ;
  }
```

19.6 Are the two program code fragments below equivalent?

```
void Function () ;

namespace X
  {
  ::Function () ;
  }
```

and:

```
namespace
  {
  void Function () ;
  }

namespace Y
  {
  ::Function () ;
  }
```

19.7 A **class** called `Base` is declared within the namespace `BaseNameSpace`:

```
namespace BaseNameSpace
  {
  class Base
    {
    private:
      int bdm ;
    public:
      Base ()
        : bdm (0) {}
    };
  }
```

Publicly derive from `Base` a **class** called `Derived` which is declared within its own **namespace**. Ensure that `Derived`'s default constructor initialises the `Base::bdm` data member.

Chapter 20

The Standard Template Library

we tend to find what we expect
K. Devlin (2000)

In this chapter we examine the Standard Template Library (STL). The STL is a comprehensive library of template functions and classes that are suppiled with all ANSI C++ compatible compilers. The objective of the STL is to provide programmers with a library of frequently used data structures and functions. In doing so a programmer's code will generally be more robust, efficient and portable than if the programmer had developed her/his own data structure or function.

The chapter begins by examining iterators, which are a generalisation of pointers and are frequently used throughout the STL. Next, function objects are introduced and the associated features of predicates, binders, negaters and member function adapters. Section 20.6 presents a comprehensive summary of the sequence algorithms, which are classified as either non-mutating or mutating. Sequence algorithms operate on data held by ordered sequences. A mutating sequence algorithm can alter either the values or order of the sequence elements, whereas a non-mutating sequence algorithm cannot alter the elements of a sequence in any way.

A big part of the STL is dedicated to containers or abstract data types. These are classified as either sequence or associative. A sequence container stores its objects in a sequential manner, whereas an associative container associates a key with each of its objects. The sequence containers examined are classes `vector` *and* `list`*, while the associative containers classes are* `set`*,* `multiset`*,* `map` *and* `multimap`*. Containers specifically designed for performing numerical calculations are also examined. These consist of classes* `numeric_limits`*,* `complex`*,* `valarray` *and the* `slice`*,* `mask` *and* `indirect_array` *classes.*

The chapter concludes by discussing adapters, class `basic_string` *and allocators. Adapters transform either a function object, iterator or container into another. Adapters are useful because they save a programmer from develping a new function object, iterator or container from scratch when it can be adapted from a similar one which already exists. Class* `basic_string` *is the fundamental string class supported by the STL and provides an extensive range of operations. Allocators relieve a programmer from having to deal directly with memory allocation when implementing container classes.*

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
© Springer-Verlag London Limited 2001

20.1 Overview of the Standard Template Library

The Standard Template Library (STL) is the latest non-language major feature to be added to the C++ programming language. The STL consists of a library of template iterators, functions and classes which are frequently used by programmers. All STL functions, containers and iterators are designed to operate in the same manner. This makes the STL easy to learn even though it is a large library.

The STL is both designed and developed in a robust and efficient manner by professional programmers. In addition, the developers of the STL use established procedures for reporting and fixing bugs. Coupled with standardisation, the STL is a solid library that is shipped with all ANSI C++ compatible libraries.

All algorithms in the STL pay particular attention to efficiency or algorithm complexity. Each algorithm is assigned a complexity index using the big-O notation: for example, $O(n)$ and $O(n^2)$ for linear and quadratic time compleixities respectively; where n denotes the number of data elements. The STL implements the currently most efficient algorithms and are typically of the order $O(1)$ for indexing a hash table, $O(n)$ for indexing an unsorted container such as a vector, $O(\log n)$ for indexing a sorted container, binary tree or heap and $O(n\log n)$ for sorting the elements of a container. Consulting algorithmic complexity can help in the decision process of selecting an algorithm or container and associated operations for a particular task.

The STL consists of the following key components: iterators, algorithms, containers, function objects, adaptors and allocators. Iterators are a generalisation of pointers and provide a consistent and generic method for accessing the elements of container objects. The STL provides extensive support for a variety of sequence and sorting algorithms. Containers in STL are categorised as either sequence or associative. The elements of sequence containers (such as vector and linked list) have an inherent order, whereas the elements of associative containers (such as set and hash tables) are unordered. Container adaptors transform one container into another container. As the name suggests, a function object is an object that acts as though it was a function and allows us to specify non-trivial operations as arguments. Allocators address the problem of different memory models for PC architecture and relieve implementors of container classes of having to deal directly with raw memory. The following sections examine each of these components of the STL separately with the objective of each section to extract the essence of each component rather than attempting to provide a complete discussion.

20.2 The STL `namespace`

The **`namespace`** of the STL is the same as the C++ standard library **`namespace`**, `std`:

```
namespace std
{
//...
}
```

To refer to an STL class, say `vector`, don't forget to specify the `std` **`namespace`**:

```
std::vector<int> v ;
//...
```

Alternatively, for convenience the using-directive can be used:

```
using namespace std ;
//...
vector<int> v ;
```

20.3 Global Functions

The STL defines numerous global template functions. The majority of these functions fall under the wing of the STL algorithms and will be discussed in Section 20.6. However, a function which will prove useful when we discuss iterators in the following section is *copy()* and is declared in ALGORITH.H:

```
template <class InputIterator, class OutputIterator>
OutputIterator copy (InputIterator first, InputIterator last,
                     OutputIterator result) ;
```

copy() copies all elements from `first` up to but not including `last` to the location referred to by `result`.

Unfortunately, support for global functions is a move away from C++ being a pure object-oriented language.

20.4 Iterators

Iterators play a key role throughout the STL, and as a result it is important first to develop a good understanding of iterators before examining the other key components of the STL. Iterators are designed to be a generalisation of pointers, and we will begin by illustrating this point by revisiting the P_SUB.CPP and P_INC.CPP files examined in Chapter 12. These two files defined functions *StrLength()*, *StrCompare()* and *StrCopy()* and are included in the file IT_PTR.CPP below:

```
// it_ptr.cpp
//...
// returns the length of a string
int StrLength (const char* string)
    {
    char* p_str = const_cast<char*>(string) ;

    while (*p_str)
         p_str++ ;
    return p_str - string ;
    }

// compares two strings
// returns 1 if str1==str2, 0 if str1!=str2
int StrCompare (const char* str1, const char* str2)
```

```
    {
    while (*str1++ == *str2++)
          {
          if (*str1 == 0)
             return 1 ;
          }
    return 0 ;
    }

// copies str2 into str1
void StrCopy (char* str1, const char* str2)
     {
     while (*str1++ = *str2++) ;
     }

// copies str2 into str1 using STL copy()
void STLStrCopy (char* str1, const char* str2)
     {
     int str2_length = StrLength (str2) ;

     std::copy (str2, str2+str2_length, str1) ;
     }

void main ()
    {
    char string1[] = "string1" ;
    char string2[] = "string2" ;
    //...

    StrCopy    (string1, string2) ;  // copy strings
    STLStrCopy (string3, string4) ;

    cout << "string1: " << string1 << endl ;
    cout << "string2: " << string2 << endl ;
    //...
    }
```

StrLength() determines the length of a string by use of a pointer to **char**, p_str, which is initialised to the operated-on string. While the characters of the string are not equal to the null terminating character p_str is incremented using the ++ operator. The characters of the string are obtained by use of the pointer dereference operator *. The length of the string is determined by use of the substraction operator - by subtracting the original value of p_str from its final value. *StrCompare()* compares two strings and uses the equality operator == in addition to the ++ and * operators. *StrCopy()* copies one string in to another string and uses the assignment operator = in addition to the ++ and * operators. Thus, in manipulating data types via pointers we have come to expect support through various operators. If you feel comfortable with pointer manipulation then the move to iterators will be relatively painless since iterators are a generalisation of pointers. General iterators similarly support operators such as the increment and decrement operators (++, --), the dereference operator (*), the logical operators (==, !=, <, <=, >, >=), the addition and subtraction operators (+, -) and the assignment operator (=).

An alternative implementation of copying a string is provided by *STLStrCopy()* and uses the `std::` *copy()* function to copy all elements from the start of string `str2` up to, but not including, the end of `str2` (i.e. the last character and not the null terminator) into string `str1`. The type of the first two arguments of *copy()* is an input iterator, whereas the type of the third argument of *copy()* is an output iterator. This use of *copy()* illustrates that the pointers to **char**, `str1` and `str2`, are equivalent to iterators.

The following section examines the class hierarchy of the iterators supported by the STL.

20.4.1 Iterator Class Herarchy

The STL supports five different kinds of iterator category, which are summarised by the standard iterator tags, ITERATOR.H:

```
struct input_iterator_tag { /* ... */ };
struct output_iterator_tag { /* ... */ };
struct forward_iterator_tag
  : public input_iterator_tag { /* ... */ };
struct bidirectional_iterator_tag
  : public forward_iterator_tag { /* ... */ };
struct random_access_iterator_tag
  : public bidirectional_iterator_tag { /* ... */ };
```

Since all iterators cannot support all operators, the iterators are classified into the categories: input, output, forward, bidirectional and random access.

Data structure `iterator` acts as the base type for all iterators and has the following declaration:

```
template <class Category, class T,
          class Distance=ptrdiff_t,
          class Pointer=T*,
          class Reference=T&>
struct iterator
  {
  typedef T         value_type ;
  typedef Distance  difference_type ;
  typedef Pointer   pointer ;
  typedef Reference reference ;
  typedef Category  iterator_category ;
  };
```

In addition to the above iterator categories there are the stream iterators. Class `istream_iterator` is derived from `iterator` for handling input streams and is therefore an input iterator:

```
template <class T, class CharT=char,
class Traits=char_traits<CharT>, class Distance=ptrdiff_t>
class istream_iterator : public iterator<input_iterator_tag,
                           T,Distance,const T*,const T&>
  {
  //...
```

```
  public:
    istream_iterator () ;
    istream_iterator (basic_istream<CharT,Traits>& s) ;
    istream_iterator (const istream_iterator& s) ;
    const T& operator * () const ;
    const T* operator ->() const ;
    istream_iterator& operator ++ () ;
    istream_iterator  operator ++ (int) ;
};
```

Also defined are overloaded non-member equality operators:

```
inline bool operator == (const istream_iterator& x,
                         const istream_iterator& y) ;
inline bool operator != (const istream_iterator& x,
                         const istream_iterator& y) ;
```

Class `istream_iterator` operates on objects of type `T`. `CharT` defines the type of characters and is by default set to **`char`**. `Traits` defines the character traits and is by default set to `char_traits`. Type `Distance` indicates the distance between objects in the stream and is set by default to `ptrdiff_t`, which is a **`typedef`** set to either **`long`** or **`int`**, depending on the memory model used. The dereference, indirect member access and increment operators are overloaded as member functions.

Input iterators can only move forward, retrieve values and perform equality and inequality comparisons, as summarised in Table 20.1.

In an analogous manner to input operators, output iterators are restricted to move forward only; they can only be dereferenced to assign a value and not to retrieve a value. Derived from `iterator` is class `ostream_iterator` for handling output streams:

Table 20.1 Operators supported by the STL iterators.

Operator	*Iterator*				
	Input	*Output*	*Forward*	*Bidirectional*	*Random access*
++	✓	✓	✓	✓	✓
--				✓	✓
* (pointer dereference)	✓ (as r-value)	✓ (as l-value)	✓	✓	✓
<					✓
<=					✓
>					✓
>=					✓
==	✓		✓	✓	✓
!=	✓		✓	✓	✓
=	✓		✓	✓	✓
+					✓
+=					✓
- (subtraction)					✓
-=					✓
[]					✓

```
template <class T, class CharT=char,
          class traits=char_traits<CharT> >
class ostream_iterator
  : public iterator<output_iterator_tag,T,
                    typename Traits::off_type,T*,T&>
  {
  //...
  public:
    ostream_iterator (basic_ostream<CharT,traits>& s) ;
    ostream_iterator (basic_ostream<CharT,traits>& s,
                      const CharT* delim) ;
    ostream_iterator (const ostream_iterator& x) ;
    ostream_iterator& operator = (const T& value) ;
    ostream_iterator& operator *  () ;
    ostream_iterator& operator ++ () ;
    ostream_iterator& operator ++ (int) ;
  };
```

and operates on type `T` only. Note that the two argument constructor accepts a `basic_ostream` object and a pointer to a character to be used as a separator between stream objects; for example `" "` and `'\n'` for specifying a space and new line between stream objects. `ostream_iterator` overloads the assignment operator for outputting `value` to the output stream.

Forward iterators are designed to traverse the elements of container classes. Forward iterators are still restricted to move in a forward direction only (in accordance with the operated-on container), but can in addition support full dereferencing and the assignment operator for both assigning and retrieving values. As an illustration of forward iterators consider the following program, which first fills a `std::list` of integers and then illustrates three different strategies for outputting the list elements to the `cout` stream:

```
// it_forw.cpp
//...
void main ()
    {
    list<int>           l ;   // list l and iterator li
    list<int>::iterator li ;

    for (int i=0; i<5; i++)   // fill l
        l.push_back (i) ;

    for (li=l.begin(); li!=l.end(); li++)  // 1st O/P
        cout << *li << " " ;
    cout << endl ;

    li = l.begin () ;  // 2nd O/P
    while (li != l.end())
          cout << *li++ << " " ;
    cout << endl ;

    ostream_iterator<int> outi (cout, " ") ;  // 3rd O/P
    copy (l.begin(),l.end(),outi) ;
    }
```

The first approach uses a **for** loop and ++ operator to iterate through the `list` elements and displays the elements by dereferencing the `list` iterator `li` and using the insertion operator <<. The second approach uses a **while** loop, the increment operator ++ to traverse the list elements, the dereference operator *, and again the insertion operator for output. The third approach alternatively uses an `ostream_iterator` iterator `outi` and the `std::copy()` function. The two argument constructor of `ostream_iterator` is used to specify that the output stream is `cout` and that each stream object is to be separated by a space. Recall from Secton 20.3 that *copy()* copies all elements of list `l` within a specified range to `outi`. Note that in this example *copy()* is operating on both `list` and `ostream` iterators, whereas in program IT_PTR.CPP *copy()* operated on type **char***, and helps to demonstrate the generic nature of the STL functions and classes. The details of the `list` member functions will be discussed shortly in Section 20.7 when we examine container classes, but it suffices to say at present that *begin()* and *end()* are member functions supported by all container classes that specify the first and last objects of a container.

Bidirectional iterators are essentially equivalent to forward iterators except that they can move in both forward or backward/reverse directions. Class `bidirectional_iterator` traverses elements in the forward direction only from *begin()* to *end()*. On the other hand, a `reverse_bidirectional_iterator` traverses elements in reverse order from *rbegin()* to *rend()*. As an illustration of reverse iterators consider the program:

```
// it_rever.cpp
//...
void main ()
    {
    list<int> l ;

    for (int i=0; i<5; i++)  // fill l
        l.push_back (i) ;

    list<int>::reverse_iterator lri ;  // O/P l
    for (lri=l.rbegin(); lri!=l.rend(); lri++)
        cout << *lri << " " ;
    }
```

with output:

```
4 3 2 1 0
```

The above program illustrates that the use of reverse iterators is equivalent to that of forward iterators (i.e. using the ++ operator to increment throughout a list), but it traverses a list from back to front rather than from front to back for a forward iterator.

Random access iterators provide full support for both traversing and accessing elements of an object; see Table 20.1 for a summary of the operators supported. To illustrate random access iterators consider the following program:

```
// it_rand.cpp
//...
void main ()
    {
    vector<int> v ;  // vector v
```

```
    for (int i=0; i<5; i++)  // fill v
        v.push_back (i) ;

    vector<int>::iterator vi ;  // iterator for v
    vi = v.begin () ;
    for (int i=0; i<5; i++)
        cout << vi[i] << " " ;
    }
```

The program defines an object v of class `std::vector` and then proceeds to fill v with elements using the member function *push_back()*. Next, a `vector` iterator is defined and initialised and the random access operator `[]` used to output the integer elements of v to `cout`.

20.5 Function Objects

Before we discuss various algorithms in the next section it is first necessary to briefly discuss function objects. Consider the following program, which iterates through a list to find all elements that are greater than 5:

```
// func_obj.cpp
//...
bool greater_than_5 (int i)
  {
  return i>5 ? true : false ;
  }

void main ()
  {
  list<int> l ;
  for (int i=0; i<10; i++)  // fill l
    l.push_back (i) ;

  list<int>::iterator li = find_if (l.begin(), l.end(),
                                    greater_than_5) ;
  if (li != l.end())
    cout << "first occurrence greater than 5: " << *li << endl ;
  }
```

The global function *greater_than_5()* returns logical-true if the argument `i` is greater than 5, else logical-false is returned. This function is passed as a logical predicate to the STL sequence algorithm function *find_if()*, which returns the first occurrence that satisfies the predicate; see later for a more detailed discussion of *find_if()*.

It is not ideal to use a separate function, such as *greater_than_5()*, for each application. Also, the particular function called is usually required to record data between function calls, which for a global function requires the definition of global variables. A more convenient, efficient and less error-prone solution is to use a member function of a class, an object of which can hold data. Thus, let's redesign the above using a function-like class called `GreaterThan`:

```
// func_obj.cpp
//...
template <class T>
class GreaterThan
  {
  private:
    T number ;
  public:
    GreaterThan (T _number=T())
      : number (_number) {}
    bool operator () (T n) { return n>number ? true : false ; }
  };

void main ()
  {
  //...
  GreaterThan<int> gt (5) ;
  li = find_if (l.begin(), l.end(), gt) ;
  //...
  }
```

Immediately we observe that the function-like class approach is more general since we can initialise an object of class `GreaterThan` to any number and are not restricted to testing against the number 5 as with the global function *`greater_than_5()`*. The STL function *`find_if()`* invokes `GreaterThan::operator()()` for each element of list `l`. The third argument of *`find_if()`* expects either a function or something that behaves like a function. By supporting **`operator`**`()()` class `GreaterThan` provides the usual function call syntax for objects that behave like functions. Such class objects are called *function-like objects* or simply *function objects*.

The STL provides two base types to assist programmers in developing function objects:

```
template <class Arg, class Result>
struct unary_function
  {
  typedef Arg    argument_type ;
  typedef Result result_type ;
  };

template <class Arg1, class Arg2, class Result>
struct binary_function
  {
  typedef Arg1   first_argument_type ;
  typedef Arg2   second_argument_type ;
  typedef Result result_type ;
  };
```

20.5.1 Predicates

A predicate is a function object that returns a Boolean value depending on whether the function arguments satisfy the predicate. The STL supports the nine predicates listed in Table 20.2 and the definition of `less` is:

Table 20.2 Predicate function objects.

Predicate	*Arity*	*Equivalent operator*
`less`	binary	`x<y`
`greater`	binary	`x>y`
`less_equal`	binary	`x<=y`
`greater_equal`	binary	`x>=y`
`equal_to`	binary	`x==y`
`not_equal_to`	binary	`x!=y`
`logical_and`	binary	`x&&y`
`logical_or`	binary	`x||y`
`logical_not`	unary	`!x`

```
template <class T>
struct less : public binary_function<T,T,bool>
  {
  bool operator () (const T& x, const T& y) const
    { return x < y ; }
  };
```

Using the `greater` predicate we can rewrite the call to *`find_if()`* in FUNC_OBJ.CPP as concisely as:

```
// func_obj.cpp
//...
li = find_if (l.begin(), l.end(),
              bind2nd(greater<int>(),5)) ;
//...
```

20.5.2 Arithmetic Function Objects

The STL makes available the standard arithmetic operators as function objects; see Table 20.3. These are useful when dealing with numeric classes.

20.5.3 Binders, Negaters and Member Function Adapters

The previous subsection illustrated the use of the `greater` predicate with the binder *`bind2nd()`*. This example illustrates a typical use of predicates and function objects in that we frequently require a new predicate or function object which is a slight variation of an existing one. To test each element of a list if it is greater than 5 requires the fixing or binding of the

Table 20.3 Arithmetic function objects.

Arithmetic function object	*Arity*	*Equivalent operator*
`plus`	binary	`x+y`
`minus`	binary	`x-y`
`multiplies`	binary	`x*y`
`divides`	binary	`x/y`
`modulus`	binary	`x%y`
`negate`	unary	`-x`

number 5 to the `greater` predicate and was performed using the *`bind2nd()`* binder. Thus, the unary predicate "greater than 5" was created from the binary predicate `greater` and the integer 5. The STL supports the following binders and negaters:

`bind2nd(y)` calls a binary function with `y` as the 2nd argument.
`bind1st(x)` calls a binary function with `x` as the 1st argument.
`not1()` negates a unary predicate.
`not2()` negates a binary predicate.

The interested reader may wish to refer to the related classes `binder2nd`, `binder1st`, `unary_negate` and `binary_negate` in FUNCTIONAL.H.

In Section 20.6.1.1 we will examine the *`for_each()`* function which traverses a sequence and calls a function for each object in the sequence. Consider calling a *`Print()`* global function for a `vector` of `Rectangle` objects:

```
void Print (Rectangle r)
  { /* ... */ }
//...
vector<Rectangle> vr;
//...
for_each (vr.begin(), vr.end(), Print) ;
```

Now suppose that *`Print()`* is a member function of class `Rectangle`. The following call to *`for_each()`* will result in a compilation error:

```
template <class T>
class Rectangle
  {
  //...
  void Print () ;
  //...
  }
//...
for_each (vr.begin(), vr.end(),
          Rectangle<int>::Print) ; // error
```

The reason this results in a compilation error is that a member function *`mf()`* needs to be called for an object p: *`p.mf()`*, whereas *`for_each()`* calls a global function *`f()`* simply as: *`f()`*.

The STL provides a solution to this problem in an analogous manner to the binder classes and functions discussed above, such as `binder2nd` and *`bind2nd()`*. Consider first the case in which it is required to call a member function that takes no arguments for elements of a container of pointers:

```
template <class R, class T>  // FUNCTIONAL.H
class mem_fun_t  : public unary_function<T*,R>
  {
  private:
    R (T::*pmf)() ;
  public:
    explicit mem_fun_t(R (T::*p)())
```

```
    : pmf(p) { }
  R operator ()(T* p) const
    { return (p->*pmf)() ; }
};

template <class R, class T>
mem_fun_t<R,T> mem_fun (R (T::*f)())
  { return mem_fun_t<R,T>(f) ; }
```

where `T` is the object type and `R` is the return type of the member function. There is an equivalent class declaration and member function for calling member functions via an object rather than through a pointer:

```
template <class R, class T>
class mem_fun_ref_t : public unary_function<T,R>
  { /* ... */ };

template <class R, class T>
mem_fun_ref_t<R,T> mem_fun_ref (R (T::*f)())
  { return mem_fun_ref_t<R,T>(f) ; }
```

Similar classes and member functions are supported to handle **const** member functions.

The call to *for_each()* to correctly invoke the *Print()* member function of `Rectangle` is now given by:

```
//...
vector<Rectangle<int> > vr ;
for_each (vr.begin(), vr.end(),
          mem_fun_ref(Rectangle<int>::Print)) ;
//...
```

To invoke one argument member functions the STL declares class `mem_fun1_t` and overloads function *mem_fun()*:

```
template <class R, class T, class A>
class mem_fun1_t : public binary_function<T*,A,R>
  {
  private:
    R (T::*pmf)(A);
  public:
    explicit mem_fun1_t(R (T::*p)(A))
      : pmf(p) {}
    R operator ()(T* p, A a) const
       { return (p->*pmf)(a) ; }
  };

  template <class R, class T, class A>
  mem_fun1_t<R,T,A> mem_fun (R (T::*f)(A))
    { return mem_fun1_t<R,T,A>(f) ; }
```

where A denotes the argument type of the member function. Class `mem_fun1_ref_t` is also declared for operating on objects rather than pointers, with support also for **const** member functions.

Algorithms such as *for_each()* which have a function argument do not differentiate between the argument being a function, a pointer to a function or a function object. It is important to binders, however, since they need to store a copy of the argument. As a result the STL defines two versions of the *ptr_fun()* function for using pointers to functions as arguments to STL algorithm functions:

```
template <class A, class R>
pointer_to_unary_function<A,R> ptr_fun (R (*f)(A)) ;

template <class A1, class A2, class R>
pointer_to_binary_function<A1,A2,R>
  ptr_fun (R (*f)(A1, A2)) ;
```

and cater for both unary and binary function objects.

20.6 Algorithms

In this section we review the key sequence (non-mutating and mutating), sorting, searching, set, heap, permutation and numeric algorithms supported by the STL.

20.6.1 Sequence Algorithms

Sequence algorithms manipulate data that consist of ordered sequences. The STL sequence algorithms consist of the two classifications: *mutating* and *non-mutating*. Mutating sequence algorithms can alter the values or the order of the sequence, whereas non-mutating algorithms treat a sequence as read-only and leave a sequence unaltered. Both types of algorithm are designed to operate on unsorted sequences. The following two subsections examine non-mutating and mutating sequence algorithms separately.

20.6.1.1 Non-Mutating Sequence Algorithms

The non-mutating sequence algorithms consist of several global functions, each of which will be examined in the following subsections.

Counting

The *count()* function searches a sequence for a specific object, `value`, and returns the number of times the object occurs in the sequence. The *count_if()* function counts the number of objects that satisfy a given predicate:

```
template <class InputIterator, class T>
typename iterator_traits<InputIterator>::difference_type
count (InputIterator first, InputIterator last,
       const T& value) ;

template <class InputIterator, class Predicate>
```

```
typename iterator_traits<InputIterator>::difference_type
count_if (InputIterator first, InputIterator last,
          Predicate pred) ;
```

The return type of both *count()* and *count_if()* is difference_type, which is a **typedef** of Distance of **struct** iterator:

```
template <class Category, class T, class Distance,/*...*/>
struct iterator
  {
  //...
  typedef Distance difference_type ;
  //...
  };
```

Examples of *count()* and *count_if()* are provided in ALGO_NM.CPP:

```
// algo_nm.cpp
//...
int array[10] ;
for (int i=0; i<10; i++)
  array[i] = i ;
array[4] = array[9] = 2 ;

ostream_iterator<int> out (cout, " ") ;
copy (array, array+10, out) ;

int count_2 = count (array, array+10, 2) ;
cout << endl << "there are " << count_2 << " twos" << endl ;

count_2 = count_if (array, array+10,
                    bind2nd(greater<int>(),2)) ;
cout << "there are " << count_2
     << " numbers greater than two" << endl ;
//...
```

An array of integers is defined and the number of integers equal to 2 are counted. Next the number of integers greater than 2 are determined using the greater predicate function object and *bind2nd()* binder. The output of the above program extract is:

```
0 1 2 3 2 5 6 7 8 2
there are 3 twos
there are 5 numbers greater than two
```

Equality

The *equal()* and *mismatch()* functions compare two sequences of equal length:

```
template <class InputIterator1, class InputIterator2>
bool equal (InputIterator1 first1, InputIterator1 last1,
            InputIterator2 first2) ;
```

```
template <class InputIterator1, class InputIterator2,
          class BinaryPredicate>
bool equal (InputIterator1 first1, InputIterator1 last1,
            InputIterator2 first2,
            BinaryPredicate binary_pred) ;

template <class InputIterator1, class InputIterator2>
pair<InputIterator1,InputIterator2>
  mismatch (InputIterator1 first1, InputIterator1 last1,
            InputIterator2 first2) ;

template <class InputIterator1, class InputIterator2,
          class BinaryPredicate>
pair<InputIterator1, InputIterator2>
  mismatch (InputIterator1 first1, InputIterator1 last1,
            InputIterator2 first2,
            BinaryPredicate binary_pred) ;
```

equal() returns a Boolean indicating whether or not two sequences are equivalent whereas *mismatch()* returns the first pair of elements (via iterators to those elements) that are not equal.

Finding

Several variants of the finding algorithm are supported by the STL:

```
template <class InputIterator, class T>
InputIterator find (InputIterator first,
                    InputIterator last, const T& value) ;

template <class InputIterator, class Predicate>
InputIterator find_if (InputIterator first,
                       InputIterator last, Predicate pred) ;

template <class ForwardIterator1, class ForwardIterator2>
ForwardIterator1 find_first_of (ForwardIterator1 first1,
ForwardIterator1 last1,
ForwardIterator2 first2, ForwardIterator2 last2) ;

template <class ForwardIterator1, class ForwardIterator2,
          class BinaryPredicate>
ForwardIterator1 find_first_of (ForwardIterator1 first1,
  ForwardIterator1 last1, ForwardIterator2 first2,
  ForwardIterator2 last2, BinaryPredicate pred) ;

template <class ForwardIterator1, class ForwardIterator2>
ForwardIterator1 find_end (ForwardIterator1 first1,
  ForwardIterator1 last1,
  ForwardIterator2 first2, ForwardIterator2 last2) ;

template <class ForwardIterator1, class ForwardIterator2,
          class BinaryPredicate>
```

```
ForwardIterator1 find_end (ForwardIterator1 first1,
  ForwardIterator1 last1, ForwardIterator2 first2,
  ForwardIterator2 last2, BinaryPredicate pred) ;

template <class ForwardIterator>
ForwardIterator adjacent_find (ForwardIterator first,
                               ForwardIterator last) ;

template <class ForwardIterator, class BinaryPredicate>
ForwardIterator adjacent_find (ForwardIterator first,
ForwardIterator last, BinaryPredicate binary_pred) ;
```

find() searches a sequence from `first` to `last` for `value` and returns an iterator referencing the first occurrence of `value`. If `value` is not found then an iterator to `last` is returned. The *find_if()* function searches a sequence for an object which satisfies the specified predicate `pred`. The *find_end()* function searches for its second sequence (`first2,last2`) as a sub-sequence of its first input sequence (`first1,last1`). If the second sequence is found, *find_end()* returns an iterator to the last match in its first sequence. *find_end()* is opposite to the *search()* function, which is discussed later. The function *find_first_of()* finds the first object of a sequence (`first1, last1`) that has a match in a second sequence (`first2,last2`). Function *adjacent_find()* searches a sequence for two adjacent objects that satisfy a specified relationship, which is equality for the first form. An illustration of *find_if()* was discussed in program FUNC_OBJ.CPP in connection with function objects and examples of the above find functions can be found in ALGO_NM.CPP:

```
// algo_nm.cpp
//...
vector<int> v ;
for (int i=0; i<5; i++)
  v.push_back (i) ;
v[2] = 5 ;
cout << "v: " ;
copy (v.begin(),v.end(),out) ;
cout << endl ;

int value (3) ;
vector<int>::iterator vi = find (v.begin(), v.end(), value) ;
if (vi != v.end())
  cout << value << " located in sequence" << endl ;

vi = find_if (v.begin(), v.end(),
              bind2nd(greater<int>(),3)) ;
if (vi != v.end())
  cout << "first occurrence greater than 3: " << *vi << endl ;

int array1[] = { 1, 2, 3 };
int array2[] = { 0, 2, 4, 6, 8 };
int* fp = find_first_of (array1, array1+3,
                         array2, array2+5) ;
if (fp != 0)
  cout << "first object of array1 with a match in array2 is: "
```

```
        << *fp << endl ;

int* ap = adjacent_find (v.begin(),v.end(),greater<int>()) ;
if (ap != 0)
  cout << "object " << *ap
       << " is greater than adjacent object "
       << *(ap+1) << endl ;
//...
```

with output:

```
v: 0 1 5 3 4
3 located in sequence
first occurrence greater than 3: 5
first object of array1 with a match in array2 is: 2
object 5 is greater than adjacent object 3
```

For Each

The *for_each()* function traverses a sequence and calls a specified function, `f`, for every object of the sequence:

```
template <class InputIterator, class Function>
Function for_each (InputIterator first,
                   InputIterator last, Function f) ;
```

The following illustrates calling the *PrintString()* function for each `string` element of `vector vs`:

```
// algo_nm.cpp
//...
void PrintString (const string& s)
  {
  cout << s << endl ;
  }

void main ()
  {
  //...
  string s1 ("one"), s2 ("two"), s3 ("three") ;
  vector<string> vs ;
  vs.push_back (s1) ; vs.push_back (s2) ; vs.push_back (s3) ;

  for_each (vs.begin(), vs.end(), PrintString) ;
  //...
  }
```

Note that *for_each()* is a non-mutating function so that it does not explicitly modify a sequence. For instance, a common misuse of *for_each()* is illustrated below in which each element of a vector of integers is multiplied by 2:

```
vector<int> v ;
```

```
//...
for_each (v.begin(), v.end(), bind2nd(multiplies<int>(),2)) ;
//....
}
```

Following the call to *for_each()* the elements of v will be left unaltered since *for_each()* simply invokes the multiplies function object.

Searching

The STL provides the two functions *search()* and *search_n()* for searching a sequence for a sub-sequence:

```
template <class ForwardIterator1, class ForwardIterator2>
ForwardIterator1 search (
  ForwardIterator1 first1,ForwardIterator1 last1,
  ForwardIterator2 first2,ForwardIterator2 last2) ;

template <class ForwardIterator1, class ForwardIterator2,
          class BinaryPredicate>
ForwardIterator1 search (
  ForwardIterator1 first1,ForwardIterator1 last1,
  ForwardIterator2 first2,ForwardIterator2 last2,
  BinaryPredicate binary_pred) ;

template <class ForwardIterator, class Size, class T>
ForwardIterator search_n (
  ForwardIterator first, ForwardIterator last,
  Size n, const T& value) ;

template <class ForwardIterator, class Size, class T,
          class BinaryPredicate>
ForwardIterator search_n (
  ForwardIterator first, ForwardIterator last,
  Size n, const T& value, BinaryPredicate pred) ;
```

search() searches for the second sequence (`first2, last2`) as a sub-sequence of the first sequence (`first1, last1`). If a match is found then an iterator is returned to the first matching element in the first sequence, otherwise the end of sequence (`last1`) is returned to indicate failure. Function *search_n()* searches a sequence for at least `n` matches of the object `value`. For example:

```
// algo_nm.cpp
//...
bool SearchString (string& s, const string& sub_s)
  {
  char* p = search (s.begin(), s.end(),
                    sub_s.begin(), sub_s.end()) ;
  if (p != s.end())
    return true ;
  return false ;
  }
```

```
void main ()
  {
  //...
  string text ("the cream of the cream") ;
  cout << "string \"of\" in text: "
       << SearchString (text, "of") << endl ;
  //...
  }
```

20.6.1.2 Mutating Sequence Algorithms

The following sub-sections examine the numerous mutating sequence algorithms of the STL.

Copying

The *copy()* function has been extensively used throughout previous sections. *copy()* copies the elements of one sequence to another in the forward direction. In addition, function *copy_backward()* copies in the reverse direction:

```
template <class InputIterator, class OutputIterator>
OutputIterator copy (InputIterator first, InputIterator last,
                     OutputIterator result) ;

template <class BidirectionalIterator1,
          class BidirectionalIterator2>
BidirectionalIterator2 copy_backward
  (BidirectionalIterator1 first, BidirectionalIterator1 last,
   BidirectionalIterator2 result) ;
```

The following program code from ALGO_M.CPP illustrates copying a `vector v` using *copy_backward()*:

```
// algo_nm.cpp
//...
vector<int> v ;
//...
vector<int> vc (v.size()) ; // copy v to vc
copy_backward (v.begin(), v.end(), vc.end()) ;
//...
```

noting that a bidirectional iterator is required for both sequences.

Fill and Generate

The filling algorithm is designed to fill all or part of a sequence with copies of a specified object, `value`. There are two versions of filling, *fill()* and *fill_n()*:

```
void fill (ForwardIterator first, ForwardIterator last,
           const T& value) ;

template <class OutputIterator, class Size, class T>
void fill_n (OutputIterator first, Size n, const T& value) ;
```

fill() assigns value to elements within the range first to last, whereas *fill_n()* assigns value to the n elements of a sequence starting from first. The filling algorithms are useful for initialising (via assignment rather than initialisation) sequences, as illustrated below:

```
// algo_m.cpp
//...
int array[10] ;
fill_n (array, 10, 1) ;
//...
```

The STL also supports the following generate functions:

```
template <class ForwardIterator, class Generator>
void generate (ForwardIterator first, ForwardIterator last,
               Generator gen) ;

template <class OutputIterator, class Size, class Generator>
void generate_n (OutputIterator first, Size n,
                 Generator gen) ;
```

generate() and *generate_n()* are equivalent to *fill()* and *fill_n()* except that rather than assigning a single value they assign values obtained by calling its function argument repeatedly. Thus, filling is a special case of the generate algorithm.

Partitioning

The partition algorithm partitions a sequence so that every element that satisfies a user-specified predicate is placed before every element that does not satisfy the predicate:

```
template <class BidirectionalIterator, class Predicate>
BidirectionalIterator partition (BidirectionalIterator first,
                                 BidirectionalIterator last,
                                 Predicate pred) ;

template <class BidirectionalIterator, class Predicate>
BidirectionalIterator stable_partition
  (BidirectionalIterator first, BidirectionalIterator last,
   Predicate pred) ;
```

The *stable_partition()* function maintains relative order among the elements that do and do not satisfy the predicate.

Random Shuffle

The random_shuffle() function rearranges the elements of a sequence so that their order is random:

```
template <class RandomAccessIterator>
void random_shuffle (RandomAccessIterator first,
                     RandomAccessIterator last) ;

template <class RandomAccessIterator,
```

```
          class RandomNumberGenerator>
void random_shuffle (RandomAccessIterator first,
                     RandomAccessIterator last,
                     RandomNumberGenerator& rand) ;
```

The first version shuffles a sequence using a uniform distribution random number generator. The second version allows an alternative different random number generator distribution to be specified via the function object `rand`.

Remove

The following set of four functions remove elements from a sequence:

```
template <class ForwardIterator, class T>
ForwardIterator remove (ForwardIterator first,
  ForwardIterator last, const T& value) ;

template <class ForwardIterator, class Predicate>
ForwardIterator remove_if (ForwardIterator first,
  ForwardIterator last, Predicate pred) ;

template <class InputIterator, class OutputIterator, class T>
OutputIterator remove_copy (InputIterator first,
  InputIterator last, OutputIterator result, const T& value) ;

template <class InputIterator, class OutputIterator,
          class Predicate>
OutputIterator remove_copy_if (InputIterator first,
  InputIterator last, OutputIterator result, Predicate pred) ;
```

`remove()` removes all occurrences of `value` within the specified sequence range (`first`, `last`). *`remove_if()`* removes elements which satisfy a user supplied predicate. The two copy functions leave the input sequence unaltered and copy the resulting sequence to `result`.

Replace

The replace algorithm replaces all occurences of objects in a sequence by other objects. There are four versions of the replace algorithm. The first function, *`replace()`*, traverses all elements in the range `first` to `last` and replaces all occurrences of `old_value` by `new_value`. Function *`replace_if()`* replaces all objects in a sequence that satisfy a specified predicate, `pred`. *`replace_copy()`* replaces elements of a sequence and produces a new sequence, `result`, which is returned upon termination. Finally, *`replace_copy_if()`* is analogous to *`replace_copy()`* provided that the specified predicate is satisfied. The declarations are as follows:

```
template <class ForwardIterator, class T>
void replace (ForwardIterator first, ForwardIterator last,
              const T& old_value, const T& new_value) ;

template <class ForwardIterator, class Predicate, class T>
void replace_if (ForwardIterator first,
                 ForwardIterator last,
```

```
                    Predicate pred, const T& new_value) ;

template <class InputIterator, class OutputIterator, class T>
OutputIterator replace_copy (InputIterator first,
  InputIterator last, OutputIterator result,
  const T& old_value, const T& new_value) ;

template <class Iterator, class OutputIterator,
          class Predicate, class T>
OutputIterator replace_copy_if (Iterator first,
  Iterator last, OutputIterator result,
  Predicate pred, const T& new_value) ;
```

The above replace functions are seen to be equivalent to those of the remove functions discussed in the previous subsection. The following extract from ALGO_M.CPP illustrates the use of *replace_if()* to replace elements of a floating-point array by 0 if they are less than 0.01:

```
// algo_m.cpp
//...
double darray[] = { 0.1256, 0.0067, 0.6731, 0.0004, 0.5582 };
//...
replace_if (darray, darray+5,
            bind2nd(less<double>(),0.01), 0.0) ;
//...
```

Reverse

The reverse algorithm reverses the elements of a sequence in the range `first` to `last`:

```
template <class BidirectionalIterator>
void reverse (BidirectionalIterator first,
              BidirectionalIterator last) ;

template <class BidirectionalIterator, class OutputIterator>
OutputIterator reverse_copy (BidirectionalIterator first,
                             BidirectionalIterator last,
                             OutputIterator result) ;
```

reverse_copy() reverses and copies the original elements to the destination sequence starting at `result`.

Rotate

The rotate algorithm rotates the elements of a sequence as if the beginning and end of the sequence were connected. There are two versions of rotate:

```
template <class ForwardIterator>
void rotate (ForwardIterator first, ForwardIterator middle,
             ForwardIterator last) ;

template <class ForwardIterator, class OutputIterator>
OutputIterator rotate_copy (ForwardIterator first,
                            ForwardIterator middle,
```

```
                                    ForwardIterator last,
                                    OutputIterator result) ;
```

rotate() rotates the sequence from `first` to `last` so that element `middle` is placed where its `first` element used to be. *rotate_copy()* rotates and copies the input sequence placing the output sequence in `result`. The following example rotates the integer elements of an array about the 5th element:

```
// algo_m.cpp
//...
int array[10] ;
for (int i=0; i<10; i++)
  array[i] = i ;
cout << "array: " ;
copy (array, array+10, out) ;
cout << endl ;

int* middle = array + 5 ;
rotate (array, middle, array+10) ;

cout << "array after rotate: " ;
copy (array, array+10, out) ;
cout << endl ;
//...
```

with output:

```
array: 0 1 2 3 4 5 6 7 8 9
array after rotate: 5 6 7 8 9 0 1 2 3 4
```

Swapping

To swap the elements of a sequence there are three versions of swap functions:

```
template <class T>
void swap (T& a, T& b) ;

template <class ForwardIterator1, class ForwardIterator2>
void iter_swap (ForwardIterator1 a, ForwardIterator2 b) ;

template <class ForwardIterator1, class ForwardIterator2>
ForwardIterator2 swap_ranges
  (ForwardIterator1 first1, ForwardIterator1 last1,
   ForwardIterator2 first2) ;
```

The first of the three functions is the standard *swap()* function and swaps objects `a` and `b` of type `T`. *iter_swap()* swaps the elements pointed to by its iterator arguments. *swap_ranges()* swaps the elements in the range `first1` to `last1` with the elements in the range of the same length beginning at `first2`. The following illustrates swapping the elements of two vectors of equal length:

```
// algo_m.cpp
//...
vector<int> u, v ;
//...
swap_ranges (v.begin(), v.end(), u.begin()) ;
//...
```

Note, however, that the container classes support their own, more efficient versions of *swap()* which should be used instead of the global swap functions.

Transform

The *transform()* function produces a transformation of the input sequence based on a user-supplied operation, and has the folowing two signatures:

```
template <class InputIterator, class OutputIterator,
          class UnaryOperation>
OutputIterator transform (InputIterator first,
  InputIterator last, OutputIterator result,
  UnaryOperation op) ;

template <class InputIterator1, class InputIterator2,
          class OutputIterator, class BinaryOperation>
OutputIterator transform (InputIterator1 first1,
  InputIterator1 last1, InputIterator2 first2,
  OutputIterator result, BinaryOperation binary_op) ;
```

The first version operates on a single sequence in the range (`first`, `last`) and requires a unary function object, `op`. The resulting sequence, `result`, is of the same length as the input sequence. The second version operates with two sequences of the same length with the second sequence starting at `first2` and requires a binary function object. For example, the following uses *transform()* to multiply the elements of two vectors of integers:

```
// algo_m.cpp
//...
vector<int> u, v, uv ;
//...
transform (u.begin(), u.end(), v.begin(),
           uv.begin(), multiplies<int>()) ;
//...
```

Unique

The *unique()* and *unique_copy()* function declarations are:

```
template <class ForwardIterator>
ForwardIterator unique (ForwardIterator first,
                        ForwardIterator last) ;

template <class ForwardIterator, class BinaryPredicate>
ForwardIterator unique (ForwardIterator first,
  ForwardIterator last, BinaryPredicate binary_pred) ;
```

```
template <class InputIterator, class OutputIterator>
OutputIterator unique_copy (InputIterator first,
  InputIterator last, OutputIterator result) ;

template <class InputIterator, class OutputIterator,
          class BinaryPredicate>
OutputIterator unique_copy (InputIterator first,
  InputIterator last, OutputIterator result,
  BinaryPredicate binary_pred) ;
```

unique() eliminates adjacent duplicates from a sequence, whereas *unique_copy()* makes a copy without duplicates. Since duplicated elements are only deleted provided they are adjacent, the unique algorithm is generally applied to sorted sequences. This is illustrated in the following example:

```
// algo_m.cpp
//...
list<string> strings, unique_strings ;
strings.push_back ("one") ; strings.push_back ("two") ;
strings.push_back ("three") ; strings.push_back ("two") ;
strings.sort () ;
unique_copy (strings.begin(), strings.end(),
             back_inserter(unique_strings)) ;
//...
```

20.6.2 Sorting and Searching Algorithms

In general, sorting and searching a sequence of elements are the most frequently required operations. As a result, the STL supports several sorting and searching optimally efficient algorithms.

20.6.2.1 Sorting

The most efficient sorting algorithms are based on the quicksort algorithm which has optimum $O(n \log n)$ efficiency. The STL sorting algorithms consist of the three functions *sort()*, *stable_sort()* and *partial_sort()* with partial sort having *_copy()* variants. Let's consider the *sort()* function first:

```
template <class RandomAccessIterator>
void sort (RandomAccessIterator first,
           RandomAccessIterator last) ;

template <class RandomAccessIterator, class Compare>
void sort (RandomAccessIterator first,
           RandomAccessIterator last, Compare comp) ;
```

The first version of *sort()* sorts the elements of a sequence from `first` to `last` in ascending order. The second form of *sort()* allows a comparison function object to be specified if an ordering other than ascending is required. The *stable_sort()* function preserves the relative order of equal elements, whereas *sort()* makes no such guarantee:

```
template <class RandomAccessIterator>
void stable_sort (RandomAccessIterator first,
                  RandomAccessIterator last) ;

template <class RandomAccessIterator, class Compare>
void stable_sort (RandomAccessIterator first,
                  RandomAccessIterator last, Compare comp) ;
```

partial_sort() sorts only a portion of a sequence. Elements only within the range `first` to `middle` are sorted:

```
template <class RandomAccessIterator>
void partial_sort (RandomAccessIterator first,
                   RandomAccessIterator middle,
                   RandomAccessIterator last) ;

template <class RandomAccessIterator, class Compare>
void partial_sort (RandomAccessIterator first,
                   RandomAccessIterator middle,
                   RandomAccessIterator last, Compare comp) ;
```

The partial sort algorithm also supports the overloaded *partial_sort_copy()* function and places the copy in the sequence pointed to by `result_first` and `result_last`:

```
template <class InputIterator, class RandomAccessIterator>
RandomAccessIterator partial_sort_copy
  (InputIterator first, InputIterator last,
   RandomAccessIterator result_first,
   RandomAccessIterator result_last) ;

template <class InputIterator, class RandomAccessIterator,
          class Compare>
RandomAccessIterator partial_sort_copy
  (InputIterator first, InputIterator last,
   RandomAccessIterator result_first,
   RandomAccessIterator result_last, Compare comp) ;
```

The following program illustrates sorting an array of integers using *sort()*:

```
// sort.cpp
//...
void main ()
  {
  int array[10] ;
  for (int i=0; i<10; i++)
    array[i] = i ;

  random_shuffle (array, array+10) ;

  ostream_iterator<int> out (cout, " ") ;
  cout << "unsorted array: " ;
```

```
    copy (array, array+10, out) ;
    cout << endl ;

    sort (array, array+10) ;
    cout << "sorted array  : " ;
    copy (array, array+10, out) ;
    //...
    }
```

with output:

```
unsorted array: 6 8 9 2 1 4 3 7 0 5
sorted array  : 0 1 2 3 4 5 6 7 8 9
```

20.6.2.2 Nth Element

The *n*th. element algorithm sorts only as far as is necessary so that the *n*th. element is in its correctly sorted position. In both versions of *nth_element()* argument nth is an iterator that refers to the position of the element in the final sorted sequence:

```
template <class RandomAccessIterator>
void nth_element (RandomAccessIterator first,
                  RandomAccessIterator nth,
                  RandomAccessIterator last) ;

template <class RandomAccessIterator, class Compare>
void nth_element (RandomAccessIterator first,
                   RandomAccessIterator nth,
                  RandomAccessIterator last, Compare comp) ;
```

The following program illustrates *nth_element()* to sort the 5th. element of integer array array:

```
// sort.cpp
//...
void main ()
  {
  //...
  random_shuffle (array, array+10) ;

  cout << "unsorted array            : " ;
  copy (array, array+10, out) ;
  cout << endl ;

  nth_element (array, array+5, array+10) ;
  cout << "nth element sorted array  : " ;
  copy (array, array+10, out) ;
  cout << endl ;
  //...
  }
```

with output:

```
unsorted array:           0 9 1 3 5 4 8 6 7 2
nth element sorted array: 0 2 1 3 4 5 6 7 8 9
```

20.6.2.3 Binary Search

If the elements of a sequence are sorted then the binary search algorithm provides an efficient (taking logarithmic time) method of searching for an element within the sequence. There are two versions of *binary_search()* which search a sequence (first, last) for value:

```
template <class ForwardIterator, class T>
bool binary_search (ForwardIterator first,
                    ForwardIterator last, const T& value) ;

template <class ForwardIterator, class T, class Compare>
bool binary_search (ForwardIterator first,
                    ForwardIterator last,
                    const T& value, Compare comp) ;
```

The second version of *binary_search()* uses a user supplied comparison function, comp.

20.6.2.4 Equal Range, Lower and Upper Bound

The *equal_range()* function searches a sorted sequence from first to last for a range of elements equal to value with and without a comaprison function, comp:

```
template <class ForwardIterator, class T>
pair<ForwardIterator, ForwardIterator>
equal_range (ForwardIterator first, ForwardIterator last,
             const T& value) ;

template <class ForwardIterator, class T, class Compare>
pair<ForwardIterator, ForwardIterator>
equal_range (ForwardIterator first, ForwardIterator last,
             const T& value, Compare comp) ;
```

The *lower_bound()* function alternatively searches a sorted sequence for the first element that is greater than or equal to value:

```
template <class ForwardIterator, class T>
ForwardIterator lower_bound (ForwardIterator first,
                             ForwardIterator last,
                             const T& value) ;
template <class ForwardIterator, class T, class Compare>
ForwardIterator lower_bound (ForwardIterator first,
                             ForwardIterator last,
                             const T& value, Compare comp) ;
```

The function *upper_bound()* is similarly supported for determining the first element that is greater than value.

The following example illustrates *equal_range()*:

```
// sort.cpp
//...
void main ()
  {
  //...
  list<int> l ;
  for (int i=0; i<10; i+=2)
    {
    l.push_back (i) ; l.push_back (i) ;
    }
  cout << "list l     : " ;
  copy (l.begin(), l.end(), out) ;
  cout << endl ;

   pair<list<int>::iterator,list<int>::iterator>
    er = equal_range (l.begin(), l.end(), 4) ;
  cout << "equal range: " ;
  copy (er.first, er.second, out) ;
  //...
  }
```

with output:

```
list l     : 0 0 2 2 4 4 6 6 8 8
equal range: 4 4
```

20.6.2.5 Merge

The *merge()* function allows us to merge two sorted sequences into a new sorted sequence or to merge two parts of a single sequence using *inplace_merge()*:

```
template <class InputIterator1, class InputIterator2,
          class OutputIterator>
OutputIterator merge (InputIterator1 first1,
                      InputIterator1 last1,
                      InputIterator2 first2,
                      InputIterator2 last2,
                      OutputIterator result) ;

template <class InputIterator1, class InputIterator2,
          class OutputIterator, class Compare>
OutputIterator merge (InputIterator1 first1,
                      InputIterator1 last1,
                      InputIterator2 first2,
                      InputIterator2 last2,
                      OutputIterator result, Compare comp) ;

template <class BidirectionalIterator>
void inplace_merge (BidirectionalIterator first,
```

```
                    BidirectionalIterator middle,
                    BidirectionalIterator last) ;

template <class BidirectionalIterator, class Compare>
void inplace_merge (BidirectionalIterator first,
                    BidirectionalIterator middle,
                    BidirectionalIterator last,
                    Compare comp) ;
```

20.6.3 Set Algorithms

The STL provides set operations for sorted sequences. The set operations supported are inclusion, union, intersection, difference and symmetric difference with each conforming to the standard mathematical definitions. Testing if all the members of one set (`first2,last2`) are in another set (`first1,last1`) is performed by *`includes()`*, with the usual support for a comparison function:

```
template <class InputIterator1, class InputIterator2>
bool includes (InputIterator1 first1, InputIterator1 last1,
               InputIterator2 first2, InputIterator2 last2) ;

template <class InputIterator1, class InputIterator2,
          class Compare>
bool includes (InputIterator1 first1, InputIterator1 last1,
               InputIterator2 first2, InputIterator2 last2,
               Compare comp) ;
```

The union of two sets is performed by function *`set_union()`*:

```
template <class InputIterator1, class InputIterator2,
          class OutputIterator>
OutputIterator set_union
  (InputIterator1 first1, InputIterator1 last1,
   InputIterator2 first2, InputIterator2 last2,
   OutputIterator result) ;

template <class InputIterator1, class InputIterator2,
          class OutputIterator, class Compare>
OutputIterator set_union
  (InputIterator1 first1, InputIterator1 last1,
   InputIterator2 first2, InputIterator2 last2,
   OutputIterator result, Compare comp) ;
```

`set_intersection()` computes the intersection of two sets:

```
template <class InputIterator1, class InputIterator2,
          class OutputIterator>
OutputIterator set_intersection
  (InputIterator1 first1, InputIterator1 last1,
   InputIterator2 first2, InputIterator2 last2,
   OutputIterator result) ;
```

```
template <class InputIterator1, class InputIterator2,
          class OutputIterator, class Compare>
OutputIterator set_intersection
  (InputIterator1 first1, InputIterator1 last1,
   InputIterator2 first2, InputIterator2 last2,
   OutputIterator result, Compare comp) ;
```

set_difference() generates a sequence of elements that are members of its first, but not second, input sequence:

```
template <class InputIterator1, class InputIterator2,
          class OutputIterator>
OutputIterator set_difference
  (InputIterator1 first1, InputIterator1 last1,
   InputIterator2 first2, InputIterator2 last2,
   OutputIterator result) ;

template <class InputIterator1, class InputIterator2,
          class OutputIterator, class Compare>
OutputIterator set_difference
  (InputIterator1 first1, InputIterator1 last1,
   InputIterator2 first2, InputIterator2 last2,
   OutputIterator result, Compare comp) ;
```

Finally, the *set_symmetric_difference()* function generates a sequence that are members of either, but not both, of the input sequences:

```
template <class InputIterator1, class InputIterator2,
          class OutputIterator>
OutputIterator set_symmetric_difference
  (InputIterator1 first1, InputIterator1 last1,
   InputIterator2 first2, InputIterator2 last2,
   OutputIterator result) ;

template <class InputIterator1, class InputIterator2,
          class OutputIterator, class Compare>
OutputIterator set_symmetric_difference
  (InputIterator1 first1, InputIterator1 last1,
   InputIterator2 first2, InputIterator2 last2,
   OutputIterator result, Compare comp) ;
```

An illustration of *set_union()* is provided below:

```
// set_algo.cpp
//...
void main ()
  {
  int seta[] = { 0, 1, 2, 3, 4  };
  int setb[] = { 2, 4, 8, 9, 10 };

  ostream_iterator<int> out (cout, " ") ;
```

```
cout << "seta: " ;
copy (seta, seta+5, out) ;
//...
int result[10] ;
int* end = set_union (seta, seta+5, setb, setb+5, result) ;
cout << "union of seta and setb: " ;
copy (result, end, out) ;
}
```

with output:

```
seta: 0 1 2 3 4
setb: 2 4 8 9 10
union of seta and setb: 0 1 2 3 4 8 9 10
```

20.6.4 Heap Algorithms

The heap algorithms in the STL consist of a binary tree in which the value of any node in the tree is greater than or equal to any of its, at most, two children. This differs slightly from a binary search tree, whose parent node value is greater than its left child node value but less than or equal to its right child node value. It follows that the largest value of a heap is held by the root node which makes heaps ideal for applications which require repeated access to the largest values of a sequence. Heaps are also an extremely efficient data structure requiring $O(\log n)$ time for both the addition and deletion of an element.

The heap functions are as follows:

```
template <class RandomAccessIterator>
void push_heap (RandomAccessIterator first,
                RandomAccessIterator last) ;

template <class RandomAccessIterator, class Compare>
void push_heap (RandomAccessIterator first,
                RandomAccessIterator last,
                Compare comp) ;

template <class RandomAccessIterator>
void pop_heap (RandomAccessIterator first,
               RandomAccessIterator last) ;

template <class RandomAccessIterator, class Compare>
void pop_heap (RandomAccessIterator first,
               RandomAccessIterator last,
               Compare comp) ;

template <class RandomAccessIterator>
void make_heap (RandomAccessIterator first,
                RandomAccessIterator last) ;

template <class RandomAccessIterator, class Compare>
void make_heap (RandomAccessIterator first,
                RandomAccessIterator last,
                Compare comp) ;
```

```
template <class RandomAccessIterator>
void sort_heap (RandomAccessIterator first,
                RandomAccessIterator last) ;

template <class RandomAccessIterator, class Compare>
void sort_heap (RandomAccessIterator first,
                RandomAccessIterator last,
                Compare comp) ;
```

Elements are added and deleted from a heap using the *push_heap()* and *pop_heap()* functions. *make_heap()* constructs a heap from the elements of a sequence. *sort_heap()* sorts the elements of a sequence represented as a heap. Upon completion the heap is destroyed and replaced by the sorted sequence. An example of making a heap is given below:

```
// heap.cpp
//...
void main ()
  {
  int array[] = { 3, 0, 1, 2, 4 };
  //...
  make_heap (array, array+5) ;
  //...
  }
```

20.6.5 Permutation Algorithms

Consider the string *abcd*. By rearranging the characters *a*, *b*, *c* and *d* we can generate 24 different and unique permutations of this string; for example *abdc*, *acbd* and so on. The number of permutations for a sequence consisting of *n* elements is *n*! The STL provides the two overloaded functions *next_permutation()* and *prev_permutation()* for generating the next and previous permutations of a sequence:

```
template <class BidirectionalIterator>
bool next_permutation (BidirectionalIterator first,
                       BidirectionalIterator last) ;

template <class BidirectionalIterator, class Compare>
bool next_permutation (BidirectionalIterator first,
                       BidirectionalIterator last,
                       Compare comp) ;

template <class BidirectionalIterator>
bool prev_permutation (BidirectionalIterator first,
                       BidirectionalIterator last) ;

template <class BidirectionalIterator, class Compare>
bool prev_permutation (BidirectionalIterator first,
                       BidirectionalIterator last,
                       Compare comp) ;
```

Each function returns a Boolean indicating whether the next or previous permutation exists. The permutations are produced in lexicographical order. Note that if the elements of a sequence are ordered then all possible permutations will be generated, otherwise permutations will be generated until no more permutations exist.

The following program illustrates using *next_permutation()* to generate all pemutations of a user entered string:

```
// perm.cpp
//...
void main ()
  {
  string s ;

  cout << "enter a string: " ;
  cin  >> s ;

  // first perform lexicographical sort on string
  sort (s.begin(), s.end()) ;

  cout << "all permutations of string: " << endl << s << '\t' ;

  int count (1) ;
  while (next_permutation (s.begin(), s.end()))
    {
    cout << s ;
    count++ ;
    count % 10 == 0 ? cout << endl : cout << '\t' ;
    }
  }
```

with some user interaction:

```
enter a string: cat
all permutations of string:
act atc cat cta tac tca
```

The program above illustrated permutations of strings based on lexicographical order. The STL also supports the function *lexicographical_compare()* for performing a lexicographical comparison of two sequences:

```
template <class InputIterator1, class InputIterator2>
bool lexicographical_compare
  (InputIterator1 first1, InputIterator1 last1,
   InputIterator2 first2, InputIterator2 last2) ;

template <class InputIterator1, class InputIterator2,
          class Compare>
bool lexicographical_compare
  (InputIterator1 first1, InputIterator1 last1,
   InputIterator2 first2, InputIterator2 last2,
   Compare comp) ;
```

20.6.6 Numeric Algorithms

If you are involved in a lot of numerical programming you may be disappointed by the support for numeric algorithms: there are no matrix classes and no linear systems solvers in the STL! However, the STL does support a few generalised numeric functions which are declared in the header file NUMERIC.H.

20.6.6.1 Accumulate

The *accumulate()* function is a generalisation of a sum of all the elements of a vector:

```
template <class InputIterator, class T>
T accumulate (InputIterator first, InputIterator last,
              T init) ;

template <class InputIterator, class T,
          class BinaryOperation>
T accumulate (InputIterator first, InputIterator last,
              T init, BinaryOperation binary_op) ;
```

The first version adds all elements of a sequence using the + operator, whereas the second version uses the supplied `binary_op` operator. The accumulation of the sequence elements is initialised to `init`.

20.6.6.2 Inner Product

Function *inner_product()* computes the inner product of two sequences of the same length:

```
template <class InputIterator1, class InputIterator2,
          class T>
T inner_product (InputIterator1 first1, InputIterator1 last1,
                 InputIterator2 first2, T init) ;

template <class InputIterator1, class InputIterator2,
          class T,
class BinaryOperation1, class BinaryOperation2>
T inner_product (InputIterator1 first1, InputIterator1 last1,
                 InputIterator2 first2, T init,
                 BinaryOperation1 binary_op1,
                 BinaryOperation2 binary_op2) ;
```

The inner product is generated by applying one operator to each pair of corresponding elements from the two sequences and then applying a second operator to all intermediate values. For example, consider the two sequences of integers [1, 2, 3, 4] and [5, 6, 7, 8]. The default operators for the first version of *inner_product()* are * and +. Thus, the intermediate sequence is [1*5, 2*6, 3*7, 4*8], the elements of which are now added, 5+12+21+32, to give a final result of 70. The second version allows the two binary operators to be explicitly specified. For example:

```
// numeric.cpp
//...
```

```
void main ()
  {
  int array1[] = { 1, 2, 3, 4 };
  int array2[] = { 5, 6, 7, 8 };
  //...
  int ip1 = inner_product (array1, array1+4, array2, 0) ;
  cout << "ip1: " << ip1 << endl ;
  int ip2 = inner_product (array1, array1+4, array2, 0,
                           plus<int>(), multiplies<int>()) ;
  cout << "ip2: " << ip2 << endl ;
  int ip3 = inner_product (array1, array1+4, array2, 1,
                           multiplies<int>(),plus<int>()) ;
  cout << "ip3: " << ip3 << endl ;
  //...
  }
```

and would produce the following output:

```
ip1: 70
ip2: 70
ip3: 5760
```

notintg that if the required outer operation is multiplication then the initial value is set to 1 and not 0.

20.6.6.3 Partial Sum and Adjacent Difference

The *partial_sum()* function calculates a running total of all elements in a sequence. The default operator is +, but an alternative operator can be specified by way of the function binary_op:

```
template <class InputIterator, class OutputIterator>
OutputIterator partial_sum (InputIterator first,
                            InputIterator last,
                            OutputIterator result) ;

template <class InputIterator, class OutputIterator,
          class BinaryOperation>
OutputIterator partial_sum (InputIterator first,
                            InputIterator last,
                            OutputIterator result,
                            BinaryOperation binary_op) ;
```

For example, given the sequence $[a,b,c,d]$ then *partial_sum()* generates the sequence $[a, a + b, a + b + c, a + b + c + d]$.

Function *adjacent_difference()* is the inverse of *partial_sum()* in that adjacent elements are substracted from each other, although an alternative operator can be specified. For example, for the sequence $[a, b, c, d]$ *adjacent_difference()* produces the sequence $[a, b - a, c - b, d - c]$. The overloaded declarations of *adjacent_difference()* are:

```
template <class InputIterator, class OutputIterator>
OutputIterator adjacent_difference
  (InputIterator first, InputIterator last,
   OutputIterator result) ;

template <class InputIterator, class OutputIterator,
          class BinaryOperation>
OutputIterator adjacent_difference
  (InputIterator first, InputIterator last,
   OutputIterator result, BinaryOperation binary_op) ;
```

20.7 Containers

The STL divides containers or abstract data types in to the two categories: *sequence* and *associative*. A sequence container stores objects in a sequential manner, such as a vector or linked-list. Objects of sequence containers are accessed either sequentially or randomly. An associative container alternatively associates a *key* to every container object and an object is accessed via its respective key. The implementation of associative containers is generally more complex and requests additional memory than sequence containers, but the benefit is that indexing of associative containers is faster.

All containers support certain operations; see Table 20.4. A default constructor and destructor are provided. The `begin()` and `end()` member functions return iterators to the first and past-the-end container objects. Member functions `size()` and `max_size()` return the current number and maximum possible number of objects in a container. The member function `empty()` returns logical-true if a container is empty, while `swap()` swaps all objects of two different containers. In addition, all containers overload the logical operators ==, !=, <, >, <=, >= and the assignment operator =. If a container can support reversible operations then it will also support the `rbegin()` and `rend()` member functions.

In addition, all containers **public**ly encapsulate the types listed in Table 20.5, noting the non-**const** and **const** versions of `reference`, `iterator` and `reverse_iterator`. These are to assist in defining iterators, determining the type of container objects and so on. The types are generally declared in terms of `Allocator` types using **typedef**; for example the types of class `vector` are of the form:

```
template <class T, class Allocator=allocator<T> >
class vector
  {
```

Table 20.4 Operations common to all containers.

Operation	*Description*
`C(), ~C()`	Constructor and destructor
`iterator begin()`	Returns an iterator referring to the first container object
`iterator end()`	Returns a past-the-end iterator
`size_type size()`	Returns the number of objects in a container
`size_type max_size()`	Returns the maximum number of objects that a container can hold
bool `empty()`	Returns logical-true if a container is empty, else logical-false
void `swap (C& c)`	Swaps all objects of the operated on container
`==, !=, <, <=, >, >=, =`	Overloaded operators

Table 20.5 Types common to all containers.

Container	*Type*	*Description*
All containers	`reference, const_reference`	Type of reference to container objects
	`iterator, const_iterator`	Type of container iterator
	`size_type`	Type to specify the number of container objects
	`difference_type`	Type of distance between container objects
	`value_type`	Type of container objects
Reversible containers	`reverse_iterator`	Type of container reverse iterator
	`const_reverse_iterator`	Type of container constant reverse iterator
Associative containers	`key_type`	Type of container key
	`key_compare`	Type of function to compare container keys
	`value_compare`	Type of function to compare container values

```
public:
  //...
  typedef typename Allocator<T>::pointer iterator ;
  typedef T                              value_type ;
  //...
};
```

The following subsections examine sequence and associative containers separately.

20.7.1 Sequence Containers

The following lists the declarations of additional operations supported by all sequence containers. Two additional constructors are provided for class C, *back()* and *front()* member functions, two versions of the *erase()* member function, three versions of the *insert()* member function and *pop* and *push* functions:

```
C (size_type n, const T& value=T()) ;
C (const_iterator first, const_iterator last) ;
reference back () ;
void      erase (iterator position) ;
void      erase (iterator first, iterator last) ;
reference front () ;
iterator  insert (iterator position, const T& x) ;
void      insert (iterator position, const_iterator first,
                                     const_iterator last) ;
void      insert (iterator position, size_type n,
                  const T& x) ;
void      pop_back () ;
void      pop_front () ;
void      push_back (const T& x) ;
void      push_front (const T& x) ;
```

The first constructor instantiates a new container object of size n with each object initialised to `value`. The second constructor instantiates a new container object whose objects are in the range `first` to `last` of an existing container object. The *back()* and *front()* member

functions return the last and first objects in a container. The first *erase()* function deletes an object at the specified index `position`, whereas the second version deletes all objects within the range `first` to `last`. Member function *insert()* inserts an object(s) before `position`. The first version inserts a copy of `x`. The second version inserts copies of objects in the range `first` to `last`. The third version inserts `n` copies of `x`. The `pop_` and `push_` member functions delete and append, respectively, objects from both the `back` and `front` of a container. Several other operations are supported by both `vector` and `list`, such as *remove()*, *reverse()*, *sort()*, *splice()* and *unqiue()* for deleting an object, reversing the order of objects, sorting in ascending order, splicing one object inside another and removing adjacent duplicates of objects.

The STL sequence container classes consist of `vector`, `deque` and `list`. Class `vector` is analogous to the template class `Vector` developed in Chapter 13 and encapsulates a contiguous array of objects. Class `deque` is similar to `vector`, except that deques support constant time insertions and deletions of objects at both the front and end of the container, with linear time insertions/deletions elsewhere. This differs from vectors, which only provide constant time inserions/deletions at the end of a container. Thus, deques are well suited to algorithms that require frequent insertions/deletions at both ends of a container. The following program helps illustrate both `vector` and `deque`:

```
// vector.cpp
//...
void main()
    {
    // vector:
    vector<int> v ;

    for (int i=0; i<5; i++)
        v.push_back(i) ;

    cout << "v: " ;
    ostream_iterator<int> outi (cout, " ") ;
    copy (v.begin(),v.end(),outi) ;
    cout << endl ;

    vector<int>::size_type size     = v.size () ;
    vector<int>::size_type max_size = v.max_size () ;
    vector<int>::reference value_ref = v[0] ;
    cout << "no. of elements     : " << size      << endl ;
    cout << "max. no. of elements: " << max_size  << endl ;
    cout << "value_ref           : " << value_ref << endl ;

    // deque:
    deque<int> dq ;
    dq.push_back (1) ;
    dq.push_front (0) ;

    cout << "dq: " ;
    copy (dq.begin(),dq.end(),outi) ;
  }
```

with output:

```
v: 0 1 2 3 4
no. of elements      : 5
max. no. of elements: 1073741823
value_ref            : 0
dq: 0 1
```

The above program also illustrates the use of the container types `size_type` and `reference`. Program IT_RAND.CPP previously illustrated the use of iterator type `vector::iterator`.

The doubly linked list class `list` is analogous to the singly linked `LinkedList` class developed in Chapter 13 except that `list` encapsulates both *next* and *previous* pointers and is therefore doubly linked. An illustration of class `list` is provided in the following program:

```
// list.cpp
//...
void main ()
    {
    list<string>           l ;
    list<string>::iterator li ;
    l.push_back ("one") ;   l.push_back ("two") ;
    l.push_back ("three") ; l.push_back ("four") ;

    // O/P l
    for (li=l.begin(); li!=l.end(); li++)
        cout << *li << " " ;
    cout << endl ;

    l.sort () ;

    // O/P sorted l
    for (li=l.begin(); li!=l.end(); li++)
        cout << *li << " " ;
  }
```

with output:

```
one two three four
four one three two
```

The above program defines a list of string objects `l` and then fills `l` with four strings using `push_back()`. The objects of `l` are then sorted in $O(n \log n)$ time using `sort()`.

20.7.2 Associative Containers

Associative containers associate a key with an object and can identify an object via its associated key. The keys are frequently integer values, but can in fact be any valid type. How the objects and their associated keys are actually stored by an associative container is the responsibility of the container and should be of no concern to a user. Internally, associative containers are stored as red–black balanced trees (class `rb_tree`), with the objects ordered. As a result, the insertion and retrieval of objects of associative containers requires logarithmic time.

The STL supports the following four associative container classes: `set`, `multiset`, `map` and `multimap`, declared and defined in the header files SET.H and MAP.H. Both `set` and `multiset` store only the keys and therefore a key in fact represents an object, whereas both `map` and `multimap` store objects and their associated keys. Both `set` and `map` support only unique keys, whereas both `multiset` and `multimap` support duplicate keys.

As with sequence containers there are several operations which all associative containers support:

```
size_type      count (const key_type& x) ;
void           erase (iterator position) ;
size_type      erase (const key_type& x) ;
void           erase (iterator first, iterator last) ;
iterator       find (const key_type& x) ;
void           insert (const_iterator first,
                       const_iterator last) ;
key_compare    key_comp () const ;
iterator       lower_bound (const key_type& x) const ;
iterator       upper_bound (const key_type& x) const ;
value_compare value_comp () const ;
```

`count()` returns the number of objects whose key equals x. Member function *`erase()`* deletes an object(s) from a container while *`find()`* returns an iterator to an object. Operation *`insert()`* inserts objects within the range `first` to `last`. Member functions *`key_comp()`* and *`value_comp()`* return comparison functions for a container. Member functions *`lower_bound()`* and *`upper_bound()`* return iterators on a range of objects equal to x.

Types `key_compare` and `value_compare` have been discussed previously in Table 21.5 and are **`typedef`**s of the `Key` and `Compare` key and comparison function types. For example, the declaration of class `set` is of the form:

```
template <class Key,
          class Compare=less<Key>,
          class Allocator=allocator<Key> >
class set
  {
  public:
    //...
    typedef Key     key_type ;
    typedef Key     value_type ;
    typedef Compare key_compare ;
    typedef Compare value_compare ;
    //...
  };
```

with identical **`typedef`**s for `multiset`. Note that the default type of the comparison function is the function object `less<Key>`. The **`typedef`**s for `map` and `multimap` are slightly different:

```
template <class Key, class T,
          class Compare=less<Key>,
          class Allocator=allocator<pair<Key,T> > >
```

```
class map
  {
  public:
    //...
    typedef key         key_type ;
    typedef pair<Key,T> value_type ;
    typedef Compare     key_compare ;
    class value_compare
      : public binary_function<value_type,value_type,bool>
      { \*...*\ };
    //...
  };
```

Data structure `pair` associates two types of objects, UTILITY.H:

```
template <class T1, class T2>
struct pair
  {
  T1 first ;
  T2 second ;
  pair (const T1& a=T1(), const T2& b=T2())
    : first (a), second (b) {}
  };
```

Let's now illustrate the associative container `set` by way of the following program:

```
// set_stl.cpp
//...
void main ()
    {
    set<int,less<int> > s ;
    s.insert (2) ; s.insert (1) ; s.insert (0) ;

    // O/P s
    ostream_iterator<int> outi (cout, " ") ;
    cout << "s: " ;
    copy (s.begin(), s.end(), outi) ;
    cout << endl ;

    // find object in s
    int i (4) ;
    set<int,less<int> >::iterator si = s.find (i) ;
    if (si != s.end())
       cout << "object: " << i << " in set" << endl ;
    else
       cout << "object: " << i << " not in set" << endl ;
    }
```

with output:

```
s: 0 1 2
```

```
object 4 not in set
```

noting that although integer objects are added to s in the order 2, 1 and 0 they are internally stored in ascending order since the comparison function object less is used in the definition of s. The program also illustrates testing if an object is held by s using the *find()* member function.

Class map is analogous to set except that it stores both an object and its associated key. Maps are frequently used as associative arrays (Section 13.6.1), as illustrated in the following program:

```
// map.cpp
//...
void main ()
    {
    // I/P file
    char     file_name[] = "NAME_AGE.DAT" ;
    ifstream name_age_file (file_name, ios::in) ;
    if (!name_age_file)
       {
       cerr << "uable to open file " << file_name ;
       exit (EXIT_SUCCESS) ;
       }

    // fill map m
    string name ;
    int    age ;
    map<string,int,less<string> > m ;
    while (name_age_file >> name >> age)
          m[name] = age ;

    // O/P m
    cout << "m: " << endl ;
    map<string,int,less<string> >::iterator mi ;
    for (mi=m.begin(); mi!=m.end(); mi++)
        cout << setw(20) << (*mi).first
             << setw(3)  << (*mi).second << endl ;

    // index m
    name = string ("melanie") ;
    cout << "age of " << name << ": " << m[name] << endl ;
    }
```

The program begins by reading the name-age data from file NAME_AGE.DAT:

```
melanie   33
christine 56
sharon    35
```

A while loop is then used to fill the map m with the name-age data using the random access operator []. The definition of m illustrates that the key type is string, the object type is **int** and the comparison function is the function object less. The contents of m are then output us-

ing an iterator, noting the use of the `pair` data members `first` and `second`. Finally, `m` is indexed using the string variable `name`. The output of the above program is:

```
m:
                christine 56
                melanie 33
                sharon 35
age of melanie: 33
```

The use of classes `multiset` and `multimap` is similar to that of `set` and `map` except for the possibility of duplicate keys.

20.7.3 Numeric Containers

In this section we summarise the STL container classes which are designed specifically for performing numerical calculations.

20.7.3.1 Class `numeric_limits`

In Chapter 4 we examined the `numeric_limits` class (declared in LIMITS.H) and its specialisations for the fundamental data types **`char`**, **`int`**, **`float`**, **`double`**, **`wchar_t`**, **`bool`** and associated modified types such as **`long double`**. Each specialisation provides all of the relevant details of its corresponding type so that the general `numeric_limits<T>` class simply declares several **`static`** constants and inline functions such as `is_signed` and *`min()`*:

```
template <class T>
class numeric_limits
  {
  public:
    static const bool is_specialized ;
    //...
    static const bool is_signed ;
    static inline T   min () ;
    //...
  };
```

Constant `is_specialised` is set to logical-true if `numeric_limits<T>` is specialised for type `T`.

20.7.3.2 Class **`complex`**

The STL class `complex` declared in COMPLEX.H supports complex arithmetic:

```
template <class T>
class complex
  {
  private:
    T re, im ;
  public:
    typedef T value_type ;
```

```
    complex (const T& re_arg=0, const T& imag_arg=0) ;
    T imag () const ;
    T real () const ;
    complex<T>& operator = (const T&) ;
    template <class X> complex<T>&
      operator = (const complex<X>& rhs) ;
    //... +=, -=, *= and /=
};
```

complex encapsulates the real and imaginary parts of a complex number, value_type, default constructor, the access member functions *imag()* and *real()* and overloaded =, +=, -= and /= operators. Class complex overloads the usual three versions of the non-member operators +, -, *, /, == and !=. For example, the overloaded + operator is:

```
complex<T> operator + (const complex<T>& lhs,
                       const complex<T>& rhs) ;
complex<T> operator + (const complex<T>& lhs, const T& rhs) ;
complex<T> operator + (const T& lhs, const complex<T>& rhs) ;
```

The member functions *abs()*, *arg()*, *conj()*, *norm()* and *polar()* are supported for returning the absolute value, argument, complex conjugate, norm and polar form respectively:

```
T          abs (const complex<T>& a) ;
T          arg (const complex<T>& a) ;
complex<T> conj (const complex<T>& a) ;
T          norm (const complex<T>& a) ;
complex<T> polar (const T& r, const T& theta=0) ;
```

The standard mathematical operations *sin()*, *cos()*, *tan()*, *sinh()*, *cosh()*, *tanh()*, *exp()*, *log()*, *log10()*, *pow()* and *sqrt()* are all supported; for example:

```
complex<T> cos (const complex<T>& a) ;
```

Both the extraction and insertion stream operators are overloaded:

```
template <class T, class CharT, class Traits>
basic_istream<CharT,Traits>& operator >> (basic_
istream<CharT,Traits>& is, complex<T>& x) ;

template <class T, class CharT, class Traits>
basic_ostream<CharT,Traits>& operator << (basic_
ostream<CharT,Traits>& os, const complex<T>& x) ;
```

A complex object is output in the format (x,y) and can be input in either of the formats x, (x) and (x,y).

Three specialisations of complex exist for the fundamental types **float**, **double** and **long double**. An illustration of complex is given in COMPLEX.CPP:

```
// complex.cpp
//...
```

```
void main ()
  {
  complex<double> a (1.0,2.0), b (3.0,4.0) ;
  cout << "a: " << a << endl ;
  cout << "b: " << b << endl ;

  complex<double> c = a + b ;
  cout << "c(=a+b): " << c << endl ;
  cout << "sqrt(a) : " << sqrt (a)  << endl ;
  cout << "pow(a,2): " << pow (a,2) << endl ;
  }
```

with output:

```
a: (1,2)
b: (3,4)
c(=a+b): (4,6)
sqrt(a): (1.27202,0.786151)
pow(a,2): (-3,4)
```

20.7.3.3 Class `valarray`

Earlier we discussed the container class `vector` for operating on a range of argument types. Although `vector` can be used for numeric types, the STL provides the `valarray`[1] vector class specifically for performing numeric vector operations. Class `valarray` is designed for performing highly efficient numeric computations, whereas `vector` is designed for ease of use and efficiency. The constructors of `valarray` are given by:

```
template <class T>  // VALARRAY.H
class valarray
  {
  public:
    typedef T value_type ;

    valarray () ;
    explicit valarray (size_t size) ;
    valarray (const T& value, size_t size) ;
    valarray (const T* pointer, size_t size) ;
    valarray (const valarray& array) ;

    valarray (const slice_array<T>&) ;
    valarray (const gslice_array<T>&) ;
    valarray (const mask_array<T>&) ;
    valarray (const indirect_array<T>&) ;
  };
```

1 If you are using the GNU/Linux g++ compiler version 2.x then `valarray` is not generally supported, although implementations do exist for particular platforms.

The types `slice_array`, `gslice_array`, `mask_array` and `indirect_array` allow a subset of a `valarray` to be specified, and are discussed in subsequent sub-sections. The following extract from VALARRAY.CPP illustrates initialising `valarray` objects:

```
// valarray.cpp
//...
valarray<int>    v0 ;
valarray<float>  v1 (1.0, 100) ;

double darray[5] = { 1.0, 2.0, 3.0, 4.0, 5.0 };
valarray<double> v2 (darray,5) ;
//...
```

Note that for both of the two argument constructors the initial value of a C-style array is before the number of elements.

The assignment and subscripting operators are overloaded as follows:

```
valarray& operator = (const valarray<T>& array) ;
valarray& operator = (const T& val) ;
valarray& operator = (const slice_array<T>& slice) ;
//... similarly for gslice_, mask_, and indirect_

T operator [] (size_t index) const ;
T& operator [] (size_t index) ;
valarray<T> operator [] (slice) const ;
slice_array<T> operator [] (slice) ;
valarray<T> operator [] (const gslice&) const ;
gslice_array<T> operator [] (const gslice&) ;
valarray<T> operator [] (const valarray<bool>&) const ;
mask_array<T> operator [] (const valarray<bool>&) ;
valarray<T> operator [] (const valarray<size_t>&) const ;
indirect_array<T> operator [] (const valarray<size_t>&) ;
```

The assignment operator allows an entire `valarray` object to be assigned to another object provided that they are both of the same size. The result of the assignment is undefined if the two `valarray` objects are of different sizes. In addition, a single value can be assigned to every element of a `valarray` object. Subscripting is via the usual integer indexing and, as with the assignment operator, via the `slice_array`, `gslice_array`, `mask_array` and `indirect_array` types. Note that for the assignment and subscripting operators no size or range checking is performed and no exceptions will be thrown, since `valarray` is optimised for efficiency.

`valarray` supports the following unary operators:

```
valarray operator + () const ;
valarray operator - () const ;
valarray operator ~ () const ;
valarray<bool> operator ! () const ;
```

and the arithmetic assignment operators +=, -=, *=, /=, %=, ^=, &=, |=, <<= and >>=. For example, += is of the form:

```
valarray& operator += (const T& val) ;
```

valarray overloads the binary operators +, -, *, /, %, &&, ||, &, |, ==, !=, <, >, <=, >=, << and >> as non-member operations. The three versions of * are:

```
valarray operator * (const valarray& , const valarray&) ;
valarray operator * (const valarray& , const T&) ;
valarray operator * (const T& , const valarray&) ;
```

The overloaded << and >> operators are shift operators and valarray does not overload the stream operators. Program VALARRAY.CPP overloads << for stream output. valarray supports the following member functions:

```
size_t size () const ;
T      sum() const ;
T      min() const ;
T      max() const ;
valarray shift (int i) const ;
valarray cshift (int i) const ;
valarray apply (T func(T)) const ;
valarray apply (T func(const T&)) const ;
void free () ;
void resize (size_t new_sz, const T val=T()) ;
```

size() returns the number of elements in a valarray object and *resize()* defines a new size to a valarray object with each element initialised to val. Member functions *sum()*, *min()* and *max()* return the sum, minimum and maximum of the elements of a vector. *free()* deletes all elements from a vector and frees the associated memory. *apply()* applies the specified function argument to every element of a valarray object (without altering the operated-on object) and the *shift()* and *cshift()* member functions return new valarray objects with the elements logically and cyclically shifted. The logical shift v1=v.shift(n) returns a valarray so that v1[i] is equivalent to v[i+n] if (i+n) is a valid index, else the default value is assigned to an element. The cyclic shift v1=v.cshift(n) returns a valarray so that v1[i] is equivalent to v[(i+n)%v.size()]. An illustration of the *shift()* and *cshift()* member functions is given below:

```
// valarray.cpp
//...
double darray[5] = { 1.0, 2.0, 3.0, 4.0, 5.0 };
valarray<double> v2 (darray,5) ;
//...
valarray<double> v2_shift  =  v2.shift (2) ;
valarray<double> v2_cshift =  v2.cshift (2) ;
cout << "shift v2 for i=2: " << v2_shift << endl ;
cout << "cyclic shift v2 for i=2: " << v2_cshift << endl ;
//...
```

with output:

```
shift v2 for i=2: [3 4 5 0 0]
cyclic shift v2 for i=2: [3 4 5 1 2]
```

20.7.3.4 Slices

Header file VALARRAY.H declares the two classes `slice` and `slice_array` for manipulating `valarray`. Class `slice` encapsulates three integers which represent the start index, the number of elements in a slice and the stride, which specifies the location of every *n*th. element, that is element `n` is at location `start+n*stride`:

```
class slice
  {
  private:
    size_t start ;
    size_t length ;
    size_t stride ;
  public:
    slice () ;
    slice (size_t start, size_t length, size_t stride) ;
    slice (const slice& sl) ;
    size_t start () const ;
    size_t size () const ;
    size_t stride () const ;
  };
```

Class `slice_array` provides a convenient way of defining a `valarray`, but internally simply makes reference to a subset of a `valarray` object through a `slice`:

```
template <class T>
class slice_array
  {
  private:
    slice_array() ;
    slice_array (const slice_array&) ;
    slice_array& operator = (const slice_array&) ;
    valarray* array ; // implementation dependent
    slice     s ;
  public:
    typedef T value_type ;
    slice get_slice () const ;
    void operator = (const valarray<T>&) const ;
    void operator = (const T&) const ;
    void operator += (const valarray<T>&) const ;
    //... similarly -=, *=, /=, %=, ^=, &=, |=, <<=, >>=
  };
```

`slice_array` prevents construction and copying by declaring the constructors and overloaded assignment operator as private. A `slice_array` results from using a slice as the subscript to a `valarray`. The following illustrates defining a `slice_array` object by way of the `valarray` subscript operator which takes as argument a slice object with start at the first element of `valarray` object `v2`, number of elements equal to `v2.size()/2` and a stride length of 2:

```
// valarray.cpp
//...
double darray[5] = { 1.0, 2.0, 3.0, 4.0, 5.0 };
valarray<double> v2 (darray,5) ;
//...
slice_array<double>& v2_even_slice =
   v2[slice(0,v2.size()/2,2)] ;
valarray<double> v2_even (v2_even_slice) ;
cout << "v2_even: " << v2_even << endl ;
//...
```

with output:

```
v2_even: [1 3]
```

The stride length of 2 effectively creates a new array with elements taken from the even indices of v2.

The STL also declares a generalised slice `gslice` which encapsulates information about *n* slices via *n* `valarray` number of elements and *n* `valarray` strides:

```
class gslice
  {
  private:
    size_t           start ;
    valarray<size_t> length ;
    valarray<size_t> stride ;
  public:
    gslice () ;
    gslice (size_t s, const valarray<size_t>& l,
            const valarray<size_t>& d) ;
    //...
  };
```

In an analogous manner to `slice_array` the STL declares a `gslice_array` class.

20.7.3.5 Masks

The STL declares class `mask_array` in VALARRAY.H which provides another way of defining an array that behaves as a `valarray` but which internally refers to a subset of a `valarray` object. A mask is a `valarray` of type **`bool`**, that is `valarray<bool>`. When a mask is used as a subscript for a `valarray`, a logical-true bit indicates that an element of `valarray` is to be considered as part of the returned `valarray`. Subscripting a `valarray` is through the `valarray<bool>` type. For example, the following illustrates that the `valarray` result has elements `b[0]`, `b[2]` and `b[4]` only:

```
// valarray.cpp
//...
double darray1[7] = { 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0 };
valarray<double> v3 (darray1,7) ;

bool b[] = { true, false, true, false, true, false, false };
```

```
valarray<bool> mask (b,7) ;
valarray<double> v3_mask = v3[mask] ;
cout << "v3_mask: " << v3_mask << endl ;
//...
```

with output:

```
v3_mask: [1 3 5]
```

20.7.3.6 Indirect Arrays

The STL declares a class called `indirect_array` in VALARRAY.H for defining a subset of a `valarray`. `indirect_array` supports exactly the same operations as `slice_array`, but where `indirect_array` differs is that it provides a means of arbitrarily subsetting and reordering a `valarray` object. For example:

```
// valarray.cpp
//...
size_t i[] = { 4, 0, 1, 3, 2 };
valarray<size_t> indirect_index (i,5) ;
valarray<double> v3_indirect = v3[indirect_index] ;
cout << "v3_indirect: " << v3_indirect << endl ;
//...
```

with output:

```
v3_indirect: [5 1 2 4 3]
```

noting that subscripting is via the type `valarray<size_t>`.

20.8 Adaptors

In Section 20.5 we discussed function adaptors for transforming one function object in to another. In this section we examine adpators for iterators and containers.

20.8.1 Iterator Adaptors

There are two main types of iterator adaptor: *reverse* and *insert*. We have previously encountered reverse iterator adaptors in program IT_REVER.CPP when we used class `reverse_iterator` to traverse the objects of a `list` from back to front. Class `reverse_iterator` achieves this by being derived from class `random_access_iterator` and overloading operators such as ++, ITERATOR.H:

```
template <class RandomAccessIterator, class T,
          class Reference=T&,
          class Distance=ptrdiff_t>
class reverse_iterator :
  public random_access_iterator<T,Distance>
```

```
{
public:
  //...
  typedef reverse_iterator<RandomAccessIterator,T,
                           Reference,Distance> self ;
  self& operator ++ () { --current ; return *this ; }
  //...
};
```

The definition of **operator**::++() illustrates that it simply decrements the encapsulated bidirectional iterator `current` rather than incrementing it. As a result `reverse_iterator` simply *adapts* exisiting operations rather than writing them from scratch.

Insert iterator adaptors have the following three variants: *back*, *front* and *insert*. Back insert iterators append objects to the end of a container. Front insert iterators insert objects at the front of a container, and insert iterators insert an object at any location inside a container. The corresponding STL classes are `back_insert_iterator`, `front_insert_iterator` and `insert_iterator` each declared in ITERATOR.H. Each class is derived from `output_iterator` and encapsulates an object of the operated-on container; for example `back_insert_iterator` is of the form:

```
template <class Container>
class back_insert_iterator : public output_iterator
  {
  protected:
    Container& container ;
  public:
    explicit back_insert_iterator (Container& c)
      : container (c) {}
    //...
  };
```

whereas class `insert_iterator` also encapsulates the specific container iterator:

```
template <class Container>
class insert_iterator : public output_iterator
  {
  protected:
    Container&                    container ;
    typename Container::iterator iter ;
  public:
    insert_iterator (Container& c,
                     typename Container::iterator i)
      : container (c), iter (i) {}
    //...
  };
```

with the constructor initialising the encapsulated container. Note that `back_insert_iterator` uses the *`push_back()`* member function of the container and is therefore restricted to the `vector`, `deque` and `list` containers. The `front_insert_iterator` iterator uses the *`push_front()`* operation and is therefore restricted to the `vector` and

deque containers. To illustrate these three types of insert iterator adaptors consider the program:

```
// in_it.cpp
//...
void main ()
  {
  // back & forward:
  vector<int> bv ;
  list<int>   fl, il ;

  for (int i=0; i<3; i++) // fill bv and fl
    {
    bv.push_back (i) ;
    fl.push_back (i) ;
    il.push_back (i) ;
    }

  back_insert_iterator<vector<int> >  bit (bv) ;
  front_insert_iterator<list<int> >   fit (fl) ;

  for (int i=3; i<6; i++)  // fill bit and fit
    {
    *bit = i ;
    *fit = i ;
    }

  ostream_iterator<int> outi (cout, " ") ;  // O/P
  copy (bv.begin(),bv.end(), outi) ; cout << endl ;
  copy (fl.begin(),fl.end(), outi) ; cout << endl ;

  // insert:
  list<int>::iterator li = il.begin () ;
  li++ ;
  insert_iterator<list<int> > it (il,li) ;

  copy(li,il.end(),it) ;

  copy(il.begin(),il.end(),outi) ;
  }
```

with output:

```
0 1 2 3 4 5
5 4 3 0 1 2
0 1 2 1 2
```

The integers 0, 1 and 2 are first pushed onto the `vector` and `list` objects `bv` and `fl`. The `back_insert_iterator` and `front_insert_iterator` objects `bit` and `fit` are then assigned the integers 3, 4 and 5 which will in turn be appended to `bv` and inserted at the front of `fl`, as illustrated by the output. The list `il` is similarly pushed the integers 0, 1 and 2. An

iterator of `list`, `li`, is then defined and both `il` and `li` used to define an `insert_iterator` object `it`. The contents of `il` are then copied to `it` which is in turn inserted in `il`. Note that `li` is incremented before the call to *`copy()`*. This is to prevent a statement of the form:

```
copy (il.begin(), il.end(), it) ;
```

which will result in an infinite loop because the first object of `il` is before the insertion point, with the result that an inserted object will never reach the end of the list.

Finally, let us examine the `raw_storage_iterator`, which is an output iterator that allows a sequence to be written into uninitialised memory. This is necessary because the standard algorithms write to a sequence, the elements of which have previously been initialised. The standard algorithms use assignment rather than copy constructors to initialise the sequence elements. Assignment can be more expensive than initialisation and `raw_storage_iterator` provides the solution of initialising rather than assigning:

```
template <class OutputIterator, class T>
class raw_storage_iterator
    : public iterator<output_iterator_tag,T,ptrdiff_t,T*,T&>
  {
  protected:
    OutputIterator iter;
  public:
    typedef OutputIterator iterator_type ;
    typedef T              element_type ;
    explicit raw_storage_iterator (OutputIterator x)
      : iter (x) {}
    raw_storage_iterator& operator * ()
      { return *this ; }
    raw_storage_iterator& operator = (const T& element)
      {
      ::new(&(*iter)) T(element) ;
      return *this ;
      }
    raw_storage_iterator& operator ++ ()
      { ++iter ; return *this ; }
    raw_storage_iterator operator ++ (int)
      {
      raw_storage_iterator tmp = *this ;
      ++iter ;
      return tmp ;
      }
  };
```

The following program illustrates `raw_storage_iterator`:

```
// raw_it.cpp
//...
class X
  {
  private:
```

```
    int data ;
  public:
    X (int _data=0)
      : data (_data) {}
   };

void main ()
  {
  // uninitialised block of memory
  X* p = (X*)malloc (sizeof(X)) ;

  raw_storage_iterator<X*,X> rsi (p) ;

  X x (0) ;

  *rsi = x ;
  }
```

20.8.2 Container Adaptors

Container adaptors transform one container in to a new container by changing the class interface. Generally, this results in the new container supporting a subset of operations of the original class. This is achieved by hiding some exisiting operations and possibly adding some new operations. The result is a class which required less implementation than writing the new class from scratch. This is analogous to the implementation of classes `Stack` and `Queue` in Chapter 13 which were derived from class `LinkedList` using the **`protected`** access specifer rather than **`public`**:

```
template <class T>
class Stack : protected LinkedList<T>
  {
  public:
    //...
    void Push (T obj) ;
    T    Pop () ;
  };
```

The STL container adaptor solution is to encapsulate the base container rather than using inheritance. For example, the declaration of class `stack` is of the form STACK.H:

```
template <class T, class Container = deque<T> >
class stack
  {
  protected:
    Container container ;
    //...
  };
```

Class `stack` adds the following **`public`** member functions and overloaded operators:

```
bool                empty () const ;
size_type           size () const ;
value_type&         top () ;
const value_type&   top () const ;
void                push (const value_type& x) ;
void                pop () ;
friend bool operator == (const stack<T,Container>& x,
                         const stack<T,Container>& y) ;
friend bool operator <  (const stack<T,Container>& x,
                         const stack<T,Container>& y) ;
```

where, for example, the definition of *push()* is:

```
void push (const value_type& x)
  {
  container.push_back (x) ;
  }
```

and illustrates that the adaptor class `stack` is simply passing the work on to the container class object `container`. As with class `Stack` the member function *push()* pushes an object on to a stack whereas *pop()* deletes or pops an object from a stack. Member function *empty()* returns logical-true if a stack is empty else logical-false, *size()* returns the number of objects in a stack and *top()* returns the object at the top of a stack, but does delete the object from the stack. The following program illustrates class `stack`:

```
// stack.cpp
//...
void main ()
    {
    stack<int,vector<int> > s ;

    s.push (0) ;
    s.push (1) ;
    s.push (2) ;
    }
```

and defines a stack of **int** objects with underlying `vector<int>` container.

In a similar fashion the STL declares the adaptor container class `queue`. The member functions and overloaded operators of `queue` are equivalent to those of `stack`, except that *top()* is replaced by *front()*:

```
value_type&         front () ;
const value_type&   front () const ;
```

with *front()* returning the object at the front of the queue but without deleting it. Program QUEUE.CPP illustrates the `queue` container class:

```
// queue.cpp
//...
void main ()
    {
```

```
using namespace std ;

queue<int,deque<int> > q ;

q.push (0) ;
q.push (1) ;
q.push (2) ;
}
```

An additional container adaptor is `priority_queue`, which implements a priority queue. Priority queues are equivalent to queues except that when a priority queue is popped it is the object with either the largest or smallest object that is removed, and this object may or may not be at the front of the queue. In other words, the object with greatest priority is removed first. Priority queues are implemented using heaps and the declaration of `priority_queue` is of the form:

```
template <class T, class Container=vector<T>,
          class Compare=less<typename
                             Container::value_type> >
class priority_queue
  { /*... */ };
```

where `T` is the object type, `Container` is the type of container that `priority_queue` is based on and is by default set to `vector`, and `Compare` is the type of comparison function and is by default set to the function object `less`. Class `priority_queue` supports the same member function names as `stack` but does not overload the == and < operators.

The following program illustrates the use of `priority_queue` by defining a priority queue container of `Person` objects. Class `Person` encapsulates a string name and integer priority which, for example, could represent the order of importance for a medical operation:

```
// p_queue.cpp
//...
// Person class: name and priority
class Person
      {
      public:
        string name ;
        int    priority ;
        Person (string _name=string(), int _priority=0)
          : name (_name), priority (_priority) {}
        bool operator < (const Person& p) const
          { return priority<p.priority ? true : false ; }
      };

// O/P Person object
inline ostream& operator << (ostream& s, const Person& p)
  { s << "name: " << p.name << ", priority: " << p.priority ; }

void main ()
    {
    priority_queue<Person,vector<Person>,less<Person> > pq ;
```

```
    pq.push (Person("jack",1)) ;
    pq.push (Person("jill",2)) ;
    pq.push (Person("john",0)) ;

    while (!pq.empty())
          {
          cout << pq.top () << endl ;
          pq.pop () ;
          }
    }
```

with output:

```
name: jill, priority: 2
name: jack, priority: 1
name: john, priority: 0
```

The `priority_queue` object `pq` operates on class `Person` and uses a `vector` container; the `less` function object is used as the comparison function, requiring the < operator to be overloaded for `Person`. The output illustrates that although the `Person` objects are pushed on to `pq` in the order *jack*, *jill* and *john*, they are prioritised when popped.

20.9 Strings

Of all the classes in the STL the `string` class is the odd one out. This is because a string is an array of characters and at first may not appear to be a template class. However, we will see that the popular `string` class is in fact a **`typedef`** of a template class called `basic_string`.

20.9.1 Character Traits

Before we examine `basic_string` we must examine character traits. The properties of a character type are defined by a specialisation of the `char_traits` **`struct`**, STRING.H:

```
template <class  CharT>
struct char_traits
  {
  typedef CharT char_type ; // character type
  static void assign (char_type& c1, const char_type& c2) ;

  typedef long int_type ;  // integer value of characters
  static char_type to_char_type (const int_type& c) ;
  static int_type  to_int_type  (const char_type& c)
  static bool      eq_int_type  (const int_type& c1,
                                 const int_type& c2) ;
  // comparisons:
  static bool eq (const char_type& c1,  char_type& c2) ;
  static bool lt (const char_type& c1, const char_type& c2) ;
```

```
    // array operations:
    static int  compare (const char_type* s1,
                         const char_type* s2, size_t n) ;
    static size_t           length (const char_type *s) ;
    static const char_type* find (const char_type* s,
                           size_t n, const char_type& a) ;
    static char_type*  move (char_type* s1,
                           const char_type* s2, size_t n) ;
    static char_type * copy (char_type *dst,
                           const char_type *src, size_t n) ;
    static char_type*  assign (char_type* s,
                            size_t n, char_type a) ;
    // I/O:
    typedef wstreamoff       off_type;   // offset in stream
    typedef mbstate_t        state_type; // multi-byte state
    typedef fpos<state_type> pos_type;   // position in stream
    static int_type   eof () ;
    static int_type   not_eof (const int_type& c) ;
    static state_type get_state (pos_type pos) ;
};
```

In order to define a valid character type for `basic_string`, a specialisation must provide all of the above types and member functions. The STL provides specialisations for character types **char** and **wchar_t**:

```
template<> struct char_traits<char>    { /* ... */ };
template<> struct char_traits<wchar_t> { /* ... */ };
```

The **typedef** `char_type` defines the character type and it must be possible to obtain an equivalent integer value for each character. The type of integer is defined by the **typedef** `int_type` and functions *`to_char_type()`* and *`to_int_type()`* perform conversions between `char_type` and `int_type`. The *`eq()`* and *`lt()`* perform equality and less than comparisons and are used by *`compare()`*, which compares characters and returns 0, <0 or >0 if the two arguments are equivalent, the first argument is lexicographically before the second argument or the first argument is lexicographically after the second argument. Member function *`move()`* copies `n` characters from `s2` to `s1` and similarly for *`copy()`* from `src` to `dst`. *`move()`* uses *`assign()`* whereas *`copy()`* uses *`memcpy()`* and is therefore more efficient. *`assign()`* assigns `n` copies of `a` into `s` using `assign(s[i],a)`. Stream function *`eof()`* returns EOF, *`not_eof()`* returns 0 if `c` is equal to EOF else returns `c`, and *`get_state()`* returns the multi-byte stream state of a character in `pos`.

20.9.2 Class `basic_string`

The standard library support for strings is based on the `basic_string` class, STRING.H:

```
template <class CharT, class traits=char_traits<CharT>, class
Allocator=allocator<CharT> >
class basic_string
  { /* ... */ };
```

The STL provides two **typedef**s for commonly used string types:

```
typedef basic_string<char>    string ;
typedef basic_string<wchar_t> wstring ;
```

The following subsections summarise the key components of `basic_string`, each of which is examined in the program file BS.CPP.

20.9.2.1 Types

Class `basic_string` makes available its types via **`typedef`**s:

```
template < /* ... */ >
class basic_string
  {
  public:
    typedef traits traits_type ;
    typedef typename traits::char_type value_type ;
    typedef Allocator allocator_type ;
    typedef typename Allocator::size_type size_type ;
    typedef typename Allocator::difference_type
                                 difference_type;
    typedef typename Allocator::reference reference ;
    typedef typename Allocator::const_reference
                                 const_reference ;
    typedef typename Allocator::pointer pointer ;
    typedef typename Allocator::const_pointer
                                 const_pointer ;
    typedef typename Allocator::pointer iterator ;
    typedef typename Allocator::const_pointer const_iterator ;
    typedef reverse_iterator<iterator> reverse_iterator ;
    typedef reverse_iterator<const_iterator>
                              const_reverse_iterator ;
    //...
  };
```

For example:

```
// bs.cpp
//...
string::const_reference character = s0[1] ;
//...
```

20.9.2.2 Constructors and Assignment

Class `basic_string` supports the following constructors:

```
template < /* ... */ >
class basic_string
  {
  public:
    explicit basic_string (const Allocator& a=Allocator()) ;
    basic_string (const basic_string& s,
```

```
                size_type pos=0, size_type n=npos,
                const Allocator& a=Allocator()) ;
  basic_string (const CharT* p, size_type n,
                const Allocator& a=Allocator()) ;
  basic_string (const CharT* p,
                const Allocator& a=Allocator()) ;
  basic_string (size_type n, CharT c,
                const Allocator& a=Allocator()) ;
  //...
};
```

and allow an object to be initialised from part of a `basic_string`, part of a C-style string, a C-style string or a sequence of characters. Note that a `basic_string` object cannot be initialised from either a single character or integer value.

The following extract from BS.CPP illustrates defining several string objects using the different constructors of `basic_string`:

```
// bs.cpp
//...
basic_string<char> bs ("basic_string") ;

string s0 ("string") ;
string s1 (s0) ;
string s2 (s0,2,3) ; // 3 characters starting from pos. 2
string s3 (10,'x') ; // 10 characters 'x'
//...
```

20.9.2.3 Iterators

In an analogous manner to other container classes `basic_string` supports the following forward and reverse iterators:

```
iterator       begin () ;
const_iterator begin () const ;
iterator       end () ;
const_iterator end () const ;
reverse_iterator       rbegin () ;
const_reverse_iterator rbegin () const ;
reverse_iterator       rend () ;
const_reverse_iterator rend () const ;
```

and can be used in conjunction with all of the STL standard algorithms. For example:

```
// bs.cpp
//...
ostream_iterator<char> outi (cout,"") ;
copy (s0.begin(), s0.end(), outi) ;
//...
```

20.9.2.4 String Operations

Class `basic_string` supports numerous operations to access and manipulate strings. To access a character of a string, either the overloaded random access operator `[]` or the *`at()`* member function can be used:

```
const_reference operator[] (size_type n) const ;
reference       operator[] (size_type n) ;
const_reference at (size_type n) const ;
reference       at (size_type) ;
```

Subscripting a `string s` via **`operator`**`: : [] ()` is not checked for out of range indexing, whereas *`at()`* is checked and throws the standard library exception `out_of_range` if we try to access outside the range `s[0]` to `s[s.length()-1]`. For example:

```
// bs.cpp
//...
try
  {
  char c0 = s0[6] ;     // unchecked
  char c1 = s0.at(6) ; // checked: throws out_of_range
  //...
  }
catch (exception& e)
  {
  cout << e.what () << endl ;
  }
//...
```

with output:

```
position beyond end of string in function: basic_
string::at(size_t)
 index: 6 is greater than max_index: 6
```

The length of a string can be obtained by the *`length()`* member function, which returns the number of characters in the string, while *`max_size()`* returns the maximum allowable number of string characters:

```
size_type length () const;
size_type max_size () const
  { return npos - sizeof(string_ref_type) - 2 ; }
```

where `string_ref_type` is an implemention dependent type of the string reference that holds the characters for `basic_string`. The maximum value of `size_type` (**`unsigned int`**) is equal to `npos`.

Comparisons on strings can be performed either by the *`compare()`* member function or the non-member overloaded operators `==`, `!=`, `<`, `<=`, `>` and `>=`. The five versions of *`compare()`* and three versions of `==` are:

```
int compare (const basic_string& s) const ;
int compare (size_type pos, size_type n,
             const basic_string& s) const ;
int compare (size_type pos, size_type n,
             const basic_string& s,
             size_type pos2, size_type n2) const ;
int compare (const CharT* p) const ;
int compare (size_type pos, size_type n,
             const CharT* p, size_type n2) const ;

bool operator == (const basic_string& lhs,
                  const basic_string& rhs) ;
bool operator == (const CharT* lhs,
                  const basic_string& rhs) ;
bool operator == (const basic_string& lhs,
                  const CharT* rhs) ;
```

The integer return value of *compare()* is equivalent to that of char_traits::*compare()* and the *strcmp()* standard library function. The functions containing an argument n compare the first n characters from pos. Each of the overloaded operators simply calls its equivalent *compare()* function.

To append characters to the end of a string, basic_string provides the *append()* member function and the overloaded += operator:

```
basic_string& append (const basic_string& s) ;
basic_string& append (const basic_string& s,
                      size_type pos, size_type n) ;
basic_string& append (const CharT* p, size_type n) ;
basic_string& append (const CharT* p) ;
basic_string& append (size_type n, CharT c) ;
basic_string& operator += (const basic_string& s) ;
basic_string& operator += (const CharT* p) ;
basic_string& operator += (CharT c) ;
```

The functions containing argument n append n characters. The *insert()* member function inserts characters before position pos:

```
basic_string& insert (size_type pos, const basic_string& s) ;
basic_string& insert (size_type pos, const basic_string& s,
                      size_type pos2, size_type n) ;
basic_string& insert (size_type pos, const CharT* p,
                      size_type n) ;
basic_string& insert (size_type pos, const CharT* p) ;
basic_string& insert (size_type pos, size_type n, CharT c) ;
iterator      insert (iterator p, CharT c) ;
```

The final overloaded function inserts character c before iterator p.

Concatenating basic_strings is performed via the non-member **operator**::+*()*:

```
basic_string operator + (const basic_string& lhs,
                         const basic_string& rhs) ;
```

```
basic_string operator + (const basic_string& lhs,
                         const CharT* rhs) ;
basic_string operator + (const CharT* lhs,
                         const basic_string& rhs) ;
basic_string operator + (const basic_string& lhs,
                         CharT rhs) ;
basic_string operator + (CharT lhs,
                         const basic_string& rhs) ;
```

and simply call *append()*.

Searching a *basic_string* is performed using the *find()* member function and its variants:

```
size_type find (const basic_string& s, size_type i=0) const ;
size_type find (const CharT* p,
                size_type i, size_type n) const ;
size_type find (const CharT* p, size_type i=0) const ;
size_type find (CharT c, size_type i=0) const ;
size_type rfind (const basic_string& s,
                 size_type i=0) const ;
//...
size_type find_first_of (const basic_string& s,
                         size_type i=0) const ;
//...
size_type find_last_of (const basic_string& s,
                        size_type i=0) const ;
//...
size_type find_first_not_of (const basic_string& s,
                             size_type i=0) const ;
//...
size_type find_last_not_of (const basic_string& s,
                            size_type i=0) const ;
//...
```

find() returns the position of a sub-string or character and searches from the front to end of the operated-on string. *rfind()* is equivalent to *find()* but searches from the end to the front of a string. *find_first_of()* and *find_last_of()* search for the first and last occurrence, while *find_first_not_of()* and *find_last_not_of()* search for the first and last occurrence of a sub-string that is not in the operated-on string. The location of a sub-string or character is indicated by `i` and `n` denotes the number of characters and not the last index of the sub-string.

Characters can be deleted from a `basic_string` using the *erase()* member function:

```
basic_string erase (size_type i=0, size_type n=npos) ;
iterator     erase (iterator it) ;
iterator     erase (iterator first, iterator last) ;
```

If no arguments are passed to *erase()* then all characters are deleted from a string and is equivalent to the *clear()* member function:

```
void clear () { erase () ; }
```

To replace a sub-string of characters then use *replace()*:

```
basic_string& replace (size_type i, size_type n,
                       const basic_string& s) ;
basic_string& replace (size_type i, size_type n,
                       const basic_string& s,
                       size_type i2, size_type n2) ;
basic_string& replace (size_type i, size_type n,
                       const CharT* p, size_type n2) ;
basic_string& replace (size_type i, size_type n,
                       const CharT* p) ;
basic_string& replace (size_type i, size_type n,
                       size_type n2, CharT c) ;
basic_string& replace (iterator i, iterator i2,
                       const basic_string& s) ;
basic_string& replace (iterator i, iterator i2,
                       const CharT* p, size_type n) ;
basic_string& replace (iterator i, iterator i2,
                       const CharT* p) ;
basic_string& replace (iterator i, iterator i2,
                       size_type n, CharT c) ;
```

For example:

```
// bs.cpp
//...
string s ("The C++ programming language") ;
s.replace (s.find("C++"),3,"Java") ;
//...
```

20.9.2.5 Conversion to C-style Strings

Class `basic_string` provides two member functions for converting a string to a C-style string (terminated with a 0):

```
const CharT* c_str () const ;
const CharT* data () const ;
```

The string returned by both of these functions is the responsibility of `basic_string`, and an attempt should not be made to delete it.

20.9.2.6 Substring

The *substr()* member function of class `basic_string` allows a sub-string to be defined in terms of a position, `i`, and length, `n`:

```
basic_string substr (size_type i=0,
                     size_type n=npos) const ;
```

For example:

```
// bs.cpp
//...
string s ("implementation") ;
string sub_string = s.substr (5,3) ;
//...
```

20.10 Allocators

Allocators[2] cater for the different memory models in PC architecture. As discussed in Section 7.16, the PC memory model consists of small, medium, compact, large and huge memory models with near and far pointers. Allocators are also designed to relieve implementors of container classes from having to deal directly with raw memory. The STL solution to memory allocation is via the `allocator` class, MEMORY.H:

```
template <class T>
class allocator
  {
  public:
    allocator () throw () ;
    template <class U>
      allocator (const allocator<U>&) throw () ;
    pointer address (reference r) const { return & r ; }
    pointer allocate (size_type n,
      allocator<void>::const_pointer hint = 0) ;
    void deallocate (pointer p, size_type) ;
    void construct (pointer p, const T& val) {new(p) T(val);}
    void destroy (T* p) { p->T::~T() ; }
    //...
  };
```

Member function *`allocate()`* allocates memory using **`operator`** `new()` and the corresponding member function *`deallocate()`* uses **`operator delete(void*)`** to deallocate memory. Thus, *`new_handler()`* may be called and `bad_alloc` thrown if there is insufficient memory available. Member function *`address()`* returns a pointer to the reference object `r`. Member function *`construct()`* invokes **`operator new`***`()`* using the placement syntax to construct an in-memory object allocated by the allocator. Member function *`destroy()`* explicitly invokes the destructor for an object which causes problems for built-in types and as a result there exist specialisations for all built-in types.

The type of the `hint` argument in *`allocate()`* is the `const_pointer` to the specialisation `allocator<void>`:

```
class allocator<void>
  {
  public:
    //...
    void* allocate (size_type n, void* p= 0) ;
```

2 That's allocators and not alligators!

```
    void  deallocate (void* p, size_type n) ;
    //...
};
```

Allocators are designed to provide programmers with a memory allocation/deallocation mechanism for the implementation of container classes. The container class declarations in Section 20.7 have illustrated the typical application of `allocator` which is used by default by all containers. For example, class `vector` is of the form:

```
template <class T, class Allocator=allocator<T> >
class vector
  {
  private:
    Allocator alloc ;  // allocator
    iterator  array ;  // pointer to elements
  public:
    // constructor
    explicit vector (size_type n, const T& val=T(),
                     const Allocator& a=Allocator())
      {
      array = alloc.allocate (n) ;
      //...
      }
  //...
  };
```

20.11 Summary

The STL is a comprehensive library and it will take you a great deal of both time and effort until you find yourself writing STL code without using a reference manual. The effort is worth it though, because the STL provides an efficient and robust solution to frequently used algorithms and containers and a consistent solution in which to traverse container objects.

In addition to the ANSI C++ standard, Standard (1998), Stroustrup (1997) includes a detailed discussion of the STL. There are several books which are devoted entirely to the STL: Musser and Saini (1996), Robson (1997), Ammerau (1997) and Austen (1999).

A good way to improve your understanding of the STL is by inspecting the source files which help reveal the design and inner workings. Since the STL is a template library, the header files will contain the source code as well as declarations. Furthermore, check to see if your compiler comes with an STL Examples directory.

Exercises

20.1 Overload the insertion operator << for `vector`. Make use of an iterator object in your implementation.

20.2 Using a function object develop a function which is passed a `vector` object `v` and object `obj` to be found in `v`:

```
vector<T>::iterator LessThanOrEqualTo (vector<T>& v,
                                       const T& obj) ;
```

The function should return the first occurrence of `obj` in `v` that is less than or equal to `obj`.

20.3 Using the *`count()`* function write a function that is passed a `list` object `l` and returns the number of objects in `l` that are equal to a supplied object, `obj`:

```
int NumberOfOccurrences (const list<T>& l, const T& obj) ;
```

20.4 Develop a function which returns the dot product of two `vector` objects using the *`inner_product()`* function.

20.5 Develop a class which can encapsulate pointers to both global and member one argument functions of the form:

```
R GlobalFunction (A) ;
R T::MemberFunction (A) ;
```

In your implementation use the STL class `mem_fun1_t<R,T,A>` declared in FUNCTIONAL.H.

20.6 Write a function which returns the reverse of a `string` object.

20.7 Declare your own allocator class which defines member functions *`allocate()`* and *`deallocate()`*. Use your allocator class in the definition of a `vector` object.

Appendix A

C++ Keywords

The following lists the keywords and reserved words of the C++ language:

asm	To declare that a block of code is to be passed to the assembler
auto	A storage class specifier that is used to define objects in a block
bool	Boolean false–true type that can hold either the **false** or **true** literals
break	Terminates a **switch** statement or a loop
case	Used specifically within a **switch** statement to specify a match for the statement's expression
catch	Specifies actions taken when an exception occurs
char	Fundamental data type that defines character objects
class	To declare a user-defined type that encapsulates data members and operations or member functions
const	To define objects whose value will not alter throughout the lifetime of program execution
continue	Transfers control to the start of a loop
default	Handles expression values in a **switch** statement that are not handled by **case**
delete	Memory deallocation operator
do	Indicates the start of a **do-while** statement in which the sub-statement is executed repeatedly until the value of the expression is logical-false
double	Fundamental data type used to define a floating-point number
else	Used specifically in **if-else** statement
enum	To declare a user-defined enumeration data type
explicit	To declare an explicit constructor
export	Allows a template definition to be *accessible from another translation unit.*
extern	An identifier specified as **extern** has external linkage to the block
false	Boolean literal of value zero
float	Fundamental data type used to define a floating-point number
for	Indicates the start of a **for** statement to achieve repetitive control
friend	A **class** or operation whose implementation can access the **private** data members of a **class**
goto	Transfer control to a specified label
if	Indicate start of an **if** statement to achieve selective control
inline	A function specifier that indicates to the compiler that inline substitution of the function body is to be preferred to the usual function call implementation

int	Fundamental data type used to define integer objects
long	A data type modifier that defines a 32-bit **int** or an extended **double**
mutable	Allows an object member to override **const**ness
namespace	Defines a scope
new	Memory allocation operator
operator	Overloads a C++ operator with a new declaration
private	Declares **class** members which are not visible outside the **class**
protected	Declares **class** members which are **private** except to derived classes
public	Declares **class** members which are visible outside the **class**
register	A storage class specifier that is an **auto** specifier, but which also indicates to the compiler that an object will be frequently used and should therefore be kept in a register
return	Returns an object to a function's caller
short	A data type modifier that defines a 16-bit **int** number
signed	A data type modifier that indicates an object's sign is to be stored in the high-order bit
sizeof	Returns the size of an object in bytes
static	The lifetime of an object defined **static** exists throughout the lifetime of program execution
struct	To declare new types that encapsulate both data and member functions
switch	Switch statement
template	Parametrised or generic type
this	A **class** pointer which points to an object or instance of the **class**
throw	Generate an exception
true	Boolean literal of value one
try	Indicates start of a block of exception handlers
typedef	Synonym for another integral or user-defined type
typeid	The **typeid** *( )* operator returns the type of its operand
typename	Within a **template** **typename** indicates that a qualified name denotes a type
union	Similar to a structure, **struct**, in that it can hold different types of data, but a union can hold only one of its members at a given time.
unsigned	A data type modifier that indicates the high-order bit is to be used for an object
using	**using** declaration and **using** directive
virtual	A function specifier that declares a member function of a **class** which will be redefined by a derived **class**
void	Absent of a type or function parameter list
volatile	Define an object which may vary in value in a way that is undetectable to the compiler
wchar_t	Wide character type
while	Start of a **while** statement and end of a **do-while** statement

Appendix B

ASCII Character Set

It appears customary to include the ASCII character set in an Appendix of a book on computer programming, so here it is[1]:

Decimal	*Hexadecimal*	*Octal*	*Binary*	*Key*	*ASCII character*
0	00	0	00000000	CTRL+2	null
1	01	1	00000001	CTRL+A	☺
2	02	2	00000010	CTRL+B	☻
3	03	3	00000011	CTRL+C	♡
4	04	4	00000100	CTRL+D	♢
5	05	5	00000101	CTRL+E	♧
6	06	6	00000110	CTRL+F	♤
7	07	7	00000111	beep	•
8	08	10	00001000	backspace	◘
9	09	11	00001001	tab	
10	0a	12	00001010	newline	■
11	0b	13	00001011	CTRL+K	♂
12	0c	14	00001100	CTRL+L	♀
13	0d	15	00001101	enter	♪
14	0e	16	00001110	CTRL+N	♫
15	0f	17	00001111	CTRL+O	¤
16	10	20	00010000	CTRL+P	.
17	11	21	00010001	CTRL+Q	,
18	12	22	00010010	CTRL+R	↕
19	13	23	00010011	CTRL+S	!!
20	14	24	00010100	CTRL+T	¶
21	15	25	00010101	CTRL+U	§
22	16	26	00010110	CTRL+V	■
23	17	27	00010111	CTRL+W	↨
24	18	30	00011000	CTRL+X	↑
25	19	31	00011001	CTRL+Y	↓
26	1a	32	00011010	CTRL+Z	→
27	1b	33	00011011	esc	←
28	1c	34	00011100	CTRL+\	@
29	1d	35	00011101	CTRL+]	↔

1 Note that characters 0–31 (inclusive) and 127 are control characters, 32–126 represent keys on the keyboard and 127–255 are the IBM extended ASCII character set, which can be displayed by pressing the ALT key and typing the decimal code of the character on the numeric keypad. The IBM extended character set is not part of the ASCII character set and as a result may not be fully portable.

Decimal	*Hexadecimal*	*Octal*	*Binary*	*Key*	*ASCII character*
30	1e	36	00011110	CTRL+6	▲
31	1f	37	00011111	CTRL+-	▼
32	20	40	00100000	spacebar	
33	21	41	00100001	!	!
34	22	42	00100010	"	"
35	23	43	00100011	#	#
36	24	44	00100100	$	$
37	25	45	00100101	%	%
38	26	46	00100110	&	&
39	27	47	00100111	'	'
40	28	50	00101000	(	(
41	29	51	00101001	)	)
42	2a	52	00101010	*	*
43	2b	53	00101011	+	+
44	2c	54	00101100	,	,
45	2d	55	00101101	-	-
46	2e	56	00101110	.	.
47	2f	57	00101111	/	/
48	30	60	00110000	0	0
49	31	61	00110001	1	1
50	32	62	00110010	2	2
51	33	63	00110011	3	3
52	34	64	00110100	4	4
53	35	65	00110101	5	5
54	36	66	00110110	6	6
55	37	67	00110111	7	7
56	38	70	00111000	8	8
57	39	71	00111001	9	9
58	3a	72	00111010	:	:
59	3b	73	00111011	;	;
60	3c	74	00111100	<	<
61	3d	75	00111101	=	=
62	3e	76	00111110	>	>
63	3f	77	00111111	?	?
64	40	100	01000000	@	@
65	41	101	01000001	A	A
66	42	102	01000010	B	B
67	43	103	01000011	C	C
68	44	104	01000100	D	D
69	45	105	01000101	E	E
70	46	106	01000110	F	F
71	47	107	01000111	G	G
72	48	110	01001000	H	H
73	49	111	01001001	I	I
74	4a	112	01001010	J	J
75	4b	113	01001011	K	K
76	4c	114	01001100	L	L
77	4d	115	01001101	M	M
78	4e	116	01001110	N	N
79	4f	117	01001111	O	O
80	50	120	01010000	P	P
81	51	121	01010001	Q	Q
82	52	122	01010010	R	R
83	53	123	01010011	S	S
84	54	124	01010100	T	T
85	55	125	01010101	U	U
86	56	126	01010110	V	V

Decimal	*Hexadecimal*	*Octal*	*Binary*	*Key*	*ASCII character*
87	57	127	01010111	W	W
88	58	130	01011000	X	X
89	59	131	01011001	Y	Y
90	5a	132	01011010	Z	Z
91	5b	133	01011011	[	[
92	5c	134	01011100	\	\
93	5d	135	01011101	]	]
94	5e	136	01011110	^	^
95	5f	137	01011111	_	_
96	60	140	01100000	'	'
97	61	141	01100001	a	a
98	62	142	01100010	b	b
99	63	143	01100011	c	c
100	64	144	01100100	d	d
101	65	145	01100101	e	e
102	66	146	01100110	f	f
103	67	147	01100111	g	g
104	68	150	01101000	h	h
105	69	151	01101001	i	i
106	6a	152	01101010	j	j
107	6b	153	01101011	k	k
108	6c	154	01101100	l	l
109	6d	155	01101101	m	m
110	6e	156	01101110	n	n
111	6f	157	01101111	o	o
112	70	160	01110000	p	p
113	71	161	01110001	q	q
114	72	162	01110010	r	r
115	73	163	01110011	s	s
116	74	164	01110100	t	t
117	75	165	01110101	u	u
118	76	166	01110110	v	v
119	77	167	01110111	w	w
120	78	170	01111000	x	x
121	79	171	01111001	y	y
122	7a	172	01111010	z	z
123	7b	173	01111011	{	{
124	7c	174	01111100		
125	7d	175	01111101	}	}
126	7e	176	01111110	~	~
127	7f	177	01111111	CTRL+¬	Δ
128	80	200	10000000	ALT+128	Ç
129	81	201	10000001	ALT+129	ü
130	82	202	10000010	ALT+130	é
131	83	203	10000011	ALT+131	â
132	84	204	10000100	ALT+132	ä
133	85	205	10000101	ALT+133	à
134	86	206	10000110	ALT+134	å
135	87	207	10000111	ALT+135	ç
136	88	210	10001000	ALT+136	ê
137	89	211	10001001	ALT+137	ë
138	8a	212	10001010	ALT+138	è
139	8b	213	10001011	ALT+139	ï
140	8c	214	10001100	ALT+140	î
141	8d	215	10001101	ALT+141	ì
142	8e	216	10001110	ALT+142	Ä
143	8f	217	10001111	ALT+143	Å

Decimal	*Hexadecimal*	*Octal*	*Binary*	*Key*	*ASCII character*
144	90	220	10010000	ALT+144	É
145	91	221	10010001	ALT+145	æ
146	92	222	10010010	ALT+146	Æ
147	93	223	10010011	ALT+147	ô
148	94	224	10010100	ALT+148	ö
149	95	225	10010101	ALT+149	ò
150	96	226	10010110	ALT+150	û
151	97	227	10010111	ALT+151	ù
152	98	230	10011000	ALT+152	
153	99	231	10011001	ALT+153	Ö
154	9a	232	10011010	ALT+154	Ü
155	9b	233	10011011	ALT+155	¢
156	9c	234	10011100	ALT+156	£
157	9d	235	10011101	ALT+157	¥
158	9e	236	10011110	ALT+158	₧
159	9f	237	10011111	ALT+159	ƒ
160	a0	240	10100000	ALT+160	á
161	a1	241	10100001	ALT+161	í
162	a2	242	10100010	ALT+162	ó
163	a3	243	10100011	ALT+163	ú
164	a4	244	10100100	ALT+164	ñ
165	a5	245	10100101	ALT+165	Ñ
166	a6	246	10100110	ALT+166	a
167	a7	247	10100111	ALT+167	o
168	a8	250	10101000	ALT+168	¿
169	a9	251	10101001	ALT+169	⌐
170	aa	252	10101010	ALT+170	¬
171	ab	253	10101011	ALT+171	½
172	ac	254	10101100	ALT+172	¼
173	ad	255	10101101	ALT+173	¡
174	ae	256	10101110	ALT+174	«
175	af	257	10101111	ALT+175	»
176	b0	260	10110000	ALT+176	░
177	b1	261	10110001	ALT+177	▒
178	b2	262	10110010	ALT+178	▓
179	b3	263	10110011	ALT+179	│
180	b4	264	10110100	ALT+180	┤
181	b5	265	10110101	ALT+181	╡
182	b6	266	10110110	ALT+182	╢
183	b7	267	10110111	ALT+183	╖
184	b8	270	10111000	ALT+184	╕
185	b9	271	10111001	ALT+185	╣
186	ba	272	10111010	ALT+186	║
187	bb	273	10111011	ALT+187	╗
188	bc	274	10111100	ALT+188	╝
189	bd	275	10111101	ALT+189	╜
190	be	276	10111110	ALT+190	╛
191	bf	277	10111111	ALT+191	┐
192	c0	300	11000000	ALT+192	└
193	c1	301	11000001	ALT+193	┴
194	c2	302	11000010	ALT+194	┬
195	c3	303	11000011	ALT+195	├
196	c4	304	11000100	ALT+196	─
197	c5	305	11000101	ALT+197	┼
198	c6	306	11000110	ALT+198	╞
199	c7	307	11000111	ALT+199	╟
200	c8	310	11001000	ALT+200	╚

Decimal	*Hexadecimal*	*Octal*	*Binary*	*Key*	*ASCII character*
201	c9	311	11001001	ALT+201	╔
202	ca	312	11001010	ALT+202	╩
203	cb	313	11001011	ALT+203	╦
204	cc	314	11001100	ALT+204	╠
205	cd	315	11001101	ALT+205	═
206	ce	316	11001110	ALT+206	╬
207	cf	317	11001111	ALT+207	╧
208	d0	320	11010000	ALT+208	╨
209	d1	321	11010001	ALT+209	╤
210	d2	322	11010010	ALT+210	╥
211	d3	323	11010011	ALT+211	╙
212	d4	324	11010100	ALT+212	╘
213	d5	325	11010101	ALT+213	╒
214	d6	326	11010110	ALT+214	╓
215	d7	327	11010111	ALT+215	╫
216	d8	330	11011000	ALT+216	╪
217	d9	331	11011001	ALT+217	┘
218	da	332	11011010	ALT+218	┌
219	db	333	11011011	ALT+219	█
220	dc	334	11011100	ALT+220	▄
221	dd	335	11011101	ALT+221	▌
222	de	336	11011110	ALT+222	▐
223	df	337	11011111	ALT+223	▀
224	e0	340	11100000	ALT+224	α
225	e1	341	11100001	ALT+225	ß
226	e2	342	11100010	ALT+226	Γ
227	e3	343	11100011	ALT+227	π
228	e4	344	11100100	ALT+228	∑
229	e5	345	11100101	ALT+229	σ
230	e6	346	11100110	ALT+230	µ
231	e7	347	11100111	ALT+231	τ
232	e8	350	11101000	ALT+232	Φ
233	e9	351	11101001	ALT+233	Θ
234	e9	351	11101010	ALT+234	Ω
235	eb	353	11101011	ALT+235	δ
236	ec	354	11101100	ALT+236	∞
237	ed	355	11101101	ALT+237	ø
238	ee	356	11101110	ALT+238	ε
239	ef	357	11101111	ALT+239	∩
240	f0	360	11110000	ALT+240	≡
241	f1	361	11110001	ALT+241	±
242	f2	362	11110010	ALT+242	≥
243	f3	363	11110011	ALT+243	≤
244	f4	364	11110100	ALT+244	⌠
245	f5	365	11110101	ALT+245	⌡
246	f6	366	11110110	ALT+246	÷
247	f7	367	11110111	ALT+247	≈
248	f8	370	11111000	ALT+248	°
249	f9	371	11111001	ALT+249	•
250	fa	372	11111010	ALT+250	·
251	fb	373	11111011	ALT+251	√
252	fc	374	11111100	ALT+252	^{n}
253	fd	375	11111101	ALT+253	^{2}
254	fe	376	11111110	ALT+254	■
255	ff	377	11111111	ALT+255	blank

Appendix **C**

Operators: Precedence, Associativity and Arity

The following list shows all of the operators in C++. The precedence, associativity and arity of each operator is given. Note that **`sizeof`**, **`new`** and **`delete`** are operators. All of the operators listed can be overloaded except for `::`, `.`, **`sizeof`**, `.*` and `?:`.

Key: L=Left, R=Right, U=Unary, B=Binary, T=Ternary, N/A=Not applicable

Operator	*Precedence*	*Associativity*	*Arity*	*Description*
`::`	17	R	U	Global scope
`::`		L	B	Class scope
`()`	16	L	N/A	Function call
`()`		L	N/A	Type construction
`[]`		L	B	Array index
`->`		L	B	Indirect member access
`.`		L	B	Direct member access
`sizeof`	15	R	U	Object size in bytes
`new`		R	U	Memory allocator
`delete`		R	U	Memory deallocator
`()`		R	B	cast
`-`		R	U	Minus sign
`+`		R	U	Plus sign
`*`		R	U	Pointer dereference
`&`		R	U	reference
`!`		R	U	Logical NOT
`~`		R	U	Bitwise NOT
`++`		R	U	increment
`--`		R	U	decrement
`.*`	14	L	B	Direct member pointer access
`->*`		L	B	Indirect member pointer access
`/`	13	L	B	division
`*`		L	B	multiplication
`%`		L	B	modulus
`+`	12	L	B	addition
`-`		L	B	subtraction
`<<`	11	L	B	Bit-shift left

Operator	*Precedence*	*Associativity*	*Arity*	*Description*
>>		L	B	Bit-shift right
<	10	L	B	Less than
>		L	B	Greater than
<=		L	B	Less than or equal to
>=		L	B	Greater than or equal to
==	9	L	B	equality
!=		L	B	inequality
&	8	L	B	Bitwise AND
^	7	L	B	Bitwise XOR
|	6	L	B	Bitwise OR
&&	5	L	B	Logical AND
||	4	L	B	Logical OR
?:	3	L	T	conditional
=	2	R	B	assignment
+=		R	B	Addition and assignment
-=		R	B	Subtraction and assignment
/=		R	B	Division and assignment
*=		R	B	Multiplication and assignment
%=		R	B	Modulus and assignment
&=		R	B	Bitwise AND and assignment
|=		R	B	Bitwise OR and assignment
^=		R	B	Bitwise XOR and assignment
<<=		R	B	Bit-shift left and assignment
>>=		R	B	Bit-shift right and assignment
,	1	L	B	Comma (sequencing)

Appendix D

Glossary

abstraction	The essential characteristics of an object which distinguish it from other objects.
array	Group of identical data types or elements.
base class	A **`class`** from which another class can inherit. Also known as a superclass.
character	Such as /, *, ?, a, A, b, ... Refer to Appendix B for a complete listing of the ASCII character set.
`class`	A **`class`** encapsulates data and functions which operate on the **`class`**'s data, defining a given structure and behaviour. A **`class`** is a type and a scope with an exact specification specified in the **`class`** declaration.
comment	A user-defined string of information placed within program code that does not influence compilation.
compile	Convert a program source code file in to an object file which contains machine language instructions.
concrete class	A **`class`** that possesses the functionality of C++'s built-in data types, such as **`int`** and **`double`**.
concurrency	The action of different objects operating simultaneously.
constant	Something that does not change.
declaration	Specifies an identifier's name and type.
definition	A declaration that allocates memory.
derived	A **`class`** that has inherited from one or more alternative classes. Also known as a subclass.
encapsulation	The formulation of separating the structure and behaviour of an abstraction into cells.
`enum`	An enumeration data type which provides mnemonic identifiers to a set of integer constants.

exception	A situation in which a program encounters an abnormal situation for which it was not designed. In C++ you can transfer control (**`throw`** an exception) to another part of a program that is designed to deal explicitly with exceptions.
execution	The process of running a program.
`friend`	A **`class`** or function that has access to the **`private`** sections of another **`class`**.
function	A standalone module that can have input or output or both.
identifier	A name given to either a constant or a variable object.
inheritance	A hierarchy among classes.
instance	An instance does things, and we can do things to an instance by sending messages. Messages are sent via member functions. An instance is also referred to as an *object*.
interface	A set of functions within a module.
keyword	Such as **`int`**, **`double`** or **`new`**. A word that the compiler already has a definition for. A keyword (or reserved word) cannot be used as the name of a data object. Keywords in C++ are in lowercase. For a list of keywords refer to Appendix A.
link	*Linking* is the process of combining object files into a single executable program.
make	Combined process of compiling and linking to generate an executable program.
member function	An operation on an object. Member functions are called *methods* or *operations* in other object-oriented programming languages.
module	A piece of program code which forms part of a larger software system. A module has a clearly defined interface. A source file is an example of a module.
object	An object does things, and we can do things to an object by sending messages. Messages are sent via member functions. An object is also referred to as an *instance*.
persistence	An object is referred to as persistent if it outlives its creator or program execution and/or an object survives a move from its original memory address.
polymorphism	A name that is attached to a variable which denotes objects of several different classes that are related through a common base **`class`**. Polymorphous – assuming many forms.
pragmatics	Rules for the good and proper use of a language.
preprocessor	The compiler's preprocessor. Preprocessor directives are prefixed with the number (or hash) sign, #.
primitives	Primary subsystems of a complex system such as base classes.

private	If a member of **class** is **private** it can only be accessed by member functions and **friend**s of the **class** in which it is declared.
protected	If a member of a **class** is **protected** it can only be accessed by member functions and **friend**s of the **class** in which it is declared or a **class** that is *derived* from this **class**.
public	If a member of a **class** is **public** it can be accessed by any function, member or not.
run-time type information	A mechanism which allows information to be obtained about an object at run-time.
scope	The visibility, availability and life of an object within a program. There are generally six different types of scope: global, local, **class**, function, **namespace** and file.
semantics	The meaning of syntactically correct constructs of a language.
source file	A file containing declarations, functions and objects. Frequently referred to as a translation unit.
STL	The C++ Standard Template Library.
structure	A user-defined declaration that encapsulates data members and member functions or operations.
syntax	The set of rules for the correct use of a language.
template	A generic **class**. Often referred to as a *parametrised type.*
translation unit	A module.
type	A **class** implements a type. A type determines what values and operations can be performed on an object. All names or objects must have a type in C++.
type checking	All objects are created with a specific type, such as **int**, **double** and `Complex*`. Type checking involves checking that the use of an object is consistent with its type. C++ is often referred to as *strongly typed.* If violations of type are detected at compilation, a language is referred to as *strongly statically typed.* Alternatively, if violations of type are allowed to pass compilation and type checking is postponed until run-time then a language is referred to as *dynamically typed.*
union	Similar to a *structure*, with the restriction that a **union** can hold only one of its members at any given time.
variable	As the name suggests, something that changes. The name of a storage location that depends on the type and size of the data that it contains.
virtual function	A base **class** member function that can be redefined by each derived **class**.

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Conclusion

computers – they'll never catch on
A. Broome, 1995

Although we are concluding our tour of the C++ programming language, I feel as though I've just begun describing C++ because of its wealth of features. Do remember that this text has been an introductory text to C++, with applications specifically oriented to computer graphics. C++ is a big programming language and has numerous applications other than graphical.

If you enjoy the C++ programming language then do continue to develop your understanding of the language and programming skills. A first step in moving from an introductory text is to read slightly more technical books on C++ and books on object-oriented programming, such as Coplien (1992). With reasonable object-oriented programming skills the next step is to begin developing better object-oriented design and analysis skills. A good first text for the C++ programmer which examines the design and analysis issues of object-oriented programming is the book *Object-Oriented Analysis and Design with Applications* (Booch, 1994). As your confidence in the object-oriented approach increases be prepared to learn other object-oriented programming languages. It is often very informative to see how programmer's tackle a given problem slightly differently with a different programming language and environment. If you do intend learning other object-oriented programming languages then can I recommend Smalltalk, Java and Python because of their similarity to C++. *Object-Oriented Programming* (Coad and Nicola, 1993) presents a fun-to-read introduction and comparison of object-oriented programming in Smalltalk and C++. Budd (1987) and Anuff (1996) provide a very good introduction to programming in Smalltalk and Java respectively. The Java language is derived from C++, but is simpler due to the removal of several C++ features, such as pointers, multiple inheritance and operator overloading. Java adds a slightly different syntax to C++ and the capabilities of automatic memory management, multithreading and the generation of architecture-neutral object files, which makes Java ideal for a range of environments. The Eiffel object-oriented programming language is derived from Simula and Ada and supports the majority of object-oriented features. Eiffel is developed by ISE and supports strong static typing and an excellent run-time graphical environment (Meyer, 1991; Thomas and Weedon, 1995). Oberon-2 is aimed at the educational community and offers a good introduction to object-oriented programming and executes on a number of platforms and operating systems. The Oberon-2 object-oriented language and Oberon operating system can be obtained free; for details of how to obtain Oberon-2 refer to Mössenböck (1993). Finally, conference proceedings, journals and the Internet are an excellent source of current thinking and development – a few of these sources are listed in Chapter 1.

Several programmers move to programming in a graphical user interface environment, such as Windows, after learning C++. Windows offers an ideal and challenging environment in which to apply your newly learned programming language. If you are interested in programming for Windows I can recommend the book *Programming Windows 3.1* (Petzold, 1992),

G. M. Seed, *An Introduction to Object-Oriented Programming in C++*
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which although it is written in C rather than C++ is nevertheless an excellent introduction to Windows programming for the C/C++ programmer. Also, *Borland C++ 4.0 Programming for Windows* (Yao, 1994) is a very enjoyable introduction to Windows programming using the Borland ObjectWindows library. Borland are also developers of the C++Builder Rapid Application Development (RAD) environment for rapid visual development of Windows GUI applications; refer to Calvert (1997) and Reisdorph and Henderson (1997). Compilers such as Borland C++ and Microsoft Visual C++ for Windows both include comprehensive class libraries (Borland's ObjectWindows library and Microsoft's foundation class library) specifically designed for Windows programming.

An increasingly important emerging technology is *Component Software*. A component is a unit of independent depolyment which has no persistent state. A collection of classes or a framework of classes are examples of components. If you have experience of the Java programming language then you may of observed that Java separates collections of classes (modules or packages) from classes. In addition, Java supports JavaBeans called 'beans' which are Java components (a set of classes and resources). For an excellent introduction to Component Software refer to Szyperski (1998).

All of the programs developed throughout the book have used sequential programming techniques. Parallel programming languages exist which are based on the C++ programming language. For example, the company KAI supports a parallel implementation of the C++ language and further details can be found on the Internet at KAI (2000).

C++ has recently undergone standardisation and as a result will change less frequently than during the pre-standardisation years. If you are new to the C++ language then I certainly recommend referring to the ANSI/ISO C++ standard, details of which can be found at Standard (1998). In addition, I recommend reading *The Draft Standard C++ Library* (Plauger, 1995), which provides a great insight into the evolution of the C++ standard library.

Several classes have been presented throughout the book which will prove useful to C++ programmers, particularly to programmers involved in graphical and geometric modelling. The classes are far from perfect, but should provide a starting block from which to develop your own data structures and geometric **`class`** libraries.

The `Point` **`class`** has played an important role throughout the latter half of the book and is an important **`class`** because of the fundamental importance of a point in three-dimensional space. A possible extension to consider when examining **`class`** `Point` is that inherent in a `Point` object is that the object exists in a Cartesian coordinate system. Alternatively, a separate coordinate system **`class`** hierarchy could be declared and an appropriate coordinate system object encapsulated in **`class`** `Point`. This would then allow a `Point` object to be defined in terms of position and respective coordinate system, which would greatly assist in the manipulation of `Point` objects in different coordinate systems. Similarly, lines and curves are essential to computational geometric modelling, and the `Line` **`class`** could be expanded to a `Line` **`class`** hierarchy which models both explicit and implicit straight and curved line objects. Shape objects, such as polygons or polyhedra, could then be defined with straight or curved edges defined explicitly in terms of a number of points or an equation. In addition, the shapes **`class`** hierarchy could be extended and generalised further to accommodate additional geometric shapes, such as general polyhedra and algebraic surfaces. Recent works which discuss computational geometry and computer graphics in C++ are O'Rourke (1994) and Laszlo (1996). However, if you consider developing your own C++ geometric **`class`** library to be too fundamental an exercise, then an alternative approach would be to consider integrating your program code with an existing geometric and solid modeller kernel such as OpenGL (2000), SVLIS (Bowyer, 1995), or ACIS (Corney, 1997), which are all written in C++.